

Public Transit Planning and Operation

Modeling, Practice and Behavior

Second Edition

Avishai Ceder



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Avishai Ceder

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and University of Auckland,
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To my late Dad, Samuel (who worked for a large bus agency as a driver and treasurer for over 30 years), and to my Mom, Anna, with wishes for many good years.

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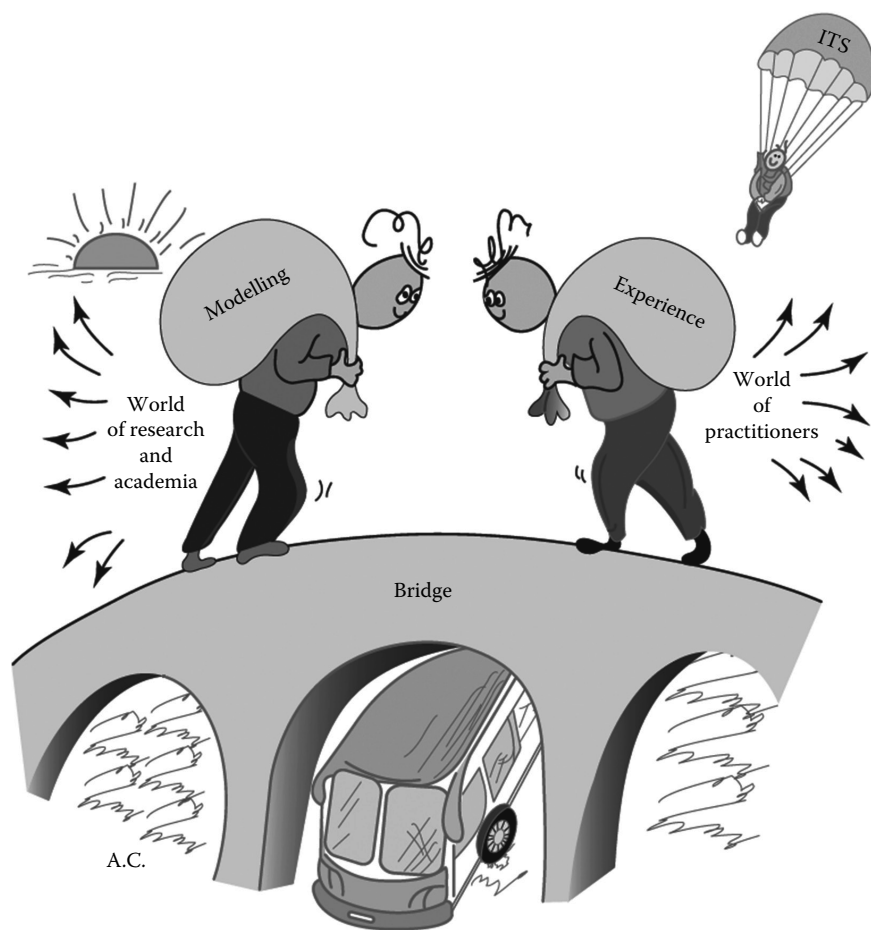
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Preface



PERSONAL MOTIVATION

The following story may serve to help understand the stimulus behind the first and this second edition of the book: A ship is sailing through a stormy ocean, and a little girl, who happens to be the captain’s daughter, is playing on the deck when all of a sudden a large wave carries her overboard into the sea. The captain, who sees this from his post, right away

orders his sailors to jump into the ocean and save the girl, but none dares for fear of risking his life. Desperately the captain turns to the passengers and asks them for help while promising that whoever saves his daughter will get anything he wants as a reward. Again, no one reacts. But then suddenly a man with a long beard who has stood by the railing lurches overboard into the sea. The sailors throw him a life preserver, and fortunately he manages to lift the little girl safely back on deck and into the arms of her father. The captain then hugs the man, who is thoroughly drenched, and says he will give him anything he wants, just name it. The hero's response: "I don't want anything. I just want to know who pushed me. ..."

What pushed me actually started between 1967 and 1971, when I was a bus driver for Egged, whose 4000 buses make it one of the largest bus companies in the world. Before gaining a bus driver's license, I had a theory about the way one should drive a bus; now I have a bus driver's license—and no theory. The second motivation for writing this book came from my consulting work at Egged from 1975 to 1985, when I was exposed daily to planning and operational problems in the public transit field. The third motivation came in 1981. I was at MIT in Boston, where, together with Professor Nigel Wilson, I was to give a new summer session course on transit operations planning. This course became an annual offering until 2003. From 2006 to date, I am delivering two short transit courses together with Professor Graham Currie of Monash University in Melbourne.

From all these foregoing activities, I internalized the following realization: experience is what you get when you are expecting something else. My teaching of transit operations planning has taken place at universities in Adelaide, Auckland, Boston, Brisbane, California, Hong Kong, Israel, Melbourne, Rome, Perth, Sydney, and Wellington. Indeed, it has been my more than 40 years of teaching, research, and hands-on experience that has pushed me to write the first and this second edition of the book and to construct it in such a way that it will help not only teachers, researchers, and students in the area, but also practitioners in the field.

PURPOSE AND INTENDED AUDIENCE

This second edition of the book is constructed from two-thirds of the first edition, published in 2007; one-third comprises updated and new material. This second edition uses the concise term *transit* to refer to public transit or public transportation or public transport, all three of which are in common use.

A major goal of this volume is to establish a sort of a bridge between the world of practitioners and the world of research and academia to narrow the gap between them. We hope that such a bridge will also lead to opportunities for collaboration and interaction in order to improve public transit service and, no less important, its image. Henry Ford once said, "Failure is only the opportunity to start all over again more intelligently." With this in mind, the book intends to introduce a few new ways of thinking about (a) already implemented and investigated transit themes while combining retrospective thoughts and cumulative experience and (b) new concepts and ideas.

One of the main features of the second edition of the book is its stand-alone (self-contained) capability, obviating the need to look back and forth at other publications for comprehending the text. At the same time, every chapter contains a "Literature Review and Further Reading" list. Practitioners may have some difficulty in comprehending the sections with mathematical notation, but hopefully they can grasp the substance of the material and its practical implications. Researchers and academic professionals may find some of the sections unnecessarily detailed, but they should be aware that the book is also aimed at

practitioners and undergraduate students, thus requiring more explanation. This work, in sum, follows the notion that (a) it is better to ask twice than to lose your way once and (b) clarity is no less important than certainty.

ORGANIZATION

Each chapter opens with a section containing information and remarks for practitioners, called “Practitioner’s Corner.” In fact, one can never tell which way the train went by looking at the track: for a practical decision, one needs more. The purpose of these “corners” is to impart guidance about sections of the chapter that are appropriate and sections that are perhaps too difficult for practitioners so as to allow the less academically inclined to flow with the book and capture its substance.

The organization of the book is described in Chapter 1. Generally speaking, five themes are addressed:

- Overview of transit planning and data collection needs (Chapters 1 and 2)
- Design and optimization of transit timetables and of vehicle and crew scheduling (Chapters 3 through 9)
- Passenger demand and assignment analysis (Chapters 10 and 11)
- Transit service, network, and route design, encompassing scheduling and behavioral elements (Chapters 12 through 17)
- Transit reliability, operational strategies, and future operations planning developments (Chapters 18 through 20)

This second edition includes significantly revised chapters (Chapters 5, 8, 12, and 15) and completely new chapters (Chapters 13, 14, and 19).

All the quantitative chapters offer exercises for practicing the methods covered; of the book’s 20 chapters, only Chapters 1, 2, and 20 are without exercises. The answers to these exercises appear at the end of the book.

The literature review of papers relevant to the topic(s) covered in a chapter appears as the last numbered section of each chapter, except for Chapters 1 and 7 (the review for which actually precedes it, in Chapter 6). The reason for this order, instead of the traditional pattern of starting a scientific article with a literature review, is to focus on each chapter’s essence from the beginning and only at the end to give the reader who may wish to broaden his or her knowledge of the particular topic a kind of annotated reference list and an extended bibliography.

REMARKS

The success of a professional book can be evaluated by the extent to which it succeeds in introducing new and improved ideas and methods. It is not only a matter of learning the book’s content; it has to do, as well, with how much the volume can inspire the reader’s imagination to think further. That concept has served as the guideline for the development of the second edition of this book.

Finally, when lecturing this transit course, I tended to use humor at times because I believe in the insight captured by the English playwright George Bernard Shaw, who once said, “When a thing is funny, search it carefully for a hidden truth.” More than once I have been

asked to employ some of this humor (including the cartoons that I have also drawn) if I ever wrote a book. I have done this to some extent, especially in the Practitioner's Corners.

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Acknowledgments

First, I would like to repeat my thanks to everyone who contributed to the first edition of this book from 2007. After all, this second edition is constructed from two-thirds of the first edition, published in 2007; one-third comprises updated and new material.

Second and foremost, many thanks are extended to both my former and my current PhD students at the University of Auckland. Over the last seven years, I have shared with these students the exploration and search for new research avenues with innovative and novel concepts and ideas. My appreciation and thanks to Dr. Subeh Chowdhury (now a staff member at the University of Auckland) for her thorough study of transit-user behavior at transfer points; Dr. Stephan Hassold (an OR analyst at Air New Zealand) for his outstanding research on timetabling and vehicle scheduling with different vehicle sizes; and Supun Perera (of Sydney, Australia), who contributed to the updated literature review and with whom I had many good and interesting conversations. My gratitude and appreciation are also extended to three of my current doctoral students for their contributions: Tao Liu, for his enthusiasm and good research on transit real-time control; Mahmood Mahmoodi Nesheli, for his hard work and advanced study on transit modeling of operational tactics; and Mahdi Amiripour, for his great interest, devotion, and solid research on transit network design problems.

Last, I offer my heartfelt thanks to my wife, Selina (Lan Shi), for her great encouragement, love, and understanding. Finally, a bouquet of affections goes to my three sons and daughters-in-law, Roy + Roni, Ohad + Liat, and Dror + Valerie, and to my seven grandchildren, Eran, Elad, Yael, Yahli, Naama, Gefen, and Jonathan, as well as to my Mom, Anna, and my brothers, Tuli and Hagai, and their families—all of whom helped me get through difficult periods when writing this second edition of the book.

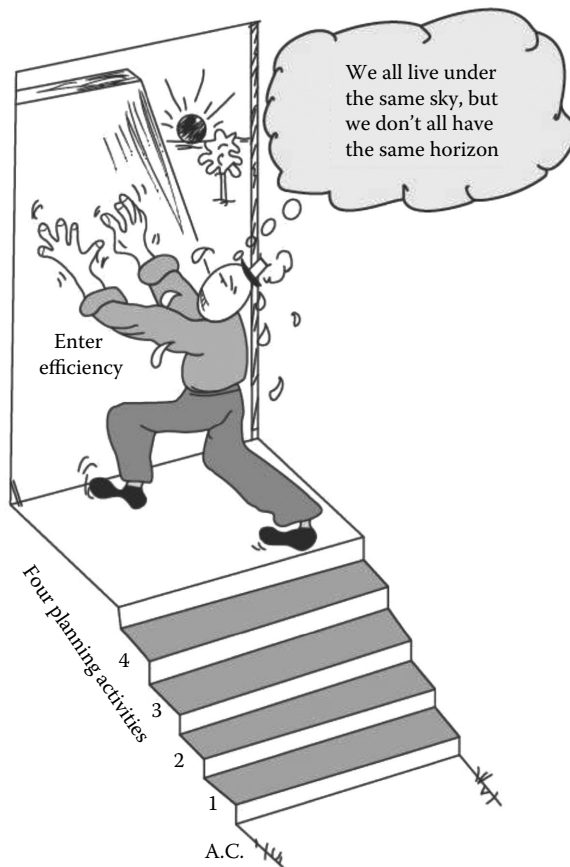
I retain, of course, sole responsibility for any errors. I would be very pleased to receive feedback.

Author



Avishai (Avi) Ceder received his PhD in 1975 at the University of California at Berkeley, and he is a professor and chair in transportation at the University of Auckland, New Zealand, and at the Technion–Israel Institute of Technology. He frequently delivers courses on public transit operations planning in Asia, Europe, Australasia, and the United States. His research and implementation interests in the subject stem from being a bus driver for four years for a large bus company.

Introduction to transit service planning



CHAPTER OUTLINE

- 1.1 Motivation
- 1.2 Operational planning decomposition process
- 1.3 Service and evaluation standards and their derivatives
- 1.4 Viability perspectives
- 1.5 Outline of other chapters
- References

PRACTITIONER'S CORNER

As stated in “Preface,” each chapter opens with information and notes to guide practitioners. This introductory chapter will show that a successful transit service is not permanent and that failure in a new service need not be fatal. The chapter contains neither mathematical formulations nor a complicated literature review.

The motivation behind the writing of this book is to bridge the world of the researchers and the world of the practitioners. The chapter first presents a real-life description of the planning process for transit operations. It then provides an overview and a critique of currently used standards and guidelines for transit service, pointing out those standards that warrant research and those for which administrative decisions are sufficient. The main components affecting the viability of a transit service from both the passenger and agency perspectives are described, as is the link between some problematic elements and the possible solutions advanced in the book.

The chapter ends with an outline of the other chapters and the links between each chapter and the main core of the transit operations planning process. Our attempt to show how to deal with and to solve in a practical manner current transit service dilemmas and confusions follows Pablo Picasso, who said: “I do not seek, I find.”

1.1 MOTIVATION

A basic premise of this volume is that using transit service should be like eating potato chips—once you start, it’s hard to stop. The fundamental question, though, is how do we design and approach such a service while knowing, because of the competition offered by cars, that an adequate transit design must provide superlative service in order to be good enough. This second edition of the book, following its first edition (Ceder, 2007), will demonstrate methods and procedures not only for improving transit design through the provision of an attractive, viable service but also for reducing its cost and increasing efficiency from the transit agency perspective.

One of the main motivations for writing the first edition of this book (Ceder, 2007) is the persistent use by transit agencies (especially in the United States) of the Bus Scheduling Manual of August 1947 (!), which was reprinted in 1982 (Rainville, 1947). The manual opens as follows: “This volume is viewed today [i.e., 1982—A.C.] as a classic reference on manual scheduling practices. It represents a comprehensive effort made over 30 years ago to pull together the state of the practice in this field, and many of those practices still remain valid today.” The manual’s cover and an inside picture of a device to measure distances appear in Figure 1.1. Those transit agencies that may still be using this manual might be asked a question: Is the treatment of passenger loads and running-time data, frequency determination, and vehicle and crew scheduling really *classic*? Or is it obsolete? The conclusion implied by the latter served as a trigger to investigate why the treatment of these elements in the manual is not classic and to submit operations planning options that exploit existing, computerized computation power and advanced modeling. On the positive side, it may be said, as it was in the 1947 manual, that some basic concepts behind the planning process for transit operations do indeed remain the same.

In 1998, the U.S. Transit Cooperative Research Program (TCRP) sponsored Report 30 (Pine et al., 1998) on transit scheduling with the idea presumably of advancing knowledge in the field for training purposes. However, Report 30 neither contains nor refers to any research on transit scheduling (despite the many studies that existed); furthermore, it does not mention a

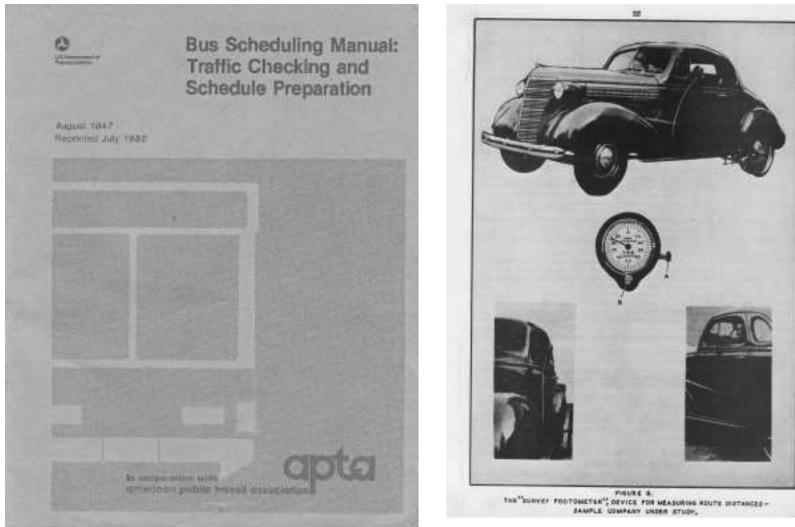


Figure 1.1 Cover and a page of the well-known 1947 *Bus Scheduling Manual*, wrongly viewed as a classic reference.

single reference. The report actually provides very basic training material for new schedulers; in concept, it is almost a replica of the 1947 manual, while emphasizing that scheduling is a craft.

All in all, the following story may sum up the motivation for writing this book: two marketing people from a European shoe factory are sent to investigate the potential sale of shoes in Africa. A few days later, faxes from each of them arrive at the manager's office. The first fax reads: "No chances, everyone is barefoot." The second fax urges: "Lots of chances, everyone is barefoot." This book will adopt the *lots of chances* view.

Another stimulus for the first edition of this book (Ceder, 2007) is a long list of standards and guidelines (see Section 1.3) for transit service planning, operations, and control that determine the quality of service seen in practice. In the event that current transit service quality has reached a satisfactory level, then these standards and guidelines are apparently effective. Nonetheless, many aspects of transit service can stand improvement, necessitating changes or replacement of some of these standards and guidelines. Like those in the 1947 manual. We can imagine that the majority of the standards and guidelines to be covered in the present volume are intuitive based or administrative based instead of being research based; hence, an opportunity is presented to develop new research tools and methods for improving some of these standards and guidelines (as discussed in Section 1.3).

This book uses the term *transit* as shorthand for public transit or public transportation or public transport, all in common use. We describe in this book transit operations planning problems, a summary of different approaches that have been proposed for their solution, and a description of new methods and approaches incorporating some of the positive aspects of prior work. In essence, the new methods and approaches are all applicable to the four major modes of transit operation: airline, railway, bus, and passenger ferry. Although the operations planning process for bus, railway, and passenger ferry is similar, that for airline has several special features, but these are not covered in this book. Moreover, railway and passenger ferry can be perceived, to some extent, as special cases of bus service because of the variety of activities and problems offered by the latter. Finally, the methods and approaches proposed are intended to be easier to implement and more sensitive to the risks of making changes than are those currently in use by practitioners and researchers.

1.2 OPERATIONAL PLANNING DECOMPOSITION PROCESS

The transit operations planning process commonly includes four basic activities, usually performed in sequence: (1) network route design, (2) timetable development, (3) vehicle scheduling, and (4) crew scheduling (Ceder and Wilson, 1986; Ceder, 2001, 2002, 2007). Figure 1.2 shows the systematic decision sequence of these four planning activities. The output of each activity positioned higher in the sequence becomes an important input for lower-level decisions. Clearly, the independence and orderliness of the separate activities exist only in the diagram; that is, decisions made further down the sequence will have some effect on higher-level decisions. It is desirable, therefore, that all four activities be planned simultaneously in order to exploit the system's capability to the greatest extent and maximize the system's productivity and efficiency. Occasionally, the sequence in Figure 1.2 is repeated; the required feedback is incorporated over time. However, since this planning process, especially for medium to large fleet sizes, is extremely cumbersome and complex, it requires separate treatment for each activity, with the outcome of one fed as an input to the next.

The quantitative treatment of the transit planning process is reflected in the welter of professional papers on these topics and in the development of numerous computer programs to automate (at least partially) these activities. In the last 30 years, a considerable amount of effort has been invested in the computerization of the four planning activities (routing, timetabling, vehicle scheduling, crew scheduling) in order to provide more efficient, controllable, and responsive schedules. The best summary of this effort, as well as of the knowledge accumulated, was presented at the 2nd through 11th international workshops on Computer-Aided Scheduling of Public Transport (CASPT); recently, this abbreviation was changed to Conference of Advanced Systems of Public Transport. The research presented at CASPT appears in the book by Ceder (2007) and the books edited by Wren (1981), Rousseau (1985), Daduna and Wren (1988), Desrochers and Rousseau (1992), Daduna et al. (1995), Wilson (1999), Voss and Daduna (2001), and Hickman et al. (2008); the research of the 10th (Leeds, United Kingdom, 2007), the 11th (Hong Kong, China, 2009), and the 12th CASPT (Santiago, Chile, 2012) appear now in proceeding volumes (not books), but the best of them appear/will appear in the new journal: *Public Transport: Planning and Operations* by Springer. To learn more of the CASPT2015 (2015), visit its website: <http://www.caspt.org>. In addition, there is a large number of software for transit scheduling. Some of the known software appears in Table 1.1.

Many planning software packages are based on static data and therefore only offer static schedules. Software needs to be developed in such a way that it includes the flexibility that is required when drawing up and continually modifying service and work schedules. Although many common aspects are involved in workforce planning, every organization has its own planning wishes and requirements. This is why the software needs to be fully customized to the specific business model and company logic. Collective labor agreement rules, legislation relating to working hours, agreed contract hours, and the individual preferences of employees can all be entered into the system. On the basis of advanced algorithms and built-in intelligence, the software is able to create an optimal schedule or support planners in the creation of a schedule. In addition, the software should offer a veritable collection of management information and analyses of deployment, availability, and planning trends. These form an optimal basis for better control of activities and greater efficiency during these activities. Moreover, as a result of its built-in intelligence, the software effortlessly handles multiresource planning, for planning people and resources with the help of a single integrated system.

These software packages concentrate primarily on the activities of vehicle and crew scheduling (activities 3, 4 in Figure 1.2) because, from the agencies' perspective, the largest single cost of providing service is generated by drivers' wages and fringe benefits. Focusing on activities 3 and 4 would seem to be the best way to reduce this cost. However, because some of

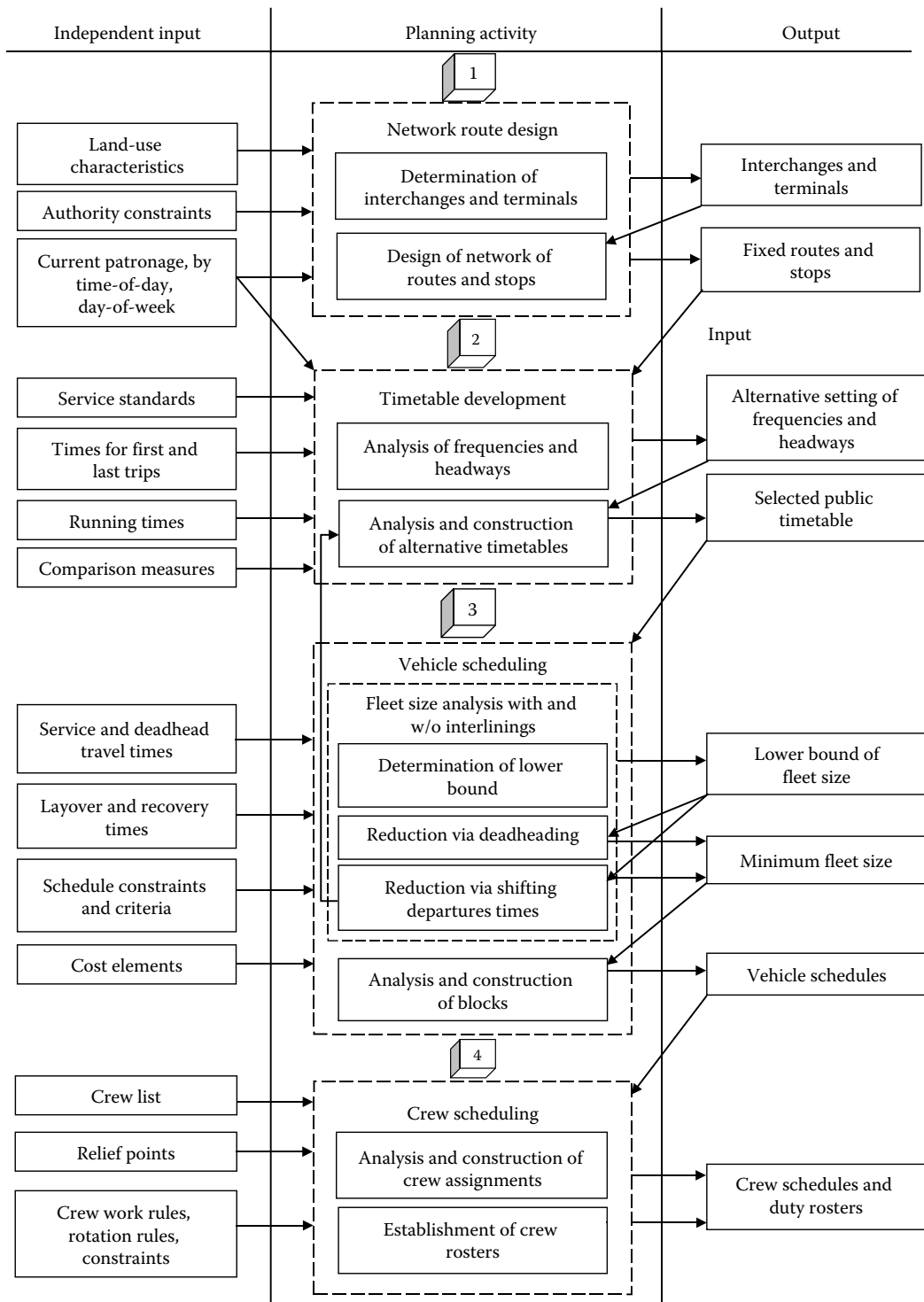


Figure 1.2 Functional diagram (system architecture) of a common transit operations planning process.

Table 1.1 Some known available transit-scheduling software

<i>Software</i>	<i>Internet address</i>	<i>Features</i>
GIRO “HASTUS”	http://www.giro.ca/en	Vehicle assignment Crew assignment
Merakas—“PIKAS”	http://www.merakas.lt/26en/	Timetable calculation Trips assignment Shifts cutting
PTV	http://www.ptv.de	Network planning Simulation
ROUTELOGIC	http://www.routelogic.com/FeaturesScheduling.html	Network planning Vehicle assignment Planning and analysis Operations management Crew assignment
Routematch	http://www.routematch.com/	Network planning Vehicle assignment Planning and analysis Operations management Crew assignment Service management
NEMSYS “Routemate”	http://www.nemsys.it/	Transportation planning Vehicle scheduling
Optibus	www.optibus.co	Real-time rescheduling Vehicle assignment Crew assignment Interactive scheduling Planning and analysis
SYSTRA	http://www.systra.com	Transportation planning Simulation
Tracsis	http://www.tracsis.com/	Demand forecasting Capacity planning Timetable development Rolling stock planning Crew scheduling
Enghouse Transportation “TranSched”	http://www.enghousetransportation.com/	Scheduling Vehicle blocking Optimized runcutting and rostering Workforce management Trip planner
Trapeze	http://www.trapezegroup.com	Network planning Vehicle assignment Planning and analysis Operations management Crew assignment

the scheduling problems in these software packages are oversimplified and decomposed into subproblems, a completely satisfactory or optimal solution is not assured, thus making room for decisions by experienced schedulers. After all, experience is what we gain when expecting something else; said another way, the exam is given first and the lesson after.

An argument in favor of automating activities 3 and 4 is that this scheduling process is extremely cumbersome and time-consuming to undertake manually. In addition to the potential for more efficient schedules, the automated process enables services to be more

controllable and more responsive. The cost and complexity of manual scheduling have served to discourage adjusting activities 1 and 2. Only with automated scheduling methods, which are becoming more widely accepted, it is feasible to focus on higher levels in the planning process. Nonetheless, a case can be made that these higher levels have received short shift by both researchers and practitioners.

The network route-design activity in Figure 1.2 is described in Chapters 12 and 15, which is also linked with Chapters 13 and 14. Planning practice in terms of transit route design focuses almost entirely on individual routes that, for one reason or another, have been identified as candidates for change. Occasionally, sets of interacting (e.g., overlapping or connecting) routes are subject to redesign, usually after a series of incremental changes to individual routes has resulted in a confusing, inefficient local system. Although it is difficult to predict the benefits that will result from redesigning any transit network without conducting a detailed assessment, it is reasonable to believe that they will be large compared with the benefits of additional efforts aimed just at problematic scheduling activities (2, 3, and 4 in Figure 1.2). The approach described mainly in Chapter 15 generates all feasible routes and transfers connecting each place (node) in the network to all others. From this vast pool of possible routes and transfers, smaller subsets are generated that maintain network connectivity. For each subset thus generated, transportation demand is met by calculating the appropriate frequency for each route. Next, prespecified optimization parameters are calculated for each subset. Based on the specific optimization parameter desired by the user, it is then possible to select the most suitable subset. This method has been designed as a tool for the planning of future transit networks as well as for the maintenance of existing networks. The method presented ensures flexibility by allowing the user either to input own data or to run the analysis automatically.

The timetable development activity in Figure 1.2 is described in Chapters 3 through 5 for establishing alternative frequencies and timetables. The aim of public timetables is to meet general public transportation demand. This demand varies during the hours of the day, the days of the week, from one season to another, and even from 1 year to another. It reflects the business, industrial, cultural, educational, social, and recreational transportation needs of the community. The purpose of this activity, then, is to set alternative timetables for each transit route in order to meet variations in public demand. Alternative timetables are determined on the basis of passenger counts, and they must comply with service frequency constraints. In Chapter 4, alternative timetables are constructed with either even headways, but not necessarily even loads on board individual vehicles at the peak-load section, or even average passenger loads on board individual vehicles, but not even headways. Average even loads on individual vehicles can be approached by relaxing the evenly spaced headways pattern (through a rearrangement of departure times). This dynamic behavior can be detected through passenger-load counts and information provided by road supervisors. The key word in the even-load cases is the ability to control the loading instead of being repeatedly exposed to an unreliable service resulting from an imbalance in loading situations.

The vehicle-scheduling activity in Figure 1.2 is described in Chapters 6 through 8 and is aimed at creating chains of trips; each is referred to as a vehicle schedule according to given timetables. This chaining process is often called vehicle blocking (a block is a sequence of revenue and nonrevenue activities for an individual vehicle). A transit trip can be planned either to transport passengers along its route or to make a deadheading trip in order to connect two service trips efficiently. The scheduler's task is to list all daily chains of trips (some deadheading) for each vehicle so as to ensure the fulfillment of both timetable and operator requirements (refueling, maintenance, etc.). The major objective of this activity is to minimize the number of vehicles required (Daduna and Paixao, 1995). Chapters 6 and 7 describe a highly informative graphical technique for the problem of finding the least

number of vehicles. The technique used is a step function, which is introduced as far back as 30 years ago as an optimization tool for minimizing the number of vehicles in a fixed-trip schedule. The step function is termed deficit function, as it represents the deficit number of vehicles required at a particular terminal in a multiterminal transit system. In Chapter 6, the fixed-schedule case is extended to include variable trip schedules, in which given shifting tolerances allow for possible shifts in departure times. This opens up an opportunity to reduce fleet size further. The deficit function, because of its graphical characteristics, has been programmed and is available on a website. In this book, the deficit function is applied and linked to the following activities: vehicle scheduling with different vehicle types (Chapter 8), the design of operational transit parking spaces (Chapter 12), network route design (Chapters 12 and 15), and short-turn design of individual and groups of routes (Chapter 16). The value of embarking on such a technique is to achieve the greatest saving in number of vehicles while complying with passenger demand. This saving is attained through a procedure incorporating a man-computer interface allowing the inclusion of practical considerations that experienced transit schedulers may wish to introduce into the schedule.

The crew scheduling activity in Figure 1.2 is described in Chapter 9. Its goal is to assign drivers according to the outcome of vehicle scheduling. This activity is often called driver-run cutting (splitting and recombining vehicle blocks into legal driver shifts or runs). This crew-assignment process must comply with some constraints, which are usually dependent on a labor contract. A brief summary is given of the conceptual analytical tools used in the modeling and software of this complex, combinatorial problem. The crew-rostering component of this activity normally refers to priority and rotation rules, rest periods, and drivers' preferences. Any transit agency wishing to utilize its resources more efficiently has to deal with problems encountered by the presence of various pay scales (regular, overtime, weekends, etc.) and with human-oriented dissatisfaction. The crew-scheduling activity is very sensitive to both internal and external factors, a factor that could easily lead to an inefficient solution.

In Table 1.2, we list the essential inputs for the transit operations planning process illustrated in Figure 1.2. These mainly independent elements are arranged by activity number. It should be emphasized that their values differ by time of day and day of week.

1.3 SERVICE AND EVALUATION STANDARDS AND THEIR DERIVATIVES

Tremendous and monotonous in transit service can be perceived as synonymous. It is best to have a monotonous, automated transit system that will always be there for the passenger, for example, a modularized personal rapid transit system conducted like a horizontal elevator. This observation serves as an introduction to the need for transit service standards and guidelines. On the one hand, standards have to do with maintaining and improving existing service levels; on the other hand, they are often a source of fiscal pressure on transit agencies. Service standards are also linked to any evaluation effort aimed at improving the efficiency, effectiveness, and productivity of a transit service. The greater these measures, the higher is the level of service that can be offered.

1.3.1 Service standards

The need for dynamically updated standards in the transit industry is described in an article by Mora and Chernicoff (2005). This need deserves attention, especially because of the rapid introduction of advanced technologies in bus and rail transit and service. While standards,

Table 1.2 Typical input elements for the three schedule-planning activities

Planning activity	Input element
Timetable development	1. Route (line) number 2. Nodes (stops and timepoints on a route) 3. Pattern (sequence of nodes on a route) 4. Average passenger loads between adjacent stops on a route 5. Load factor (desired number of passengers on board the transit vehicle) 6. Policy headway (the inverse of the minimum frequency standard) 7. Vehicle type 8. Vehicle capacity 9. Average running time (travel time between stops/timepoints on a route)
Vehicle scheduling	1. Trip recovery-time tolerances (maximum and minimum time to be prepared for next trip) 2. Trip departure-time tolerances (maximum departure delay and maximum advance departure) 3. List of garages (names and locations) 4. List of trip start and end locations 5. Average deadhead times from garage locations to each trip start location (pull-outs) 6. Average deadhead times from trip end locations to garage locations (pull-ins) 7. Average deadhead time matrix between all trip end and start locations (by time-of-day)
Crew scheduling	1. Relief-point location (stops, trip start and end points, garages) 2. Average travel times between relief points 3. Trip-layover time (minimum and maximum rest times between two adjacent trips) 4. Type of duty (early, late, split, full, tripper, etc.) 5. Duty length (maximum spread time) 6. Number of vehicle changes on duty 7. Meal breaks 8. Duty composition 9. Other work rules 10. List of drivers by name and type (e.g., part-time, full-time, seniority) 11. Driver priority and equality rules 12. One-day-on, one-day-off work pattern

regulations, and best practices are in particular justified for supporting safety and security applications, they are also crucial for creating satisfactory transit service.

More than 50% of the transit agencies in the United States (METRO, 1984; TCRP Report, 1995) and in Europe (QUATTRO, 1998) employ formal standards in service planning. The three reports discussing standards and guidelines in the U.S. bus industry (METRO, 1984; TCRP Report, 1995) and in transit service in Europe (QUATTRO, 1998) are based solely on surveys conducted among transit agencies, for example, 109 of 345 agencies responded in the 1984 report and 11 of 297 agencies in the 1995 report. The objectives of each survey were predominately based on the *what* type of questions and only a few on the *how* type of question. (For example, what service standards are currently used? What data are collected and used to evaluate the various performance criteria utilized? How is it collected?) The surveys related only to the effectiveness of transit service evaluation efforts. They were conducted on the basis of how the agency perceived effectiveness (10 and 5 gradation scores in 1984 and 1995, respectively) in improving (1) service delivery, (2) equity, (3) ridership, and (4) operating costs. The highest ratings (i.e., standards perceived to be more effective) were given to (1) and (4), while (3) received the lowest score in 1984 and (2) the lowest in 1995.

The main standards currently utilized can be partitioned into two categories: (1) route design and (2) service design. These can be divided further into route level and network level standards and into planning level and monitoring level standards. The resulting 20 standards appear in Table 1.3 by category, group, number, standard name, typical criteria

Table 1.3 Available service standards and their typical criteria ranges

Category	Group	#	Standard item	Typical criterion range ^a	Remarks
Route design	Route level	1.	Route length	Max 40–100 min one-way	Longer length for larger agencies
		2.	Stop spacing	120–400 m (urban area)	Depends on pop. density, land use
		3.	Route directness ^b	Upper limit on deviation from car's shortest path, of 20%–50%	Seeking higher productivity on deviated segments
		4.	Short turn ^b	At peak period only	Aim at reducing operations cost
	Network level	5.	Route coverage	Min 800–1000 m route spacing (urban area); Max 400–800 m walk to stop	Min 50%–95% of residents to have below Max walk distance to stop
		6.	Route overlapping	Overlapping allowed only on approaches to CBD	Avoiding confusion and balancing route dispersion
		7.	Route structure ^b	Max of 2–3 branches per route/ loops around terminals	Reducing confusion by different route #
		8.	Route connectivity	Min of 1–3 routes that intersect a given route (transfer points)	Especially for a new route in an existing network
Service design	Planning level	9.	Span of service	Min operating hours by day-of-week, between 5–6 a.m. and 10 p.m. to 2 a.m.	Later ends for larger agencies
		10.	Load (crowding) level	Max load factor 125%–150% of seat capacity at peak hours/ segments; 100% at off-peak	Higher load factors (150%–175%) for short-hand service (Shuttle, Feeder)
		11.	Standees ^b	Max 50% of # of seats	Depends on the interior vehicles configuration
		12.	Headway, upper limit ^b	Max (policy) headway, 15–30 min at peak period; otherwise, 20–60 min	Varies by type of service and day-of-week
		13.	Headway, lower limit	Min headway, 2–3 min	More use in smaller agencies
	Monitoring level	14.	Transfers	Max of 1–3 transfers for any O-D	Larger agencies permit more transfers
		15.	Passenger shelters ^b	Min of 65–100 daily boarding pass.	Attention to locations of elderly, and centers, such as hospital
		16.	Schedule adherence ^b	Min 80% “on time” (0–5 min behind schedule) at peak period, 90% otherwise	This guideline is usually relaxed for short headway
		17.	Timed transfer	Max of 3–8 min vehicle wait at transfer point	More use in smaller agencies
		18.	Missed trips ^b	Min 90%–95% of scheduled trips are OK	Missed trips can also not comply with trip reliability criterion

(Continued)

Table 1.3 (Continued) Available service standards and their typical criteria ranges

Category	Group	#	Standard item	Typical criterion range ^a	Remarks
		19.	Passenger safety ^b	Max 6–10 pass. accidents per 10 ⁶ pass.; Max. 4–8 accidents per 1.6·10 ⁵ vehicle-km	Depends on updated safety data
		20.	Public complaints	Limits on # of complaints per driver/pass./time period	Public comments and complaints always received

^a Reflects data mainly from the United States.

^b Standards commonly in use.

range, and remarks. The figure, like elsewhere in the book, uses abbreviations for compactness of presentation: passengers (pass.), population (pop.), minimum (min), maximum (max), meter (m), origin–destination (O-D), and central business district (CBD). Table 1.3 shows typical bounds for each standard, which can be used informally as a guideline. In “Remarks,” we provide extra information for each standard that will enhance its implementation characteristics.

Although the standards in Table 1.3 are survey based, using informative questions, we think that two, more elaborated and important questions are missing: (1) What are the specific purposes of a given standard? (2) Are the purposes in (1) fully and optimally attained? In answering these two questions, the transit agency may find out that the purposes are not clear enough to warrant a standard. Furthermore, even for well-defined purposes, the transit agency may realize that answering questions requires some search/research. The latter case indeed deserves well-founded research to justify the use of standards in terms of their range. We attempt in Table 1.4 to outline, for each standard, whether it is necessary to undergo research or whether it is sufficient to determine its criteria administratively. We found that 6 standards necessitated research, 3 standards could be administrative based, and research findings could be input for administrative decisions for the remaining 11 standards.

Under the research-based column in Table 1.4, we listed the recommended objective for 17 of the 20 standards. Under the administrative-based column, we inserted the recommended criteria for 15 of the 20 standards. Finally, under the remarks column, we introduced some helpful notes for each analysis recommended.

This book illustrates some of the required research findings and methodologies for certain important service standards. Chapter 3 demonstrates standards 10 (load-crowding level), 11 (standees), and 12 (headway, upper limit). Chapters 5 and 13 show standard 17 (timed transfer). Chapters 12 and 15 show standards 1 (route length), 3 (route directness), and 15 (transfers). Chapter 16 illustrates standard 4 (short turn), and Chapter 18 illustrates standard 16 (schedule adherence).

1.3.2 Evaluation standards

An assessment of ridership productivity and financial performance of any transit agency largely relies on five variables, determined on a route basis: (a) vehicle-hours, (b) vehicle-kilometers, (c) passenger measures, (d) revenue, and (e) operating cost. These five variables form the base for seven economic and productivity standards that are in use in the United States and Europe (METRO, 1984; TCRP Report, 1995; QUATTRO, 1998): (1) passengers per vehicle-hour, (2) passengers per vehicle-kilometer, (3) passengers per trip, (4) cost per passenger, (5) cost-recovery ratio, (6) subsidy per passenger, and (7) relative performance.

Table 1.4 Suitability of standards to be determined by research results, administrative decision, or both

#	Standard item	Research-based suitability	Administrative-based suitability	Remarks
1.	Route length	Max utilization	—	User and operator variables
2.	Stop spacing	Max utilization	—	User and operator variables
3.	Route direction ^a	Demand sensitivity to route deviation	Max % deviation from car's shortest path	Demand contains potential users
4.	Short turn ^a	Max vehicle saving with Min short turns	—	Operator and user variables
5.	Route coverage	—	Max walking distance to stop for majority of residents	Adequate route spacing will result
6.	Route overlapping	Max utilization	—	To comply with O-D demand
7.	Route structure ^a	Max utilization	—	To comply with O-D demand
8.	Route connectivity ^a	Max utilization	—	To allow for all O-D movements
9.	Span of service	—	Min operation hours, by day-of-week	Can be extended through lowering the frequency
10.	Load (crowding) Level	Determine acceptable criteria	Max load at the max load-point, by time-of-day	Flexible range for certain loading figures
11.	Standees ^a	Determine acceptable criteria	Max standees as % of seats for each vehicle type	Flexible range for certain loading figures
12.	Headway upper limit ^a	Determine criteria	Max (policy) headway, by time-of-day	Depends on previous and next headways
13.	Headway lower limit	Determine criteria	Min headway, by time-of-day	Depends on size of vehicles
14.	Transfers	Determine acceptable criteria	Max # of transfers for a given O-D	Depends on transfer smoothness and waiting time
15.	Passenger shelters ^a	Determine criteria	Min daily boarding to warrant shelter	Site specific (% of elderly, handicapped, etc.)
16.	Schedule adherence ^a	Determine criteria through modeling	Min <i>on-time</i> performance	Depends on real-time information provided
17.	Time transfer	Determine acceptable criteria	Max waiting time at transfer points	Depends on real-time information provided
18.	Missed trips ^a	Determine criteria	Max allowed % of missed trips	Vehicle breakdown separate from reliability problems
19.	Passenger safety ^a	Determine acceptable criteria	Max pass.-accidents per pass. and km driven	Depends on safety data
20.	Public complaints	—	Max # of complaints per driver/pass./time period	Unimportant separate from serious complaints

^a Standards commonly in use.

The main evaluation standards currently utilized can be divided into two categories: (1) passenger based and (2) cost based. The first relates to ridership productivity criteria and the second to financial criteria. We inserted these seven standards in Table 1.5 by category, number, standard name, typical criteria range, and remarks. We added two more abbreviations, for vehicles (veh) and for kilometers (km). Whereas the service standards are

Table 1.5 List of evaluation standards and their typical criteria range

Category	#	Standard item	Typical criteria ranges ^a	Remarks
Passenger based	1.	Pass. per ^b Veh-hour (PVH)	Min of 8–40 PVH; Min 50%–100% systemwide PVH average	Depends on type of service, time of day, and day of week
	2.	Pass. per ^b Veh-km (PVK)	Min of 0.6–1.5 PVK; Min 60%–80% systemwide PVK average	Depends on type of service, time of day, and day of week
	3.	Pass. per trip	Min 5–15 riders per trip; Min 15 passengers average load for all routes	Min average load on express trip, where this standard is most useful, is 20–30 passengers
Cost based	4.	Cost per passenger	Max 1.4 of system average	This standard is often included in a composite score with other criteria
	5.	Cost–recovery ratio ^b	Min 0.15–0.30 ratio; Min 1.0 ratio for express-type services	Larger agencies differentiate between types of services and use a composite score
	6.	Subsidy per passenger ^b	Based on revenue per pass. with min 25%–33% of system average	Used as a difference between cost and revenue, it is simpler than cost to explain to the public
	7.	Relative performance	Min 10%–20% across all routes of a composite productivity score	Route performance is measured against that of other routes

^a Reflects data mainly from the United States.

^b Standards commonly in use.

self-explanatory, the evaluation standards need interpretation. Standard 1 in Table 1.5, pass./veh-hour, is the most widely used productivity criterion, mainly because of the fact that the operating budget is paid out on an hourly basis. It is based, to the greatest extent possible, on unlinked passenger trips (i.e., each boarding adds one to the amount of passengers) and service (revenue) hours. Standard 2, pass./veh-km, reflects the number of riders boarding a vehicle along a unit of distance rather than a unit of time; it, too, is based on unlinked passenger trips and on service kilometers. Standard 3, pass./trip, is the number of boarding passengers per single (one-way) trip; its advantage lies in its simplicity. Standard 4, cost/pass., is a financial criterion attempting to ascertain the productivity of a route. Standard 5, cost–recovery ratio, is the ratio between direct operating costs (wages, benefits, and maintenance costs) and the share recovered by the fares paid by the route’s riders. Standard 6, subsidy/pass., is usually the difference between cost/pass. and revenue/pass. The revenue/pass. criterion reflects different fares and is used as a measure for one of the comparisons of routes. The last standard in Table 1.5 is the relative performance of a route compared to other routes usually having the same characteristics, such as its percentile rank in either an overall ranking of system routes or in a group of routes associated with the same type of service. The exact measure of this ranking varies across transit systems. Table 1.5 shows typical criteria ranges for each evaluation standard that can be used informally or in a more formal manner. “Remarks” provides extra information on each standard that will enhance implementation simplicity.

1.4 VIABILITY PERSPECTIVES

Following our discussion of the framework and derivatives of service and evaluation standards that focus on tools to set adequate and improved transit service, we will now present a more general view of how the basic transit goals should be approached. Current practice shows

that transit agencies are experiencing increasing fiscal pressures caused by a decline in transit patronage, increased operational costs, and decreased government support. In response to this trend, transit agencies have more often been reexamining the manner in which their limited resources are allocated. We shall show here that this concentration on savings should not overlook some amenities (Ceder, 2002, 2007) that are concerned with the viability of service.

The decline in transit patronage is the result of two main factors: (a) poor level of service and (b) better competitors. New roads, bridges, and tunnels serve the automobile and to some extent the railroad, whereas the investment in transit enhancements has been at a relatively much lower level. On one hand, there is no need to promote transit service in a free market environment; on the other hand, transit has the best land-transportation safety record, and it can relieve some traffic congestion as well as help preserve the environment.

Transit viability can be looked at through the perspectives of the passenger and the agency. Orderliness is the key for success. In Figure 1.3, we show three cases when passenger will use transit service: (1) when there is no alternative, (2) when the service offered is acceptable, and

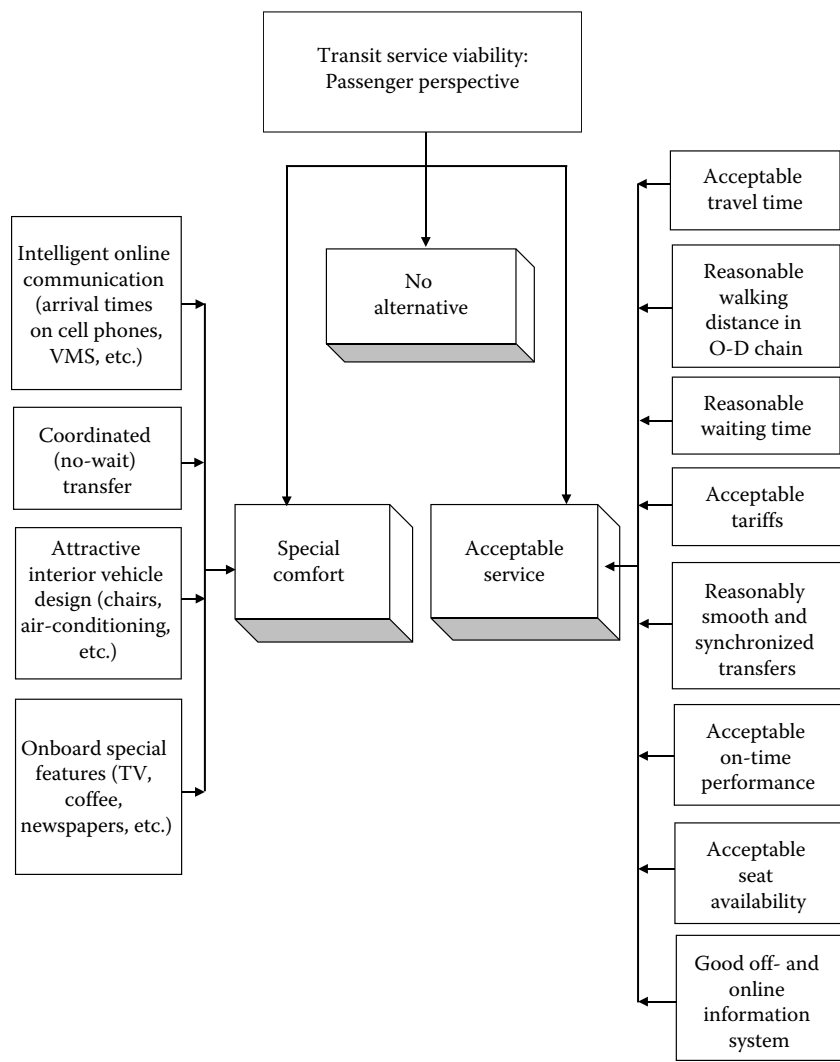


Figure 1.3 Viability of transit service from passenger perspectives.

(3) when the service offers more comfort than does the automobile. Acceptable services, as shown in Figure 1.3, rely basically on reasonable and acceptable travel time, walking distance to and from the transit stop, waiting time, tariff, transfers, and timetable adherence, as well as seat availability and reliability and availability of information. Overall walking distance in the O-D or door-to-door chain and comfortable transfers are central points in understanding passenger travel behavior, which has to do with the integration of transit service end points and the passenger's origin and destination points. Comfortable transfers (vehicle to vehicle, using either the same transit mode or different modes) ought to be smooth (escalator, same platform, etc.) and synchronized (a match between arrival and departure times as stated by online information). Case (3) in Figure 1.3 refers to passengers who regularly use their automobile, but would switch to transit if given special comfort features. Such features can include intelligent online communication, coordinated transfers without waiting, attractive transit vehicle interior design (à la advanced aircraft), and onboard services (again à la aircraft), none of which can be obtained in an ordinary automobile.

Figure 1.4 presents the agency perspective for viability. The provision of subsidy and commercial concessions is self-explanatory; however, the approaches to attain increased patronage and revenue and a reduction in operating costs deserve some elaboration. Increased patronage and revenue can be realized through remedying the current transit service illness, that is, obtaining a better match between service and demand, reliable service, coordinated network (à la a network of routes in a good metro system), and improved comfort and information. This book presents some remedies for a better match between service and demand in Chapter 4, for solving reliability problems in Chapter 18, and for coordination

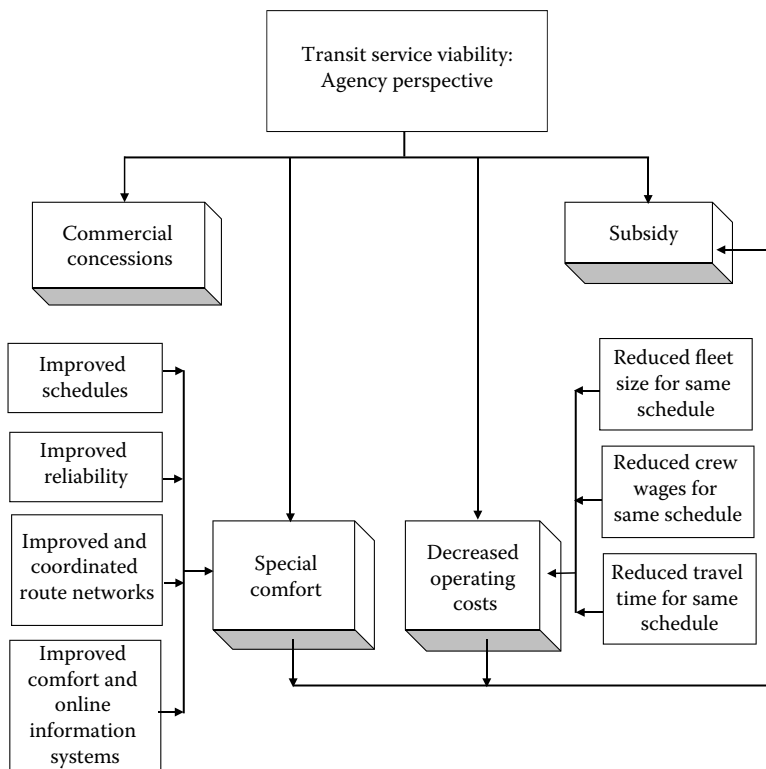


Figure 1.4 Viability of transit service from agency perspectives.

among transit routes in Chapters 5, 13, and 14. Decreased operating costs can be attained by reducing fleet size (less capital cost), total drivers' wages, and the average travel time in the transit network system without changing the service (timetables) offered. We will introduce techniques and methods to reduce fleet size in Chapters 6 through 8, and 12, to reduce total drivers' wages in Chapter 9, and to reduce travel time through a better match between service and demand in Chapter 4.

I.5 OUTLINE OF OTHER CHAPTERS

This introductory chapter has emphasized four major activities, as seen schematically in Figure 1.2, as the core of transit planning. Figure 1.5 describes the remaining themes of the book around these four activities; pertaining to the first edition of this book (Ceder, 2007), it includes significantly revised Chapters 5, 8, 12, and 15 and completely new Chapters 13, 14, and 19. Each block in Figure 1.5 lists the chapter(s) dealing with the content specified. As we see it, the appetite to change transit service across the board ought to start with a small board.

Chapter 2 covers transit data collection systems including manual and automated techniques, automatic vehicle monitoring sampling considerations, and some notes on passenger surveys. Chapter 3, which is part of the second activity in Figure 1.5, analyzes passenger-load and running-time data and four methods of frequency, and Chapters 4 and 5 presents advanced methods of constructing timetables. One of these (in Chapter 5) creates timetables with even passenger loads at different max load points for individual vehicles as opposed to a single max load point (across all vehicles in 1 h). Chapter 5 ties in with the second planning activity in Figure 1.5. A second method, in Chapter 5, allows an examination of optimal timetables for a reduced fleet for a network of transit routes. Such a problem can arise in a practical context when the fleet size is reduced because of vehicle-age attrition and budget

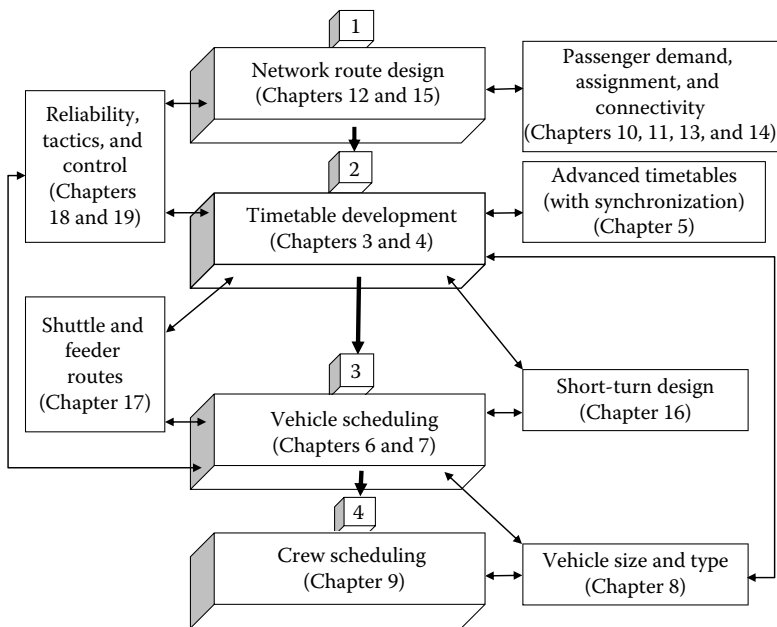


Figure 1.5 Relationship of the main four planning activities to the remaining book themes. Note: Data collection (Chapter 2) and future transit (Chapter 20) are general and technology based and, therefore, related to all planning activities.

policy decisions. A third method, in Chapter 5, describes optimal procedures for designing timetables with maximum simultaneous arrivals at given stops. This synchronization solution is based on a given range of the headways determined.

Chapter 8 introduces different vehicle types with scheduling requirements, examines various scheduling scenarios, and presents new methods to create the best schedules with vehicle-type constraints. In addition, Chapter 8 describes different models to determine the required size of a vehicle. This chapter ties in with both the second and third activity in Figure 1.5. Chapter 9, which is the fourth activity in Figure 1.5, demonstrates some of the optimization concepts behind the problem of assigning crew (mainly drivers) to vehicle schedules with the minimum cost involved. Following a review of some existing models, we present a method to create a good framework for optimal duties. This chapter interacts to some extent with procedures described in the previous chapter for different vehicle types.

Chapters 10 and 11 provide overview on passenger demand, route choice, and assignment. In Chapter 10, we give an overview of the factors affecting passenger demand and their sensitivity (elasticity) and methods designed to predict passenger demand; we also show estimation methods for predicting O-D matrices and how best to forecast ridership. Chapter 11 presents passengers' dilemma in route choice among alternative routes; the route choice is then incorporated into transit assignment modeling at the network level. The modeling in Chapters 10 and 11 interacts with the first activity in Figure 1.5.

Chapters 12 through 15 constitute the first planning activity outlined in Figure 1.5. Chapter 12 presents basic service design elements for different purposes and introduces a new method for dealing optimally with stop placement on streets with the consideration of topography. In addition, this chapter includes a theoretical approach of optimal stop location on a network of roads. Chapter 12 introduces also a new method to determine optimally operational parking limitations. This on-street bound on maximum parking spaces calls for solutions to avoid traffic congestion at some route end points. Chapters 13 and 14 are new chapters, in this second edition, brought up to enhance the importance of the need for coordination and connectivity. Chapter 13 presents new measures and analysis for transit connectivity, transfer based, including a method to detect bottlenecks of coordination in networks of transit routes. Chapter 14 delivers the behavioral aspects of transit coordination and connectivity; it is based on surveys and behavioral modeling showing how people appraise routes with transfers, with some advantages, compared with routes without transfers. Chapter 15 establishes objective functions for designing a network of transit routes while complying with passenger, agency, and government perspectives. This chapter delivers a complete solution for the creation of transit routes while taking into account the activities of vehicle scheduling and timetabling. Chapter 15 describes also some metaheuristic methods to attain a reliable transit network including a scenario in which the demand is varied across seasons. Overall these four chapters, Chapters 12 through 15, construct routes and transfers, assigning O-D demand and initial frequencies, and provide optimal criteria and best route network solutions for decision makers.

Chapter 16 defines a cost-effectiveness approach for operating single routes with short turns (not all trips start and end at the end points of the route). Employing some of the developments resulting from the second and third activities in Figure 1.5, this chapter provides an optimal method to maintain the required level of service while minimizing the fleet size through creating timetables with minimum short-turn trips and minimum required vehicles. Chapter 17 presents the need for and the concept behind an integrated transit system, with an emphasis on shorthand transit service (shuttle and feeder routes). This integrated service is presented as an attractive, reliable, rapid, smooth, and synchronized system. Ten different routing strategies are developed for bus shuttle and feeder routes while using a simulation

tool for analysis along with real-time communication possibilities. Chapter 17 interacts with the second and third activities in Figure 1.5.

Chapters 18 and 19, especially the latter, focus on new technology principles to assist in improving transit operations planning. Chapter 18 discusses the essential part of transit reliability and its impact on operations planning. It covers the subjects of variability of concern among passengers and in the agency, the bus-bunching phenomenon, methods to improve reliability, the calculating of passenger waiting time and vehicle running time, and features and benefits of automatic vehicle location (AVL) systems. Naturally, Chapter 18 interacts with elements of the first three activities in Figure 1.5. In Chapter 19, we present a bridge between new technologies and transit operations using in particular operational tactics, the concept of flexible routing based on distributed computing, and electronic user-operator communication. The book ends with an outline of some future research needs.

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Answers to Exercises 691

1 Chapter 1: Introduction to Transit Service Planning

As stated in "Preface," each chapter opens with information and notes to guide practitioners.

This introductory chapter will show that a successful transit service is not permanent and that

failure in a new service need not be fatal. The chapter contains neither mathematical formula

tions nor a complicated literature review. The motivation behind the writing of this book is to bridge the world of the researchers and

the world of the practitioners. The chapter first presents a real-life description of the planning

process for transit operations. It then provides an overview and a critique of currently used

standards and guidelines for transit service, pointing out those standards that warrant research

and those for which administrative decisions are sufficient. The main components affecting the

viability of a transit service from both the passenger and agency perspectives are described, as

is the link between some problematic elements and the possible solutions advanced in the book. The chapter ends with an outline of the other chapters and the links between each chapter

and the main core of the transit operations planning process. Our attempt to show how to deal

with and to solve in a practical manner current transit service dilemmas and confusions follows

Pablo Picasso, who said: "I do not seek, I find."

1.1 MOTIVATION

A basic premise of this volume is that using transit service should be like eating potato

chips—once you start, it's hard to stop. The fundamental question, though, is how do we

design and approach such a service while knowing, because of the competition offered by

cars, that an adequate transit design must provide superlative service in order to be good

enough. This second edition of the book, following its first edition (Ceder, 2007), will demonstrate

methods and procedures not only for improving transit design through the provision

of an attractive, viable service but also for reducing its cost and increasing efficiency

from the transit agency perspective.

One of the main motivations for writing the first edition of this book (Ceder, 2007) is

the persistent use by transit agencies (especially in the United States) of the Bus Scheduling

Manual of August 1947 (!), which was reprinted in 1982 (Rainville, 1947). The manual

opens as follows: “This volume is viewed today [i.e., 1982–A.C.] as a classic reference on

manual scheduling practices. It represents a comprehensive effort made over 30 years ago to

pull together the state of the practice in this field, and many of those practices still remain

valid today.” The manual's cover and an inside picture of a device to measure distances

appear in Figure 1.1. Those transit agencies that may still be using this manual might be

asked a question: Is the treatment of p

assenger loads and running-time data, frequency

determination, and vehicle and crew scheduling
really classic? Or is it obsolete? The con

clusion implied by the latter served as a trigger to
investigate why the treatment of these

elements in the manual is not classic and to
submit operations planning options that exploit

existing, computerized computation power and a
advanced modeling. On the positive side, it

may be said, as it was in the 1947 manual,
that some basic concepts behind the planning

process for transit operations do indeed remain the same.

In 1998, the U.S. Transit Cooperative Research Program
(TCRP) sponsored Report 30 (Pine

et al., 1998) on transit scheduling with the
idea presumably of advancing knowledge in the

field for training purposes. However, Report 30
neither contains nor refers to any research on

transit scheduling (despite the many studies that
have existed); furthermore, it does not mention a

single reference. The report actually provides very basic
training material for new schedulers; in

concept, it is almost a replica of the 1947 manual, while
emphasizing that scheduling is a craft.

All in all, the following story may sum up the motivation
for writing this book: two

marketing people from a European shoe factory are sent to
investigate the potential sale of

shoes in Africa. A few days later, faxes from each of them
arrive at the manager's office. The

rst fax reads: "No chances, everyone is barefoot." The second fax urges: "Lots of chances,

everyone is barefoot." This book will adopt the lots of chances view.

Another stimulus for the first edition of this book (Ceder, 2007) is a long list of standards

and guidelines (see Section 1.3) for transit service planning, operations, and control that

determine the quality of service seen in practice. In the event that current transit service

quality has reached a satisfactory level, then these standards and guidelines are apparently

effective. Nonetheless, many aspects of transit service can stand improvement, necessitating

changes or replacement of some of these standards and guidelines. Like those in the 1947

manual. We can imagine that the majority of the standards and guidelines to be covered

in the present volume are intuitive based or administrative based instead of being research

based; hence, an opportunity is presented to develop new research tools and methods for

improving some of these standards and guidelines (as discussed in Section 1.3).

This book uses the term transit as shorthand for public transit or public transportation

or public transport, all in common use. We describe in this book transit operations plan

ning problems, a summary of different approaches that have been proposed for their solu

tion, and a description of new methods and approaches incorporating some of the positive

aspects of prior work. In essence, the new methods and approaches are all applicable to the

four major modes of transit operation: airline, railway, bus, and passenger ferry. Although

the operations planning process for bus, railway, and passenger ferry is similar, that for

airline has several special features, but these are not covered in this book. Moreover, rail

way and passenger ferry can be perceived, to some extent, as special cases of bus service

because of the variety of activities and problems offered by the latter. Finally, the methods

and approaches proposed are intended to be easier to implement and more sensitive to the

risks of making changes than are those currently in use by practitioners and researchers.

Figure 1.1 Cover and a page of the well-known 1947 Bus Scheduling Manual, wrongly viewed as a classic reference.

1.2 OPERATIONAL PLANNING DECOMPOSITION PROCESS

The transit operations planning process commonly includes four basic activities, usually

performed in sequence: (1) network route design, (2) timetable development, (3) vehicle

scheduling, and (4) crew scheduling (Ceder and Wilson, 1986; Ceder, 2001, 2002, 2007).

Figure 1.2 shows the systematic decision sequence of these four planning activities. The out

put of each activity positioned higher in the sequence becomes an important input for lower

level decisions. Clearly, the independence and orderliness of the separate activities exist only

in the diagram; that is, decisions made further down the sequence will have some effect on

higher-level decisions. It is desirable, therefore, that all four activities be planned simultaneously

neously in order to exploit the system's capability to the greatest extent and maximize the

system's productivity and efficiency. Occasionally, the sequence in Figure 1.2 is repeated;

the required feedback is incorporated over time. However, since this planning process, espe-

cially for medium to large fleet sizes, is extremely cumbersome and complex, it requires

separate treatment for each activity, with the outcome of one fed as an input to the next.

The quantitative treatment of the transit planning process is reflected in the welter of profes-

sional papers on these topics and in the development of numerous computer programs to automate

mate (at least partially) these activities. In the last 30 years, a considerable amount of effort has

been invested in the computerization of the four planning activities (routing, timetabling, vehicle

scheduling, crew scheduling) in order to provide more efficient, controllable, and responsive

schedules. The best summary of this effort, as well as of the knowledge accumulated, was

presented at the 2nd through 11th international workshops on Computer-Aided Scheduling of

Public Transport (CASPT); recently, this abbreviation was changed to Conference of Advanced

Systems of Public Transport. The research p

resented at C ASPT appears in the book by Ceder

(2007) and the books edited by Wren (1981), Rousseau (1985), Daduna and Wren (1988),

Desrochers and Rousseau (1992), Daduna et al. (1995), Wilson (1999), Voss and Daduna

(2001), and Hickman et al. (2008); the research of the 10th (Leeds, United Kingdom, 2007),

the 11th (Hong Kong, China, 2009), and the 12th C ASPT (Santiago, Chile, 2012) appear now

in proceeding volumes (not books), but the best of them appear/will appear in the new journal:

Public Transport: Planning and Operations by Springer. To learn more of the C ASPT2015

(2015), visit its website: <http://www.caspt.org>. In addition, there is a large number of software

for transit scheduling. Some of the known software appears in Table 1.1.

Many planning software packages are based on static data and therefore only offer static

schedules. Software needs to be developed in such a way that it includes the flexibility that is

required when drawing up and continually modifying service and work schedules. Although

many common aspects are involved in workforce planning, every organization has its own

planning wishes and requirements. This is why the software needs to be fully customized to

the specific business model and company logic. Collective labor agreement rules, legislation

can all be entered into the system. On the basis of advanced algorithms and built-in i

ntelli

gence, the software is able to create an optimal schedule or support planners in the creation of

a schedule. In addition, the software should offer a veritable collection of management infor

mation and analyses of deployment, availability, and planning trends. These form an optimal

basis for better control of activities and greater efficiency during these activities. Moreover, as

a result of its built-in intelligence, the software effortlessly handles multi-resource planning,

for planning people and resources with the help of a single integrated system.

These software packages concentrate primarily on the activities of vehicle and crew sched

uling (activities 3, 4 in Figure 1.2) because, from the agencies' perspective, the largest single

cost of providing service is generated by drivers' wages and fringe benefits. Focusing on activities

ities 3 and 4 would seem to be the best way to reduce this cost. However, because some of 2.1 Vehicle scheduling Timetable development 4 Crew scheduling Independent input Planning activity Output Land-use characteristics Authority constraints Current patronage, by time-of-day, day-of-week Service standards Times for first and last trips Running times Comparison measures Service and deadhead travel times Layover and recovery times Schedule constraints and criteria Cost elements Crew list Relief points Crew work rules, rotation rules, constraints Network route design Determination of interchanges and terminals Design of network of routes and stops Analysis of frequencies and headways Analysis and construction of alternative timetables Analysis and construction of crew assignments Establishment of crew rosters Fleet size analysis with and w/o interlinings Determination of lower bound Reduction via deadheading

Reduction via shifting departures times Analysis and construction of blocks Interchanges and terminals Fixed routes and stops Alternative setting of frequencies and headways Selected public timetable Lower bound of fleet size Minimum fleet size Vehicle schedules Crew schedules and duty rosters Input 3

Figure 1.2 Functional diagram (system architecture) of a common transit operations planning process.

the scheduling problems in these software packages are oversimplified and decomposed into

subproblems, a completely satisfactory or optimal solution is not assured, thus making room

for decisions by experienced schedulers. After all, experience is what we gain when expecting

something else; said another way, the examination is given first and the lesson after.

An argument in favor of automating activities 3 and 4 is that this scheduling process

is extremely cumbersome and time-consuming to undertake manually. In addition to the

potential for more efficient schedules, the automated process enables services to be more

Table 1.1 Some known available transit-scheduling software

Software	Internet address	Features
GIRO "HASTUS"	http://www.giro.ca/en	Vehicle assignment Crew assignment
Merakas-"PIKAS"	http://www.merakas.lt/26en/	Timetable calculation Trips assignment Shifts cutting
PTV	http://www.ptv.de	Network planning Simulation
ROUTELOGIC	http://www.routelogic.com/FeaturesScheduling.html	Network planning Vehicle assignment Planning and analysis Operations management Crew assignment
Routematch	http://www.routematch.com/	Network planning Vehicle assignment Planning and analysis Operations

management Crew assignment Service management

NEMSYS “Routemate” <http://www.nemsys.it/> Transportation planning Vehicle scheduling

Optibus www.optibus.co Real-time rescheduling Vehicle assignment Crew assignment Interactive scheduling Planning and analysis

SYSTRA <http://www.systra.com> Transportation planning Simulation

Tracsis <http://www.tracsis.com/> Demand forecasting Capacity planning Timetable development Rolling stock planning Crew scheduling

Enghouse Transportation

“TranSched” <http://www.enghousetransportation.com/> Scheduling Vehicle blocking Optimized runcutting and rostering Workforce management Trip planner

Trapeze <http://www.trapezegroup.com> Network planning Vehicle assignment Planning and analysis Operations management Crew assignment

controllable and more responsive. The cost and complexity of manual scheduling have

served to discourage adjusting activities and only with automated scheduling methods,

which are becoming more widely accepted, it is feasible to focus on higher levels in the plan

ning process. Nonetheless, a case can be made that these higher levels have received short

shift by both researchers and practitioners.

The network route-design activity in Figure 1.2 is described in Chapters 12 and 15,

which is also linked with Chapters 13 and 14. Planning practice in terms of transit route

design focuses almost entirely on individual routes that, for one reason or another, have

been identified as candidates for change. Occasionally, sets of interacting (e.g., overlapping

or connecting) routes are subject to redesign, usually after a series of incremental changes

to individual routes has resulted in a confusing, inefficient local system. Although it is difficult

to predict the benefits that will result from redesigning a city transit network without

conducting a detailed assessment, it is reasonable to believe that they will be large compared

with the benefits of additional efforts aimed just at problematic scheduling activities (2, 3,

and 4 in Figure 1.2). The approach described mainly in Chapter 15 generates all feasible

routes and transfers connecting each place (node) in the network to all others. From this vast

pool of possible routes and transfers, smaller subsets are generated that maintain network

connectivity. For each subset thus generated, transportation demand is met by calculating

the appropriate frequency for each route. Next, specified optimization parameters are

calculated for each subset. Based on the specific optimization parameter desired by the user,

it is then possible to select the most suitable subset. This method has been designed as a

tool for the planning of future transit networks as well as for the maintenance of existing

networks. The method presented ensures flexibility by allowing the user either to i

input or when

data or to run the analysis automatically.

The timetable development activity in Figure 1.2 is described in Chapters 3 through 5 for

establishing alternative frequencies and timetables. The aim of public timetables is to meet

general public transportation demand. This demand varies during the hours of the day, the

days of the week, from one season to another, and even from 1 year to another. It reflects

the business, industrial, cultural, educational, social, and recreational transportation needs

of the community. The purpose of this activity, then, is to set alternative timetables for

each transit route in order to meet variations in public demand. Alternative timetables are

determined on the basis of passenger counts, and they must comply with service frequency

constraints. In Chapter 4, alternative timetables are constructed with either even headways,

but not necessarily even loads on board individual vehicles at the peak-load section, or even

average passenger loads on board individual vehicles, but not even headways. Average even

loads on individual vehicles can be approached by relaxing the evenly spaced headways pat-

tern (through a rearrangement of departure times). This dynamic behavior can be detected

through passenger-load counts and information provided by road supervisors. The key word

in the even-load cases is the ability to control the loading instead of being repeatedly exposed

to an unreliable service resulting from an imbalance in loading situations.

The vehicle-scheduling activity in Figure 1.2 is described in Chapters 6 through 8 and is

aimed at creating chains of trips; each is referred to as a vehicle schedule according to given

timetables. This chaining process is often called vehicle blocking (a block is a sequence of

revenue and nonrevenue activities for an individual vehicle). A transit trip can be planned

either to transport passengers along its route or to make a deadheading trip in order to

connect two service trips efficiently. The scheduler's task is to list all daily chains of trips

(some deadheading) for each vehicle so as to ensure the fulfillment of both timetable and

operator requirements (refueling, maintenance, etc.). The major objective of this activity is

to minimize the number of vehicles required (Daduna and Pao, 1995). Chapters 6 and

7 describe a highly informative graphical technique for the problem of finding the least

number of vehicles. The technique used is a step function, which is introduced as far back

as 30 years ago as an optimization tool for minimizing the number of vehicles in a fixed-trip

schedule. The step function is termed deficit function, as it represents the deficit number

of vehicles required at a particular terminal in a multiterminal transit system. In

Chapter 6, the fixed-schedule case is extended to include variable trip schedules, in which

given shifting tolerances allow for possible shifts in departure times. This opens up an

opportunity to reduce fleet size further. The deficit function, because of its graphical characteristics, has been programmed and is available on a website. In this book, the deficit function

is applied and linked to the following activities: vehicle scheduling with different vehicle

types (Chapter 8), the design of operational transit parking spaces (Chapter 12), network

route design (Chapters 12 and 15), and short-turn design of individual and groups of routes

(Chapter 16). The value of embarking on such a technique is to achieve the greatest saving in

number of vehicles while complying with passenger demand. This saving is attained through

a procedure incorporating a man-computer interface allowing the inclusion of practical

considerations that experienced transit schedulers may wish to introduce into the schedule.

The crew scheduling activity in Figure 1.2 is described in Chapter 9. Its goal is to assign

drivers according to the outcome of vehicle scheduling. This activity is often called driver

run cutting (splitting and recombining vehicle blocks into legal driver shifts or runs). This

crew-assignment process must comply with some c

constraints, which are usually dependent

on a labor contract. A brief summary is given of the conceptual analytical tools used in

the modeling and software of this complex, combinatorial problem. The crew-rostering

component of this activity normally refers to priority and rotation rules, rest periods, and

to deal with problems encountered by the presence of various pay scales (regular, overtime,

weekends, etc.) and with human-oriented dissatisfaction. The crew-scheduling activity is

very sensitive to both internal and external factors, a factor that could easily lead to an ineff

icient solution.

In Table 1.2, we list the essential inputs for the transit operations planning process illustrated

in Figure 1.2. These mainly independent elements are arranged by activity number. It

should be emphasized that their values differ by time of day and day of week.

1.3 SERVICE AND EVALUATION STANDARDS AND THEIR DERIVATIVES

Tremendous and monotonous in transit service can be perceived as synonymous. It is best

to have a monotonous, automated transit system that will always be there for the passen

ger, for example, a modularized personal rapid transit system conducted like a horizontal

elevator. This observation serves as an introduction to the need for transit service standards

and guidelines. On the one hand, standards have to do with maintaining and improving

existing service levels; on the other hand, they are often a source of fiscal pressure on transit

agencies. Service standards are also linked to any evaluation effort aimed at improving the

efficiency, effectiveness, and productivity of a transit service. The greater these measures,

the higher is the level of service that can be offered.

1.3.1 Service standards

The need for dynamically updated standards in the transit industry is described in an article

by Morra and Chericoff (2005). This need deserves attention, especially because of the rapid

introduction of advanced technologies in bus and rail transit and service. While standards,

regulations, and best practices are in particular justified for supporting safety and security

applications, they are also crucial for creating satisfactory transit service.

More than 50% of the transit agencies in the United States (METRO, 1984; TCRP Report,

1995) and in Europe (QUATTRO, 1998) employ formal standards in service planning.

The three reports discussing standards and guidelines in the U.S. bus industry (METRO, 1984;

TCRP Report, 1995) and in transit service in Europe (QUATTRO, 1998) are based solely on

surveys conducted among transit agencies, for example, 109 of 345 agencies responded in the

1984 report and 11 of 297 agencies in the 1995 report. The objectives of each survey were pre

dominately based on the what type of questions and only a few on the how type of question.

(For example, what service standards are currently used? What data are collected and used to

evaluate the various performance criteria utilized? How is it collected?) The surveys related

only to the effectiveness of transit service evaluation efforts. They were conducted on the

basis of how the agency perceived effectiveness (10 and 5 graduation scores in 1984 and 1995,

respectively) in improving (1) service delivery, (2) equity, (3) ridership, and (4) operating costs.

The highest ratings (i.e., standards perceived to be more effective) were given to (1) and (4),

while (3) received the lowest score in 1984 and (2) the lowest in 1995.

The main standards currently utilized can be partitioned into two categories: (1) route

design and (2) service design. These can be divided further into route level and network

level standards and into planning level and monitoring level standards. The resulting 20

standards appear in Table 1.3 by category, group, number, standard name, typical criteria

Table 1.2 Typical input elements for the three schedule-planning activities

Planning activity Input element

Timetable

development 1. Route (line) number 2. Nodes (stops and timepoints on a route) 3. Pattern (sequence of nodes on a

route) 4. Average passenger loads between adjacent stops on a route 5. Load factor (desired number of passengers on board the transit vehicle) 6. Policy headway (the inverse of the minimum frequency standard) 7. Vehicle type 8. Vehicle capacity 9. Average running time (travel time between stops/timepoints on a route)

Vehicle

scheduling 1. Trip recovery-time tolerances (maximum and minimum time to be prepared for next trip) 2. Trip departure-time tolerances (maximum departure delay and maximum advance departure) 3. List of garages (names and locations) 4. List of trip start and end locations 5. Average deadhead times from garage locations to each trip start location (pull-outs) 6. Average deadhead times from trip end locations to garage locations (pull-ins) 7. Average deadhead time matrix between all trip end and start locations (by time-of-day)

Crew

scheduling 1. Relief-point location (stops, trip start and end points, garages) 2. Average travel times between relief points 3. Trip-layover time (minimum and maximum rest times between two adjacent trips) 4. Type of duty (early, late, split, full, tripper, etc.) 5. Duty length (maximum spread time) 6. Number of vehicle changes on duty 7. Meal breaks 8. Duty composition 9. Other work rules 10. List of drivers by name and type (e.g., part-time, full-time, seniority) 11. Driver priority and equality rules 12. One-day-on, one-day-off work pattern

Table 1.3 Available service standards and their typical criteria ranges

Category	Group #	Standard item	Typical criterion range	Remarks
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Route

design	Route level	1. Route length	Max 40-100 min one-way	
		Longer length for larger agencies	2. Stop spacing	120-400 m (urban area) Depends on pop. density, land use
		3. Route directness	b Upper limit on deviation from car's shortest path, of 20%-50%	Seeking higher productivity on deviated segments
		4. Short turn	b At peak period only	Aim at reducing operations cost
	Network level	5. Route coverage	Min 800-1000 m route spacing (urban area); Max 400-800 m walk to stop	Min 50%-95% of residents to have below Max

walk distance to stop 6. Route overlapping Overlapping allowed only on approaches to CBD Avoiding confusion and balancing route dispersion 7. Route structure b Max of 2-3 branches per route/ loops around terminals Reducing confusion by different route # 8. Route connectivity Min of 1-3 routes that intersect a given route (transfer points) Especially for a new route in an existing network

Service

design Planning level 9. Span of service Min operating hours by day-of-week, between 5-6 a.m. and 10 p.m. to 2 a.m. Later ends for larger agencies 10. Load (crowding) level Max load factor 125%-150% of seat capacity at peak hours/ segments; 100% at off-peak Higher load factors (150%-175%) for short-hand service (Shuttle, Feeder) 11. Standees b Max 50% of # of seats Depends on the interior vehicles configuration 12. Headway, upper limit b Max (policy) headway, 15-30 min at peak period; otherwise, 20-60 min Varies by type of service and day-of-week 13. Headway, lower limit Min headway, 2-3 min More use in smaller agencies 14. Transfers Max of 1-3 transfers for any O-D Larger agencies permit more transfers 15. Passenger shelters b Min of 65-100 daily boarding pass. Attention to locations of elderly, and centers, such as hospital Monitoring level 16. Schedule adherence b Min 80% "on time" (0-5 min behind schedule) at peak period, 90% otherwise This guideline is usually relaxed for short headway 17. Timed transfer Max of 3-8 min vehicle wait at transfer point More use in smaller agencies 18. Missed trips b Min 90%-95% of scheduled trips are OK Missed trips can also not comply with trip reliability criterion (Continued)

range, and remarks. The figure, like elsewhere in the book, uses abbreviations for com

pactness o f p resentation: p assengers (pass.), p opulation (pop.), m inimum (min), m aximum

(max), me ter (m), o rigin-destination (O-D), a nd c entral b usiness d istrict (CBD). T able 1 .3

s hows t ypical b ounds f or e ach s tandard, w hich c an b e u sed i nformally a s a g uideline. I n

" Remarks," w e p rovide e xtra i nformation f or e ach s tandard t hat w ill e nhance i ts i mplemen

tation characteristics.

Although the standards in Table 1.3 are survey based, using informative questions, we

think that two, more elaborated and important questions are missing: (1) What are the specific

purposes of a given standard? (2) Are the purposes in (1) fully and optimally attained?

In answering these two questions, the transit agency may find out that the purposes are

not clear enough to warrant a standard. Furthermore, even for well-defined purposes, the

transit agency may realize that answering questions requires some search/research. The

latter case indeed deserves well-founded research to justify the use of standards in terms of

their range. We attempt in Table 1.4 to outline, for each standard, whether it is necessary

to undergo research or whether it is sufficient to determine its criteria administratively. We

found that 6 standards necessitated research, 3 standards could be administrative based,

and research findings could be input for administrative decisions for the remaining 11

standards.

Under the research-based column in Table 1.4, we listed the recommended objective for 17

of the 20 standards. Under the administrative-based column, we inserted the recommended

criteria for 15 of the 20 standards. Finally, under the remarks column, we introduced some

helpful notes for each analysis recommended.

This book illustrates some of the required research findings and methodologies for certain

important service standards. Chapter 3 demonstrates standards 10 (load-crowding level),

11 (standees), and 12 (headway, upper limit). Chapters 5 and 13 show standard 17 (timed

transfer). Chapters 12 and 15 show standards 1 (route length), 3 (route directness), and

15 (transfers). Chapter 16 illustrates standard 4 (short turn), and Chapter 18 illustrates

standard 16 (schedule adherence).

1.3.2 Evaluation standards

An assessment of ridership productivity and financial performance of any transit agency largely

relies on several variables, determined on a route basis: (a) vehicle-hours, (b) vehicle-kilometers,

(c) passenger measures, (d) revenue, and (e) operating cost. These variables form the base

for several economic and productivity standards that are in use in the United States and Europe

(METRO, 1984; TCRP Report, 1995; QUATTRO, 1998): (1) passengers per vehicle-hour,

(2) passengers per vehicle-kilometer, (3) passengers per trip, (4) cost per passenger, (5) cost-

recovery ratio, (6) subsidy per passenger, and (7) relative performance.

Table 1.3 (Continued) Available service standards and their typical criteria ranges

Category	Group #	Standard item	Typical criterion range	Remarks
19. Passenger safety	b	Max 6-10 pass. accidents per 10 ⁶ pass.;	Max. 4-8 accidents per 1.6 · 10 ⁵	

vehicle-km Depends on updated safety data 20. Public complaints Limits on # of complaints per driver/pass./time period Public comments and complaints always received

a Reflects data mainly from the United States.

b Standards commonly in use.

The main evaluation standards currently utilized can be divided into two categories:

(1) passenger based and (2) cost based. The first relates to ridership productivity criteria

and the second to financial criteria. We inserted these seven standards in Table 1.5 by category,

number, standard name, typical criteria range, and remarks. We added two more

abbreviations, for vehicles (veh) and for kilometers (km). Whereas the service standards are

Table 1.4 Suitability of standards to be determined by research results, administrative decision, or both

#	Standard item	Research-based suitability	Administrative-based suitability	Remarks
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1.	Route length	Max utilization – User and operator variables		
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2.	Stop spacing	Max utilization – User and operator variables		
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3.	Route direction	a Demand sensitivity to route deviation Max % deviation from car's shortest path Demand contains potential users		
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4.	Short turn	a Max vehicle saving with Min short turns – Operator and user variables		
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5.	Route coverage	– Max walking distance to stop for majority of residents Adequate route spacing will result		
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6.	Route overlapping	Max utilization – To comply with O-D demand		
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7. Route structure a Max utilization – To comply with O-D demand
8. Route connectivity a Max utilization – To allow for all O-D movements
9. Span of service – Min operation hours, by day-of-week
Can be extended through lowering the frequency
10. Load (crowding) Level Determine acceptable criteria
Max load at the max load-point, by time-of-day Flexible
range for certain loading figures
11. Standees a Determine acceptable criteria Max standees
as % of seats for each vehicle type Flexible range for
certain loading figures
12. Headway upper limit a Determine criteria Max (policy)
headway, by time-of-day Depends on previous and next
headways
13. Headway lower limit Determine criteria Min headway, by
time-of-day Depends on size of vehicles
14. Transfers Determine acceptable criteria Max # of
transfers for a given O-D Depends on transfer smoothness
and waiting time
15. Passenger shelters a Determine criteria Min daily
boarding to warrant shelter Site specific (% of elderly,
handicapped, etc.)
16. Schedule adherence a Determine criteria through
modeling Min on-time performance Depends on real-time
information provided
17. Time transfer Determine acceptable criteria Max
waiting time at transfer points Depends on real-time
information provided
18. Missed trips a Determine criteria Max allowed % of
missed trips Vehicle breakdown separate from reliability
problems
19. Passenger safety a Determine acceptable criteria Max
pass.-accidents per pass. and km driven Depends on safety
data
20. Public complaints – Max # of complaints per
driver/pass./time period Unimportant separate from serious

complaints

a Standards commonly in use.

self-explanatory, the evaluation standards need interpretation. Standard 1 in Table 1.5, pass./

veh-hour, is the most widely used productivity criterion, mainly because of the fact that the

operating budget is paid out on an hourly basis. It is based, to the greatest extent possible, on

unlinked passenger trips (i.e., each boarding adds one to the amount of passengers) and is

revenue) hours. Standard 2, pass./veh-km, reflects the number of riders boarding a vehi

cle along a unit of distance rather than a unit of time; it, too, is based on unlinked passenger

trips and on service kilometers. Standard 3, pass./trip, is the number of boarding passengers

per single (one-way) trip; its advantage lies in its simplicity. Standard 4, cost/pass., is a

special criterion attempting to ascertain the productivity of a route. Standard 5, cost-recovery

ratio, is the ratio between direct operating costs (wages, benefits, and maintenance costs) and

the share recovered by the fares paid by the route's riders. Standard 6, subsidy/pass., is usually

the difference between cost/pass. and revenue/pass. The revenue/pass. criterion reflects differ

ent fares and is used as a measure for one of the comparisons of routes. The last standard in

Table 1.5 is the relative performance of a route compared to other routes usually having the

same characteristics, such as its percentile rank in either an overall ranking of system routes

or in a group of routes associated with the same type of service. The exact measure of this

ranking varies across transit systems. Table 1.5 shows typical criteria ranges for each evaluation

standard that can be used informally or in a more formal manner. "Remarks" provides

extra information on each standard that will enhance implementation simplicity.

1.4 VIABILITY PERSPECTIVES

Following our discussion of the framework and derivatives of service and evaluation standards

that focus on tools to set adequate and improved transit service, we will now present a more

general view of how the basic transit goals should be approached. Current practice shows

Table 1.5 List of evaluation standards and their typical criteria range

Category	#	Standard item	Typical criteria ranges	Remarks
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Passenger				
	1.	Pass. per b Veh-hour (PVH)	Min of 8-40 PVH; Min 50%-100% systemwide PVH average	Depends on type of service, time of day, and day of week
	2.	Pass. per b Veh-km (PVK)	Min of 0.6-1.5 PVK; Min 60%-80% systemwide PVK average	Depends on type of service, time of day, and day of week
	3.	Pass. per trip	Min 5-15 riders per trip; Min 15 passengers average load for all routes	Min average load on express trip, where this standard is most useful, is 20-30 passengers

Cost

based 4. Cost per passenger Max 1.4 of system average This standard is often included in a composite score with other criteria 5. Cost-recovery ratio b Min 0.15-0.30 ratio; Min 1.0 ratio for expresstype services Larger agencies differentiate between types of services and use a composite score 6. Subsidy per passenger b Based on revenue per pass. with min 25%-33% of system average Used as a difference between cost and revenue, it is simpler than cost to explain to the public 7. Relative performance Min 10%-20% across all routes of a composite productivity score Route performance is measured against that of other routes

a Reflects data mainly from the United States.

b Standards commonly in use.

that transit agencies are experiencing increasing fiscal pressures caused by a decline in transit

patronage, increased operational costs, and decreased government support. In response to this

trend, transit agencies have more often been reexamining the manner in which their limited

resources are allocated. We shall show here that this concentration on savings should not

overlook some amenities (Ceder, 2002, 2007) that are concerned with the viability of service.

The decline in transit patronage is the result of two main factors: (a) poor level of service

and (b) better competitors. New roads, bridges, and tunnels serve the automobile and to

some extent the railroad, whereas the investment in transit enhancements has been at a

relatively much lower level. Nonetheless, there is no need to promote transit service i

n a

free market environment; on the other hand, transit has the best land-transportation safety

record, and it can relieve some traffic congestion as well as help preserve the environment.

Transit viability can be looked at through the perspectives of the passenger and the agency.

Orderliness is the key for success. In Figure 1.3, we show three cases when passenger will use

transit service: (1) when there is no alternative, (2) when the service offered is acceptable, and Transit service viability: Passenger perspective No alternative Special comfort Acceptable service Intelligent online communication (arrival times on cell phones, VMS, etc.) Coordinated (no-wait) transfer Attractive interior vehicle design (chairs, air-conditioning, etc.) Onboard special features (TV, coffee, newspapers, etc.) Acceptable travel time Reasonable walking distance in O-D chain Reasonable waiting time Acceptable tariffs Reasonably smooth and synchronized transfers Acceptable on-time performance Acceptable seat availability Good off- and online information system

Figure 1.3 Viability of transit service from passenger perspectives.

(3) when the service offers more comfort than does the automobile. Acceptable services, as

shown in Figure 1.3, rely basically on reasonable and acceptable travel time, walking distance

to and from the transit stop, waiting time, tariff, transfers, and timetable adherence, as well

as seat availability and reliability and availability of information. Overall walking distance in

the O-D or door-to-door chain and comfortable

transfers a central points in understanding

passenger travel behavior, which has to do with the integration of transit service end points

and the passenger's origin and destination points. Comfortable transfers (vehicle to vehicle,

using either the same transit mode or different modes) ought to be smooth (escalator, same

platform, etc.) and synchronized (a match between arrival and departure times as stated

by online information). Case (3) in Figure 1.3 refers to passengers who regularly use their

automobile, but would switch to transit if given special comfort features. Such features can

include intelligent online communication, coordinated transfers without waiting, attractive

transit vehicle interior design (ala advanced aircraft), and onboard services (again ala air

craft), none of which can be obtained in an ordinary automobile.

Figure 1.4 presents the agency perspective for viability. The provision of subsidy and com

mercial concessions is self-explanatory; however, the approaches to attain increased patron

age and revenue and a reduction in operating costs deserve some elaboration. Increased

patronage and revenue can be realized through remedying the current transit service illness,

that is, obtaining a better match between service and demand, reliable service, coordinated

network (ala a network of routes in a good metro system), and improved comfort and

information. This book presents some remedies for a better match between service and

demand in Chapter 4, for solving reliability problems in Chapter 18, and for coordination. Transit service viability: Agency perspective Subsidy Special comfort Decreased operating costs Improved schedules Improved reliability Improved and coordinated route networks Improved comfort and online information systems Reduced fleet size for same schedule Reduced crew wages for same schedule Reduced travel time for same schedule Commercial concessions

Figure 1.4 Viability of transit service from agency perspectives.

among transit routes in Chapters 5, 13, and 14. Decreased operating costs can be attained

by reducing fleet size (less capital cost), total drivers' wages, and the average travel time in

the transit network system without changing the service (timetables) offered. We will introduce

techniques and methods to reduce fleet size in Chapters 6 through 8, and 12, to reduce

total drivers' wages in Chapter 9, and to reduce travel time through a better match between

service and demand in Chapter 4.

1.5 OUTLINE OF OTHER CHAPTERS

This introductory chapter has emphasized four major activities, as seen schematically in

Figure 1.2, as the core of transit planning. Figure 1.5 describes the remaining themes of the

book around these four activities; pertaining to the first edition of this book (Ceder, 2007), it

includes significantly revised Chapters 5, 8, 1

2, and 15 and completely new Chapters 13, 14,

and 19. Each block in Figure 1.5 lists the chapter(s) dealing with the contents specified. As we

see it, the appetite to change transit service across the board ought to start with a small board.

Chapter 2 covers transit data collection systems including manual and automated techniques

techniques, automatic vehicle monitoring sampling considerations, and some notes on passenger

surveys. Chapter 3, which is part of the second activity in Figure 1.5, analyzes passenger

load and running-time data and four methods of frequency, and Chapters 4 and 5 presents

advanced methods of constructing timetables. One of these (in Chapter 5) creates timetables

with even passenger loads at different maximum load points for individual vehicles as opposed to

a single maximum load point (across all vehicles in 1 h). Chapter 5 ties in with the second planning

activity in Figure 1.5. A second method, in Chapter 5, allows an examination of optimal

timetables for a reduced fleet for a network of transit routes. Such a problem can arise in a

practical context when the fleet size is reduced because of vehicle-age attrition and budget 2 Reliability, tactics, and control (Chapters 18 and 19) Passenger demand, assignment, and connectivity (Chapters 10, 11, 13, and 14) Short-turn design (Chapter 16) Vehicle size and type (Chapter 8) Shuttle and feeder routes (Chapter 17) Advanced timetables (with synchronization) (Chapter 5) 4 1 3 Crew scheduling (Chapter 9) Vehicle scheduling (Chapters 6 and 7) Timetable development (Chapters 3 and 4) Network

route design (Chapters 12 and 15)

Figure 1.5 Relationship of the main four planning activities to the remaining book themes. Note: Data collection (Chapter 2) and future transit (Chapter 20) are general and technology based and, therefore, related to all planning activities.

policy decisions. A third method, in Chapter 5, describes optimal procedures for designing

timetables with maximum simultaneous arrivals at given stops. This synchronization solu

tion is based on a given range of the headways determined.

Chapter 8 introduces different vehicle types with scheduling requirements, examines

various scheduling scenarios, and presents new methods to create the best schedules with

vehicle-type constraints. In addition, Chapter 8 describes different models to determine

the required size of a vehicle. This chapter ties in with both the second and third activity

in Figure 1.5. Chapter 9, which is the fourth activity in Figure 1.5, demonstrates some

of the optimization concepts behind the problem of assigning crew (mainly drivers) to

vehicle schedules with the minimum cost involved. Following a review of some existing

models, we present a method to create a good framework for optimal duties. This chapter

interacts to some extent with procedures described in the previous chapter for different

vehicle types.

Chapters 10 and 11 provide overview on passenger demand, route choice, and assign

ment. In Chapter 10, we give an overview of

the factors affecting passenger demand and

their sensitivity (elasticity) and methods designed to predict passenger demand; we also

show estimation methods for predicting O-D matrices and how best to forecast ridership.

Chapter 11 presents passengers' dilemma in route choice among alternative routes; the route

choice is then incorporated into transit assignment modeling at the network level. The model

elaborated in Chapters 10 and 11 interacts with the first activity in Figure 1.5.

Chapters 12 through 15 constitute the first planning activity outlined in Figure 1.5.

Chapter 12 presents basic service design elements for different purposes and introduces a

new method for dealing optimally with stop placement on streets with the consideration of

topography. In addition, this chapter includes a theoretical approach of optimal stop location

on a network of roads. Chapter 12 introduces a new method to determine optimally

operational parking limitations. This on-street bound on maximum parking spaces

calls for solutions to avoid traffic congestion at some route end points. Chapters 13 and 14

are new chapters, in this second edition, brought up to enhance the importance of the need

for coordination and connectivity. Chapter 13 presents new measures and analysis for transit

connectivity, transfer based, including a method to detect bottlenecks of coordination in

networks of transit routes. Chapter 14 delivers the behavioral aspects of transit coordination

and connectivity; it is based on surveys and behavioral modeling showing how people

appraise routes with transfers, with some advantages, compared with routes without trans

fers. Chapter 15 establishes objective functions for designing a network of transit routes

while complying with passenger, agency, and government perspectives. This chapter delivers

a complete solution for the creation of transit routes while taking into account the activities

of vehicles scheduling and timetabling. Chapter 15 describes also some metaheuristic meth

ods to attain a reliable transit network including a scenario in which the demand is varied

across seasons. Overall these four chapters, Chapters 12 through 15, construct routes and

transfers, assigning OD demand and initial frequencies, and provide optimal criteria and

best route network solutions for decision makers.

Chapter 16 defines a cost-effectiveness approach for operating single routes with short

turns (not all trips start and end at the endpoints of the route). Employing some of the devel

opments resulting from the second and third activities in Figure 1.5, this chapter provides

an optimal method to maintain the required level of service while minimizing the fleet size

through creating timetables with minimum short-turn trips and minimum required vehicles.

Chapter 17 presents the need for and the concept behind an integrated transit system, with

an emphasis on short-haul transit service (shuttle and feeder routes). This integrated service

is presented as a n attractive, reliable, rapid, smooth, and synchronized system. Then different

routing strategies are developed for bus shuttle and feeder routes while using a simulation

tool for analysis along with real-time communication possibilities. Chapter 17 interacts with

the second and third activities in Figure 1.5.

Chapters 18 and 19, especially the latter, focus on new technology principles to assist in

improving transit operations planning. Chapter 18 discusses the essential part of transit reliability

and its impact on operations planning. It covers the subjects of variability of concern

among passengers and in the agency, the bus-bunching phenomenon, methods to improve

reliability, the calculating of passenger waiting time and vehicle running time, and features

and benefits of automatic vehicle location (AVL) systems. Naturally, Chapter 18 interacts

with elements of the first three activities in Figure 1.5. In Chapter 19, we present a bridge

between new technologies and transit operations using in particular operational tactics, the

concept of flexible routing based on distributed computing, and electronic user-operator

communication. The book ends with an outline of some future research needs.

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2 Chapter 2: Data Requirements and Collection

This chapter offers a number of potential benefits to transit agencies in the areas of data acquisition,

data merits, and data utilization. The major objective is to help develop a comprehensive,

statistically based data collection plan that will enable transit agencies to collect proper data and

the right amount in a cost-effective manner. The chapter starts with the three core keys representing the objectives of this book: (1) to

know, (2) to plan and decide wisely, and (3) to operate in a smart manner. Achieving the first key,

through the gathering of comprehensive data, is the objective of this chapter. The first section

describes the various data collection techniques and methods, including definitions and inter

pretations. The second section continues with the linkage of the data collection method, the

results expected from the data analysis, and enhanced service elements gained from the results.

These two sections of the chapter are most suitable for practitioners, who usually understand

the difficulties and sometimes even conflicts involved in real-life data collection as is well repre

sented in the following real anecdote. While trying to maximize their comfort, drivers are more sophisticated than one can imagine.

The following anecdote concerns gaining extra planned travel time by arriving early and having

extended layover time. The checkers in a bus agency who boarded the vehicle used to announce

to the driver that they were measuring average travel time with a rate factor. If the bus is too

slow (i.e., encountering more overtaking vehicles than it overtook), the rate is below 10, otherwise more than 10. One driver found out when this check was going to take place and asked

members of his family to drive their cars at a slow pace just in front of his bus in order for him

to earn a rate of below 10 and to extend the planned travel time. The third and longest section of this chapter introduces basic statistical tools that are

employed in almost every phase of the data collection undertaking. There is no prerequisite

for understanding the statistical tools, though practitioners may want to skip the mathematical

formulations and concentrate on the examples. The chapter ends with a literature review and recommendations for further reading. Overall,

the data collection effort should be handled professionally, with a precise understanding of

the purpose of each data element. Finally, since we want not only high-quality data, but not in

excessive amounts, it is good to recall what Galileo said: "Science proceeds more by what it has

learned to ignore than what it takes into account."

2.1 INTRODUCTION

Current practice in transit agencies shows that sufficient data seldom exist for service operations

planning. Manual data collection efforts are costly and, consequently, must be used

sparingly. Automated data collection systems, though growing rapidly, are not yet perfectly

linked to the requirements of planning data. Extraction of data from an automated system,

which is ostensibly a simple task, may turn out in actuality to be rather complex. Having

too much data is often a bad as having too little. The only concept we can't trust is that in

any collection of data, the more most obviously correct beyond all need for checking is the

mistake. At the same time, the data are essential for responding to basic passenger needs,

namely, the route: where is the closest stop, what time should I be at the stop, and more. The

data are certainly crucial, too, for responding to the operations planning needs of each train

station: How can the network of routes, stops, and terminals be improved? How can each

route be improved? What is the best timetable to deliver? How can fleet size be minimized

while maintaining the same level of service? How can crew cost be minimized without ser-

vice changes? No doubt, a common element for all transit agencies is their pursuit of data to

aid in improving the efficiency, productivity, and effectiveness of their systems.

This chapter demonstrates data requirements and applications for enhancing transit

service productivity and efficiency. Fundamentally, there are three keys for achieving a suc-

cessful transit service: (a) gathering and comprehending adequate data, (b) using the data

collected for intelligent planning and decisions, and (c) employing the plans and decisions

for a stately conducted operations and control.

These three keys appear schematically in

Figure 2.1. This chapter, which introduces the framework for a training the rest key, consists

of four parts: First, different data collection techniques; second, the data requirements, with

attention paid to their coordination; third, basic tools in statistics needed to comprehend

the mechanism of data analysis; and fourth, a literature review and list for further reading.

2.2 DATA COLLECTION TECHNIQUES

We can divide transit data collection techniques required for operations planning into three

categories: (1) manual-based methods, (2) automated-based methods, and (3) automatic

vehicle location (AVL)-based methods. Fundamentally, there are only two types of methods,

manual and automated. However, AVL or automatic vehicle monitoring (AVM) systems

furnish more accurate information, especially in time and space, than other item-specific

automated methods and, therefore, can be looked at as a separate category. The AVL-based

technique will be covered in Chapter 17 in connection with linking the technique to rem

edies for reliability problems.

Five primary techniques for collecting transit data may be identified: point check, ride

check, deadhead check, passenger survey, and population surveys. To gather and comprehend data To plan and decide intelligently Successful transit system To operate astutely

ree core keys

Figure 2.1 The three important keys for a successful transit system.

2.2.1 Point check

Point check is usually described as counts and measurements performed by a checker sta

tioned at a transit stop. The stops selected i
s virtually the maximum (peak) load point, at w
high

the transit vehicle departing this stop has,
on average, the maximum onboard load a cross

all route segments. A route segment is defined
as a section of the route between two a
d adjacent

stops. For each vehicle passing the stop, t
he point check usually contains load counts, a
rrival

and departure times, and vehicle and route identifications.
Other point-check locations than

the peak stop are more (multiple) peak p
oints, end points, and strategic points. M
ultiple

point checks accommodate situations in which t
here are simultaneously several peak points

and a situation of long routes and branching
routes, in which a branching route is one

that stretches a long the base route and adds
a certain branch. End point checks accom

modate running-time measurements and, if a
pplicable, record farebox readings. Strategic

point checks are useful for item-specific c
hecks, such as at major transfer points (
measuring

transfer time and successful vehicle meetings),
major activity centers (observing passenger

behavior in selecting a competitor mode), and n

ew neighborhoods (measuring changes in

passenger demand).

2.2.2 Ride check

Ride check refers to counts and measurements performed by either a checker riding the tran-

sit vehicle along the entire route or an automated instrument (hence, replacing the human

checker). The ride check contains mainly on-and-off passenger counts, from which one can

derive the onboard passenger load for each route segment, arrival and departure times for

each stop, and sometimes item-specific surveys or measurements (vehicle running speed,

boarding by fare category, gender of passengers, and baggage size) and record farebox read-

ings. The common automated instrument for ride checks, called an automated passenger coun-

ter (APC), can perform the main ride-check tasks. It cannot, however, replace the checker

in counting boarding by fare category and in surveying passengers. A special ride check is

one performed by the operator (driver). It usually involves the interaction of the driver and

a farebox, with the driver inserting into the machine information related to fare category

and origin-destination (O-D) per passenger. The action, however, increases the dwell time

at the stop.

Figure 2.2 depicts passenger demand as interpreted by hourly passenger load, using both

point-check and ride-check methods. This is a vital data element affecting operations plan-

ning. We can obtain from point check only the load for the peak segment; with ride check,

we can illustrate the entire load profile by either time or space. Figure 2.2 will be used as

space-based (passenger-km) illustrations in the next chapter, which is concerned with a

determination of vehicle frequencies and headways. The time-based (passenger-hour) load

profile will be used in Chapters 13 and 14, which deal with route design service and network.

Figure 2.2 shows the location of the maximal load point (stop), from which 320 passengers are

transported during given hour, along the maximal (peak) load segment. In addition, Figure 2.2

demonstrates the different visual appearances of the space- and time-load profiles. Whereas

the x-axis of the space-based profile is fixed for a given route direction, the x-axis of a time

based profile is set according to average values.

2.2.3 Deadhead check

Deadhead check refers to the average vehicle running time between an arrival point on one

route and a departure point on another route. This deadheading time is required in a transit

system with interlining routes. It is measured mainly by agency cars traveling along the

shortest path (in time) between the two route end points. This shortest path varies by time

of day, day of week, and type of day.

2.2.4 Passenger survey

Transit passenger surveys are conducted in essence while directly confronting the passen-

gers. The known survey methods of this type are onboard, at stop, at terminal, and mail

back (postage-free forms). The most common is the onboard survey. All the surveys are

carried out by agency checkers or drivers or by specially trained staff who either distribute

forms to all or ask the questions in person. One way to increase the response incentive is Passenger load at hour j Passenger load at hour j Average running time (h) Passenger-hour 320 215 170 198 161 Passenger load at hour j Distance traveled (km) Passenger-km 320 215 170 198 161 Ride check Max load point Distance traveled (km) 320 Point check Stop

Figure 2.2 Main information obtained for operations planning through point check and ride check.

to hand out a symbolic gift, for example, a nice pen or key holder. Such a token may open up

more opportunities for cooperation as the gesture is appreciated. Basically, there are general

surveys and special purpose surveys. In a general survey, we obtain multitype information,

such as O-D, access and egress modes and distances, trip purpose, routes selected on a trip,

fare paid, type of payment, frequency of use by time of day, and socioeconomic and atti-

tude elements. Special purpose surveys aim at eliciting only one type of information, such

as O-D, opinion of service changes, transfer

activities, pass-holder usage, attitude toward

possible fare changes, or the proportion of different fare types (adult, student, free passes,

transfer, and special fare).

2.2.5 Population survey

Generally, the population surveys are conducted at the regional level at home, shop, or work

place. These surveys are usually interview based, involving both transit users and nonusers,

in order to capture public attitudes and opinions about transit service changes (including the

impact on household location decision), fare changes, and transportation, traffic, and land

use projects. The population interviews of users and nonusers also address the vital issues

of potential ridership, market segmentation, market opportunities, and suggestions for new

transit initiatives.

Further description of some of the data collection techniques can be found in the transit

data collection manual (UMTA, 1985). With this background of transit data collection

techniques and methods, we can proceed to answer two important questions: (1) What data

are required for enhancing specific service components? (2) How can we best exploit and

analyze the data? The next section covers these issues.

2.3 DATA REQUIREMENTS

This section focuses on the need for certain

data elements in order to provide an opportunity

for achieving service enhancements. Generally, detailed information on passenger demand

and service characteristics is not available at the route level. Without this information, however

ever, the efficient deployment of transit service commensurate with demand is impossible.

This is an example of why we emphasize that important data elements be acquired so as to

(1) obtain badly needed data and drop insignificant data and (2) gain an overview of the entire

data requirements for attaining a reduced-cost data collection system. Since a cost appraisal

of the data collection effort is ostensibly high, let us emphasize Einstein's saying: "Everything

should be made as simple as possible, but not simpler."

There is a common thread of data needs across all transit agencies. This thread exists as

long as agencies share these objectives: (a) improving service and operations, (b) improv-

ing productivity and efficiency by better matching supply and demand, (c) improving levels

of service through increased reliability as a result of better control and response, and (d)

reducing data gathering, processing, and reporting costs. Broadly speaking, the data items

gathered by the techniques described earlier are useful for one or more aspects of route and

service design, scheduling, information system, m-

marketing, demand allocation, monitoring

management, and external reporting. This section will show, however, only the data elements

ments pertaining to operations planning.

Figure 2.3 portrays in flowchart form three operations planning categories: an adequate

data collection method, the results required from a data analysis, and relevant, enhanced

service elements. Classification of data collection methods appears in the previous section.

tion. Figure 2.3 refers to three planning levels: regional, group of routes, and route. At the Regional level Group-of-routes level Route level
Data collection method Result of data analysis* Regional
O-D matrices and market segmentation of entire
population, by mode travel Developing new service (routes
and terminals) Transit O-D matrices with amount of O-D
transfers Transit O-D and transfer surveys (at stops, on
board, mailback) Developing new routes, changing route
structure, realignment of terminals and stops Route O-D
matrices by stops, approximate passenger load and
waiting times Route O-D surveys (on board, farebox,
mailback) Developing new (changes in) express and local
routes, short-turn and zonal services Boarding and
alighting counts, running times, dwell times, passenger
load by route segment, on-time performance, departure and
arrival times Ride checks (on board, APC) Passenger load
at route peak points Point checks (at peak points)
Deadhead O-D shortest paths, deadhead running times
Deadhead checks (by car) Revenue by fare category Revenue
counts (farebox) * By time of day, day of week, type of
day. Updating the design of vehicle scheduling,
short-turn, and zonal services Productivity assessment
Updating vehicle size, frequency, and departure times
Updating vehicle size, frequency, and departure, running,
layover, and recovery times Enhanced service elements
Survey of regional O-D characteristics, and attitudes of
entire population (at houses, shops, plants, work)

Figure 2.3 Data collection methods and resultant analysis and service elements.

regional level, we focus on developing new or improved transit service by means of routes

and terminals. To accomplish this, we need the entire regional population O-D matrices and

market segmentation by mode of travel. Chapters 10 through 15 cover some aspects of this

planning undertaking. At the group-of-routes level, we focus solely on transit service with

the objective of creating new and improved routes, stops, and terminals. This differs from

the regional level by the treatment of a specific group of routes characterized by their own

O-D matrices and transfer activities.

At the route level in Figure 2.3, there is the bulk of enhanced service elements. We start

with route O-D matrices, approximate passenger loads, and waiting times in order to develop

or improve express and local routes, as well as short-turn and zonal services. The analysis

of ride-check data yields the next set of enhanced elements, which includes the evaluation of

vehicle size, determination of frequencies and headways, and the construction of timetables

with their accompanying parameters. The third set of enhanced elements is based on point

check data analysis for updating vehicle sizes, for example, mini, standard, or articulated

buses, frequencies and headways based on peak point counts, and their derived timetables.

The fourth set of elements relies on O-D average running-time information pertaining

to deadheading trips, together with a description of O-D shortest paths. The last s

et of

enhanced elements, which is revenue based, assesses productivity measures. The procedures

and modeling to attain these route-level enhanced service elements appear in Chapters 3

through 8 and 11 through 18, that is to say, in the majority of the chapters in this book.

Altogether, there are 16 crucial data elements required for transit operations planning.

The lower part of Figure 2.4 lists these 16 data elements. Figure 2.4 presents the abstract

relationships among the four main planning activities depicted in the center of the figure and

the remaining themes of this book, citing the relevant data elements. This figure virtually

follows the illustration in Figure 1.5, but with an emphasis on the data elements.

Two vital issues must accompany any data collection system. The first is the sample size

required; the second is how often the data should be collected. It is obvious that not all

transit trips or transit passengers can be observed in the data collection process. Since transit

agencies usually collect only a fraction of the data, it is uncertain how well the sample data

represent the entire population. The next section provides the fundamental statistical tools

that can broadly cover the sample size issue.

The second issue is the matter of how often

should data be collected. We recognize that

transit service exists in a dynamic environment, in which changes in passenger demand occur

regularly. Transit agency road inspectors and supervisors are supposed to deliver information

on noticeable changes that would warrant a new data collection effort. Unfortunately, we can

not rely on the drivers to do so, since by nature, they try to maximize their comfort and will

not report changes that may put more of a burden on their shoulders (e.g., reduced passenger

load—lots of empty seats—that justifies trip cancellations). Essentially, there are data items that

are recommended to be collected quarterly, and others annually or every few years. The data

elements numbered 8–13 in Figure 2.4 are those that usually need to be collected quarterly.

It should be noted that the statistical properties and characteristics of each data element—for

example, its variability—can shed light on the how often it is as well as seen in the next section.

2.4 BASIC STATISTICAL TOOLS

Statistical techniques play an important role in achieving the objectives of proper transit

data acquisition, handling, and interpretation. The Merriam-Webster Collegiate Dictionary

(2000) defines statistics as follows: “(1) a branch of mathematics dealing with a collection,

analysis, interpretation of masses of numerical data; (2) a collection of quantitative data.”

Basically, the two definitions possess common elements implying a collection of data with

inference or inference making as the objective. A large body of data is called population,

and the subset selected from it is a sample. In any transit data collection system, the objective

of the statistics is to infer (estimate or decide) something about the population, based on

the information contained in a sample.

2.4.1 Histogram and distribution

Consider a study whose aim is to determine important variables affecting headways between

transit vehicles moving on the same route and in the same direction. Among these variables

are existing passenger demand, potential or hidden passenger demand, and a minimum level Key to data elements: 1. Regional O-D matrices 2. Market segmentation by travel mode 3. Transit O-D matrices 4. Transfer counts 5. Route O-D matrices by stop 6. Approximate vehicle loads 7. Approximate passenger waiting times 8. Passenger boarding and alighting counts 9. Running times 10. On-time performance 11. Vehicle loads by route segment (between stops) 12. Actual departure and arrival times 13. Vehicle loads at route peak points 14. Deadhead O-D shortest paths 15. Deadhead running times 16. Revenue by fare category Data: 9, 14, 15 Vehicle scheduling (Chapters 6 and 7) Data: 1, 2, 3, 4, 6, 9, 16 Network route design (Chapters 12 and 15) Data: 7, 8, 9, 10, 11, 12 Timetable development (Chapters 3 and 4) Crew scheduling (Chapter 9) Data: 1, 2, 3, 4, 14, 15 Shuttle and feeder routes (Chapter 17) Data: 7, 8, 9, 10, 11, 12 Reliability and control (Chapters 18, and 19) Data: 5, 9, 10, 11, 12, 14, 15 Short-turn design (Chapter 16) Data: 9, 11, 13, 14, 15 Vehicle size and type (Chapter 8) Data: 5, 9, 10, 11, 12 Timetable synchronization (Chapter 5) Data: 1, 2, 3, 4, 5 Passenger demand and assignment (Chapters 10, 11, 13, and 14) Data: 14, 15

Figure 2.4 Data requirements by book chapter themes.

of service regardless of demand (see standard 12 in Tables 1.3 and 1.4). The transit planner

will wish to measure these variables with the objective of utilizing the information in the

sample to infer the approximate relationship of the variables with headways and to measure

the strength of this relationship. Certainly, the objective will be to determine optimum con

ditions for setting the best headways from different perspectives, that of the passenger, the

agency, or the government (community).

Any set of measurements can be characterized by a frequency histogram—a graph with

an axis subdivided into intervals of equal width, with rectangles over each interval. The

height of these rectangles is proportional to the fraction of the total number of measure

ments in each interval. For example, in order to characterize 10 measurements of transit

vehicle running times (in minutes)—say, 36, 52, 38, 44, 48, 39, 63, 28, 36, and 55 min—we

could divide the xaxis into 10 min intervals as is shown in the upper part of Figure 2.5.

We wish in this example to obtain information on the form of the distribution of the data,

which is usually mound shaped. The larger the amount of data, the greater will be the number

of intervals that can be included and s

till present a satisfactory picture of the data. In

addition, we can decide that if a measurement falls on a point of division, it will belong to

the interval on its left on the x-axis. 20 30 40 50 60
 7010 0 1 2 3 4 5 20 30 40 50 60 7010 0 Running time (min) R
 e l a t i v e f r e q u e n c y R e l a t i v e f r e q u e
 n c y

Figure 2.5 Relative frequency histogram and distribution of the running times in the example.

If we can assume that the measurements were selected at random from the original set,

we can move one step further to a probabilistic interpretation. The probability that a mea

surement would fall in a given interval is proportional to the area under the histogram that

lies over the interval. For example, the probability of selecting a running-time measurement

over the 20-40 min interval in Figure 2.5 is 0.5. This interpretation applies to the distribu

tion of any set of measurements, including populations. If the lower part of Figure 2.5 gives

the frequency distribution of running times (of a given transit route, direction of travel, and

time period), the probability that the running time will lie in the 20-40 interval is propor

tional to the shaded area under the distribution curve. In statistical terms, the distribution

function for a random variable running time is denoted $F(t')$ and given by $F(t') = P(T' \leq t')$

for $-\infty < t' < \infty$, where $P(T' \leq t')$ is the probability of T' being less than or equal to the value

of t' and T' is continuous. The function $F(t')$ is called the distribution function (or cumula

tive distribution function), and its derivative (if existing) is $f(t') = dF(t')/dt'$. The function $f(t')$

is called a probability density function (pdf) for the random variable T' . The distribution

that appears in the lower part of Figure 2.5 is $f(t')$.

Since many similar histograms could be formed from the same set of measurements, we need

to rigorously define quantities with which to measure the sample data, n . The most common

measure of a central tendency used in statistics is the arithmetic mean or average, the latter

being the term we use in this book. The most common measure of variability used in statistics is

the variance, which is the function of deviations of the sample measurements from their average.

The square root of the variance is called standard deviation. These three measures—variance,

deviation, and standard deviation—appear in expressions (2.1) and (2.2) for a set of n measured

responses $x_1, x_2, \dots, x_n$, where $\bar{x}$, s^2 , and s are the average, variance, and standard deviation,

respectively. The corresponding three measures of population are denoted μ , σ^2 , and σ : $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ (2.1) $s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ (2.2)

For the example of Figure 2.5, we obtain $\bar{x} = 43.9 \text{ min}$, $s^2 = 98.578 \text{ min}^2$, and $s = 9.929 \text{ min}$.

It may be noted that many distributions of transit data (as is virtually all other data in

real life) are mound shaped, which can be approximated by a bell-shaped frequency dis

tribution. This is known as a normal curve, with a normal density function, and may be

expressed as follows: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $-\infty < x < \infty$, $-\infty < \mu < \infty$, $\sigma > 0$ (2.3)

where e is the base of a natural logarithm ($= 2.7182818 \dots$) π is the ratio of the length of the circle to its diameter ($= 3.14159 \dots$) μ and σ are the average and standard deviation, respectively, and hence locate the center of the distribution and measure its spread

Data possessing the normal type of distribution have the following empirical rule: $\mu \pm i\sigma$ con

tains approximately 68%, 95%, and 99.7% of the measurements for $i = 1, 2, 3$, respectively.

In practical terms, we find that most random variables observed in nature lie within 1, 2, or

3 standard deviations of their average, with probabilities in the vicinity of 0.68, 0.95, and

0.997, respectively.

2.4.2 Estimation

We will now return to the objective of statistics—making an inference about the popula

tion, based on information contained in a sample. Usually, this inference is about one

or more population parameters. For example, we might wish to estimate the fraction

of trips that would have running times of less than or equal to 20 min. Let T' equal

the running time; the parameter of interest is $P(T' \leq 20)$, which is the area under the

pdf (lower part of Figure 2.5) over the interval $t' \leq 20$. Suppose we wish to estimate the

average of T' and to use the sample in Figure 2.5, with its average estimate of 43.9 min.

One way to measure the goodness of this e

estimation is in terms of the distance between

the estimate and its real (unknown) population value. This distance, which varies in a

random manner in repeated sampling, is called the error of estimation, and we certainly

would like it to be as small as possible. If this error is given in percentages, it is often

called precision ($\pm$ given %). If it is in actual values (minutes for T'), it is often called

tolerance ($\pm$ given minutes).

Assuming that we want this average running time of 43.9 min to have a precision of

$\pm 1.0\%$, we then ask, "What is the probability that the average of T' will fall in the interval

between 39.51 and 48.29 min?" This probability is called confidence level or confidence

coefficient, and the interval is called confidence interval. The objective here is to find the

tolerance that generates a narrow interval that encloses the average of T' with a high con-

fidence level (probability) of usually 0.95, or 95%. A standard normal random variable

with a normal distribution is expressed in Equation 2.3 is usually denoted Z . If we want

to find a confidence interval that possesses a confidence coefficient equal to $(1 - \alpha)$ for Z ,

we can use the following probability statement: $P(-Z_{\alpha/2} \leq Z \leq Z_{\alpha/2}) = 1 - \alpha$ (2.4)

where P is the probability $Z_{\alpha/2}$ and $-Z_{\alpha/2}$ are two tail end values of the standard normal distribution (the probability in between the tails is $1 - \alpha$)

For instance, Z has a 0.95 probability ($\alpha = 0.05$) of being between -1.96 and $+1.96$ (based on

normal distribution tables). In this case, Equation 2.4 becomes $P(-1.96 < Z < 1.96) = 0.95$.

That is, $Z_{1-\alpha/2} = Z_{1-0.05/2} = Z_{0.975} = 1.96$. More interpretations and description of this subject

can be found in numerous books on statistics, such as Washington et al. (2003).

The transit data collection manual (UMTA, 1985) contains a reference to the U.S. national

database specification. In this specification, the precision is $\pm 1.0\%$, and the confidence level

is at 95% for reporting annual passenger-boarding counts and passenger miles. This is a

system-wide requirement by the Urban Mass Transit Administration (UMTA) for an aver

age weekday and for Saturday and Sunday.

Other useful measures of the error of estimation described in the transit data collec

tion manual (UMTA, 1985) are absolute tolerance (AT) and absolute equivalent tolerance

(AET). AT and AET represent the error of estimation of a proportion of observations

rather than the average of the data. The UMTA manual states as an example that sched

ule adherence is the proportion of trips leaving in the categories early, on time, and late.

For instance, an estimated proportion of on-time performance may be 0.8 with $AT = \pm 0.1$.

However, when the estimated proportion approaches 0 or 1.0, the AT resulting from a

given sample size is small compared to the AT value for an estimated proportion in the

vicinity of 0.5. Thus, the second measure, AET, is established as the absolute (equivalent)

tolerance that can be attained with the following formula (UMTA, 1985) if the estimated

proportion is 0.5: $AET = 0.5AT \sqrt{p(1-p)}$ (2.5)

where p is the expected proportion.

2.4.3 Selecting sample size

The sampling procedure affects the quantity of information per measurement. This procedure,

together with the sample size, n , controls the total amount of relevant information

in the sample. Start with the simplest sampling situation, in which random sampling is

conducted among a relatively large population. For an independent, continuous variable X ,

we define the average or the arithmetic mean as the expected value, $E(X)$, with a variance

of $V(X)$. It is known (see, e.g., Washington et al., 2003) that for $x_1, x_2, \dots, x_n$ independent

random variables (or trials of an experiment), with $E(x_i) = \mu$ and $V(x_i) = \sigma^2$, we will obtain

$E(X) = \mu$ and $V(X)/n = \sigma^2/n$, where X is distributed with an average of μ and a standard deviation of σ .

$\sigma / \sqrt{n}$.

Suppose we wish to estimate the average vehicle load at peak hours μ' , and we wish the tolerance

(error of estimation) to be fewer than eight passengers, with a confidence level (probability)

of 95 %. This is for a given route and travel direction during weekdays. Since approximately

95% of the sample averages, in terms of the number of trips to be sampled, will fall within $2 \times \sigma$

of μ' in repeated sampling, we can ask that $2 \times \sigma$ equal eight passengers, or $2 / n \times \sigma = 8$,

from which we obtain the sample size $n = \sigma^2 / 16$. Certainly, we do not know the population

standard deviation, σ ; hence, we must use an estimate from a previous sample or knowledge of

the range in which the measurement will fall. An empirical rule is that this range is approxi

mately 4σ , and in this way, we will find σ . For our example, suppose the range is 50 passengers

(average load during peak hours lies between 20 and 70 passengers). Thus, $n = (50/4)^2 / 16 =$

9.77. Accordingly, we will sample 10 trips during the peak hours of a given route and direc

tion of travels spread out over several weekdays. We would then be reasonably certain (with a

probability of 95%) that our estimate will lie within $2 \times \sigma = 8$ passengers of the true average of

the vehicle load. The general sample size formula for this case is as follows: $n \geq (T_0/2)^2 / 2 \times \alpha \times \sigma^2$ (2.6)

where $2 \times 1-\alpha/2$ is the $(1-\alpha/2)$ percentile normal deviate and T_0 is the tolerance. Based on the

empirical rule cited earlier, we can use the formula in a practical situation as $n \geq (T_0)^2 / 2 \times \sigma^2$ (2.7)

where s is the estimate of σ .

In effect, we would expect in the example used that the tolerance would be fewer than

eight passengers. According to the empirical rule, there is a probability of 68% that the

tolerance is less than $\sigma \times 4$ passengers. Note that these probabilities of 95% and 68% are

inexact, because σ was approximated.

2.4.4 Practical sample size for origin-destination survey

The importance of O-D data in improving transit service is apparent from Figure 2.3 for all

operations planning levels. O-D matrices ought to be constructed for developing new transit

routes, terminals, and stops, changing route structure and strategies, and amending time

tables and vehicle and crew schedules. Therefore, this section will be devoted to a practical

method of finding an adequate sample size for O-D surveys.

While sampling from where people want to go and to where (by time of day, purpose of

trip, etc.), we enter a process of interaction between events from a statistical perspective. We

want to survey N people, with the outcome of each falling into one of the O-D cells (each

cell is a specific O-D). The experiment of such N independent trials is called a multinomial

experiment (generalization of the binomial experiment) and has a useful probability (mul

tinomial) distribution for discrete random variables. The proper O-D matrix is, in fact, a

contingency table based on a multinomial distribution, in which contingency means the

relationship between O-D classifications. That is, there are $B_1, B_2, \dots, B_k$ possible O-D

relationships with the corresponding $\theta_1, \theta_2, \dots, \theta_k$ probabilities, such that $\sum_{i=1}^k \theta_i = 1$. We are

interested in knowing the probability that, in N trials, B

1 will be observed $\times 1$ times, B 2 will

be observed $\times 2$ times, ..., and B k will be observed $\times k$ times where $\sum_i i \times N$. Because of inde

pendency, the following is obtained: $P(B_1) P(B_2) P(B_3) \dots P(B_k) = \theta_1^{\times 1} \theta_2^{\times 2} \dots \theta_k^{\times k} = \dots = \theta_1^{\times 1} \theta_2^{\times 2} \dots \theta_k^{\times k}$, and $P(x_1, x_2, \dots, x_k) = \frac{N!}{x_1! x_2! \dots x_k!} \theta_1^{x_1} \theta_2^{x_2} \dots \theta_k^{x_k}$ (multinomial distribution) (2.8)

where P represents probability.

Consider now x_i as a binomial variable with average $N\theta_i$ and variance $N\theta_i(1 - \theta_i)(1 - n/N)$,

where n is the sample size. For $n \ll N$, the fraction n/N approaches zero. Further, for

large n, say over 50, x_i can be considered normally distributed as in Equation 2.3: $x_i \sim$

$\text{Normal}[\mu = n\theta_i, \sigma^2 = n\theta_i(1 - \theta_i)]$.

Let $p_i = x_i/n$ represent a group i proportion within the population. Thus, $p_i \sim \text{Normal}$

$[\theta_i, \theta_i(1 - \theta_i)/n]$. The sample size in Equations 2.6 and 2.7 yields $n \geq \frac{z^2 \sum p_i(1 - p_i)}{\alpha}$, where s,

the estimate of σ , can be represented by $p_i(1 - p_i)$. Since p_i is the group proportion, $p_i(1 - p_i)$

can never go beyond 0.25 (the case in which $p_i = 0.5$). Hence, it can be used as an upper

bound for the sample size. The required sample size, therefore, is $n \geq \frac{z^2 \sum p_i(1 - p_i)}{\alpha}$, (2.9)

In cases in which we cannot consider $n \ll N$, we can as a practical step use the following

expression for no sample size: $n_o = n/(1 - n/N)$

We can now establish a stepwise procedure for choosing the sample size:

1. Ratio estimation in each cell of the O-D matrix. That is, what is the ratio, R, between the number of passengers traveling from a given origin to a given destination and

the population (which ratio can be estimated by performing a small pretest)?

2. Determination of the required precision in each cell of the O-D matrix.

3. Determination of the confidence interval.

4. Determination of the base ratio for the sample size selection procedure. It should be noted that the selection of the smallest ratio in the O-D matrix will result in the largest sample size, n_m . However, an average ratio will result in a smaller sample size than n_m , but with reduced confidence intervals for cells having lower ratios than the average.

5. Selection of the sample size based on given precision tables (see example in the following text).

The end of this chapter presents the precision table for commonly used sample sizes for an

O-D survey: 500, 1,000, 2,000, 5,000, and 10,000. For example, there are in Table 2.1

three origins and three destinations, a, b, and c, and the estimated ratios, R , between

the number of passengers for each O-D cell and the true population. We then assume

for all the cells a precision of $T0_i / p_i = \pm 10\%$ and a confidence interval of 95%. The base

ratio R is selected to be the average, or close to the average, of all the ratios in Table 2.1,

$R_{12} = 0.76$ ().

Table 2.2 extracts the relevant portion of the entire precision table at the end of the

chapter and emphasizes the precisions for $R_{12} = 0.76$ for all commonly used n . That is,

For $n = 500$, the precision is $\pm 3.74\%$.

For $n = 1,000$, the precision is $\pm 1.678\%$.

For $n = 2,000$, the precision is $\pm 1.187\%$.

For $n = 5,000$, the precision is $\pm 7.51\%$.

For $n = 10,000$, the precision is $\pm 5.31\%$.

It may be seen that none of the precision levels is $\pm 10\%$. In order to attain greater precision,

we can decide on $\pm 7.51\%$ (the closest to $\pm 10\%$ from below) and a sample of 5000 people.

Consequently, for $n = 5000$, and based on Table 2.2, the precision is ± 12.08 for the cell with

$R = 0.05\%$.

For 0.10, the precision is $\pm 8.32\%$.

For 0.15, the precision is $\pm 6.60\%$.

For 0.20, the precision is $\pm 5.54\%$.

These cell-based precisions are also emphasized in Table 2.2. Finally, more transit-related

sample size formulae appear in the transit data collection manual UMTA (1985). Table 2.1 Example of ratios between each origin-destination cell and the true population From To a b c a - 0.05 0.20 b 0.15 - 0.10 c 0.15 0.05 -

2.5 LITERATURE REVIEW AND FURTHER READING

This section reviews papers that deal with the collection of transit data, focusing on data on

service performance and ridership. These data are necessary for operations planning in the

short run and for network design in the long run. The collection of general travel behavior

this review.

According to the transit data collection design manual (UMTA, 1985), the route-specific

data needed for effective basic operation p

lanning include total boardings, loads at key

points, running time, schedule adherence, revenues, and some passenger characteristics. At

the design level, data include passenger origins and destinations, fare category distribution,

boardings and alightings by stop, transfers between routes, passenger attitudes, and passen

gers' travel behavior. The manual divides the techniques for collecting transit data into four

groups, according to the identity and location of the surveyor: ride checks, which are taken

on board; point checks, taken by an observer standing at the roadside; driver checks, with

the data collected by the driver; and automated checks, in which the information is collected

by machines. Eight different types of counts and readings are described: on/off passenger

counts, counts of boardings only, passenger-load counts, farebox readings, revenue counts,

transfer counts, O-D stop counts, and passenger surveys. The manual also presents sam

pling methods, techniques for sample size determination, calculation of conversion factors

to enable survey results to represent a whole population, and data accuracy determination. Table 2.2 Precision table for the example R = Ratio 95% Confidence interval n = Sample size 500 1,000 2,000 5,000 10,000 0.01 87.21 61.67 43.61 23.58 19.50 0.02 61.36 43.39 30.68 19.40 13.72 0.03 49.84 35.24 24.92 15.76 11.14 0.04 42.94 30.36 21.4 13.58 9.60 0.05 38.21 27.02 19.10 12.08 8.54 0.06 34.69 24.53 17.35 10.97 7.76 0.07 31.95 22.59 15.97 10.10 7.14 0.08 29.72 21.02 14.86 9.40 6.65 0.09 27.87 19.71 13.94 8.81 6.23 0.10 26.30 18.59 13.15 8.32 5.88 0.11 24.93 17.63 12.47 7.88 5.58 0.12 23.74 16.78 11.87 7.51 5.31 0.13 22.68 16.03 11.34 7.17 5.07 0.14 21.72 15.36 10.86 6.87 4.86 0.15 20.87 14.75 10.43 6.60 4.67 0.16 20.08 14.20 10.04 6.35 4.49 0.17 19.37 13.70 9.68 6.12 4.33 0.18 18.71 13.23 9.35 5.92 4.18

0.19 18.10 12.80 9.05 5.72 4.05 0.20 17.53 12.40 8.77 5.54
3.92

The review of transit data collection techniques by
Booz-Allen and Hamilton (1985)

documents the most common collection techniques.
Nine leading, manual collection techniques

techniques are described: ride checks, point c
checks, boarding counts, farebox readings, revenue

counts, speed and delay studies, running-time c
checks, transfer counts, and stop studies.

The review describes each technique and p
rocedures for its application, as well as the
data

items collected by the technique. Instructions are given on
how to aggregate and manipu

late the data.

Bamford et al. (1984) assert that onboard p
assenger information can be gathered either

by using self-completed questionnaires or by a
n interviewer. They examine the benefits of

a data collection method based on a very short
questionnaire that can be self-completed

by means of rub-off, within no more than 1 min and without
the use of a pen. They claim

that this method is cheaper than most other
onboard surveying methods and yields higher

response rates.

A modular approach to onboard automatic data c
ollection systems (NCTRD report,

1986) defines standard requirements for such s
ystems. The authors argue that these sys

tems offer advantages over manual systems in many transit
applications: improved data

processing time, lower collection costs, and better quality data. They describe a standard

microprocessor-based system that provides an electronic means of gathering passenger,

fare, and scheduled data. Step-by-step guidelines for selecting and implementing the data

collection system are presented, as is a discussion of the cost of such a system. The paper

also enumerates the requirements for equipping a bus with an automatic data collection

system and details a technique for determining the necessary number of equipped buses,

which works out to almost 10% of the fleet. Perhaps the most important conclusion derived

from equipping a bus fleet with automatic data collection devices is the need for coordina-

tion with the existing dispatching procedures. The cooperation of dispatchers and drivers,

particularly in regard to modifications of dispatching practices, is crucial.

Further, McCollom (1987) focuses on the calculation of conversion factors, which are

used for converting survey results (number of boardings, lightings, peak load, revenue, etc.)

into representative values. The authors discuss statistical aspects of the conversion factor

approach, including sample size estimation and determination of accuracy measures. They

outline several scenarios concerning available input data, since preliminary information

about mean values is not always available. An optimal sampling plan to minimize costs is

offered.

Macbirar (1989) d iscusses d ata r etrieval f rom e
lectronic t icket m achines. T hese m achines

c onstitute a p otential s ource o f i nformation n ot
o nly o n fa res a nd r evenues b ut a lso o n p
as

senger characteristics that change with fare, origins and
destinations, and riding times. The

p aper d iscusses i n d etail t he i mportance o f k
eeping a s imple, w ell-arranged e numeration

s ystem f or t rips, s tops, e tc., w hich i s n
ecessary f or d ocumenting t he d ata. T he p aper a
lso

d iscusses d ata n eeds f or r eimbursing c
oncession-fare s chemes, t hat i s, t he w ay o
perators

r eceive compensation for being forced to provide reduced
prices for speci c sectors.

Weinstein and Albom (1998) discuss techniques for
collecting data on qualitative features

o f t ransit p erformance t hat a re d if cult t o m
easure, s uch a s s tation/vehicle cl eanliness. T hey

p resent a methodology, including rating criteria, for
estimating these features.

Richardson (1998) d escribes a s urvey t echnique t
hat h as b een u sed t o m easure t icket

usage r ates. T his i nformation sh ould a ssist i n
t he a llocation o f fa rebox r evenue t o o perators

of a private transit system. Because of the
unrepresentative sampling method, a principal

c omponent o f t he m ethodology p resented i s a w
eighting p rocedure, b ased o n d ata p artially

c ollected b eforehand. A c ombination o f q
uestionnaires a nd o bservational s urveys i s u sed
i n

order to obtain accurate bias estimates.

Navarrete (1999) describes a methodology for using digital assistants instead of paper

records when inspectors collect field data. Criteria for the efficiency of the digital data collection

collection device are developed; among them are ease of data entry and retrieval, utilization of

existing technologies, and effective handwriting recognition. A suitable device is chosen and

tested. According to field tests, the most difficult problem encountered is the learning curve

for the handwriting-recognition program.

Barua et al. (2001) discuss the use of geographical information systems (GIS) for transit

OD data analysis following an onboard transit survey. GIS is used not just for geocoding

origin and destination locations but also as a tool for checking survey data quality and

for validating the data analysis. A suggested methodology for weighing survey results and

converting them into a representative trip matrix is based on spatial criteria. The GIS-based

approach for validating the quality of the results includes a comparison of transit link volume

with data from passenger counts and a parallel validation of subway transit demand

characteristics.

Some of the literature in the field of data collection presents the features of specific software

or hardware systems, without detailing methodologies.

For example, Barnes and

Urbanik (1990) describe a software for a computerized data collection process. Rossetti

(1996) reports on a transit-monitoring system based on radio frequency identification that

integrates AVL with automatic passenger counting. Other papers on AVL and AVM are

discussed in Chapters 18 and 19.

Morency et al. (2007) have analyzed the potential of smart card data for measuring

the variability of urban public transit network use. A discussion on the organization of

data using an object-oriented approach is presented, and then the measures of spatial and

temporal variability of transit use for various types of card are defined and estimated

using the datasets presented. Data-mining techniques have also been occupied to identify

transit use cycles and homogenous days and weeks of travel among card segments and

at various times of the year. This study shows that smart card data have the potential

to give a continuous profile of transit use by various types of cards. The aforementioned

text has been concluded using well-formatted data that were obtained with the help of an

object-oriented approach to identify all the objects of the system. From the experiments,

the authors were able to conclude that it is possible to observe regularity indicators by

using raw information on boardings only, even though little individual information was

available.

Wilson et al. (2009) describe two applications of an automated data collection system using

and passenger behavior, which have been developed jointly between Massachusetts Institute

of Technology (MIT) and the Chicago Transit Authority (CTA), using CTA's automatic fare

collection (AFC) and AVL systems. The two applications put forth by the authors illustrate

the ability of engaging inference methods to estimate the passenger O-D travel patterns and

rail path choice behavior without using any survey data. The authors conclude by suggesting

possible extensions to the proposed approaches, which will ultimately enable comprehensive

coverage of multimodal public transport systems. Such work will eventually provide valuable

insights on the multidimensional performance aspects of the public transit services and

its impact on passenger travel behaviors.

Mandelzys and Hellinga (2010) present a methodology for identifying bus stops that are

unable to meet schedule adherence performance standards and the factors causing in-

adequate performance. The methodology is designed to be automated and therefore can be

applied efficiently to AVL/APC data. The analysis presented in this paper can be ultimately

conducted for an entire transit network to identify those bus stops that are not meeting

the expected performance standards. The authors suggest that the use of this proposed

method will enable transit agencies to more efficiently

identify service quality issues and

their respective root causes. Furthermore, this information, in conjunction with other

visualization techniques, can provide useful insights for transit network planners to make

effective decisions at a low cost.

Pelletier et al. (2011) present a review of the smart card data use in public transit. The

authors emphasize the importance of smart card AFC systems on collecting large quantities

of very detailed data on onboard transactions. These data can be very useful to transit planners

from the day-to-day operation of the transit system to the strategic long-term planning

of the network. This review covers several aspects of smart card data use in the public

transit context by first presenting the available technologies and legal issues related to the

dissemination of smart card data, data storage, and encryption. Then, the various uses of

the data at three levels of management are described: strategic (long-term planning), tactical

(service adjustments and network development), and operational (ridership statistics and

performance indicators). A number of issues such as data/journey validation, economic feasibility,

security, new modeling and analysis methods, and technological improvements are identified

as potential future challenges for the smart card transit operators. Notwithstanding privacy

concerns, the authors state that smart card data

data can provide a rich continuous source of

data, which can be used to understand the behavior of transit users, compare planned and

implemented schedules, undertake adjustments to the systematic schedules, and apply the

survival models to ridership.

Lee and Hickman (2014) use the extensive farecard transaction data for deriving useful

information about transit passenger behavior, namely, trip purpose or activity. They illustrate

how the farecard data can be used to infer trip purpose and to reveal travel patterns

in an urban area. A case study is used to demonstrate the process of trip purpose inference

based on farecard data from Metro Transit in the Minneapolis-St. Paul metropolitan area.

The proposed approach only concerns a small set of transit users, although in conjunction

with O-D pairs, it is possible to combine a unbiased source of transit demand data with a

inferred trip purpose—which is typically a difficult task to undertake through traditional

surveys. The authors state that future research efforts should focus on validation of the data

with onboard surveys and subsequent interpretations to gain further insights to the transit

users' trip purposes.

Munizaga and Palma (2012) present a methodology for estimating a public transport

O-D matrix from smartcard and GPS data for Santiago, Chile. The proposed method is

applied to two 1-week datasets obtained for different time periods. From the data available,

detailed information about the time and position of boarding public transportation are

obtained, and an estimation of time and position of alighting for over 80% of the boarding

transactions is generated. The results are available at any desired time-space disaggregation.

After some postprocessing and after incorporating expansion factors to account for unobserved

served trips, public transport O-D matrices are built. The main contributions of this paper

are moving from a trip segment perspective to a trip O-D estimation and using generalized

time rather than physical distance to estimate alighting point for transfers and destinations.

Even though some information are missing, such that evasion and trip segments in non-

integrated transport modes and validation with exogenous data needs to be conducted, the

magnitude of the achieved success rates suggests that this new source of information has as a

great potential and can actually replace an important part of the large O-D surveys.

Table 2.3 summarizes the main features and characteristics of the literature reviewed by

categories. Having presented the overview of the first key in Figure 2.1, we will proceed in

the next 14 chapters to deal with the s

second key: how to plan and decide intelligently. The

third key will be discussed in Chapters 17 and 18.

Table 2.3 Summary of the main features of the literature reviewed, by category

Source	Type of data collection	Data collection method	Contribution of the paper	Includes methodology for weighting survey results?	Includes discussion of data accuracy?
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Transit Data

Collection Design

Manual (1985)	Various	Various	Review	Yes	Yes
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Review of Transit

Data Collection

Techniques (1985)	Various	Various	Review	Yes	Yes
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Bamford et al.

(1984)	Various	On board questionnaires	Technical features of the questionnaire	No	No
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Modular Approach

to On Board

Automatic Data

Collection

Systems (1986)	Various	On board automatic	Definition of system requirements	No	No
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Furth and

McCollom (1987)	Various	Any method	Analytical method	Yes	Yes
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Macbirar (1989)	Various	On board automatic (through ticket machine)	Review/ discussion	No	No
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Weinstein and

Albom (1998) Qualitative data, such as cleanliness
 Observation (on board or at station) Analytical method
 No Yes

Richardson (1998) Ticket-usage rates On board,
 combination of questionnaires and observations
 Analytical method Yes Yes

Navarrete (1999) Various Digital assistants New technology
 No No

Barua et al. (2001) Passenger origin- destination Any
 (focus is on analysis method) Analytical method Yes Yes

Barnes and

Urbanik (1990) Various On board automatic New technology
 No No

Rossetti (1996) Various Combination of AVL and onboard
 automatic counters New technology No No

Morency et al.

(2007) Variability of urban public transit network use
 Smart card Review/ discussion No Yes (Continued)

Precision table for five survey sample sizes Precision table
 90% Confidence interval 95% Confidence interval 99% Confidence
 interval n

R 500 1,000 2,000 5,000 10,000 500 1,000 2,000 5,000 10,000
 500 1,000 2,000 5,000 10,000

0.01 73.19 51.75 36.60 23.15 16.37 87.21 61.67 43.61 27.58
 19.50 114.62 81.05 57.31 36.25 25.63

0.02 51.49 36.41 25.75 16.28 11.51 61.36 43.39 30.68 19.40
 13.72 80.64 57.02 40.32 25.50 18.03

0.03 41.83 29.58 20.91 13.23 9.35 49.84 35.24 24.92 15.76
 11.14 65.50 46.32 32.75 20.71 14.65

0.04 36.04 25.48 18.02 11.40 8.06 42.94 30.36 21.47 13.58
 9.60 56.43 39.90 28.22 17.85 12.62

0.05 32.06 22.67 16.03 10.14 7.17 38.21 27.02 19.10 12.08
 8.54 50.21 35.51 25.11 15.88 11.23

0.06 29.12 20.59 14.56 9.21 6.51 34.69 24.53 17.35 10.97
7.76 45.60 32.24 22.80 14.42 10.20

0.07 26.81 18.96 13.41 8.48 6.00 31.95 22.59 15.97 10.10
7.14 41.99 29.69 20.99 13.28 9.39

0.08 24.95 17.64 12.47 7.89 5.58 29.72 21.02 14.86 9.40
6.65 39.06 27.62 19.53 12.35 8.74

0.09 23.39 16.54 11.70 7.40 5.23 27.87 19.71 13.94 8.81
6.23 36.63 25.90 18.31 11.58 8.19

0.10 22.07 15.60 11.03 6.98 4.93 26.30 18.59 13.15 8.32
5.88 34.56 24.44 17.28 10.93 7.73

0.11 20.92 14.80 10.46 6.62 4.68 24.93 17.63 12.47 7.88
5.58 32.77 23.17 16.38 10.36 7.33

0.12 19.92 14.09 9.96 6.30 4.45 23.74 16.78 11.87 7.51 5.31
31.19 22.06 15.60 9.86 6.98

0.13 19.03 13.46 9.51 6.02 4.26 22.68 16.03 11.34 7.17 5.07
29.80 21.07 14.90 9.42 6.66

0.14 18.23 12.89 9.12 5.77 4.08 21.72 15.36 10.86 6.87 4.86
28.55 20.19 14.28 9.03 6.38

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0.16 16.85 11.92 8.43 5.33 3.77 20.08 14.20 10.04 6.35 4.49
26.39 18.66 13.20 8.35 5.90

0.17 16.25 11.49 8.13 5.14 3.63 19.37 13.70 9.68 6.12 4.33
25.45 18.00 12.73 8.05 5.69

0.18 15.70 11.10 7.85 4.96 3.51 18.71 13.23 9.35 5.92 4.18
24.59 17.39 12.29 7.78 5.50

0.19 15.19 10.74 7.59 4.80 3.40 18.10 12.80 9.05 5.72 4.05
23.78 16.82 11.89 7.52 5.32

0.20 14.71 10.40 7.36 4.65 3.29 17.53 12.40 8.77 5.54 3.92
23.04 16.29 11.52 7.29 5.15

0.21 14.27 10.09 7.13 4.51 3.19 17.00 12.02 8.50 5.38 3.80
22.34 15.80 11.17 7.07 5.00

0.22 13.85 9.79 6.93 4.38 3.10 16.50 11.67 8.25 5.22 3.69
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21.08 14.90 10.54 6.67 4.71

0.24 13.09 9.26 6.55 4.14 2.93 15.60 11.03 7.80 4.93 3.49
20.50 14.50 10.25 6.48 4.58 (Continued)

Table 2.3 (Continued) Summary of the main features of the literature reviewed, by category

Source Type of data collection Data collection method
Contribution of the paper Includes methodology for
weighting survey results? Includes discussion of data
accuracy?

Wilson et al.

(2009) Various On board automatic Analytical method No No

Mandelzys and

Hellinga (2010) Various Automatic vehicle location and
automatic passenger counting (APC) Analytical method No
No

Pelletier et al.

(2011) Various Smart card Review No No

Lee and Hickman

(2014) Passenger-trip purpose or activity Farecard
transactions Analytical method No Yes

Munizaga and

Palma (2012) Passenger origin- destination Smart card and
GPS Analytical method No No

Precision table for five survey sample sizes Precision table
90% Confidence interval 95% Confidence interval 99% Confidence
interval n

R 500 1,000 2,000 5,000 10,000 500 1,000 2,000 5,000 10,000
500 1,000 2,000 5,000 10,000

0.25 12.74 9.01 6.37 4.03 2.85 15.18 10.74 7.59 4.80 3.39
19.95 14.11 9.98 6.31 4.46

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18.94 13.39 9.47 5.99 4.24

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18.02 12.75 9.01 5.70 4.03

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16.05 11.35 8.02 5.08 3.59

0.35 10.02 7.09 5.01 3.17 2.24 11.94 8.45 5.97 3.78 2.67
15.70 11.10 7.85 4.96 3.51

0.36 9.81 6.94 4.90 3.10 2.19 11.69 8.26 5.84 3.70 2.61
15.36 10.86 7.68 4.86 3.43

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0.38 9.40 6.64 4.70 2.97 2.10 11.20 7.92 5.60 3.54 2.50
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0.39 9.20 6.51 4.60 2.91 2.06 10.96 7.75 5.48 3.47 2.45
14.41 10.19 7.20 4.56 3.22

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5.46 3.86 2.44 1.73

0.70 4.82 3.41 2.41 1.52 1.08 5.74 4.06 2.87 1.81 1.28 7.54
5.33 3.77 2.38 1.69 (Continued)

Precision table for five survey sample sizes Precision table
90% Confidence interval 95% Confidence interval 99% Confidence
interval n

R 500 1,000 2,000 5,000 10,000 500 1,000 2,000 5,000 10,000
500 1,000 2,000 5,000 10,000

0.71 4.70 3.32 2.35 1.49 1.05 5.60 3.96 2.80 1.77 1.25 7.36
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1.43 1.01 0.64 0.45

0.98 1.05 0.74 0.53 0.33 0.23 1.25 0.89 0.63 0.40 0.28 1.65
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0.99 0.74 0.52 0.37 0.23 0.17 0.88 0.62 0.44 0.28 0.20 1.16
0.82 0.58 0.37 0.26

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3 Chapter 3: Frequency and Headway Determination

This chapter addresses the topic of determining frequencies on transit routes, a problem that

must receive attention, either explicitly or implicitly, several times a year. The chapter provides

insights and a solution for intelligently integrating resource saving and an effective level of

service. The basic premise here is that such an integration is achievable. The importance of ridership information has led transit agencies to introduce automated

surveillance techniques or, alternatively, to increase the amount of manually collected data.

Naturally, the transit agencies are expected to gain useful information for operations planning by

obtaining more accurate passenger counts. This chapter describes and analyzes several appropriate

data collection approaches in order to set frequencies and headways efficiently. Four different

methods are presented for deriving frequency, two based on point-check (max load) data and

two on ride-check (load-profile) data. A ride check provides more complete information than

does a point check, but at greater cost; there is a question whether the additional information

gained justifies the expense. The four methods, all of which are based on available load profiles,

provide the planner with adequate guidance in selecting the type of data collection procedure.

In addition, the planner or scheduler can evaluate the minimum expected vehicle runs when the

load standard is eased and avoid overcrowding (in an average sense) at the same time. The chapter starts by describing the significance of proper frequency calculation, followed

by a presentation and interpretation of the four methods. The fourth section established a

criterion, the profile-based load, for selecting either the point-check or the ride-check data

collection technique. In Section 3.5, two examples demonstrate the four methods, one of the

examples employing real-life, heavy bus-route data. The chapter ends with a literature review

and exercises for practicing the methods. The majority of the chapter is suitable for practitio

ners; however, the end of Sections 3.2 and 3.4 may be skipped to some extent. They can be

replaced by the procedure depicted, in flowchart form, in Figure 3.10 in Section 3.5. Dealing with data collection to affect the number of vehicle runs brings to mind the follow

ing real-life anecdote. A chief planner in a bus agency once observed that there was approved

extra, but unnecessary, runs on one route. He then found out that this run was performed at

the end of a duty day in order to bring the driver close to his home. The chief planner immedi

ately ordered the local planner to eliminate this run. Six months later, the chief planner learned

to his surprise that this run still existed. Angrily, he then called the local planner and asked for

an explanation, to which the latter replied: "Oh, I told this driver that on the day the checker is

there, not to make this run." We can wrap up this corner with some pithy advice and admonition: the chains of traditional

methods are too weak to be felt until they are too strong to be broken. In transit operations

planning, therefore, there should always be a window for flexibility, new initiatives, and changes.

3.1 INTRODUCTION

One of the major factors in determining transit service is the selection of the most suitable

frequency (vehicles/hour) for each route in the system, by time of day, day of week, and day

type. This chapter provides an initial focus on the subject of the second key in Figure 2.1,

intelligent decision making when planning transit service. This second key will accompany

us through Chapter 16.

Figure 1.2 showed that frequency determination was in essence of paramount importance

for creating transit timetables. Furth and Wilson (1981) wrote that transit agencies typically

used service standards as the basis for setting frequencies, while combining this action with

experience, judgment, and passenger counts. The service standards in normal use appear

in Table 1.2: crowding level, allowed maximum standees, and upper (policy) and lower

limits on headways. Table 1.3 emphasizes that these standards cannot be rules of thumb;

they must be based on determined criteria. This is especially true because of the high cost

involved in providing higher frequencies, and that may not always be necessary. Prudent

transit management requires a balance between increasing frequency and the cost of its

implementation. Often, we observe that some transit agencies, following the belief that last

week's frequency is good for this week, do not adjust their service to fluctuating demand. To

those with this philosophy, we can only say: it is better to wear out by changing than to rust

out by believing that what you have is good enough.

This chapter presents different methods that allow not only for efficient frequency setting

but also for a sensitivity analysis of possible changes. Basically, the objectives of this chapter

are twofold: one, to set vehicle frequencies in order to maintain adequate service quality and

minimize the number of required vehicle runs; two, to construct a new valuation tool that will

efficiently allocate the cost of gathering appropriate passenger-load data at the route level.

The main input data for this chapter will be point-check and ride-check data. These

types of passenger counts are explicitly described in Section 2.2 and Figure 2.2; they will

be combined in the analysis with crowding level and minimum frequency (inverse of policy

headway) standards. It is common for load-profile data to be gathered annually or every

few years along the entire length of the transit route (ride check). Usually, the most recent

passenger-load information will come from one or more selected stops along that part of the

route where the vehicle carries its heaviest loads (point check). Point-check information is

routinely surveyed several times a year for the purpose of possible schedule revisions, which

can range from completely new timetables for new or revised routes to daily adjustments to

accommodate changes in Central Business District (CBD) and industrial area working hours

and school-dismissal times. A ride check provides more complete information than a point

check but is more expensive because of the need for either additional checkers to provide

the required data or an automated surveillance system (e.g., automatic passenger counter

[APC], AVL). The question is whether the additional information justifies the expense of

gathering it. This chapter explores ways in which a transit agency can use the old profile to

determine which method, ride check or point check, is more appropriate and less costly for

collecting the new data.

Following Furth and Wilson (1981), transit agencies use methods to set headways that

are commonly based on existing service standards of crowding level and policy headway.

These standards are based on two requirements: (1) adequate space will be provided to meet

passenger demand and (2) an upper bound value is placed on the headways to assure a minimum

frequency of service. The first requirement is a appropriate for heavily traveled route

hours (e.g., peak period) and the second for lightly traveled hours. The first requirement

is usually met by a widely used peak-load factor method (point check), which is similar to

the max load procedure—both will be explained in Sections 3.2 and 3.3, respectively. The

second requirement is met by policy headway, which usually does not exceed 60 min and is

restricted in some cases to 30 min. Occasionally, a lower-bound value is set on the headway,

based on productivity or revenue/cost measures. There are also mathematical programming

techniques to approach the problems of route design and service frequency simultaneously.

Such a technique was adopted by Furth and Wilson (1981) to find the appropriate head

way that would maximize social benefit, subject to the constraints of total subsidy, fleet

size, and bus-occupancy levels. Nevertheless, mathematical programming models are hardly

employed, since they cannot incorporate practical operational considerations in the optimi

zation analysis. A further review of the literature appears at the end of the chapter.

This chapter consists of three major parts. First, two methods are presented to derive

vehicle frequencies based on point-check (max load) data. Second, two further methods are

proposed, using ride-check (load-profile) data. Third, a criterion is established for determin

ing the appropriateness of each data collection method. All four methods are applied—and

analyzed—on a route basis, following Ceder (1984). These three parts are succeeded by two

examples (one real life): a literature review and exercises.

3.2 MAX LOAD (POINT-CHECK) METHODS

One of the basic objectives in the provision of transit service is to ensure adequate space to

accommodate the maximum number of onboard passengers along the entire route over a

given time period. Let us denote the time period (usually an hour) as j . Based on the peak

load factor concept, the number of vehicles required for period j is $F_j = \frac{P_{mj}}{c_j \cdot \gamma_j}$ (3.1)

where P_{mj} is the average maximum number of passengers (max load) observed on board in period j , c_j represents the capacity of a vehicle (number of seats plus the maximum allowable standees), γ_j is the load factor during period j , $0 < \gamma_j \leq 1.0$

For convenience, let us refer to the product $\gamma_j \cdot c_j$ as d_{oj} , the desired occupancy on the vehicle

at period j . The standard γ_j can be set so that d_{oj} is equal to a desired fraction of the capacity

(e.g., d_{oj} = number of seats). It should be noted here that if P_{mj} is based on a series of measure

ments, one can take its variability into account. This can be done by replacing the average

value in Equation 3.1 with $P_{b.mj} + S_{pj}$, where b is a predetermined constant and S_{pj} is the

standard deviation associated with P_{mj} .

The max load data are usually collected by a trained checker, who stands and counts

at the transit stop located at the beginning of the max load section(s). This stop is usually

determined from the old ride-check data or from the information given by a mobile super

visor. Often, the checkers are told to count at only one stop for the entire day instead of

switching among different max load points, depending on period j . Certainly, it is less costly

to position a checker at one stop than to have several checkers switching among stops. Given

that a checker is assigned to one stop, that which apparently is the heaviest daily load point

along the route, we can establish the so-called Method 1 for determining the frequency

associated with this single stop $F_{P_d}, F_{j=1,2}, q_{ij}, m_{dj}, o_{mj} = \dots = \max (3.2) P_{\max} P_{md_i} S_{ij} j = 1, q_{ij} = \dots = \epsilon = \dots = \sum \sum P_{P_{md_j}} i \cdot j =$

where F_{mj} is the minimum required frequency (reciprocal of policy headway) for period j ; there are q time periods S represents the set of all route stops including the last stop $i \cdot j$ is the daily max load point P_{ij} is a defined statistical measure (simple average or average plus standard deviation) of the total number of passengers on board all the vehicles departing stop i during period j

The terms P_{md_j} and P_{md} are used for the (average) observed load at the daily max load point

at time j and the total load observed at this point, respectively.

Figure 3.1 exhibits an example of passenger counts (Example 1) along a 10 km route with

six stops between 6:00 and 11:00 a.m. The second column in the table in this figure presents

the distances, in km, between each stop. The desired occupancy and minimum frequency

are same for all hours, and hence their time period subscript is dropped: $d_o = 50$ passengers

and $F_m = 3$ vehicles, respectively. The set of stops S

includes $\forall$ i is, $j = 1, 2, \dots, 5$, each period

of 1 h being associated with a given column. The last column in the table represents $\sum_{j=1}^5 i_j P_{ij}$

in which each entry in the table is P_{ij} (an average value across several checks). Thus, i^* is

the third stop with $P_{md} = 1740$, and P_{mdj} in Equation 3.2 refers only to those entries in the

third row.

The second point-check method, or Method 2, is based on the max load observed in each

time period. That is, $F_{Pd}, F_{j1,2}, \dots, F_{j2j} = \max_{i \in S} P_{ij} = \dots \max$ (3.3)

where $P_{mj} = \max_{i \in S} P_{ij}$, which stands for the maximum observed load (across all stops) in

each period j .

In the table in Figure 3.1, the values of P_{mj} are circled, and a rectangle is placed around P_{md} .

Figure 3.1 also illustrates passenger counts in three dimensions (load, distance, and period),

from which the hourly max load is observed for the first 3 h. The results of Equations 3.1

and 3.2 applied to Example 1 appear in Table 3.1 for both frequency (F_{kj}) and headways (H_{kj})

rounded to the nearest integer, where $k = 1, 2$. The only nonrounded headway is $H_{kj} = 7.5$ min,

since it fits the so-called clock headways: these have the feature of creating timetables that

repeat themselves every hour, starting on the hour. Practically speaking, $H_{kj} = 7.5$ can be

implemented in an even-headway timetable by alternating between $H_{kj} = 7$ and $H_{kj} = 8$.

We will also retain nonrounded F_{kj} s and show in the next chapter how to use these deter

mined values for constructing timetables without even headways. Stop # Distance (km) to next stop Average observed load (passengers), by hour
 Total load (passengers) 6:00- 7:00 7:00- 8:00 8:00- 9:00
 9:00- 10:00 10:00- 11:00 1 2 50 136 245 250 95 776 2 1 100
 510 310 208 122 1250 3 1.5 400 420 400 320 200 1740 4 3 135
 335 350 166 220 1206 5 2.5 32 210 300 78 105 725 Notes: (1)
 Route length is 10 km, and stop #6 is the last stop
 (2) For all hours, $d_o = 50$, $c = 90$ passengers, $F_m = 3$
 veh/h Average passenger load 11:00
 Distance traveled (km) 500 400 300 200 100 0 6:00 7:00 8:00
 9:00 10:00 2 3 4 5 6 Hourly max load Stop # 1

Figure 3.1 Example 1: Five-hour load profiles, with indications of hourly and daily max load points.

Table 3.1 Frequency and headway results for Example 1 in Figure 3.1, according to Methods 1 and 2 Method 1 (daily max load point) Method 2 (hourly max load point)

Period j F_{1j} (veh/h) H_{1j} (min) F_{2j} (veh/h) H_{2j} (min)

6:00-7:00 8.0 7.5 8.0 7.5

7:00-8:00 8.4 7 10.2 6

8:00-9:00 8.0 7.5 8.0 7.5

9:00-10:00 6.4 9 6.4 9

10:00-11:00 4.0 15 4.4 14

Although Method 1 has an advantage over Method 2 through cost saving in data gathering,

it cannot be traded off with accuracy of the results. That is, it is less costly and more

convenient to retain a checker at one transit stop throughout the entire working day than

to assign the same checker or others to a different stop at every period j. Consequently,

we will now concentrate on a comparison of the two methods: if the difference is statistically

not significant, then the routine use of Method 1 will be preferred. This comparison

may be carried out by the chi-square statistic, χ^2 , for testing the hypothesis concerning the

population variances of the data supplied for each method. We assume that the random

sample of passenger counts comes from a normal distribution (see Section 2.4 in this book

for basic statistics), with averages μ_1 , μ_2 and variances σ_1^2 , σ_2^2 , for Methods 1 and 2, respec

tively. The hypothesis that $\sigma_1^2 = \sigma_2^2$ will be examined, using the χ^2 test while knowing that

σ_2^2 is the theoretical (expected) variance with which it is to be compared. The statistic χ^2 ,

$\chi^2 = (n - 1) s^2 / \sigma_2^2$ has a chi-square probability distribution with $(n - 1)$ degrees of freedom,

where s^2 is computed from the sample. If $\sigma_1^2 > \sigma_2^2$, both s^2 and χ^2 will be larger than expected.

The hypothesis will be rejected if $\chi^2 > \chi_{\alpha}^2$, where χ_{α}^2 is chosen so that $P(\chi^2 > \chi_{\alpha}^2) = \alpha$ (see

Section 2.4); the χ_{α}^2 tables can be found, for example, in the website <http://www.stat.lsu.edu/>

[exstweb/statlab/Tables/TABLES98-Chi.html](http://www.statlab/Tables/TABLES98-Chi.html), (LSU Experimental Statistics, 2015).

It should be noted that there is often a misjudgment of statistical data when chi-square

tests are incorrectly utilized (Ceder and Dressler, 1980). For example, it is incorrect to com

pare the frequency or headway results given by the two methods. Instead, the actual obser

vations should be compared, because any transformation of the actual units distorts the

χ^2 test. If the observed (O) and the expected (E) data are multiplied by k , then the calculated

χ^2 is equal to k times the corrected χ^2 value: $\Sigma \frac{(O - E)^2}{E} = \Sigma \frac{(kO - kE)^2}{kE}$

$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$

In our

case, the observed items are the passenger counts of Method 1 and the expected items are

the counts of Method 2, using the following

calculation: $\chi^2 = \sum_{j=1}^q \frac{(P_j - E_j)^2}{E_j}$ $P = [420, 200, 510, 220]$
 $E = [510, 220, 420, 220]$ (3.4)

where i is the daily maximum load point.

A comparison of the two methods can now be applied to Example 1 using Equation

3.4. This yields $\chi^2 = (420 - 510)^2 / 510 + (200 - 220)^2 / 220 = 17.7$; for $\alpha = 0.05$, we obtain

$\chi^2_{\alpha} = 9.49$. from the chi-square table, with a degree of freedom of four. Hence, the possibility

of assigning a checker only to stop 3 of Example 1 must be rejected, and we can skip over

the Method 1 results.

3.3 LOAD-PROFILE (RIDE-CHECK) METHODS

The data collected by ride check enable the planner to observe the load variability among

the transit stops, or what is termed the load profile. Usually, a recurrent, unsatisfactory

distribution of loads will suggest the need for possible improvements in route design. The

most common operational strategy resulting from observing the various loads is short turn

ing (shortlining). A start-ahead and/or turn-back point(s) after the start and/or before the

end of the route may be chosen, creating a new route that overlaps the existing route. This

short-turn design problem is covered in Chapter 15. Other route design-related actions

using load data are route splitting and route shortening,

both of which are dealt with in

Chapter 13. This section will use the ride-check data for creating more alternatives to derive

adequate frequencies, while assuming that the route remains same. Nevertheless, we know

that in practice, the redesign of an existing route is not an activity often undertaken by

transit agencies.

Two examples of load profiles are illustrated in Figure 3.2. These profiles are extracted

from Example 1 in Figure 3.1 for the first and third hours. It may be noted that in most

available transit-scheduling software (see, e.g., Section 1.2), these load profiles are plotted with

respect to each stop without relating the x-axis to any scale. A more appropriate way to plot

the loads is to establish a passenger-load profile with respect to the distance traveled from

the departure stop to the end of the route. It is also possible to replace the (deterministic) dis

tance by the average running time; in the latter case, however, it is desirable for the running

time to be characterized by low and persistent variations. These plots furnish the important

evaluation measures of passenger-km and passenger-hour as is also shown in Figure 3.2.

Let us observe the area marked by dashed lines in Figure 3.2. If a straight line is drawn

a cross the load profile where the number of passengers is equal to the observed average

hourly max load, then the area below this line but above the load profile is a measure of

superfluous productivity. When Method 2 is used to derive the headways, this area represents Passenger load at hour 6:00 - 7:00 Distance traveled (km) 400 50 32 135 100 2715 Empty space (seat)-km 2 1 1.5 3 km 2.5 100 200 300 400 0 1285 passenger-km Passenger load at hour 8:00 - 9:00 400 245 300 350 310 800 Empty space (seat)-km 100 200 300 400 0 Max load point 3200 passenger-km L=Route length=10 km

Figure 3.2 Two load profiles from Example 1 with the same 8-vehicle frequency, but with different passenger- and empty space-km.

empty space-kilometers. Furthermore, if d_{oj} in Method 2 is equal to the number of seats-

often, this is the desired occupancy or load factor used-then this measure is empty-seat

kilometers. In light of this measure of unproductive service, we can see in Figure 3.2 that

the 8:00-9:00 load profile is more than twice as productive as the 6:00-7:00 profile, though

both have the same (max load point-based) frequency. We can now use the additional infor

mation supplied by the load profile to overcome the problem exhibited in Figure 3.2 when

using Method 2. This can be done by introducing frequency determination methods based

on passenger-km rather than on a max load measure. The first load-profile method consid

ers a lower-bound level on the frequency or an upper bound on the headway, given the same

vehicle capacity constraint. We call this Method 3, and it is expressed as follows:
$$F_{max} A d L, P c F 3 j j o j m j m j = \cdot \dots \dots \dots, (3.5) A$$

$$P_{ij} = \frac{1}{L_{ij}} \sum_{i \in S} \sum_{j \in S} l_{ij} = \frac{1}{L_{ij}} \sum_{i \in S} \sum_{j \in S} l_{ij},$$

where l_{ij} is the distance between stop i and the next stop $(i + 1)$. A_{ij} is the area in passenger-km under the load profile during time period j . L_{ij} is the route length.

The other notations were previously defined in Equations 3.1 through 3.3.

One way to look at Method 3 is to view the ratio A_{ij}/L_{ij} as an average representative of

the load P_{ij} (regardless of its statistical definition), as opposed to the max load (P_{mj}) in

Method 2. Method 3 guarantees, on the average basis of P_{ij} , that the onboard passengers at

the max load route segment will not experience crowding above the given vehicle capacity

constraint. This method is appropriate for cases in which the planner wishes to know the

number of vehicle runs (frequency) expected, while relaxing the desired occupancy stan-

dard constraint and, at the same time, avoiding situations in which passengers are unable

to board the vehicle in an average sense. Using the results of Method 3 allows planners

to handle (1) demand changes without increasing the available number of vehicles, (2)

situations in which some vehicles are needed elsewhere (e.g., breakdown and maintenance

problems or emergencies), and (3) occasions when there are fewer drivers than usual (e.g.,

owing to budget cuts or problems with the drivers' union). On the other hand, Method 3

can result in unpleasant travel for an extended distance in which the load (occupancy) is

above d_{oj} .

To eliminate or control the possibility of such an undesirable phenomenon, we introduce

another method, called Method 4. This Method 4 establishes a level-of-service consider

ation by restricting the total portion of the route length having loads greater than the desired

occupancy. Method 4 takes the explicit form
$$A d L, P c F 4j j o j m j m j = \cdot \beta \beta \beta \beta \beta \beta, (3.6)$$
 subject to (s.t.) $l L, i i I j j \in \Sigma \leq \cdot \beta$

where mathematically $I j = \{i:(P i j / F j) > d o j \}$; in other words, $I j$ is the set of all stops i in time

period j , such that the load $P i j$ exceeds the product of $d o j$ times the frequency $F 4j$, and βj is the

allowable portion of the route length at period j in which $P i j$ can exceed the product $F 4j \cdot d o j$.

The other notations in Equation 3.6 were previously defined. By controlling parameter βj , it

is possible to establish a level-of-service criterion. We should note that for $\beta j = 0$ and $\beta j = 1.0$,

Method 4 converges to Methods 2 and 3, respectively.

The load profile of Example 1 (see Figure 3.1) is presented in Figure 3.3 pertaining to the

hour 9:00-10:00. The considered load level associated with Method 3 is simply the area

under the load profile, divided by $L = 10$ km, or 188.1 passengers in this case. This is a

average load profile. However, all loads between stops 1 and 4 (4.5 km), which stretch out

for 45% of the route length, exceed this average. To avoid the load exceeding the desired

profile for more than a predetermined percentage of the route length, Method 4 can be

introduced. If this percentage is set at 40% of the route length, the considered load will

be 208 passengers, thus allowing only the stretch between stops 1 and 2 (2 km) and that

between stops 3 and 4 (1.5 km) to have this excess load. We term this situation Method 4

(40%). Setting the percentage to 20% results in an average of 250 passengers; in the case of

10% (1 km), the considered load level converges to the Method 2 average of 320 passengers.

Figure 3.4 illustrates the fundamental trade-off between the load profile and maximum load con-

cepts. We will show it for the case of Method 4 (20%) and the 9:00-10:00 h. Based on Figure 3.1

data and Equation 3.6, with $\beta_j = 0.2$, we attain $F_4 = \max(250/50, 320/90, 3) = 5$ veh/h from

raising the considered load level from 188.1 to 250 passengers.

The vehicle assigned departures will, in an average sense, carry $320/5 = 64$ passengers

between stops 3 and 4 (14 more than the desired load of 50). Nonetheless, this excess load

for 1.5 is traded off with $(320-250) \cdot 0.5 = 59.5$ empty space-km as is shown in Figure 3.4.

This trade-off can be interpreted economically and perhaps affect the ticket tariff.

In case that the calculated frequency in Equations 3.5 and 3.6 is the result of P_{mj}

/c, then

the considered load will be determined by the product $(P_{mj}/c) \cdot d_{oj}$. Table 3.2 shows the results

for Methods 3 and 4, in which the percentage of route length allowed to have an excess load

in Method 4 is set at 10%, 20%, and 30%.

We may observe that in the Method 3 results in Table 3.2, the first hour relies on P_{mj}/c or,

specifically, $400/90 = 4.44$ veh/h. When we turn to Method 4 for this first hour, the results Passenger load at hour 9:00 – 10:00
Distance (km) 0 250 208 320 1 2 3 4 5 6 7 8 9 10 78 188.1
Passenger-km Method 2, Method 4 (10%) Method 4 (40%)
Considered load level by method 166 Method 4 (20%) Method 3 (=188.1)

Figure 3.3 Load profile from Example 1 between 9:00 and 10:00 with considered load levels for three methods and Method 4 standing for 10%, 20%, and 40% of the route length.

of Method 2 for 10% are attained, since the max load stretches along more than 10% of the

route length. For 20% and 30%, the vehicle capacity constraint still governs. In the second

hour, 7:00–8:00, the following is obtained for Method 3: $F_3 = \max(2942/50 \cdot 10, 510/90, 3) =$

5.88 veh/h. Continuing in the second hour for Method 4 (10%), the average max load of

294.2 passengers rises to 420, resulting in $F_4 = 8.40$ veh/h. Table 3.2 continues to be fulfilled

in the same manner. It should be noted that, as in Table 3.1, all the headways are rounded

to their nearest integer.

Although we aim at a resource saving using Methods 3 and 4,

there is a question as to

whether this saving justifies the additional expense involved in using ride check as opposed

to point check. The next section attempts to answer this question by constructing a crite

riterion suggesting when to use the point check or, otherwise, the ride-check data collection

technique.

3.4 CRITERION FOR SELECTING POINT CHECK OR RIDE CHECK

This section will test an assumption that particular load-profile characteristics suggest the

data collection technique to be used. The basic idea is to provide the transit agency with

adequate preliminary guidance in selecting the type of technique based on old load profiles.

The assumption will be investigated whether a relatively flat profile suggests the use of a Passenger load
 at hour 9 : 00 - 10 : 00 Distance (km) 0 250 208 320
 166 78 1 2 3 4 5 6 7 8 9 10 Trade-off 100 200 300 0 =L 1881
 Passenger-km

Figure 3.4 Load profile from Example 1 (9:00-10:00) using Method 4 (20%), with an indication of trade-off between this method and Method 2 (more crowding in return for less empty space-km).

Table 3.2 Frequency and headway results for Example 1 in Figure 3.1 for Methods 3 and 4 Method 4 Method 3 10% 20% 30%

Period j F 3j (veh/h) H 3j (min) F 4j (veh/h) H 4j (min) F 4j (veh/h) H 4j (min) F 4j (veh/h) H 4j (min)

6:00-7:00 4.44 14 8.00 7.5 4.44 14 4.44 14

7:00-8:00 5.88 10 8.40 7 8.40 7 6.70 9

8:00-9:00 6.40 9 8.00 7.5 7.00 9 7.00 9

9:00-10:00 3.72 16 6.40 9 5.00 12 5.00 12

10:00-11:00 3.07 20 4.40 14 4.40 14 4.00 15

point-check procedure (Method 1 or 2) or whether a
ride-check procedure (Method 3 or 4)

would be more appropriate.

One property of the load profile is its density, p . This is
the observed measure of total

passenger-km (i.e., total ridership over the route),
divided by the product of the length of the

route and its max load. The product, in the denominator, is
the passenger-km that would be

observed if the max load existed across all stops. Thus,
the load-profile density for hour j , p_j , is $p_j = \frac{A_j}{L \cdot M_j}$ (3.7)

The load-profile density is used to examine profile
characteristics. High values of p indicate a

relatively flat profile, whereas low values of p
indicate load variability among the route stops.

One way to explore load-profile density is to ap-
proximate the observed shapes of profile

curves through a mathematical model. The lognormal mo-
del will be selected for this pur-

pose, since it provides a family of curves that can
be controlled by varying the two pa-ram-

eters, μ and σ . The lognormal model takes the
form $f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$ (3.8)

The equation satisfying the optimum conditions, $df(x)/dx =$
 0 , is $x e^{(\ln x - \mu)^2 / 2\sigma^2} = -\mu \sigma$, in which Equation

3.8 reaches maximum. In our case, the maximum coincides
with the max load. This con-

tinuous model can only approximate some of the observed
load profiles, since it has only one

peak and represents monotonically increasing and decreasing

functions before and after this

peak, respectively. Nonetheless, this model is useful for observing some general differences

between the ride-check and point-check methods.

In order to compare the methods, $f(x)$ will be used as a normalized load (the load divided

by the max load) and x as a normalized distance (the distance from the initial route stop

divided by the length of the route). At a given time interval of 1 h, j , the considered max load

is $P_{mj} = 650$ passengers. Given that $d_{oj} = 65$, $c = 100$, and $F_m = 2$, the determined frequency

and headway for both Methods 1 and 2 are $F_j = 10$ and $H_j = 6$. By applying this information

to Methods 3 and 4, and using a variety of lognormal curves, we will obtain the frequen

cies and headways shown in Table 3.3. The results in this table are arranged in order of

increasing density. For Method 3, the capacity constraint determines the values of F and H

up to and including $p = 0.64$ and up to different p values (if any) for Method 4. Examples of

the lognormal normalized curves are shown in Figures 3.5 and 3.6, the latter showing the

termination of the frequency for one curve ($\mu = -1.5$, $\sigma = 1.0$). Given that the normalized

route length is 1.0, then the different percentages (for this length) in Figure 3.6 are set for the

Method 4 criterion, with an explicit calculation example for the 10% case.

From Table 3.3, it appears that, for Method 3, the ride-check data result in the same

rounded headway as the point-check data for $p >$

0.84. For Method 4, the ride-check and

point-check information tend to yield the same headways for $p > 0.34$ (10% case), $p > 0.50$

(20% case), and $p > 0.68$ (30% case).

In practice, the transit agency wishes to save vehicle runs and, eventually, to perform the

matching between demand and supply with fewer vehicles. As we will show in Chapters 4

and 6, different headway values do not necessarily save vehicle runs or reduce the required

fleet size. However, the analysis of the profile-density measure can be used by the transit

agency as a preliminary check before embarking on a more comprehensive analysis. The

following are practical observations: (1) for densities below 0.5, savings are likely to result

by gathering the load-profile information and using ride-check methods (alternatively, the

profile can be examined with such low p values for short-turn strategies, to be discussed

in Chapter 14); (2) for densities between 0.5 and 0.85, actual comparisons can be made

between the point-check and ride-check methods, along with Table 3.3 frequencies (F),

rounded headways (H), and density (p) for different load-profile configurations (lognormal

model based), using Methods 3 and 4 further saving considerations (for constructing time

tables and vehicle schedules); and (3) for densities above 0.85, it is likely that the majority of

the required information for the headway c

alculation can be obtained from a point-check

procedure (either Method 1 or 2). A further simplification and explicit practical criterion is to

use Methods 3 and 4 for $p \leq 0.5$, and Method 1 and 2 otherwise. It may be argued that for

the range $0.5 < p \leq 0.85$, the use of load-profile methods cannot produce significant gains

over the maximum load methods, although theoretically it may be justified. This argument is supported by the relatively small amount of passenger-km in the trade-off situation exhibited

in Figure 3.4.

3.5 CONCLUSION (TWO EXAMPLES)

Another simple example, which we call Example 2, will now ascertain fully the procedures

employed. This example will be followed by a real-life example that includes the results of

Table 3.3 Frequencies (F), rounded headways (H), and density (p) for different load-profile configurations (lognormal model based), using Methods 3 and 4

Lognormal

model

parameters Profile density Method 4 Method 3 10% 20% 30%

$\mu \sigma p F H F H F H F H$

-1.0 0.2 0.18 6.50 a 9 7.60 7 6.50 a 9 6.50 a 9

-1.5 0.5 0.25 6.50 a 9 8.46 7 6.50 a 9 6.50 a 9

-1.5 1.0 0.32 6.50 a 9 8.36 7 6.50 a 9 6.50 a 9

-1.0 1.5 0.34 6.50 a 9 7.55 7 6.50 a 9 6.50 a 9

-1.0 1.2 0.43 6.50 a 9 9.00 6 b 7.05 8 6.50 a 9

-1.0 0.6 0.44 6.50 a 9 9.50 6 b 8.10 7 6.50 a 9

-0.8 0.5 0.47 6.50 a 9 9.55 6 b 8.40 7 6.50 a 9

-1.0 0.9 0.48 6.50 a 9 9.45 6 b 8.05 7 7.05 8

-0.4 1.5 0.50 6.50 a 9 9.05 6 b 7.35 8 6.50 9

0.0 0.5 0.56 6.50 a 9 9.92 6 b 9.67 6 b 9.27 9

-0.5 0.5 0.57 6.50 a 9 9.76 6 b 9.11 6 b 8.16 6

-0.4 0.5 0.59 6.50 a 9 9.81 6 b 9.31 6 b 8.46 7

-0.4 1.2 0.62 6.50 a 9 9.65 6 b 8.85 6 b 7.80 7

-0.5 1.0 0.64 6.50 a 9 9.79 6 b 9.04 6 b 8.19 7

-0.4 0.8 0.68 6.77 8 9.87 6 b 9.42 6 b 8.72 7

0.5 1.5 0.75 7.46 8 9.86 6 b 9.36 6 b 8.76 6 b

0.0 1.0 0.76 7.63 7 9.93 6 b 9.68 6 b 9.23 6 b

0.5 1.0 0.78 7.77 7 9.97 6 b 9.87 6 b 9.72 6 b

1.0 1.5 0.84 8.41 7 9.96 6 b 9.76 6 b 9.46 6 b

1.5 1.5 0.87 8.72 6 9.97 6 b 9.92 6 b 9.82 6 b

Note: For Methods 1 and 2, $F = 10$, $H = 6$, where $d_o = 65$,
 $c = 100$.

a Whenever $F = 6.5$, $H = 9$, the capacity constraint is met.

b Whenever $H = 6$, ride check and point check yield the
same headway.

the four methods. This section will end by portraying the
methodology proposed in low

chart form, with an orientation toward practice.

Example 2, which is presented and illustrated
in Figure 3.7, will also be used through

out Chapters 4 and 5. The basic required input
consists of (1) distance between stops,

(2) desired occupancy per hour (for every p

lanned vehicle type), (3) minimum frequency

per hour (could be the inverse of policy headway), (4)
number of observed (scheduled

usually) vehicles each hour, (5) observed load
(an average value or a consideration of its

variability) between each two adjacent stops for
or each hour, and (6) vehicle capacity (for

every planned vehicle type). A 2 h operation period (06:00-08:00) is chosen for simplicity ($\mu = -1.5$, $\sigma = 1.0$) $0.84 P = 650$ $0.84 = 546$, $F_4(10\%) = \max$
 $0.59 P = 384$, $F = 6.5$ 546 650 100 65 $0.39 P = 254$, $F = 6.5$ 0.33
 $P = 208$, $F = 6.5$ 10% 20% 30% 35% (of entire length) Normalized
distance distance route length 0.2 0.4 0.6 0.8 1.0 0.1 0.2
0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0, 2, = 8.4 Normalized
loaded load max load

Figure 3.6 An approximated load profile based on the
lognormal model, with indications of different (route)
length percentages associated with Method 4. $(-1.5, 1.0)$
 $(-0.5, 1.0)$ $p = 0.75$ $p = 0.34$ $p = 0.87$ $(-1.0, 1.5)$ $(0.5, 1.5)$
 $(1.5, 1.5) = (\mu, \sigma)$ 0.2 0.4 0.6 0.8 1.0 0.1 0.2 0.3 0.4 0.5
0.6 0.7 0.8 0.9 1.0 $p = 0.32$ $p = 0.64$ Normalized distance
distance route length Normalized loaded load max load

Figure 3.5 Five approximated load profiles based on the
lognormal model with normalized scales.

in this example. Furthermore, as a matter of decision, all
vehicles depart on the hour per

taining to this next hour.

Let us recall the assumption that the observed
loads in each hour are based on a uni

form passenger-arrival rate (demand). That is, the
number of passengers carried by the first

vehicle in each hour is divided proportionally
to reflect the demand in two time intervals:

(1) the start of the hour and the departure time of the
first vehicle that hour and the (2) last

departure time in the previous hour and the start of the

considered hour.

Hence, vehicles departing on the hour carry a demand from the previous hour. Figure 3.8

interprets this proportionality for the 7:15 departure; in this figure, vehicle profiles are

shown for each observed vehicle. We assume that the load on the 7:15 vehicle contains $2/5$

of the 6:00-7:00 demand (10 min from 6:50 to 7:00 and 15 min from 7:00 to 7:15). This

proportion is marked in Figure 3.8 and inserted in the table of Figure 3.7 under average load

by hour.	Stop #	Distance (km)	to next stop	Average observed load (passenger), by scheduled departure time
Average load, by hour* Total hourly load (passengers)				
6:15	6:50	7:15	7:35	7:50
6:00-7:00	7:00-8:00	1	2	22
25	50	60	45	67
135	202	2	1.5	52
40	90	87	75	128
216	344	3	4.5	35
65	85	44	83	134
178	312	Number of observed scheduled vehicles		
2	2	Calculated Desired occupancy		
50	60	Minimum frequency		
1	2	Vehicle capacity (seats + allowed standees)		
90	90	Area under the load profile (passenger-km)		
929	1395	Average hourly load (passengers)		
Distance (km)				
0	1	2	3	4
5	6	7	8	100
177	216	128	135	67
6:00-7:00	174.38	(7:00-8:00)	116.12	(6:00-7:00)
Average load level				
200	* See calculation in Figure 3.8 under the 7:15 departure.			
7:00-8:00	134			

Figure 3.7 Example 2: three-hour data for frequency and headway determination by each of the four methods.

We can see from the table in Figure 3.7 that the daily max load point is stop 2, with

stops 3 and 2 being the hourly max points for the 6:00-7:00 and the 7:00-8:00 h, respec

tively. The load profiles are depicted on the same scales in Figure 3.7. The last column,

last row, and “average load by hour” column in Figure 3.7 are calculated items. The dif

ference between Method 1 and 2 is simply between the set of

counts (128, 216) and (134,

216), respectively. Using Equation 3.4, we obtain $\chi^2 = 0.269$, where $\chi^2 = 3.84$ for $\alpha = 0.05$

with a single degree of freedom. Hence, Method 1 can substitute for Method 2 for future Average hourly load (passengers) 1 2 3 4 5 6 7 8 20 40 60 80 100 0 20 40 60 80 0 20 40 60 80 100 0 7:50 7:15 6:50 6:15 Departure 22 52 35 65 25 40 50 30 90 54 51 85 60 87 44 45 75 83 Distance (km) 20 40 60 80 100 0 20 40 60 80 0 7:35 Pertains to 6:00-7:00 demand Average vehicle peak load

Figure 3.8 Example 2 data by vehicle load, showing the demand (passenger-km) that pertains to the 6:00- 7:00 period, assuming uniform passenger-arrival rates.

passenger counts. Moreover, by using Equation 3.7 and Figure 3.7, we obtain $p_{6-7} = 0.87$

and $p_{7-8} = 0.81$, meaning that the max load-point data collection procedure is acceptable.

The results of Example 2 for all four methods are shown in Table 3.4 along with their calcu

lations. Equations 3.2 and 3.3 may be applied for Methods 1 and 2, respectively, and Equations

3.5 and 3.6 for Methods 3 and 4, respectively. Table 3.4 considers, for Method 4, the constraint

that no more than 20% of the route length is allowed to have an excess load, provided that

the considered load level for this excess load is equal to or below the hourly max load divided

by vehicle capacity. The headways in Table 3.4 are rounded off to their nearest integer. The

frequencies in this table will be used in the next chapter (using the same example) as noninteger

numbers while attempting to construct alternative (and automated) timetables.

For a real-life situation, we will demonstrate the procedures developed when using ride-check

data from an old Los Angeles Metro (previously called SCRTD) bus route. This route, 217, was

considered a heavy route and was characterized by 60 stops. All the trips on this route cross the

daily max load point. The complete northbound ride-check input for route 217 appears in Table

3.5. The content of this table is comprehensive: it includes the distances (in km) to the next stop

and the stop name in the first and second columns, respectively; the last column represents the

total load for the whole day. The entries in Table 3.5 are based on rounded average values fol

lowing several checks. Vehicle capacity, required for Methods 3 and 4, is 80 passengers.

An automated program is constructed for the four methods to be commensurate with

Equations 3.2, 3.3, 3.5, and 3.6. The intermediate result of this program, which concerns

max load information, is illustrated in Table 3.6. The daily max load point is the Fairfax/

Rosewood stop, with a total of 4413 observed passengers over the entire day. The hourly max

load points for each hour appear in Table 3.6. Occasionally, there are multiple-stop results;

for example, at 10:00-11:00 and 22:00-23:00. The results of the load-profile density mea

sure reveal that all $p < 0.5$; hence, the ride-check count is the data collection technique appropriate

for this route. The frequency and the midway results for route 217 are summarized in

Table 3.7 for all four methods. The chi-square comparison between Method 1 and 2 for every

hourly overall hours, using $\alpha = 0.05$, shows that the hypothesis concerning equal methods is

rejected; therefore, the hourly maximum load data collection technique should be applied for this

dataset. Nonetheless, as we mentioned before, the ride check is the appropriate technique.

The full results, in Table 3.7, are presented solely for comparison purposes; they show that

from 6:00 to 7:00 and after 19:00, route 217 relies on its minimum frequency. Methods 2

and 3 serve as upper and lower bounds on the frequency, respectively; and Method 4 ranges

between the two, depending on the percentage of route length allowed to have excess load.

A graphical comparison of the frequency results of Methods 2, 3, and 4 (20%) for both

directions of route 217 is presented in Figure 3.9. This figure also contains the actual observed

Table 3.4 Frequency (F) and rounded headway (H) results for Example 2, by each of the four methods Method 1 Method 2 Method 3 Method 4 (20%)

Hour F H F H F H F H

6:00–7:00 2.56 23 2.68 22 2.32 26 2.68 22

7:00–8:00 3.60 17 3.60 17 2.91 21 2.95 20

Calculation

6:00–7:00 $\max \{ \frac{128}{50}, \frac{1}{2}, \frac{56}{60}, \frac{1}{2}, \frac{134}{90}, \frac{1}{2}, \frac{134}{90}, \frac{1}{2} \} = \max \{ \frac{134}{90}, \frac{1}{2}, \frac{134}{90}, \frac{1}{2}, \frac{134}{90}, \frac{1}{2}, \frac{134}{90}, \frac{1}{2} \} = \frac{134}{90}$

7:00–8:00 $\max \{ \frac{216}{60}, \frac{2}{3}, \frac{60}{60}, \frac{2}{3}, \frac{216}{90}, \frac{2}{3}, \frac{216}{90}, \frac{2}{3} \} = \max \{ \frac{216}{90}, \frac{2}{3}, \frac{216}{90}, \frac{2}{3}, \frac{216}{90}, \frac{2}{3}, \frac{216}{90}, \frac{2}{3} \} = \frac{216}{90}$

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0014:00-15:0015:00-16:0016:00-
17:0017:00-18:0018:00-19:0019:
00-20:0020:00-21:0021:00-22:00
22:00-23:0023:00-24:0024:00-25
:00#ofbusesobserved61088878881
0896533222Minimumfrequency(veh
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ccupancy(passengers)6070706050
50505050607070606050505050Di
st.(km)StopnameTotal0.16Adams/
Washingt1327115241111641091623
188232513080.26Fairfax/Adams21
681483916272681619243823123725
15030.27Fairfax/Washingt228916
048223229152320354023153865159
60.26Fairfax/Apple251011635126
342916232340452414386516370.27
Fairfax/Venice2911218353374337
22263460513219389527650.27Fair
fax/VeniceF3712421784524451383
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rome40131192895249544238607765
35198119739810.24Fairfax/Pickf
ord451511959448505442406078673
51991297310180.24Fairfax/Satur
n48167211975850554539607363311
891397310560.24Fairfax/Pico562
172461168378849573941059749301
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49853311412159415840.00Fairfax
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Olympic/Ogden59332278152123991
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517114412414118114214915113473
4017151912622690.24Fairfax/8th
St6635733017515012814518816415

6170158774218161913623780.29Fairfax/Wilshire5436334923628825
035435634239738234319091402436
18841210.29Fairfax/6thSt553693
512382912573633653473983863431
899441273418841740.29Fairfax/D
rexel5437635523429125838137835
03983913391919342273418942190.
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623534094593772189859444420942
890.35Fairfax/Oakwood523913512
372712823153663524164673702201
0657485122943830.34Fairfax/Ros
ewood4937032624628829232737636
741848137121410257505120844130
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gh4411217315523824629135834440
64113531919151454819835840.35F
airfax/SantaMo3390140148154174
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9429310.35Fairfax/Fountain3592
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1 : 0 0 - 1 2 : 0 0 1 2 : 0 0 - 1 3 : 0 0 1 3 : 0 0 - 1 4 :

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17:0017:00-18:0018:00-19:0019:
00-20:0020:00-21:0021:00-22:00
22:00-23:0023:00-24:0024:00-25
:00# of buses observed61088878881
0896533222Minimum frequency (veh
/h)2222222222222222Desired o
ccupancy (passengers)6070706050
50505050607070606050505050Di
st. (km) Stop name Total0.29Fairfa
x/Sunset3270951321271441722492
102702802641448044484820424330
.29Sunset/Genesee3169901331251
431702482062692872591468042484
720424170.29Sunset/St Stanley3071
921351271351712482062602882531
457743464818423970.29Sunset/Ga
rdner3173851321171241652402102
492742411388343464517423170.29
Sunset/Martel32758213111011616
422820823626322812882404745174
22360.29Sunset/Poinsett3373771
271181151712201932322602221298
138554618422120.30LaBrea/Sunse
t36657012211310616321117722225
22181328639544517221300.18LaBr
ea/Hollywood366268118116981561
991722132422061278136564216120
450.18Hollywood/Sycamore366567
118116104155192165208225194122
7838564517120020.18Hollywood/O
range3262661121121071521761661
972041791137536544317119040.18
Hollywood/Highland193349849094
130147154185146146886533553516
215710.18Hollywood/LasPalm1626
457783891201261401681411398558
31553415214500.18Hollywood/Whi
tley15224262677611192116144129
116684327533113212290.18Hollyw
ood/Wilcox14214054626610584108
13111998583818442713211020.19H
ollywood/Cahuenga1120364461669
080961231078550401641251311005
0.19Hollywood/Ivar916333350577
36184111988447351429241018690.
19Hollywood/Vine31419173139353
3688773674725142417916230.19Ar
gyle/Hollywood4201720313532446
991836647201624151006440.19Arg

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y l e / Y u c c a 3 2 0 1 7 1 9 3 0 2 8 3 2 4 4 6 8 8 1 7 9
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1 1 1 0 1 7 8 6 0 4 5 1 0 . 1 1 G o w e r / F r a n k l i n
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1 2 8 9 1 5 1 2 2 5 4 9 2 1 2 4 7 4 6 3 0 0 0 2 1 9 0 . 1 8
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5 4 8 1 9 1 9 5 2 5 3 0 0 0 2 0 1 0 . 1 8 B e a c h w o o d
/ S c e n i c 2 1 0 1 1 1 0 7 5 1 3 1 1 1 6 4 4 1 7 1 8 4 2
5 3 0 0 0 1 7 8 0 . 1 6 B e a c h w o o d / T e m p l e H 2
9 1 0 9 7 4 1 0 9 1 4 4 4 1 4 1 3 3 2 3 3 0 0 0 1 5 6 0 . 1
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(scheduled) frequency. We can easily see that the observed frequencies in both directions rep

resent a n e xcessive a mount o f b us r uns. T he t ransit a gency p rovided t he i nput, i ncluding t he

d esired o ccupancy, s et f orth b y t he p lanners. T he m ethod u sed b y t he a gency's p lanning d epart

ment o verlapped w ith M ethod 2 . T he o bserved f requency, t herefore, p oses a n a pparent e nigma:

w hy d oes t he s cheduled f requency n ot o verlap, o r a t l east c ome c lose t o, t he r esults o f M ethod 2?

More in-depth investigation can provide an answer, which has to do with the objective function

o f t he d rivers w ho a ttempt t o m aximize t heir c omfort. W hen a b us r un i s o verloaded r epeat

edly, t he d river r equests r einforcement; t hat i s, a n a dded r un. W hen a b us r un i s a lmost e mpty,

silence becomes golden. With time, the number of bus runs only increases, and this is the main

reason behind the phenomenon exhibited in Figure 3.9. Incidentally, the agency's schedulers

changed their schedule following the presentation of the Figure 3.9 results. Nonetheless, the actual

data support the use of Method 4 (20%) from a resource-saving perspective. It is interesting to

note that Method 4 (20%) results in a much lower frequency than does Method 2, particularly

in the southbound direction. The lower bound on the frequency needed to accommodate the

passenger load while neglecting the load factor (desired occupancy) is exhibited by the results

of Method 3. In most hours, it is as much as half the observed frequency. Table 3.6 Max load information for LA route 217 northbound, using Methods 1 and 2 Time period Max load point Max load (passengers) 06:00-07:00 Fairfax / San Vice 70 07:00-08:00 Fairfax / 3rd St 401 08:00-09:00 Fairfax / 3rd St 370 09:00-10:00 Fairfax / Rosewood 246 10:00-11:00 Fairfax / 6th St 291 Fairfax / Drexel 11:00-12:00 Fairfax / Rosewood 292 12:00-13:00 Fairfax / Drexel 381 13:00-14:00 Fairfax / 1st St 385 14:00-15:00 Fairfax / 1st St 373 15:00-16:00 Fairfax / Melrose 427 16:00-17:00 Fairfax / Rosewood 481 17:00-18:00 Fairfax / Beverly 377 18:00-19:00 Fairfax / Oakwood 220 19:00-20:00 Fairfax / Oakwood 106 20:00-21:00 Fairfax / Beverly 59 21:00-22:00 La Brea / Hollywood 56 Hollywood / Sycamore 22:00-23:00 Fairfax / Oakwood 51 Fairfax / Rosewood Fairfax / Santa Monica 23:00-24:00 Fairfax / Oakwood 22 24:00-25:00 Fairfax / Drexel 9 Fairfax / 3rd St Fairfax / 1st St Fairfax / Beverly Fairfax / Oakwood All day Fairfax / Rosewood 4413

Finally, the procedures described and interpreted in this chapter will be demonstrated, using

a flowchart (Figure 3.10), which provides a pragmatic overview leading to the use of either

maximum load-point method (Method 1 or 2) or load-profile
Method 4. The method chosen has a

direct impact on the technique required for data
acquisition. From a practitioner's perspective,

we may note that the noticeable importance of
ridership information has led transit a
gencies

to introduce automated surveillance techniques mixed with
manually collected data.

We are also aware of the current c
onsiderations by many transit agencies worldwide

that want to implement APCs. Such systems a
re based on (1) infrared beam interruption,

(2) pressure-sensitive stairwell mats, (3) c
ounts by weighing the load on the vehicle, a
nd

(4) ultrasonic beam interruption. Naturally, the
transit agencies expected to gain useful infor

mation for operations planning by obtaining more accurate
passenger counts. However, as

we mentioned earlier, there is always a point
at which the additional accuracy is not w
orth

the accompanying costs of data collection and
analysis. Finally, in light of the methods p
ro

posed, as opposed to existing methods in use,
the following rule may apply: forgetting is a
s

important as remembering in the practical use of what is
right to do.

3.6 LITERATURE REVIEW AND FURTHER READING

Only papers that focus on the determination of
frequencies or headways with a linkage to the

[illegible]

23:00-24:00 2.00 30 2.00 30 2.00 30 2.00 30 2.00 30 2.00 30

24:00-25:00 2.00 30 2.00 30 2.00 30 2.00 30 2.00 30 2.00 30

Furth and Wilson (1981) mention four common approaches for determining the head

ways of transit routes: policy headways, which are not directly derived from passenger

demand; headways determined according to peak passenger load and vehicle capacity; head

ways designed such that the revenue/cost ratio will not exceed a preset value; and headways

designed to achieve a desired ratio of passenger miles or vehicle miles per hour. An optimization

(nonlinear programming) model and a solution algorithm are developed for setting fre

quencies in a given route network. The model assumes that the demand on each line is elastic;

that is, sensitive to frequency changes. However, there are no demand relationships between

different routes. The model allocates a available buses to routes, subject to subsidy, fleet size,

and vehicle-loading-level constraints. Multiple day periods are taken into account. The

frequencies determined depend on fares, subsidies, fleet size, and the value of waiting time.

Koutsopoulos et al. (1985) develop a programming problem for determining frequen

cies in a transit network with a demand that varies a long daily periods. Operating costs

and travel times are assumed to be time varying, as well. The optimization is subject to

subsidy, fleet size, and vehicle capacity constraints. The model is not based on the common

assumption of a half-headway average waiting time, but on a more sophisticated waiting

time submodel; for example, the situation in which some passengers cannot board the rest

arriving bus because of crowding is considered. Another submodel measures passenger

inconvenience, which is included in the total cost that the model aims to minimize. Since the Frequency (veh/h) Frequency (veh/h) 12 10 8 6 4 2 0 Observed Method 2 Method 3 Method 4 (20%) Northbound 5:00 7:00 9:00 11:00 13:00 15:00 17:00 19:00 21:00 23:00 Time of day Southbound 12 10 8 6 4 2 0

Figure 3.9 Comparison of frequency results of Methods 2, 3, and 4 (20%) and the observed frequency of route 217 in Los Angeles.

solution procedure of the nonlinear problem presented may be complicated, the authors

propose a simplified version of the initial formulation, in which the daily periods are divided

into subperiods, during which headways are constant. The simplified problem can be solved

by, for example, using linear programming.

LeBlanc (1988) introduced a model for determining frequencies, using a modal-split

assignment programming model with distinct transit routes. The author shows how to refine

conventional modal-split assignment models to include features of the proposed methodology.

Multimodal considerations, such as the proportion of transit riders and the influence of

the determined frequencies on road congestion, are taken into account.

Banks (1990) presents a model for setting headways in a transit route system. A case

in which route demands vary with frequency is compared to a fixed-demand case; in the

frequency-dependent demand case, there is no interline dependency, and demand functions

are assumed to be unknown. An optimal solution is discussed in an unconstrained case and

in several cases with subsidy, fleet size, and capacity constraints. In the unconstrained case,

a complete analytical solution is presented, showing that the optimal headway is propor-

tional to the square root of operating cost and of travel time. It is also inversely proportional

to the square root of waiting time and cost and of the number of passengers. No analytical

expressions for optimal headway are developed for the constrained cases.

Although Khasnabis and Rudraraju (1997) do not propose a methodology for frequency

determination, they use simulation experiments to show that preemption strategies that are Use Method 2 for frequency determination Ride-check passenger counts Construction of load profiles (for each hour) Calculation of load profile density ρ (for each hour) Calculation of frequency using Method 3 and Method 4 (for various predetermined route lengths, %) Comparison of passenger counts associated with Method 1 and Method 2 using χ^2 test ($\alpha=0.05$) Retain Method 3 results as lower bound on the frequency, and select the % route length allowed to have excess load for Method 4 $p \leq 0.5$ Methods 1 and 2 alike Use Method 1 for frequency determination Use Method 4 with selected (route length) %

Yes Yes No No

Figure 3.10 Summary in flowchart form of procedures for determining frequency.

used in signalized intersections along the bus route are an important factor that should be

taken into consideration when setting service headway.

Wirasinghe (2003) examines the validity of a traditional method, formulated by Newell, for

determining frequencies. According to this method, frequency is proportional to the square

root of the arrival rate of passengers if the vehicles are sufficiently large and to the arrival rate,

otherwise. According to a modification of this method, suggested by Newell as well, optimal

frequency is also proportional to the square root of the ratio of the value of the passengers'

waiting time to the cost of dispatching a vehicle. The validity of Newell's formulae is examined

in several situations, based on assumptions that are different from the original: uniform head

way during off-peak periods, policy headways, alternative waiting-time assumptions, stochastic

demand, and many-to-many demand. Wirasinghe concludes that the original square root

policy proposed by Newell is applicable, with some modifications, under most conditions.

Kim et al. (2009) determine the optimal bus service frequency by considering bus operation

costs to the service supplier, passenger demand, and so forth. Optimal bus frequency

has been determined, and a timetable for each bus stop has been created by applying the

demand- and travel time-responsive model or the demand- and travel time-responsive model

for critical scheduling areas. Furthermore, a real-time bus headway control model has been

developed, which enables the transit operators to monitor and assess punctuality of a given

service. This model is based on real-time event data, collected from buses in service, such as

bus stop departures and arrivals for buses operating on a line-based timetable, constructed

for each bus stop. Subsequently, this information has been converted to traffic cost per bus

stop, from which the service punctuality could be determined. The model's ability to evaluate

the punctuality of operation made it possible to transmit headway control instructions

(when to decelerate or accelerate) to a bus driver via an onboard unit. Such an approach

allows transmission of headway adjustment instructions, which could reduce traffic cost

per bus stop by aiming at key bottleneck bus stops—where punctuality issues have been

experienced. A model verification process was implemented by using data collected from a

bus management system and integrated transit fare card system for the bus route in Seoul.

To evaluate the reliability and uncertainty of the optimal solution, a sensitivity analysis was

implemented for the various parameters and assumptions used in the models.

Ceder et al. (2013a) describe a methodology to approach even-headway and even-load time

tables by utilizing different available bus sizes. The composition and size of the fleet of buses can

be either given or with a flexibility of introducing new sizes. Two objectives were set. First, minimize

the deviation of the determined headways from a desired even headway. Second, minimize

the deviation of the observed passenger loads from a desired even-load level of the buses at the

maximum-load point. Two heuristics have been developed to attain these objectives: (1) determination

of the best headway with respect to minimizing the load discrepancy from an even-load

level on all buses at the maximum load point and (2) construction of the best timetable to approach

planned even headway and even load at the maximum load point, with the selected headway of (1)

and based on a range of alternative timetables represented in a Pareto frontier. The methodology

has been applied to a case study in Auckland, New Zealand. Overall, the algorithms developed

provide significantly better results than the commonly used timetable. The case study, with even

headway, shows that the approach of this work can reduce the observed 3.8% load discrepancy,

from a desired load at the maximum load point to a discrepancy between only 0%-15% when utilizing

ing properly a variable sizes of buses. This result is attained together with only between 0% and

7% relative time deviation of departure times from those of even-headway timetable.

Ceder et al. (2013b) also present the creation of bus timetables using different vehicle sizes,

but with the use of simulation. This work attempts to bridge between the timetables that are

designed with even headways and timetables with even passenger loads on the vehicles at

the maximum-load points, but with uneven headways. The analysis contains both of these

strategies through the use of the incorporation of a mixed fleet size, running in conjunction.

The timetables have been constructed using two key concepts: assigning capacity and shift

ing departure times. The methodology for the creation of timetables was applied to a real-life

example from Auckland, New Zealand. Several instances of timetables have been created and

tested using simulation. The simulation showed encouraging results for all cases. Comparing

the proposed timetable to a currently operated one, the waiting time of passengers could

be reduced by 15%. At the same time, the number of empty-seat hours can be significantly

improved by 47%, while the number of standee hours did not increase dramatically.

Table 3.8 summarizes by different categories the main features and characteristics of

the literature reviewed. Although we are nished with the four methods described in

Table 3.8 Characteristics of the literature covered in Section 3.6

Source Demand features Constraints Other features
Variables in the frequency formula derived

Furth and Wilson

(1981) Elastic frequency, but without interline influence;
multiperiod Subsidy, fleet size, vehicle-loading level
Fares, subsidy, fleet size, value of waiting time.

Koutsopoulos

et al. (1985) Multiperiod Subsidy, fleet size, vehicle
capacity Waiting-time submodel, passenger inconvenience
submodel

LeBlanc (1988) Multimodal considerations

Banks (1990) Elastic/Fixed cases examined Subsidy, fleet
size, vehicle capacity (several cases examined,
including unconstrained cases) Operating cost, travel
time, value of waiting time, demand level (in the
unconstrained case).

Khasnabis and

Rudraraju (1997) Emphasis on the need-to-consider signal
preemption strategies

Mirasinghe (2003) Among the cases discussed: stochastic
demand, one-to-many demand, many-to-many demand
Alternative waiting-time assumptions Operating cost,
value of waiting time, passenger arrival rate.

Kim et al. (2009) Microscopic demand Subsidy, fleet size,
bus-occupancy levels Bus operation costs to the service.
Supplier and passenger demand.

Ceder et al.

(2013a) Based on ride-check data Number of buses of each
size Heuristic algorithm; sensitivity analysis Multisize
vehicles.

Ceder et al.

associated with a maximum of 20% of the route length having an excess load.

(b) Compare the results of the four methods, and draw conclusions.

(c) What is the trade-off between excess load (in passenger-km) using Methods 3 and 4 (20%) and the reduced empty space-km for each hour?

(d) If the cost of a single empty space-km is 3 ¢/km (operations cost), what is the range of a reduced single tariff per km for route segments suffering excess load that yield a total saving (cost associated with reduced empty space-km) larger than the total reduced (farebox) income?

3.2 Given a transit route and 2 h data with a bus capacity of 80 passengers. The basic ride-check data are shown in the following:

Stop # Distance to next stop (km) Period and departure times 06:00-07:00 07:00-08:00 06:15 06:45 07:10 07:40

1 2 22 25 50 60

2 3 52 40 60 75

3 3 35 65 80 45

Desired occupancy (passengers) 40 50

Minimum frequency (veh/h) 2 3

(a) Determine the hourly max loads and max load points.

(b) Determine the frequency for each hour, using Method 2.

(c) Determine the frequency using Method 4 (40%), in which a maximum of 40% of the route length has excess load.

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4 Chapter 4: Timetable Development

The aim of this chapter is to create and present approaches and procedures for deriv

ing prudent public timetables. These timetables and their compliance mirror the quality

of the transit service provided. Hence, vehicles departing too early or ahead of schedule

need to be restrained, just as those leaving late must be scheduled or rescheduled to be

on time. Because of existing problems of transit reliability, there is a need to improve the

correspondence of vehicle departure times with passenger demand instead of assuming that

passengers will adjust themselves to given timetables (excluding situations characterized by

short headways). The following riddle may shed some light on the substance of adequate,

and accurate, timetables. Two identical shopping malls, M and K, are joined by a road connecting them. A bus departs

every 10 min (headway) from each of the malls to the other mall (M to K and K to M); the travel

time in each direction is the same. A bus stop is located at exactly the midpoint between the

two malls. Passengers arrive randomly at this bus stop and take the first bus to arrive, regard

less of the mall to which it is headed. Observations at this bus stop reveal that 9 out of every

10 passengers board the bus that goes from M to K. Is this possible? If so, how? (The answer

appears at the end of this Practitioner's corner.) The chapter starts with an explication of the advantage of having optional or alternative

public timetables. This feature gives rise to improved service quality and resource saving, using

derived comparison measures that constitute a basis for comparing the alternative timetables.

A procedure for calculating departure times is provided, based on even headways and a smooth

transition between time periods (usually hours), which allows for even average loads on the

last and first vehicles in each period. Thereafter, another procedure is presented for calculating

departure times based on even average passenger loads (as required by the standard), rather

than on even headways, at the hourly max load point. The chapter considers four frequency

determination methods and gives examples (one of them is real life) used in Chapter 3, along

with possible practical decisions on headways and number of departures. The chapter ends with

a literature review and practical exercises. Practitioners are encouraged to read the entire chapter, but to concentrate on the

examples and their figures while studying the two procedures, rather than attempting to

fully comprehend the methodological arguments. The overall analysis of the optional time

tables also contains a direct planner/scheduler intervention in setting frequencies. Such an

intervention is required in situations known only to the planner and does not rely on pas

senger loads. We can now return to the riddle. Indeed, the situation described can be realized if the

timetables from M to K and K to M are set to differ by 1 min difference. Specifically, the bus

departs from M to K at 7:00, 7:10, 7:20, and so on, whereas the bus leaves from K to M at 7:01,

7:11, 7:21, etc. Ostensibly, the average waiting time for the M to K bus is nine times that for the

opposite direction; proportionally, therefore, more passengers will be on hand for the M to K

bus. This example shows how important it is to set timetables adequately. Situations similar to

that of the riddle, but for a network of routes, undoubtedly imply an unreliable transit service.

Improvement in creating timetables can offer a remedy.

4.1 INTRODUCTION

Public transit timetables constitute the most profound bridge between the transit agency

(and/or the community) and the passengers seeking a reliable service. Inadequate and/or

inaccurate timetables not only confuse the passengers but also reinforce the bad image of

public transit as a whole. There is a saying: "A stitch in time would have confused Einstein."

Along these lines, one can say that many stitches in a transit timetable would confuse those

(the passengers) who want to use the service.

The assumption that passengers will adjust themselves to a given timetable (with head

ways of, say, longer than 10 min), instead of planners' adjusting the timetable to passenger

demand, constitutes a major source of unreliable service. When passenger demand is not

met, transit vehicles slow down (i.e., increase dwell time), traveling behind schedule and

entering the inevitable process of further slowing down. This situation will eventually lead

to the known bunching phenomenon with the vehicles that follow. In contrast, a situation of

overestimating demand may result in transit vehicles running a head of time. Neither situa-

tion is observed when the service is frequent and characterized by a low variance in headway

distribution.

The products of the derived frequencies and headways given in Chapter 3 yield the

timetables for the public, for the drivers, and for the supervisors and inspectors. Once

the timetables are constructed, as we see in Figure 1.2, it is possible to initiate the task

of scheduling vehicles and crews for the previously determined trips. Naturally, the tran-

sit agency wishes to utilize its resources more efficiently by minimizing the number of

required vehicles and crew costs. To accomplish this, the planner (or an automated pro-

cedure) examines optional timetables during the vehicle and crew assignment processes.

Optional timetables are derived by shifting departure times and/or reducing the number of

departures without prudent consideration of the load profile. Often, because of the uncer-

tainty involved, some planners/schedulers prefer to shift departure times in small incre-

ments instead of adjusting them to the data-based demand-supply point. This can follow

the rule that if you put aside a small amount for savings every day, you will be s-

surprised to

learn how little was accumulated in a year. Although in all, it is desirable to extend the analysis

of deriving appropriate headways to an evaluation of optional timetables in conjunction

with the required resources.

This chapter, and the next one as well, will continue with an explanation of the second

and planning activity in Figure 1.2, timetable development. The basic challenge is how to

improve the correspondence of vehicle departure times with fluctuating passenger demand.

It is known that passenger demand varies even within the space of 1 h. This dynamic

behavior can be detected through passenger-load counts and information provided by road

inspectors. A more balanced load timetable is achievable by adjusting departure times. The

two procedures presented in this chapter, and one in the next, can be applied to both single

and interlining transit routes, and they can be carried out in a n automated manner. While

the first procedure yields departure times (a timetable) for vehicles with even scheduled

headways, the second procedure produces departure times for vehicles having even average

loads at the hourly maximum load point. A third procedure, discussed in Chapter 5, refers to

balanced (even) loads on individual vehicles at their (individual) peak-load segments. The

key point here is to be able to control the loading, instead of repeatedly exposing p

assengers

to unreliable service resulting from an imbalance in loading situations.

This chapter consists of three major parts. First, a spectrum of optional timetables is

exhibited, along with utilization measures, thus enabling a comparison of various timetables.

Second, a procedure is presented for constructing departure times in which the headways

are evenly spaced. In this context, smoothing techniques are developed in the transition segments

between adjacent time periods, usually hourly segments. Third, another procedure is

offered, this one for a case in which the headways are allowed to be unevenly spaced; here,

the departure times are shifted so as to obtain uniform average loads at the hourly maximum

load point, instead of even headways. All the procedures are applied on a route basis; they

are similar, corrected, and improved over those presented earlier by Ceder (1986, 2003).

A literature review and exercises follow these three parts.

4.2 OBJECTIVES, OPTIONAL TIMETABLES, AND COMPARISON MEASURES

A cost-effective and efficient transit timetable embodies a compromise between passenger

comfort and service cost. A good match between vehicle supply and passenger demand

occurs when transit schedules are constructed so that the observed passenger demand is

accommodated while the number of vehicles used is minimized. This approach helps in

minimizing the agency cost in terms of drivers' wages and capital costs required to purchase

vehicles. This cost-effective concept led us to establish the objectives in creating public

transit timetables:

1. Evaluate optional timetables in terms of required resources.
2. Improve the correspondence of vehicle departure times with passenger demand while minimizing resources.
3. Permit, in the timetable construction procedure, direct vehicle-frequency changes for possible exceptions (known to the planner/scheduler) that do not rely on passenger demand data.
4. Allow the construction of timetables with headway-smoothing techniques (similar to those performed manually) in the transition segments between adjacent time periods.
5. Integrate different frequency-setting methods and different timetable construction procedures.

This chapter attempts to show how to fulfill these objectives. The next chapter continues

this attempt.

4.2.1 Elements in practice

Different transit agencies use different scheduling strategies based on their own experi

ence. Consequently, it is unlikely that two independent transit agencies will use exactly

the same scheduling procedures, at least at the level of detail. In addition, even within

the same transit agency, planners may use different scheduling procedures for different

groups of routes. Therefore, we believe that

when developing computerized procedures,

there is need to supply the schedulers with schedule options, along with an interpretation

and explanation of each option. Undoubtedly, it is desirable that one of the options should

coincide with the scheduler's manual/semimanual procedure. In this way, the scheduler

will be in a position not only to expedite manual/semimanual tasks but also to compare

procedures and methods in regard to the trade-off between the passengers' comfort and

the operating cost.

The number of vehicle runs determined by the timetable and eventually the number of vehi

cles required are sensitive to the procedure that the scheduler uses to construct the departure

times. Some transit agencies routinely round off the frequency F_j to the next highest integer

and then calculate the appropriate headways for the time period. By doing so, they increase the

number of daily departures beyond what is needed to appropriately match demand with sup

ply. Such a procedure may result in nonproductive runs (many empty seat-km). For example,

Table 3.7, in Chapter 3, lists the number of required daily departures $\sum_{j=1}^J F_j$ for Los Angeles

County Metropolitan Transportation Authority (LACMTA) route 217: 86.52 and 78.11 for

Methods 2 and 4 (20% case), respectively. When the quantity F_j is rounded up, we obtain,

respectively, 9.2 and 8.5 daily departures for these two methods. Obviously, by rounding off

F_j to the next highest integer, the level of passenger comfort is increased but at the expense of

unnecessary operating costs. However, in some cases, the rounding-off procedure may be useful

if P_{ij} (see Equation 3.2) is used as an average load when the load variance is high. In this

case (provided that additional runs are made by rounding off F_j upward), possible overcrowd

ing situations may be reduced as opposed to increasing average empty seat-km. Nonetheless,

to overcome the problem of highly variable loads, we can use a statistical load measure that

considers its variance as input to a frequency method; this was explained under Equation 3.2.

4.2.2 Optional timetables

The five objectives enumerated in the previous text and current timetable construction pro

cedures provide the basis to establish a spectrum of optional timetables. Three categories of

options may be identified: (1) type of timetable, (2) method or combination of methods for

setting frequencies, and (3) special requests. These three groups of options are illustrated

in Figure 4.1. A selected path in this figure provides a single timetable. Hence, there are a

variety of timetable options.

The first category in Figure 4.1 concerns alternative types of timetables. The even headway

type simply means constant time intervals between adjacent departures in each time period,

or the case of evenly spaced headways. Even average load refers to unevenly spaced head

ways in each time period, but the observed passenger loads at the hourly max load point

are similar on all vehicles. A second type of timetables entails situations in which even

headways will result in significantly uneven loads. Such uneven-load circumstances occur,

for example, a round work and school dismissal times, but they may in fact occur on many

other occasions. Figure 4.1 shows that the average even load can be managed either at the

hourly max load point (even loads on all vehicles at that point) or at each individual vehicle's

max load point. The average even load at the hourly max load point type is discussed in this

chapter, and the other type in the next chapter.

In the second category of options, it is possible to select different frequency or headway

setting methods. This category allows for the selection of one method or for combinations

of methods for different time periods. The methods considered, and indicated in Figure 4.1,

are the two-point check and the two-ride check, both described in Chapter 3. In addition,

there might be procedures used by the planner/scheduler that are not based on data but on

observations made by road supervisors and inspectors or

other sources of information.

In the third category of selections, we allow for special scheduling requests. One characteristic

of existing timetables is the competition of departure times, usually every hour. These

easy-to-memorize departure times are based on so-called clock headways: 6, 7.5, 10, 12, 15,

20, 30, 40, 45, and 60 min. Obviously, headways less than or equal to 5 min are not thought

to influence the timing of passenger arrivals at a transit stop. The clock headway is obtained

by rounding the derived headway down to the nearest of these clock values. Consequently,

similar to the rounding off upward of frequencies, clock headways require a higher number

of departures than what is actually necessary to meet the demand.

A second possible special request is to allow the scheduler to predetermine the total number

of vehicle departures during any time period. This request is most useful in crises,

when the agency needs to supply a working timetable for a no-operation based on tightly

limited resources (vehicles and/or crews). By controlling the total number of departures

while complying with other requests, the scheduler achieves better results than by simply

dropping departures without any systematic procedure. Furthermore, there might be cases

in which the agency would like to increase the level of service by allowing more departures

in the belief that passenger demand can be increased by providing improved (more frequent)

service. Certainly, this special request can also be approached through varying the desired

occupancy (load factor) standard; however, this option can be a compulsory standard.

Finally, it should be emphasized that not all the paths concerning clock headways in

Figure 4.1 are meaningful. The selection of the even average load type of timetable cannot

be performed if there is a clock-headway constraint. Moreover, the number of departures

cannot be predetermined for clock headways because of the specific time restrictions on

those headways. Even headway Max load (point check) methods Load profile (ride check) methods Intuitive or experience-based determination Daily Max load (Method 1) Hourly Max load (Method 2) With level-of-service criterion (Method 4) Public timetable (list of departure times per time point) Predetermined number of departures Clock headways Without level-of-service criterion (Method 3) Even average load Type of timetable Method for setting frequencies Special requests * Subject covered in Chapter 5. At individual vehicle Max load point* At hourly Max load point

Figure 4.1 Optional public timetables.

4.2.3 Comparison measures

With computerized timetable construction, the transit agency can assess a variety of

optional timetables rather than being limited to examining one or a few. Two interrelated

measures may be useful for the agency to compare optional timetables: (1) number of

required runs (departures) and (2) required single-route fleet size.

The first comparison measure, total number of departures, can serve as an indicator of

the number of vehicles required and also whether or not it is possible to save vehicle runs.

The second comparison measure refers to each route separately and provides an estimate of

the required fleet size at the route level. In a large transit agency, an efficient arrangement of

vehicle blocks includes interlining (switching a vehicle from one route to another) and dead

heading trips. Hence, fleet size is not determined at the route level but at the network level (see

Chapters 7 and 8). The second comparison measure, however, is based on a simple formula

derived by Salzborn (1972) for a continuous time function and explicitly shown by Ceder

(1984) for discrete time points. This formula states that if T is the round-trip time, including

layover and turnaround time, then the minimum fleet size is the largest number of vehicles

departing at any time interval during T . This value is adequate for a single route with a coin

ciding departure and arrival location. Consequently, the second comparison measure can be

used for each direction separately as well as for both directions when selecting the maximum

of two derived values. This single-route fleet size formula will be elaborated in Chapter 7.

4.2.4 Anchoring the timetable to a single time point

A public timetable usually consists of lists of vehicle departure times at all route stops or at

selected stops called time points. Occasionally, this public timetable is given at just a single

point—the route departure point or a major boarding stop. The running time between adjacent

time points may vary from one time period to another, based on ride-check information.

In essence, the timetable can initially be constructed at only one point, and be referred

to as such, and then extended forward and backward using the average running-time information.

That is, in order to ensure an appropriate transit service to meet the variations in

passenger demand, it suffices to construct the timetable at one point. This observation is

stated in the following proposition:

Proposition:

For a route-based transit timetable consisting of more than one time point, the association

of maximum observed load in each time period with only a single time point ensures that the

average vehicle load on each route segment is less than or equal to the desired occupancy.

Proof and interpretation:

For simplicity's sake, we refer to the single time point as the route departure point. The

derived frequency is greater than or equal to the maximum required frequency (across all

route segments) in each time period. The p

roof frequency determined by Method 2 was

provided in Chapter 3; however, this can easily be converted to, and shown by, any of the

four methods. As a note, when only the daily maximum load point is considered, this does not

necessarily imply that the observed maximum load in each period occurs at that point. We can

then treat the problems similarly to the 3D time-load-distance representation in Figure 4.2,

based on Example 2 (which appears in Figure 3.7 in Chapter 3). That is, the shaded two-part

area in Figure 4.2 describes the maximum loads of Example 2 in each hour: 134 (between 06:00

and 07:00) and 216 passengers (07:00-08:00). Calculation of the frequency at distance = 0

in Figure 4.2 ensures compliance with Method 2.

Since Example 2 is used throughout this chapter, Table 4.1 presents its entire input data

required for constructing optional timetables. Table 4.1 also contains information on (1) T , the

time required for the same vehicle, at the same departure point, to pick up another departure

and (2) the calculated data on an individual vehicle basis. For the sake of clarity, Figure 4.3

shows five time-space vehicle trajectories of Example 2. Because of possible different average

running times in each period, the headways at each stop do not necessarily coincide as seen

in Figure 4.3. However, since the hourly maximum load is used for calculating the frequency, the

timetable, set only at one time point, reflects the maximum number of required vehicles at

the observed hourly max load point. There is, then, a question of whether the determined fre

quency at the observed hourly max load point complies with the desired occupancy constraint. 150 200 250 134 216 5 6
8:00 Time Distance from hourly max load point (km) Load
(passengers)

Figure 4.2 Three-dimensional illustration of Example 2 load profiles positioned relative to the distance from the hourly max load point.

Table 4.1 Part of the Example 2 (Figure 3.7) data and calculated complementary data required for illustrating the optional timetable procedure

Time period Departure time Average load (passengers) at the max load point, by method number Area under the load profile (passenger-km) Frequency (veh/h) and headway (min, in parenthesis), by method number 1 2 1 2 4 (20%)

6:00-7:00 06:00 a - - 929 2.56 (23) 2.68 (22) 2.68 (22)
06:15 52 35 06:50 40 65

7:00-8:00 07:15 90 90 1395 3.60 (17) 3.60 (17) 2.95 (20)
07:35 87 87 07:50 75 75

Added data: T (round-trip time + layover and turnaround time) = 60 min.

a First predetermined trip.

It is important to note that the running-time information must rely on the fact that in an

average sense, vehicles do not overtake each other. A average running times should be determined

not only from ride-check data but also from the requirement that the first vehicle to

depart cannot be the second to arrive at any time point. Thus, the time-space trajectories in

Figure 4.3 cannot cross each other in an average

(deterministic) context.

Based on possible differences in actual average headways at each stop, the associated time

span at the observed hourly maximum load point, covering all trips across the time period at the

route's departure point, can be shorter or longer than the time span of the route's departure

point. For example, we see in Figure 4.3 that the time span of the last three departures,

35 min (7:15-7:50), is larger than the time span of these trips at the hourly maximum load point

of period II, stop 3. Fortunately in either case, what governs the frequency calculation is

the observed maximum load. Hence, by construction, the result is that as long as the time-space

trajectories do not cross one another, the requested frequency at each stop must be less than

or equal to the frequency determined at the route's departure point.

Finally, it may be noted that the proposition in the previous text can also be applied to

Methods 1, 3, and 4—each with its own loading standards and constraints.

4.3 EVEN HEADWAYS WITH SMOOTH TRANSITIONS

One characteristic of existing timetables is the repetition of the same headway in each time

period. However, a problem facing the scheduler in creating these timetables is how to

set departure times in the transition segments between adjacent time periods. This section

addresses the issue. 1 3 5 7 2 4 6 0 Time Period I 8 6:00
 22 52 35 65 40 25 85 44 83 90 87 50 60 45 7:00 8:00 Period
 II Space Segment passenger load 1 2 3 4 :15 :50 :15 :35
 :50 75

Figure 4.3 Time-space trajectories of all vehicles in Example 2, including segment loads.

4.3.1 Underlying principle

A common headway-smoothing rule in the transition between time periods is to use an

average headway. Many transit agencies employ this simple rule, but it may be shown that

it can result in either undesirable overcrowding or underutilization. For example, consider

two time periods, 06:00–07:00 and 07:00–08:00, in which the first vehicle is predetermined

to depart at 06:00. In the first time period, the desired occupancy is 50 passengers and

in the second 70 passengers. The observed maximum demand to be considered in these

periods is 120 and 80 passengers, respectively. These observed loads at a single point

are based on the uniform passenger-arrival-rate assumption. The determined frequencies

are $120/50 = 2.4$ vehicles and $80/70 = 1.2$ vehicles for the two respective periods, and

their associated headways are 2.5 and 5 min, respectively. If one uses the common average

headway rule, the transition headway is $(2.5 + 5)/2 = 3.75$ min; hence, the timetable is set to

0 6:00, 0 6:25, 0 6:50, 0 7:05, 0 7:10, 0 7:15, ... ,
0 7:55, and 0 8:00. By assuming a uniform pas-

senger arrival rate, the first period contributes
to the vehicle departing at 0 7:05 the a-
verage

amount of $(10/25) \cdot 50 = 20$ passengers at the m-
ax load point; the second period contributes

$(5/5) \cdot 70 = 70$ passengers. Consequently, the e-
xpected load at the maximum load point is $20 +$

$70 = 90$, a figure representing a average o-
vercrowding over the desired 70 passengers a-
fter

7:00. Certainly, the uniform arrival-rate a-
ssumption does not hold in reality. However, i-
n

some real-life situation (e.g., after work and
school dismissals), the observed demand in

5 min can be more than three times the o-
bserved demand during the previous 10 min, as

is the case in this example. In order to
overcome this undesirable situation, the f-
ollowing

principle, suggested by Ceder (2003), may be
employed.

Principle 4.1:

Establish a curve representing the cumulative (noninteger) frequency determined versus

time. Move horizontally for each departure u-
ntil intersecting the cumulative curve, and

then vertically; this will result in the required
departure time.

Proposition 4.1:

Principle 4.1 provides the required evenly spaced headways
with a transition load approach

ing the average desired occupancies of d_{oj} and $d_{o(j+1)}$ for two consecutive time periods,

j and $j + 1$.

Proof: Figure 4.4 illustrates Principle 4.1 using the frequency results from Example 2

(see Figure 3.7) appearing in Table 3.4 (Chapter 3). Since the slopes of the lines are 2.68

and 3.60 for $j = 1$ and $j = 2$, respectively, the resultant headways are those required. The

transition load is the load associated with the 7:05 departure, which consists of a arriv

ing passengers during 16 min for $j = 1$ and of arriving passengers during 5 min for $j = 2$.

Therefore, $(16/22) \cdot 50 + (5/17) \cdot 60 = 54$ approximately. This transition load is not the exact

average between $d_{o1} = 50$ and $d_{o2} = 60$, since departures are made in integer minutes. That

is, the exact determined departure after 7:00 is $(3 - 2.68) \cdot 60 / 3.60 = 5.33$ min. Inserting

this value, instead of the 5 min mentioned earlier, yields a value that is closer to the

exact average. Basically, the proportions considered satisfy the proof by construction of

Proposition 4.1. 2 4 6 8 3 5 7 1 6:00 7:00 Cumulative frequency (veh/h) Cumulative frequency (veh/h) Method 1 Method 4 (20%) Method 3 Different departure times for sixth departure by method selected Time 7:41 7:46 7:55 2 4 6 8 3 5 7 1 6:00 6:44 7:05 7:22 6:22 7:39 7:56 Predetermined first departure Method 2 1+2.68 1+2.68+3.6 8:00 8:00 7:00

Figure 4.4 Determination of departure times for evenly spaced headways and for all methods in Example 2, including a smoothing process between periods.

4.3.2 Interpretation using Example 2

The upper part of Figure 4.4 exhibits the resulting six departures for the even-headway

procedure, using frequency setting of Method 2, whereby the determined frequencies are

kept noninteger as in Table 3.4. The lower part of Figure 4.4 shows the cumulative frequency

results for Methods 1, 3, and 4 (20%). For the sake of clarity, the sixth departure is

determined to be 7:41, 7:46, and 7:56 according to Methods 1, 4, (20%) and 3, respectively.

Overall, Principle 4.1 allows for saving some unnecessary vehicle runs and also stabilizes the

average load during the transition segment between time periods.

The resultant departure times at the route departure point—it can be any other time

point—appear in Table 4.2 using the even-headway procedure. This table presents ve

optional timetables (at a single time point), using three frequency-setting methods, one of

which, Method 2, is also combined with special requests. These ve timetables are accom

panied by the two comparison measures: total departures and minimum single-route leet

size. The first two timetables in Table 4.2 are based on Figure 4.4; the three others d

eserve

explanation.

One option in selecting optional timetables (see Figure 4.1) is to use different frequency

setting methods for different time periods. In this way, the transit agency can examine

the effect of these settings on vehicle (resource) requirements during peak and off-peak

periods, for example, examining the use of Method 4 during peak periods, in which the

need for more vehicles is highest, and Method 2 for off-peak periods. In the third option

in Table 4.2, Method 1 is applied to the first time period and Method 4 (20% case) to the

second (say, peak) period. The timetable construction procedure combines, in the transition

between hours, the leftover vehicle demand utilizing Method 1 with the frequency required

by Method 4. That is, using Table 3.4 results, $F = 2.56$ according to Method 1, from 6:00

to 7:00, and $F = 2.95$ according to Method 4, from 7:00 to 8:00. The last vehicle according

to Method 1, therefore, departs at 6:46, and the leftover vehicle demand is 14 ($2.56/60$) =

0.597. Hence, $1 - 0.597 = 0.403$ is the applicable vehicle demand for Method 4, and this

yields $0.403(60/2.95) = 8.2$ min, which is rounded off to 8 min past 7:00. The remaining

departures after 7:08 follow $H = 20$ of Method 4 (20%).

Table 4.2 Five derived sets of departure times using Example 2 for even headways, by method or combination of methods selected

Timetable

characteristics

(even headways) Method 2 Method 4 (20%) Method 1 for first hour, Method 4 (20%) for second hour Method 2 with

clock headway Method 2 with predetermined number (=5)
of departures

Departure times at

anchored time point

(6:00-8:00) 6:00 a 6:00 a 6:00 a 6:00 a 6:00 a 6:22 6:22
6:23 6:20 6:35 6:44 6:44 6:46 6:40 7:08 7:05 7:06 7:08 7:00
7:34 7:22 7:26 7:28 7:15 8:00 7:39 7:46 7:48 7:30 7:56 7:45
8:00

Total number of

departures 7 6 6 8 5 (as requested)

Minimum single-route

fleet size (T = 60 min) 4 3 3 4 3

a First predetermined trip.

The fourth timetable option in Table 4.2 is based on Method
2 with clock headways. In

this case, we simply round down the headway
determined by Method 2 to its nearest clock

headway. Thus, from 6:00 to 7:00, $H = 22$ (
see Table 3.4), which is rounded to 20, and

$H = 17$, from 7:00 to 8:00, which is rounded down to 15.

The last timetable option is based on vehicle
predetermined departures (including the first

6:00 departure) while using Method 2. The total number of
required vehicles using Method

2 (excluding 6:00) is $2.68 + 3.60 = 6.28$. Since
only four departures have to be constructed,

the frequencies are modified proportionally by the
ratio $4/6.28 = 0.637$. The procedure

continues the same way as that without the
special request. That is, $F = 1.71 (= 2$
 $.68 \cdot 0.637)$

with $H = 3.5$ for $6:00-7:00$, and $F = 2.29$ with $H = 2.6$ for $7:00-8:00$. Certainly, if the

demand remains same, the scheduler should recognize the potential risk of overcrowding

when restricting the total number of departures. Nevertheless, the purpose of this special

request is to have a systematic computerized procedure to manage both crisis situations

(limited resources) and situations in which additional passenger demand can possibly be

attracted (i.e., requesting more departures than calculated).

This explication of Table 4.2 ends with the derivation of the single-route fleet size

using the round-trip time information of $T = 60$ min. The derivation of this comparison

measure will be illustrated for the first option in Table 4.2 (Method 2). Adding 60 min

to the first 6:00 departure reveals that the vehicle associated with this departure can

perform the 7:05 departure. Hence, three vehicles are required for this step (to perform

6:00, 6:22, and 6:44 departures). By continuing to add 60 to 6:22, this will enable

the vehicle to pick up the 7:22 departure (again, three vehicles are required). Adding

60 to 6:44 shows that the next pickup is executed by the 7:56 departure, resulting in

four independent departures (6:44, 7:05, 7:22, and 7:39). Continuing to count the number

of departures in each 60 min window leads to finding the maximum number, which

is the minimum single-route fleet size required. For the first option in Table 4.2, this

number of stints is four.

4.4 HEADWAYS WITH EVEN AVERAGE LOADS

This section opens with the following premise: transit managers/planners/schedulers who

believe that problems related to attracting more transit users and reliability problems are

rooted in the ocean of even-headway timetables should be told that these problems know

how to swim. In other words, even-headway timetables do not necessarily deliver the mer-

chandise (satisfactory transit service) to the customer (passengers).

We have already noted that passenger demand varies even within a single time period,

hence resulting from even headways in an imbalanced load on individual vehicles at the hourly

maximum load point. On heavy-load routes and short headways, the even-headway timetable suf-

fices. However, in the course of reducing reliability problems, we may occasionally prefer

to use the even-load instead of the even-headway procedure. Moreover, the availability of

automatic passenger counts provides a framework in which to investigate systematically

the variation in passenger demand. With the anticipated vast amount of passenger-load

data, we can then better match vehicle departure times with variable demand. Two proce-

dures carry out this endeavor. The first, a

addressed in this section, deals with even average

load on individual vehicles at the hourly (or daily) maximum load point. The second procedure,

addressed in the next chapter, ensures an even average load at each individual vehicle maximum

load point. In this section, the procedures described by Ceder (1986, 2003) will be corrected

and improved.

4.4.1 Underlying principle

A simple example is presented here to illustrate the underlying load-balancing problem.

Consider an evenly spaced headway timetable in which vehicles depart every 20 min

between 07:00 and 08:00, that is, at 07:20, 07:40, and 08:00. The observed load data consistently show that the second vehicle, which

departs at 07:40, has significantly more passengers than the third vehicle. The observed (average) maximum load during this 60 min period

is 150 passengers, and the desired occupancy is 50 passengers. Hence, using Method 2,

three vehicles are required to serve the demand as in the case of the evenly spaced headways

timetable. The average observed loads at the hourly maximum load point on the three vehicles

are 50, 70, and 30 passengers, respectively. Given that these average loads are consistent,

then the transit agency can adjust the departure times so that each vehicle has a balanced

load of 50 passengers on the average at the hourly maximum load point. The assumption of a

uniform passenger-arrival rate results in $70/20 = 3.5$ passengers/min between 07:20 and

07:40 and $30/20 = 1.5$ passengers/min between 07:40 and 08:00. If the departure time of

the second vehicle is shifted by X min backward (i.e., an early departure), then the equation

$3.5X = 70 - 50$ yields the balanced schedule, with $X = 5.7 \approx 6$ min, or departures at

07:20, 07:34, and 08:00. The third departure will add the difference of 20 passengers at

the hourly maximum load point. The even-headway setting assures enough vehicles to accommodate

moderate hourly demand, but it cannot guarantee balanced loads for each vehicle at the

peak point. In order to avoid this imbalanced situation, the following principles should be

exploited.

Principle 4.2:

Construct a curve representing the cumulative loads observed on individual vehicles at the

hourly maximum load points. Move horizontally per each day for all j , until intersecting the

cumulative-load curve, and then vertically; this results in the required departure times.

Proposition 4.2:

Principle 4.2 results in departure times such that the average maximum load on individual vehicles

at the hourly j th max load point approaches the desired occupancy d_{oj} .

Proof: Figure 4.5 illustrates Principle 4.2 for the Example 2 problem appearing in Figure 3.7

and Table 4.1. Method 2 will be used in the upper part of Figure 4.5 in which the derived

departure times are unevenly spaced to obtain even loads at stop 3 for $j = 1$ and at stop 2

for $j = 2$. These even loads are constructed on the cumulative curve to approach $d_{o1} = 50$

and $d_{o2} = 60$. If we assume a uniform passenger-arrival rate between each two observed

departures, it can be shown that the load (at stop 3) of the first derived departure (6:23)

consists of the arrival rate between 6:00 and 6:15 ($35/15 = 2.33$) and the rate between 6:15

and 6:50 ($65/35 = 1.86$). Thus, $2.33 \cdot 15 + 1.86 \cdot 8 \approx 50$. In the transition between $j = 1$ and

$j = 2$ (in the upper part of Figure 4.5), the value of $d_2 = 60$ is considered, since the resultant

departure comes after 7:00. The load of the vehicle departing at 7:07 at its hourly max load

point, stop 2, is simply $17 \cdot (90/25) = 61.2$ from rounding off the departure time to the nearest

integer. That is, $(10 + y) \cdot (90/25) = 60$ results in $y = 6.67$ min. This completes the proof

by construction of Proposition 4.2. 6:15 7:156:50
7:35 7:50 100 35 190 277 352 400 300 200 100 Method 2 6:00
6:50 7:07 7:00 7:22 6:23 7:36 7:48 C u m u l a t i v e l o

a d (p a s s e n g e r s) 8:00 60 50 8:00 300 200 100
 6:15 7:15 6:50 7:35 7:50 6:00 8:00 6:52 7:13 7:00 Time
 7:30 6:14 269 182 344 =73.22 2.95 3.60 60 52 7:45 92 Method
 1 (6:00-7:00) Method 4 (20%) (7:00-8:00) 400 73.22 50 C u m
 u l a t i v e l o a d (p a s s e n g e r s) 8:00

Figure 4.5 Determination of Example 2 departure times with even load at the hourly max load point using Method 2 and at the daily max load point using combination of Methods 1 and 4 (20%).

4.4.2 Interpretation of even load using Example 2

In its upper part, Figure 4.5 shows the resulting seven departures for the even-load proce

dure, using the loads observed at the hourly (Method 2) max load points (as in Table 4.1).

A cumulative curve of straight lines can be drawn from observed departure times and max

loads (35 + 65 + 90 + ...). The slope of each line will represent the uniform arrival rate. For

example, the arrival rate between the 6:50 and 7:15 departures is $90/25 = 3.6$ passengers/

min. The time on the curve associated with the first desired occupancy, $d_0 = 50$ passengers,

is 06:23, and the second is 6:50; this means that the second departure is unchanged, since

$35 + 65 = 50 + 50$. We then can check the cumulative-load level of 150 and learn that its

associated time on the curve is after 7:00. Hence, $d_2 = 60$ will be applied for the rest of the

example. The cumulative-load curve also ensures that the first vehicle to depart in any time

period will accommodate the desired occupancy assigned to that period. The only exception

in this process occurs when the determined frequency is based on the minimum frequency

standard of Equation 3.3. In such a case, we may disregard

even loads for a given time

period j and switch to even headways, since virtually no overcrowding is expected. However,

if we still want to maintain the even-load situation for the minimum frequency case, then d_{oj}

must be replaced according to the formula shown in the following section.

The lower part of Figure 4.5 shows the derivation of an even average load timetable at the

daily max load point as opposed to the hourly max load point described for Method 2. In

this example, we will combine the results of Method 1 (6:00-7:00) with Method 4 using the

20% criterion (7:00-8:00). The cumulative-load curve is based on the loads observed only

at stop 2 (daily max load point) and appearing in Table 4.1 and Figure 3.7. While the desired

occupancy for Method 1 is the same as for Method 2 ($d_{o1} = 50$), the value of d_{o2} needs to be

commensurate with the frequency result given by Method 4 (20%). Thus, we can calculate

the substitution for d_{o2} simply in proportional terms: d_{i1} and d_{i2} $F_{ij} P_{mj} / d_{ij} o_{ij} m_{ij} o_{ij} = \dots$, , , , 3 4 1 2 (4.1)

where d_{ij} is the adjusted desired occupancy i is the frequency determination method number j is the time period F_{ij} , P_{mj} , and d_{oj} are the frequency, maximum observed load, and desired occupancy, respectively as defined in Chapter 3

Equation 4.1 also holds for situations in which the determined frequency is the minimum

frequency standard for all methods. This standard is the result of Equations 3.3 and 3.5,

that is, in the case of Method 2, $F_{2j} = \max(P_{mj} / d_{oj}, F_{mj}) = F_{mj}$, where F_{mj} is the minimum fre

quency in time period j .

Based on Equation 4.1, and using Example 2, Method 4 (20%) yields the following:

$d_{4,2} = 60(3.60/2.95) = 73.22$. This adjusted quantity is used in Figure 4.5 for constructing the

timetable from the route's departure point. The results of the two timetables of Figure 4.5 and

of other optional timetables for Example 2 are shown in four columns in Table 4.3.

When applying the load-profile-based (frequency) methods, we can expect higher than

the desired loads at the max load points as a trade-off for less empty space-km. The sec

ond column of Table 4.3 presents the resulting timetable at the hourly max load points

according to Method 4 (20%). This timetable is derived according to Principle 4.2, with the

same cumulative-load curve as in the upper part of Figure 4.5, but an adjusted d_{ij} based on

Equation 4.1. Since the frequencies given by Methods 2 and 4 (20%) for the first 6:00–7:00

period coincide, $d_{i1} = 50$ for both. For the 7:00–8:00 period, $d_{i2} = 73.22$, with both d_{i1} and

d_{i2} used for the hourly max loads. That is, the first departure after 7:00 will be y min from

6:50 with a load of 73.22; hence, $y = 73.22 \cdot 25/90 = 20.33 \approx 20$, thereby determining the

6:50 + 20 = 7:10 departure.

The last timetable option in Table 4.3 is based on predetermined departures (including

the first 6:00 departure) while using Method 2. The desired occupancy is modified

proportionally in the same manner as was done in Section 4.3.2, by the ratio $4/6.28 =$

0.637 . Consequently, the two periods in Example 2 will use the load quantities $50/0.637 =$

78.5 and $60/0.637 = 94.2$, respectively, instead of $d_01 = 50$ and $d_02 = 60$. With these quanti-

ties, Principle 4.2 will be applied on the upper cumulative curve of Figure 4.5 to arrive at

the results. It should be understood that the special request of a clock headway cannot be

applied to even-load procedures. Obviously, timetables with clock headways cannot incor-

porate uneven headways.

The comparison measures in Table 4.3 are derived in the same manner as for the even

headway procedure as explained in Section 4.3.2. The comparison measures of number of

departures and single-route fleet sizes should be virtually similar for the even-headway and

even-load procedures. Nonetheless, because of different usages of the end of the schedule

(8:00), this expected similarity almost disappears. The next section, using LACMTA real

life data, will show that this similarity indeed exists, as expected, since both types of proce-

dures rely on the same frequencies.

4.5 AUTOMATION, TEST RUNS, AND CONCLUSION

The outcome of Chapters 3 and 4 can be jointly automated. Practically speaking, a set

of computer programs was created that perform (1) conversion from transit agencies

to a adequate input les, (2) setting o f f
 requencies and headways b y f our m ethods, a nd

Table 4.3 Four derived sets of departure times using
 Example 2 for an even average load (at a single time
 point), by method or combination of methods selected

Timetable characteristics

(even average load) Method 2 Method 4 (20%) Method 1
 (6:00-7:00), Method 4 (20%) (7:00-8:00) Method 2 with
 predetermined number (=5) of departures

Point where average

even load is expected Hourly max load Hourly max load
 Daily max load Hourly max load

Departure times at

anchored time point

(6:00-8:00) 6:00 a 6:00 a 6:00 a 6:00 a 6:23 6:23 6:14 6:38
 6:50 6:50 6:52 7:10 7:07 7:10 7:13 7:33 7:22 7:28 7:30 7:52
 7:36 7:44 7:45 7:48 7:58 8:00 8:00

Total number of

departures 8 7 7 5 (as requested)

Minimum single-route

leet size (T = 60 min) 5 4 4 3

a First predetermined trip.

(3) construction of a public timetable at all route time
 points. We are aware of the need, in

p ractice, t o a ctually t est o ptional t imetables
 p rior t o d rawing a d efinite c onclusion. O ften,

t he f ollowing s aying i s t rue: l ogic i s a s
 ystematic m ethod o f c oming t o t he w rong c
 onclu

sion w ith c onfidence. H ence, o nly r eal t esting,
 u sing a b efore a nd a fter s tudy, c an r eveal t
 he

magnitude—if at all—of the improvement.

Figure 4.6 exhibits a summary of the procedures that can take place in any possible auto

mation. This flowchart is fed by the data required before embarking on frequency-setting

and timetable construction procedures. Compliance with the third objective of this chapter

(see Section 4.2) is assured, and the possibility is furnished of direct frequency insertion and

the manual construction of timetables.

For a real-life example and test runs, the ride-check data from the old Los Angeles

(LA) Metro route 217 (in Los Angeles), which has already been described and employed

in Chapter 3, were used. Route 217 was characterized by 60 stops and 8 time points in

each direction. In our set of programs, the user can request various optional timetables. Execute the four frequency methods Scheduler's own frequency determination Execute frequency Methods 1 and 2 Use Method 1 Use Method 2 Determine combination of runs: (1) Type of timetable (2) Frequency method by time period (3) Predetermined departures (4) Clock headways Execute timetable construction program Public timetable More runs ? Construct timetable manually (special occasions) Yes Yes Yes No No No Timetable construction Ride check available ? Frequency setting Continue manually ? Methods 1 and 2 alike ? No Yes Compare optional timetables When applicable Passenger loads, service standards, and route and vehicle input data

Figure 4.6 Overall automated procedure, in flowchart form, for optional timetable development.

according to the combinations indicated in Figure 4.6. For each computer run, the user

simply keypunches requests as follows:

1. Type of timetable—"1" for evenly spaced headways and "2" for even average maximum load (at hourly maximum load point)
2. Number of method to be used (among the inserted frequency-setting methods)
3. For each used method, the user specifies
 - a. Method number
 - b. The time period number in which to start using the method
 - c. The last time-period number in which to use the method in the combination being considered (i.e., the same method can be used several times for different time periods, but each combination must be specified)
4. Clock headway—"0" for not required and "1" for required
5. Predetermined number of departures – "0" for none and Given number of departures for using the constraint

Eighteen different combinations of runs for each direction of travel are shown in Table 4.4

as the results of the comparison measures for each run. Tables 4.5 through 4.7 display the

results of a variety of runs at the northbound departure point of route 217. Table 4.5 shows

a comparison of Method 2 and 4 (20%) results for both even-headway and even-load procedures.

In Table 4.6, there are two types of comparisons: between Method 2 and 4 (20%)

for even clock headways and between even-headway and even-load procedures using a com-

bination of Method 2 and 4 (20%). Table 4.7, as well, presents two types of comparisons:

between predetermined 70 and 140 departures u

sing the even-load procedure and between

Method 1 and Method 3 results using the even-headway procedure.

As was noted earlier, the comparison measures across all the real-life results in Tables 4.5

through 4.7 are the same for the two types of timetables (even headway, even load). Where

they differ is in the frequency methods selected and the special requests (clock headway,

predetermined number of departures). See, for example, the comparison measures for the

lower and upper bounds of route 217 northbound. Specifically, Method 3 delivers a lower

bound of 67 and 11 for the number of departures and the single-route fleet size, respectively.

This contrasts with the upper bound, given by Method 2 and a clock headway, of 89 departures

and 16 vehicles required. Finally, for the sake of clarity, Table 4.8 presents a complete

timetable (across all eight time points), this time for route 217 southbound, using Method 4

(20%) and an even-headway procedure.

Admittedly, the large number and variety of timetables may complicate the decision

process at the transit agency. However, it offers an opportunity to examine different time

table and frequency scenarios rapidly. We anticipate that the skilled scheduler/planner

will select only a few options for comparison while recognizing the full potential of the

procedures.

In conclusion, the consequence of this chapter can be described generally in the light

of the various study objectives set forth in Section 4.2. Using passenger-load data, the procedures

developed provide optional timetables in terms of vehicle departure times at all

specified time points. Each timetable is accompanied by two comparison measures. These

evaluation measures fulfill the first objective. One is set of options when selecting the procedure

to construct the timetables is referred to as balancing passenger loads on individual

vehicles while allowing unevenly spaced headways. The underlying approach in that set of

options is to shift the departure times of individual vehicles so as to obtain even average

loads while relaxing the even-headway requirement. The latter accomplishes the second

objective.

The third objective of the study is to allow for direct vehicle-frequency changes. This is achieved

in the process described by Figure 4.6, whereby the schedulers can either interject their own set of

intuitive or experience-based frequencies or they can substitute some of the derived frequencies.

In this way, possible scheduling exceptions (e.g., special passenger demand because of a sports

event) could be inserted. The common manual process in creating timetables often encounters a

problem in smoothing the headways in the transition segments between adjacent time periods.

In both types of timetables that have been described (even headway and even load), the procedure

developed ensures, in an average sense, the fulfillment of the desired occupancy standard.

Table 4.4 Combination of requests for test runs using Los Angeles route 217 ride-check data

Run

number Direction of travel: South (S) and North (N)
 Type of timetable: 1 = Even Headway; 2 = Even Load Method
 # Clock headway: 1 = With; 0 = Without Predetermined
 number of departures Results of comparison measures
 Number of departures Minimum fleet size

1 N 1 1 0 0 84 15

2 N 1 2 0 0 87 15

3 N 1 3 0 0 67 11

4 N 1 4 0 0 78 14

5 N 1 2, 4 a 0 0 85 15

6 N 1 2 1 0 80 16

7 N 1 4 1 0 78 14

8 N 1 2 0 70 70 12

9 N 1 2 0 140 140 24

10 N 2 1 0 0 84 15

11 N 2 2 0 0 87 15

12 N 2 3 0 0 67 11

13 N 2 4 0 0 78 14

14 N 2 2, 4 a 0 0 85 15

15 N 2 2 0 70 70 12
 16 N 2 2 0 140 140 24
 17 S 1 1 0 0 88 19
 18 S 1 2 0 0 91 19
 19 S 1 3 0 0 69 13
 20 S 1 4 0 0 73 13
 21 S 1 2, 4 a 0 0 89 19
 22 S 1 2 1 0 91 20
 23 S 1 4 1 0 74 14
 24 S 1 2 0 70 70 15
 25 S 1 2 0 140 140 30
 26 S 2 1 0 0 88 19
 27 S 2 2 0 0 91 20
 28 S 2 3 0 0 69 13
 29 S 2 4 0 0 73 13
 30 S 2 2, 4 a 0 0 89 20
 31 S 2 2 0 70 70 15
 32 S 2 2 0 140 140 30

a Method 2 is used for off-peak periods # (1), (4, 5, 6, 7, 8, 9, 10, 11) and (14, 15, 16, 17, 18, 19); Method 4 is used for peak

periods # (2, 3) and (12, 13)—a total of five combinations.

Table 4.5 Comparison of Methods 2 and 4 (20% criterion) using the two types of timetables for Los Angeles route 217 northbound

Characteristics Even headways, Method 2 Even headways, Method 4 (20%)

Departure

times at route

departure

point 06:00 09:31 12:11 14:10 16:15 19:28 06:00 10:04 12:54
15:04 17:36 23:59 06:30 09:45 12:19 14:18 16:24 19:58 06:30
10:16 13:03 15:13 17:48 00:30 07:00 10:00 12:27 14:26 16:33
20:28 07:00 10:28 13:12 15:22 18:01 01:00 07:11 10:10 12:35
14:34 16:42 20:58 07:12 10:41 13:20 15:31 18:19 07:21 10:21
12:43 14:42 16:50 21:29 07:24 10:53 13:29 15:40 18:38 07:32
10:31 12:51 14:50 16:59 21:59 07:36 11:05 13:37 15:49 18:57
07:42 10:42 12:59 14:58 17:10 22:29 07:48 11:17 13:46 15:58
19:25 07:53 10:52 13:07 15:07 17:21 22:59 08:00 11:30 13:54
16:08 19:56 08:04 11:02 13:15 15:15 17:33 23:29 08:13 11:42
14:03 16:18 20:26 08:15 11:13 13:22 15:24 17:44 24:00 08:26
11:54 14:11 16:28 20:57 08:27 11:23 13:30 15:32 17:55 00:30
08:39 12:05 14:20 16:38 21:27 08:38 11:33 13:38 15:41 18:09
01:00 08:51 12:15 14:29 16:49 21:57 08:49 11:44 13:46 15:49
18:26 09:07 12:25 14:38 16:59 22:28 09:01 11:54 13:54 15:58
18:42 09:26 12:34 14:46 17:11 22:58 09:16 12:03 14:02 16:06
18:59 09:46 12:44 14:55 17:23 23:29

Total number of

departures 87 78

Minimum

single-route

-eet size 15 14 Even average load, Method 2 Even average
load, Method 4 (20%)

Departure

times at route

departure

point 06:00 09:25 12:07 14:08 16:14 19:28 06:00 10:03 12:51
15:04 17:32 23:59 06:38 09:39 12:14 14:15 16:26 19:58 06:36
10:17 13:03 15:15 17:51 00:12 07:03 10:00 12:25 14:29 16:33
20:23 07:03 10:31 13:07 15:20 18:02 01:00 07:23 10:12 12:33
14:35 16:43 20:59 07:21 10:41 13:15 15:31 18:17 07:35 10:24
12:40 14:44 16:49 21:34 07:35 10:54 13:24 15:37 18:36 07:42
10:35 12:49 14:53 16:59 21:58 07:44 11:06 13:33 15:46 18:55
07:50 10:44 12:58 14:59 17:09 22:26 07:52 11:16 13:46 15:58
19:26 07:56 10:54 13:05 15:08 17:19 22:59 08:00 11:30 13:54
16:07 19:56 08:02 11:04 13:10 15:16 17:27 23:17 08:07 11:39

14:02	16:20	20:19	08:10	11:13	13:17	15:23	17:46	23:59	08:17
11:53	14:08	16:30	20:57	08:21	11:20	13:25	15:34	17:56	00:13
08:33	12:04	14:16	16:42	21:26	08:36	11:32	13:34	15:40	18:12
01:00	08:44	12:12	14:30	16:49	21:57	08:44	11:43	13:44	15:48
18:25	09:03	12:24	14:37	16:59	22:26	09:01	11:54	13:53	15:58
18:41	09:20	12:34	14:48	17:10	22:58	09:11	12:03	14:02	16:05
18:57	09:37	12:41	14:55	17:20	23:16				

Total number of

departures 87 78

Minimum

single-route

fleet size 15 14

Table 4.6 Comparison of Methods 2 and 4 (20% criterion) using the two types of timetables and of the two types of timetables for a combination of these methods, for Los Angeles route 217 northbound

Characteristics Even clock headways, Method 2 Even clock headways, Method 4 (20%)

Departure

times at route

departure

point	06:00	09:30	12:07	14:00	15:52	18:45	06:00	10:00	12:50
	15:00	17:36	00:00	06:30	09:45	12:15	14:07	16:00	19:00
	06:30	10:12	13:00	15:10	17:48	00:30	07:00	10:00	12:22
	14:15	16:10	19:30	07:00	10:24	13:07	15:20	18:00	01:00
	07:10	10:10	12:30	14:22	16:20	20:00	07:12	10:36	13:15
	15:30	18:20	07:20	10:20	12:37	14:30	16:30	20:30	07:24
	10:48	13:22	15:40	18:40	07:30	10:30	12:45	14:37	16:40
	21:00	07:36	11:00	13:30	15:50	19:00	07:40	10:40	12:52
	14:45	16:50	21:30	07:48	11:12	13:37	16:00	19:30	07:50
	10:50	13:00	14:52	17:00	22:00	08:00	11:24	13:45	16:10
	20:00	08:00	11:00	13:07	15:00	17:12	22:30	08:12	11:36
	13:52	16:20	20:30	08:12	11:10	13:15	15:07	17:24	23:00
	08:24	11:48	14:00	16:30	21:00	08:24	11:20	13:22	15:15
	17:36	23:30	08:36	12:00	14:10	16:40	21:30	08:36	11:30
	13:30	15:22	17:48	00:00	08:48	12:10	14:20	16:50	22:00
	08:48	11:40	13:37	15:30	18:00	00:30	09:00	12:20	14:30
	17:00	22:30	09:00	11:50	13:45	15:37	18:15	01:00	09:20
	12:30	14:40	17:12	23:00	09:15	12:00	13:52	15:45	18:30
	09:40	12:40	14:50	17:24	23:30				

Total number of

departures 89 78

Minimum

single-route

fleet size 16 14 Even headways, combination of Methods
2 and 4 (20%) Even average load, combination of Methods
2 and 4 (20%)

Departure

times at route

departure

point 06:00 09:48 12:20 14:19 16:25 20:28 06:00 09:43 12:16
14:17 16:27 20:23 06:30 10:02 12:28 14:27 16:34 20:59 06:38
10:02 12:26 14:30 16:33 20:59 07:00 10:12 12:36 14:35 16:43
21:29 07:01 10:15 12:34 14:36 16:44 21:35 07:12 10:22 12:44
14:43 16:51 21:59 07:21 10:27 12:41 14:46 16:50 21:58 07:24
10:33 12:52 14:51 17:00 22:29 07:35 10:36 12:50 14:54 17:00
22:27 07:36 10:43 13:00 15:00 17:13 22:59 07:43 10:46 13:00
15:00 17:11 22:59 07:47 10:54 13:08 15:08 17:25 23:29 07:52
10:55 13:05 15:11 17:22 23:17 07:59 11:04 13:16 15:16 17:37
00:00 08:00 11:07 13:11 15:16 17:35 00:00 08:12 11:14 13:24
15:25 17:50 00:30 08:07 11:14 13:18 15:24 17:53 00:13 08:25
11:25 13:31 15:33 18:03 01:00 08:16 11:21 13:26 15:35 18:07
01:00 08:38 11:35 13:39 15:42 18:22 08:33 11:34 13:35 15:41
18:20 08:50 11:45 13:47 15:50 18:40 08:43 11:45 13:46 15:50
18:38 09:04 11:56 13:55 15:59 18:59 09:02 11:56 13:54 15:59
18:58 09:18 12:05 14:03 16:08 19:28 09:14 12:03 14:02 16:06
19:29 09:33 12:12 14:11 16:16 19:58 09:27 12:08 14:09 16:16
19:58

Total number of

departures 85 85

Minimum

single-route

fleet size 15 15

Table 4.7 Comparison of timetables associated with a
predetermined number of departures and of timetables using
Methods 1 and 3 for Los Angeles route 217 northbound

Characteristics Even average load, 70 departures, Method
2 Even average load, 140 departures, Method 2

Departure

times at route

departure

point 06:00 10:57 14:07 17:21 06:00 08:37 10:56 12:35 14:05
15:38 17:18 20:57 06:49 11:11 14:16 17:36 06:32 08:42 11:02
12:40 14:09 15:42 17:22 21:20 07:13 11:19 14:31 17:54 06:49
08:49 11:10 12:46 14:14 15:47 17:28 21:39 07:32 11:34 14:39
18:12 06:55 09:01 11:14 12:51 14:23 15:54 17:39 21:52 07:42
11:48 14:52 18:30 07:12 09:06 11:18 12:57 14:29 15:59 17:48
22:11 07:51 12:01 15:00 18:49 07:24 09:15 11:22 13:02 14:33
16:04 17:55 22:28 07:59 12:07 15:15 19:21 07:31 09:23 11:32
13:05 14:36 16:08 18:03 22:45 08:08 12:16 15:19 19:59 07:38
09:32 11:39 13:08 14:41 16:14 18:13 23:04 08:21 12:28 15:33
20:30 07:42 09:41 11:46 13:12 14:49 16:23 18:20 23:14 08:38
12:37 15:41 21:18 07:47 09:55 11:53 13:16 14:54 16:28 18:31
23:39 08:50 12:48 15:53 21:51 07:51 10:03 11:59 13:21 14:57
16:32 18:41 00:02 09:06 13:00 16:02 22:27 07:55 10:11 12:03
13:27 15:01 16:37 18:51 00:09 09:24 13:06 16:12 23:04 07:58
10:19 12:06 13:32 15:08 16:44 19:06 00:32 09:44 13:14 16:27
23:38 08:03 10:27 12:09 13:38 15:15 16:47 19:24 01:00 10:04
13:24 16:35 00:09 08:07 10:33 12:14 13:45 15:16 16:53 19:44
10:20 13:35 16:46 01:00 08:12 10:38 12:21 13:50 15:21 16:59
20:01 10:34 13:48 16:58 08:19 10:44 12:27 13:55 15:29 17:04
20:15 10:46 13:58 17:10 08:31 10:51 12:31 14:01 15:34 17:12
20:32

Total number of

departures 70 (as requested) 140 (as requested)

Minimum

single-route

feet size 12 24 Even headways, Method 1 Even headways,
Method 3

Departure

times at route

departure

point 06:00 09:32 12:06 14:06 16:04 18:41 06:00 09:50 13:21
 16:11 19:55 06:30 09:47 12:15 14:14 16:13 18:58 06:30 10:09
 13:34 16:21 20:26 07:00 10:01 12:24 14:22 16:22 19:27 07:00
 10:25 13:46 16:31 20:56 07:12 10:12 12:33 14:30 16:31 19:57
 07:13 10:42 13:59 16:41 21:27 07:23 10:22 12:43 14:39 16:40
 20:28 07:25 10:59 14:12 16:51 21:57 07:35 10:33 12:52 14:47
 16:48 20:58 07:37 11:16 14:25 17:02 22:28 07:46 10:43 13:01
 14:55 16:57 21:28 07:49 11:32 14:38 17:15 22:58 07:57 10:54
 13:09 15:03 17:08 21:58 08:01 11:49 14:51 17:28 23:29 08:10
 11:04 13:17 15:12 17:19 22:29 08:14 12:04 15:04 17:41 23:59
 08:23 11:14 13:25 15:21 17:31 22:59 08:28 12:17 15:15 17:54
 00:30 08:36 11:25 13:33 15:29 17:42 23:29 08:41 12:30 15:27
 18:12 01:00 08:49 11:35 13:41 15:38 17:53 00:00 08:54 12:43
 15:38 18:34 09:02 11:45 13:49 15:47 18:07 00:30 09:11 12:56
 15:49 18:56 09:17 11:56 13:57 15:56 18:24 01:00 09:31 13:08
 16:01 19:25

Total number of

departures 84 67

Minimum

single-route

feet size 15 11

Table 4.8 Complete timetable with even headway, Method 4 (20%), at all time points of route 217 southbound in Los Angeles

Departure # Beachwood, Westshire Gower, Franklin
 Hollywood, Vine La Brea, Sunset Fairfax, Santa Monica
 Fairfax, Beverly Fairfax, Olympic Adams, Washington

1 05:37 05:41 05:44 05:50 05:56 06:00 06:05 06:11

2 06:07 06:11 06:14 06:20 06:26 06:30 06:36 06:42

3 06:33 06:38 06:41 06:47 06:55 07:00 07:06 07:13

4 06:46 06:51 06:54 07:01 07:08 07:13 07:19 07:26

5 06:59 07:04 07:07 07:14 07:21 07:26 07:32 07:39

6 07:12 07:17 07:20 07:27 07:35 07:40 07:46 07:52

7 07:25 07:30 07:33 07:40 07:48 07:53 07:59 08:05

8 07:35 07:40 07:43 07:51 08:00 08:05 08:11 08:19

9 07:47 07:52 07:55 08:03 08:12 08:17 08:23 08:31
10 07:59 08:04 08:07 08:15 08:24 08:29 08:35 08:43
11 08:11 08:16 08:19 08:27 08:36 08:41 08:47 08:55
12 08:23 08:28 08:31 08:39 08:48 08:53 08:59 09:07
13 08:34 08:39 08:42 08:50 08:59 09:04 09:10 09:18
14 08:45 08:50 08:53 09:01 09:10 09:15 09:21 09:29
15 08:57 09:02 09:05 09:13 09:22 09:27 09:33 09:41
16 09:08 09:13 09:16 09:24 09:33 09:38 09:44 09:52
17 09:19 09:24 09:27 09:35 09:44 09:49 09:55 10:03
18 09:30 09:35 09:38 09:46 09:55 10:00 10:07 10:15
19 09:40 09:45 09:48 09:56 10:05 10:10 10:17 10:25
20 09:50 09:55 09:58 10:06 10:15 10:21 10:27 10:36
21 10:00 10:05 10:08 10:16 10:25 10:31 10:37 10:46
22 10:11 10:16 10:19 10:27 10:36 10:41 10:48 10:56
23 10:21 10:26 10:29 10:37 10:46 10:51 10:58 11:06
24 10:31 10:36 10:39 10:47 10:56 11:02 11:09 11:18
25 10:42 10:47 10:50 10:58 11:07 11:13 11:20 11:29
26 10:53 10:58 11:01 11:09 11:18 11:24 11:31 11:40
27 11:04 11:09 11:12 11:20 11:29 11:35 11:42 11:51
28 11:15 11:20 11:23 11:31 11:40 11:46 11:53 12:02
29 11:26 11:31 11:34 11:42 11:51 11:57 12:04 12:13
30 11:38 11:43 11:46 11:54 12:03 12:09 12:16 12:27
31 11:50 11:55 11:58 12:06 12:15 12:21 12:28 12:39
32 12:02 12:07 12:10 12:18 12:27 12:33 12:40 12:52
33 12:15 12:20 12:23 12:31 12:40 12:46 12:53 13:04

34 12:27 12:32 12:35 12:43 12:52 12:58 13:05 13:16

35 12:37 12:43 12:46 12:54 13:03 13:09 13:16 13:25

36 12:48 12:54 12:56 13:05 13:14 13:20 13:27 13:36

37 12:59 13:04 13:07 13:16 13:25 13:31 13:38 13:47

38 13:10 13:15 13:18 13:26 13:35 13:41 13:48 13:57

39 13:21 13:26 13:29 13:37 13:46 13:52 13:59 14:08

40 13:29 13:34 13:37 13:47 13:57 14:03 14:11 14:20

41 13:38 13:43 13:46 13:56 14:06 14:12 14:20 14:29

42 13:47 13:52 13:55 14:05 14:15 14:21 14:29 14:38 (Continued)

This basically fulfills the smoothing necessity expressed in the fourth objective. The fifth and last

objective is attained by allowing the transit agency to request a selection of different frequencies

setting methods for different time periods. The scheduler/planner can then select for peak periods

those methods that are more sensitive to resource saving (e.g., Method 4) and for off-peak

periods methods that are more sensitive to passenger comfort (e.g., Method 2).

4.6 LITERATURE REVIEW AND FURTHER READING

The problem of finding the best dispatching policy for transit vehicles on fixed routes

has a direct impact on constructing timetables. This dispatching-policy problem, which

has been dealt with quite extensively in the literature, can be categorized into four

Table 4.8 (Continued) Complete timetable with even

headway, Method 4 (20%), at all time points of route 217 southbound in Los Angeles

Departure # Beachwood, Westshire Gower, Franklin Hollywood, Vine La Brea, Sunset Fairfax, Santa Monica Fairfax, Beverly Fairfax, Olympic Adams, Washington

43	13:57	14:02	14:05	14:15	14:25	14:31	14:39	14:48
44	14:06	14:11	14:14	14:24	14:34	14:40	14:48	14:57
45	14:16	14:21	14:24	14:34	14:44	14:50	14:58	15:07
46	14:25	14:30	14:33	14:43	14:53	14:59	15:07	15:16
47	14:34	14:39	14:42	14:52	15:02	15:08	15:16	15:25
48	14:42	14:47	14:50	15:00	15:10	15:16	15:24	15:33
49	14:51	14:56	14:59	15:09	15:19	15:25	15:33	15:42
50	14:59	15:04	15:07	15:17	15:27	15:33	15:41	15:50
51	15:08	15:13	15:16	15:26	15:36	15:42	15:50	15:59
52	15:16	15:21	15:24	15:34	15:44	15:50	15:58	16:07
53	15:25	15:30	15:33	15:43	15:53	15:59	16:07	16:16
54	15:41	15:46	15:49	15:59	16:09	16:15	16:23	16:32
55	15:58	16:03	16:06	16:16	16:26	16:32	16:40	16:49
56	16:15	16:20	16:23	16:33	16:43	16:49	16:57	17:06
57	16:35	16:40	16:43	16:53	17:03	17:09	17:17	17:26
58	17:00	17:04	17:07	17:17	17:27	17:33	17:41	17:50
59	17:24	17:29	17:32	17:42	17:52	17:58	18:06	18:15
60	17:57	18:01	18:04	18:13	18:22	18:27	18:34	18:43
61	18:27	18:31	18:34	18:43	18:52	18:57	19:04	19:13
62	18:59	19:04	19:07	19:15	19:23	19:28	19:34	19:42
63	19:29	19:34	19:37	19:45	19:53	19:58	20:04	20:12
64	20:00	20:05	20:08	20:16	20:23	20:28	20:34	20:42

65 20:30 20:35 20:38 20:46 20:53 20:58 21:04 21:12

66 21:01 21:06 21:09 21:17 21:24 21:29 21:35 21:43

67 21:31 21:36 21:39 21:47 21:54 21:59 22:05 22:13

68 22:01 22:06 22:09 22:17 22:24 22:29 22:35 02:43

69 22:31 22:36 22:39 22:47 22:54 22:59 23:05 23:13

70 23:01 23:06 23:09 23:17 23:24 23:29 23:35 23:43

71 23:32 23:37 23:40 23:48 23:55 00:00 00:06 00:14

72 00:02 00:07 00:10 00:18 00:25 00:30 00:36 00:44

73 00:32 00:37 00:40 00:48 00:55 01:00 01:06 01:14

groups: (1) models for an idealized transit system, (2) simulation models, (3) mathematical

cal programming models, and (4) data-based models.

The first group, idealized transit systems, was investigated by, for example, Newell (1971),

Osana and Newell (1972), Hurdle (1973), Wiersinghe (1990, 2003), and De Palma and

Lindsey (2001). Newell (1971) assumed a given passenger-arrival rate as a smooth function

of time, with the objective of minimizing total passenger waiting time. He showed

analytically that the frequency of transit vehicles with large capacities (in order to serve

all waiting passengers) and the number of passengers served per vehicle each varied with

time approximately as the square root of the arrival rate of passengers. Osana and Newell

(1972) developed control strategies for either holding back a transit vehicle or dispatching

it immediately, based on a given number of vehicles, random round-trip travel times with

known distribution functions, and uniform passenger-arrival rates with a minimum waiting

time objective. Using dynamic programming, they found that the optimal strategy for two

vehicles and a small coefficient of variation of trip time retained nearly equally spaced

times. Hurdle (1973), investigating a similar problem, used a continuum of

model to derive an optimal dispatching policy while attempting to minimize the total cost

of passenger waiting time and vehicle operation.

Wirasinghe (1990, 2003) examined and extended Newell's dispatching policy while con-

sidering the cost components initially used by Newell (1973). Wirasinghe considered the

average value of a unit waiting time per passenger (C_1) and the cost of dispatching a vehicle

(C_2) to show that the passenger-arrival rate in Newell's square root formula is multiplied by

($C_1/2C_2$). Wirasinghe also showed how to derive the equations of total mean cost per unit of

time by using both uniform headway policy and Newell's variable-dispatching policy.

De Palma and Lindsey (2001) developed a method for designing an optimal timetable for a

single line with only two stations. The method is suitable for a situation in which each rider

has a precise time in which the person wants to travel; traveling earlier or later than desired

increases the total cost. The objective is to minimize riders' total schedule-delay costs. Two

units of delay equally. The second case assumes several rider groups, with different levels

of delay costs assigned to riders from the different groups. In addition, the researchers com

pared two models: a line model, in which passengers travel times are uniformly distributed

over part of the day and trips cannot be rescheduled between days, and a circle model, in

which passengers travel times are uniformly distributed over the full 24 h day and trips can be

rescheduled between days. Optimal timetables are derived for each of the models.

In the second group, simulation models were studied by, for example, Marlin et al. (1988),

Adamski (1998), and Dessouky et al. (1999). Marlin et al. (1988) developed a simulation model

for dispatching transit vehicles every day. They checked the feasibility of the results and used

mathematical programming for vehicle assignments in an interactive computer-support system.

Adamski (1998) employed a simulation model for real-time dispatching control of transit vehicles

while attempting to increase the reliability of service in terms of on-time performance. His

simulation implemented optimal stochastic control with linear feedback. The use of intelligent

transportation systems was applied by Dessouky et al. (1999) in a study of bus dispatching at

timed transfer points. The researchers used a s

imulation analysis to show that the benefit of

knowing the location of the bus was most significant when the bus was experiencing a significant

delay, especially when there was a small number of connecting buses at the transfers point.

Mathematical programming methods, the third group for determining frequencies and

timetables, have been proposed by Furth and Wilson (1981), Koutsopoulos et al. (1985),

Ceder and Stern (1984), Eberlein et al. (1998), Gallo and Di-Miele (2001), and Peeters and

Kroon (2001). Furth and Wilson sought to maximize the net social benefit, consisting of

ridership benefit and waiting-time saving, subject to constraints on total subsidy, fleet size,

and passenger-load levels. Koutsopoulos et al. extended this formulation by incorporat

ing crowding-discomfort costs into the objective function and treating the time-dependent

character of transit demand and performance. Their initial problem consisted of a nonlinear

optimization program relaxed by linear approximations. Ceder and Stern addressed the

problem with an integer programming formulation and a heuristic person-computer inter

active procedure. Their approach focuses on reconstructing timetables when the available

vehicle fleet size is restricted. Eberlein et al. (1998) studied a special dispatching problem

for the purpose of introducing deadheading trips in high-frequency transit operations. They

solved their dispatching strategy optimally; they also determined the number of stops that

could be skipped in order to minimize total passenger cost in the system.

Gallo and Di-Miele (2001) produced a model for the special case of dispatching buses from

parking depots. Their model is based on the decomposition of generalized assignments and

design, noncrossing, and matching subproblems. It can be extended to a case in which there is

an overlap between arrival and departure vehicles. Peters and Kroon (2001) present a procedure for planning an optimal cyclic railway

timetable, that is, a timetable in which trains leave

at the same minute every hour. The problem is represented through a constraint graph, in which

each node is an event that needs to be scheduled; cycles are examined according to a calculation

of tensions and potentials. The model is formulated as a mixed-integer nonlinear program with

the objectives of minimizing passenger time, maximizing timetable robustness, and minimizing

the number of required trains. A solution procedure is suggested, by which the nonlinear part

of the formulation is transformed into a mixed-integer linear problem that is an approximation

of the original problem; further actions are taken in order to reduce the number of constraints.

Louwerse and Huisman (2012) consider the problem

of adjusting a railway timetable in

case of large disruptions. They present integer programming formulations based on event

activity networks for the situations of a partial and a complete blockade. The disruption

measures included, to convert the normal timetable into a disposition timetable, delaying

trains, cancelling trains, and reversing trains at stations adjacent to the blockade. The model

also determines which trains need to be cancelled and defines the disposition timetable of

the trains that are not cancelled. The authors also present how the models can be modified

in case a regular disposition timetable is preferred. Two real-world cases are used to test

the models presented. The results show a balance between the number of trains that can

be operated in the disposition timetable and the maximum delay that is allowed. Also, the

authors show the effect of limited start inventories at the number of trains that can be operated.

By presenting also the regular disposition timetable, a trade-off is offered between a

disposition timetable that is more regular or one that deviates less from the normal timetable.

For all instances, the time needed to compute the disposition timetable is at most a few

seconds, which is acceptable in real-time disruption management.

Hassold and Ceder (2012) demonstrate how to model

make public-transit services more attractive

by using two simultaneous objectives: minimizing the average estimated passenger waiting

time and minimizing the discrepancy from a desired occupancy level on the vehicles. The

first objective aims at improving the service and attract more users, and the second objective

ensures economical operation. A network-based procedure is used to create timetables with

multiple vehicle types to solve this bi-objective problem. In addition, the authors suggest that this

methodology allows the synchronization of the arrival times with the departure times of other

modes and services to enable easier and more convenient transfers. The methodology developed

is applied to a case study in Auckland, New Zealand, where more than 46% of passenger

waiting time was achieved while, at the same time, attaining an acceptable passenger load on

all vehicles. Moreover, for each of the desired arrival times, which can be either given as a list

or determined by a fixed headway, the timetables show departures that arrive within a certain

tolerance at the interchange point. The suggested methodology demonstrates that a significant

improvement is attainable in terms of efficiency and use of resources where, at the same time,

this improvement can be combined with reducing the waiting time for passengers.

Hassold and Ceder (2013) present an extension

to a previously introduced multiobjective

methodology to create timetables using multiple vehicle types. The extension has been made to

base the creation of timetables on the data at different maximum load points. The methodol

ogy used allows evaluating the timetables according to multiple criteria. It has been applied to

a case study from a major bus line in Auckland, New Zealand. The Pareto frontiers presented

in this study indicate that the timetables created using the hourly and daily maximum load

points are very limited in their range to achieve different objective function combinations.

Moreover, they violate capacity constraints and show a number of standees on the vehicles.

The authors state that comparison of the results for the hourly and daily max load points

restricts conclusions to be derived for a universal recommendation. However, in this study,

the hourly max load point timetables consistently indicate a significantly higher number of

standee minutes, so that the daily max load point method should be preferred in most of the

cases in the case study presented. Although none of the solutions from the individual max

load point timetables dominate the other solutions, the authors say that it is the only method

that ensures that the constraints are held at all times. Moreover, it is the only method that

allows more flexibility to reach certain objective value combinations if required.

Ceder et al. (2013a) continue the study of different available bus sizes and describe a new

heuristic-based methodology to approach even-headway and even-load timetables. Ceder

et al. (2013b) also present the creation of bus timetables using different vehicle sizes but

with the use of simulation. Both of these articles are reviewed in the previous chapter and

are elaborated in the next chapter.

In the fourth and last group, the data-based models described in this chapter are based

on Ceder (1986, 2003). Advanced models pertaining to this group will be presented in the

next chapter.

EXERCISES

4.1 During a morning time period of 90 min (07:00-08:30), ride-check data were collected on a given bus route (average across several days). The route is 7 km long, and there is only one stop (B) between the first (departure) stop (A) and last (arrival) stop (C), the distance from A to B being 3 km. The average of the data gathered and information on three trips are as follows:

Departure time	Average load (passengers) when departing the stop	Desired occupancy (passengers)	Minimum frequency (veh/period)
(07:00-08:30)			
Bus capacity (passengers)	A	B	07:20 50 25 50 2 80 07:55 70 49 08:30 40 30

Given also a predetermined departure time of 07:00,

(a) Construct a new even-load timetable at point A using (i) Method 3 and (ii) Method 4 (30%) for frequency determination, with Method 4 having a new excess load along a maximum of 30% of its route length

(b) Repeat (a) the aforementioned step, but with a bus capacity of 60 (instead of 80) for all trips

4.2 Based on the data and results of Exercise 3.1 (Chapter 3),

(a) Construct evenly spaced headway timetables using Method 2 and 4 (20%) at the route departure point

(b) Construct clock-headway timetables using Method 2 and 4 (20%) at the route departure point

(c) Find the comparison measures for each of the four timetables in (a) and (b) given that the complete round-trip time is 45 min, that is, total number of departures and minimum required single-route fleet size

4.3 Based on the data and results of Exercise 3.2 (Chapter 3),

(a) Construct an even average load timetable according to Method 2

(b) Use the results of (a) to answer the following: (i) Are the loads at each stop, on each vehicle, in each hour, the same? (ii) If the individual loads are not same, at which stop and for what bus will the average load exceed the desired occupancy?

4.4 Using the following data and a requirement that the average max load on the last vehicle between 07:00 and 08:00 be the same as on the first vehicle between 08:00 and 09:00, or $44 = (64 + 24)/2$ passengers, Time Average max load (passengers) Desired occupancy (passengers) Passengers' arrival-rate pattern Minimum required frequency (veh/h) 07:00-08:00 240 64 Uniform 2 08:00-09:00 180 24 Uniform 4

(a) Construct even-headway timetable at the max load point (the same for both hours), assuming that passenger demand starts at 07:00; include the first departure after 09:00, assuming that demand after 09:00 is same as that from 08:00-09:00

(b) What departure times and average loads are derived by Method 2 using an evenheadway p

procedure in the transition between the 2 h ? E
explain differences from (a) and draw conclusions

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5 Chapter 5: Advanced Timetables: Optimization and Synchronization

The term advanced in the title of this chapter has already surrendered its level of computation.

Indeed not all the sections may interest practitioners. Nonetheless, this chapter and the next

present a practical framework and tools for improving and optimizing the construction and set

ting of transit timetables. Practitioners can, almost effortlessly, study the procedures with the

aid of graphical schemes, though the process of constructing timetables themselves requires

some mathematical treatment. The present chapter contains four main, independent parts. First, it picks up where Chapter 4

left off, with the construction of timetables, while assuring an even average max load at each

vehicle's max load point. This contrasts with Chapter 4's even average max load at a specific

time point, usually the hourly max load point. Second, the possibility of mixed fleet sizes is

introduced, in timetabling, to attain (1) even headways and even average load and (2) minimum

empty-seat km and minimum passenger wait time. Then, Section 5.5 presents the principles of

optimization to aid the remaining parts of this chapter. The third part covers optimal timetables,

in which passenger-crowding situations are minimized and are derived for a reduced fleet size.

In other words, an answer is given to the question: What are the best timetable frequencies

that can be assigned for a given fleet size when trying to minimize expected overcrowding situa

tions? The fourth main part of this chapter addresses and

formulates the problem of generating

headway-based timetables, in which the number of simultaneous vehicle arrivals at connection

(transfer) points is maximized. This is a maximum synchronization problem in time and space

that is meant to enable the transfer of passengers from one route to another with minimum

waiting time at the transfer points. Heuristic procedures (i.e., procedures or algorithms that

do not result for sure in an optimum solution) are proffered, in which the setting of departure

times follows efficient rules. These results are then improved (i.e., increasing the number of

meetings between vehicles on different routes) by allowing a shift in departure time within given

boundaries. This improvement uses defined efficient rules imbedded in the second heuristic

procedure. We provide detailed examples to illustrate the two proposed procedures explicitly.

These examples follow the two procedures step-by-step and can help practitioners gain a better

grasp of the analysis and results. Overall, Sections 5.7 asserts that, basically, a noncoordinated

transit network is simply the unfolding of a miscalculation. After reading Section 5.1, practitioners are encouraged to follow carefully the implementation

of Example 2, used initially in Chapters 3 and 4 in the case of even loads at different vehicle-max

load points. This implementation takes place in Section 5.2.3, which includes an interpretation

and explanation of the procedure developed. Section 5.3 attempts the creation of timetables

featuring both even headways and even loads; this is, in particular, relevant for a fleet consisting

of different vehicle sizes or if a new fleet is to be purchased. In addition, practitioners may be

interested in Sections 5.6.1 and 5.6.3, and possibly 5.6.4, as they discuss this special problem

of optimal timetables for a fixed fleet size. Practitioners are encouraged to read Section 5.7 to

get a better understanding of what a synchronized service means. It is also useful to solve the

exercises at the end of the chapter by means of the procedures described. The following is a short story about people who do not know how to keep up with a time

table. Three persons arrive at a train station and ask for the next train to some destination.

The conductor in the booth replies that the train has just left and the next one will leave in 1 h.

The three look at one another and say: "Let's go and have a beer, we have time." When they

return, the conductor tells them that they missed the train by 2 min and they will have to wait

1 h again. The three go off again to have another beer. For the third time, they miss the train, by

1 min. They are then told that the next train will be the last one of the day and will be leaving

in 90 min, so they better be on time. Ninety minutes, the three nod, and go to have more beer.

Arriving back at the station, they see the train starting to pull out. All three run after it, two

succeeding to hang on while the third person remains standing and laughing on the platform.

The conductor asks him: "Why are you laughing?" He replies: "You see the two people hanging

onto the train; they just wanted to escort me to the train." We hope in this chapter to offer

some planning tools to help passengers to be on time and also to put to rest the following saying:

"The only way to catch a bus/train is to miss the one before it."

5.1 INTRODUCTION

Although the even load procedure described in Chapter 4 ensures even average loads of jobs

at the job hourly max load point, it does not guarantee that the average load for individual

vehicles at other stops will not exceed jobs; therefore, the result may be overcrowding. In other

words, the hourly max load point represents an average peak point at job, whereas the max

load point for an individual vehicle can come at another stop and exceed jobs. For a given

time period, then, each vehicle may have a different max load point and a different observed

average load across the entire vehicle route. The purpose of the rest part of this chapter is to

derive a timetable such that, on average, all vehicles will have even loads (equal to the desired

occupancy) at the max load stop for each vehicle. The adjustments in the timetable are not

intended for highly frequent urban services, in which the headway may be less than, say,

10 min or an hourly frequency of about six vehicles or more. The objective of the rest part of

this chapter (based on Ceder, 2001 and Ceder and Philibert, 2014) is to construct a timetable

so as to avoid passenger-overcrowding situations (i.e., loads greater than d_{oj}).

The second part of this chapter introduces another variable, namely, the size of vehicles,

to attain timetables that result with both even headways and average even load. Certainly,

the implication of different vehicle sizes refers also to the component of vehicle scheduling,

and this will be dealt with in Chapter 8 . This new approach has the advantage of having

even headway timetables with balanced passenger loads resulting in an improved reliability

of service.

The third part of the chapter will examine optimal timetables for a network of transit

routes with reduced fleet size. In a practical context, this problem can arise when the fleet size

is reduced because of vehicle-age attrition and budget policy decisions. Although the next

two chapters cover vehicle scheduling and minimization of fleet size, they assume that the

frequencies of departure times are given. The purpose of the second part of this chapter is to

construct timetables while assuring that (1) the difference between the observed maximum load at

hour j and d_{oj} is minimized for all routes and (2) the given (available) fleet size is sufficient.

This is a problem of optimal multiterminal timetable construction for different fixed sizes of

a vehicle fleet. The formulation used in this part follows Ceder and Stern (1984).

The fourth and last main part of this chapter focuses on the problem of how to construct a

timetable with a maximum number of simultaneous vehicle arrivals at connection (transfer)

points. From the user perspective, a global approach to establishing these timetables usually

considers the minimization of travel and waiting (and possibly walking) times. This, then,

becomes a transit-network scheduling problem, which utilizes OD (Origin-Destination) data.

We assume, as is done in practice, that there is an existing transit network of routes with a

certain passenger demand according to the time of the day. Maximum synchronization is a

rather important objective from both the agency's and the user's perspectives. The process

described in this chapter follows Ceder et al. (2001) and Ceder and Tal (2001).

5.2 EVEN MAX LOAD ON INDIVIDUAL VEHICLES

This section continues from Sections 4.3 and 4.4, but employs a procedure to assure a ver-

age even load at each vehicle's maximum load point. This is in contrast to Principle 4.2 and

Proposition 4.2 (Section 4.4.1), in which even loads are practicable only at the hourly maximum

load point. To have an even desired load at the (individual) vehicle's maximum load is an addi-

tional notion in the attempt to further reduce overcrowding; the procedure can be imple

mented in certain transit operations scenarios by route and time period. We will continue

to use Example 2 as it appeared in Figure 3.7 and Table 4.1. Recently, this procedure was

published by Ceder and Philibert (2014).

5.2.1 Underlying principle

The results of applying Proposition 4.2 on Example 2 are shown in Figure 4.5 and Table 4.3.

The upper part of Figure 4.5 and the first column of Table 4.3 show the derived departure

times, based on even load at Stop 3 (6:00-7:00) and Stop 2 (7:00-8:00), using Method 2

for frequency determination. Take, for example, the resultant departure at 7:48 (Figure 4.5

and Table 4.3, using Method 2), with 60 average passengers on board at Stop 2 (hourly maximum

load point). This (desired) load of 60 is derived by construction. However, if we check the

average load of this 7:48 departure at Stop 3, we will find a larger load than 60. That is,

from Figure 3.7, the average passenger-arrival rate between 7:35 and 7:50 is $8 \frac{3}{15} = 5.53$

passenger per minute. Hence, between 7:36 and 7:48 (the results based on Proposition 4.2),

this rate holds and will yield $12 \cdot 5.53 \approx 66$ passengers on the 7:48 bus at Stop 3. In order

to overcome such undesirable overcrowding (above the desired load of 60), the following

principle is employed:

Principle 5.3

Construct a cumulative passenger-load curve at each stop (except for the arrival point) moving

horizontally per each d_j (desired occupancy at time period j) for all j on each curve until

each of the cumulative-load curves intersects, and then vertically to establish a departure

time for each curve. The required departure time is the minimum time across all curves.

Using the last determined departure time, set the corresponding loads across all curves; add

to these loads d_j for the next d_j (in the transition between time periods) and move horizon

and vertically, as in the first step, to derive the next departure time. Repeat until the

end of the time span.

Proposition 5.3

Principle 5.3 results in departure times such that the observed average maximum load on individual

vehicles approaches the desired occupancy d_j .

Proof: Proposition 5.3 is proved, as in Chapter 4, by construction. Fortunately Principle 5.3,

similar to Principles 4.1 and 4.2, can be graphically interpreted to ease the proof. Figure

5.1 illustrates Principle 5.3 for Example 2 in Figure 3.7. Figure 5.2 is an enlarged part of

Figure 5.1 for the beginning times at Stops 2 and 3. Figure 5.1 shows the load profiles

of

the vehicle departures and three cumulative-load curves at three stops. The curves at Stops 2

and 3 are shifted by 8 and 14 min, respectively, to allow for an equal time basis at the

route's departure point. At the initialization, the value of $d_0 = 50$ is coordinated with the Cumulative passenger load 200 100 6:15 7:15:50 7:35 7:50 22 7:00 47 97 157 202 300 200 100 300 200 100 7:00 7:00 7:54 7:43 6:41 182 92 52 269 344 229 185 100 35 33 +50 21 +50 Stop 1 departure Stop 2 ($t - 8$) ($t - 14$) 276 50 142 202 226 60 t Time/time-shifted 336 6:00 0 6:00 0 6:00 0 1 2 3 4 5 6 7 8 100 20 100 7:50 7:15 6:50 6:15 22 52 35 65 25 40 50 90 85 60 87 44 45 75 83 Distance (km) 20 40 60 80 0 20 40 60 80 0 40 60 80 0 20 40 60 80 100 0 20 40 60 80 0 7:35 Average vehicle peak load (8 min) (6 min) (16 min) Passenger load 6:14 7:04 7:20 7:29 312 Stop 3

Figure 5.1 Interpretation of Principle 5.3 for Example 2 (see also enlarged Figure 5.2).

three cumulative-load curves to obtain: 6:51.5 at Stop 1, 6:14 at Stop 2, and 6:41 at Stop 3.

Although this is clearly seen in Figure 5.3 for Stops 2 and 3, the determination at Stop 1

is based on an arrival rate of 2 passengers per minute ($=50/25$) and an additional load of

3 beyond the observed load of 47 at 6:50. According to Principle 5.3, the minimum time

among the three is selected: the departure at 6:14 (emphasized in the figures). This means

that the scheduled 6:15 departure is shifted backward by 1 min to have an average of 50

instead of 52 passengers at Stop 2. Then, $d_0 = 50$ is added to the value of the cumulative

load curves at 6:14 to attain three more candidate departure times: 7:02 (see following note),

6:52, and 6:41 for Stops 1, 2, and 3, respectively. Hence, 6:41 is selected. If the candidate

time is beyond the boundary of the time period (7:00), another $d_{o2} = 60$, should be

applied; therefore, 7:02 becomes 7:07 for Stop 1. The procedure continues with $d_{o2} = 60$ and

results in 7:04 as the third departure having an average even 60-passenger load at Stop 2 and

four more departures as illustrated in Figure 5.1. This completes the proof-by-construction

of Proposition 5.3. Time (shifted) 6:15 7:156:50 200
200 100 6:52 7:00 6:41 6:14 7:04 182 92 52 100 35 33 +50
Stop 2 $t - 8$ $t - 14$ 142 202 +60 100 50 +60 185 Cumulative
passenger load +60 6:00 0 6:00 0 6:23 50 Stop 3 82

Figure 5.2 Enlarged part for Stops 2 and 3 of the graphical interpretation of the algorithm balance shown in Figure 5.1.

The basic assumptions of Proposition 5.3 are as follows:

(1) the change of departure times

will not affect the arrival pattern of passengers and (2) the change of departure times (with

the same frequency) will not affect passenger demand. These assumptions are not critical

to the procedure described, since changes in passenger-arrival patterns and demand will be

captured frequently using updated data.

5.2.2 Minimum frequency standard

The procedure described by Principle 5.3, similar to Principle 4.2, does not guarantee that

the minimum frequency standard (the inverse of the headway policy), F_{mj} , for each time

interval, j , will always be met. Usually, F_{mj} is necessary at the beginning and at the end of the

day. Therefore, during the process of Principle 5.3, F_{mj} needs to be checked in the

transition

between time intervals—that is, whenever a new derived departure is in advance of the time

boundary. Using N_j , which is the number of departures derived by Principle 5.3 during time

interval $j - 1$, F_{mj} is checked in the following manner:

1. Check for each j , $N_j > F_{mj}$?; if yes—END, otherwise continue $j = 1, 2, \dots, q$.

2. Calculate the alternative desired occupancy, d_{om} , for the minimum frequency situation: $d_{om} = \max_{i=1,2,\dots,n} [L_i(t_j) - L_i(t_{j-1})]/(F_{mj} + 1)$, and return to the procedure based on Principle 5.3 at the $(j - 1)$ th transition time, where here a represents, t_j is the j th transition time (between time periods), and $L_i(t_j)$ is the cumulative-load curve at Stop i at t_j ; change d_{oj} to d_{om} and go to (1).

The procedure described by steps (1) and (2) ensures that the minimum frequency criterion

will always be met. If the procedure described by Principle 5.3 results in time interval j

with a frequency of less than F_{mj} , then the maximum difference in passenger loads on the

cumulative-load curve between t_j and t_{j-1} determines a new desired occupancy, d_{om} . This d_{om}

then replaces d_{oj} until the derivation of the first departure time at interval $j + 1$. Difference from desired max load (passengers) 10 30 -10 7:00 8:00 6:00 40 20 0 Desired occupancy 50 passengers 60 passengers Even load at hourly max load point Observed data Evenly spaced headways Even load at individual max load point 6:41 7:04 7:20 7:29 7:43 Time 7:54 6:14

Figure 5.3 Differences between individual vehicle's max load and the desired occupancy according to the procedure selected in Example 2.

There are two more comments that should be mentioned about the procedure described

by Principle 5.3 and the minimum frequency case:

1. If a different size of vehicle (than the one for which d_{oj} is set) is considered for the determination of a given departure, d_{oj} can be changed in Principle 5.3. This may be the case for an excessive load, which may result in too short a headway, or for the opposite case of a large amount of empty seat-km, resulting in F_{mj} . Both cases can be observed from the cumulative-load curves or load profiles.
2. If use of a headway criterion policy (or the inverse of F_{mj} for each j) is preferred, then there will be an extra check in the procedure for F_{mj} in (b) for each derived departure time k at j : if $h_{kj} > 1/F_{mj}$, set $h_{kj} = 1/F_{mj}$ and continue with the procedure described by Principle 5.3, where h_{kj} is the headway between the k th and $(k - 1)$ th departures, otherwise-END.

5.2.3 Comparison

The comparison between the observed data of Example 2 and the results of the procedures

using Principles 4.1, 4.2, and 5.3 are summarized in Table 5.1 and illustrated in Figure 5.3.

Table 5.1 Departure times for Example 2 route-departure points (three even headways and even load procedures), accompanied by their associated max load and its starting stop, using frequencies in Method 2

Trip characteristics associated with

route-departure point Observed data Procedures Evenly spaced headways Even average load at hourly max load point Even average load at individual max load point

1st departure Depart. time 6:15 6:22 6:23 6:14 max load 52 60 61 50 max load point Stop 2 Stop 2 Stop 2 Stop 2

2nd 6:50 6:44 6:50 6:41 65 41 50 50 Stop 3 Stop 3 Stop 3 Stop 3

3rd 7:15 7:05 7:07 7:04 90 61 61 60 Stop 2 Stop 2 Stop 2 Stop 2

4th 7:35 7:22 7:22 7:20 87 66 59 60 Stop 2 Stop 2 Stop 2
Stop 2

5th 7:50 7:39 7:36 7:29 83 77 62 60 Stop 3 Stop 2 Stop 2
Stop 2

6th – 7:56 7:48 7:43 94 66 60 Stop 3 Stop 3 Stop 3

7th – – 8:00 7:54 66 60 Stop 3 Stop 3

a Subsequent to a predetermined 6:00 departure.

Table 5.1 shows the associated individual average max load
under each departure; its cor

responding stop appears under max load.

Figure 5.3 shows the diversity of the i
ndividual max loads across all the procedures
and

the observed ones. Certainly, this comparison a
pplies only to the specific Example 2, as it

will vary from one situation to another. In
situations in which the hourly max load p
oint

usually coincides with the individual max load, the results
of Principle 4.2 will be close to

those of Principle 5.3 as is the case of
Example 2. In the examples that were p
resented by

Ceder (2001), the differences are more significant.

5.2.4 Case study of even headway and the two even load procedures

This section is based on Ceder and Philibert (2014) in
which real data from Auckland, New

Zealand, are used for the comparison between
even headway and even load at hourly max

load point and even load at individual max
load timetables. Two types of analyses are c
on

sidered: (1) use of calculated stop-based demand pattern exhibited by the data and (2) use of

demand derived by simulation. In other words, Analysis (1) examines passenger loads on

the vehicles for each of the three timetables derived and Analysis (2) examines the same, but

introduces stochastic consideration of travel time with different passenger-boarding rates

for the three timetables.

Consider the even headway procedure as EHP, the even load procedure based on hourly

max loads as E LHP, and the even load procedure based on individual-vehicle max loads

as ELIP. The data were collected for 3 weeks on 2012 by both intelligent transportation

systems (ITS) (GPS-based) information and checkers. The three timetables by the three

procedures were calculated utilizing MATLAB®. The values of the desired occupancy

and minimum frequency standards follow those of the Auckland company, that is, $d_{ou} = 37$

(all day), $F_{mj} = 4$ (6 a.m.–10 p.m.), and $F_{mj} = 2$ (10 p.m.–midnight).

The results of Analysis (1), without simulation, and Analysis (2) with simulation, are

shown in Table 5.2 and Figures 5.4 and 5.5, respectively. These table and figures refer to

the description of overcrowding at the max load point of each vehicle (each departure) in

comparison with the desired occupancy. Table 5.2 shows the percentage of departures fac

ing overcrowding by each procedure and Figure 5.4 of the load differences of Analysis (1).

W e n o t e t h a t A n a l y s i s (2) e m p l o y s s i m u l a t i o n (V i s s i m R e l e a s e 5 . 4 0) f o r t h e e x a m i n a t i o n o f

o v e r c r o w d i n g s i t u a t i o n s w i t h t h e u s e o f v a r i a t i o n s o f t h e a v e r a g e s o f p a s s e n g e r - b o a r d i n g

r a t e s a n d t r a v e l t i m e s , a s i s s h o w n i n F i g u r e 5 . 5 . T h e r e s u l t s o f b o t h a n a l y s e s c l e a r l y d e m o n

s t r a t e t h a t u t i l i z i n g E L H P l e a d s t o o v e r c r o w d e d v e h i c l e s . H o w e v e r , t h e d i f f e r e n c e b e t w e e n t h e

r e s u l t s o f E L H P a n d E L I P i s m u c h l e s s t h a n w i t h E H P . T h e s t o c h a s t i c i m p a c t o f v a r i a t i o n s

o f p a s s e n g e r - b o a r d i n g r a t e s a n d t r a v e l t i m e s c e r t a i n l y c a u s e i n c r e a s e i n t h e o v e r c r o w d i n g

p e r c e n t a g e s .

A n a l y s i s (2) e n a b l e s t h e c o m p a r i s o n b e t w e e n t h e s c h e d u l e d a n d s i m u l a t e d d e p a r t u r e t i m e s

a t e a c h s t o p . T h i s c o m p a r i s o n o f s c h e d u l e a d h e r e n c e h a s a d i r e c t i m p l i c a t i o n o f t h e r e l i a b i l i t y

Table 5.2 Results of overcrowding by Analysis (1) compared with Analysis (2) Characteristic EHP(%)	
ELHP(%)	ELIP(%)
Percent of overcrowded vehicles at the max load point using calculation (Analysis (1))	16.1
Percent of overcrowded vehicles at the max load point using simulation (Analysis (2))	19.1

o f s e r v i c e , o r i n o t h e r w o r d s , o n t h e i m a g e o f t h e t r a n s i t s e r v i c e . F i g u r e 5 . 6 i l l u s t r a t e s t h e s e

d i f f e r e n c e s . C a r e f u l o b s e r v a t i o n o f t h e r e s u l t s o f F i g u r e 5 . 7 , a n d a n a n a l y s i s o f t h e s t a n d a r d

d e v i a t i o n s o f t h e s e d i f f e r e n c e s , s h o w s t h a t t h e E L J P r e s u l t s i n t h e m i n i m u m d i s c r e p a n c y i n

m o s t o f t h e s t o p s , t h u s w i t h t h e b e s t r e l i a b i l i t y e x p e c t e d .

5.3 B O T H— E V E N H E A D W A Y S W I T H E V E N A V E R A G E L O A D S

This section continues from Sections 4.3 and 4.4 in which timetables with even headways

and even average (passenger) loads, respectively, were introduced. The even headway feature

reduces the flexibility of the scheduler to accommodate fluctuations of demand within

a given time period. This lack of flexibility may result in undesirable operational scenarios

Figure 5.4

Differences between the Max and desired loads

for individual vehicles using Analysis (1) without

simulation. -20 -40 Difference from desired

max load (passengers) 0 20 40 60 80 6:00

8:24 10:48 13:12 15:36 18:00 20:24 22:48 Departure time

EHP ELHP ELIP

Figure 5.5

Differences between the Max and desired loads

for individual vehicles using Analysis (1) without

simulation. -20 -40 Difference from desired

max load (passengers) 0 20 40 60 80 6:00

8:24 10:48 13:12 15:36 18:00 20:24 22:48 Departure time EHP ELHP ELIP

for individual vehicles using Analysis (2) with variations of boarding rates and travel times.

such as overcrowding or vehicles running almost empty. Uneven loads lead to either pas

senger discomfort, in the case of overcrowding, or uneconomical and energy inefficient

operation of the vehicles in the latter case (Potter, 2003). However, the even load timetables

can lead to long and exceedingly irregular headways and thus increase waiting times for pas

sengers who arrive randomly. The time passengers spend at the bus stop waiting for a bus to

arrive is largely recognized as a major factor of satisfaction of the transit user. The waiting

time interpretation and formulation is described in Section 18.4.

This section introduces an approach to overcome the disadvantages of both even head

way and even load by using different types and sizes of vehicles. Figure 5.7 illustrates how

multiple vehicle sizes can be used to achieve such timetables. Nonetheless, in practice, this

is not always possible. Hence, the quality of the timetables is based on two criteria: (1) load

discrepancy on the vehicles from a desirable load and (2) time discrepancy from a desirable

headway. The process described follows Ceder et al. (2013b).

5.3.1 Graphical approach

The focus of the following analysis is to determine the appropriate headway and departure

times so as to come near to even headway and even load timetables. This is done by subse

quent heuristics.

A range of feasible headways for multiple vehicle sizes are calculated using the inverse

of the frequency based on Method 2 or on Equation 3.2; this is to find the most suitable

headway. It is done by calculating a lower bound, H_{min} , and an upper bound, H_{max} ,

for the headway by using the smallest and largest vehicle sizes, respectively. Any headway

outside this range would result in either underutilized service or overcrowding situation.

Given a length of a time period T_j , $j = 1, 2, \dots, q$, a desired occupancy d_{okj} for vehicle size S_k , 8355 0 0.5 1 1.5 2 2.5 89068902 8911 8915 8919 8597 1407 8594 8433 8429 8425 8421 8417 8413 8409 8405 8401 8503 7151 7147 7058 8900 8904 8909 EHP ELHP ELIP

D i

e

r e

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e d

a n

d

s i m

u l

a t

e d

d

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e (

m

i n

) 8913 8917 8921 7894 7896 8435 8431 8427 Stop ID 8423 8419
8415 8411 8407 8403 8505 8501 7149 7089

Figure 5.6 Results of scheduled vs. simulated headways
using Analysis (2). (b) (a) 7 14 21 2810 50 100 150 Load=35
Load=50 Load=37 Load=18 C u m u l a t i v e p a s s e n g e
r l o a d Even-headway departure times using one vehicle
size (50 seats) Time (in min) 3025201510 50 100 150 Load=50
Load=50 Load=50 13 C u m u l a t i v e p a s s e n g e r l
o a d Even-load departure times using one vehicle size (50
seats) Time (in min) (c) 30252010 15 Load=50 Load=75
Load=25 50 100 150 5 Time (in min) C u m u l a t i v e p a
s s e n g e r l o a d Even-headway departure times using
three vehicle sizes (25, 50, and 75 seats)

Figure 5.7 Illustration of departure times with their

corresponding passenger loads in three cases with (a) even headways (b) even load, and (c) both, by three vehicle sizes.

$k = 1, 2, \dots, K$, and period j , and $h_{\max}$ the inverse of $F_{\min}$, the range for the best headway,

B_{hw} , is formulated in the following equation: $H_{\min} \leq B_{hw} \leq H_{\max}$ $\min_{j \in J} \{ \sum_{i \in I} d_{ij} \} \leq \sum_{i \in I} \lambda_i \leq \min_{j \in J} \{ \sum_{i \in I} d_{ikj} \}$ (5.1)

where

d_{ij} and d_{ikj} are, respectively, the desired occupancy of the smallest and the largest vehicles for period j

$h_{\min}$ is the minimum headway, from a productivity perspective

5.3.1.1 Heuristic procedure for finding the headway

Let the observed max load of passengers, traversing the max load segment, during period

T_j be P_{mj} . The following procedure (Ceder et al., 2013b) describes how to determine the

preferred (best) headway B_{hw} . Two criteria are used: LD and %LD, the total of the load

discrepancy and the total of the relative load discrepancy, respectively. The load discrep

ancy, designated $Disc(t)$ for a departure at t , is the difference between the expected load

on a given vehicle size and its d_{ikj} at the max load point. The relative load discrepancy,

designated $RLD(t)$ for a departure at t , is the percentages of discrepancy from d_{ikj} of a given

vehicle size.

The best headway can be either based on the load-discrepancy criterion, designated

B_{hwLD} , or on the relative load-discrepancy criterion, designated $B_{hw\%LD}$. The best head

way is determined from the minimum between its related total load discrepancy $LD(BhwLD)$

and its related total relative load discrepancy $\%LD(BhwLD)$. In practice, timetables are pro

vided in integer minutes. Thus, all time values used such as the headway, increments, and

shifts of departure times, are integers. Figure 5.8 summarizes the procedure to find the best

headway; this procedure is called Algorithm 5.1.

5.3.1.2 Heuristic procedure for creating timetables

Based on Algorithm 5.1, the expected passenger load for a vehicle departing at time t is des

ignated $L(t)$, and it is based on the cumulative-load curve at the max load point. Following

is a heuristic procedure to create timetables based on the previously determined preferred

headway. The service hours are assumed to be split into q adjacent periods T_j , $j = 1, 2, \dots, q$,

where T_j , starts at time t_{js} and ends at t_{je} . In this procedure, vehicles are assigned departure

times based on even headways while complying with the requirement to handle the observed

max load of passengers.

The procedure to create the departure times is called Algorithm 5.2. To enable a compari

son base, the last departure of any timetable, dep , is modified to coincide with the length

of T_j . The transition between subsequent periods is done by modifying their end and start

times. In case that the last departure does not coincide with the end of the period and would

fall into the next period, the departure is cancelled and the end of the current period is set

to the time of the last departure. Following a change of the end of period T_j and the start of

T_{j+1} , Algorithm 5.1 can be executed again using the modified period T_{j+1} to determine the

best headway.

A flowchart of Algorithm 5.2 is given in Figure 5.9; this figure also highlights some macro

steps such as the Next t -function to determine the next departure time. Those macrosteps

represent a range of different actions and outcomes, determined by a certain strategy chosen Initial input: -Set of vehicle sizes $\{S_k\}$, $k = 1, 2, \dots, K$, and desired occupancies d_{okj} -Cumulative load profiles at max load point - P_{mj} time period T_j $hw = H_{min}$ $t + hw = T_j$? Final output: Best headway with respect to -Load discrepancy: B_{hw} LD such that $LD(B_{hw} LD) = \min_{hw} [LD(hw)]$ -Relative load discrepancy (%): $B_{hw} \%LD$ such that $LD(B_{hw} \%LD) = \min_{hw} [\%LD(hw)]$ $hw = hw + 1$ $t = t + hw$ Record: t , $L(t)$, S_k , $Disc(t) = RLD(t) = 100$ No Yes Yes $H_{min} = \max d_{okj} \times T_j P_{mj}$, h_{min} $H_{max} = \min d_{okj} \times T_j P_{mj}$, h_{max} $t = 0 + hw$ Compute load $L(t)$ $hw = H_{max}$? $L(t) - d_{okj}$ $L(t) - d_{okj} = \min_i$ Find S_k such that $() L(t) - d_{okj}$ $L(t) - d_{okj} d_{okj} LD(hw) = L(t) - d_{okj} () \%LD(hw) = N(hw)$ $RLD(t) \Sigma$ Record: hw , $LD(hw)$, $\%LD(hw)$ No

Figure 5.8 Flowchart of Algorithm 5.1.

to deal with the choices of vehicle size. The following three main strategies are considered

in the first macrostep to assign vehicle size.

Strategy C1: Minimizing the size of the bus by assigning the largest bus size amongst all available sizes such that its seat capacity is less than or equal to the average observed (hence expected) passenger load, at the max load point. That is, for a departure at t with an expected load of $L(t)$, $d_{o(k-1)j} \leq L(t) < d_{okj}$, where $d_{o(k-1)j}$ and d_{okj} are the desired Initial input: -Set of vehicles S -Load profile -Desired headway hw -Time periods T_j , $j = 1, 2, \dots, q$ $t = t_{je}$? Stop: Algorithm 5.2 ends successfully

$t = t_{(j+1)s}$ Record: t_{dep} , $L(t_{dep})$, S_k , $Disc(t_{dep}) = L(t_{dep}) - d_{okj}$ Yes Yes No $t = t_{js}$ Compute load $L(t)_j =$


```

q? Assign bus: S k Output Timetable with bus assignment
for T j Set departure time: t dep Establish comparability
Next t Legend of Macro Step Next t > t je ? j = q? No Yes
Yes Set t je = t, and t (j+1)s = t + 1 Run Algorithm 5.1
for T j+1 No No

```

Figure 5.9 Flowchart of Algorithm 5.2. occupancies of two sequential (w.r.t. size) available vehicle sizes; $d_o(k-1)j$ will then be the desired occupancy of the assigned vehicle. This may imply overcrowding on certain vehicles.

Strategy C2: Maximizing the size of the vehicle by assigning the smallest vehicle size amongst all available sizes such that its seat capacity is greater than or equal to the average observed (hence expected) passenger load, at the max load point. That is, for a departure at time t with an expected load of $L(t)$, $d_o(k-1)j < L(t) \leq d_{okj}$; d_{okj} is the desired occupancy of the assigned vehicle size k in period j . This may imply empty running seats on certain vehicles.

Strategy C3: Selecting the vehicle whose size is closest to the average observed load, per vehicle, at the max load point. That is, for a departure at time t with an expected load of $L(t)$, select the size S_k such that $|d_{okj} - L(t)|$ is minimal for all k . This can result in either overcrowding or running empty seats.

Figure 5.10 illustrates the three strategies on a cumulative observed-load curve of individual

vehicles at the max load point. This curve follows the method described in Section 4.4 in

which for a given desired load, the departure times are established by moving horizontally

with increments of the desired load, intersecting the curve, and vertically to find the departure

time on the time axis. In the example of Figure 5.10, the examined departure is at

$t = 24$ (in minutes after the beginning of the time period). The previous departure relates

to

$t = 15$ with 25 passengers (pax) on board at the maximum load point. The load associated with

the examined departure is $L(24) = 60 - 25 = 35$ pax. Vehicle sizes available are of $S_1 = 25$

and $S_2 = 40$ seat capacity. For Strategy C1, vehicle S_1 is selected and vehicle S_2 is selected

for Strategies C2 and C3.

The three strategies allow for the creation of timetables with even headways, but the

results could be with uneven loads even when utilizing different vehicle sizes. Hence, there

is a benefit of shifting departure times away from the even headway to approach a better

balance of onboard passenger load. Figure 5.11 illustrates an example in which there are

departures at $t = 15$ and $t = 22$ with even headway of 7 min. However, $L(22) = 30$ pax and

the assigned vehicle has a seat capacity of 35 pax. The shifting to the right in Figure 5.11 24 0 10 20 Cumulative load (passengers) 30 40 50 60 70 0 5 10 15 20 25 30 Time (min) 40 pax 35 pax load 25 pax C1 C3 C2

Figure 5.10 Strategies for selecting the size of vehicle.

(see arrow) will make the headway 8 min, but will provide a more efficient service without

harming, in an average sense, the quality of service.

Shifting tolerance can be introduced to maintain the headways as even as possible.

Following the calculation of the departure time by the allowed shifting, or without shift

ing, the next departure examined will be determined by

applying the Next t function of

Algorithm 5.2 in Figure 5.9. Assuming that an initial departure at t_0 is being shifted by a

small time increment Δt to $t_0 + \Delta t$, the Next t function can have two possible outcomes.

First, the Next t follows the original timetable before shifting and sets the next departure

to $t_0 + hw$. We note that in certain cases Δt might be larger than the desired headway hw

to have the next departure as $t_0 + m \cdot hw$ with m being the first integer to comply with $\Delta t <$

$m \cdot hw$. Second, the Next t strictly adheres to the desired headway with the next departure

time examined be $t_0 + \Delta t + hw$.

Combining the two options for Next t with different shifting policies allows for the con

to predetermine which strategy is the best for the given scenario. In this case, different

scenarios should be tested and examined. Five basic methods can be defined and termed as

A, B, C, D, and E based on the setting shown in Table 5.3. Each of these methods can use

vehicle selection strategies C1, C2, or C3, thus 15 combinations for constructing a set of

feasible timetables can be examined. Note that setting up the limited shift to different values

can increase the number of possibilities; in Ceder et al. (2013) seven, not five, methods are

shown including these possibilities. Table 5.3 shows all combinations. 22010 Cumulative load (passengers) 20 30 40 50 60 70 80 90 0 5 10 15 20 25 Time (min) 30 pax 35 pax

Figure 5.11 Shifting procedure.

Table 5.3 All combinations used in the selection of best timetable Method

Strategy A B C D E

Bus-selection strategy C1, C2, or C3 C1, C2, or C3 C1, C2, or C3 C1, C2, or C3 C1, C2, or C3

Setting departure

time No shift Unlimited shift Unlimited shift Limited shift Limited shift

Next t function NA $t + \Delta t + hw$ $t + hw$ $t + hw$ $t + \Delta t + hw$

5.3.1.3 Evaluation of solutions

The timetable solutions of all combinations are compared with indicators of expected

cost, the level of service, the headways, and the resulting waiting times. For simplicity, it is

assumed that vehicle size S_k is the desired occupancy level d_{okj} for all periods j , which is usu

ally, but not necessarily, the number of seats.

5.3.1.3.1 Cost indicator

The cost of implementation cannot be determined without knowing the schedule of vehicles.

However, the number of departures of each vehicle type (size-dependent) can provide an

indication of the cost. Generally speaking, the larger the number of departures in a time

table, the higher the number of vehicles expected. Moreover, the relative number of depar

tures associated with each vehicle size may provide an idea about the required composition

of the fleet of vehicles.

5.3.1.3.2 Level-of-service indicators

Level-of-service measures are associated with the headways and the deviation of the max

load from the standard desired load level. Because the headways are based on the max load

observed, the measures used are load-discrepancy dependent. Four measures are used in

the analysis: (1) total load discrepancy, (2) average relative (%) load discrepancy, (3) average

percentage of standees, and (4) level of service.

5.3.1.3.2.1 TOTAL LOAD DISCREPANCY

The load discrepancy refers to the difference between the average number of onboard pas

sengers at the max load point and the desired occupancy for a given vehicle size. It is $Disc(t) =$

$L(t) - d_{okj}$ for a bus of size S_k in period j and $L(t)$ passengers at the max load point; if posi

tive, it represents overcrowding or average number of standees, otherwise underutilization

or average number of empty seats at the heaviest load point. The total load discrepancy for

a timetable T with N departures at times t_i , $i = 1, 2, \dots, N$, is $LD_T = \frac{1}{N} \sum_{i=1}^N (L(t_i) - d_{okj})$ (5.2)

5.3.1.3.2.2 AVERAGE RELATIVE LOAD DISCREPANCY

In contrast to $Disc(t)$, the relative load discrepancy is the ratio between $Disc(t)$ and the

desired occupancy of vehicle S_k in percentage. That is, given that the desired occupancy is

the number of seats, a load discrepancy of 5 passengers in a vehicle with 75 seats has a

relative load discrepancy of 6.7% compared with 25% for a vehicle size of 20 seats. Thus,

it is in percentage $RLD(t) = 100((L(t) - d_{okj})/d_{okj})$.

The average relative load discrepancy for a

timetable T_t with N departures at times t_i is

$$\%LD(T_t, RLD_t, N) = \frac{1}{N} \sum_{i=1}^N (5.3)$$

5.3.1.3.2.3 AVERAGE PERCENTAGE OF STANDEES

Equation 5.3 provides the average relative load discrepancy in absolute terms, but does not

give any insight of the nature of the discrepancy. The average percentage of standees can

add further information to enable drawing more solid conclusions. The average percentage

of standees for a timetable T_t with N departures at times t_i is

$$\% \text{ Stand}(T_t, RLD_t, N) = \frac{1}{N} \sum_{i=1}^N (5.4)$$

5.3.1.3.2.4 AVERAGE LEVEL OF SERVICE

The average level of service (LOS) refers to the loading situation of all departures at the daily

max load point. Given a timetable T_t with N departures at times t_i and vehicles of sizes S_{ki}

(with a desired occupancy of $d_o(k_i)$) carrying $L(t_i)$ passengers at the max load segment, for

$i = 1, 2, \dots, N$ (assuming theoretically that for each departure another size of a bus can be

assigned), the average level of service is defined as

$$LOS = \frac{1}{N} \sum_{i=1}^N \frac{L(t_i)}{d_o(k_i) S_{ki}} (5.5)$$

That is, the desired occupancy level at every departure provides an LOS of 100%. Passengers

experience overcrowding for LOS greater than 100% and vehicles are underutilized for LOS less

than 100%. This indicator represents only an average figure, thus cannot be used on its own.

5.3.1.3.3 Time indicators

The quality of the timetable with respect to the headway is

determined by applying the fol

lowing i ndicators.

5.3.1.3.4 Average headway

Computation o f t he a verage h eadway o f t he nal
t imetable.

5.3.1.3.5 Expected waiting time

The expected waiting time, of randomly arriving passengers,
is dependent on the variance of

the headway as is shown in Section 18.4. Because of the
coupling of regularity and the aver

age headway criteria, this is a better indicator than the
average headway. However, to make

different t imetables c omparable t he c onsideration
o f b oth i ndicators n eeds t o t ake p lace.

5.3.1.3.6 Time-discrepancy ratio

The time-discrepancy ratio (TDR) is deñned as the ratio
between the expected waiting time

and the average headway. This normalizes the waiting time
and makes different timetables

comparable. It provides information about the evenness of
the headways. For simplicity this

indicator is set between 0% and 50%, where 0% presents the
perfect even headway timetable

without any time discrepancy and 50% the case in which the
headway is exponentially dis

tributed. This TDR indicator is termed %TDR and shown in
the following equation: $\%TDR = \frac{W_t}{h_w} \frac{\text{Var } h_w}{h_w} = \frac{\text{Var } h_w}{h_w^2} \times 100$ (5.6)

Further details and a case study of this even headway and
even load graphical-based approach

appear in Ceder et al. (2013). The following different
approach is based on the optimiza

tion of the timetables to attain at the same time (called multiobjective approach) minimum

empty-seat km and minimum passenger waiting time. The interpretation of optimization

and its principles appear in Section 5.4; those who are unfamiliar with optimization are

advised to read Section 5.5 before Section 5.4.

5.4 TIMETABLES BASED ON A MULTIOBJECTIVE APPROACH

The methodology described in this section is based on Hassold and Ceder (2012). It is a

network-based approach using a multiobjective label correcting algorithm to find the set of

dominant solutions considering minimizing a penalty for running empty seats (ES) on the

vehicles and minimizing the estimated passenger waiting time (WT).

Figure 5.12 illustrates a sample network of the problem for a given bus line. The network

consists of nodes representing feasible departure times of buses and directed arcs connecting

certain nodes, which represent feasible sequences of departures. The source S and sink T are

the beginning and end of the operating period of the bus service. The service is using mul

tipl vehicle types/sizes, represented by different arcs in the network. A dotted line between

departures D2 and D5 implies that D5 is performed by a standard bus, if a path is selected

from S to T via this arc. For another path, for example, a path containing D3 the departure

D5 is handled by a different vehicle type, a minibus. Moreover, each arc has two cost ele

ments associated with the objectives of the problem. These costs represent waiting times for

passengers arriving between the two departures, and a penalty for empty seats. The WT

is calculated as passenger waiting minutes, namely, the sum of passenger waiting times; in

the example it is assumed that passengers are arriving randomly at a uniform arrival rate Arc costs: Waiting time (WT) Standard bus Minibus Articulated bus Empty seat penalty (ES) D1 D2 D3 D4 D5 DT Time t 1 5 T 2 3 (WT / ES) 4 S

Figure 5.12 Sample network.

though any demand function can be considered. ES is the empty seats on the bus multiplied

by the distance traveled. In Figure 5.12, the arc D 2-T and D 4-T share the same end node and

same demand pattern. Because the headway D 2-T is greater than for D 4-T, the total cost

WT will be greater for D 2-T. However, both arcs represent the same vehicle type such that

the higher occupancy for D2-T results in a smaller ES than for D4-T.

The procedure of creating the network for a transit line is illustrated in Figure 5.13. The

cumulated demand function $D(t)$ is given for each stop of the line. Naturally, the demand

functions shown vary along the line. In addition, the procedure considers the dependent

travel times between stops $T(t)$ visualized by different slopes in the connecting arcs between

two nodes. It is to note that these travel times can vary significantly by the time of the day. In

the example of Figure 5.13, a greater travel time is causing the headway between departure

1 and 2 to be shorter at Stop 2 than at Stop n. This also impacts the waiting times for arriving

passengers at Stop n. The nodes on the left side of Figure 5.13 represent departure times

at the first stop and each sequence of connected nodes has to satisfy the capacity constraint

of the vehicle at all individual stops; otherwise the selected departure time is invalid. In

addition, there are minimum and maximum headway constraints providing a certain time

window for each departure time to take place. The choice between different vehicle types

(minibus, standard bus, and articulated bus) results in different feasible departure-time

ranges for each vehicle type as is illustrated in Figure 5.14. The solution algorithm starts at

the source node S and finds the feasible departure times for each given vehicle type that are

then added to the network as individual nodes. These nodes (see Figure 5.12) are connected

to S by a directed arc with the information about the type of vehicle used and the associated

costs. Separate arcs are inserted for nodes reached by different vehicle types. The algorithm

continues from the next (earliest) node in the

This is in fact a network flow formulation (

see Appendix 6.A) of the shortest path problem

with the objective function in Equation 5.7 aiming at minimizing the sum of the costs for all

arcs used. Equations 5.8 and 5.9 make sure that the inbound flows equal the outbound flows

at every node and that every arc can be used either exactly once or not at all. Nodes S generate

ates and Node T absorbs one unit of flow, thus these two nodes are exceptions. Standard

shortest path algorithms, such as Dijkstra's (1959) can be used to find the timetable with

the least waiting time or the least empty-seat km. See Appendix 9.A for these algorithms. Selected headway Time t Time t

Maximum

headway Maximum headway Capacity constraint Capacity constraint Minimum headway

Minimum

headway Minimum occupancy level Minibus Standard bus Articulated bus

Departure 1 Stop 1 TT 12 (t) Stop 2 D(t) D(t) t t Demand D(t) at stop 1 Demand D(t) at stop 2 Second departure

Figure 5.14 Headway ranges for different vehicle types.

However, we are dealing with two objectives at the same time and, thus, we face a biob

jective problem. Equation 5.10 shows the biobjective formulation, and to this we add the

constraints of Equations 5.8 and 5.9: $\min z(x) \quad z_c x \quad z_c x$
 $i j \quad i j \quad A \quad i j \quad i j \quad A = = = \begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 \end{matrix} \in \Sigma \Sigma 1 \quad 1 \quad 2 \quad 2$

(,) (,) (5.10)

The objective function vector for a very feasible solution in X consists of two elements, one

for each objective. In contrast to the single objective case this makes it impossible to compare

solutions with a simple greater/equal/smaller than relation. To do so the following,

Definition BI is introduced:

Definition BI: A solution $x \in X$ with objective function values $z(x)_1$ and $z(x)_2$ is called efficient or

dominant if there is no other solution x' with $z_1(x') \leq z_1(x)$ while at the same time $z_2(x') < z_2(x)$.

Hence every x' that satisfies either of the two equations dominates solution x .

Based on Definition BI, the different solutions can now be compared with. If a solution x'

dominates solution x , then solution x' is better and solution x can be discarded. It implies

that a multitude of dominant solutions can exist; this set of dominant solutions is referred

to in the following as Pareto frontier.

The problem of creating timetables that consider both waiting time and empty seat percentage

can be solved using a biobjective label-correcting algorithm that works like its single

objective counterparts; the main difference being that one node can carry a set of dominant

labels instead of only one value. Based on the comparison of solution strategies for biobjective

shortest path problems by R. E. Bryant (1989), we use the approach of Brumby

Smith and Shier (1989) as follows.

The algorithm starts by labeling the source S with its set of labels consisting only of one

element $Labels(S) = \{(\emptyset, \emptyset)\}$. The sets of labels are extended to the succeeding nodes, which

means a particular node j carries all labels of its preceding nodes i , increased by the cost of

the incoming arcs (i, j) . However, only dominant labels are added to the set of labels; dominated

labels are discarded. This operation is called merging; it is the most time-consuming

step of the algorithm. Using ordered labels allows a computational effort of $O(|L| + |M|)$

when merging label sets L and M . See Section 5.4 for understanding the $O(\cdot)$ notation. Every

time the algorithm changes a label set the set is marked for follow-up. Marked sets are

processed in the order the changes occurred until the set of marked nodes is empty and the

algorithm ends as shown in the following description.

```
Algorithm: Bi-Objective Label correcting followUp={} //set
of nodes to check again, in order of entering Labels(i)=∅
//starting with an empty set of labels at each node
Label(S)={(∅,∅)} //adding label to node S append S to
followUp//making S the rst node to consider while followUp
is nonempty do remove rst node i from follow Up for
all outgoing arcs (i,j) do //extend the set of labels to
all nodes merge (Labels(i) + cij, Labels(j))// extend
the label set from i to j if (changes in Labels(j) AND j ∉
followUp) then append j to follow Up endif end for end
while
```

After the algorithm terminates successfully, the label set at T contains the lengths of all

the shortest paths between S and T . The corresponding paths can then be found by back

tracking through the network. The algorithm then results with a set of Pareto-optimal time

tables exhibited, for example, in Figure 5.15; the data used for Figure 5.15 are explicated in

Hassold and Ceder (2012). That is, there are no better timetables in terms of $[ES, WT]$ pairs

than those belonging to the Pareto-optimal set shown by the curve in Figure 5.15; any bet

ter, but impossible, timetable will have a solution to the left and/or below the Pareto frontier

curve. The alternative solutions with different $[ES, WT]$ pairs, along the curve, enable the

transit planner to decide on the desirable timetable for implementation. This multiobjec

tive approach also provides benefits for the vehicle-scheduling phase when using different

vehicle sizes; see Chapter 8 for details. Finally, the next section introduces the principles of

optimization to improve the comprehension of this section and of Sections 5.6 through 5.8

as well as of other chapters.

5.5 OPTIMIZATION, OPERATIONS RESEARCH, AND COMPLEXITY

This section introduces the notion of optimization, using operations research (OR) prin

ciples. Optimization usually means finding the best solution to some problem from a set of

alternatives. Following Ceder (1999), OR means a scientific approach to decision-making

or formulating problems so that they can be solved quantitatively. Hence, optimization involves

dealing with mathematical models of situations or phenomena that exist in the real world.

The development of powerful computers has had a very strong impact on the motivation to

develop complex, sophisticated OR models.

The basic elements in OR models include the following: (1) Variables (decision, policy,

independent, or dependent variables); these are quantities that can be manipulated to achieve

some desired objective or outcome. (2) Objective function (profit, revenue, or cost function);

this is a measure of effectiveness (efficiency, productivity, or utility value) associated with
0 500 1000 Empty seat penalty (ES) 1500
2000 2500 0 5,000 10,000 15,000 20,000 Waiting time (WT)

Figure 5.15 Pareto frontier of optimal solutions compared with the current timetable (gray square).

some particular combination of the variables and is a function that is to be optimized (mini

mized or maximized). (3) Constraints (feasibility conditions); these are equations or inequalities

that the variables must satisfy in addition to providing a minimum or maximum value

of the objective function.

Mathematically speaking, the variables are often represented as $x_1, x_2, \dots, x_n$ or the vector

of the variables $x = (x_1, x_2, \dots, x_n)$. The objective function, as a single-value function, is

represented as $z = f(x_1, x_2, \dots, x_n)$, or $z =$

$f(x)$. The constraints are also represented as a func

tion of x , $g_v(x)$, with v as a constraint number. The general variable optimization problem,

with m constraints, is to minimize/maximize $z = f(x)$ subject to
(s.t.) $g_v(x) \geq \leq = (\text{ }) (\text{ }) \{ , , \} , v = 1, 2, \dots, m$
(5.11)

It is apparent that $\min[f(x)] = -\max[-f(x)]$, and hence, each minimization problem can be

transformed as a maximization problem and vice versa. Notations that correspond to OR and are

used in the formulations in this book contain the symbols $\forall$ (for all), $\in$ (belong to), $\notin$ (not

belong to), and $\emptyset$ (empty set).

Occasionally, we cannot prove that optimality is attained, nor can we prove that opti

mality cannot be attained in all those cases. In these situations, the OR model is termed

a heuristic procedure. In certain cases, a mathematical formulation may possibly be too

complex to be handled even by a advanced computers, thereby leading to the intentional

construction of a heuristic procedure. This is actually the case with timetable synchronization

tion, to be dealt with in due course.

A framework for developing a mathematical OR model of a system's problem is shown

in Figure 5.16 for a deterministic (average-based) study. It relies on a nine-step procedure.

OR analysts should first define the system's (organization's) problem while searching for its

actual objectives, then look for the entire system's limitations and constraints. The third

step, in Figure 5.16, translates the rest two steps into a formulation similar to Equation

5.11. Once the formulation is correctly established, a check can be made of the existence

of feasible solutions or solutions that comply fully with all the constraints. If feasibility

does not exist, return to the rest two steps for possible changes and/or relaxation of some

constraints. Otherwise, select a software package or develop an algorithm to solve the OR

formulation automatically. The sixth step will involve examining the important issue of

the complexity (described in the following) of the algorithm to verify whether indeed the

system's problem can be solved by a computer within a reasonable amount of time. If the

algorithm is too complex, thus requiring too long a running time, a simplified heuristic

procedure may be constructed to solve the problem. Otherwise, a check will have to be

conducted of the sensitivity of the problem's objective value to different input data. The

step appropriately presents the results along with the sensitivity analysis and the problem's

limitations. In the course of a heuristic procedure, it becomes apparent that optimality can

not be guaranteed.

One important characteristic of an OR model/procedure/algorithm is the efficiency with

which the formulation may be run by all the

the various computing resources required. The

solution time is usually checked for the worst-case scenario or worst-case complexity,

which is well described by Ahuja et al. (1993). It becomes commonplace to use the notation

$O(f(k, q))$, a notation of $f(k, q)$, for expressing the running-time complexity function,

which is dependent on the input parameters k and q , where $f(k, q)$ is the function of

those two parameters. For example, if the running time is $f(k, q) = (10k + 0.01k^3) \cdot q^5$, then

for all $k \geq 100$ (as we always look for the worst case), the second term in the parenthesis

dominates, and the complexity of this function is $O(k^3 \cdot q^5)$. The $O(\cdot)$ notation provides,

for sufficiently large input values, an indication of the number of elementary mathematical

operations required. Following Ahuja et al. (1993), the complexity of OR optimization

algorithms is often interpreted as efficient if its worst case is bounded by a polynomial function

of the input parameters; for example, algorithms with $O(k^3 + q^5)$, $O(kq + q^3 \log(k^2))$.

Nonetheless, the algorithm found in many combinatorial optimization problems is of the

type called exponential-time algorithm with, for instance, $O(k!)$, $O(3^q)$, $O(k \log(q))$. Some of

these hard-to-solve problems belong to a class known as NP-Complete (nonpolynomial

complete). Certainly, it is a challenge to d

develop a polynomial-time algorithm for a problem
 Unacceptable Acceptable 7 Perform a sensitivity analysis
 using various ranges of the input data 8 Present the
 solution, its sensitivity analysis results, and its
 limitations 5 Select or create a solution algorithm 2
 Explore and find the system's limitations and constraints
 3 Formulate an optimization (OR) model 1 Explore and define
 the system's objectives No feasible solution Solutions
 exist Construct a good heuristic procedure Optimality is
 not guaranteed 4 Examine the existence and range of
 feasible bounded solutions 6 Examine the algorithm's
 complexity and assess its potential to solve the given
 system

Figure 5.16 Schematic framework of an OR process to solve a deterministic optimization problem analytically.

in this class. In these NP-complete problems, we may depart from a completely proven opti-

mization solution and begin developing a heuristic procedure.

The previous Section 5.4, the next two sections, as well as several subsequent chapters,

will use the OR optimization concept as a framework for achieving the best results.

5.6 MINIMUM PASSENGER-CROWDING TIMETABLES FOR A FIXED VEHICLE FLEET

The problem addressed in this section is that of optimal multiroute timetable construction

for a transit agency. The motivation for this task lies in the challenge facing schedulers/planners

in reconstructing timetables when the available vehicle fleet is reduced. The purpose of

this section is to present the formulation to solve the problem optimally by using OR tools

(i.e., integer programming). Because of the complexity involved in solving integer programs

ming formulation, a heuristic approach designed for a person-computer interactive procedure will also be proffered in the next chapter.

5.6.1 Problem description

More concretely, the problem is as follows: Given a fixed fleet size (number of vehicles) and

passenger load at the maximum load segment between a set of terminals (predetermined routes), what

is the frequency of vehicle departures that will maximize some measure of passenger satisfaction?

A fixed set of terminals and a fixed planning horizon are given.

The time horizon for each terminal is partitioned into a fixed number of time periods

(usually hours, but not necessarily equal). For each time period j , $j = 1, 2, \dots, q$ and route, the

maximum load is given. The number of vehicle departures, F_{jk} , for a given terminal k and for each

time period j must lie within a bounded range, the two standards of the lower and upper

bounds of the frequency (see Figure 1.4). The lower bound is based on a predetermined

policy headway and the upper bound on a minimum passenger-loading factor. The objective

is to minimize the sum of all service measures that might be interpreted as a minimum

passenger-disservice cost, or crowding, at the maximum load segments.

5.6.2 Formulation

Let $d(k, t)$ represent the total number of departures less the total number of trip arrivals

at terminal k up to and including time t . The maximum value of $d(k, t)$ over the schedule

horizon $[T_1, T_2]$ is designated $D(k)$. If we denote the set of all terminals as T , the sum of

$D(k) \forall k \in T$ is equal to the minimum number of vehicles required to service the set T .

This is known as the fleet size formula. It was independently derived by Bartlett (1957),

Gertsbach and Gurevich (1977), and Salzborn (1972). Mathematically, for a given xed

schedule: $N \geq \max_{k \in T} \sum_{t \in [T_1, T_2]} d(k, t)$ (5.12)

where N is the minimum number of vehicles to service the set T . Further interpretation and

analysis using Equation 5.12 appears in Chapter 6.

Let T_k be the set of all departure times from terminal k . Let $(j, k_1, k_2) \in T_k$ represent a

possible trip bundle departing during period j from terminal k_1 to terminal k_2 .

Define a 0-1 variable x_{j,k_1,k_2} if F departures are selected during period j from terminal F , $j = 1, 2, \dots, T$ to otherwise $x_{j,k_1,k_2} = 0$.

where F is an index running from $L(\cdot)$ to $U(\cdot)$; that is, $F = L(\cdot), L(\cdot) + 1, L(\cdot) + 2, \dots, U(\cdot) - 1, U(\cdot)$.

Using the definition of Equation 3.2, let $P_m(\cdot)$ be the max load and $d_o(\cdot)$ the desired

occupancy—both definitions are associated with period j and the route from terminal k_1

to k_2 . The crowding measure (i.e., cost) associated with x_{j,k_1,k_2} is defined as $c_{j,k_1,k_2} = \max[P_m(\cdot) - d_o(\cdot), 0]$ (5.13)

and the total crowding defining the objective function, Z : $Z = \sum_{j \in T} \sum_{k_1 \in T} \sum_{k_2 \in T} c_{j,k_1,k_2} x_{j,k_1,k_2}$

The mathematical programming formulation, which is inspired from Salzborn's note (1972)

on fleet routing models for transportation systems is stated as follows: minimize $\sum_{j \in J} \sum_{k \in K} c_{jk} x_{jk}$ (5.14)

s.t.

Bundle departure constraints: $\sum_{j \in J} x_{jk} = 1 \quad \forall k \in K$ (5.15)

Assigned vehicle bounds: the net number of departures less arrivals that occur before or at t at terminal k determined by the value of x_{jk} , $t \in T$ (5.16)

where $D(k)$ is the maximum number of vehicles assigned to terminal k .

Resource constraint (total fleet size): $\sum_{k \in K} D(k) \leq N$ (5.17)

where N is the total (fixed and given) fleet size.

Variable constraints: $x_{jk} \in \{0, 1\}$ (5.18)

In the overall formulation, $D(k)$ will be an integer in any feasible solution.

Constraint (5.15) ensures that only one bundle of departures is selected for a given

terminal pair (route) and time period. Constraint (5.16) ensures that the number of vehicles

using a given terminal k up to time t does not exceed the number of vehicles, $D(k)$,

assigned to terminal k . The left-hand side of constraint (5.16) can be represented as a

linear function of the x_{jk} variable (as will be demonstrated in the example in the next

section). Not all departure times need be c

onsidered, as some lead to redundant equations

tions; this can occur when there is a sequence of departures times unbroken by intervening

arrivals. Constraint (5.17) indicates that the sum of vehicles assigned to all terminals is no

greater than the given fleet size N_0 . Finally, constraint (5.8) defines $x_{F(i)}$ as a binary and

the boundary of $D(k)$.

The optimal solution $x_{D \cup F(i)}, (i) \in \{1, \dots, k\}$ yields both the assignment of vehicles to terminals,

$D(k)$ and the optimal number of departures within each time period. Thus, an optimal

timetable may be constructed from $x_{D \cup F(i)}$.

5.6.3 Example

In order to gain further understanding of the underlying structure of the mathematical

formulation, an example consisting of two terminals and two time periods will now be presented.

It follows Ceder and Stern (1984). The basic input data of the example problem,

along with the $0-1$ variables and their associated cost, $c_{F(i)}$, are indicated in Table 5.4. For

each trip bundle (i) , the upper and lower bounds on the number of departures are shown

in Table 5.3. For example: $L(1, b, a) = 2$ and $U(1, b, a) = 3$. The indicated cost for each (i)

is calculated in accordance with Equation 5.13; that is, $\max(P_m - F \cdot d_o, 0)$. There are three

$0-1$ variables for the first trip bundle $(1, a, b)$ and two variables for every other trip bundle

information given in Tables 5.4 and 5.5. In constraints (ii), each possible combination of

the net number of departures for a given terminal is restricted so as not to exceed the

number of vehicles assigned to that terminal. For example, the first constraint in (ii) refers

to $0 \leq t \leq 4$ and the second to $0 \leq t \leq 6$, in regard to the net number of departures in terminal

a. For $0 \leq t \leq 6$ in terminal a, we take three possible departures of x_3 , two departures

of x_2 , and one departure of x_1 and x_5 , as opposed to two arrivals of x_7 and one arrival of x_6 .

The known OR package MPSX has been used to solve this simple example for $N_0 = 7, 6, 5, 4,$

$3, 2$. It is possible to obtain the solutions by relaxing the integrality constraint on the x 's and, if

necessary, round off any fractions to the nearest integer. The results are presented in Table 5.6.

Note that the right-hand side of N_0 may be used to directly obtain a curve of fleet size versus

minimum cost in a single computer run. This trade-off is shown as the solid line in Figure 5.17.

In the next section, the problem is resolved by allowing small departure times between vehicles.

5.6.4 Variable scheduling

Practical vehicle scheduling often involves shifting departure times in order to better match vehi

cle assignment with a given set of trips. This practice is mentioned in Chapter 1 and described in

Chapter 8. In this case, the departure times

are allowed to vary over prescribed limits.

Table 5.5 Departure and arrival times for each set of departures

Departure terminal a b

Arrival terminal b a

Variable $x_1 \times x_2 \times x_3 \times x_4 \times x_5 \times x_6 \times x_7 \times x_8 \times x_9$

Departure time 4 2 4 1% 2% 4 8 6 8 2 4 1% 2% 4 8 6 8

Arrival time 8 6 8 5% 6% 8 12 10 12 5 7 4% 5% 7 11 9 11

A precomputation analysis can be carried out of the bounds on the number of departures.

The steps of this analysis are shown in flowchart form in Figure 5.18. The functions $d(k, t)$

are first constructed for the minimum and maximum number of departures, $L(\cdot)$ and $U(\cdot)$,

for all the trip bundles $(\cdot)$. The lower and upper bounds on the fleet size are then determined

as N_L and N_U , respectively. Before executing the integer program, two tests should be per-

formed: (1) feasibility and (2) appropriateness of the $U(\cdot)$ solution. These two tests are shown

in Figure 5.15. Applying this precomputation stage (without shifting departure times) to the

example problem obviates the need to execute the integer programming for three fleet size

values $N_0 = 7, 6$, and 2 , since $N_L = 3$ and $N_U = 6$.

Assume that the departure-time tolerance for the example problem, described in Tables 5.4

and 5.5, is $\Delta(\cdot) = \pm(1/3)$ time units for all trip bundles. The analysis of $d(k, t)$, carried out

For cases (2) and (3), the first and last constraints in (2) are changed into $2 \ 2 \ 3 \ 2 \ 2 \ 2 \ 3 \ 2 \ 1 \ 7 \ 6 \ 3 \ 2 \ 1 \ 8 \ 9 \times \times \times D$
 $\times \times \times \times \times \times \times D + + \leq + - - + + \leq () () a \ b$ Construct
 $d(k, t)$ for Initial input: Fixed fleet size, No; Number of

the importance of eliminating a large number of transfer points because of their diverse

effect on the user. From the operating-cost perspective, a trade-off no doubt exists between

this elimination and the efficiency of the transit route network. This trade-off is depicted

schematically in Figure 5.19.

In order to allow for an adequate or compromised level of service, the planners/schedulers

face a synchronization task to ensure maximum smooth transfers involving the switching

Table 5.7 Optimal results for the example variable scheduling problem

Case Fleet size no. Sets of departures in the solution, x_i
 = 1 D(a) D(b) Minimum cost

(I) $\geq 5 \times 3, x_5, x_7, x_9 \ 2 \ 3 \ 0 \ 4 \times 2, x_5, x_7, x_9$
 $1 \ 3 \ 15 \ 3 \times 1, x_5, x_7, x_8 \ 0 \ 3 \ 103$

(II) and (III) $\geq 5 \times 3, x_5, x_7, x_9 \ 2 \ 3 \ 0 \ 4 \times 2, x_5,$
 $x_6, x_9 \ 2 \ 2 \ 45 \ 3 \times 1, x_5, x_6, x_8 \ 1 \ 2 \ 133$ The lower
 the level of service The higher the number of transfers
 The more costeffective the transit network operation The
 higher the number of simultaneous arrivals The higher
 the level of service The higher the importance of maximum
 synchronization (max simultaneous arrivals) Compromised
 level of service

Figure 5.19 Trade-offs dependent on transit-passenger transfers.

of passengers from one route to another without waiting time. This task is extensive, but

minimizes the waiting time for passengers who require connections. The result is that the

planner/scheduler creates a more attractive transit system, one that generates the opportu-

nity for increasing the number of riders. Figure 5.16 illustrates how the concept of synchro-

nized transfers mitigates any adverse impact on the level of service while achieving a more

cost-effective transit network.

Actually, synchronization is one of the most difficult, but vital, scheduling tasks. A poor

transfer can cause the user to stop using the transit service. It is analogous to the Chinese

proverb: "Two leaps [in our case, two trips with a transfer] per chasm are fatal." Nevertheless,

synchronization is addressed almost intuitively. Generally speaking, the planner/scheduler

attempts to fix the departure times in the timetable while conforming to three elements: (1)

required frequency, (2) efficient assignment of trips to a single vehicle chain, and (3) maxi

mum synchronization of arrivals. The second element will be dealt with in Chapters 6 and 7.

This chapter will present an efficient mathematical procedure to help achieve maximum

synchronization; the procedure can be employed as a useful tool for creating timetables. The

process described in this chapter follows Ceder et al. (2001) and Ceder and Tal (2001).

5.7.1 Formulating an OR model for synchronization

Chapters 4 and 5 established optional timetables in conjunction with adequate frequency and

satisfactory loads for a single transit route. When two or more routes interact, particularly at

connection (transfer) points, the issue of coordination becomes critical. In order to gain a better

approach to coordination at connection points, there is a need to relax some of the rigorously

defined timetable parameters. The parameter chosen that can offer a measure of flexibility is the

headway. Fortunately, only certain routes, at certain time periods, are able to be included in this

optimal search for reducing waiting time at transfer points. This section follows the meaning of

optimization as a framework for mathematical treatment, as described earlier in this chapter,

and formulates the problem of maximizing simultaneous vehicle arrivals at transfer points.

The following OR model, based on Ceder et al. (2001), is presented in five parts: (1) nota

tion of known (given) data, (2) decision variables, (3) objective function, (4) constraints, and

(5) assumptions and notes.

5.7.1.1 Notation of known (given) data

The given transit network is presented by $G = \{A, N\}$

where A is a set of directed arcs representing the traveling path of transit routes N is a set of transfer nodes in network G

The problem data are the following:

T = planning horizon (departure times can be set in the interval $[0, T]$, which is a discrete interval)

M = number of transit routes in the network

N = number of transfer nodes in the network

$H_{min k}$ = minimum headway (agency's requirements) between two adjacent departures on route k ($1 \leq k \leq M$) $H_{max k}$ = maximum headway (policy headway) permitted between two adjacent bus departures on route k ($1 \leq k \leq M$) F_k = number of departures to be scheduled for route k during the interval $[0, T]$ ($1 \leq k \leq M$) T_{kj} = travel time from the starting point of route k to node j ($1 \leq k \leq M, 1 \leq j \leq N$) Travel times are considered to be deterministic and can be referred to as mean travel times

5.7.1.2 Decision variables

1. X_{ik} represents the i th departure time on route k ($1 \leq i \leq F_k$).

2. Z_{ikjqn} is a binary variable that yields the value 1 if the vehicle of the i th departure on route k meets the vehicle of the j th departure of route q at node n ; otherwise, it yields the value 0.

Let $A_{kq} = \{n: 1 \leq n \leq N, T_{kn} \geq 0, T_{qn} \geq 0\}$.

5.7.1.3 Objective function $\text{Max } \sum_{k \in K} \sum_{i=1}^{F_k} \sum_{j=1}^{F_q} \sum_{n \in A_{kq}} Z_{ikjqn} (F_q - k + M_i - F_k + M_q)$ (5.19)

5.7.1.4 Constraints $X_{H_k} + M_k - k_1 \leq \max$, (5.20) $X_{T_k} + M_{F_k} - k \leq 1$ (5.21) $H_k + X_{H_k} + M_i - F_k \leq \min$, (5.22) $\sum_{k \in K} \sum_{i=1}^{F_k} \sum_{j=1}^{F_q} \sum_{n \in A_{kq}} Z_{ikjqn} (X_{T_k} - X_{T_q}) \leq 0$, (5.23)

Constraint (5.20) ensures that the first departure time will not begin beyond the maximum

headway from the start of the time horizon, while constraint (5.21) ensures that the

last departure is executed within the planning horizon. Constraint (5.22) indicates the

headway limits, and constraint (5.23) defines the binary variable of the objective function.

5.7.1.5 Assumptions and notes

1. It is assumed that the first departure on each route k must take place in the interval $[0, H_{\max k}]$.

2. The problem is impractical unless the following constraints hold for each k :

(a) $H_k + H_k - k_{\max} \geq \min$ (5.24)

(b) $T_{H_k} \geq -(F_k - 1) \min$ (5.25)

(c) $T_{F_k} + H_k \leq \max$ (5.26)

3. The case of a route k that does not pass through a node j is represented by $T_{kj} = -1$.

The OR model given earlier can be simplified by defining a variable, Y_{kq} , representing the

overall number of simultaneous arrivals of vehicles on route k with vehicles on route q . The

model is changed to $\text{Max } Y_{kq} \quad q, k \in M \quad k \neq q$ (5.27) s.t. $Y_{kq} \leq X_{ik} + T_{kn} - X_{jq} - T_{qn} \quad j \in F_i, n \in A_{kq}$ (5.28)

Constraints (5.20) through (5.22) remain unchanged.

The last formulation represents a nonlinear programming problem. It can be reformulated

as a mixed-integer linear programming (MIP) problem, which can be solved (up to

certain sizes) by several software packages. The nonlinear constraint is (5.28). Let D_{ijkq}

denote a binary variable (defined over the same domain as Z_{ijkqn}) and B a large number

($B = T + \text{Max}_{i,j} T_{ij}$). The constraint in (5.28) is exchanged with the following constraints: $B - D_{ijkq} \leq X_{ik} + T_{kn} - X_{jq} - T_{qn} \quad j \in F_i, n \in A_{kq}$ (5.29) $B - D_{ijkq} \leq X_{ik} + T_{kn} - X_{jq} - T_{qn} \quad j \in F_i, n \in A_{kq}$ (5.30) $Y_{kq} \leq \sum_{i,j} D_{ijkq} + 1$ (5.31)

If $X_{ik} + T_{kn} = X_{jq} + T_{qn}$, there is a simultaneous arrival of the vehicle of the i th departure on

route k with the vehicle of the j th departure of route q at node n . The variable D_{ijkq} can yield

the value 0 and Y_{kq} is increased by one, according to (5.31).

If $X_{ik} + T_{kn} \neq X_{jq} + T_{qn}$, the arrivals do not coincide, and D_{ijkq} must yield the value in

order to satisfy constraints (5.29) and (5.30). The number of simultaneous arrivals between

vehicles of routes k and q (Y_{kq}) is not increased in (5.31).

The following is an upper bound on the number of possible simultaneous arrivals in a given

transit network: $2 * \text{minimum}(F_k, F_q) = \sum_{k,q \in M, k \neq q} Y_{kq}$

$+ = - \sum \sum \sum ,) 11 1 (5.32)$

The number of integer variables in an MIP problem is generally a good index of its complexity

(represented by the computerized processing time required). The variable D_{ijkq} represents

the simultaneous arrival of the vehicle of the i th departure of route k and the vehicle of the

j th departure of route q at node n . This means an integer variable for every combination

of two trips on different routes that intersect at node n . Let F be $\max(F_k)$; the number of

integer variables in the worst case is $O(NM^2 F^2)$, which is a very large number. However,

in a more realistic setting, the number is $O(M^2 F^2)$, where N can be replaced by the average

number of nodes commonly shared by any two routes.

The problem formulated by (5.27) through (5.31) is a large, mixed-integer, linear

programming problem. Running small network examples (five routes, five nodes), using

general algebraic modeling system (GAMS) software on a PC requires hours, even days.

This time-consuming process provided the motivation to develop heuristic procedures

that would solve such problems within a reasonable time. Two heuristic procedures were

implemented in Turbo Pascal, and many examples were checked and compared to the

optimal solutions obtained using the GAMS. The first procedure is based on the selection

of nodes within the network. In each step, the next node is selected, provided that

not all the departure times have been determined for that node. After the departure time

is resolved, all its corresponding arrival times are set. The second procedure attempts

to improve the solution through possible shifts in departure times. The following two

sections present these two heuristic procedures.

5.7.2 Synchro-1 procedure

Definition 5.1:

A node is defined as possible if

1. At least one transit route that passes through the node, and not all the departure times for that route are set
2. More simultaneous arrivals can be created at the node

Definition 5.2:

A node is defined as new if no arrival times have been set for it.

The Synchro-1 procedure uses several components, as shown in the flowchart in Figure

5.20. Its steps are as follows.

Step 1: Initialization; check whether the problem is feasible and set the data structure. Mark all nodes as possible.

Step 2: Select the next node, N_0 , from the possible nodes.

Step 3: If N_0 is new, perform component FIRST, otherwise perform component MIDDLE.

Step 4: If there is any possible node, go to Step 2; if there are any more routes, perform component CHOOSE and go to Step 2, otherwise stop.

Step 1 contains a check of whether the problem is feasible; if so, toward that end, two data

s structures are built:

1. A structure called route for each transit route i , which includes $H_{\min i}$, $H_{\max i}$, F_i , the number of nodes the route passes through, and the departure times that have already been set.
2. A structure called node for each node n , which includes two elements: (1) the number of routes passing through the node and (2) the route with the maximum traveling time to the node; it also includes the number of simultaneous arrivals at the node at each time point in interval $[0, T + \text{MaxT}_{ij}]$, where the max is over all i, j .

All the nodes are marked possible.

In Step 2, the next node, N_0 , is selected from among the possible nodes. Node N_0 must

satisfy the following conditions:

1. The number of different vehicle-arrival times at the node is at its maximum; in such a node, there is a greater probability that another vehicle departure can be set so that it will arrive at N_0 by any one of the (already set) arrival times.
2. Among the nodes satisfying the preceding condition, N_0 is that through which a maximum number of routes pass; in such a node, there is a greater potential for simultaneous arrivals.
3. Among the nodes satisfying the preceding condition, N_0 minimizes the maximum travel time of all routes from their origin to the node (after the departure times of vehicles are set in order to meet at N_0 , the potential for simultaneous arrivals at distant nodes still exists).

A distinction is made in Step 3 between a new node and another node. For a new node, the

component FIRST attempts to set the departure times of vehicles that pass through it, such

that the vehicles will arrive at the node at the earliest

time possible and simultaneous arrivals

will continue to be created at the node according to the $H_{\min}$ and $H_{\max}$ of the routes. For

example, transit vehicles on Routes 2 and 3 arrive simultaneously at a certain node at time t_0 ;

if the next departure time for each of these routes can be set at a fixed difference, d minutes,

from the last departure time of the route [for all $i = 1, 2$, and 3 , $\max(H_{\min i}) \leq d \leq$

$(H_{\max i})$], there will be additional simultaneous arrivals at the node at time $t_0 + d$. Component

FIRST finds the minimum d possible. If parameter d cannot be set ($H_{\min i} > H_{\max j}$), the next

departure times of vehicles on these routes will not be resolved in this step.

For a node that is not new, an attempt is made to set the vehicle-departure times on routes

passing through it, such that the vehicles will arrive at the node at the earliest of all arrival

times already set in that node. If no more simultaneous arrivals are available at the node, the

node is marked as not possible. This component is called MIDDLE. Initialization Select the next node (NO) Is NO a new node? FIRST MIDDLE End Begin Are there any more routes? Yes Yes Yes No No No CHOOSE Are there more possible nodes?

Figure 5.20 Synchro-1 procedure flowchart.

Step 4 tests whether any more nodes are possible. If not, there may be routes on which

not all the vehicle departure times were set. In such cases, the route that passes through

the maximum number of nodes is chosen, and its next departure time is set by using the

difference $H_{\min i}$ from the last departure. In this way, the procedure sets additional vehicle

arrivals for the maximum number of nodes possible. All the nodes through which route i

passes are marked possible, and the procedure returns to Step 2. This component is called

CHOOSE.

The complexity of the Synchro-1 procedure is, in the worst case, $O(N^2)$, where N is

the number of transfer nodes, M is the number of routes, T is the planning horizon, and F is

the maximum number of scheduled departure times across all routes.

5.7.3 Synchro-2 procedure

The results of Synchro-1 could be improved by allowing a shifting of departure times in the

timetable obtained by the procedure. That is, for each node and for each two time points,

t_1 and t_2 ($t_1 < t_2$) at which vehicles arrive at the node, an attempt can be made to introduce a

shift in all the vehicle departure times for vehicles arriving at the node at t_1 so that they will

arrive at time t_2 . If this succeeds, the timetable is changed accordingly, and the number of

simultaneous arrivals increases. It should be noted that in order to shift a single d

departure

time, the following must be checked:

1. After the shifting is done, the constraints on H_{min} and H_{max} must still hold. Otherwise, additional departure times on the route should be shifted.
2. As a result of shifting the departure time of vehicle i , its arrival time for all nodes through which it passes is changed. Therefore, the departure time of each vehicle that arrives simultaneously with vehicle i at any node must also be shifted. For each shift, the constraints on H_{min} and H_{max} must be checked, and so on. This process is recursive, and an attempt to shift a single departure time may cause the shifting of all network departure times.
3. A shift of Δt minutes in the departure time of vehicle i may result in changing other departure times, which ultimately will lead to changing the departure time of vehicle i by more than Δt . A check must eliminate this situation.

The shifting procedure can be added to the Synchrono-1 procedure with two components,

MERGE and MOVE, and a process TRY-MOVE as shown in Figure 5.21. MERGE MOVE TRY-MOVE

Figure 5.21 Synchrono-2 procedure flowchart.

5.7.3.1 Component MERGE

Component MERGE identifies two departure times, t_1 and t_2 ($t_1 < t_2$), for a given node NO

and delivers three elements to component MOVE: the transit routes arriving at t_1 and t_2 and

the required shift $t_2 - t_1$.

5.7.3.2 Component MOVE

The MOVE component attempts to shift t_1 by $\Delta t = t_2 - t_1$, where t_2 is changed by route

number. This component also contains the number of vehicles that need to be shifted, a

nd

a vehicle array with all possible vehicles that need to be checked for shifting. Each vehicle

is identified by route number and vehicle code. Component MOVE uses the process TRY

MOVE, which allows for individual shifts to be checked.

TRY-MOVE is a recursive process that attempts to shift the departure time of a given

vehicle and indicates whether MOVE is successful or false. This TRY-MOVE process checks

whether the required new departure time is within $[0, T]$ and whether or not the resultant

headway exceeds H_{max} .

5.7.3.3 Synchro-1 and Synchro-2 combined

Synchro-1 and Synchro-2 can be combined to shift Synchro-1 results. Further, the first selected

node N_0 can change N times: $N_0 = 1, 2, \dots, N$; that is, in order to run Synchro-1 N times.

Such N time runs may be required when the criterion of Synchro-1 for selecting N_0 cannot be

justified; for example, in a case in which the node with the maximum routes passing through it

is far from the origin. These N runs over different N_0 s can also be combined with Synchro-2.

Example 3 in the following examines four variations: (1) Synchro-1, (2) N runs of Synchro-1

(first node varies), (3) Synchro-2, and (4) Shifts in the results of N runs of Synchro-1.

The complexity (running time as a function of network size parameters) in the worst

case) of each variation is certainly not the same. Synchrono-1 has the lowest complexity,

$O(N^2 T^2)$, followed by N runs of Synchrono-1, $O(N^2 T^2 M^2)$, which is less than the complex

ity of Synchrono-2, $O(N^2 T^3 F^2 M^2)$. The most complex run is the one with N changes in the first

node while running Synchrono-2, $O(N^3 T^3 F^2 M^2)$.

5.7.4 Examples

Three examples of transit networks will now be described, following Ceder et al. (2001) and

Ceder and Tal (2001). Two of the examples are represented in detail for the sake of clarity. An

optimal procedure with the linear programming (LP) software GAMS was applied to each

example for comparison purposes.

Detailed Example 1

Figure 5.22 presents a simple network that combines two transfer points with two routes.

The numbers on the arcs are average planned travel times (in minutes). A demonstration of

Synchrono-1 follows.

Step 1: Building the route data structure I Hmin i Hmax i F i No. of nodes I 5 15 4 2 II 8 20 3 2 and the node data structure: Node n No. of routes Route with Max T in 1 2 II 2 2 II The data comply with the three constraints (5.24) through (5.26).

Step 2: Three criteria must be selected: N_0 with maximum arrival time, maximum routes crossing, and minimum of all routes' maximum travel time (min-max criterion) from origin to N_0 . Thus, the number of departure (arrival) times equals 0 of

for both nodes. The number of crossing routes equals 2 for both nodes. The maximum travel time for node 2 is $\text{Max}(17, 27) = 27$; for node 1, it is $\text{Max}(7, 12) = 12$; the minimum of the two is 12. The selected NO is, therefore, node 1.

Step 3: The earliest time possible is set for node 1. Continuing in this node, the synchronization is based on $\text{Max}(5, 8) \leq d \leq \text{minimum}(15, 20) \rightarrow d_{\min} = 8$. This is the FIRST component, which provides the following results: Departure time Meeting time Route I Route II 5 0 12 13 8 20 21 16 28 where meeting refers to a simultaneous arrival. The number of departures for route II complies with $F_2 = 3$.

Step 4: Because node 2 has yet to be examined, Step 2 is selected.

Step 3: Because node 2 is not new, component MIDDLE is applied. $F_1 = 4 > 3$ (currently created). Thus, one more departure time is set for route I based on the already created departure times for route II (at 0, 8, 16) so that the synchronization is made at node 2. The result is the additional departure time for route I (at 26, which meets the route II vehicle's departure at 16).

Step 4: No more routes are left to examine, and the algorithm ends with the final results as follows: 2 Origin Route I Route II 7 12 10 15 1

Figure 5.22 Example 1 basic network. Departure time Meeting time Total meetings Route I Route II At node 1 At node 2
5 0 12 13 8 20 4 21 16 28 26 43

The optimal and heuristic procedures coincide in this simple example.

Detailed Example 2

Figure 5.23 presents the second example with a case of $H_{\min i} > H_{\max j}$ for a two-route,

four-node network. The numbers on the arcs are average planned travel times (in minutes).

Both Synchro-1 and Synchro-2 may be used for this example.

The data for example 2 are as follows: i H_{min} i H_{max} i F
 i T Route I 6 10 4 30 Route II 3 5 6

in which $H_{min} 1 > H_{max} 2$.

Using Synchro-1

Step 1: In the route structure data, the number of nodes through which each route passes is four. Node data structure: Node n No. of routes
 Route with Max T in 1 2 II 2 2 II 3 2 II 4 2 I, II

Step 2: NO is selected such that (1) the number of departure (arrival) times equals 0 for all nodes, (2) the number of crossing routes equals 2 for all nodes, and (3) the route's maximum travel times for the nodes are 10, 16, 20, and 23, respectively.

The minimum is 10, so NO = 1. 2 Route I Route II 5 1 3 4
 10 4 4 3 7 6 7

Figure 5.23 Example 2 basic network.

Step 3: Component FIRST is applied. The first meeting time possible at node 1 is 10. The component cannot set the parameter d . Therefore, procedure FIRST results in only the first departure time for each route. That is, departure times 3 and 0 for routes I and II, respectively, to meet at 10.

Step 4: With more possible nodes, Synchro-1 returns to step 2.

Step 2: Number of arrival times is 1, 2, 2, and 2, respectively, for nodes 1, 2, 3, and 4. NO = 2 (min-max travel time is 16).

Step 3: Component MIDDLE sets a new meeting at node 2, at time 27, by setting the third departure time for route I to 15 and the fourth departure time for route II to 11.

Steps 2, 3: NO = 3. Component MIDDLE sets new meetings at node 3, at times 28 and 34, by setting the second departure time for route I to 9, the third departure time for route II to 8, and the fifth departure for route II to 14.

Steps 2, 3: NO = 4. No meetings are possible.

Synchro-1 continues to perform component MIDDLE until it ends with the following

Results: Departure time Meeting time at node 1 Meeting time at node 2 Meeting time at node 3 Total meetings
Route I Route II 3 0 10 5 9 3 28 15 8 27 34 21 11 28 14 18

The meeting time is shown in the same row as its associated departure for route I.

Using Synchro-2

Component MERGE and MOVE check with the possible shifting of TRY-MOVE. The only

successful shifting is the second departure for route I from 3 to 5. The new results are as

follows: Departure time Meeting time at node 1 Meeting time at node 2 Meeting time at node 3 Total meetings
Route I Route II 3 0 10 9 5 21 28 6 15 8 27 34 21 11 28 14 18

The optimal solution is six meetings, coinciding with the solution given by Synchro-2.

Example 3

Figure 5.24 exhibits the third example, in which the numbers on the arcs are average planned

travel times (in minutes).

The following data apply for Figure 5.24: H_{min} H_{max} F_i T Route I 3 8 20 combinations (see Table 6.1) Route II 4 7 20 Route III 5 12

This third example was examined with four variations of procedures: Synchro-1, Synchro-1

with N changes of first node, Synchro-2, and Synchro-2 with N changes of first node. The

results of these four variations appear in Table 5.8, including a comparison of the optimal

results. Table 5.7 shows 20 combinations of frequencies F_1 , F_2 , and F_3 for routes I, II,

and III, respectively.

The results of Table 5.8 are as follows: (

1) for the most complex variation, with S
ynchro-2

and N changes at first node (of S ynchro-1), t

here are 18 (out of 20) optimal results; (

2)

S ynchro-1 and S ynchro-1 with N runs result in

13 identical results (in other words, the

decision about the first node in Synchro-1 was the best
selection in 13 of the 20 cases);

and (3) based on example 3 , it is not o
bvious that S ynchro-2 provides better results t
han

S ynchro-1 + N runs.

5.7.5 Real-life example

The study by Ceder et al. (2001) selected a real-life
synchronization problem in Israel for

testing S ynchro-1 and S ynchro-2. The real-life
network, shown in Figure 5.25, consists of

a main road that has three major transfer n
odes. There are seven bus routes traveling in

each direction on this network and a total o
f 14 routes that meet at three nodes. (

Route

numbers appear in Figure 5.25.) The time s
pan for the synchronization problem was 3 h

during morning rush hours. The best results (using all four
variations) were 240 meetings

using S ynchro-1 with N changes for the first s
elected node. The real-life example could not

be checked against the optimal solution because of its
large size. 1 2 3 8 3 7 4 5 2 Route I Route II Route III

Figure 5.24 Example 3 basic network.

The approach presented in this section can serve as a useful tool to assist the trans

sitioner scheduler in one of the most difficult scheduling tasks. The problem of a successful

connection is reflected in the following definition of a missed transfer: the same thing happens

opened today that happened yesterday, but usually to different passengers. Thus, the more

successful connections that can be created, the better and more reliable will be the service. Table 5.8 Maximum number of meetings produced by 4 variations of Synchro-1 (S1) and Synchro-2 (S2) with 20 combinations of vehicle frequencies

Combination no.	F 1	F 2	F 3	Procedure variation	Optimal	S1	S1 + N	runs	S2	S2 + N	runs (with shifts)
1	1	1	1	1	2	2	2	2	2	1	1
2	1	3	3	3	3	4	1	2	2	3	3
3	3	4	4	7	2	2	1	3	3	3	4
4	3	3	4	4	8	2	2	2	4	4	5
5	5	5	5	9	3	1	1	2	3	2	3
6	3	10	3	1	2	3	3	4	4	11	3
7	3	3	1	4	4	5	5	14	3	3	2
8	3	2	3	4	5	6	7	15	3	1	3
9	3	2	3	4	5	6	7	17	3	4	4
10	7	7	17	3	4	4	7	10	10	10	18
11	4	4	4	8	8	11	11	11	11	11	11
12	19	5	5	5	8	8	8	8	20	6	6
13	5	4	7	4	7	7	8	9	10	11	12
14	1	2	3	11	11	11	1	2	3	4	5
15	6	7	12	13	14	8	9	11	1	2	4
16	1	10	3	8	9	2	4	4	9	2	5

of bus routes

Figure 5.25 A real-life example of bus routes that need synchronization at three points.

This planning tool needs to be complemented with the on-time performance of transit

vehicles. The latter is expected to be realized when using new automatic vehicle location

(AVL) and communications technologies. All together, the procedures described in this

section may offer a step forward in changing the image of transit services and, ultimately,

in view of increasing traffic congestion, lead to increased ridership. The next section pro

vides a literature review of the research conducted in this area.

5.8 L I T E R A T U R E R E V I E W O N S Y N C H R O N I Z A T I O N A N D F U R T H E R R E A D I N G

The advanced timetabling related literature is already covered in the previous chapter. This

section is about the literature of transit synchronization. The need to transfer between

routes generates a major cause of discomfort for transit users. Designing routes and schedules

with a minimum amount of waiting time during a transfer may decrease the level

of inconvenience. This section reviews methodologies for the design of such synchronized

transit services.

Rapp and Gehner (1976) describe a graphic, interactive tool for minimizing transfer

delays in a transit network. Optimal system-wide timetable coordination, under operating

cost constraints, is achieved through determining deviations from the original departure

times. The diversity of headways along different routes and the interdependence of layover

times in terminals are taken into account. Seeking an optimal solution involves an iterative

assignment of passenger demand on the route network.

Salzborn (1980) presents a methodology for scheduling a system that includes several

urban feeder lines and one interurban line. The latter passes by a few terminals and its

timetable is coordinated with the arrival times of passengers in all terminals. Two separate

s scheduling processes are described, one for the feeder lines and one for the interurban line.

The objective is to minimize both passenger delays and the number of required vehicles.

The decision variables consist of time slots that are assigned to line departures from the

terminals.

Although Nelson et al. (1981) do not focus on the development of a design methodology,

their paper mentions four different transfer-design strategies: simple timed transfer,

pulse scheduling, lineup, and neighborhood pulse. The simple timed transfer strategy means

adjusting timetables such that two buses will arrive at a station at the same time. The pulse

scheduling strategy requires network-wide actions that include route adjustments, head

ways, layover times, and various control strategies. The two other strategies are similar to

pulse scheduling, but their implementation is limited either to off-peak hours (lineup strategy)

or to a small geographical area (neighborhood pulse strategy).

Kyte et al. (1982) present the process of building a route network in which a main trunk

line passes through a series of transit centers. The process includes a determination of clock

headways; that is, headways that form the same schedule every hour, with the objective of

coordinating transfer times at the transit centers.

Schneider et al. (1984) provide a detailed list of criteria for choosing a proper site for the

location of a timed-transfer transit center.

Hall (1985) develops a model for schedule coordination at a single transit terminal

between a set of feeder routes and the line that they feed. The travel time on each of the

routes is assumed to be subject to random delay. The optimized variable is the slack time

between feeder arrivals and the main-line departure.

Abkowitz et al. (1987) relate to the simple case of two bus routes that meet at one spot.

Their analysis aims at determining the conditions under which time transfer provides a

improved service level compared with unscheduled transferring. A computer program

is developed that simulates four transfer strategies: unscheduled transfers, scheduled

transfers without vehicle waiting, scheduled transfers in which the lower-frequency bus

is held until the higher-frequency vehicle arrives, and scheduled transfers in which both

buses are held until a transfer event occurs. Mathematical expressions for waiting times

are developed for each of the four cases. Simulation results of the four cases are ana

lyzed to arrive at a generalization of situations for which each of the transfer strategies

is suitable.

Klemt and Stemme (1988) formulate an integer-programming problem that attempts to

minimize transfer times in a network, represented as a graph. The authors describe rules for

the construction of a route graph on which the solution procedure is based.

Voss (1990) discusses two cases of a schedule-synchronization problem formulated as

programming problem. The first case is similar to the one presented by Klemm and Stemme;

namely, a relaxed version of a quadratic assignment problem. The transit routes in this case

meet at a given set of transfer points. The second case, a modification of the first one, represents

a transit system in which different routes partly use the same streets or tracks. Solution

approaches are discussed and initial feasible solutions generated; they are then improved by

using a Tabu search algorithm.

Desilets and Rousseau (1990) propose a model that is basically equivalent to Klemm and

Stemme, but they also include an extended discussion of the definition of the cost that

the model aims at minimizing. Several alternative cost definitions are suggested, such as

definition that focuses on service reliability and a definition based on choosing a group of

connections for which synchronization is especially desirable. The paper also describes a

heuristic solution based on an initial random solution and on a local search technique for

solution improvement.

Lee and Schonfeld (1991) introduce two models for optimizing the slack time between

atrain line and a bus line that interchange passengers at adjoining terminals. The model

is analytical, but is solved numerically or by using simulation. The first model

assumes that train-arrival times are always the same, whereas bus arrivals are accord

ing to a given probability function. In the second model, both train and bus arrivals

are probabilistic. Transfer-cost functions are developed, showing relationships between

slack times and headways, transfer volumes, passenger-time values, bus operating costs,

and the variability of bus and train arrivals. Schedule coordination between the two

routes is found to be not worth attempting when the standard deviation of arrivals

exceeds certain levels.

Adamski (1993) develops several transfer-optimization models. The first is a static model

for a single transfer point, under the assumption of random vehicle travel time. Optimal

timetable offsets are calculated with the objective of minimizing passenger disutility, which

is a function of waiting time. Four alternative forms of the disutility function are examined.

Cases with and without a holding control are analyzed. In the second model, optimal offset

times are determined in the case of different lines that use a common road segment. The

third model that Adamski develops is a dynamic model for real-time transfer optimization.

A measure of off-schedule deviations is used to determine the value of control variables,

such as holding times at control points.

Daduna and Voss (1993) discuss the optimization of transfers, using a quadratic semias-

signed problem formulated as a programming problem. The authors focus on heuristic

solution procedures rather than on a new synchronization methodology. A regret heuristic

is proposed for finding an initial solution. Tabu search and simulated annealing strategies

for improving the solution are described.

Adamski and Chmiel (1997) formulate two schedule-synchronization problems,

which they solve by using genetic algorithms; the first for the synchronization problem

of transfers at transfer points and the second for the synchronization of different

transit lines that travel a common road segment. Deterministic and stochastic cases are

compared. Decision variables are offsets entered into timetables in order to minimize

waiting times.

Clever (1997) discusses the differences between systems with a single-mode transfer cen-

ter and systems with multimodal, multicenter transfer timing. Characteristics of both types

of networks are presented and compared. Among the aspects discussed are service sym-

metry, relations between service and passenger d-

emand, minimal headway, and economic

aspects.

Shih et al. (1998) present a heuristic model for the design of a coordinated network

with transfer centers. The routes are not predetermined, but an output of the model.

In addition, this model determines the appropriate vehicle size for each route. The

proposed design process is composed of four major procedures. The first procedure gen

erates a set of routes. In the second procedure, the network is analyzed and frequencies

are set; this procedure incorporates a trip-assignment model. In the third procedure,

transfer-center locations are determined. The last procedure involves network improve

ments. Algorithms are described qualitatively, using flowcharts for each of the four

procedures.

Becker and Spielberg (1999) present principles for designing and implementing a multi

center-based timed-transfer network. To construct such a network, a prime headway is

chosen, and all headways in the network are set to be multiples of this headway. Transfer

centers are located in such a manner that the distances between them are also multiples

of the prime headway. Other adjustments are made in order to improve network coordi

nation, such as connecting routes at common ends and connecting feeder routes to the

transfer centers.

Maxwell (1999) discusses four transfer strategies: (1) close headways with short waiting

times, so that synchronization is not needed; (2) main-line trunk with timed-transfer branches

lines; (3) a single hub with spoke lines, so that all transfers occur at the same spot; and (4)

several transfer hubs with fixed-interval timed transfers. Advantages and disadvantages of

each strategy are discussed and compared. A methodology for using the fourth strategy is

presented in detail. Transfer points on a symmetrical route are shown to be located in spots

that divide the route into an integer amount of equal segments. It is also shown that other

locations can be chosen if speeds are adjusted accordingly. The use of a schedule map and

symmetrical train graphs as analysis tools is demonstrated.

Ibarra-Rojas and Rios-Solis (2012) studied the bus network of Monterrey, Mexico,

which is a large bus network where passenger transfers must be favored, almost evenly

spaced departures are sought, and bus bunching of different lines must be avoided. The

authors formulate the timetabling problem of this network with the objective of maxi-

mizing the number of synchronizations to facilitate passenger transfers and avoid bus

bunching along the network. These synchronizations are defined as the arrivals of two

trips with a separation time within a time window to make a

flexible formulation. This

flexibility is identified as a critical aspect for the bus network, since travel times vary

because of reasons such as driver's speed, traffic congestion, and accidents. By proving

that the subject problem is NP-hard, the authors answer a 10-year-old open question

about the NP-hardness of similar problems present in literature. Next, the study presents

an analysis of the structural properties of the feasible solution space of the model.

This analysis leads to a preprocessing stage that eliminates numerous decision variables

and constraints. Moreover, this preprocessing defines feasible synchronization and

arrival time windows that are used in a new metaheuristic algorithm. Finally, this study

states that empirical experimentation shows that the proposed algorithm obtains high

quality solutions for real-size instances in less than 1 min.

Ibarra-Rojas et al. (2012) focus on the synchronization bus timetable problem (SBT)

that allows to maximize passenger transfers, avoids bus bunching, and includes flexibility

that is crucial for many transit networks in Latin America. The authors state that a time

tabling solution that is not close to the optimum has strong repercussions in the vehicle

and crew scheduling problems since sequential resolution approaches are often required

for solving the entire planning process. As such, the authors develop families of valid

inequalities that drastically strengthen the integer linear programming model of SBT. Three

of them manage to bound the number of synchronizations while the other two are lifting

ones. Experimental results presented in this study indicate that the enhanced mixed-integer

linear programming yields high quality solutions (less than 1 % of relative deviation from

the optimal solution) in minutes for large instances based on real transit networks (200 buses

lines and 40 synchronization nodes).

Ceder et al. (2013) analyzed how to use selected operational tactics in public-transport

networks for increasing the actual occurrence of scheduled transfers. A model is developed

to determine the impact of instructing vehicles to either hold at or skip certain stops, on the

total passenger travel time and the number of simultaneous transfers. The model is com-

posed of two components. First, a simulation of public-transport network examines the

two tactics for maximizing the number of transfers. Second, an ILOG optimization model

is used for optimal determination of the combination of the two tactics to achieve the maxi-

mum number of simultaneous transfers. A bus network has been created, as a case study, in

Auckland, New Zealand, to verify the impact of the model's application. The results show

that applying online operational tactics dramatically improves the frequency of simultane-

ous transfers by more than 100%. The authors conclude that the proposed concept has large

potential for increasing the efficiency and attractiveness of public-transport networks that

involve scheduled transfers. Further studies on this type of synchronization using tactics

appear in Chapter 19.

It should be noted that several guidelines for the design of the implementation of syn

chronized transit systems are mentioned repeatedly in most of the papers reviewed. Some of

these guidelines are as follows:

- For an efficient synchronized system, route design and schedule design should be effected simultaneously.
- The success of timed-transfer services relies heavily on the use of real-time control. Schedule adherence has a crucial role.
- The effort in introducing synchronized services may not be worthwhile when frequencies are relatively high. Most papers mention a 30 min headway as an example of a headway that justifies transfer timing.

The main characteristics of the models reviewed are described in Table 5.9.

Table 5.9 Characteristics of works related to transit synchronization covered in Section 5.7

Source	Network structure/ transferring strategies	Location of synchronized transfer	Special features
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Abkowitz et al. (1987)	Two bus routes that meet at one spot. Four transfer strategies: unscheduled, scheduled without waiting, scheduled when the less-frequent bus waits, scheduled when both buses wait	A single meeting point	
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Adamski (1993)	1st model: several routes that meet at a terminal. 2nd model: several routes that traverse a common road segment. 3rd model: real-time control		
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strategies 1st model: terminals. 2nd model: a road segment along which several routes pass. 3rd model: control points Random vehicle travel time; several objectivefunction alternatives in the 1st model

Adamski and Chmiel

(1997) 1st model: several routes that meet at a terminal. 2nd model: several routes that traverse a common road segment 1st model: terminals. 2nd model: a road segment along which several routes pass Deterministic and stochastic cases

Becker and Spielberg

(1999) Any route structure Multiple centers

Ceder et al. (2001) and

Ceder and Tal (2001) Any route structure All meeting points Several headway-design options and several objective-function alternatives

Ceder et al. (2013) Any route structure Potential to apply online real-time tactics in the real world

Clever (1997) Any route structure 1st option: single transfer center. 2nd option: multiple centers

Daduna and Voss (1993) Any route structure All meeting points

Desilets and Rousseau

(1990) Any route structure All meeting points Several objective function alternatives

Hall (1985) Several feeder lines and one main line A single terminal Random delay

Ibarra-Rojas and

Rios-Solis (2012) Lines with sub-lines that share a common route segment but have variations at the beginning and at the end of the routes At bunching nodes and transfer nodes

Ibarra-Rojas et al.

(2012) Any route structure Case (a): At bunching nodes.

Case (b): At transfer nodes

Klemmt and Stemme

(1988) Any route structure All meeting points (Continued)

EXERCISES

5.1 Ride-checked data, shown in the following table, were collected on a given bus route (average across several days). There are two stops (B and C) between the first (route departure) stop (A) and the last (route arrival) stop (D). The data cover a 2 h morning period (6:00–8:00). Average travel times between stops are as follows: 8 min (A to B), 14 min (B to C), and 22 min (C to D). Minimum frequency is 2 vehicles/h for both hours. Using Method 2 for frequency determination, find and compare the departure times obtained by (a) the existing (given) data, (b) the even headway procedure, (c) the even load procedure at the hourly maximum load point, and (d) the even load procedure at the individual vehicle maximum load point. In this comparison, provide the average maximum load and its associated maximum load point for each departure and procedure used. Note: only one departure having a desired occupancy of 50 passengers will be derived after 8:00.

Table 5.9 (Continued) Characteristics of works related to transit synchronization covered in Section 5.7

Source	Network structure/ transferring strategies	Location of synchronized transfer	Special features
--------	--	-----------------------------------	------------------

Kyte et al. (1982)	A main trunk line with complementary services	Several centers through which the main line passes	
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Lee and Schonfeld

(1991)	A train line and a bus line that meet at one spot	A single meeting point	Two models with different assumptions regarding delay variability
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Maxwell (1999)	Four transfer strategies: close headways without synchronization, main-line trunk with timed-transfer branch lines, a single hub and spoke, several transfer hubs with fixed-interval timed transfers		
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Nelson et al. (1981) Four transfer strategies: simple timed transfer, pulse scheduling, line-up, neighborhood pulse

Rapp and Gehner (1976) Any route structure All meeting points

Salzborn (1980) Several feeder lines and one main line
Several centers through which the main line passes
Objective includes minimizing the number of required vehicles

Schneider et al. (1984) Detailed discussion of transfer-center location

Shih et al. (1998) Any route structure (routes are generated by the model) Multiple centers Transfer-center locations and optimal vehicle sizes are determined

Voss (1990) Any route structure 1st case: meeting points.
2nd case: road segments along which several routes pass
Departure time (at A) Average load (passengers) when departing stop Desired occupancy (passengers)
A B C 6:00 25 18 12 6:35 65 72 48 40 7:10 67 82 35 7:45 84 33 47 60 8:00 75 41 57 50

5.2 The simple network given combines two transfer points with three routes. The numbers on the arcs are travel times (average in minutes). 1 2 3 5 7 8 4 5
Route II Route I Route III
Origin The data for the network: i H_{min} i H_{max} i F i T
Route I 7 13 2 20 Route II 7 15 2 Route III – 20 1
Note that there is no H_{min} for Route III, since only one vehicle is assigned. Find the number of meetings, their upper-bound Z^* , and their associated departure and meeting times using S ynchro-1 and S ynchro-2. S uggestion: R un a n o ptimization s oftware for optimal (maximum number of meetings) results.

5.3 The network given combines two transfer points with four routes. The numbers on the arcs are travel times (average in minutes). 5 4 2 12 12
Route II Route I Route III Route IV 1 2 3 4 5 5 15 10 7 9
The data for the network: i H_{min} i H_{max} i F i T
Route I 5 10 3 30 Route II 6 10 3 Route III 7 8 3 Route IV 10 15 3
Find the number of meetings, their upper-bound Z^* , and their associated departure and meeting times using S ynchro-1 and S ynchro-2. S uggestion: R un a n o ptimization s oftware for optimal (maximum number of meetings) results.

5.4 S ynchro-1 a nd S ynchro-2 a re i ntended t o m aximize t he n umber o f s imultaneous a rrivals (meetings). G iven t he a verage n umber o f p assengers t hat n eeds t o b e t ransferred b etween r outes, c onstruct a h euristic p rocedure t hat, i nstead o f m aximizing t he n umber o f m eetings, w ill m aximize t he n umber o f p assenger m eetings (each p assenger t ransfer, w ithout w aiting t ime, c ounts a s 1). T he n ew o bjective f unction w eighs e ach p assenger t ransfer t he same.

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6 Chapter 6: Vehicle Scheduling I: Fixed Schedules

Given demand and coordinated timetables, the next phase is to establish chains of daily trips

or vehicle blocs. Each chain or bloc constitutes a vehicle's schedule for a day. The problem of

minimizing the number of chains and, at the same time, fulfilling agency requirements (refueling,

maintenance, etc.) is complex and cumbersome for medium and large-scale transit agencies.

This chapter provides an overview of the problem, outlines specific solution procedures and

tools, and describes some of the experience with a single transit agency. The chapter contains five main parts following an introductory section. Section 6.2 exhibits

a simple method for ascertaining the minimum number of vehicles required for a given single

route without interlining. Section 6.3 presents exact mathematical-programming solutions for

the case of multiroute vehicle scheduling. Practitioners can skip the math formulation here but

should concentrate on the experience of a 4000-bus agency that led to the adoption of differ

ent approaches (described in the fourth part). In Section 6.4, the problem is converted to a

known network-flow problem for enriching the quantitative interpretation of the underlying

required analysis. Practitioners can skip this third part. The fourth part, Section 6.5, is the core

of the chapter, proffering a graphical person-computer interactive approach based on a step

function called deficit function. This approach provides a highly informative graphical technique

that is simple to interact with and use. Practical

suggestions can be interjected by the scheduler

followed immediately by describing the effects of the suggestions on the vehicle's schedule.

Section 6.6, the fifth part, suggests a treatment for an often imperative postulate-maintaining

a balance in the number of vehicles starting and ending at a depot. This part shows the formula

tion of both balancing depots and complying with limitations on the number of overnight spaces

at depots or parking facilities. The chapter ends with literature review and exercises. Practitioners may skip all of Section 6.4 and Appendix 6.A, as well as the math treatment

(formulation) in Sections 6.3, 6.5, and 6.6. They should, though, pay special attention to the

examples and procedures depicted in Figures 6.6 through 6.12. The subjects of vehicle and crew scheduling, the latter to be dealt with in Chapter 10, do

receive attention in available software. This is not to say that those agencies using the soft

ware are fully happy with the results. Nonetheless, the existing competition among the various

software companies is healthy and brings to mind the following story. A large branch of Chase

Manhattan Bank in New York displayed a sign with the company's then slogan: "You have a

friend at Chase." Bank Leumi Le-Israel (BLL) decided to open one of its large branches adjacent

to this branch of Chase. To compete with it, BLL also put up a huge sign: "With BLL you have a

FAMILY." Three weeks later, Chase responded with another sign: "With such a family you really

need a FRIEND."

6.1 INTRODUCTION

Vehicles, especially buses, are often shifted from one route to another (interlining) within

a large-scale transit system, which frequently has to perform deadheading (DH) trips

in order to operate a given timetable with the minimum vehicles required. As was noted in

explaining Figure 1.2, it is desirable to analyze simultaneously the procedures for constructing

timetables and for scheduling vehicles. However, because of the complexity of this analysis,

the two procedures are treated separately. This chapter has two major foci: (1) to describe the

task of vehicle scheduling and possible operations research (OR) solutions for both a single

transit route and a network of routes and (2) to propose a graphical technique that is easy to

interact with and that responds to practical concerns.

The motivation for the second focus arises from a problematic component of Egged

Israel's national bus carrier cooperative, which is presently engaged in scheduling about

4000 buses over some 2000 routes. Egged operates over an extensive geographical

network composed of urban, suburban, regional, and intercity routes and performs

an average of 50,000 daily trips—one of the world's largest schedules. Egged's crucial

component of scheduling buses to trips was performed manually by about 60 schedul

ers using Gantt charts. A scheduler's duty was to list all daily chains (some DH) for a

bus, ensuring the fulfillment of the timetable and the operator's requirements (refueling,

maintenance, etc.). However, because of frequent changes in the schedule and frequent

additional imposed trips, the skilled schedulers were not capable of handling the bus

scheduling tasks efficiently. Consequently, the Egg Management decided to test a fully

computerized system. Their experience with this system is shown in Section 6.3. The

experience led to the development of an informative graphical method, to be explicated

in Sections 6.5 and 6.6.

In Adelaide, Australia, the newspaper headline on April 14, 2000, shown in Figure 6.1,

captures our eyes. We might think that a reverse headline, "Fewer Buses, No Waiting,"

would be the prudent one deserving a scoop. This chapter seeks ways of minimizing the

number of vehicles and, at the same time, maintaining the best level of service set forth by

the decided timetable.

Approaches to vehicle scheduling worldwide range from primitive decision-making to

computerized mathematical-programming techniques. The latter is reviewed in Section 6.3.

As for primitive approaches, an example was observed in Santo Domingo. The drivers for

a 40-bus company would gather very early each morning and create vehicle schedules like

it was a lottery. That is, all trips in the timetable were represented by small pieces of papers

(each trip with a different number), and the drivers simply picked up their piece of paper

containing the chain of trips to be performed that day as though they were drawing a lucky

number from a hat. Some drivers, of course, went back to sleep.

A common practice in vehicle scheduling is to use time-space diagrams similar to the

one from Europe appearing in Figure 6.2. Each line in the diagram represents a trip moving

over time (x-axis) at the same average commercial speed represented by the slope of the line.

Although many schedulers became accustomed to this description, it is cumbersome, if not

impossible, to use these diagrams to make changes and improvements in the scheduling.

It is impossible to use them for interlining because the space (y-axis) refers only to a single

route, as can be seen in the figure. It is also difficult to use different average speeds for different

route segments, in which the lines in the time-space diagram can cross one another;

this is not to mention the inconvenience of using these diagrams manually for inserting D H

trips and/or shifting departure times. These limitations of the time-space diagram caused

us to look more closely into a more appealing approach—the one presented in Sections 6.5

and 6.6.

6.2 FLEET SIZE REQUIRED FOR A SINGLE ROUTE

This section considers the case in which interlinings and DH trips are not allowed and each

route operates separately. Given the average round-trip time and chosen layover time, the

minimum fleet size for a radial route can be found according to the formula derived by

Salzborn (1972). Specifically, let T_r be the average round-trip time, including layover and

turn-around times, of a radial route r (departure and arrival points are same). The minimum

fleet size is equal to the largest number of vehicles that departs within T_r .

Although Salzborn's modeling provides the base for fleet size calculation, it relies on

three assumptions that do not hold up in practice: (1) vehicle departure rate is a continu-

ous function of time, (2) T_r is the same throughout the period under consideration, and

(3) route r is a radial route starting at a major point (e.g., CBD). In practice, departure times

are discrete (see Chapters 4 and 5), average trip time is usually dependent on time-of-day,

and a single transit route usually has different timetables for each direction of travel. For

that reason, we will now broaden Salzborn's model to account for practical operations

planning.

Let route r have two end points: a and b . Let T_{ria} and T_{rjb} be the average trip time on

route r for vehicles departing at t_{ia} and t_{jb} from a and b , respectively, including layover time

at their respective arrival points. Let n_{ia} be the number of departures from a between t_{ia} , in

which departure ia is included, and $t_{i'a}$ in which departure $i'a$ is excluded. Thus, ia arrives

to terminal b , then continues with trip jb , the latter being the first feasible departure from

Figure 6.1 Main headline of an Adelaide newspaper on April 14, 2000. Wouldn't a real scoop be the reverse: "Fewer Buses, No Waiting"?

b to a at a time greater than or equal to the time $t_{ia} + T_{ria}$; $t_{i'a}$ is the first feasible departure

from a to b at a time greater than or equal to $t_{jb} + T_{rjb}$. Similarly n_{jb} may be defined for a

trip j from b .

Lemma 6.1

In the case of no DH trips, n_{ia} departures must be performed by different vehicles at a ,

and n_{jb} must be performed by different vehicles at b , for all ia and jb in the timetables

of r .

Proof: The proof is actually based on a contradiction. Let us assume that the same vehicle

can perform two departures included in n_{ia} at a . However, in order to complete a full round

trip, including layover times, this vehicle can only pick up the $i'a$ departure at a , which is

not included (by definition) in n_{ia} ; hence, it is impossible for the same vehicle to execute two

departures within n_{ia} .

Theorem 6.1

In the case with no interlining (between routes) and no DH trips, the minimum fleet size

required for route r is $N \max\{\max_i n_{ia}, \max_j n_{jb}\}$ (6.1)

Proof: Based on Lemma 6.1, $\max_i n_{ia}$ and $\max_j n_{jb}$ represent the minimum number of vehicles

required to execute the timetables at a and b , respectively. Consequently, the minimum fleet

size for r is the greater of the two.

Figure 6.2 Typical time-space diagram of transit trips used to describe and change vehicle scheduling in many transit agencies.

An example of deriving the required fleet size for a single transit route r is shown in

Figure 6.3. In this figure, a single average travel time $T_{ria} = T_{rjb} = 15$ min is used through

out the timetable for both directions of r . The timetables contain 12 departures at b and

10 at a . The calculations for n_{ia} and n_{jb} are shown by arrows—starting with the departures

at a for n_{ia} (using $T_{ria} = 15$), and starting at b for n_{jb} . The solid line in Figure 6.3 represents

the direction from the starting time to the first feasible connection (after 15 min), and the

dashed line in the opposite direction links to the first feasible connection (also after 15 min)

from the starting point. This leads to a determination of both n_{ia} and n_{jb} , and eventually

the minimum fleet size, $N_{min\ r} = 5$, according to Equation 6.1. It should be mentioned that

the same T_{ria} and T_{rjb} are used throughout the example only for the sake of simplicity. Varied

T_{ria} and T_{rjb} can be utilized in the same manner for each departure.

Vehicle chains (blocks) can be constructed by

using the FIFO (First-in, first-out) rule. That

is, a block will start at a depot for the first assigned scheduled trip, and then will make

the first feasible connection with a departure (based on the route's timetable) at the other

end point of the route, and so on. The block usually ends with a trip back to the depot. Calculating $n_{ia} T_{rja} = 15 \text{ min}$ 6:00 6:15 6:30 6:45 7:00 7:10 7:20 7:25 7:40 8:00 5:00 5:30 6:00 6:30 6:50 7:05 7:10 7:15 7:20 7:30 7:40 8:00 4 Max n_{ia} 5 Max n_{jb} 4 Nr min 3 2 3 3 4 4 3 2 2 – Timetable at a Timetable at b n_{ia} 6:00 6:15 6:30 6:45 7:00 7:10 7:20 7:25 7:40 8:00 5:00 5:30 6:00 6:30 6:50 7:05 7:10 7:15 7:20 7:30 7:40 8:00 Timetable at a Timetable at b n_{jb} 3 2 1 2 5 5 4 4 3 3 2 – Calculating $n_{jb} T_{rjb} = 15 \text{ min}$ Max(4, 5)=5

Figure 6.3 Example of derivation of single-route fleet size with no deadheading trips.

The trips to and from the depot are often DH trips. In the example illustrated in Figure 6.3,

the vehicle blocks can be constructed starting with the first departure (5:00) at b and using the

FIFO (First-feasible connection) rule, then deleting the departures selected and continuing

with another block until all departures are used. At the start of each step (at b), a check

is made to see whether the next (in time) departure can be connected to an earlier unused

departure at a and, if so, whether this connection can be allowed. The five blocks, therefore,

are as follows: [5:00(at b) –6:00(a)–6:30(b)–6:45(a)–7:05(b)–7:20(a)–7:40(b)–8:00(a)];

[5:30(b)–6:15(a)–6:50(b)–7:10(a)–7:30(b)]; [6:00(b)–6:30(a)–7:10(b)–7:25(a)–8:00(b)];

[7:00(a)–7:15(b)–7:40(a)]; [7:20(b)]. A new earlier c

connection, linking the 7:15 departure at b

to the 7:00 departure at a is possible only
in the fourth block. The preceding FIFO process

can certainly start at a as well. Note that the last block
has only one trip; the five blocks

can undergo changes, including swapping trips,
between blocks. Each block can start and

end at a depot or can be used as a segment
in a larger block.

Finally, when DH trips between the ends of two
routes and/or slightly shifting depar

ture times are allowed, it is more complex to
use the formulation developed earlier.

Instead, the solution can be found using the graphical
method presented in Section 6.5

and in Chapter 7.

6.3 EXAMPLE OF AN EXACT SOLUTION FOR MULTIROUTE VEHICLE SCHEDULING

Practical vehicle scheduling usually involves a network
of transit routes with interlining

between routes. The switch from one route to
another can be executed either by a service trip

when the two routes share the same end point
(terminal) or by a DH (empty) trip. The
most

problematic part of the vehicle-scheduling task
for a network of routes is to minimize the

number of vehicles required to carry out the
timetables. It is basically a cost-minimization

problem. This section provides an example of the
implementation of one solution.

The problem of scheduling vehicles in a m

ultiterminal (multiroute) scenario is known

as the Multi-Depot Vehicle Scheduling Problem (MDVSP). This problem is complex

(NP-complete, which is explained in Section 5.5), and considerable effort is devoted to solving

it in an exact way. A review and description of some exact solutions can be found in

Drososiers et al. (1995), Daduna and Paixao (1995), and in the literature review at the end

of this chapter.

An example of the formulation of the MDVSP is as follows:

Objective function $\text{Min } c_{ij} y_{ij} \quad i, j = 1, \dots, n+1$ (6.2)

Subject to $y_{ij} \in \{0, 1\}$ and $y_{i,n+1} = \sum_{j=1}^n y_{ij} = 1 \quad i = 1, \dots, n$
 $\sum_{i=1}^n y_{i,n+1} = \sum_{j=1}^n y_{ij} \quad j = 1, \dots, n$

Finally, there are also constraints ensuring that trip end i can link to trip start j , and that y_{ij}

is integer and greater than or equal to zero, where

i is the end of a trip at time t_i ,

j is the start of a trip at time t_j , $y_{1,1}$ ending is a connection to start 0, otherwise $y_{ij} = 0$

For $i = n+1$, then $y_{n+1,j} = 1$ if a depot supplies a vehicle for the j th trip. For $j = n+1$, then

$y_{i,n+1} = 1$ if, after end of the i th trip, the vehicle returns to a depot and $y_{n+1,n+1}$ = number of

vehicles remaining unused at a depot. The cost function c_{ij} takes the form: $c_{ij} = K + L E_{ij}$;
 $E_{ij} = 1$ if $t_j - t_i \leq 0$, $E_{ij} = 0$ otherwise;
 $i, j = 1, \dots, n+1$ (6.3)

where

K is the saving incurred by reducing the fleet size by one vehicle

L_{ij} is the DH cost from event i to j E_{ij} is the cost of a driver's idle time between i and j

This formulation, appearing in a similar form in Gavish et al. (1978), covers the chaining of

vehicles in a sequential order from the depot to the transit routes, alternating with idle time

and DH trips, and back to the depot. In Dorot et al. (see Section 5.5), this is a zero-one integer

programming problem in that it can be converted to a large-scale assignment problem. The

objectives are to minimize fleet requirements (minimize linkages of routes) and minimize DH

and idle time (maximize vehicle and driver utilization). In addition, the assignment of vehicles

from the depots to the vehicle schedule generated in the foregoing chaining process can be for

mulated as a "transportation problem," as this is well known in the OR literature. The objectives

of the assignment stage are to minimize the driving cost from the depot to the first trip on the

schedule, and from the last trip to a depot.

The formulation presented partially in Equations 6.2 and 6.3 was applied by Eshed (the

Israeli national bus carrier). As mentioned earlier, Eshed has about 4 000 buses and tried to

gain more efficient schedules handling based on Equations 6.2 and 6.3. Eshed examined the

algorithm based on several sets of bus-scheduling data. It is worth noting that the program

is capable of solving bus-scheduling problems involving up to 2500 bus trips. The full test

results and discussion of this examination were reported by Ceder and Gonen (1980). For

example, the results of five sets are shown in Figure 6.4. Only the DH and idle times of the

test data set are not available (indicated by N/A). These examples were selected for peak

hour bus schedules, and at a first look the results seemed quite attractive. However, when

there was an attempt to implement the results, it became evident that several significant

constraints remained unfulfilled. Applying these constraints results in a solution that is con-

siderably different from that obtained by the optimal method. In England, as in most transit

agencies, each bus is used by the same drivers. During the manual scheduling of buses to

trips, it was necessary for the scheduler to consider some of the drivers' constraints, as well,

although that is of secondary importance at this stage.

In summary, the major limitations of the programmable algorithm were that it could not

consider the following: (1) integration of more than 2500 trips, (2) availability of a adequate

bus type for each trip, (3) need for bus

refueling, (4) need for driver's meals, and (5) location

and availability of drivers. The Eggged schedulers agreed that using software as a black box

was as blind as opposed to understanding the software and interacting with it, which becomes

a real eye-opener. In order to satisfy these limitations, it was decided to search for a procedure

that would allow the inclusion of practical considerations that experienced schedulers

	800	900	1000	1300	1400	1500	500	400	300	200	150	Fleet size required
Dead heading time (h)	291	260	215	206	275	253	362	340	457	427	484	454
	1200	1100	100	80	60	40	20					
	43.1	75.9	52.5	27.7	19.7	61.2	52.7	32.4	20.6	93.9	70.3	N/A
	0	800	900	1300	1400	1500	130	110	90	70	50	
Idle time (h)	883	930	1008	1100	1200	1417						
Manual Optimization												
Number of trips in zone examined	92.8	129.1										
	76.2	89.6	44.2	30.9	50.6	47.0	69.5	62.6	N/A	1000	1200	1100

Figure 6.4 Comparison of manual and optimization (programmable) solutions for vehicle-scheduling planning at a large bus agency.

might wish to introduce into the schedule. This procedure, based on the deficit-function

graphical method, is described in Section 6.5.

6.4 MAX-FLOW TECHNIQUE FOR FIXED VEHICLE SCHEDULING

This section shows that the procedure for solving the minimum fleet size problem has its roots

in the classic work of Ford and Fulkerson (1962). The approach described is based on Ceder

(1978) and Stern and Ceder (1983a) using network-flow techniques well known in the OR field.

6.4.1 Vehicle-scheduling task

Consider a schedule (set of timetables) of required transit trips in which each trip is defined

by a starting terminal, starting time, ending terminal, and ending time. The problem is to

find the minimum number of vehicles (the fleet size) that can carry out all the trips in the

schedule. As this is most efficiently done if single vehicles are allowed to service several trips

in succession, a second part of the problem is to find the set of trips assigned to each vehicle

in the fleet. If trip j immediately follows i , then (a) the starting time of trip j must be greater

than or equal to the ending time of trip i and (b) the starting and ending terminal of j and i

must be identical. If the second condition is not met, some systems allow the vehicle to

return empty from i to j . If the travel time for this empty trip can be completed before the starting

time of trip j , then both trips may appear in the same chain and be assigned the same vehicle.

Bus operations offer an example of transportation systems that often carry out such empty

or DH trips. On the other hand, airlines companies rarely, if ever, "deadhead" their aircraft

because of the high cost of running empty.

6.4.2 Formulation and transformation of the Max-flow problem

A trip-joining array for S may be constructed by associating the i th row with the arrival

event of the i th trip, and the j th column

with the departure event of the j th trip. Cell (i, j) will

be admissible if i and j can feasibly be joined. Otherwise, (i, j) will be an inadmissible cell.

Let x_{ij} be a 0-1 variable associated with cell (i, j) and I be the set of required trips; consider,

then, the following problem:

Problem P1 $\text{Max } \sum_{(i,j) \in I} x_{ij} \quad (6.4)$

Subject to: $\sum_{(i,j) \in I} x_{ij} \leq 1, \quad (6.5) \quad \sum_{(i,j) \in I} x_{ij} \leq 1, \quad (6.6) \quad x_{ij} \in \{0,1\} \text{ all } (i,j) \text{ admissible}$
 $x_{ij} = 0 \text{ all } (i,j) \text{ inadmissible}$
 $x_{ij} \in \{0,1\} \quad (6.7)$

A solution with $x_{ij} = 1$ indicates that trips i and j are joined. The objective function maxi

mizes the number of such joinings. Constraint (6.5) ensures that each trip may be joined

with, at most, one successor trip. Similarly, constraint (6.6) indicates that each trip may be

joined with, at most, one predecessor trip. The following theorem states that maximizing

(6.4) in Problem P1 is tantamount to minimizing the number of chains for a trip schedule

of size n .

Theorem 6.2

Let N and n denote the number of chains and trips, respectively. Then, $M \leq n - M$.

Proof of Theorem 6.2 can be found in Ford and Fulkerson (1962).

The Problem P1 is equivalent to a special arrangement of the maximum-flow (Max-flow)

problem. The Max-flow algorithm that solves the vehicle-scheduling problem with DH trips

is called an augmenting-path algorithm. It appears, in an extensive treatment, in the classic

book by Ford and Fulkerson (1962). A complete description of the augmenting-path algo-

rithm, which will be used for further applications in this volume, is shown in Appendix 6.A

at the end of this chapter.

The vehicle-scheduling problem can be transformed to a unit-capacity bipartite network,

in which the solution time has the complexity of $O(n^{1/2}m)$ with n nodes (departure times)

and m arcs. A bipartite network is a network in which the set of nodes is partitioned into

two subsets of nodes such that directed arcs exist only between (not within) the two subsets.

This solution time, first shown by Even and Tarjan (1975), was explicated by Ahuja et al.

(1993). The complexity function is explained in Section 5.5.

We will use a simple numerical example as an explanatory tool to describe and inter-

pret the construction and solution of the vehicle-scheduling problem transformed into a

Max-flow network problem. Consider the three-terminal problem defined by the data in

Tables 6.1 and 6.2. The data in Table 6.1 are transformed into the upper part of the network

work-flow representation in Figure 6.5, which has two dummy nodes: a source node s and

a sink node t . The nodes, being connected from s , are the arrival times of the example, with

an indication, in parenthesis, of the arrival terminal. The nodes connected to t are the departure times, with an indication of the departure terminal. Feasible connections between the

arrival and departure times, utilizing the DH data in Table 6.2, establish the arcs between

the left-side and right-side nodes, based on Equation 6.4. Each arc capacity represents the

number of connections that can "flow" through the arc. In our case, there is only a unit

capacity assigned to each arc, because only one connection (if any) between a given arrival

time and terminal and a given departure time and terminal is possible. The more flow

Table 6.1 Trip schedule S for the example problem

Trip number (i) Departure terminal (p_i) Departure time (t_i) Arrival terminal (q_i) Arrival time (τ_i)

1 b 6:00 b 6:30

2 a 7:05 c 8:05

3 c 7:10 a 8:00

4 b 8:30 a 9:20

5 a 9:00 b 9:45 Table 6.2 Average DH travel-time (minutes) matrix for the example in Table 6.1

Arrival terminal Departure terminal a b c a 0 30 50 b 35 0 45 c 45 40 0

Network-flow representation Final residual network flow:

solution 1a Last iteration in the residual network flow:

solution 1b Solution 1a (connection) paths: $s-6:30 (b)-7:05$

$(a)-t s-8:05 (c)-9:00 (a)-t s-8:00 (a)-8:30 (b)-t$ Solution

1b (connection) paths: $s-6:30 (b)-8:30 (b)-t s-8:05$

$(c)-9:00 (a)-t s-8:00 (a)-8:30 (b)-6:30 (b)-7:05 (a)-t$

Using reverse arc $6:30 (b) 8:05 (c) 8:00 (a) 9:20 (a) 9:45$

(b) 6:00 (b) 7:05 (a) 7:10 (c) 8:30 (b) 9:00 (a) s t 6:30
 (b) 8:05 (c) 8:00 (a) 9:20 (a) 9:45 (b) 6:00 (b) 7:05 (a)
 7:10 (c) 8:30 (b) 9:00 (a) s t 6:30 (b) 8:05 (c) 8:00 (a)
 9:20 (a) 9:45 (b) 6:00 (b) 7:05 (a) 7:10 (c) 8:30 (b) 9:00
 (a) s t Arrival time (terminal) Departure time (terminal)
 Minimum s-t cut Capacity = 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 Minimum s-t cut Flows () and capacities () on all
 arcs = 1

Figure 6.5 Solutions 1a and 1b of the vehicle-scheduling example using the augmenting-path algorithm appearing in Appendix 6.A.

created, the fewer chains will be required as stated by Theorem 6.2. The objective function

2 1 i n P roblem P 1 e quals t he ! ow t o b e c
 reated a t s a nd a bsorbed a t t. T he m aximum !
 ow o f

f our i n F igure 6 .5 m eans t hat a ll t rips c
 an b e h andled b y a s ingle c hain, w ith f our l
 inkages

b etween arrivals and departures.

The a ugmenting-path a lgorithm d escribed i n A
 ppendix 6 .A i s a pplied f or t he e xample

problem. We look for the minimum-arc path between s and t,
 assign a flow = 1, and

c reate a r esidual n etwork i n w hich a ll a rcs u
 tilized h ave a r everse a rc (dashed l ine i n t
 he

t wo r esidual n etworks i n F igure 6 .5). A ll c
 apacities a nd f lows i n t he r esidual n etworks

h ave a u nit v alue. W e t hen s earch f or t he n
 ext m inimum-arc p ath b etween s a nd t,

a ssigning t o i t f low = 1 o n t he r esidual n
 etwork, a nd c onstruct a n u pdated r esidual

network. The process continues until no s-t path is found.
 Two solutions, 1a and 1b,

l eading t o t he s ame o ptimal r esult (Max-flow
 = 3) a re s hown i n F igure 6 .5. I n s olution

1 b, we show a case in which a reverse arc is used in the augmenting-path algorithm.

Based on Theorem 6.A.4 (in the appendix), $\text{Max-flow} = \text{minimum s-t cut}$ in the origi-

nal network-flow. This minimum s-t cut, depicted in Appendix 6.A, is shown in the

upper part of Figure 6.5.

The result of the example is $\text{Max-flow} = \text{Max } Z = 3$, and, following Theorem 6.2, Min

$N = 5 - 3 = 2$ chains. That is, the timetable in Table 6.1 can be carried out by a minimum of

two vehicles having the connections shown in Figure 6.5. Explicitly the two blocs, by their

trip number in Table 6.1, are [1-2-5] and [3-4]. These two blocs have three DH trips for

connecting arrival and departure terminals: b-a and c-a (in the first bloc) and a-b (in the

second bloc), with the total of, respectively, $35 + 45 + 30 = 110$ min DH time.

6.5 DE FICIT-FUNCTION MODEL WITH DEADHEADING TRIP INSERTION

The minimum fleet size problem may be referred to with or without DH trips. When DH

is allowed, we can reach the counterintuitive result of decreasing the required resources

(fleet size) by introducing more work into the system (adding DH trips). This approach

assumes that the capital cost of saving a vehicle far outweighs the cost of a any increased

operational cost (driver and vehicle travel c

ost) imposed by the introduction of DHTrips.

This section offers a highly informative, graphical person-computer interactive technique

based on a step function, called deficit function (DF), which is simple to use and interact

with. The lesson that is learned is that it is easier to deadhead vehicles or shift their departure

time (see next chapter) when the problem is seen graphically, even if it is almost

mission impossible. Perhaps the inspiration for this approach is the saying: "It is easier for

an elephant to pass through the eye of a needle if it is lightly greased." Following is explanation

of background on a step-function approach, described by Ceder and Stern (1981) and

Ceder (2003), for allocating the minimum number of vehicles to a given timetable.

6.5.1 Definitions and notations

Let $I = \{i: i = 1, \dots, n\}$ denote a set of required trips. The trips are conducted between a set of

terminals $K = \{k: k = 1, \dots, q\}$, each trip to be serviced by a single vehicle, and each vehicle

able to service any trip. Each trip can be represented as a 4-tuple (i, j, p, q) , where $i, j \in I$ is the trip in which the

ordered elements denote departure terminal, departure (start) time, arrival terminal, and

arrival (end) time. It is assumed that each trip lies within a schedule horizon $[T_1, T_2]$, that

is, $T(t) \leq T(s) \leq 2$. The set of all trips $S(t) \subseteq K$ is $S(t) = \{i \in K : t_i \leq t\}$. p constitutes the time

table. Two trips, i and j , may be serviced sequentially (feasibly joined) by the same vehicle if

and only if (a) $t_i \leq t_j$ and (b) $q_i = p_j$.

If i is feasibly joined to j , then i is said to be the predecessor of j , and j the successor

of i . A sequence of trips $i_1, i_2, \dots, i_w$ ordered in such a way that each adjacent pair of

trips satisfies (a) and (b) is called a chain or block. It follows that a chain is a set of trips

that can be serviced by a single vehicle. A set of chains in which each trip is included

in exactly once is said to constitute a vehicle schedule. The problem of finding the

minimum number of chains for a fixed schedule is defined as the minimum fleet size

problem.

Let us define a DH trip as a nonempty trip from some terminal p to some terminal q in time

$\tau(p, q)$. If it is permissible to introduce DH trips into the schedule, then conditions (a) and (b)

for the feasible joining of two trips, i, j , may be replaced by the following: $t_i \leq t_j + \tau(p, q)$ (6.8)

Now let us introduce a deficit-function-based model, which will be utilized and employed

throughout most of the remaining chapters of the book.

A deficit function (DF) is a step function defined on a cross the schedule horizon that

increases by one at the time of each trip departure and decreases by one at the time of each

trip arrival. This step function is called a DF because it represents the deficit number of

vehicles required at a particular terminal in a multiterminal transit system. To construct a

set of DFs, the only information needed is a timetable of required trips. The main advantage

of the DF is its visual nature. Let $d(k, t, S)$ denote the DF for terminal k at time t for

schedule S . The value of $d(k, t, S)$ represents the total number of departures minus the total

number of trip arrivals at terminal k , up to and including time t . The maximum value of

$d(k, t, S)$ over the schedule horizon $[T_1, T_2]$, designated $D(k, S)$, depicts the deficit number

of vehicles required at k .

The DF notations are presented in Figure 6.6, in which $[T_1, T_2] = [5:00, 8:30]$. It is

possible to partition the schedule horizon of $d(k, t)$ into a sequence of alternating high

and maximum intervals $(t_0, t_1), (t_1, t_2), \dots, (t_{j-1}, t_j), (t_j, t_{j+1}), \dots, (t_{n-1}, t_n)$ where $t_0 = T_1$ and $t_n = T_2$. Note that S will be

deleted when it is clear which underlying schedule is being considered. Maximum intervals

are $t_0, t_1, t_2, \dots, t_{j-1}, t_j, t_{j+1}, \dots, t_{n-1}, t_n$

where $t_0, t_1, t_2, \dots, t_{j-1}, t_j, t_{j+1}, \dots, t_{n-1}, t_n$ define the intervals of time over which $d(k, t)$ takes on its maxi

maximum value. Index j represents the j th maximum intervals from the left; $n(k)$ represents the

total number of maximal intervals in $d(k, t)$, where s_{jk} is the departure time for a trip leaving

terminal k and e_{jk} is the time of arrival at terminal k for this trip. The one exception occurs

when the DFR reaches its maximum value at $M_n(k)$ and is not followed by an arrival, in which

case $e_n(k) = T_2$.

A hollow interval H_{jk} , $j = 0, 1, 2, \dots, n(k)$ is defined as the interval between two maximum

intervals: this includes the first hollow, from T_1 to the first maximum interval, $H_{T_1 1 k} = [,]$;

and the last hollow, which is from the last interval to T_2 , $H_{e_n(k) n(k) k} = [,]$. k follows may

contain only one point; if this case is not on the schedule horizon boundaries (T_1 or T_2), the

graphical representation of $d(k, t)$ is emphasized by a clear dot.

The sum of all DFRs over k is defined as the overall DFR, $g(t) = \sum_k d(k, t)$. This function

$g(t)$ represents the number of trips that are simultaneously in operation, that is, a count,

from a bird's-eye view at time t , of the number of transit vehicles in actual service over

the entire transit network of routes. The maximum value of $g(t)$, $G(S)$, is exploited for

determination of the lower bound on the fleet size (see Chapter 7). An example of a

two-terminal operation, a fixed schedule of trips, and the corresponding set of DFRs and

notations is illustrated in Figure 6.6.

6.5.2 Fleet size formula

Determining the minimum fleet size, $D(S)$, from the set of DFs is simple enough—one merely

adds up the deficits of all the terminals. In the example in Figure 6.6 without DH trips, $D(S) =$

$D(a) + D(b) = 4$. This result was apparently derived independently by Bartlett (1957), Szalborn

(1972, 1974), and Gertsbach and Gurevich (1977). It is formally stated as Theorem 6.3.

3 2 1 0 T 1 s a 2 e a 1 = H a 0 M a 1 M a 2 2 1 0 -1 H b 0
H b 1 H b 2 M b 2 e b 1 e b 2 5:00 6:00 7:00 8:00 5:00 6:00
7:00 8:00 a a a a a b b b a b b b b 1 6 2 3 5 4 7 4 3 2 1
0 H a 1 Fixed schedule S Deficit function at a $d(a, t, S)$ Deficit function at b $d(b, t, S)$ Sum of all deficit
functions (# of vehicles in service) $g(t, S)$ $D(S) = 4$
 $D(b, S) = 1$ $D(a, S) = 3$ $G(S) = 4$ Component Maximum value
Detailed example Time Time H a 2 s b 1 s b 2 s a 1 e a 2 T
2 M b 1 T 1 T 2

Figure 6.6 Illustration of two-terminal fixed schedule with associated deficit functions and their sum, including notations and definitions.

Theorem 6.3

If, for a set of terminals K and a fixed set of required trips I , all trips start and end within

the schedule horizon $[T_1, T_2]$ and no DH insertions are allowed, then the minimum number

of vehicles required to service all trips in I is equal to the sum of all the maximum deficits: $\min N = \sum_{k \in K} \max_{t \in [T_1, T_2]} d(k, t, S)$ (6.9)

Proof: Let F_k be the number of vehicles present in terminal k at the start of the schedule

horizon T_1 ; let $s(k, t)$ and $e(k, t)$ be the cumulative number of trips starting and ending at k

from T_1 up to and including time t . The number of vehicles remaining at k at time $t \geq T_1$ is

$$F_k = s(k, t) + e(k, t).$$

In order to service all trips leaving k , the preceding expression must be nonnegative, that

is, $F_k \geq s(k, t) - e(k, t)$, $T_1 \leq t \leq T_2$. The minimum number of vehicles required at k is then

$$\text{equal to the maximum deficit at } k: M_k \text{ in } F_k = \max_t [s(k, t) - e(k, t)] = \max_t d(k, t). \text{ Hence, the}$$

minimum number of vehicles required for all terminals in the system is equal to the total

$$\text{deficit } \min N = \min F = D(S). \quad K \quad K = \sum_k = \sum_k = \sum_k e_k k k k()$$

6.5.3 DH trip insertion: Effect and initial fleet size lower bound

This section follows Ceder and Stern (1981). A DH trip is an empty trip between two

terminals that is usually inserted into the schedule in order to (1) ensure that the schedule

is balanced at the start and end of the day, (2) transfer a vehicle from one terminal where

it is not needed to another where it is needed to service a required trip, and (3) refuel or

undergo maintenance. This section will consider case (2), and Section 6.6 will comment on

case (3). We start by asking where and when are such trips needed. Usually, a trip schedule

received from operating personnel includes such DH trips, and it is easy to apply the fleet

size formula to determine the minimum fleet size, followed by the FIFO rule to construct

each vehicle's schedule. The assumption is that the trip schedule S has been purged of all DH

trips, leaving only required trips. From this point, the question of how to insert DH trips

into the schedule in order to further reduce the fleet size will be examined. At first, it seems

counter-intuitive that this can be achieved, since it implies that increased work (adding trips

to the schedule) can be carried out with decreased resources (fewer vehicles). This section

will show through an examination of the effect of such DH trip insertions on DFs that this

is indeed possible.

Consider the example in Figure 6.7. In its present configuration, according to the fleet size

formula, four vehicles are required at terminal a, 0 at terminal b, and 1 at terminal c for a

fleet size of five. The dashed arrows in the figure represent the insertion of DH 1 trip from

b to a and DH 2 from c to b. After the introduction of these DH trips into the schedule, the

DFs at all three terminals are shown updated by the dotted lines. The net effect is a reduction

in fleet size by one unit at terminal a. It is interesting to examine the particular circumstances under which this reduction was achieved.

After adding a arrival point in the first

half of terminal a before s1a, the maximal

interval when using DH 1 is reduced by one unit,

causing a unit decrease in the deficit at a.
This arrival point becomes, therefore, e1a.
Since

the DH 1 departure point is added in the middle hollow
of terminal b, at e1b, it is necessary to

introduce a second DH trip, which will arrive
at the start of the second maximum interval

of b. Fortunately, this DH 2 trip departs from the
last hollow of c, where it could no longer
6:00 6:20 6:40
7:00 7:20 a a a a a c b c b b c DH 1 DH 2 $D(c) = 1$ $D(b) = 0$
 $d(a, t) g(t) d(b, t) d(c, t)$ Fixed schedule
e d u l e DH 1 DH 2 Deficit function after DH insertion $\tau(p, q) = 20$ min, $p, q = a, b, c$ Time 3 4 5 2 1 0 3 4 2 1 0 2 1 0
-1 2 1 0 -1 6:00 6:20 6:40 7:00 7:20 $G = 3$ 4 $D(a) = 4$ 3

Figure 6.7 Description of six-trip, two-terminal example
in which the fleet size is reduced by one using a chain of
two DH trips (URDHC) and in which $g(t)$ is changed.

affect the deficit at c. In general, it is possible to have
a string of DH trips to reduce the fleet

size by one unit: one initiator trip and the others
compensating trips.

All successful DH trip chains follow a common
pattern. The initial DH trip is introduced

to arrive in the first hollow of a terminal in
which a reduction is desired. This DH trip must

depart from some hollow of a other terminal.
Moving to the end of this hollow, a other DH
H

trip is inserted, such that its arrival epoch will
compensate for the departure epoch added by

the first DH trip. Each time a DH trip is inserted
(from p to q) to arrive at the end of a hollow

H i

q from the start of a hollow $H_j p$, it must pass a feasibility test, that is, $e, s_j i p q p q+ \leq + \tau() . 1$. If the

inequality is true with $<$, then there will be some slack time, during which the DH trip can be

shifted. Let this slack time be defined as δ $\tau p q k q p p q = - + + s e , i j [()]$. In practice, if the DH time

plus the slack time are greater than or equal to the average service travel time, then a service

trip may replace the DH trip. In this way, an additional service trip is introduced, thereby

resulting in higher frequency (i.e., an improved level of service) at usually the same operational

cost. Figure 6.8 illustrates an in-trip schedule with three DH times required to reduce D(b) by

one. In this example, DH 2 can be converted to a service trip in which $\delta_{ac} = 30$ min.

The process ends when a final hollow of some terminal q is reached (i.e., $H H i n q q q = ()$), after

which no compensation is necessary. It is possible to arrive at a point where no feasible com

pensating DH trips can be inserted, in which case the procedure terminates or one may back

track to the arrival point of the last DH trip added and try to replace it with another. This

procedure results in a sequence of DH trips known as a unit reduction deadheading chain

(URDHC) if it ends successfully (i.e., if it reduces the fleet size by a unit amount). A URDHC

is a sequence of DH trips of the form $(k_1, DH_1, k_2, DH_2, \dots, k_j, DH_j, k_{j+1}, DH_w, k_{w+1})$, where

DH_j is a DH trip from terminal k_j to k_{j+1} . The DH trips are inserted into the DFs from

low to high, with DH_1 arriving at the first hollow H_2 of $d(k_2, t)$ and DH_w departing from

the last hollow of $d(k_w, t)$. The procedure inserts URDHCs until no more can be found or a

lower bound on the minimum fleet size is reached. The examples of Figures 6.7 and 6.8 include

a single URDHC with two and three DH trips, respectively.

Obviously, the continued reward for such a search must stop, and the following “Initial

Lower Bound Theorem” provides a condition when it is sufficient to continue this search. The

initial, and rather intuitive, lower bound on the fleet size is stated in the following text.

Further improvements in the fleet size lower bound appear in Chapter 7.

Theorem 6.4

(a) For a set of terminals K and a trip schedule S , the maximum number of trips in simultaneous operation provides a lower bound on the minimum fleet size
$$G \geq \min_{t \in T} \left(\sum_{k \in K} d(k, t) \right) \quad (6.10)$$

(b) If $G = M$ in N , it is impossible to reduce the minimum fleet size through the introduction of DH trips.

Proof

(i) In general it is known that
$$\sum_{k \in K} d(k, t) \leq \sum_{k \in K} d(k, t) \quad \forall t \in T$$

From the definition of $g(t)$ and (6.9), $\max () [,] t T T g$
 $t \text{ Min } N \in \leq 1 \text{ 2 DH travel time between all terminals} = 30$
 min (Service time=60 min) Before (5 FIFO chains): [1-7],
 [2], [3-4], [5-6], [8-9] After (4 FIFO chains): [1-DH 1
 -8-9], [2-DH 3 -6], [3-DH 2 -7], [5-4] DH 1 DH 2 DH 3 D(a)
 $= 2 \text{ D(c)} = 1 \text{ d (c , t) d (b , t) d (a , t)}$ 6:00 6:30
 7:00 7:30 8:00 8:30 9:00 9:30 c a b c b a a c b a a b c a
 b c 3 1 2 7 4 6 5 9 8 c b b a a c DH 1 DH 3 DH 2 F i x e d
 s c h e d u l e D e a d h e a d i n g t r i p s Change to a
 service trip? Time 6:00 6:30 7:30 8:00 8:30 9:00 9:30 a c 0
 1 2 0 1 2 -1 0 1 2 1 D(b) = 2 7:00

Figure 6.8 Example of a URDHC with three DH trips in which
 DH 2 can be converted to a service trip.

(ii) Let S represent the original schedule of given trips.
 Add some trip i starting at $t(s)$ and ending at $t(e)$
 to the schedule (this can be a DH trip). Call the new schedule (with i) S' . The
 symbol S or S' is added to the argument of t
 he d e c i t a n d o v e r a l l D F s t o i n d i c a t e t h e s e t o f

Since $g(t, S)$ for $t(t(s), t(e))$ $g(t, S)$ for $t(t(s),$
 $() , [] , [' = \# + \in 1 t(e) g(t, S) g(t, S) t t] \max ()$
 $\max \# \# \# \geq'$

$$G(S') \geq G(S)$$

From (i) $\text{Min } N(S') \geq G(S')$ and using the assumption in (b)

$$\text{Min } N(S) = G(S)$$

Therefore,

$$\text{Min } N(S') \geq \text{Min } N(S)$$

This theorem is quite useful, as it enables
 one to understand the situations, based on the

decisions, in which we can be sure that no
 reduction in fleet size can be (further) achieved.

Figures 6.6 and 6.7 show and indicate both
 $g(t)$ and G . Figure 6.7 also demonstrates the

impact of DH insertion on the value of G .
 The result of Theorem 6.4 determines the ex

tent

to which we can expect (in a maximal successful case) to reduce a fleet size through the

introduction of DH trips, compared to the minimum fleet size required when DH trip insertions are not allowed.

6.5.4 Heuristic algorithm for a DH trip insertion

The results of the prior section provide a number of clues for the insertion of DH trips into

the schedule that can form the basis of a heuristic algorithm to reduce fleet size. As much

freedom of choice exists in the selection of DH trips at any point in the chain construction

procedure, the algorithm can be programmed so that it may be employed in a conversational

(manual mode), person-machine mode. This allows the practical considerations of experi-

enced schedulers to be brought into the process. In addition, the algorithm may operate in a

purely automatic manner, that is, selections of DH trips being made on the basis of several

criteria, to be discussed later in the text. The primary inputs required for the heuristic pro-

cedure are (1) the initial set of trips, S^0 , defined over a set of terminals K , and (2) the

travel time matrix for potential DH trips between each pair of terminals in the schedule.

The output of the algorithm is a new trip schedule that

includes the set of DH trips inserted

and the number of vehicles required at each terminal. This constitutes the fleet size. Finally

DF information is also available. In order to construct the trip schedule of each vehicle in the

fleet, a second phase is required. This can be done by applying the chain construction rules

described in the next section.

A general framework of the heuristic algorithm is shown in a flow diagram form, fol

lowing Ceder and Stern (1981), in Figure 6.9a and b. The parentheses { } denote a set

of elements. The result of the lower-bound fleet-reduction theorem suggests that $G(S_0)$

should be examined before trying to insert DH trips into the schedule. This is called a

lower-bound termination test in Figure 6.9a. Note that after DH trip insertions are made,

$G(S_0)$ may be increased as shown in the example of Figure 6.6. It is thus important to

evoke the lower-bound termination test after each DH chain insertion. Nonetheless in

Chapter 7, $G(S)$ is improved (can be increased) to $G''(S'')$; hence, the lower-bound termination

test will have $G''(S'')$, a non-updated value, instead of $G(S)$ to compare with $D(S)$.

In the latter case, the updated $G(S)$ is kept in order to know the number of vehicles in

service at any time. Initial input: Trip schedule S_0 , Set of terminals K , DH time matrix. Set $S=S_0$, $W=\emptyset$

Computation of deficit function information: $\{d(k,t,S)\}$,
 $D(S)$, $g(t,S)$, $G(S)$ Lower bound termination test* $G(S)$
 $=D(S)$? Update terminals examined. Set $W=W+u$ C B (a) Reduce
a vehicle at u . Add set of DH trips $\{DH\}$ to the
schedule: $S=S+\{DH\}$, $W=\emptyset$ Select next terminal u , and
examine for possible reduction in number of vehicles
using rule NT No No Yes Unit reduction DH chain found
from terminal u ? A All terminals examined $W=K$? STOP Yes
No Yes DH cost > vehicle saving cost ? No Yes * $G(S)$ is
improved in Chapter 7 to $G''(S'')$; the improved boundary
takes place in this test

Figure 6.9 (a) Flow diagram of the DH insertion heuristic algorithm. (Continued)

The next step in Figure 6.9a is an attempt to reduce the fleet size at some terminal in

the schedule. The selection of the terminal is made by using rule NT (next terminal). This

may be done manually on the basis of depot capacity violations, that is, the present fleet

size required at a terminal exceeds the capacity of the terminal. A default rule selects the

terminal whose first follow is the longest. The rationale here is to try to offer the largest

best opportunity for the insertion of the first DH trip. This selection should, of course, be

made from terminals that have not been previously examined; they should not be made

from terminals previously examined for U RDHCs with unsuccessful results. In cases of

multiple longest-first follows, the selection is based on a terminal whose overall maxi

mum region (from the start of the first maximum interval to the end of the last one) is

the shortest. Set $q=u$, $\{DH\}=\emptyset$ Attempt to insert DH trip to arrive at terminal q before time s_q Determine set of

feasible candidate departure points for DH trip from v to q
 $\{e_v\}$ A departure point e_p selected? Is e_p in a
last hollow? Add DH trip (p,q) to set $\{DH\}$ A URDHC from u
has not been found A URDHC has been found B C (b) Determine
next terminal; insert a compensating DH trip. Set $q=p$ Find
end of hollow s_p $R=0$ Manual $R=1$ MTT $R=2$ FSH $R=3$ FEH A
Select departure point using rule: R No Yes No Yes List
maximum e_{kj} , s_{kj} intervals for all terminals k K

Figure 6.9 (Continued) (b) Part of the flow diagram in (a),
in which the URDHC subroutine is illustrated.

Once a terminal, u , is selected, the algorithm searches for
a URDHC from that terminal.

This is done in the sub-routine URDHC, which
appears with a dashed line in Figure 6.9b.

If a URDHC is not found from u , this terminal is added to the set W of terminals examined.

If all terminals have been examined ($W = K$),
then the algorithm stops. Otherwise, a new

terminal is selected and the search for a URDHC continues. If a URDHC has been found

from u , in practice another test takes place. This concerns
the DH cost involved as compared

with the cost saving of a single vehicle; the comparison is usually computed by accountants.

If the DH cost is higher than the saving cost, the URDHC is cancelled. Otherwise, the set of

DH trips $\{DH\}$ is added to the previous schedule; all DF information is updated for this new

schedule, and a new iteration initiated. The algorithm must terminate after a finite number

of steps because the fleet size in each iteration is reduced by one unit until either it reaches

its lower bound or a URDHC cannot be found.

Since the search for a U RDHC is conducted

many times in the algorithm, it has been placed in a sub-routine, called U RDHC, which will

now be described.

The sub-routine U RDHC in Figure 6.9b starts by setting a dummy computer variable

q to u. The number taken on by q represents the terminal for which the next DH trip is to be

inserted. This trip is to arrive before time $s_{j+1} q$, that is, before the end of the hollow $H_j q$ (in the

case of $q = u$, $H_j q u = \emptyset$). The computer then determines a set of feasible candidate departure

points $\{e_v\}$ from the set of terminals $V = \{u\}$, $q \in V$. These points are identified by the start of

hollows that lie to the right of the last hollow examined. That is, the selection of one of the

candidate trips is made using rule R. When $R = \emptyset$, the computer allows this selection to be

inserted manually into a conversational mode. Otherwise, an automatic selection is made by

the computer using one of the following criteria:

R = 1: The minimum DH trip time (MTT). Insert the candidate DH trip with the least travel time.

R = 2: The furthest start of a hollow (FSH). Insert a candidate DH trip whose hollow starts furthest to the right.

R = 3: The farthest end of a hollow (FEH). Insert a candidate DH trip whose hollow ends furthest to the right.

Figure 6.10 shows the effect of each criterion

on three examples. For each example, the value

of the total travel time of the DH trips in the URDHC is found, and the minimum value

across all criteria circled. These examples show that the minimum DH travel time associated

with the URDHC can be found by any of the criterion; no one criterion dominates the others

for this performance measure.

If a DH trip cannot be selected and the completion of a URDHC is blocked, the algorithm

backs up to the last candidate list with the last selection deleted; it then proceeds to select

another DH trip from the candidate list. In the manual mode, the scheduler may choose a

new terminal and initiate a new search for a URDHC instead of backing up. The option

is also available in the automatic mode to forego the backup feature when blocks occur in

order to reduce excessive computation time. If the backup feature is allowed to continue, one

may return to the initial hollow of terminal u, after which a new terminal may be selected

to reinitiate the search for a URDHC. When the full backup feature is used, the procedure

will find the optimal solution, which is equivalent to Ford and Fulkerson's (1962) Max

flow procedure, which is explained in Section 6.4. If a final hollow is reached, a URDHC

has been found, and it is returned to the main program for introduction into the schedule.

Otherwise, the search for a compensating trip is repeated until one trip is found that departs

from a last hollow.

6.5.5 Constructing vehicle schedules (chains/blocks)

At the end of the heuristic algorithm, all trips, including the DH trips, are chained for

constructing the vehicle schedule (blocks). Two rules can be applied for creating the chains:

FFO and a chain extraction procedure described by Gertsbach and Gurevich (1977). The

FFO rules imply links the arrival time of a trip to the nearest departure time of another trip

(at the same terminal); it continues to create a schedule until no connection can be made.

The trips considered are deleted, and the process continues. Total DH time for given rule R (min) MTT FSH
 FEH 50 60 60 70 70 60 80 70 80 6:00 6:30 7:00 7:30 8:00 8:30
 9:00 9:30 d(e, t) d(c, t) d(b, t) d(a, t) d(e, t) d(c, t)
 d(b, t) d(a, t) d(e, t) d(c, t) d(b, t) d(a, t) Time 6:00
 6:30 7:00 7:30 8:00 8:30 9:00 9:30 a b c e a 0 20 30 40 b
 10 0 30 20 c 30 30 0 50 e 20 20 30 0 DH matrix (min)
 Deficit functions and DH trip insertion rules Key: DH
 chain using R for minimum DH time Other DH chains

Figure 6.10 The effect on total deadheading time of three alternative heuristic rules for selecting candidate DH trip-departure terminals.

The chain extraction procedure allows an arrival-departure connection for any pair

within a given hollow (on each DF). The pairs considered are deleted, and the procedure

continues. Figure 6.11 illustrates one D F at k . This $d(k, t)$ has four follows, $H_j k$, $j = 0, 1, 2, 3$,

with $H_1 k$ having arrivals of Trips 1, 2, and 3 and departures of Trips 4, 5, and 6. Figure

6.11a re the F IFO connections (within this follow) as well as other alternatives; in all, the

minimum fleet size at k , $D(k)$, is maintained. In addition, two full F IFO chains are shown in

Figure 6.8 before and after DH insertion procedure.

6.6 D EPO T-CONSTRAINED VEHICLE SCHEDULING

This section combines the description of previous sections with the methodology advanced

by Stern and Ceder (1983a). A vehicle schedule (chain/block) is said to be balanced if for

every terminal k , the number of vehicles starting their schedules from k equals the number

of vehicles ending their schedules at k (not necessarily the same vehicles). Otherwise, the

vehicle schedule is unbalanced. In practice, transit agencies seek this balanced condition at

terminals with overnight parking (usually called depots).

6.6.1 Deficit function formulation

An example of a balanced schedule appears in Figure 6.12 for a solution without DH trips,

using the data of Table 6.1. The solution indicates that exactly one vehicle chain starts and

ends at each of the three terminals (albeit not the same chain for each terminal). However,

a solution with DH trips, which saved a v

vehicle at terminal a , resulted in an unbalanced

schedule (see lower part of Figure 6.12). A D
 H trip from a to c after 9:20 will balance the

schedule. To determine whether a schedule is balanced, it
 is not necessary to carry out

the actual chain constructions. The assessment
 may be made through an examination of

the DFs of the schedules. If $v_s(k)$ and $v_e(k)$
 are defined as the number of vehicles required

at the start and remaining at the end of
 the schedule horizon, respectively, for terminal
 k ,

then the schedule is balanced if $v_s(k) = v_e(k)$ for all k . Max $d(k, t) = D(k)$ 1 2 3 4 6 5 8 6 4 2
 $\emptyset$ T 1 T 2 Time FIFO set of connections: [(1-4), (2-5),
 (3-6)] Other sets of connections: [(1-4), (2-6), (3-5)],
 [(1-5), (2-6), (3-4)], [(1-6), (2-4), (3-5)], [(1-5),
 (2-4), (3-6)], [(1-6), (2-5), (3-4)] H k $\emptyset$ H k 1 H k 2 H k
 $3 d(k, t)$

Figure 6.11 Example of creating trips connections within
 one hollow, $H_1 k$, using the FIFO rule and all other
 possibilities while maintaining the minimum fleet size
 attained.

Theorem 6.5

A vehicle schedule is balanced if and only if
 the value of each DF at the end of the
 schedule

horizon is zero, that is,

$$d(k, T_2) = 0, \quad k \in K$$

Proof: $D(k) - d(k, T_2)$ is the net number of arrivals
 (arrivals minus departures) from the end

of the last maximum interval. As these arrivals
 are not followed by departures in any
 chain

construction, they represent the number of chains at k , that is, $v_e(k) = D(k) - d(k, T)$.

Furthermore, it follows from Theorem 6.5 that $v_s(k) = D(k)$. The requirement from a balanced

scheduled schedule, $v_s(k) = v_e(k)$, implies $d(k, T) = 0$. $d(c, t) d(b, t) d(a, t) d(c, t)$ Unbalanced (revised deficit functions result in two chains, [1-DH 1 -2- DH 3 -5], [3-DH 2 -4] unbalanced at a and c) 10:00 Depotbalanced condition $b c c b a a b 1 4$ Balanced (three chains, [1-4], [2], [3-5], with one at the start and one at the end at each terminal) 6:00 7:00 9:00 10:00 8:00 Fixed schedule $d(a, t) d(b, t)$ 6:00 7:00 8:00 $D(c) = 1 D(b) = 1$ DH 3 DH 2 DH 1 Time 10:00 9:00 8:00 7:00 Time 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0 6:00 9:00 3 2 5 $b a a D(a)=1 0$

Figure 6.12 Deficit-function solution, with depot-balanced evidence, for the example in Tables 6.1 and 6.2.

The conclusions drawn from Theorem 6.5 are as follows: (1) if $d(k, T) < 0$, then a surplus

of $d(k, T)$ vehicles is present at k at the end of the day; (2) if $d(k, T) > 0$, then a shortage of

$d(k, T)$ vehicles occurs at k at the end of the day; and (3) the following identity is true under

a system that conserves vehicles: $d(k, T) = d(k, T) + \sum_{k \in K_1} d(k, T) - \sum_{k \in K_2} d(k, T)$ (6.11)

in which K_1 and K_2 represent the sets of shortage and surplus terminals, respectively.

One may notice, however, that in order to balance the schedule, it may not be enough to

satisfy practical situations in which limitations exist on the number of overnight depots or

parking facilities. Let $Q(k)$ represent such a limitation at terminal k . Then a feasible schedule

must also satisfy the depot constraints:

$v_s(k) \leq Q(k), v_e(k) \leq Q(k)$ for all k

If the schedule is balanced, this reduces to $D(k) = Q(k)$, and the depot-constrained balance

schedule problem may be stated in DF form as follows:

Problem P2

Insert as many DH trips as necessary into the schedule of required trips in order to $\text{Min} \sum_{k \in K} D(k)^2 = \epsilon \sum_{k \in K} Q(k)^2$ (6.12)

Subject to: $d(\cdot, T) \leq K$ (6.13)

$D(k) \leq Q(k), k \in K$ (6.14)

Alternatively, using the identities of the proof of Theorem 6.5, the problem may be restated as $\text{Min} \sum_{k \in K} v_s(k)^2 = \epsilon \sum_{k \in K} Q(k)^2$ (6.15)

Subject to:

$v_s(k) - v_e(k) = 0, k \in K$ (6.16)

$0 \leq v_s(k) \leq Q(k), k \in K$ (6.17)

$0 \leq v_e(k) \leq Q(k), k \in K$ (6.18)

In this case, either (6.17) or (6.18) is redundant, but is still included in order to facilitate the

development of the next section.

6.6.2 Mathematical programming formulation

In order to extend Problem P1 (Section 6.4.2) to include depot and vehicle-balancing

constraints, a number of dummy overnight trips may be defined that travel backward in

time. For each terminal k , let $n+k$ represent an undisclosed number of trips of the form

$(k, n+k, n+k, k)$, $n \in \{1, 2, \dots, K\}$ such that $T(n+k) \leq T(k)$. The times $n+k$ and $n+k$ represent the

latest time at which a vehicle can arrive at

depot k after servicing its last required trip
and

the earliest time a vehicle can leave depot k
to service its first required trip, respectively.
Such

times may be stipulated in a driver-union c
ontract. The following additional variables are

now introduced:

$Y_{n+k,j} = 1$ if a vehicle departs for depot k (after
completing dummy trip $n + k$) to service j ,

its first required trip; otherwise, equals 0.

$Y_{i,n+k} = 1$ if a vehicle services its last required trip
as trip i before arriving at depot k to park

for the night (start dummy trip $n + k$); otherwise equals 0.

z_{kk} is an integer variable whose value represents the
number of unused spaces at depot k

($0 \leq z_{kk} \leq Q(k)$, $k \in K$).

Let the travel time of the morning and e
vening DHTrips from and to depot k be d
efined

as $\tau(n + k, p_j)$ and $\tau(q_i, n + k)$, respectively. The
joining $(n + k, j)$ will be considered admissible

if $t(n + k)$ for $j \in K$. e

$n + j \leq j_k k p_k + \epsilon \in \tau,)$, 0 otherwise, $(n + k, j)$
will be inadmissible, and $Y_{n+k,j} = 0$.

The joining $(i, n + k)$ is admissible if $t(i, n$
 t for $i \in K$. e $i \leq n + + \leq \epsilon \in \tau q_k k k)$, 0 otherwise,
 $(i, n + k)$

is inadmissible, and $Y_{i,n+k} = 0$. Note that $(n + k,$
 $j)$ and $(i, n + k)$ are always admissible if

$p_j = k$ and $q_i = k$, respectively. Let A represent the
set of admissible joinings of the preceding

type in addition to those between required t

of (6.22) and (6.23) represent the following:

$\sum_{j \in I_{n+}} y_{jk}$ is the number of vehicles that start their daily schedule from depot k

$\sum_{i \in I_{1,n+}} y_{ik}$ is the number of vehicles that end their daily schedule at depot k

Both terms are also equal to the number of occupied overnight parking spaces (used capacity)

at depot k .

Theorem 6.6

Problems P2 and P3 are equivalent. The solution to Problem P3 minimizes fleet size, subject

to depot-capacity and depot-balance constraints.

Proof: Make the following substitutions in Problem P3: y_{jk} vs K_{n+} , $j \in I_k$, $k \in \Sigma = \{1, \dots, n+1\}$, y_{ik} vs $K_{1,n+}$, $i \in I_k$, $k \in \Sigma = \{1, \dots, n+1\}$,

Any solution satisfying (6.22) and (6.23) satisfies the balance constraint, since subtracting

(6.23) from (6.22) for each k yields Equation 6.16:

$$v_s(k) - v_e(k) = 0, \quad k \in K$$

Constraints (6.22) and (6.23) are also equivalent to the depot constraints (6.17) and (6.18)

when the nonnegative slack variables z_{kk} are dropped. Summing up Equation 6.21 for all j

shows that the objective functions for both problems are equivalent, that is, $\max x_n \max y_n \min v_s(\cdot) = \sum_{j \in I_{n+}} y_{jk} - \sum_{i \in I_{1,n+}} y_{ik} = \sum_{k \in K} (\sum_{j \in I_{n+}} y_{jk} - \sum_{i \in I_{1,n+}} y_{ik}) = \sum_{k \in K} (v_s(k) - v_e(k)) = 0$

It should be noted that Problem P3 is a capacitated transportation problem, known in

the OR eld. As such, it has a bipartite graph representation similar to the example in

Figure 6.5, but without s, t . The set of the first-column (supply) nodes are in one-to-one cor

correspondence with the arrival epochs of trips $i = 1, 2, \dots, n$ and $n+1, n+2, \dots, n+k, \dots,$

$n+q$. Similarly, the set of second-column (demand) nodes are in one-to-one correspondence

with the departure epochs of each of the required and dummy trips. Since the supplies and

demands are integers, the solution will be integer upon relaxing the integer requirements on

the variables. Any of the standard transportation algorithms (see, e.g., Ahuja et al., 1993)

may be used to solve Problem P3. However, it is possible that Problem P3 is infeasible,

since not every possible arc is admissible from the supply nodes to the demand nodes.

Feasibility conditions are stated in the multiterminal, supply-demand theorem found in

Ford and Fulkerson (1962).

6.7 LITERATURE REVIEW AND FURTHER READING

Vehicle scheduling refers to the problem of determining the optimal allocation of vehicles to

carry out all the trips in a given transit timetable. A chain of trips is assigned to each vehicle

although some of them may be DHO or empty trips in order to reach optimality. The number

of feasible solutions to this problem is extremely high, especially in the case in which the

vehicles are based in multiple depots. Much of the focus of the literature on scheduling procedures is, therefore, on computational issues.

Dell Amico et al. (1993) developed several heuristic formulations, based on a shortest

path problem, that seek to minimize the number of required vehicles in a multiple-depot

scheduling. The algorithm presented is performed in stages, in each of which the duty of a

new vehicle is determined. In each such stage, a set of forbidden arcs is defined, and then a

feasible circuit through the network is sought that does not use any of the forbidden arcs.

Computational efficiency is obtained by searching for the shortest path across a subset of all

arcs in the network, rather than searching the entire network. Several modifications to the

basic algorithms are offered that save computer time by substituting parts of the full problem

with problems of a reduced size. These modifications include, for instance, solving the

reassignment of trips as a single-depot problem, an attempt to swap parts of duty segments,

and an internal reassignment of trips within each pair of vehicles associated with different

depots.

Löbel (1998, 1999) discussed the MDVSP and its relaxation into a linear programming

formulation that can be tackled using the b

ranch-and-cut method. A special multicommodity

flow formulation is presented, which, unlike most other such formulations, is not arc

oriented. A column-generation solution technique is developed, called Lagrangian pricing; it

is based on two different Lagrangian relaxations. Heuristics are used within the procedure

to determine the upper and lower bounds of the solution, but the final solution is proved to

be the real optimum.

Kuan and Rahin (1999) described a non object-oriented approach for bus scheduling,

based on the VAMPIRES algorithm for iterative improvement of the solution presented by

Smith and Wren in 1981. A key feature of VAMPIRES is the attempt to swap links at each

stage of the solution in order to improve the current schedule; Kuan and Rahin improved

this feature by refining the swapping criteria. In addition, the methodology presented

introduces a hierarchical classification of auxiliary activities: trip, layover, relocation,

invalid layover, invalid relocation, depot return, depot start, depot end. This classification

scheme for vehicle activities enables planners to improve the current solution more

efficiently.

Mesquita and Pao (1999) used a tree-search procedure, based on a multicommodity

network flow formulation, to obtain an exact solution for the multidepot vehicle-scheduling

problem. The methodology employs two different types of decision variables. The first type

describes connections between trips in order to obtain the vehicle blocks, and the other

relates to the assignment of trips to depots. The procedure includes creating a more com

pact, multicommodity network flow formulation that contains just one type of variable

and a smaller amount of constraints, which are then solved using a branch-and-bound

algorithm.

Banihashemi and Haghani (2000) and Haghani and Banihashemi (2002) focused on the

solvability of real-world, large-scale, MDVSPs. The case presented includes additional con

straints on route time in order to account for realistic operational restrictions such as fuel

consumption. The authors proposed a formulation of the problem and the constraints, as

well as an exact solution algorithm. In addition, they described several heuristic solution

procedures. Among the differences between the exact approach and the heuristics is the

replacement of each incorrect block of trips with a legal block in each iteration of the heu

ristics. Applications of the procedures in large cities are shown to require a reduction in the

number of variables and constraints. Techniques for reducing the size of the problem are

introduced, using such modifications as converting the problem into a series of single-depot

problems.

Freling et al. (2001b) discussed the case of single-depot with identical vehicles, concen

trating on q uasi-assignment formulations and a auction algorithms. A q uasi-assignment is a

reduced-size, linear problem in which some of the nodes and their corresponding arcs are

not considered. A n auction algorithm is a n iterative procedure in which n either the primal

or the dual costs are obliged to show a n improvement after each iteration. The authors

proposed four different algorithms and compared their performance: an existing auction

algorithm for the symmetric assignment problem; a n ew auction algorithm for the q uasi

assignment problem; a n alternative, two-phase, a symmetric assignment formulation (valid

in a special case), in which vehicle blocks are determined first and combined afterwards; a n d

a core-oriented approach for reducing the problem size.

Freling et al. (2001a) and Huisman et al. (2005) presented an integrated approach for

vehicle and crew scheduling for a single bus route. The two problems are first de

termined separately; the vehicle-scheduling problem is formulated as a network-flow problem, in which

each path represents a feasible vehicle schedule, and each node a trip. In the combined ver

sion, the network problem is incorporated into the same program with a set partitioning

formulation of the crew-scheduling problem (see Chapter 10).

Haase et al. (2001) formulated a rather problem that incorporated both crew and vehicle

scheduling. For vehicle scheduling, the case of a single depot with a homogeneous fleet is

considered. The crew-scheduling problem (see Chapter 10) is a set partitioning formulation

that includes side constraints for the bus itineraries; these constraints guarantee that a n

optimal vehicle assignment can be derived afterward.

Haghani et al. (2003) compared three vehicle-scheduling models: one multiple-depot (pre

sented by Banhashemi and Haghani, 2000) and two single-depot formulations, which are

special cases of the multiple-depot problem. The analysis showed that a single-depot vehicle

scheduling model performed better under certain conditions. A sensitivity analysis with

respect to some important parameters is also performed; the results indicated that the travel

speed in the DH trip was a very influential parameter.

Huisman et al. (2004) proposed a dynamic formulation of the multidepot vehicle-schedul

ing problem. The traditional, static vehicle-scheduling problem assumes that travel times are

a fixed input that enters the solution procedure only once; the dynamic formulation relaxes

this assumption by solving a sequence of optimization problems for shorter periods. The

dynamic approach enables an analysis based on other objectives except for the traditional

minimization of the number of vehicles, that is, by minimizing the number of trips start

ing late and minimizing the overall cost of delays. The authors showed that a solution that

required only a slight increase in the number of vehicles could also satisfy the minimum

late starts and minimum delay-cost objectives. To solve the dynamic problem, a “cluster

re-schedule” heuristic was used; it started with a static problem in which trips were assigned

to depots, and then it solved many dynamic single-depot problems. The optimization itself

was formulated through standard mathematical programming in a way that could use stan-

dard software.

Pepin et al. (2009) compare the performance of several different heuristic approaches

for solving the MDVSP. The approaches considered include a truncated branch-and-cut

method, a Lagrangian heuristic, a truncated column generation method, a large neighbor-

hood search (LNS) heuristic using truncated column generation for neighborhood evalu-

ation, and a tabu search heuristic. The first three methods are adaptations of existing

methods, while the last two are new in the context of this problem. The presented com-

parisons reveal that the column generation heuristic produces the highest quality solution

given sufficient computational time and stability. In contrast, the LNS method, which

utilizes truncated column generation for neighborhood evaluation, provides a faster solu-

tion without adversely compromising the solution quality. The comparisons and results

illustrate that using mathematical programming tools in conjunction with a metaheuristic

framework can offer a well-balanced solution quality and computational efficiency for the

MDVSP.

Steinzen et al. (2010) discuss the integrated vehicle- and crew-scheduling problem in pub

lic transit with multiple depots. The authors present a new modeling approach that is based

on a time-space network representation of the underlying vehicle and crew-scheduling

problem. The suggested approach is based on a Lagrangian heuristic in conjunction with

column generation, where the pricing problem is formulated as a resource-constrained

shortest-path problem. It has been illustrated that this proposed network representation

for the pricing problem produces favorable results compared to the other approaches avail

able in literature. The results presented in this study indicate that medium-sized instances,

with a round 640 trips and 4 depots, can be solved efficiently. The presented method out

performs other methods from the literature, in terms of the solution quality and compu

tational time.

Naumann et al. (2011) present a new stochastic programming approach for robust vehicle

scheduling in bus transit. The presented approach uses typical disruption scenarios during

the optimization to minimize the expected sum of planned costs and costs caused by dis

ruptions. The schedule is represented as a time-space network with a link connecting arcs to

enable independent penalization of every connection between two consecutive service trips.

The results indicate that stochastic programming for the vehicle-scheduling problem with

disruptions leads to solutions that are of higher quality in terms of total expected costs

compared to a simple approach with fixed buffer times. Notwithstanding the consideration

of complete delay-propagation, the presented stochastic programming approach finds the

optimal solution for the given data, which outperforms the simple approach. The authors

have also created a set of Pareto-optimal solutions in terms of maximum robustness and

minimum planned costs. Considering the entire delay propagation, the stochastic program

ming solutions still provide solutions that are of higher quality in terms of total expected

costs compared to the simple approach adding fixed buffer times. It is therefore concluded

that despite the increased computational complexity of the approach presented, small and

medium-sized real-world instances can be solved.

Lin and Kwan (2012) propose a two-phase approach for the train unit scheduling prob-

lem. The first phase assigns and sequences train trips to train units temporarily ignoring

some infrastructure details at train stations. The solution of the first phase is near-operable.

The second phase focuses on satisfying the remaining constraints with respect to detailed

station layouts. Real-world scenarios such as compatibility among reaction types and time

allowances for coupling/decoupling activities are also considered, so that the solutions

would be fully operable. The first phase is modeled as a large-scale integer multicom

modity network flow (MCNF) problem. A branch-and-price integer linear programming

approach is proposed. The authors state that the preliminary experiments have shown

that the approach would not yield solutions in reasonable time for problem instances

exceeding about 1,000 train trips. It is stated in this study that the train company col

laborating in this research operates over 2,400 train trips on a typical weekday with 10

different types of train units in use. Hence, a two-phase approach is proposed where the

problem is first addressed from a network wide perspective, subsequently focusing on ner

details at the individual train station level. A heuristic has been designed by the authors

for hybridization of the first phase solver by compacting the problem instance to a much

smaller size, before the integer linear programming (ILP) solver is applied. The process is

iterative with evolving compaction based on the results from the previous iteration thereby

converging to near optimal results. The second phase is modeled as a multidimensional

matching problem with a mixed integer linear programming (MILP) formulation. Finally,

the authors acknowledge that further testing and refinement of the models with feedback

from the railway company are ongoing.

Borndörfer et al. (2012) provide a generic formulation for rolling stock planning problems

in the context of intercity passenger traffic. The main contributions are a graph theoretic

cal model and a mixed-integer-programming formulation that integrate all main require

ments of the considered vehicle-rotation-planning problem (VRPP). The authors show that

it is possible to solve this model for real-world instances using modern algorithms and

computers. The results from this study show that the expected behavior of algorithms that try to

solve the VRPP. The authors conclude that the train composition and regularity aspects are

not too hard to tackle while the maintenance constraints and moreover capacity constraints

increase the complexity of the VRPP significantly. The computational study presented in

this paper demonstrates that the solution approach developed can be used to produce high

quality solutions for large-scale vehicle-rotation-planning problems.

Main features of the works reviewed are illustrated in Table 6.3.

EXERCISES

6.1 Given a bus system consisting of only two cyclical routes (performing round trips) departing from and arriving at the same terminal, the mean trip time for both routes is given as 3 h. The headways for each route are determined based on passenger demand.

These headways are given by hour of day in the following table: Route 1 Route 2

Hour of day	Headway (min)	Hour of day	Headway (min)
6 a.m.-9 a.m.	7.5	6 a.m.-8 a.m.	6
9 a.m.-12 noon	12	8 a.m.-12 noon	7.5
12 noon-2 p.m.	10	12 noon-3 p.m.	10
2 p.m.-4 p.m.	15	3 p.m.-6 p.m.	7.5
4 p.m.-6 p.m.	6	6 p.m.-11 p.m.	10
6 p.m.-11 p.m.	10		

6 p.m.-11 p.m. 10 What is the minimum fleet size required for this bus system? Do not use DF. Comment on the generalization of your approach. Note: Buses may be switched from one route to another if necessary.

6.2 Given the following eight-trip, four-terminal timetable and D H matrix, find the minimum number of vehicles required to carry out the timetable, including their chains (blocks), by using the Max-flow augmenting-path algorithm.

Table 6.3 Characteristics of work covered in Section 6.7 concerning literature review and further reading

Source	Depots	Exact or heuristic	Special features
Dell Amico et al. (1993)	Multiple	Heuristic	
Löbel (1998, 1999)	Multiple	Both	
Kuan and Rahin (1999)	Single	Heuristic	Object-oriented program
Mesquita and Paixao (1999)	Multiple	Exact	
Banihashemi and Haghani (2000)			
Haghani and Banihashemi (2002)	Multiple	Both	
Freling et al. (2001b)	Single	Heuristic	
Freling et al. (2001a), Huisman et al. (2005)	Only one route	Exact	Vehicle and crew scheduling

Haase et al. (2001) Single Exact Vehicle and crew scheduling

Haghani et al. (2003) Multiple and single Heuristic
Comparison of three formulations

Huisman et al. (2004) Multiple Heuristic Dynamic scheduling

Pepin et al. (2009) Multiple and single Heuristic
Comparison of five different heuristic approaches

Steinzen et al. (2010) Multiple Heuristic

Naumann et al. (2011) Multiple Exact Stochastic programming

Lin and Kwan (2012) Multiple Heuristic

Borndörfer et al. (2012) Multiple Heuristic Trip number (i)
Departure terminal Departure time Arrival terminal Arrival
time 1 a 7:00 b 7:25 2 d 7:00 a 7:35 3 b 7:15 c 8:15 4 c
7:30 d 8:35 5 a 7:35 d 8:00 6 d 8:00 a 8:30 7 a 8:35 d 9:05
8 c 9:05 d 10:00 Average DH travel time (minutes) matrix
Arrival terminal Departure terminal a b c d a 0 15 30 25 b
20 0 40 30 c 25 35 0 45 d 20 25 35 0

6.3 The following problem uses the DF solution approach. A
schedule S is given between 5:00 (T_1) and 9:30 (T_2),
with two terminals (a, b) and one depot. An eight-trip
schedule is shown in the following table. Note that trips 6
and 7 are conducted at the same time and in the same direction
because of specific passenger demand. The DH average travel
time for all trips between a and b (both directions of travel)
is 30 min, and there are no DH trips between either a or b and
the depot (both directions). The DH trips are to be assigned
to the latest time possible.

(i) Construct the DFs for terminals a and b only; find the
minimum number of vehicles required at terminals a and b using
DH trip-insertion procedures between the two terminals.

(ii) Construct the sum function $g(t)$ for the entire
schedule S ; is $G(S)$ the minimum fleet size required for S ?
Explain.

(iii) Find the minimum fleet size required for S and construct
vehicle schedules (chains, blocks) using the FIFO rule. Trip
number Departure terminal Departure time Arrival terminal
Arrival time 1 Depot 5:00 b

6:00 2 Depot 5:30 a 6:30 3 a 6:00 b 6:30 4 Depot 6:40 a
 8:00 5 a 7:00 Depot 9:00 6 a 7:30 Depot 9:30 7 a 7:30 Depot
 9:30 8 b 8:30 a 9:30 Note: Further exercises involving DFs,
 shifting departure times, and DH-insertion procedures
 appear in the Exercises sections of Chapters 7, 8, and 16.

APPENDIX 6.A: MAXIMUM-FLOW (MAX-FLOW) PROBLEM

The following description is based on Ceder (1978). Further
 reading can be found in the book

by Ahuja et al. (1993). The general augmenting-path
 algorithm described can solve the vehicle

scheduling problem. This appendix supplements material for
 Section 6.4, and Chapter 13.

6.A.1 Definitions

In the Max-flow problem, one considers a directed graph
 (network) $G = \{N, A\}$, with a single

node s as source and another node t as a sink. Each arc $(i,$
 $j)$ has a capacity $c(i, j)$ = the maxi

mum flow that can be traversed from i to j . This can be
 associated with a flow of people,

vehicles, trains, bits of information, dollars, water, etc.
 An s - t flow is the amount of the flow

that leaves s and arrives at t , provided that:

(a) $0 \leq f(i, j) \leq c(i, j) \forall (i, j) \in A$,

(b) And to maintain flow conservation, $\sum_{j \in N} f(i, j) - \sum_{j \in N} f(j, i) = 0$ for all $i \in N$, $i \neq s, t$, where $f(i, j)$ is the arc (i, j) 's flow.

Lemma 6.A.1

Let f be an s - t flow on $G = \{N, A\}$. Then: $f(s, N) - f(N, s)$
 $= f(N, t) - f(t, N)$, where $f(U, V)$ is

the flow between two subsets, U and V , such that $(U, V) \in A$
 $f(U, V) = \sum_{(i, j) \in A} f(i, j)$

$\sum_{(i, j) \in A} f(i, j)$

Proof: $f(N, N) = f(s, N) - f(N, s) = f(N, t) - f(t, N)$
 $= \sum_{i \in N, i \neq s, t} (f(i, N) - f(N, i)) = \sum_{i \in N, i \neq s, t} (f(i, N) - f(N, i))$
 $= \sum_{i \in N, i \neq s, t} (f(i, N) - f(N, i)) = \sum_{i \in N, i \neq s, t} (f(i, N) - f(N, i))$

Since the flow is conserved at each node, $f(i, N) = f(N, i)$
 $\forall i \in N, i \neq s, t$. Therefore, we

obtain $V(f) = f(s, N) - f(N, s) = f(N, t) - f(t, N)$, where $V(f)$ is called the value of the flow.
 4, 3 2 5 4 1 3 2, 1 3, 3 2, 2 1, 1 1, 0 3, 1 s
 t

Example:

The first number on each arc (see network on the right) represents its capacity and the

second the flow. In this example, the s-t flow = 3 and can be measured at node 1 or 5.

s-t cut: If in a directed graph $G = (N, A)$ the set of nodes N is partitioned into two sets,

X and X^c (i.e., $X \cup X^c = N$ and $X \cap X^c = \emptyset$), such that $s \in X$ and $t \in X^c$, then $(X, X^c) \in \mathcal{C}$ is the set of

all arcs in A connecting nodes in X to nodes in X^c is called the s-t cut. The capacity of the

cut is denoted as $c(X, X^c)$.

Example: In the preceding example network: $(X, X^c) = \{(1, 4), (3, 4), (3, 5)\}$, $c(X, X^c) = 5 = 3 + 2 = \{1, 2, 3, 4, 5\}$
 $(X, X^c) = \{(1, 2), (3, 4), (3, 5)\}$, $c(X, X^c) = 0 = 0$

Lemma 6.1.2 $V(f) = c(X, X^c) - f(X, X^c)$

Proof: $V(f) = f(s, N) - f(N, s) = f(s, X) - f(X, s) + f(X, X^c) - f(X^c, X) + f(X^c, s) - f(s, X^c)$
 $= f(s, X) - f(X, s) + f(X, X^c) - f(X^c, X) + f(X^c, s) - f(s, X^c)$
 $= f(s, X) - f(X, s) + f(X, X^c) - f(X^c, X) + f(X^c, s) - f(s, X^c)$
 $= f(s, X) - f(X, s) + f(X, X^c) - f(X^c, X) + f(X^c, s) - f(s, X^c)$

Theorem 6.1.1

For any s-t flow f and any s-t cut (X, X^c) :
 $V(f) \leq c(X, X^c)$

Proof: For every f and (X, X^c) : $(X, X^c) \in \mathcal{C}$ and $(X, X^c) \in \mathcal{C}$
 $\forall f \in \mathcal{F}$

Therefore, by Lemma 6.A.2 $\forall (f) (X, X) (X, X) c$
 $(X, X) \cdot f = - \leq f$

Theorem 6.A.1 leads to the fundamental result in
network-flow theory: a flow exists that

is equal to the minimum capacity cut; hence, it is the
maximum flow. The latter claim will be

shown following the description of the maximum-flow
(Max-flow) algorithm in the next section.

6.A.2 Augmenting-path algorithm

For a given $G = \{N, A\}$ with $c(i, j) \forall (i, j) \in A$, and some
s-t flow f (it is possible to start with

$f(i, j) = 0 \forall (i, j) \in A$):

Step 1: Construct a residual network $G(f) = \{N, A(f)\}$ in
which each arc is labeled with num

ber $\alpha(i, j)$ as follows:

a For $f(i, j) < c(i, j)$, then $(i, j) \in A(f)$ and $\alpha(i, j) =$
 $c(i, j) - f(i, j)$; (i, j) is called a forward arc.

b For $f(i, j) > 0$, then $(j, i) \in A(f)$ and
 $\alpha(j, i) = f(i, j)$; (j, i) is called a r
everse arc.

Step 2: If there is no path from s to t in $G(f)$, then
terminate; f is the maximum s-t flow

(see Theorem 6.A.3).

Step 3: Let P be the shortest path (least number of arcs),
and let $\alpha = \min_{(i, j) \in P} \alpha(i, j)$. Define a

new flow f' on G (see Theorem 6.A.2):

a If (i, j) is a forward arc on P , let $f'(i, j) = f(i, j)$
 $+ \alpha$

b If (i, j) is a reverse arc on P , let $f'(i, j) = f(i, j)$
 $- \alpha$

c If (i, j) does not belong to P , let $f'(i, j) = f(i, j)$

Step 4: Let $f := f'$; go to Step 1.

6.A.3 Labeling procedure to find minimum-arc paths

In Step 3 of the augmentation algorithm, we seek to find the shortest path from s to

t . (An arbitrary path can also be found in Step 3, but usually with more computations than the

shortest path. The latter has a finite number, and normally fewer computations; hence,

it is a n improved version). The labeling procedure is simple: at each stage, every node

$i \in N$ is in one of three states: (i) unlabeled (indicated by a blank), (ii) labeled but not

scanned (indicated by a single label, $l(i)$), and (iii) labeled and scanned (indicated by $l(i)$

and marked by $\checkmark$).

The procedure:

Step L 1 : Label the node s as $l(s) = s$.

Step L 2 : If node t is labeled, stop; $[l(t), t]$ is the last arc on the shortest path from s to t . $[l(l(t)), l(t)]$ is the next-to-last arc, and so on until reaching node s .

Step L 3 : If all labeled nodes are scanned, stop; no path exists from s to t .

Step L 4 : If all labeled, but non-scanned nodes i , use the "first labeled, first scanned" policy. Label each unlabeled node j , provided $(j, i) \in A$, with $l(j) = i$. Mark i as a scanned node with a $\checkmark$.

Step L 5 : Go to Step L 2.

6.A.4 Example of the augmenting-path algorithm

A network $G = \{N, A\}$ is given in which the first number on each arc represents the capacity,

and the second the flow. Residual network, $G(f)$ $P = \{(1, 2), (2, 3), (3, 5)\}$; $\alpha = 3$ Initial flow

First iteration

```

1 2 3 4 5 6, 0 4, 0 3, 0 2, 0 2, 0 2, 0 3, 0 7, 0 1 2 3 4 5
6 4 3 2 2 2 3 7

```

In order to demonstrate the labeling procedure, we will show how one can find P in the

first iteration (it is used similarly in all other iterations). Iteration Node First Second Third Fourth $s = 1$
 $1 = l(1)$ 1 ✓ 1 ✓ 1 ✓ 2 $1 = l(2)$ 1 ✓ 1 ✓ 3 $2 = l(3)$ 2 ✓ 4 $2 = l(4)$ 2 $t = 5$ $3 = l(5)$

The shortest path is then reconstructed backward: $5 \rightarrow 3 \rightarrow 2 \rightarrow 1$ (see P earlier). Residual network, $G(f)$ $P = \{(1, 2), (2, 4), (4, 3), (3, 5)\}; \alpha = 2$ New flow

```

1 2 3 4 5
6, 3 4, 0 3, 0 2, 0 2, 0 2, 0 3, 3 7, 3 1 2 3 4 3 4 3 2 2 2
3 4 3 5 3

```

2nd iteration 4, 2 Because there is no more augmented s - t path

3rd iteration 1 2 3 4 5 New (final-max) flow=5 1 2 3
Residual network, $G(f)$ 6, 5 3, 0 2, 0 2, 0 2, 2 3, 3 7, 5 5
2 3 2 2 3 5 1 5 2 2 4 2

The final and optimum result is Max-flow = 5. The bottleneck (minimum cut; see

Theorems 6.A.3 and 6.A.4) consists of this Max-flow and is emphasized by a heavy gray

line in the last residual network.

Note: When an augmenting path that includes a reverse arc (j^*, i^*) is detected, we can use

it for increasing the s - t flow. However, according to Step 3(b) of the algorithm, we have to

subtract the value of α from the previous flow on (i^*, j^*) in constructing the new flow.

6.A.5 Additional theorems

The following theorems assure that the augmenting-path algorithm indeed solves the

Max-flow problem.

Theorem 6.A.2

(a) The function f' (see Step 3) is an s-t (i.e., $0 \leq f'(i, j) \leq c(i, j) \forall (i, j) \in A$ and $f'(i, N) = f'(N, i) \forall i \in N, i \neq s, t$)

(b) $V(f') > V(f)$

Proof: (a) The values of $\alpha(i, j)$ and $\alpha(j, i)$ are the maximum amount of flow with which we

can increase or decrease $f(i, j)$, $(i, j) \in A$, respectively. The function f' is different from f only

for $(i, j) \in P =$ augmenting path. Since $\alpha = \min_{(i, j) \in P} \alpha(i, j) = \alpha(q, k)$, then $f'(q, k) \leq c(q, k)$ and

the capacity constraints hold for all other arcs in the augmenting path. Hence, $0 \leq f'(i, j) \leq$

$c(i, j)$. The conservation-flow required retains because for every node i along P (except s, t),

the number of arcs entering i = number of arcs leaving i ;
for (i, j) as a forward arc $f''(i, j) =$

$f(i, j) + \alpha$; for (i, j) as a reverse arc $f''(i, j) = f(i, j) - \alpha$.

(b) Similar to the arguments in (a), it is easy to show that:

$f''(s, N) - f''(N, s) = f(s, N) - f(N, s) + \alpha$
and since $\alpha > 0 \rightarrow V(f') > V(f)$

Theorem 6.A.2 demonstrates the feasibility criterion in part (a) and the improvement crite

riterion in part (b). The optimality criterion is shown in Theorem 6.A.3.

Theorem 6.A.3

If the augmenting-path algorithm is terminated with a flow f^* (see Step 2), then

s.t. $f(i,j) \leq c(i,j)$ for all $(i,j) \in A$, $f(i,j) \geq 0$ for all $(i,j) \in A$.

This LP has $|N|$ conservation constraints (for all $i \in N$), $|A|$ capacity constraints (one for each

arc in G), and $|A|$ nonnegativity constraints.

The LP formulation enables us to solve the Max-flow problem by the simplex method

(or one of its variants—see OR literature). However, we note that the coefficient matrix of

the conservation constraints has a very special structure (this matrix is called the incidence

matrix of G'). The rows and columns in this matrix correspond to the nodes and arcs in G' ,

respectively. Every column of this matrix contains exactly two nonzero elements: 1 and -1.

Because of this special structure of the matrix, it is possible to reduce significantly the com

putational effort required by the simplex method.

6.A.7 Useful extensions

Undirected arcs

An undirected arc (i, j) with capacity $c(i, j)$ can be replaced by a pair of directed arcs (i, j)

and (j, i) , each with a capacity of $c(i, j)$.

Unlimited capacities

Some arcs may have unlimited capacities, and the augmenting-path algorithm might end

with a finite flow. Therefore, in this case we calculate primarily max s-t paths $\{ \min \alpha \}$.

Nodes with capacities

It is possible to establish an upper bound on the amount of flow traversing given nodes.

In this case, the node i with capacity $c(i)$ is replaced by two new nodes i' and i'' with a

arc having the capacity $c(i) = c(i', i'')$ in between. i is replaced by i' and i'' with a

Lower bounds on the flow

For cases with a lower bound on the flow (e.g., assuring minimum flow to justify the permit

sion to use the arc), the flow constraints are changed to $l(i, j) \leq f(i, j) \leq c(i, j)$, where $l(i, j)$ is the

lower bound on the flow on arc (i, j) . This problem can be solved by converting the original

network into an artificial network (satisfying both upper and lower bounds) called a trans

shipment scheme. Its solution appears in OR literature, such as in Ahuja et al. (1993).

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7 Chapter 7: Vehicle Scheduling II: Variable Schedules

This chapter extends the deficit-function (DF) approach to include possible modifications in

the creation and editing of trip timetables and vehicle schedules (blocks). In addition to dead

heading (DH) trip insertions, schedulers may consider a variable, instead of a fixed schedule,

that is, shifting departure times (SDTs) based on some acceptable tolerances in the process of

minimizing fleet size. Admittedly, this shifting effort is a very time-consuming task and usually

not done in a systematic manner. This chapter will show how to use the graphical person-

machine interactive approach to assist vehicle schedulers in selecting the most efficient shifts (in

trip-departure times), with and without the inclusion of DH trip insertions. To that end, we can

say that this will not be hard work or involve complex software that will wear out the scheduler.

It is the knowledge that a small change can do the job. The chapter contains six main parts, following an introductory section. Section 7.2 presents

DF procedures for fixed schedules in order to find rapidly the lower bound on the fleet size

without creating vehicle schedules. This lower-bound calculation can help decision makers in

evaluating the number of vehicles required for any given change. Section 7.3 introduces the

concept of maximum advance (early) and delay (late) from the scheduled departure time,

given shifting tolerances (early, late) for each trip. Left (early) and right (late) shift limits are

established, including the possibility of simultaneous left

and right shifts in opposite directions.

Section 7.4 provides an analysis and method of deriving the fleet size lower bound when SDTs

are allowed within their tolerances. Section 7.5 exhibits procedures in which a single or chain

(multiple) of shifting is required to reduce the fleet size, with and without its integration with

DH trip insertions. Section 7.6 illustrates the process of implementation through an example in

which variable schedules are used to construct vehicle blocks in a large bus agency. Section 7.7

looks into the effect of shifting scheduled departure times on even-load timetable. A criterion

for maximum allowable change in the even load is introduced. The chapter ends with exercises. Although this chapter contains math notations and proofs of derived conclusions, all sections

are appropriate for practitioners. We especially recommend looking at the examples, illustrated

in figures, to capture the interactive person-machine process. We end this corner, as we do

almost customarily, with a story. A scheduler who couldn't insert DH trips because of only a

few minutes difference and who made an effort not to increase the number of buses went to

his supervisor. "Guess what solution I found (shifting)?" he asked. The supervisor told him that

he reminded him that a new father once told his friend: "I have a new born baby, guess what?"

"A boy," replied the friend. "Guess again" said the father. "A girl." The new father exclaimed:

"Who told you?"

7.1 INTRODUCTION

Chapter 6 dealt with fixed schedules, in which departure times could not be changed. In prac

tical transit scheduling, however, schedulers should attempt to allocate vehicles in the most

efficient manner possible, including the employment of small shifts in departure times. For

example, in Egged, Israel's national bus carrier, the schedulers consider a variable instead of a

fixed schedule in addition to DH trip insertions. That is, they multiply shifting trip-departure times

(SDTs), based on some acceptable tolerances. Although confident of their work efficiency, the

Egged schedulers agree that variable scheduling with DH trip insertions is a very complex

process, not systematic, and a very time-consuming task; it can turn a good schedule into a

soured one. This situation is analogue to the following truth. If you put a spoonful of wine

(an efficient chain of trips) into a barrel of sewage (problematic schedule), you get sewage.

If you put a spoonful of sewage (a chain of trips impossible to execute) in a barrel full of wine

(implemental schedules), you still get sewage (problematic schedule).

This chapter presents an extension of the DF concept in an effort to provide an interactive

computer technique to assist the scheduler in the planning task. Usually, only the timetable

construction and vehicle-scheduling (generation of chains or blocks) portions of the transit

planning process are demonstrable. Given a multiterminal trip schedule, the object is to

minimize the required fleet size, subject to a number of constraints, including trip-departure

time tolerances and insertable DH trips. Once the fleet size is determined, a number of alternative

native block constructions may be generated.

Another important feature, and one that will open this chapter, is the derivation of a fleet

size lower bound for both fixed and variable trip schedules. The DF lower-bound approach

introduced can be used across almost all operational planning components. The importance

of knowing how many vehicles are actually required applies to many transit-design and

planning problems. Five such problems are as follows: (1) vehicle scheduling with different

vehicle types (see Chapter 8), (2) crew scheduling (see Chapter 9), (3) design of operational

transit-parking spaces (see Chapter 12), (4) design of a new transit network or redesign of

an existing one (see Chapter 15), and (5) design of efficient short-turn trips (see Chapter 16).

Finally, Appendix 7.A, of this chapter, contains an example of vehicle-scheduling software

including its key features and expected savings. The software is one of those presented

in Table 1.1.

7.2 FLEET SIZE LOWER BOUND FOR FIXED SCHEDULES

7.2.1 Overview and example

The initial lower bound on the fleet size with DH trip insertions was found by Ceder and

Stern (1981) to be the sum of all DFs, G

(S), as shown in Figures 6.6 and 6.7. An improved

lower bound was established later by Stern and Ceder (1983), based on extending each

trip's arrival time to the time of the first feasible departure of a trip to which it may be

linked or to the end of the night time horizon. Further, lower-bound improvements for

fixed and variable schedules appear in Ceder (2002, 2005). The direct calculation of the

fleet size lower bound enables schedulers and transit decision makers to ascertain more

promptly how much the fleet size can be reduced by DH trip insertions and allowing shifts

in departure times.

Figure 7.1 presents a line-trip example with four terminals (a, b, c, and d). Table 7.1 shows

the data required for the simple example used in this section and Sections 7.3 and 7.4 for

demonstrating the lower-bound methods. Four DFs are constructed along with the overall

DF. According to the next terminal (NT) procedure (see Chapter 6), terminal d (whose first

allow is the longest) is selected for a possible reduction in D(d). The DH insertion process

depicted in Figure 6.9a and b continues using the criterion $R = 2$. The first unit reduction

DH chain (URDHC) is $DH_1 + DH_2$, and the second DH_3 . The result is that $D(c)$ and $D(d)$

are reduced from 1 to 0 and from 2 to 1, respectively; hence, $N = D(S) = 5$ and G is 1

increased

from 3 to 4 using three inserted DH trips. The five FIFO-based blocks are as follows: [1-5],

[2-DH 1-7], [3-DH 2-9], [4-DH 3-6], and [8].

7.2.2 Stronger fleet size lower bound

While Stern and Ceder (1983) extended each unlinked trip's departure time (i.e., one that

cannot be linked to any trip's arrival time) to both T_1 and T_2 , Ceder (2002, 2005) proved

that a new extension only to T_2 was sufficient. The extension to the time of the first feasible

departure time of a trip with which it may be linked, or to T_2 , results in an S' schedule and

an overall DF, $g'(t, S')$, with its maximum value $G'(S')$.

While S' is being created, it is possible that several trip-arrival points are extended forward

to the same departure point that is their first feasible connection. However, in the

solution of the minimum fleet size problem, only one of these extensions will be linked to

the single departure point. This observation provides an opportunity to look into further block construction.
2 6 5 1 3 4 7 c d b 9 c d 8 Fixed schedule $d(a, t) d(b, t) D(a) = 3 d(d, t) 4 3 2 1 0 D(b) = 1 3 2 1 0 d(c, t) 3 2 1 0 -1 3 2 1 0 -1 g(t) 4 3 2 1 0 6:00 6:20 6:40 7:00 7:20 7:40 8:00 8:20 8:40 9:00 6:00 6:20 6:40 7:00 7:20 7:40 8:00 8:20 8:40 9:00$ Time Time DH 2 DH 3 DH 1 $G = 3 4 D(b) = 2 1 D(c) = 1 0$

Figure 7.1 Nine-trip example with DH trip insertions for reducing fleet size.

artificial extensions of certain trip-arrival points without violating the generalization of

requiring all possible combinations for maintaining the fleet size at its lower bound.

Figure 7.2 illustrates three cases of multiple extensions to the same departure point. Case (i)

shows two extensions, Trips 1 and 2, both with the same arrival point b, which is their rest

feasible connection at point a of Trip 3. Because only one of the two trips will be connected

to Trip 3, the question is, "Which one can be extended further?" It is clear that Trip 1 has

better DH chances to be connected to Trip 4 than to Trip 2 because of its longer DH time.

Hence, Trip 1 can be further extended (second extension) to the start of Trip 4 if it is feasible.

Case (ii), Figure 7.2, shows that Trips 1 and 2 do not end at the same point and that Trip 4

has different points than in Case (i). The argument of Case (i) cannot hold here, since the DH

time differs between each two different points. In this case, the second feasible connection

for Trip 1 is T2. By using the Case (i) argument, one can then create three possible chains, [1],

[2-3], and [4], instead of two chains, [1-3] and [2-4]. Case (iii) shows an opposite situation

to that of Case (ii), with multiple extensions from different arrival points. If we link, in Case

(iii), Trip 1 (longest DH time to the common departure point) and Trip 3 and extend Trip 2

to Trip 4, we have another multiple extension case like Case (i), this one concerning the start

of Trip 4 (linked to Trips 2 and 3). Following the Case (ii) argument, Trip 3 will be linked to

Trip 4, and Trip 2 will have its third extension. This results in three possible chains, [1-3-4],

[2], and [5], instead of two, [1-5] and [2-3-4]. Cases (ii) and (iii) show why it is impossible

to apply any general rule to a multiple extension of different arrival epochs. Consequently,

further improvement of $G'(S')$ can be made only for Case (i) situations.

The following is the procedure for finding a stronger fleet size lower bound:

1. Establish S' .

2. Select a case in which more than one extension is linked to the same departure time t_{sjk} of trip j at terminal k . If none more such cases-stop. Otherwise, select a group (two or more) of extensions with the same scheduled arrival terminal, u , and apply the following steps:

a. Find a trip t that fulfills $\min_{i \in E_{sj}} t_i \forall e - (t)_{uk} u$, where E_u = set of all trips arriving at u and extended to t_{sjk} and t_i is the arrival epoch of trip i at terminal u .

b. Perform the second feasible extension for all trips $i \in E_u$, except the one selected in Step 2a. Go to Step 2. Table 7.1 Input data for the problem illustrated in Figure 7.1

Trip no.	Departure terminal	Departure time	Arrival terminal	Arrival time
1	a	6:00	c	6:30
2	a	6:20	b	6:50
3	b	6:40	a	7:10
4	a	7:00	a	7:20
5	c	7:10	a	7:30
6	c	7:40	a	8:10
7	d	7:50	d	8:10

Deadheading (DH) trips Between terminals DH time (same for both directions) (min)

and c 30 8 d 8:00 c 8:30 9 b 8:30 d 9:00 b and d 30 c and d 20

Using this procedure, define the overall DF of the extended S' schedule by $g''(t, S'')$ with

the maximum value $G''(S'')$. The following theorem and its proof establish that $G''(S'')$ is a

stronger lower bound than $G'(S')$.

Theorem 7.1

Let $N_o(S)$ be the minimum fleet size for S with DH insertions. Let $G'(S')$ and $G''(S'')$ be the

maximum value of the overall DF for S' and S'' , respectively. Then,

(i) $G''(S'') \geq G'(S')$.

(ii) $G''(S'') \leq N_o(S)$.
 a c a a b b c c 1 3 2 b c c a a a 2
 3 1 4 4 cb To this epoch, Trip 1 has better DH connection
 chances than Trip 2 T 2 1st 2nd Feasible connection 3rd b
 b a a 2 3 4 cb ca 1 ac 5 (b) (c) (a) Time
 8:158:007:457:307:157:006:456:306:156:00
 8:158:007:457:307:157:006:456:306:156:00 Time

Figure 7.2 Part (a) shows why one should select the Trip 2 extension, Part (b) shows that the argument in (a) cannot be used in case of multiple connections from different terminals, and Part (c) shows another case in which multiple connections cannot be applied for constructing the lower bound.

Proof

(i) The new overall DF, $g''(t, S'')$, has more extensions than $g'(t, S')$; that is, $g''(t, S'') \geq g'(t, S')$. Therefore, $G''(S'') \geq G'(S')$.

(ii) According to the definition of S'' , at any time t in which $g''(t, S'') = G''(S'')$, there exist $G''(S'') - g'(t, S')$ trip extensions over S' . The additional extensions in S'' represent multiple extensions (second, third, etc.), given that each extended trip is associated with another trip having the same arrival epoch and terminal and has only one extension. In the optimal chain solution, a departure time t_{s*} may or may not be linked to its nearest feasible arrival epoch (t_{e*}) across all

other points representing the same arrival terminal. Linkage to $t e *$ complies with the procedure to construct S'' . Otherwise, $t e *$ in S'' is further extended either to another trip or to T_2 , while $t s *$ is linked to $t t e ** e * <$. We should note that $t e **$ is linked to $t s **$ when using the procedure described. Because $t e *$ to $t s *$ is the shortest link, the additional extension of $t e *$ cannot be linked to a trip that starts before $t s **$ (otherwise, $t e **$ too will be linked to it and not to $t s *$). Therefore, the additional extension of $t e *$ in the optimal chain solution, $N o(S)$, results in a greater overlap between trips (when constructing $g''(t, S'')$). Hence, $G''(S'') \leq N o(S)$.

Figure 7.3 presents the schedule of Figure 7.1, with S' in its upper part, S'' in its middle

part, and three overall DFs— $g(t, S)$, $g'(t, S')$, and $g''(t, S'')$ —in the lower part. For S' , it

may be observed that Trips 3, 4, and 5 are extended to the same departure point as Trip 6

from the same arrival terminal a . According to the procedure for constructing S'' , the extension

of Trip 5 is selected, and Trips 3 and 4 are further extended to the departure time of

Trip 9. These additional extensions create another multiple connection associated with Trips

3 and 4, in which Trip 4 is the selected extension and Trip 3 is further (third time) extended.

The initial lower bound is $G = 3$, the first improved lower bound is $G' = 4$, and the proposed

improved lower bound is $G'' = 5$, which happens to be the optimal solution (see Figure 7.1).

7.3 VARIABLE TRIP DEPARTURE TIMES

A small amount of shifting in scheduled departure times becomes almost common in practice

when attempting to minimize fleet size or the number of vehicles required. This section presents

ents methods, mostly according to the D F, to realize a variable trip schedule in an efficient

manner. It may be recalled that Chapters 3 and 4 provided methods to improve the corre-

spondence of vehicle departure times with passenger demand. The construction of timetables

in Chapters 4 and 5 was based on either even headways or even average loads, entailing

situations in which even headways result in uneven passenger loads. SDTs, therefore, may

unbalance these desirable features in the timetable while favoring resource (vehicle) savings.

Nonetheless, the last section of this chapter introduces a method to eliminate, at least to some

extent, the possibility of too drastic changes in the even-load timetable requirement. Similar to

the subject order of Chapter 6, we will start this section with reference to a single transit route

and then continue with a minimum fleet size analysis for a network of routes.

7.3.1 Single-route minimum fleet size

Section 6.2 describes a method for ascertaining the minimum fleet size required for a given

single route without interlining. We will now extend this method to account for possible

shifting in departure times for given backward and forward shifting tolerances (in minutes)

for each trip. This additional flexibility, which is employed in practice, can reduce the fleet

size required as the primary objective. A secondary objective is to minimize the length of the

shifting within their given tolerances.

In practice, departure times are shifted without any systematic method. Shifting tolerances

are usually determined by rule of thumb although it makes sense to correlate them

with the headways between departures. A proposed method appears in Table 7.2, in which

the length of the shifting tolerance is headway dependent. That is to say, the longer the

headway, the shorter is the tolerance. If the shifting is backward, the preceding headway is

considered as H ; if it is forward, the next headway is considered. $a_4 g(t, S) = 4 \ 3 \ 2 \ 1 \ 0 \ g'(t, S') = 4 \ 3 \ 2 \ 1 \ 0 \ b \ c \ c \ a \ d \ a \ a \ a \ b \ a \ d \ 2 \ 6 \ 5 \ 1 \ 3 \ 7 \ c \ d \ b \ 9 \ c \ d \ 8$ Schedule $S' : 6:00 \ 6:20 \ 6:40 \ 7:00 \ 7:20 \ 7:40 \ 8:00 \ 8:20 \ 8:40 \ 9:00$ $6:00 \ 6:20 \ 6:40 \ 7:00 \ 7:20 \ 7:40 \ 8:00 \ 8:20 \ 8:40 \ 9:00$ Time $g''(t, S'') = 5 \ 4 \ 3 \ 2 \ 1 \ 0 \ G(S) = 3 \ a \ 4 \ b \ c \ c \ a \ d \ a \ a \ a \ b \ a \ d \ 2 \ 6 \ 5 \ 1 \ 3 \ 7 \ c \ d \ b \ 9 \ c \ d \ 8 \ G'(S') = 4 \ G''(S'') = 5$ Key: 1st feasible connection Multiple connections from same terminal 2nd feasible connection 3rd feasible connection Extension selected Schedule S''

Figure 7.3 Lower-bound determination using the example shown in Figure 7.1, with the first and second improvement procedures.

A new process needs to be designed for applying the shifting capability of departure times

on single routes. This process simply attempts, through possible shifting of the relevant

departure times, to reduce the minimum fleet size required. We will use the same notation for

route r as in Section 6.2, but the symbol r is deleted, because it is clear which underlying route

is being referred to. Thus, a and b are the end points; T_{ia} and T_{jb} are the average trip times

on the route for vehicles departing at t_{ia} and t_{jb} from a and b , respectively, including layover

time at their respective arrival points; and n_{ia} is the number of departures from a between t_{ia} ,

in which departure ia is included, and $t_{i'a}$, in which departure $i'a$ is excluded. Trip ia arrives

at terminal b , then continues with trip jb , the latter being the first feasible departure from b

to a at a time greater than or equal to the time $t_{ia} + T_{ia}$; and $t_{i'a}$ is the first feasible departure

from a to b at a time greater than or equal to $t_{jb} + T_{jb}$. Similar notations are defined for a trip

starting from b .

Let $[t_{ia} - \Delta_{ia}(-), t_{ia} + \Delta_{ia}(+)]$ be the tolerance time interval of the departure time of trip ia , in

which $\Delta_{ia}(-)$ = maximum advance of the trip's scheduled departure time (the case of an early

departure) and $\Delta_{ia}(+)$ = maximum delay from the scheduled departure time (the case of a late

departure). Note that $t_{ik} + \Delta_{ik}(+) < t_{(i+1)k}$ and $t_{ik} - \Delta_{ik}(-) > t_{(i-1)k}$, for all $k \in K$. The minimum fleet

size, $N_{\min}$, is then attained by construction, using the procedure illustrated in a flow diagram

in Figure 7.4a and b. The procedure described is the case of Equation 6.1, in which $N_{\min} =$

$\max i n_{ia}$. For the case in which $N_{\min} = \max j n_{jb}$ (determined by a trip starting from b), the same

procedure is applied, but with b replacing a and j replacing i . The procedure identifies the

departure ia (or one of a few) referring to $N_{\min} = n_{ia}$; then it attempts through shifting t_{ia} to

arrive

at $t(j-1)b$, and most importantly to arrive before or at $t(i+1)a$. If the process manages

to reduce n_{ia} by one or more units, it looks for the next $n_{ia} = N_{\min}$ or $n_{jb} = N_{\min}$ to continue.

A successful process is that in which $N_{\min}$ is reduced. In addition, the procedure depicted in

Figure 7.4a and b minimizes the length of S DTs, except for the shifting of the first departure, t_{ia} .

The interpretation of the shifting procedure may be assisted by the example in Figure

6.3. Here, $N_{\min} = 5$, resulting from the fifth and sixth departures from b. When b replaces

a and j replaces i in Figure 7.4a and b, we can then use the procedure described and start

with the 6:50b departure. Given $\Delta 6:50b(-) = 5$ min, then $\Delta 7:00a(+) = \Delta 7:20b(+) = \Delta 7:15b(+) = \Delta 7:10b(+) =$

3 min. The first check in the Figure 6.3 example results in shifting 6:50-6:45 from b having

$\Delta 6:45b(-) = 0$ min. Then from a, the first feasible connection is at 7:03 (including a forward

tolerance). The second check, $7:00 \geq 6:45 + 15$, results in setting the departure time from a at

7:00 with $\Delta 7:00a(-) = 0$. The third check, $7:00 + 15 \leq 7:20 + 3$, leads to finding the first feasible

connection to 7:00 to be from a. That is, $\min [7:20 + 3, 7:15 + 3, 7:10 + 3] \geq 7:15$ is 7:18. In

the fourth check, $t(j-1)b = 7:15$. Hence, $n_{6:45b} = 3$, instead of the previously $n_{6:50b} = 5$.

We now move to the 7:05 departure from b, in which $n_{7:05b} = 5$. Given $\Delta 7:05b(-) = 0$ min

and $\Delta 7:10a(+) = \Delta 7:20b(-) = \Delta 7:30b(+) = 5$ min. In the first check, an early departure from a is

impossible. The second check, $7:20 - 0 + 15 \leq 7:30 + 5$, results in setting $t_{(j'-1)b} = 7:35$ and

$n_{7:05b} = 4$. The result is a multicase of $N_{min} = n_{ia} = n_{jb} = 4$, but without any further possibility

of improvement, because $n_{7:05b}$ cannot be further reduced; this case is based on the procedure

constructed in Figure 7.4a and b. Table 7.2 Shifting tolerances as headway dependent Headway (H, in min)
 Percentage of H for tolerance determination (%) Tolerance length as H dependent (min) <10 50 0.5H 10-20 40 0.4H 21-40 30 0.3H >40 20 0.2H

7.3.2 Variable scheduling using deficit functions

The transit scheduler who employs shifting in trip-departure times is not always aware of

the consequences that could arise from these shifts. This section develops a formal algo-

rithm to handle the complexities of SDTs. The algorithm is intended for both automatic and

man-computer conversational modes.

According to the definitions in Section 6.5.1, s_{jk} and e_{jk} are the start and end of the j th

maximal interval, M_{jk} , $j = 1, 2, \dots, n(k)$, at terminal k , $k \in K$, and N is the number of vehicles

(chains, blocks) determined. If the length of M_{jk} is denoted as $M_{es_{jj}k} = -$, then s_{jk} and e_{jk} are

associated with t_{si} and $t_{ei'}$, respectively. That is, s_{jk} refers to the departure time of a trip desig-

nated by i and e_{jk} to the arrival time of a trip designated by i' (where i, i' can be selected from

several trips that depart at time s_{jk} and arrive at e_{jk} , respectively). The shifting terms defined

in the foregoing section will also be used.

ing effect of S DTs. In this example, $\Delta i(+) = \Delta i(-) = 10$ min. Hence, it is possible at the outset to

reduce $D(b)$ by one unit through the shifts of t_3 to the right and t_4 to the left. However, these

shifts increase $D(a)$, and the net saving is zero. Consequently, another iteration is needed

in which t_7 is shifted to the right. Only then can a total saving be obtained of one vehicle.

Given the desire to reduce a maximal interval M_{jk} by shifting a maximum of two trips, let us ask: Yes? $t(i'-v)a \geq t_{jb} - \Delta j_b(-) + T_{jb}$. Set $t(i'-1)a \geq t_{jb} - \Delta j_b(-) + T_{jb}$. n_{ia} is reduced to n_{ia} new. Yes (b) Update relevant t_{ia} , $\Delta i_a(-)$, t_{jb} , $\Delta j_b(-)$, $t(j-w)b$, $\Delta(j-w)b(+)$, $\Delta(j-w)b(-)$, $t(i'-v)a$, $\Delta(i'-v)a(+)$, $\Delta(i'-v)a(-)$, and select $N_{min} = n_{ia} > n_{ia}$ new. For $N_{min} = n_{jb} > n_{ia}$ new change symbols $\text{Max } j \text{ } N_{ia}$, $\text{Max } j \text{ } N_{jb} > n_{ia}$ new n_{ia} new $= N_{min}$ (reduced)? No Find $t(i'-v)a$, $v = 1, 2, \dots$, to comply with: $\geq t_{jb} - \Delta j_b(-) + T_{jb}$. $t(i'-v)a + \Delta(i'-v)a(+)$. $\text{Min } v$. Set $t(i'-1)a = t(i'-1)a - \text{min } T$. $(i'-v)a - t_{jb} + \Delta j_b(-) - T_{jb}$, $\Delta(i''-v)a(-)$.

Figure 7.4 (Continued) (b) flow diagram continued from (a).

consider the following three cases: (A) shift only trip i to the right, (B) shift only trip i' to the

left, and (C) shift both trips i and i' in opposite directions (see Figure 6.6 for definitions used).

7.3.2.1 Case A: Right-shift limit

Let trip i (between terminals k and m), which starts at t_{sk} , $s_{ij} = k$, arrive at hollow H_{q1-m} (preced

ing M_{qm}) at time t_{ei} . Shift trip i to the right as close to e_{jk} as possible without increasing the

maximal interval M_{qm} or without exceeding $\Delta i(+)$. Let this right-shift limit be defined as $\delta(\min, \dots, \dots) = -\{ \dots \} + \dots$ $s_{tm} = e_{ij} = m$ $k \Delta$ (7.1) $b \ c \ c \ a \ a \ a \ a \ b \ a$ $d \ 2 \ 6 \ 5 \ 1 \ 3 \ 4 \ 7 \ c \ d \ b \ 9 \ c \ d \ 8$ Fixed schedule $d(a, t) d(b, t) D(a) = 3 \ 2 \ d(d, t) 4 \ 3 \ 2 \ 1 \ 0 D(b) = 1 \ 3 \ 2 \ 1 \ 0 d(c, t) 3 \ 2 \ 1 \ 0 -1 \ 3 \ 2 \ 1 \ 0 -1 g(t) 4 \ 3 \ 2 \ 1 \ 0$ Time 6:00 6:20 6:40 7:00 7:20 7:40 8:20 8:40 9:00 6:00 6:20 6:40

7:00 7:20 7:40 8:20 8:40 9:00 Time G=3 D(b) =2 1 0
D(c) = 1 0 DH 2 DH 3 DH 1

Figure 7.5 The nine-trip example (of Figure 7.1), first with shifting and second with DH trip insertion, for reducing fleet size.

If $\delta = -s + q + e + m$, then the shift has reached $M + q + m$ and any further right shift will increase $D(m)$.

If $\delta(+) = \Delta i(+)$, then the shift is stopped by the tolerance limit of trip i . If $\delta(+) = M + j + k$, then a

successful shift has occurred, and $D(k)$ is reduced by one.

7.3.2.2 Case B: Left-shift limit

Let trip i' (between terminals m and k) depart from hollow $H + q - 1$ at time $t + s + i'$. Shift trip i' to

the left as close to $s + j + k$ as possible, without increasing the maximal interval $M + q - 1$ or without

exceeding $\Delta' - i(-)$. Let this left-shift limit be defined as $\delta(\min, () = - \{ \} ' - ' -) + e + M + s + i + q + j + 1 + m + k + \Delta$ (7.2)

If $\delta(- = - ' -) + e + s + i + q + 1 + m$, then the shift has reached $M + q - 1$, and any further shift left will increase

$D(m)$. If $\delta(() = - ' -) + i + \Delta$, then the shift is stopped by the tolerance limit of Trip i' . If $\delta() = M + j + k$,

then $D(k)$ is reduced by one.

7.3.2.3 Case C: Shift both trips

Without loss of generality, $D(k)$ can be reduced by shifting both trips i and i' in opposite

directions. Assume that the procedure starts with an attempt to shift trip i to the right and

is unsuccessful, $\delta(+ <) + M + j + k$. Now perform Case B, with the length of $' = - + M + M + j + j + k + k + \delta()$ reduced

from $M + j + k$. Similarly, the procedure can start with Case B and continue with Case A.

These three cases can be incorporated into a formal SDT algorithm. Two points should

be noted: (1) If a unit reduction shifting chain (URSC) is allowed, as in Figure 7.6, the first b

Schedule Time Time Time 6:00 6:30 7:00 7:30 8:00: :30
 6:00 6:30 7:00 7:30 8:00 :30 6:00 6:30 7:00 7:30 8:00 :30 3
 2 1 0 2 1 0 Initial schedule and deficit functions 1st
 shifting iteration 2nd shifting iteration $d(a, t)$ $d(b, t)$ $D(a) = 1$ $D(b) = 3$ $D(a) = 2$ $D(a) = 1$ $D(b) = 2$ $D(b) = 2$ b b
 b b b b b a a a a a a a 2 3 1 6 7 b b b b b b a a a a a a a
 2 3 1 4 5 6 7 b b b b b b a a a a a a a 2 3 1 4 5 6 7 4 5

Figure 7.6 Example of two shifting iterations to reduce the required fleet size from 4 to 3.

criterion in Equations 7.1 and 7.2 is dropped (i.e., $s \leq t$ and $t \leq s$); the process for URSC

is finite, as we always move toward T_1 and/or T_2 and stop there (for an unsuccessful URSC).

(2) A criterion for minimum shifting can be established (i.e., a least square of the shifting

length) in order to assure minimum deviation from scheduled departure times (Case C is

affected by such a criterion). The next section analyzes fleet size lower bound with SDTs;

this will be followed by a section on SDT algorithms.

7.4 FLEET SIZE LOWER BOUND FOR VARIABLE SCHEDULES

In regard to a lower bound on the fleet size, with allowed shifts in trip-departure time, it is

certain that where S can be shifted in $g(t, S)$, the lower bound will be equal to or less than

$G(S)$. The main part of this section covers an algorithm that was devised for giving the

opportunity to reach the lower bound through the combination of shifting and DH trips.

This procedure follows Cedar (2002, 2005) but with

further adjustments and revisions. In

addition, the end of the section presents a special lower-bound case in which only shifting,

but not DH trips, is permitted.

7.4.1 Lower bound using shifting and DH trips

The algorithm for the lower bound consists of three phases. Phase A is an LB-SHIFT

procedure for reducing $G(S)$. Phase B is an LB-DH&SHIFT procedure for constructing

a new $g'(t, S')$ with shifting considerations, and Phase C is the construction of a new

$g''(t, S'')$. Phase A involves the construction of a new temporary schedule, with shifts

to be termed S_{sf} , which will end with $G(S_{sf}) \leq G(S)$. Phase B constructs a new tem

porary schedule with shifts combined with DH trips, to be termed $'S_{sf}$, that will end

with $G('S_{sf}) \leq G(S_{sf}) \leq G(S')$. Note that $G(S_{sf})$ is an extended schedule that follows the

construction of $g'(t, S')$ in Section 8.2 but with an S_{sf} , in which no shifts are made

in constructing $g'(t, S_{sf})$. Finally, Phase C applies the procedure for constructing $g''(t, S'')$

in Section 7.2 to $'S_{sf}$ without any additional shifts. All notations used here are applicable

to the procedures that follow.

7.4.1.1 Phase A: LB-SHIFT procedure

Step 0 (initialization): Define $M_j = [s_j, e_j]$, $j = 1, 2, \dots, n(g)$, and $M_j = e_j - s_j$ as the j th maximum interval and its length, respectively, of $g(t, S_{sf})$. Let $s_j = t_{s_i}$, $e_j = t_{e_{i'}}$, and $S = S_{sf}$, in which s_j refers to the departure time of a trip designated by i and e_j to the arrival time of a trip designated by i' . Set $j = 0$.

Step 1 (selecting the next maximum interval): Let $j = j + 1$; if $j > n(g)$, stop; otherwise, continue.

Step 2 (feasibility check): If $M_j(i(+)) \leq + + ' - \Delta \Delta$, continue; otherwise, go to Step 1.

Step 3 (right shift): Compute $\delta(\min\{, \} (+) + = +) M_j$ for $g(t, S_{sf})$; if $\delta(+) = M_j$, go to Step 5; otherwise, continue.

Step 4 (left shift): Compute $\delta(\min\{, \} (-) - = ' - \Delta i$ for $g(t, S_{sf})$; if $\delta(-) = M_j$, go to Step 5; otherwise, if $\Delta' - i(+) \geq M_j - \Delta i(+)$ (Case C for $g(t, S_{sf})$ in Section 8.3), set $\delta(-) = M_j - \Delta i(+)$, $\delta(+) = \Delta i(+)$, and go to Step 5. Otherwise, go to Step 1.

Step 5 (reduce the lower bound): Update S_{sf} and $g(t, S_{sf})$ with the confirmed shifts; go to Step 1.

7.4.1.2 Phase B: LB-DH&SHIFT procedure

Step 0 (initialization): Construct $g'(t, S_{sf})$ from $g(t, S_{sf})$. Define $M_j = [s_j, e_j]$, $j = 1, 2, \dots$, $n(g')$ and $' = ' - ' M_j$ as the j th maximum interval and its length, respectively, of $g'(t, S_{sf})$. Let $s_j = t_{si}$, $e_j = t_{ei}'$, and $S_{sf} = S_{sf}$, in which s_j refers to the departure time of a trip designated by i and e_j to the arrival time of a trip designated by i' . Set $j = 0$.

Step 1 (selecting the next maximum interval): Let $j = j + 1$; if $j > n(g')$, stop; otherwise, continue.

Step 2 (right-shift check): If trip i of $s_j = t_{si}$ can be shifted by $\Delta i(+)$ (was not shifted in Phase A) and results in an increase in $G'(S_{sf})$, go to Step 4; otherwise, continue.

Step 3 (artificial shift): Let t_{si} be shifted to the right by $\Delta i(+)$ and included in $' S_{sf}$; set $t_{si} = t_{si} + \Delta i(+)$.

Step 4 (trip extensions with shifts): Let E be the set of all trips in which their artificial extension overlaps with M_j , $i \in E$; examine each trip u , $u \in E$, to determine whether its first feasible linkage point (departure time of a trip with which it may be linked) is t_{si} by performing a shift to the left ($\leq \Delta u(-)$); if no linkage exists, stop; no improvement can be made. Otherwise, continue.

Step 5 (new shifts and extensions): Perform for all $u \in E$ that can be shifted to the left (and linked to t) the new shifts and extensions to be included in $'S_{sf}$; go to Step 1.

If Phase B indeed improves the lower bound, $'G(S_{sf}) < 'G(S)$, the foregoing $'S_{sf}$ is then subject

to Phase C. The latter has further extensions of arrival epochs if more than a single extension exists from the same terminal; this results in

construction $''S_{sf}$ from $'S_{sf}$ according to the

analysis of the construction of S'' in Section 7.2.

To better comprehend Phases A and B, the example problem of Figures 7.1 and 7.3 is presented.

This problem undergoes these phases in Figure 7.7, with $\Delta i(+) = \Delta i(-) = 5$ min for all $i \in S$.

Figure 7.7 shows the schedule and shifting required to construct $g(t, S_{sf})$ and $g'(t, 'S_{sf})$. The process of constructing $g(t, S_{sf})$ exhibits Phase A, in which

two trips are shifted in opposite directions, Trips 7 and 8, to allow for $G(S_{sf}) = 2 < G(S) = 3$; this change is represented by a dotted

line. The function $g'(t, 'S_{sf})$ exhibits Phase B; the dotted line in $g'(t, 'S_{sf})$ shows that the departure

time of Trip 5 is the start of the only maximum interval that exists in $g'(t, S_{sf})$ following the two

shifts in Phase A. Shifting Trip 5 to the right by 5 min will open up the possibility for Trips 2

and 3 to be extended to Trip 5 while being shifted to the left by 5 min, instead of extended as

it was previously to Trip 6. This is not the case with Trip 4, whose original extension overlaps,

too, with the maximum interval. Finally, it should be observed that by shifting Trip 5 to the

right, it can no longer be extended to Trip 6, and its new extension is to Trip 9. However, this

new extension does not create a new maximum interval (see Step 2 in Section 7.4.1.2).

The final Phase C applies the procedure to attain $G''(\tau, S_{sf})$ of the function $g''(t, S_{sf})$. That

is, to start with schedule τ, S_{sf} and allow further extensions of arrival points if more than a

single extension exists from the same terminal. In the example shown in Figure 7.7, Phase C

cannot be implemented. Assuming, however, that Trip 3 starts and ends at same terminal b,

then Trip 2 could be further extended to Trip 6 (the shortest extension of Trip 3 is the one

selected to remain). Nonetheless, the latter extension of Trip 2, based on the earlier assumption,

will not increase $G'(\tau, S_{sf}) = 3$. This final lower bound of three is the optimal number

of chains (blocks) that can be derived in the example problem. This compares with five

chains with only a DH trip insertion. The shifting of departure times in the LB-SHIFT and

LB-DH&SHIFT procedures can be used in the example to create the optimal three trip

chains: [1-4-6], [3(shifted)-5(shifted)-9], and [2-7(shifted)-8(shifted)], in which the shift

of Trip 2 in Figure 7.7 is not required, because it links to Trip 7 not Trip 5.

A special scheduling case occurs when only shifting, but not a DH trip insertion, is permitted.

In practice, this situation arises in transit systems without interlining and in which

passenger demand does not differ greatly in either

direction of the route (e.g., a rail opera

tion). The lower-bound algorithm for this case follows the LB-DH&SHIFT procedure, but

without a DH extension consideration. That is, each trip's arrival time is extended to the

time of the first feasible departure time of a trip to which it may be linked, by having the

same arrival and departure terminal as the extension, or to the end of the finite time horizon.

4 3 2 1 0 $g'(t, S s'f)$
 $g(t, S s f)$ a 4 b c c a d a a a b a d 2 6 5 1 3 7
c d b 9 c d 8 Schedule S sf 6:00 6:20 6:40 7:00 7:20 7:40
8:00 8:20 8:40 9:00 6:00 6:20 6:40 7:00 7:20 7:40 8:00 8:20
8:40 9:00 Time a 4 b c c a d a a a b a d 2 6 5 1 3 7 c d
b 9 c d 8 Time $G(S sf) = 3$ 2 $G'(S s'f) = 4$ 3 4 3 2 1 0
Key: Part of $g(t, S sf)$ and $g'(t, S s'f)$ before phase (b)
shifts Shift direction and magnitude Schedule S s'f

Figure 7.7 Lower-bound determination using the example in Figure 7.1 with LB-SHIFT and LB-DH&SHIFT procedures.

Let $' S sf$ and $'' S sf$ be the extended schedules that are similar to $' S sf$ and $'' S sf$, but with a possible

extension to link only the same terminals or to the end of the time horizon. We then obtain

$' ' \geq ' ' G(S G(S sf sf))$ and $'' ' \geq '' '' G(S G(S sf sf))$ and of course $'' ' \geq ' ' G(S G(S sf sf))$. Figure 7.8 uses the example

in Figure 7.1 for constructing $' S sf$, which has the same results as $'' S sf$. In this example, therefore,

$'' ' = ' ' = G(S G(S sf sf)) = 6$.

7.5 FLEET-REDUCTION PROCEDURES

This section will describe some of the considerations in incorporating the SDT algorithm

into the fleet-reduction procedures. The primary inputs required for the heuristic procedures

are the following:

1. An initial set of fixed trips, S_o , defined over

a set of terminals K

2. A tolerance matrix of trip-departure times

3. A travel time matrix of potential DH trips
(if applicable)

The output of the algorithm will be a new vehicle schedule,
including the shifts of

trip-departure times, the set of DH trips inserted (if
allowed), and the required number of

vehicles at each terminal. The trips assigned to each
vehicle can be constructed in a second

phase using the FIFO rule on the chain
extraction procedure explained in Section 6.5.5. a
4 b c c a d a a a b a d 2 6 5 1 3 7 c d 9 c d 6:00 6:20
6:40 7:00 7:20 7:40 8:00 8:20 8:40 9:00 Time $S s' f = S s''$
 $f S c h e d u l e$ 6:00 6:20 6:40 7:00 7:20 7:40 8:00 8:20
8:40 9:00 Time 6 5 4 3 2 1 0 $G'(S s' f) = G''(S s'' f) = 6 g'$
($t, S s' f$) Key: Part of $g(t, S)$ Shift direction and
magnitude b 8

Figure 7.8 Lower-bound determination using the example in
Figure 7.1 with the LB-EXT SHIFT procedure.

7.5.1 SDT algorithm

Step 0 (initialization): Let E represent the set of
unexamined terminals; set $E = K$.

Step 1 (select the next terminal): If $E = \emptyset$, stop.
Otherwise, select a terminal k from E and remove it; u
pdate $E = E - \{k\}$; $j = 1$.

Step 2 (select the next maximum interval): Let
 $j = j + 1$; if $j > n(k)$, go to Step 6.

Step 3 (feasibility check): If $M_{jk} \leq \Delta i(+) + \Delta$
 $' - i()$, continue; otherwise, go to Step 1.

Step 4 (right shift): Compute $\delta(+)$; if $\delta(+)$
 $= M_{jk}$, go to Step 6; otherwise, continue.

Step 5 (left shift): Compute $\delta(-)$; and if $\delta(-) = M_{jk}$, or
 $M_{jk} - + \leq -\delta \delta() ()$ (see Case C in Section 7.3), g
o to Step 6; otherwise, go to Step 1.

Step 6 (reduce the fleet): Perform all shifts and update the DF; if $G(S_{sf}) = D(S)$, or $G''(S_{sf}) = D(S)$ in the case with DH trips, stop. Otherwise, go to Step 1.

The foregoing traces the basic structure of the SDT algorithm. In the operational algorithm,

the order in which terminals and maximum intervals are selected for examination is determined

by various heuristic priority rules. As noted at the end of Section 7.3, a criterion

for minimum shifting can be established and will affect Steps 3–5 in the SDT algorithm.

Chaining shifts, as shown by the example of Figure 7.6, constitutes a second-level phase

examination of all the terminals. This is not described here out of an effort to simplify

understanding of the underlying basic principles of the algorithm.

Another possibility is to consider variable trip-departure times along with the DH trip

insertion procedure. In that case, the feasibility requirement shown in Equation 6.8 for a

feasible joining of two trips, i and j , is changed to $t_i - t_j - \tau_{q_i p_i} + \tau_{q_j p_j} \leq \Delta$ (7.3)

in which $\tau(q_i, p_i)$ is the DH time of trip i from terminal q to terminal p .

7.5.2 Order of subroutines

In the variable-scheduling problem, two major subroutines are identified in order to reduce

the fleet size: (a) SDT within their tolerances and (b) searching for a URDHC. In some prac

tical cases, DH trips are not allowed and only an SDT algorithm is utilized. The URDHC

procedure could be made both with and without consideration of the SDT subroutines. The

following is a discussion of these subroutines.

The selection of the SDT subroutine needs to be examined primarily in light of the maximum

possible saving of vehicles. Figure 7.9 illustrates an example of the terminals using

the DF representation. The first column, URDHC Start, begins with the URDHC procedure

and ends with a modified URDHC (mixed with the SDT) procedure. The second column,

SDT Start, uses first the SDT procedure and second the modified URDHC procedure. In this

example, $\Delta i(+) = \Delta i(-) = 5$ min for all trips i in the schedule; the DH trip time for all possible

DH trips is 10 min. It is also assumed that the departure times for trips shifted to the left and

the arrival times for trips shifted to the right are outside the timescale for the SDT procedure.

The first-column solution succeeds in reducing $D(a)$ from 4 to 2 in Step 1 of Figure 7.9.

In Step 2, $D(b)$ is reduced by one through four shifts of trip-departure times and two DH

trips. The final schedule can be constructed from the DFs in the last step, in which $D(S)$,

the fleet size, has been reduced from 11 to 8 vehicles. In the second-column solution (SDT

Start), $D(b)$ is reduced from 4 to 3 by four trip-departure time shifts. Then in Step 2, $D(a)$ is

reduced from 4 required vehicles to 2 through two shifts of the trip-departure time and two

DH trips. The final schedule has $D(S) = 8$.

Note that in the final DF configuration in both columns of

Figure 7.9, no further reduc

tions in fleet size can be achieved based on the formal SDT algorithm and on the feasibility

requirement for DH trip insertion indicated in Equation 7.3. In consideration of the second

any objectives, it appears that the second column has priority over the first column (four

shifts and two DH trips as against four shifts and six DH trips, respectively). Intuitively, it

seems worthwhile to start with the SDT procedure. However, the next example shows that

this is not always the case. $D(a) = 2$ $D(b) = 3$ $D(c) = 3$ Final
 7:00 Step 1 Step 2 $d(a, t)$ $d(a, t)$ $d(a, t)$ d
 (b, t) $d(b, t)$ $d(b, t)$ $d(c, t)$ $d(c, t)$ d
 (c, t) $D(a) = 2$ $D(b) = 3$ $D(c) = 3$ URDHC start SDT start
 Time Time 5:00 5:30 6:00 6:30 7:00 5:00 5:30 6:00 6:30 4 3
 2 1 0 4 3 2 1 0 3 2 1 0 4 3 2 1 0 3 2 1 0 4 3 2 1
 0 4 3 2 1 0 3 2 1 0

Figure 7.9 Two solutions for reducing fleet size; the first starts with the URDHC subroutine and the second with the SDT subroutine; both continue with the modified URDHC procedure.

Figure 7.10 presents two examples with shift tolerances and a DH trip time as in the example

in Figure 7.9. In Example 1, $d(a, t)$ and $d(b, t)$ are the same as in Figure 7.9, but a small change is

introduced in $d(c, t)$. The trip, which departs at 6:20, moves to depart at 6:10 from terminal c.

In Example 1, only two vehicles are saved (see Figure 7.10) by starting with the URDHC procedure.

This is in comparison with saving of three vehicles if we begin with the SDT procedure.

(see the better solution indicated under Example 1 in Figure 7.10). In Example 2, $d(a, t)$ and

$d(b, t)$ are the same as in Figure 7.9; and in $d(c, t)$, one trip arrives at 5:35 instead of 5:30. In this η

Better

solution Example 2 (Shifting first is worst case)

Step 1

Step 2 $d(a, t)$ $d(a, t)$ $d(b, t)$ $d(b, t)$ $d(c, t)$ Example 1 (DH first is worst case) Time 5:00 5:30 6:00 6:30 7:00 5:00 5:30 6:00 6:30 7:00 $D(a) = 4$ 3 $D(b) = 3$ $D(c) = 3$ μ η μ $D(a) = 2$ $D(b) = 4$ $D(c) = 3$ Use same shifting first and then DH trip insertions as in Steps 1 and 2 in the "SDT Start" solution in Figure 7.8; this results in $D(a) = 2$, $D(b) = 3$, $D(c) = 3$ $D(S) = 8$ Use same DH trip insertions first and then shifting as in Steps 1 and 2 in "URDHC Start" solution in Figure 7.8 except for the 5:30 shift at c; this results in $D(a) = 2$, $D(b) = 3$, $D(c) = 3$ $D(S) = 8$ 4 3 2 1 0 4 3 2 1 0 3 2 1 0 4 3 2 1 0 4 3 2 1 0 3 2 1 0 $d(c, t)$

Figure 7.10 Two examples demonstrating no explicit priority for starting with either the URDHC or the SDT procedure.

example, we relax the assumption that the SDT procedure influences departure and arrival

times outside the time scale (though this assumption holds in Example 1).

In Step 1 of Example 2, in Figure 7.10, four shifts are illustrated. The shifts that are

denoted by μ and η (forward shifts) affect the D 's at terminals a and b, respectively. Shift μ

refers to a trip between terminals b and a with a trip time of 30 min, and shift η refers to a

round trip from terminal b with a length of 10 min. Thus, a DH trip between terminals c

and also reduces D (a) by one. Consequently, we can see in Step 2 (Example 2) in Figure 7.10

that by starting with the SDT procedure, it is possible to save only two vehicles at terminals

band a. This compares to a saving of three vehicles when applying the better solution indicated

cated in Example 2 in Figure 7.10.

A DH departure has the effect of increasing the departure terminals' DF and also of

reducing the deficit of some maximum intervals at their destination terminals (either by

reducing the fleet requirement or by compensating for an earlier DH trip departure).

In this way, maximum intervals are leveled and expanded at both the departure and

the arrival terminals. This reduces the opportunities for a further reduction of the fleet

size through the SDT procedure. On the other hand, similar consequences are observed

when starting with the shifting of trip-departure times. That is, the SDT procedure generates

ally reduces the opportunities for a further reduction of the fleet size through DH trips

(though some shifts might open up new opportunities because of a larger time interval for

a DH trip insertion).

7.5.3 Description of the procedure

For the fleet-reduction procedure, when DH trip insertion is allowed, we use a heuristic rule

and start with the SDT subroutine, followed by the modified URDHC (mixed with SDT)

subroutine. This rule can represent to some extent the viewpoint of the transit agency that

wishes to minimize operational costs.

The basic fleet-reduction procedure that contains only the SDT algorithm is presented in

the flow diagram in Figure 7.11 and is designed for a person-computer interactive system.

It does not interact with the DH trip-insertion process. The selection of a terminal u can be

made by the scheduler by inspecting the DFs on a graphical display. The search for a reduction

in the fleet requirement at terminal u (see Figure 7.10) can be performed manually by

the scheduler or by procedures based on the formal SDT algorithm. If a URSC is found, all

DF information is updated for this new schedule, and a new iteration initiated. The SDT sub

routine continues until either the lower-bound test is successful or all terminals are examined.

The modified URDHC (mixed with SDT) subroutine, shown schematically in a flow dia

gram in Figure 7.12, is based on the URDHC procedure described in Section 6.5.4. The

upper part of the flow diagram is the SDT subroutine described in Figure 7.11. In the lower

part of Figure 7.12, the lower-bound determination suggests that $G''(S, S_f)$ should be examined

before trying to insert DH trips into the schedule. In this modified procedure, the feasibility

requirement for DH trip insertion is based on Equation 7.3.

Another point that should be

mentioned is that DH trips added to schedule S should include a shifting tolerance for the

next iterations of the SD T and modified URDHC procedures. Finally, if a URDHC has been

found, the DH chain cost involved is compared with the saving cost of a single vehicle. If the

DH cost is higher than the saving cost, the URDHC is cancelled. Otherwise, the set of DH

trips, designated {DH}, is added to the previous schedule, and all DF information, including

shifting, is updated for a new schedule. The modified URDHC subroutine continues with

updated S until $G''(S) = D(S)$ or until all terminals are examined.

7.6 EXPERIENCES WITH BUS SCHEDULES

Before sharing some experiences, it is advised to look at Appendix 7.A. This appendix con

tains an example of vehicle-scheduling software including its key features and expected

saving. The software is one of those presented in Table 1.1. Appendix 7.A is specially demon

strating the saving one can expect using software of vehicle scheduling.

As mentioned in Section 7.1, fundamental problems in designing variable schedules arise

from practice. An example of a transit agency that faced these problems is Eged, Israel's

national bus carrier, which was discussed in Section 6.3. The Eged fleet of about 4,000

buses operates over a spread-out national network with some 2,000 routes; it performs a n

average of 50,000 daily trips, among them 12,000 DH trips. Bus schedules are produced by

about 60 schedulers using Gantt charts through a trial-and-error approach. The need for a

quicker response to timetable changes has led the Eged management to investigate the use

of a fully computerized system. As is explained in Section 6.3, attempts to implement the Update terminals examined. Set $Q=Q+q$ Initial input: Set of terminals K ; trip schedule S_0 ; set of tolerances for trip-departure times. Set $S=S_0$, $Q=\Phi$ STOP Yes Yes No Yes No No Select next terminal q to examine for a possible reduction of $D(q,S)$ by shifting trip-departure times within their tolerances Computation of deficit function information: $\{d(k,t,S)\}$, $D(S)$, $g(t,S)$, $G(S)$, $G(S_{sf})$ Lower-bound termination test $G(S_{sf})=D(S)$? All terminals examined $Q=K$? Reduce a vehicle at q incorporate the new trip-departure times into S to form the schedule S_q ; set $S=S_q$; Unit reduction, by shifts, of $D(q,S)$ found?

Figure 7.11 Flow diagram of a fleet-reduction procedure involving only shifting of departure times (SDT algorithm). Initial input: Set of terminals K ; trip schedule S_0 ; set of tolerances for trip-departure times; DH time matrix. Set $S=S_0$, $Q=\Phi$ Computation of deficit function information: $\{d(k,t,S)\}$, $D(S)$, $g(t,S)$, $G(S)$, $G(S_{sf})$, $G(S_{sf})$ Select next terminal q to examine for a possible reduction of $D(q,S)$ by shifting trip-departure times within their tolerances Lower-bound termination test $G(S_{sf})=D(S)$? Unit reduction, by shifts, of $D(q,S)$ found? All terminals examined, $Q=K$? Update terminals examined. Set $Q=Q+q$ STOP Reduce a vehicle at q ; incorporate the new trip-departure times into S to form the schedule S_q ; set $S=S_q$ Yes Yes No Yes No Set $Q=\Phi$ Update terminals examined; set $Q=Q+p$ Select next terminal p , to examine for a possible reduction of number of vehicles using DH trip insertion incorporated with shifting trip-departure time Lower-bound termination test $G(S_{sf})=D(S)$? No No All terminals examined, $Q=K$? Yes No Yes No Yes DH cost > vehicle saving cost? Unit reduction DH chain found at terminal p ? No Yes Reduce a vehicle at q ; add set of DH trips and shifts $\{DH+SH\}$ to the schedule: $S=S+\{DH+SH\}$ Set $Q=\Phi$

Figure 7.12 Flow diagram of fleet-reduction procedure involving shifting of departure times and DH insertions

(modified URDHC subroutine).

computer-generated schedule have failed because of an inability to meet a number of neces

sary practical constraints.

It was, therefore, decided to continue the search for an approach that would combine the

advantages of modern electronic computers while, at the same time, allow the scheduler

to make his or her own contribution to the scheduling task. Because of its visual nature,

a DF approach was selected for use with a person-machine interactive system. The imple

mentation of the DF approach was introduced gradually so that the schedulers could gain

confidence in this approach and reach the conclusion that this method was very useful for

increasing the speed and accuracy of the scheduling tasks.

Two simple, real-life examples are described in the succeeding text to demonstrate the

implementation stage at Egged. In the first example, illustrated in Figure 7.13, the schedul

ers claimed at first that it was impossible to further reduce fleet size from their Gantt chart

scheduling results. Figure 7.13 shows only a small part of the Gantt chart and the corre

sponding DFs, those that are undergoing change. The schedulers allow for an acceptable

shift in trip-departure time for all trips by
 $\Delta d(+) = \Delta d(-) = 3 \text{ min.}$ Bus block no. 1 2 3 4 5
6 Deficit function Bus block no. 1 2 3 4 5 Deficit
function Part of Gantt chart Part of Gantt chart $d(k$
, $t)$ $d(k, t)$ 6 4 2 0 6 4 2 0 B A D C F E H G J I B A D
C F E H G J I A B C D G J $D(k) = 4$ $D(k) = 3$ 7:00 7:06 7:12
7:18 7:24 7:36 7:30 7:36 7:30 7:42 7:48 7:54 7:00 7:06 7:12

7:18 7:24 7:42 7:48 7:54 A f t e r B e f o r e Time Time

Figure 7.13 Demonstration of the superiority of deficit-function representation over the Gantt chart approach.

By illustrating $d(k, t)$, however, we saved one bus by shifting six trips, each by 3 min.

Before the changes in Figure 7.13, trips are identified in the Gantt chart by letters. This is

for the sake of clarity when referring to shifts and for reconstructing the Gantt chart chains

after the changes. From Figure 7.13, the problem appears easy to handle. However, before

the changes in part of Figure 7.13, only 6 of 52 portions of the bus blocks were shown, and

those 6 rows were spread among the 46 other rows in the Gantt chart.

Following this demonstration, the schedulers were still not wholly convinced. They argued

that with a little more effort on their part, they too could have saved the bus as in Figure 7.13.

Therefore, a more complex example was decided on, as shown in Figure 7.14. This second

example refers to an afternoon schedule for two Egged branches: Ramle, terminal k, and

Lod, terminal m. On the left-hand side of Figure 7.14, only trips that involved changes are

exhibited in the before-and-after Gantt chart representation; trips are designated by letters.

On the right-hand side, the DFs of the complete schedule are illustrated, including trips not

shown in the left-hand side. The schedulers again claimed that no further reductions could

be achieved from the $D(k) + D(m) = 57 + 19 = 76$ fleet size requirement. The given informa

tion was that $\Delta i(+) = \Delta i(-) = 2$ min and that the DH trip time between the k and m terminals

(both directions) was 7 min.

As seen in Figure 7.14, 6 shifts in trip-departure times and a single DH trip were required

in order to save 2 buses and to reduce the total fleet requirement to 74 buses. It was only

after this second demonstration that the schedulers began to take a serious interest in the

DF model. This was particularly due to its simplicity and visual nature. The schedulers

expressed a positive feeling about the valuable aid of this gradual approach. B A D C F G E H J I
 K DH(a b) d (a , t) d (b , t) D(b) = 18 D(a) = 56
 56 54 52 50 18 16 14 12 10 Time 16:45 16:40 16:45 16:50
 16:55 17:00 17:05 17:10 16:45 16:40 16:45 16:50 16:55 17:00
 17:05 17:10 16:45 16:40 16:45 16:50 16:55 17:00 17:05 17:10
 16:45 16:40 16:45 16:50 16:55 17:00 17:05 17:10 d (a , t)
 d (b , t) 56 54 52 50 18 16 14 12 10 D(a) = 57 D(b)=19 A
 A' B' C' E' G' DH Time Bus block no. Terminal
 1 a Terminal b A B D C E F G H I J K A' B' D' C' E'
 F' G' 1 2 3 4 5 6 1 2 3 4 Bus block no. Terminal
 1 a Terminal b A' B' D' C' F' E' DH(a b)G' 1 2
 3 1 2 3 4 5 Part of deficit-function Part of Gantt chart

A f

t e

r

B e

f o

r e

Figure 7.14 Two-terminal case, in which two buses are saved through shifting departure times and modifying URDHC procedures.

7.7 EXAMINATION AND CONSIDERATION OF EVEN-LOAD TIMETABLES

We noted in Section 7.3 that the construction of timetables in Chapters 4 and 5 was based on

either even headways or even average loads, entailing situations in which even headways resulted

in uneven passenger loads. Even average load can be formulated either at the hourly max load

point or at the max load point of individual vehicles (Ceder and P hilibert, 2014). SDTs, therefore,

may unbalance these desirable features in the timetable while favoring cost (vehicle) saving. This

section introduces the possibility of establishing a load-tolerance criterion for even-load timetables.

Following the notation in Section 5.2, let $S L_i(t)$ be the slope of $L_i(t)$ at t , in which $L_i(t)$ is

the cumulative load curve at stop i , $i = 1, 2, \dots, n$; $\Delta d \tau$ is a given positive tolerance (in pas

sengers) of the desired occupancy $d \tau$ at time interval $\tau = 1, 2, \dots, v$; $t T_{1j} * q_j q = - \tau$ is a depar

ture time determined by Principle 5.3 in Section 5.2 (even load on individual vehicles) at

stop q , in which $t q_j$ is the j th candidate departure time from stop q , $j = 1, 2, \dots, m$; and $T q \tau$

is the average service travel time between the departure terminal $q = 1$ and stop q when $t q_j$

is at interval τ (note that by definition, $T_{1\tau} = 0$), and $\Delta j \tau(-)$ and $\Delta j \tau(+)$ are given positive toler

ances (in minutes) for maximum shifting $t_{1j} *$ to the left (early departure) and to the right (late

departure) for each interval τ , $j = 1, 2, \dots, m$; $\tau = 1, 2,$

..., V .

When d_i is added a long $L_i(t)$, the minimum time that intersects one of the cumulative load

curves (see Principle 5.3 in Section 5.2) is determined at stop q . Hence, $d_i(t) = L_i(t) - L_i(t - \Delta\tau) - 1$, * (7.4)

The shifts in departure times made by the DF model are defined as follows: $t_j - t_{j-1} = - \Delta\tau$ (7.5) $t_j - t_{j-1} = + \Delta\tau$

for all $j = 1, 2, \dots, m$, and relevant τ for t_{j-1} and t_j .

In order to avoid excessive average loads beyond $d_i + \Delta d_i$ at each trip's critical point, two

criteria may be established for early and late departures. What follows is a formal derivation

of the early-departure criterion, while a similar derivation for the late-departure criterion is

presented as an exercise at the end of the chapter.

7.7.1 Early departure criterion

According to the foregoing definitions, $L_i(t) = L_i(t - \Delta\tau) + 1$ (7.6)

where

$L_i(t) = L_i(t - \Delta\tau) + 1$ for $i = 1, 2, \dots, n$ and t_{j-1} belongs to τ

In case the slope is changed within a $\Delta\tau$ shift for any stop i , Equation 7.6 should consider two

(or more) decreased portions, each related to a different slope and its associated part of $\Delta\tau$.

The loads at t_{j-1} can be expressed as $L_i(t) = L_i(t - \Delta\tau) + 1$ (7.7)

These loads for the new t_{j1} - departure across all stops are based on Principle 3 (Section 5.2)

in which the desired occupancy $d\tau$ is attained for t_{j1} only at q and the load is less than $d\tau$ at

all other stops. Using a $\Delta j\tau(-)$ shift will further reduce these loads. However, the loads at each

stop if for the departure times adjacent to t_{j1} - at $t_{j1-1}, *$ will increase the loads if the shifting takes

place. This increase is $\Delta j_{ij} SL_{t\tau}(\tau) * (\tau) - 1$ for all i and relevant τ , or it is the sum of portions of the

slope that are changed within $\Delta j\tau(-)$.

The increased new loads at $t_{j1-1}, *$ need to be checked against $d\tau + \Delta d\tau$ across all stops. This

check is applied for the maximum load increase, and hence the early-departure criterion for

accepting $\Delta j\tau(-)$ is as follows: $\max_{i,j} L_{t\tau}(\tau) - L_{t\tau}(\tau) \leq +1$ (7.8)

where

$t_{j1-1}, *$ belongs to interval τ

$L_{t\tau}(\tau) - 1$ is obtained by Equation 7.6

Figure 7.15 illustrates a simple example of Principle 3 in Section 5.2. Given a transit line

$A \rightarrow B \rightarrow C$, with average travel times of 15 min between A and B ; three departures, at

6:15, 6:45, and 7:10; and a desired occupancy of 50 passengers. The average observed

onboard loads on the 6:15 vehicle are 30 passengers at stop A and 65 at stop B ; on the

6:45 vehicle, 80 and 35 passengers at A and

B, respectively; and on the 7:10 vehicle,

25 and 80 passengers at A and B, respectively.
Figure 7.15 shows $L_A(t)$ and $L_B(t - 15)$

as the cumulative load curves of the three vehicles, in which the curve at B is shifted by

15 min to allow for an equal time basis (at the route's departure point) in the analysis.

The procedure explicated in Principle 5.3 in Section 5.2 results in individual even-load

departures at 6:11, 6:31, and 6:56. This procedure is indicated by the lines with arrows

in Figure 7.15.

Figure 7.16 presents the example of Figure 7.15 with a $\Delta j\tau(-)$ shift in part (a) of the figure

and a $\Delta j\tau(+)$ shift in part (b), utilizing $\Delta d\tau = 10$ passengers for both parts. In part (a), the

shifting is $\Delta j\tau(-) = 3$ min. The solid lines (with arrows) show graphically how to determine

the new loads on the (new) 6:28 and 6:56 departures at both stops A and B. The slope at A

between 6:28 and 6:31 is $8/3$ and times 3 min; this results in an average load of eight more

passengers on the 6:56 departure and eight fewer on the 6:28 departure. At stop B, the slope

is $7/6$, and the change in load is 3.5 passengers. The criterion in Equation 7.8 considers the

load at A with $120 - 72 + 8 = 56$ and at B with $134 - 84 + 3.5 = 53.5$ passengers for the 6:56

departure. The maximum of the two is 56; because $\Delta d\tau = 10$ (the average load can reach

60 passengers), the value of $\Delta j\tau(-) = 3$ min is accepted.

Part (b) of Figure 7.16 presents an example of $\Delta j\tau(+) = 3$ min for the 6:31 departure. The

solid lines (with arrows) show graphically how to determine the new loads on the (new) 6:34

and 6:56 departures at both A and B. The slope at A is the same as for part (a) of Figure 7.16

and results in a difference of 8 passengers, while the outcome of the slope at B is 3.5 passengers

sengers. The increased load at A is $50 + 8 = 58$, and at B is $84 + 3.5 = 87.5$ passengers

for the new 6:34 departure. Consequently, and similar to the early-departure case, the shift

of $\Delta j\tau(-) = 3$ min is accepted.

This section provided a procedure to integrate two operational planning components

(timetabling and vehicle scheduling) based on the DFM model and given tolerances.

The outcome may be a set of efficient schedules from both the passenger and operator

perspectives. 180 22 200 150 50 0 200 150 100 50 0 84 135
Cumulative passenger load 120 122 72 6:15 6:45
7:10-Departure 110 30 6:00 6:22.5 6:58 (2) 6:31 6:11 6:56
(1) (3) (t-15) t 100 65 6:00 6:20 6:40 7:00 7:20 134 Route:
Travel time: Observed departures: 6:15, 6:45, 7:10
Desired occupancy: 50 passengers A B C Time (shifted)
Time STOP A STOP B 15 min

Figure 7.15 Example using the even max load procedure on individual vehicles (see Section 5.2). 30 110 135 200 150 0
65 100 New (more) load on 6:56 120 New (less) load on
6:34 New (less) load on 6:34 New (more) load on 6:56 (1)

6:11 (3) 6:56 6:15 Cumulative passenger load 6:45
 7:10-Departure 200 150 50 0 22 134 180 t 6:00 (2) 6:34 (t -
 15) 6:00 6:20 6:40 7:00 7:20 Time (shifted) Time STOP B
 6:31 STOP A 84 72 New (less) load on 6:28 200 150 0 200
 150 100 50 0 New (less) load on 6:28 New (more) load on
 6:56 New (more) load on 6:56 135 110 30 t 6:00 100 180 65
 120 72 22 134 84 Cumulative passenger load 6:15 6:45
 7:10-Departure Time (shifted) (t - 15) 6:00 6:20 7:00 6:20
 (1) 6:11 (3) 6:56 (2) STOP B Time STOP A 6:28 6:31 6:40 (a)
 (b) $\Delta d \tau = 10$ pass.

Figure 7.16 Effect on even max load when shifting the 6:31 departure backward (a) and forward (b) by 3 min.

EXERCISES

7.1 Given:

- A network with vertices a, b, c, d, e —shown in the figure of the following text.
- A table with trip number, departure terminal, arrival terminal, average travel times (and a total of 47 trips).
- A table with average travel times (including layover times) between terminals (same in both directions); the numbers in parentheses are the DH times between each terminal.
- Tolerances for possible departure-time shifting are ± 5 min for all trips.

Perform the vehicle-scheduling task with these data with the aim of minimizing fleet

size using the following three approaches:

- Only shifting trip-departure times
- Only DH trip insertion
- Both shifting trip times and DH trip insertion

For each of the three approaches, the final outcome should embody the following:

- A final DF for each terminal, marked by changes made
- Chains of trips (blocks, vehicle schedule) using • FIFO approach • Within a hollow approach (example of

other possible chains)

Basic route network a b c d e

Average travel times (DH times in parentheses), in minutes,
for both directions of travel a a b b c c d d e 50 (40) 70
(45) 60 (40) 25 (20) 40 (30) 55 (40) 30 (25) 35 (25) 20 25
35 (20) e

Trip schedules Trip # Departure terminal Arrival
terminal Departure time Trip # Departure terminal
Arrival terminal Departure time 1 d d 6:30 25 b c 7:30
2 b a 6:30 26 c c 7:30 3 a d 6:30 27 d a 7:35 4 e a 6:30
28 e b 7:35 5 c c 6:30 29 b d 7:45 6 b a 6:45 30 b a 7:45
7 d c 6:45 31 a c 7:45 8 e a 6:50 32 a e 7:50 9 b a 6:55
33 d a 7:55 10 d d 7:00 34 d b 8:00 11 a d 7:00 35 a b 8:00
12 b a 7:00 36 c a 8:00 13 b d 7:00 37 c d 8:00 14 a b 7:00
38 a d 8:00 15 c a 7:00 39 c b 8:05 16 c b 7:05 40 e a 8:10
17 a c 7:10 41 b a 8:15 18 c e 7:25 42 b a 8:30 19 e c 7:25
43 b e 8:30 20 d d 7:30 44 c e 8:30 21 b e 7:30 45 e c 8:30
22 b a 7:30 46 e b 8:30 23 a b 7:30 47 a b 8:35 24 e a 7:30

7.2 Given the following is their trip schedule (where each
route has the same schedule),

c oordinated with the train schedule: Shuttle route 1
Shuttle route 1 Trip # Departure time Arrival time Trip #
Departure time Arrival time 1 6:30 7:05 6 7:55 8:15 2
6:45 7:25 7 6:45 7:35 3 7:00 7:28 8 7:00 7:50 4 7:20 7:50
9 7:20 8:00 5 7:40 8:05 10 7:50 8:15

(1) Find the minimum fleet size required to execute the
entire trip schedule, using (1) the modeling of single
routes and (2) the DF model.

(2) Repeat (1) with shifting possibilities in trip-
departure times, in which for all trips, the
forward shifting tolerance ($\Delta t(+)$) is 3 min and
the backward shifting tolerance ($\Delta t(-)$) is 6 min.

(3) Construct the DF after the shifting; can the
shifted schedule have an adverse effect on
the Shuttle 2 route operation?

(4) What is the improved lower bound of the trip schedule?
Is it needed?

(5) Apply the FIFO rule to create vehicle chains (blocks),
using the shifted schedule determined in (2).

7.3 Find the three levels of lower bound— $G(S)$, $G'(S')$, and $G''(S'')$ —for the 10-trip schedule,

given as follows: Trip # Departure terminal Departure time Arrival terminal Arrival time Deadheading (DH) trips Between terminals DH time (same for both directions) (min)

1	b	6:00	c	6:30	a and b	60
2	a	6:10	a	6:50	3 d	6:10
3	b	7:10	a and c	30	4 b	7:00
4	b	7:30	5 c	7:10	b	7:30
5	a and d	30	6 c	7:40	b	8:10
6	b	8:10	7 d	7:50	a	8:30
7	b and c	10	8 a	8:00	a	8:30
8	9 a	8:30	d	9:10	b and d	50
9	10 b	9:00	c	9:20	c and d	20

7.4 For the 10-trip schedule in Exercise 7.3, consider SDTs by 5 min for both forward and

backward shifts ($\Delta i(+) = \Delta i(-) = 5$) for all trips.

(1) Find $G'(S \text{ sf }')$ and $G''(S \text{ sf }'')$ by constructing $g(t, S \text{ sf }')$ and $g''(t, S \text{ sf }'')$.

(2) Find $G(S \text{ sf })$, $G'(S \text{ sf }')$, $G''(S \text{ sf }'')$ using the LB-SHIFT and LB-DH&SHIFT procedures.

7.5 Derive a load-tolerance criterion for even-load timetables for the case of a late departure.

End the derivation with a formulation similar to Equation 7.8.

APPENDIX 7. A: EXAMPLE OF VEHICLE-SCHEDULING SOFTWARE

There is no doubt more and more planning/scheduling software packages will have to

include a real-time component for adjustments and changes. Table 1.1 lists some of the

known software available with their Internet address. The software, called Optibus, spe

cializes in real-time rescheduling and thus will be briefly described, in this appendix, as an

example of such software.

Optibus is an algorithm-based Software as a Service (SaaS) platform optimizing sched

uling of vehicles and crew in real time. Figure 7.A.1 of this system exposes a web inter

face in which the scheduler can generate optimized schedules while taking into account the

incorporate his or her expertise in the generated schedule while receiving recommendations

and feedback from the optimization engine in an interactive manner. This interactive system

handles simultaneously both vehicle and crew scheduling.

7.A.1 Key features

The key features of the Optibus system are related to (1) fast generation of optimized vehicle

and crew schedule while taking into consideration more than a few constraints and prefer

ences, such as the following:

- Multiple vehicle types and their hierarchy
- Multiple depots, with different vehicle capacities, with the option to create schedules that return to the starting depot
- Crew-work regulation and labor constraints
- Similarity with existing schedules and between days of week

(2) Optimal allocation of vehicle parking and service depots to minimize costs; (3) an interac

tive Gantt-based interface (see Figure 7.A.1) visualizing the schedule and allows for sched

uler's by adjustments; (4) real-time rescheduling integrated with automatic vehicle location

(AVL) and automatic passenger counter (APC) systems to allow for anticipation of unex

pected deviation from the original schedule and changes of passengers demand performed

minimizing costs; (5) generating different detailed

reports for analysis, planning, and deci

sions to take place (see, for instance, the illustration in Figure 7.A.2 of the number of vehicles

required by the optimal solution [of a given depot] every hour of the day with the upper and

lower curves representing urban and interurban fleet requirements, respectively); and (6) use

of computer-based components such as RESTful API and reusable HTTP for integrating/

embedding the planning with other management needs such as a control center.

Figure 7.A.1 Interactive vehicle-scheduling optimization process.

7.A.2 Expected saving

Two experiments, done by the Optibus software, provide a rough idea of the saving expected.

The comparison, shown in the succeeding text, in the first table is based on 975,713 yearly

service trips with an accumulated distance of 19,865,410 km. Source Idle kilometers (thousands) Idle trip hours (thousands) Standby hours (thousands) Available solution 3422 94 239 Optibus outcome 2,678 (i.e., -22%) 78 (i.e., -17%) 186 (i.e., -22%)

The second table shows an experiment done at a single operational branch with a different

set of schedules comprised of 1,100 daily service trips. Source Single day idle kilometers Single day nonservice paid hours (idle + standby) Existing figures 3056 294 Optibus outcome 2545 (i.e., -17%) 240 (i.e., -18%)

Ceder, A. (2002). A step function for improving transit operations planning using fixed and variable scheduling. In Transportation and Traffic Theory (M.A.P. Taylor, ed.), pp. 1-21, Elsevier Science, Oxford, U.K.

Ceder, A. (2005). Estimation of fleet size for variable bus schedules. Transportation Research Record, 1903, 3-10.
5:00

Hour

0

10 18 27 30 30 30 23 20 28 30 29 29 29 29 25 21 21 21 20 20
38 47 43 48 50 49 46 46 43 50 49 44 36 31 42 49 55 55 16 1
28 4 3

20

30

40

50

60

Count of vehicles 6:00 7:00 8:00 9:00 10:00 11:00 12:00
13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00
23:00 0:00 1:00 2:00

Figure 7.A.2 Example of a generated report of optimal urban (upper curve) and interurban (lower curve) vehicle requirements, by time of day.

Ceder, A. and Philibert, L. (2014). Transit timetables resulting in even max load on individual vehicles. IEEE Transactions on Intelligent Transportation Systems, 15(6), 2605-2614.

Ceder, A. and Stern, H.I. (1981). Deficit function bus scheduling with deadheading trip insertion for fleet size reduction. Transportation Science, 15, 338-363.

Stern, H.I. and Ceder, A. (1983). An improved lower bound to the minimum fleet size problem. Transportation Science, 17, 471-477.

8 Chapter 8: Vehicle Type and Size Considerations in Vehicle Scheduling

This chapter contains three main, almost independent parts. First, it addresses the problem of

how to allocate vehicles efficiently for carrying out all the trips in a given transit timetable, while

taking into account the association between the characteristics of each trip (urban, peripheral,

intercity, etc.), and its required vehicle type. Second, a methodology is introduced also for

assigning vehicles of different types, but with the consideration of optimal timetables described

in Section 5.4. Third, it provides an overview and tools for the determination of and a practical

decision concerning transit vehicle size. Both parts are directly related to the prominent act of

purchasing a vehicle or a fleet of vehicles. Ostensibly when dealing with vehicle type there is need for more operational details. Paying

attention to details is similar to answering the following riddle. (Do so without using a calcula

tor.) You are driving a bus from terminal a to terminal b. At the first stop, 10 people get on the

bus and 2 get off. At the second stop, 12 get on and 7 get off. At the third stop, 5 people get on

and 10 get off. At the fourth stop, 11 people get off and 1 gets on. You then arrive at terminal b.

What is the name of the driver? (See end of this corner.)

The chapter consists of six sections, following the introduction. Section 8.2 provides an

overview of literature review and describes optimization framework for the vehicle-scheduling

problem, in which categories of vehicle types are arranged in decreasing order of vehicle cost

and comfort level. The assignment of vehicles to trips using DH (deadheading) and shifting

procedures within the optimization framework is discussed in Section 8.3; the objective is to

minimize the total purchasing cost. Section 8.4 presents two detailed examples (one of which is

real life) of the optimization procedures developed. Section 8.5 introduces a method enabling,

in assigning vehicles, to stipulate the use of a particular vehicle type for a trip or to allow for

a substitution either by a larger vehicle or by a combination of smaller vehicles with the same

or higher total capacity; it is also linked with the selection of optimal timetable. Section 8.6

provides basic tools for the determination of vehicle size, including a known square-root for

mula. Section 8.7 continues, more in depth, with the overview of the tools and formulation for

vehicle-size determination. The chapter ends with exercises. There is no doubt that the successful deployment of optimal or adequate vehicle types

requires a considerable amount of analysis. After all, success comes before work only in the

dictionary. Finally, as to the riddle, it said: "You are driving a bus"

8.1 INTRODUCTION

The vehicle-scheduling task described in Chapters 6 and 7 considers only one type of transit

vehicle. In practice, however, more than one type is used; for example, a bus operation may

employ minibuses, articulated and double-decker buses, and standard buses with varying

degrees of comfort and different numbers of seats.

Commonly, the consideration of vehicle

type in transit operations planning involves two considerations: first, determining the suitability

of optimal vehicle size; second, choosing vehicles with different comfort levels, depending

on trip characteristics. Certainly, a multicriteria effort may treat both considerations

simultaneously, but this is seldom done in practice. The issue of what vehicle type to consider

arises when purchasing a vehicle or a fleet of vehicles, an undertaking that is not performed

frequently. There is a saying that good judgment comes from experience, and experience

comes from bad judgment. Another is that experience is not what is happening to us, but what

we do with what is happening to us. In our case, the experience accumulated focuses on the

need for an analysis framework in order to decide how many and which type of vehicle to

purchase. This chapter attempts to introduce (a) a cost-effective framework for choosing

a vehicle's comfort level, (b) a method for optimal vehicle assignment integrated with the

selection of optimal timetable, and (c) the basic tools for analyzing optimal vehicle size.

The purpose of the first part of the chapter is to address the vehicle-scheduling problem, while

taking into account the association between the characteristics of each trip (urban, peripheral,

inter-city, etc.) and the vehicle type r

required for the particular trip. This means complying

with a certain level of service for that trip: degree of comfort, seat availability, and other

operational features. The purpose of the second part is to introduce a methodology for optimal

vehicle scheduling using multiple vehicle types. The methodology is based on a minimum-cost

network flow model with an input of a set of Pareto-optimal timetables (see Section 5.4) for

individual transit lines. The purpose of the third part of the chapter (Sections 8.6 and 8.7) is

to review a formulation for trade-offs between vehicle size and operational variables that can

be used as a tool for the determination of optimal vehicle size for a given frequency of service.

8.2 OVERVIEW AND OPTIMIZATION FRAMEWORK

Section 8.7 is the main literature review part of the chapter, but it focuses only on vehicle

size and not on vehicle scheduling with different sizes. Thus, in what follows is a brief over

view of optimization-related literature of what is known as multiple vehicle types vehicle

scheduling problem (MVT-VSP).

Bunte and Klier (2009) provide a detailed literature review and discuss the model

ing approaches for different kinds of vehicle-scheduling problems. In the MVT-VSP, vehicle

types have different capacities and might have different fixed and variable cost. According

to Lenstra and Kan (1981), the problem is NP-complete (see Section 5.5) for the single depot

case. There have been numerous approaches published, for example, El-Azmi (1985), all of

which consider multiple vehicle types. A common approach by Bodin et al. (1983), used also

by Costa et al. (1995), includes the use of a multigraph with a subnetwork for each vehicle

type. However, if there are further restrictions such as certain trips requiring a certain vehi

cle type, the model has to be further extended. This was considered by Forbes et al. (1994),

Löbel (1997), and Ceder (2011). It was also included in a time-space network approach by

Kliewer et al. (2006). Finally, Kliewer et al. (2012) consider a multiple depot case for the

simultaneous vehicle and crew scheduling with time windows. A time-space network formulation

is used and shifting of trips is explicitly modeled using additional arcs in the network.

This and the two following sections use the notation and procedures of Definit

(DF) described in Chapters 6 and 7. The problem addressed concerns the assignment of

vehicles to trips using DH and shifting procedures, in which categories of types are arranged

in decreasing order of vehicle cost and comfort. The process described follows Ceder (1995,

2011). Section 8.5 provides another approach, not DF based, that uses the optimal timetable

developed in Section 5.4.

The problem, entitled the vehicle-type-scheduling problem (VTSP), is based on given set S of

trips (schedule) and set M of vehicle types. The set M is arranged in decreasing order of vehicle

cost so that if $u \in M$ is listed above $v \in M$, it means that $c_u > c_v$, where c_u, c_v are the costs

involved in employing vehicle types u and v , respectively. Each trip $i \in S$ can be carried out by

vehicle type $u \in M$ or by other types listed prior to u in the aforementioned order of M .

The problem can be formulated as a cost-flow network problem, in which each trip i is

a node and an arc connects two trips i if, and only if, it is possible to link them in a time

sequence with and without DH connections. On each arc (i, j) , there is a capacity of one

unit and an assigned cost C_{ij} . If the cost of the lower-level vehicle type associated with

trip i is higher than the cost of the vehicle type (even if of a lower level) required for trip j ,

then $C_{ij} = c_i$. That is, $C_{ij} = \max(c_i, c_j)$. The use of such a formulation was implemented by

Costa et al. (1995), who employed three categories of solutions: (a) a multicommodity network

flow; (b) a single-depot vehicle-scheduling problem; and (c) a set-partitioning problem

with side constraints. The mixed-integer p

programming of these problems is known to be

NP-complete (see Section 5.5) as may be seen, for example, in Bertossi et al. (1987). The

math-formulation concepts for the third category are further explained in Chapter 9.

Because of the complexity involved in reaching an optimal solution for a large number

of trips in S , a heuristic method is considered a more practical approach. The heuristic pro-

cedure developed is called the V TSP algorithm. It begins by establishing lower and upper

bounds on fleet size. The upper bound is attained by creating different DFs, each associated

with a certain vehicle type $u \in M$, which includes only the trips whose lower-level required

vehicle type is u . Certainly, this scheduling solution reflects a high cost, caused by the large

number of vehicles demanded. The lower bound on fleet size is attained by using only

one vehicle type: the most luxurious one with the highest cost that can clearly carry out any

trip in the timetable. Between these bounds on fleet size, the procedure searches for the best

solution, based on the properties and characteristics of DF theory.

This optimization framework is illustrated in Figure 8.1, with (C_1, N_1) and (C_2, N_2) rep-

resenting the lower- and upper-boundary solutions, respectively. The following are added

notations to DF theory (described in Chapters 6 and 7):

M is the set of all vehicle types u , $u = 1, 2, \dots, m$, arranged in decreasing order of vehicle cost

S_u is the set of required trips (schedule) for vehicle types u , $u = 1, 2, \dots, m$

$S_{u=1:m} = \bigcup_u S_u$, which is the combined schedule of all trips

t_{su}

i is the start time of trip $i \in S_u$ of type $u \in M$

t_{eu}

i is the end time of trip $i \in S_u$ of type $u \in M$

$\Delta u i()^+$ is the maximum delay tolerance from the scheduled departure time of trip $i \in S$ of type $u \in M$

$\Delta u i()^-$ is the maximum advance tolerance of the trip scheduled departure time of trip $i \in S$ of type $u \in M$
 Cost Number of vehicles One vehicle type only (most luxurious) Each trip has its own vehicle type (treated separately) $C_2 \leq C_1 \leq C_{opt} \leq N_1$ (Lower bound) $N_{opt} \leq N_2$ (Upper bound)

Figure 8.1 Trade-off between cost of purchasing vehicles and number of vehicles.

$d_u(k, t, S_u)$, $D(k, S_u)$, $D(S_u)$, $[s_{eiu} i_{uk} k,]$, $g(t, S_u)$, $G(S_u)$ = refer to the DF definitions (see Chapters 6 and 7), but only with respect to trips of type $u \in M$

c_u is the cost involved in employing a vehicle of type $u \in M$

N_1 is the minimum number of vehicles required to service all trips in S , using a single vehicle of type $u = 1$

n_{2u} is the minimum number of vehicles required to service all trips in S_u , $u \in M$

$N_{num} = n_{u=1} + n_{u=2} + \dots + n_{u=m} = \Sigma =$ sum of all minimum numbers of vehicles required when treating each type separately

$C_1 = c_1 N_1$ is the total cost involved in employing N_1 vehicles of type 1 (most luxurious)

$C_{cnu} = \sum_{u=1}^m c_u n_u$ is the total cost involved in employing N_2 vehicles (for each type separately)

$C_{cnu} = \sum_{u=1}^m c_u n_u$ is the objective function (total cost) following algorithm VTSP, with n_u vehicles required of type u for all $u \in M$

$N = \sum_{u=1}^m n_u$ is the total number of vehicles incurring cost C

8.3 DF BASED PROCEDURE FOR VEHICLE-TYPE SCHEDULING PROBLEM

The algorithm VTSP developed is heuristic in nature while incorporating all DF components.

Because of the graphical features associated with DF theory, the algorithm can be

applied in an interactive manner or in an automatic mode, along with the possibility of

examining its intermediate steps. The following is a general description of algorithm VTSP

in a stepwise manner.

8.3.1 Algorithm VTSP

Step 0: Arrange the set of vehicle types M in decreasing order of vehicle cost (so that if $u \in M$ is listed above $v \in M$, it means that $c_u \geq c_v$).

Step 1: Solve the problem as a single-vehicle-type problem using DF theory, including the DH and shifting procedures (see Figures 6.9a and b and 7.12), to obtain N_1 vehicles, considered as type 1, with a total cost of C_1 .

Step 2: Divide the trips by their associated type and apply the DF methodology with the DH and shifting procedures for each type separately. Add up the number of vehicles derived to obtain the total of N_2 vehicles with a total cost C_2 .

Step 3: If $N_1 = N_2$, stop. Use the solution in Step 2.

Step 4: Consider $d_u(k, t)$ as in Step 2 for all $k \in K$ and $u \in M$.

Step 5: Perform the U RSC (shifting only) procedure for shifting departure times within their tolerances (see Figure 8.11).

Step 6: Find a URDHC (see Figures 6.9a and 6.12), such that each DH trip (with possible shifting) in this chain fulfills condition (a) and/or (b): (a) The DH trip is from DF of vehicle type u to DF of type v , such that $u \leq v$, meaning that $c_u \geq c_v$ (see Proposition 8.2). (b) The URDHC aims at saving a vehicle of type w and $-c_w + \sum_{i \in E} \delta_i \leq \forall \delta_i \in \mathbb{R}, \delta_i \geq 0$ in which the set E is composed of all DH trips included in the URDHC; each DH trip is from a DF of vehicle type r to a DF of vehicle type q , in which $q < r$ (see Proposition 8.3). If no URDHC can be found, stop.

Step 7: Examine whether the total cost of the URDHC (DH cost) is less than the cost of saving one vehicle (of the type considered). If it is not, delete this possibility and go to Step 6. Otherwise, update $d_u(k, t)$ for all $k \in K$ and $u \in M$.

Step 8: Apply the improved lower-bound check. If $D(S) = G''(S)$, stop; otherwise, go to Step 5.

Among the eight steps of an algorithm V TSP, the conditions specified in Step 3 and particularly

Step 6 deserve further attention. The following four propositions clarify and interpret these

conditions:

Proposition 8.1 (for Step 3): If $N_1 = N_2$, then $C_2 \leq C_1$.

Proof: Given $M \geq 2$ and $c_1 \geq c_2 \geq \dots \geq c_m$, the proof is straightforward because $C_N = C_N + C_N$, and $N_2 = \sum_{j=1}^m 2j = \sum_{j=1}^m 2j = \sum_{j=1}^m 2j$,

Proposition 8.2 (for Step 6 a): Any DH trip c

connection from a D F of vehicle type u to a D F

of type v , such that $u \leq v$ within any URDHC, does not increase C .

Proof: Any DH trip from u to v will link, in the national vehicle chain (block), an arrival epoch

at u and a departure epoch at v , including possible idle times at u and v . Because $c_u \geq c_v$,

this DH trip connection cannot lead to an upgrade of the vehicle type; therefore, it cannot

increase the objective function C .

Proposition 8.3 (for Step 6 b): Any URDHC that aims at saving a vehicle of type w does not

increase C if $-c_w + \sum_{q \in E} c_q \leq \epsilon$ for all $q, r \in C(\cdot)$, $\emptyset$ where the set E is composed of all the DH trips included

in the URDHC; each DH trip is from a D F of vehicle type r to a D F of type q in which $q < r$.

Proof: In the URDHC, there may be DH trips that comply with Propositions 8.1 and 8.2

and, therefore, do not increase C . Each of the other DH trips is from a D F of vehicle type r

to a D F of type q , in which $q < r$, that is, $c_q \geq c_r$. This DH trip connection may upgrade the

vehicle type (carrying out this DH trip) from r to q . That is, the result may be a saving of a

vehicle of type w , along with several vehicle-type upgrades. In order to ensure that C does

not increase, the condition set forth is that the cost saving of vehicle type w is g

reater than

or equal to the additional cost required by the possible upgrades.

Algorithm VTSP is shown in Figure 8.2 in two parts. It starts in Figure 8.2a by calculating

N_1 and C_1 , and then N_2 and C_2 . If $N_1 = N_2$, we stop with $N = N_2$ and $C = C_2$, indicating that

each trip is carried out by its own designated type. Otherwise, the algorithm continues by

using Figure 8.2b, in which Step 6, with its two conditions, is utilized. The algorithm ends

once all terminals are examined.

8.3.2 Sensitivity analysis

The VTSP presents a vehicle-scheduling problem in which the objective is to minimize the

cost involved; primarily, it aims at vehicle-purchase cost although it also affects

operational cost. Often, we want to know the result of the solution when changes occur in

vehicle cost. Alternatively, we may want to know the extent to which c_u for all $u \in M$ can

change and still have the same scheduling solution. Certainly, changes in vehicle cost by type

can shift the location of $u \in M$ in decreasing order of vehicle cost. The examination of the

solution for these cost-change possibilities is usually called sensitivity analysis.

The first observation in algorithm VTSP is that as long as

the arrangement in M of the

decreasing order of vehicle cost is preserved, the scheduling solution will not change. That

is, any change of c_u —it can be of more than one vehicle type—that will not change the order

of vehicle cost will result (except in special cases) in a change of C , but not in the blocks

attained. The proof of this observation is straightforward because of unchanged situations

applied to conditions (a) and (b) in Step 6 in algorithm VTSP. No Set $u=1$ for all $u \in M$ B No No Set $u = \{1, 2, \dots, m\}$ for all $u \in M$ Stop. $N=N-2$ $C=C-2$ N, C Stop. Fleet-reduction procedures; involved are shifting departure times and DH trip insertions (see Figure 8.12 re terminals q and p) All terminals examined $Q=K$? No No No Yes Yes Yes (a) $n_u = N - D(S_u)$ $N_u C_u n_u u u C_u = A$ Initial input: K ; T ; DH time matrix; S_u for all $u \in M$; set of vehicle types M ; set of tolerances for trip-departure times; vehicle cost by type. Set $Q=\emptyset$ $N-1$ found? $N-2$ found? $N-1$ found? $N-1 = D(S) C-1 = c-1$. $N-1$ $N-2$ found? $N-1 = N-2$? Yes Yes Yes $n_{2u} = u u D(S_u) N-2 = n_{2u} C-2 = c_u n_{2u}$ Computation of deficit-function information: $\{d(K, t, S_u)\}$, $D(S_u)$, $g(t, S_u)$, $G(S_u)$, $G''(S_u)$ for all relevant $S_u = S, S', S'' S'' sf sf$,

Figure 8.2 (a) Flow diagram of the VTSP algorithm.
(Continued)

The second observation concerns changes in c_u in which the order of vehicle costs is

also changed. Following the cost change(s), three cases will be analyzed: (1) a DH trip that

fulfills condition (a) in Step 6 of algorithm VTSP, changing from a DF of vehicle type u

to a DF of type v in which $c_u < c_v$; (2) a DH trip that fulfills condition (b), thus becoming

fulfilled condition (a) in Step 6 of algorithm VTSP; and (3) a chain of DH trips that fulfills

condition (b) in Step 6 of algorithm VTSP. Case (1) will not change the scheduling solu

tion—it will only change the value of C —because the former fulfilled condition (a) now

becomes fulfilled condition (b). That is, saving a vehicle of type v and converting vehicle

type u to v results in $-(c_v + \Delta v) + (c_v + \Delta v) - c_u < 0$, in which Δv is the increase in c_v that

yields $c_u < c_v$. Case (2) will also not change the scheduling solution because of the existence

of condition (a). In case (3), condition (b) in Step 6 of algorithm VTSP needs to be checked.

If fulfilled, the scheduling solution remains the same; otherwise, algorithm VTSP needs to

be executed again.

8.4 EXAMPLES OF THE DF-BASED VTSP

Three examples are presented in this section for comprehending the underlying principles of

the VTSP algorithm. The first two are detailed examples, with and without shifting depart

ure times; the third conveys the results of a real-life example. B Reduce a vehicle at q or p ; form S_q and/or add set of DH trips $\{DH\}$ to the schedule: $S = S_q + \{DH\}$ or $S = S + \{DH\}$. Set $Q = \emptyset$ A No No Yes Yes Yes Yes No (b) No URDHC saves a vehicle of type w , and $(c_w + c_v) \leq 0$? $-c_w + N_1$ found? N_2 found? DH trips fulfill condition (a) in step 6 $(c_u \geq c_v)$? for all $q, r \in E$

Figure 8.2 (Continued) (b) Continuation of the flow diagram of the VTSP algorithm.

8.4.1 Detailed Example 1

Example 1, which is illustrated in Figure 8.3, consists of 8 trips, 3 terminals (a, b, c), and

3 types of vehicles, with the cost of 12, 5, and 3 cost units, respectively. Figure 8.3a presents

the simple network of the routes, in which the DH travel time between each two terminals

is 20 min and shifts in departure times are not allowed.
The timetable and trip travel times

are shown in Figure 8.3b according to vehicle type. The DFs of Step 1 of algorithm VTSP

for Example 1 are depicted in Figure 8.3c; all trips, it should be recalled, are served by the

same vehicle type (type 1). For inserting a DH trip, the NT rule (the first hollow is the lon

gest) is applied; this results in the selection of terminal b (for the DH insertion procedure,

see Section 6.5.4). The URDHC procedure with $R = 2$ (furthest start of a hollow) then

results in three DH trips, in which DH 2 is used for maintaining the level of $D(a)$. Figure 8.4

constructs $g'(t, S') = g''(t, S'')$, which determines $G'(S') = G''(S'') = 2$. Thus, Step 1 stops when

$D(S) = N - 1 = 2$. Two vehicle chains are then created, using the FIFO [1-3-DH 1 -5-7] and

[2-4-DH 2 -6-DH 3 -8], and the total cost is $C - 1 = 24$.

Algorithm VTSP continues in Step 2, in which vehicle types are treated separately. Figure 8.5,

which illustrates this step, is marked by (d) to show that this step follows Figure 8.3c.

The maximum DF of types 1 and 2 are reduced by one, using DH 1 and DH 2, respectively; DH 1 Type 2 ($c - 2 = 5$) Type 3 ($c - 3 = 3$) Type 1 ($c - 1 = 12$) b c a (a) Time 6:00 6:20 6:40 7:00 7:20 7:40 8:00 8:20 8:40 9:00 9:20 9:40 Time d (c , t) d (b , t) $D(c) = 0 \ 0 \ -1 \ -2 \ -3$ d (a , t) $D(a) = 1$ DH 2 DH 3 $D(b)=3 \ 2 \ 1$ 6:00 6:20 6:40 7:00 7:20 7:40 8:00 8:20 8:40 9:00 9:20 9:40 b b a a b c c c a a c a b c a 7 8 6 5 3 4 2 1 3 2 1 0 2 1 0 (c) (b) b

Figure 8.3 Example 1: (a) network of routes, vehicle types, and cost; (b) eight-trip schedule; (c) DFs with DH insertions for a single vehicle. 6:00 (d) -1 0 1 -1 0 1 -1 0 1 -1 0 1 -2 -1 0 0 1 2 6:20 6:40 7:00 7:20 7:40 8:00 Time d 3 (c , t) d 3 (a , t) d 2 (b , t) d 2 (a , t) d 1

(c , t) d 1 (b , t) 8:20 8:40 9:00 9:20 9:40 6:00 6:20
 6:40 7:00 7:20 7:40 8:00 Time D(b, S 1) = 2 1 D(c, S 1) =
 0 D(a, S 2) = 1 D(b, S 2) = 1 0 D(a, S 3) = 1 D(c, S 3)
 = 1 8:20 8:40 9:00 9:20 9:40 DH 2 DH 1

Figure 8.5 Example 1 (continued from Figure 8.3): (d) DFs and DH insertions for each vehicle type. 6:00 0 1 2 3 $g'(t) = g''(t)$ $G'(S') = G''(S'') = 2$ b 1 2 3 4 5 7 8 6 b b c a b a a a c c b a c a c 6:20 6:40 7:00 7:20 7:40 8:00 Time 8:20 8:40 9:00 9:20 9:40

Figure 8.4 Determination of the lower bound corresponding to Step 1 of algorithm VTSP for Example 1.

the number of type 3 vehicles remains same. Thus, $N_2 = 1 + 1 + 2 = 4$, and the four following

chains are derived by using the FIFO rule: [1-DH 1 -4] (vehicle type 1), [2-5-DH 2 -8] (vehicle

type 2), [3-7], and [6] (vehicles of type 3); this results in a total cost of $C_2 = 23$.

The next step in algorithm VTSP compares $N_1 = 2$ with $N_2 = 4$, and then moves to the

relevant Step 6 because of not allowing shifting departure times. Figure 8.6 illustrates the

process of Step 6, using its two conditions. Note that S_u will be deleted (as in Figure 8.6)

when it is clear which underlying vehicle type is being considered. This step again applies

the NT rule and the URDHC procedure with $R = 2$ (furthest start of a hollow), but this

time (especially condition (a)) with the possibility of inserting any DH trip from a DF with

a more expensive vehicle type to a DF with a less expensive type. Following the 'low dia

gram in Figure 8.2, the first terminal selected is b, based on $d_2(b, t)$, from which DH 1 is

determined from terminal a. The DFs are then updated and the next terminal is again b, but

related to $d_1(b, t)$; DH 2 is inserted from terminal c .
We continue with the next selected termi

nal c , based on $d_3(c, t)$; however, no DH trip can be
inserted into its maximum-interval start

ing point, including the check for both conditions (a) and
(b) of Step 6. Thus, a is selected

next, based on $d_3(a, t)$, and DH 3 is inserted to arrive
from c , based on the updated $d_1(c, t)$.

This terminates Step 6 and results in the three following
(FIFO) chains: [1-DH 2 -4-DH 3 -6]

(vehicle type 1), [2-5-DH 1 -8] (vehicle type 2), and [3-7]
(vehicle type 3), with a total cost of

$12 + 5 + 3 = 20$. Time Time $D(b)=2$ 1 6:00 (e) -1 0 1 -1 0 1
-1 0 1 -1 0 1 -2 -1 0 0 1 2 6:20 6:40 7:00 7:20 7:40 8:00
8:20 8:40 9:00 9:20 9:40 6:00 6:20 DH 2 $d_1(b, t)$ $d_1(c, t)$ $d_2(a, t)$ $d_2(b, t)$ $d_3(a, t)$ $d_3(c, t)$ DH 1 DH 3 6:40 7:00 7:20 7:40 8:00 8:20 8:40 9:00 9:20
 $D(c) = 0$ $D(a) = 0$ $D(b)=1$ 0 $D(c) = 1$ $D(b)=1$ 0 9:40

Figure 8.6 Example 1 (continued from Figures 8.3 and 8.5):
(e) DFs and DH insertions for optimal solution.

For the sake of simplicity, we assume that a check of Step
7 of algorithm VTSP allows for

the latter solution. Moving to Step 8, one may then see
that $D(S) > G''(S'')$, but the process

stops by reiterating Step 6, because no URDHC can be found.
At this stage, we may con

clude that the 3-chain (blocks) solution with $C = 20$ is
the best one attained.

8.4.2 Detailed Example 2

The example demonstrated in this section exploits shifts in
departure times and condition

(b) in Step 6 of algorithm VTSP. Example 2, illustrated in
parts (a) and (b) in Figure 8.7,

consists of 14 trips, 3 terminals, and 3 types of vehicles,
with costs of 10, 3, and 2, respec

tively. Figure 8.7c shows the fleet-reduction procedure, using both DH trip insertion and

shifting departure times. For simplicity, all DH travel times between the three terminals are

15 min, $\Delta u_i^+ = \Delta u_i^- = 15$ min, for all $u \in M$, and priority in shifting departure times is given

to late departures.

The DF analysis with only type 1, in Figure 8.7c, results in one shift, enabling a reduction of

one vehicle; the lower-bound determination, shown in Figure 8.8, includes possible shifts of

trip times combined with feasible extensions. The first analysis terminates with $N_1 = 6$ (lower

bound) and $C_1 = 60$. The analysis proceeds with finding $N_2 = 7$ and $C_2 = 2 \cdot 10 + 4 \cdot 8 + 1 \cdot 6 = 58$, $ba\ ac\ ab\ a\ b\ b\ b\ c\ a\ c\ c\ b\ c\ a\ a\ a\ a\ a\ a\ c\ c\ c\ b\ c\ b\ 1\ 2\ 3\ 4\ 12\ 11\ 5\ 6\ 7\ 8\ 9\ 10$ Type 1 ($c_1 = 10$) Type 2 Type 3 ($c_1 = 6$) 13 14 6:00 6:30 7:00 7:30 8:00 8:30 9:00 9:30 $D(c) = 2\ D(b)=4\ D(a) = 1$ 0 10:00 $d(c, t)\ d(b, t)\ d(a, t)$ 6:30 6:00 7:00 7:30 8:00 8:30 Time -1 0 1 -1 0 1 2 0 1 2 3 4 9:00 9:30 10:00 (a) (b) (c)

Figure 8.7 Example 2: (a) Basic input of a network of routes, vehicle types, and cost, (b) complete 14-trip schedule, and (c) DFs with Step 1 of algorithm VTSP.

in which $n_{21} = 2$, $n_{22} = 4$, and $n_{23} = 1$, using Step 2 of algorithm VTSP. This is illustrated with

one shift in departure time and one DH trip in Figure 8.9. Following the flow diagram in

Figure 8.2a, we can now check whether $N_1 = N_2$ to obtain that $N_1 < N_2$; then continuing with

Steps 4 and 5 of the algorithm, resulting in a single shift forward (late departure) of trip 10

at 7:30 as is shown in Figure 8.10. Because $D(S_{sf}) = 8 \neq G''(S_{sf}) = 6$, the algorithm advances

to a search of URDHC, including possible shifts in

departure times. The NT rule determines

terminal c of type 2 as a candidate location for reducing a vehicle; this results in DH 1 fol

lowing condition (a) of Step 6 of the algorithm. However, in order to avoid the increase of

$D(a)$ in $d_1(a, t)$, another DH trip is found, DH 2, but this time using condition (b) of Step 6 of

the algorithm. The search for another URDHC allows for DH 3 and DH 4, using conditions

(b) and (a), respectively, of Step 6 of the algorithm. The procedure then stops because the sum

of all $D(k) = 2 + 2 + 2 = 6$, $k = a, b, c$, equals the lower bound attained.

Unlike Example 1, the types of vehicles corresponding to the resultant six blocks cannot

be derived by the sum $D(k)$, $k = a, b, c$, associated with each type. First, the blocks need to

be constructed using the inserted DH trips; second, the type of each block is determined by $b\ c\ b\ c\ a\ b\ c\ c\ a\ a\ a\ b\ b\ b\ c\ c\ c\ a\ a\ a\ c\ a\ b\ a\ c\ a\ a\ c\ g(t, S\ s\ f)g'(t, S\ s'f) = g''(t, S\ s''f)$

	4	10	1	2	3	6	7	8	9	11	12	13
14	6:00	6:30	7:00	7:30	8:00	8:30	9:00	9:30	10:00	Time	6:00	
	6:30	7:00	7:30	8:00	8:30	9:00	9:30	10:00	7	6	5	4
	6	5	4	3	2	1	0	7				
	6	5	4	3	2	1	0	$G(S\ s\ f)$	$= 7$	6	$G'(S\ s'f)$	$= G''(S\ s''f) = 7$
											6	5

Figure 8.8 Determination of the lower bound corresponding to Step 1 of algorithm VTSP for Example 2.

its highest (most expensive) type trip. Following the FIFO rule, for instance, the first three

blocks are of type 1: [1-9], [3-DH 1 -8], and [5-DH 2 -12] because there is at least one trip of

this highest type in each block (trips 8 and 5 are of types 2 and 3, respectively). The second

three blocks are of type 2: [2-DH 3 -7-11-DH 4 -14], [6-13], [4-10] (trips 2 and 14 are of type 3).

Thus, in the best solution found, $n_1 = 3$, $n_2 = 3$, n_3

= 0, $N = 6$, and $C = 54$, compared with

$C_1 = 60$ and $C_2 = 58$.

8.4.3 A real-life example

Algorithm VTSP was used to examine a real-life scheduling problem. The problem selected

pertains to the Egged bus company, which has three different bus routes departing from a

main terminal in Haifa. The three routes are shown schematically in Figure 8.11a: inter

city, peripheral, and urban. The daily timetable of the intercity route is characterized by

18 departures, 120 min being the average travel time and 105 min the average DH time

between terminals A and B in each direction. The peripheral route has 22 daily departures,

with 45 min the average travel time between terminals A and C, 24 min the average DH time

between A and C, and 36 min between C and D. The urban route has 24 daily departures,

with 30 min the average travel time and 15 min the average DH time between A and D. The

relative costs of the intercity, peripheral, and urban vehicles are 1.6, 1.3, and 1.0, respec

tively. The allowed shifts are $\Delta u_i(+) = \Delta u_i(-) = 3$ min for all trips i and vehicle type u .

The VTSP algorithm results in the optimal solution shown in Figure 8.11b, with $C =$

19.4 and 14 vehicles required: 7 intercity, 4 peripheral, and 3 urban vehicles, respectively. 6:00 6:30 7:00 7:30 8:00 8:30 9:00 9:30 10:00 10:30 $d_1(a, t) d_2(a, t) d_2(c, t) d_1(b, t) D(a) = 0 D(b) = 2 D(c) = 3 d_3(b, t) d_3(c, t) D(c) = 0$ DH 1 0 -1 2 2 1 0 3 2 1 0 Time 6:00 (d) 6:30 7:00 7:30 8:00 8:30 9:00 9:30 10:00 10:30 2 1 0 -1 2 1 0 0 -1 -2 $D(a) = 2 D(b) = 2 D(c) = 1$

Figure 8.9 Example 2 (continued from Figure 8.7): (d) DFs describing Step 2 of algorithm VTSP.

Steps 1 and 2 of algorithm VTSP result in $C_1 = 22.4$ (14 intercity vehicles) and $C_2 = 22$ (7

intercity, 6 peripheral, and 3 urban vehicles), respectively. This outcome of the algorithm

is circled in Figure 8.11b. The results of the heuristic method suggest that algorithm VTSP

can be used, too, for large transit agencies, ensuring an efficient allocation of the different

vehicles to trips while reducing the cost involved to a minimum level. The use of DF may be

subject to the interference of the schedulers in the process whenever they think it justifiable.

8.5 MIN-COST FLOW PROCEDURE FOR MULTI-TYPES VEHICLE-SCHEDULING PROBLEM

This section is based on the study by Hassold and Ceder (2014a) with a methodology using

minimum-cost network flow model. The input is set of Pareto-optimal timetables explicated

in Section 5.4 for individual transit lines.

8.5.1 Description of concepts

In the optimization-related literature review of MTV-VSP, in Section 8.2, the problems are

solved by dividing the vehicle-scheduling problem into sub-problems—one for each vehicle

type—and optimize it independently. However, it does not necessarily lead to the best global

solution. This is shown by an example depicted (with notations) in Figure 8.12, in which

trips between Stations A and B are to be covered. 6:00 (e)
d 3 (c , t) d 3 (b , t) d 2 (c , t) d 2 (a , t) d 1
(b , t) d 1 (a , t) -1 -2 0 0 1 2 0 1 2 0 -1 1 2 0 1 2
-2 -1 0 3 6:30 7:00 7:30 8:00 8:30 Time 9:00 9:30 10:00

6:00 6:30 7:00 7:30 8:00 8:30 9:00 9:30 10:00 D(a) = 0
D(b)=2 D(c) = 0 DH 4 DH 3 DH 1 DH 2 D(b)=2 D(a) = 2 1 0
D(c) = 3 2

Figure 8.10 Example 2 (continued from Figures 8.7 and 8.9):
(e) DFs describing Steps 5 and 6 of algorithm VTSP.

Interaction between the sub-problems in Figure 8.12 can certainly reduce the number of

vehicles required while complying with the capacity constraints. This is somewhat similar

to what was done in Sections 8.3 and 8.4 using the DF principles. Moreover, optimality can

be attained using a set of Pareto efficient timetables following the analysis of Section 5.4.

This possibility of selecting a slightly different, yet optimal, timetable to improve the vehicle

schedule is an added value to the whole process. This approach is illustrated in Figure 8.13

with trips for the two routes shown in Figure 8.13a. There is only one timetable for Route 1

while there are two slightly different timetables for Route 2. Figure 8.13b and c shows possible

vehicle schedules based on the selection of Timetable 1 and Timetable 2, respectively.

Three vehicles are required for Timetable 2, one standard-bus and two minibuses. Two

vehicles are required for Timetable 1, one standard-bus and one minibus. Thus, the fleet

size can be reduced by selecting Timetable 1 for Route 2. In this solution, the capacity of Terminal B Terminal C 24 Departures C D =1.0 22 Departures C C =1.3 18 Departures C B =1.6
Peripheral route Urban route Intercity route Main

terminal A (a) 14 15 23 22 21 20 19 One vehicle type
Treated separately Optimal solution Relative cost
Number of vehicles 22.4 19.4 16 (b)

Figure 8.11 (a) Schematic description of a real-life, four-terminal, three-vehicles type example and (b) trade-off situation between total number of vehicles and relative cost associated with five disparate solutions.

the individual trips remains roughly the same enabling same level of passenger comfort.

The following methodology is founded on these two observations with the objective to

minimize the total fleet size and operating cost.

8.5.2 Layers of network used

Following the concepts of Figures 8.12 and 8.13, we use network G that consists of set of

nodes N and set of arcs A , as a network with flows of vehicles (see a similar approach used in

Section 6.4 for vehicle scheduling). For the sake of clarity, G is separated into two compat

ible layers, the logical layer G_L and the transportation layer G_T . Figure 8.14 illustrates these

two layers. In addition, Table 8.1 introduces notations and definitions used in describing

the solution process; Table 8.1 follows the notations of Hassold and Ceder (2014a), and not

necessarily those used in the sections given earlier. A B Depot Depot Time M A A B B M S (a) Depot Depot Time 2 2 3 3 A B B S A M 1 1 BMA (b) B B MA A 1 and 2 Depot Time 2 2 2M BMA 1 1 Depot (c)

Figure 8.12 Vehicle requirements for a simple transit network. (a) An example of time-space network for a timetable to serve Stations A and B consisting of two minibus trips (M) and one standard-bus trip (S). (b) Optimal solution for the network of (a) using separate problems for each vehicle type with arcs 1-3 denoting vehicle IDs. (c) Optimal solution for the network of (a) allowing for a replacement of a standard-bus by two

minibuses (2M) given that the capacity of two minibuses is greater than or equal to that of one standard-bus.

Figure 8.14 shows that the logical layer contains the routes and the corresponding timetables,

but does not represent the actual vehicle flows. The transportation layer represents the actual

vehicle flows on the network. A flow of one unit represents the operation of a single vehicle

on the associated arc. The transportation and logical layers consist of the following elements:

Circulation-arcs: Arcs added in the transportation layer between the sink node and the source node of the same vehicle type. The number of vehicles required can be determined by the flow on the circulation-arcs.

Departure and arrival nodes: Each such node stores the information of departure or arrival times of trips. The set of arrival nodes is called N_A the set of departure nodes is N_D ; $N_A, N_D \subset N$. These nodes are contained in both layers.

Logical-arcs: Arcs of the logical layer that establish a logical connection between trips belonging to the same timetable. For a chosen timetable, a flow of one unit is transported along the logical-arcs and along trip-arcs as shown in Figure 8.14.

Logical-circulation-arcs: Arcs that create a feasible network structure; these arcs are a certain type of logical-arcs connecting the source node S_1 and sink node T_1 . A lower bound (LB) and upper bound (UB) of one unit limits the number of selected timetables for a single line to be exactly one.

A A A A S M M M M S S MB B B B Start
Route 1 Route 2-1 Route 2-2 B A A A End Time B B A B (a) A
Depot Depot 1 1 2 2 S M S M M M S M B B B B B A A A A 2
2 1 A B A B 1 Time(b) A A A M M S S S B A A Depot Depot 3 3
3 A M B 1 11 1 2 2 B B B B 2 M M B B A A Time(c)

Figure 8.13 Fleet size reduction by using alternative timetables. (a) An example of time-space network illustrating two routes: route 1 (from A to B) with a single timetable, and route 2 (from B to A) with two timetables; vehicle types are denoted by M (minibus) and S (standard-bus). (b) Vehicle schedule for the network of (a) based on Timetable 1 with arcs 1-2 denoting vehicle

IDs; in this solution two vehicles are required. (c) Vehicle schedule for the network of (a) based on Timetable 2 with arcs 1-3 denoting vehicle IDs; in this solution three vehicles are required.

Table 8.1 Notations and definitions

General Network Layers

Tt Timetable G Network G L Logical layer

m Trip $\in$ Tt N Set of nodes G T Transportation layer

d m Departure time of trip m A Set of arcs A L Set of logical arcs

a m Arrival time of trip m N A Set of arrival nodes A T Set of transportation arcs

s m First stop of trip m N D Set of departure nodes A L m Logical arcs for trip m

e m Last stop of trip m S $\emptyset$ Source node A T m Transportation arcs for trip m

L Set of bus-lines l S l Source node for line l A T k Transportation layer for vehicle type k

VT Set of bus-types k T $\emptyset$ Sink node A T k m Transportation arcs for trip m vehicle type k

v k Capacity of bus-types k T l Sink node for line l Others:

TT uv Travel-time between station u and v LB l ij Lower bound Lower bound arc ij MS Minibus Standard-bus

m- α -n Compatibility relation: $a_m + TT_{uv}(t) \leq s_n$ UB l ij Upper bound Upper bound arc ij c k c ij Capital cost for bus-type k Cost for arc ij Circulation-Pull-in/Pull-out-arcs Transportation-layer Logical-layer Logical-circulation-arc Trip-Arc Logical-Arc Transportation-arc (DH) Waiting-arc

Figure 8.14 Illustration of the separate network layers: transportation layer and logical layer.

Pull-in/pull-out-arcs: Special transportation-arcs representing the movement of a vehicle from a depot to its first departure location (pull-out), or back to the depot (pull-in). These arcs are part

of the transportation layer.

Source and sink nodes of lines: A sub-network for each line $l \in L$ is formed between S_l and T_l . These arcs belong to the logical layer.

Transportation-arcs: Arcs running in parallel to each trip-arc between departure and arrival nodes and represent the actual flow of vehicles. The transportation-arcs can also connect an arrival node, at one location, with a departure node at another location; such a connection is then referred to as deadheading-arc (DH) implying a trip of an empty vehicle. The set of transportation-arcs will be referred to as $A_T \subset A$ and is part of the transportation layer.

Trip-arcs: Each trip-arc within the logical layer represents a trip, for which the flow can be either 1 if it is part of a selected timetable, or zero otherwise. Trip-arcs belong to the set of logical-arcs $A_L \subset A$.

Waiting-arcs: These arcs are added if a vehicle is waiting for the next trip at one location. These arcs, of the transportation layer, are special transportation-arcs.

Each timetable T represents a sequence of trips with given departure times d_m , first

station/stop s_m and arrival times a_m at the last station/stop e_m . Moreover, the designated

vehicle type $k \in VT$ for this trip is given with its size v_k . With these basic definitions, the

network-based minimum cost flow problem is described in the next section.

8.5.3 Min-cost flow formulation

The min-cost flow formulation is shown by Equations 8.1 through 8.8 following Hassold

$$\text{and Ceder (2014a). } \min c \cdot x \quad \text{s.t. } x_{ij} - x_{ji} = b_{ij} \quad \forall (i,j) \in \Sigma^+ \quad (8.1)$$

$$\text{s.t. } x_{ij} - x_{ji} = b_{ij} \quad \forall (i,j) \in \Sigma^+ \quad (8.2)$$

$$\text{(8.2) } x_{ij} - x_{ji} = b_{ij} \quad \forall (i,j) \in \Sigma^+ \quad (8.3)$$

$$\text{(8.3) } x_{ij} - x_{ji} = b_{ij} \quad \forall (i,j) \in \Sigma^+ \quad (8.4)$$

$$= \sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \geq \sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \quad (8.4)$$

$$\sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \leq \sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \quad (8.5)$$

$$\sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \leq \sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \quad (8.6)$$

$$\sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \leq \sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \quad (8.7)$$

$$\sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \leq \sum_{i \in V} \sum_{j \in V} k_{ij} x_{ij} \quad (8.8)$$

In this formulation, cap_m is the capacity of the vehicle originally scheduled for trip m .

The objective function (8.1) minimizes the cost of the flow on all arcs plus a element e ,

explained later. Constraint (8.2) presents the standard flow conservation constraint, that is,

in-flow equals out-flow for each node. Moreover, it is necessary to ensure that the logical

and transportation flows are not mixed. The flow conservation constraint for the logical

arcs is presented in (8.3). Further restrictions on the transportation flow are required by

Constraint (8.4), which prevent the flows of different vehicle types to be mixed. Constraint

(8.3) together with Constraint (8.4) do not make the first flow conservation Constraint (8.2)

obsolete because the circulation-arcs have not been assigned to any subset of arcs and also

have not been considered yet. Constraint (8.5) guarantees that the capacity of the vehicles

assigned to a trip is greater than or equal the originally planned capacity.

Hassold and Ceder (2014a) show also how to use a alternative timetable (as per Section

5.4) and how to deal with vehicles substitution following the description in Figure 8.12. An

element e is introduced in Constraint (8.6) to represent empty seats on the vehicles served

ing transportation-arcs for trip m . This will equal to zero in case that of no substitution of

vehicles. Adding e to the objective function, with the possibility to assign it a weight, firstly

prevents the use of transportation-arcs, and secondly avoids the assignment of overly large

capacity to a trip if not required. The substitution of vehicles can reduce the total number of

vehicles and thus reduce the cost of the vehicle schedule. However, in case that a vehicle type

K_1 with capacity v_{K_1} is operated by a vehicle of type K_2 with capacity v_{K_2} $v_{K_2} > v_{K_1}$, there will be

more empty seats on the vehicle than originally calculated. Therefore, the actual penalty for

empty-seats time for this timetable has to be recalculated and considered in the final evaluation

of the solutions. The remaining Constraints (8.7) and (8.8) ensure the compliance with

the lower boundaries l_{ij} (lower bound) and u_{ij} (upper bound) and also of the flow (representing

vehicles) being an integer value.

The min-cost formulation assumes that the operator is indifferent between the given

timetables and does not have any preference toward either even headway or a reduction in

empty-seat-kilometer. It is shown by Hassold and Ceder (2014a) that this model can be used

for the cases with alternative timetables, and with vehicle substitution.

Finally, Hassold and Ceder (2014a) demonstrated that a substitution of vehicles is bene

cial and can lead to significant cost reductions in the range of more than 27%. The method

ology of this section was applied to a real-life case study in Auckland, New Zealand, with

a saving of more than 15% of the cost of fleet compared with vehicle schedules that are

provided by standard models. In addition, Hassold and Ceder (2014b) examined the ben

efits attained from an energy-efficiency perspective. Their results show that using timetables

with multiple vehicle types can increase passenger occupancy of the vehicles by 19%, hence

reducing total energy consumption. These results are achieved without any increase of the

total passenger waiting time and fleet size.

8.6 VEHICLE SIZE DETERMINATION

There is doubt that there is a connection between different vehicle types and the size of vehicle

types used in public transit services. In the foregoing section, it was assumed that all vehicle

types possessed a similar size so that the frequencies and timetables determined remained

unchanged. However, practice often entails making a decision about vehicle size and the

amount available for purchasing. That is, a bus fleet can consist of vehicles of the same size,

to hereby simplifying operations planning, or have a mix of sizes; that is, minibuses, standard

buses, articulated/double-decker buses, etc. This section provides some preliminary analytical

tools for evaluating trade-offs between different vehicle sizes and for reviewing the

models developed.

8.6.1 Example: A standard bus versus two minibuses

Given a bus route operated by standard 50-seat buses, can the standard bus be exchanged

more cost effectively and viably for 2 minibuses with 25 seats each? The rest line of thought

is directed to the question: What operational strategy should be conducted for the two

minibuses? Basically, two operational alternatives can be drawn: (1) Let the two minibuses

depart at the same time, with each minibus picking up and dropping off passengers at each

other stop. This will result in reducing travel time and, in parallel, retaining the timetable

(frequency). (2) Let the minibus headway be half that of the standard bus and let each

minibus pick up and drop off at each stop. This alternative will result in reducing passenger

waiting time and, in parallel, maintaining average travel time.

The following basic analysis assumes that for a alternative (1) we have a long bus route (end

effects ignored) that is a typical commuter collector—for example, to the CBD—and the

ridership is unchanged. The assumptions for alternative (2) are that the average waiting time

is half the headway (deterministic case; on headway distribution see Chapter 18), that the

probability of no one being at the stop is the same for both the minibus and the standard

bus, and that the ridership is fixed.

For an example of alternative (1), the following data are used: standard bus (SB) operating

cost = \$ 15/h, minibus (MB) operating cost = \$ 12/h, SB travel speed denoted by μ miles/h

$= 3/4$ MB travel speed, P = average number of passengers on board SB, and the value of a

passenger's riding time is \$ (1/3)/h. Hence, SB operating cost per μ miles = $15 \times 1 = \$ 15$,

and 2 MB operating cost per μ miles = $2(12 \times 3/4) = \$ 18$. The problem arises: Who is

going to pay the additional cost for the 2 MB (\$3 per μ miles)? A possible answer is the

passenger because their riding time is reduced. A saving in travel time by MB per μ miles

is $1 - 3/4 = 1/4$ h. The value of the total saving is $1/4 \cdot P \cdot 1/3 = P/12$. Requiring that the saving

be greater than the additional cost results in $P/12 > \$3$ or $P > 36$ passengers. In other words,

a two-minibus operation is preferable in alternative (1) for average loads greater than 36 on

the standard bus.

For an example of alternative (2), the f

following data are used: S B and M B operating

cost and P are the same as in the foregoing example, S B headway = 10 min, M B headway =

5 min, and the value of waiting time per passenger = \$ 1/h. The number of hourly onboard

passengers is $6P$, and the saving in average waiting time when using M B is $10/2 - 5/2 =$

2.5 min per passenger. Total hourly saving, therefore, is $6P \cdot 2.5/60 \cdot 1 = \$(1/4)P$, to be viable

must be greater than the additional M B hourly operating cost. That is, $(1/4)P > 2 \cdot 12 - 15$,

which results, as in the example for alternative (1), in $P > 36$ passengers.

8.6.2 Vehicle size square root formula

A simple derivation of transit vehicle size accentuates a trade-off between vehicle capacity cost and passenger waiting-time cost. We assume that operating cost per vehicle-hour

is the same (independent of vehicle size and load carried), average waiting time is half the

headway, riding time is independent of vehicle size, and travel time and stopping time are

independent.

The following notation is used:

Z is the desired average occupancy in the max load segment,

which corresponds to vehicle size

C_b is the operating cost per vehicle-hour

P is the number of average hourly passengers carried in the max load segment

C_w is the value of hourly waiting time

The hourly frequency, utilizing Method 2 in Chapter 3, is $P/2$, and the headway (in hours)

is Z/P . Hence, the total operating cost is $C_b \cdot P/2$, and the passenger waiting-time cost is

$\frac{1}{2} Z/P \cdot P \cdot C_w = \frac{1}{2} Z C_w$. The overall cost, therefore, is $C_b \cdot P/2 + \frac{1}{2} Z C_w$.

The optimal vehicle size, Z^* , is attained by minimizing the overall cost using the first

derivative of the overall cost with respect to $Z = 0$. This results in the following formula: $Z^* = \sqrt{2 C_b P / C_w}$ (8.9)

Equation 8.9, called the square root formula, signifies, generally speaking, that the optimal

vehicle size varies according to the square root of the number of passengers carried. In addition,

this optimal size is sensitive to changes in C_b and C_w . The next section reviews more

advanced vehicle-size modeling.

8.7 OPTIMAL TRANSIT VEHICLE SIZE: LITERATURE REVIEW

The following notation is used throughout this section (unless mentioned otherwise):

Z is the optimal vehicle size

a is the constant of operational cost function (vehicle operational cost as a function of size, $a + bZ$)

b is the slope of operational cost function

w is the value of passenger waiting time

x is the value of access/walking time

r is the value of riding/in-vehicle time

L is the route length

P is the total passenger flow

Q is the peak passenger flow

V is the vehicle speed

ϕ is the maximum allowed load factor

In some cases, parameters are given in a different notation in order to enable a convenient

comparison.

An early reference to bus-size optimization was made by Jansson (1980), who paid signi-

cant attention to the cost of operating similar services at peak and off-peak during daylight.

The following formula for optimal bus size was developed: $Z = \frac{1}{2} hJQ Ew + tQ E_r + \frac{1}{2} (\frac{1}{\phi} \phi a b(\beta^2))$ (8.10)

where h is the running and transitional time per km J is the average trip length β is the ratio of mean flow rate for the whole day to the peak-hour flow E is the extent of peak + off-peak periods per day t is the boarding and alighting time per passenger

Calculation of the optimal bus size, given various levels of passenger flows, leads to the con-

clusion that most buses in a actual operation are too big.

Gwilliam et al. (1985) describe a simple formula for an optimal bus size: $Z = \frac{2}{P} a$ (8.11)

Based on assigning common values of a and P to their formula, the authors claim that oper-

ating much smaller buses than those commonly used is not justified.

A more sophisticated model that aims at determining optimal bus size is provided by

Oldfield and Bly (1988). Unlike most other models, this one assumes elastic demand; that

is, demand that changes with passenger-trip cost. The model

also takes into account the

influence of changes in bus demand on road congestion; it further assumes that in-vehicle

trip time increases with average bus load because of boarding times. Significant attention

is given to the effect of capacity constraints on average passenger waiting time in cases in

which passengers are unable to board the first bus that arrives because it is full. Complex

expressions are introduced explaining the dependence of headway on bus size. The expres

sion developed for optimal bus size is the following:
$$Z_{AY} Y W (1) 2 n P K U R 2 i i / i g = + + + + \beta \beta \beta \beta - a 1 2 \beta \quad (8.12)$$

where K is the number of bus-km provided U is the average cost of walking time to/from bus stop R is the cost of time spent in the bus n is the time-value constant that takes into account the extra time that a big bus spends at stops owing to a large number of boardings g is the elasticity with respect to generalized cost (positive) A , β , Y , i , W are the constants (expressions for calculating them are developed in the paper)

Assigning common values of urban bus services in the United Kingdom, assuming no

subsidies, the authors find that the optimal bus size lies between 55 and 65 seats. Under

various other conditions, the model suggests that the optimum size at typical urban levels

of demand will not be fewer than about 40 seats. Although this model is very sensitive

to various phenomena that influence passenger demand and operator costs, it seems that

the preliminary modeling and calibrating efforts required for using it are too intensive for

practical use.

the route system after the determination of optimal vehicle size and frequency for each

route. Each iteration of the transit assignment yields corrected values of $(Q_k)_{\max}$ and P_k

so that new optimal vehicle sizes can be calculated. The iterative process does not seek

to minimize system-wide cost, but to determine the optimal bus size for each separate

line. Implementation of the procedure is illustrated using data from a transit network in

Austin, Texas. Of 40 bus routes, results suggest that 37 have an optimal bus size of below

25 seats.

A model developed by Lee et al. (1995) attempts to find the optimal bus size not only for

each route but also for each period of day so that more than one bus size can be used on

one route. The model also tries to determine the conditions under which it is better to use

one bus size or, alternatively, a mixed-size fleet. The bus size that gives the minimum total

operator and user cost on one line is determined as follows: $Z_{aLQ} wVP = Z_2$ (8.15)

If only one bus size is operated on all routes during all periods, the system-wide optimal size

is found as follows: $Z_{aLQ} V wP Q_r r_t r_t t m r r_t r_t t m r = \sum \sum Z_{11} / n$ (8.16)

where r is the route index n is the number of routes t is the time-period index m is the number of time periods

If two bus sizes (Z_1 and Z_2) are used, the following test can determine which is better for

each route: $Q_L PV wZ_2 a Z_1 Z_2 =$ (8.17)

If the left-hand side is greater than the right-hand side, bigger buses should be used. If the

right-hand side is greater, the operation of smaller buses is justified. For cases in which Z_1

and Z_2 are not given, the paper describes an algorithm for determining the two bus sizes

that give minimum cost. The algorithm is illustrated in a simple four-route network with a

two-period demand. The optimal fleet is concluded to consist of 15 buses with 33 seats and

29 buses with 20 seats. The total fleet size required for a network operation with this mix

is smaller than that needed for operating the same route system with buses all of the same

size. It is also shown that under the conditions of the given example, a mixed-fleet operation

is preferable if the ratio of peak demand to off-peak demand is more than 1.92. If there is no

significant demand variation between periods of the day, then the operation of a uniform bus

size is preferable.

Gronau (2000) presents this model for determining the optimal bus size for a specific

route: $Z = t_0 r/P + t_1 (r) + \alpha_0 + \alpha_1 + \alpha_2 = \alpha + \beta \lambda + \beta ()$ (8.18)

where $\alpha_0 + \alpha_1 + \alpha_2$ is the distance-related operating cost $\beta_0 + \beta_1 + \beta_2$ is the time-related operating cost $t_0 + t_1 + t_2$ is the bus travel time λ is the ratio of the value of waiting time to the value of vehicle time, divided by the ratio of the headway to the waiting time (expected value: 1.5-2)

This model, introduced as a basis for a series of mathematical developments, aims at exam

ining the option of using two bus sizes on one route that serves passengers with different

values of time. The usefulness of this option is investigated in detail.

Tisato (2000) analyzes the variation of public transit subsidy levels among several con

straint cases with respect to bus size and load factor. Four cases are analyzed: fixed-load

factor with variable bus size; fixed bus size with variable load factor; both fixed; and both

variable. Conditions for maximum economic surplus are determined for each of the cases. In

the case in which bus size is the only variable, the expression derived for optimal bus size is $Z_{ALPA} w = 2 \frac{1}{25} \frac{1}{65} \frac{15}{()} \cdot \phi \phi$ (8.19)

where A is the average passenger-trip length, divided by bus-trip length.

In addition, it is shown how optimal bus size changes with a varying target-load factor.

A subgroup of vehicle-size optimization tools consists of the formulation of bus-sizing

models as part of a comparison between fixed-route and flexible-route services. Chang and

Schonfeld (1991), who make such a comparison, argue that a rectangular service area should

be connected to an adjacent transportation terminal. If a fixed-route service is provided, the

optimal bus size is the following: $Z_{a g S D L} x w V = \frac{1}{8} \frac{2}{2} \frac{2}{2} \frac{1}{3} /$ (8.20)

where g is the access speed S is the length of service area D is the passenger-demand density in service area

If the bus route is flexible, the optimal size is $Z_{ua L D w k V b r T} = + \frac{1}{3} \frac{3}{3} \frac{2}{2} \frac{1}{5} \frac{2}{()} / /$ (8.21)

where u is the average number of passengers per pickup point $L T$ is the equivalent-line haul distance k is the constant (estimated value described in the paper)

investigates the impacts of travel time variability and crowding discomfort on the optimal bus

frequency and bus size for a highly congested corridor with cars, buses, and walking as

travel alternatives. The goal of this research was not directly about bus size, but to study

the optimal pricing structure of both cars and buses. Tirachini et al. (2014) propose a social

welfare maximization framework using a known mean-variance model with the use of an

empirical relationship between the mean and standard deviation of travel times. They con

clude, regarding bus sizes, that both travel time variability and crowding discomfort leads

to higher optimal bus sizes.

8.7.1 Literature review conclusions

The following may be concluded from the review of the literature on optimal transit vehicle size:

- According to most of the models, an increase in the value of passenger time decreases the optimal vehicle size. Vehicle size in some of the models is proportionally opposite to the square root of the time value.
- There is no consistency among models concerning which trip-time elements influence the optimal vehicle size. Each element (access, waiting in-vehicle) appears as a variable in some models, but no one element appears in all of them. Table 8.2 Summary of the characteristics of the models reviewed
Comparisons subject Source Jansson (1980) Guillaume et al. (1985) Oldfield and Bly (1988) Jansson (1993) Shih and Mahmassani (1994) Lee et al. (1995) Gronau (2000) Tisato (2000) Chang and Schonfeld (1991) Chien et al. (2001) Principles and assumptions: Elastic routed demand, with/without a fixed, systemwide, demand constraint No No Yes, withou

t No Yes , with No No No No No The model ena
ble the use of any demand function D
emand in - uences road congestion No
No Yes No No No No No No No In vehicle t
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ormula quoted here) No No No D if f e r e
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d eld and Bly (1988) Jansson (1993)
Shih and Mahmassani (1994) Lee et a
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00) Chang and Schonfeld (1991) Chie
netal. (2001) Total seats per hou
r (product of vehicles size and frequ
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Fixed cases are examined Fixed Fixed F
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to calculate the optimal vehicle size other than using data that are readily available to most transit agencies. The model presented by O'Neil and Bly is different in that it requires much more sophisticated data. Similarly, the model developed by Jansson (1993) requires detailed information about passenger demand.

EXERCISES

8.1 Given a set of terminals, a, b, c , and d ; four types of vehicles with their unit cost c_i ;

$i = 1, 2, 3, 4$ (in monetary units); DH travel-time matrix; and a fixed schedule of trips.

(a) Find the number of vehicles for each type required to obtain the minimum total cost; use algorithm VTSP.

(b) Find the solution to (a) after changing the costs of type I and type II from 8.5 and 5 to 10 and 4 cost units, respectively; use algorithm VTSP, but think about steps that there is no need to repeat. Trip number Type of vehicle
Departure time Departure terminal Arrival time Arrival terminal
1 I 6:00 a 6:30 c 2 II 6:20 a 6:50 b 3 IV 6:20 b 6:50 c 4 III 6:40 d 7:10 a 5 I 7:10 a 7:40 a 6 II 7:10 b 7:40 a 7 III 7:20 d 7:50 a 8 I 7:40 c 8:10 a 9 IV 7:50 d 8:10 b

10 IV 8:00 b 8:30 c

11 IV 8:10 b 8:30 d

12 III 8:20 a 8:50 d DH travel time (min) a b c d a 0 10 10 10 b 10 0 10 10 c 10 10 0 20 d 10 10 20 0 a b c d Type I (c 1 = 10) Type II (c 2 = 8.5) Type III (c 3 = 5) Type IV (c 4 = 4)

8.2 Solve Example 1 in Section 8.4.1 using the following three new sets of input cost values; use algorithm VTSP, but think about steps that there is no need to repeat:

(a) $c_1 = 10, c_2 = 5, c_3 = 3$

(b) $c_1 = 10, c_2 = 11, c_3 = 6$

(c) $c_1 = 12, c_2 = 8, c_3 = 9$

8.3 Given a set of terminals, a, b, c ; three types of vehicles with unit costs c_i ; $i = 1, 2, 3$ (in monetary units); DH travel-time matrix; shifting tolerance of $-15 \text{ min} \leq \Delta \leq 15 \text{ min}$; and a fixed schedule of trips. Find the number of vehicles for each type required to obtain the minimum total cost; use algorithm VTSP. DH travel time (min) $a \ b \ c$ $a \ 0 \ 30 \ 30 \ b \ 30 \ 0 \ 30 \ c \ 30 \ 30 \ 0$ a Type II ($c_2 = 4$) Type I ($c_1 = 5$) Type III ($c_3 = 3$) $b \ c$ Trip number Type of vehicle Departure time Departure terminal Arrival time Arrival terminal 1 I 6:00 a 7:00 c 2 II 6:00 a 7:00 b 3 III 6:00 b 7:00 c 4 III 6:00 b 7:00 c 5 II 6:30 a 7:30 b 6 II 6:30 b 7:30 a 7 III 7:00 c 8:00 b 8 II 7:00 b 8:00 a 9 I 8:00 a 9:00 a

10 III 8:00 c 9:00 b

11 I 8:00 c 9:00 a

12 I 8:30 c 9:30 a

13 II 8:30 a 9:30 b

14 II 9:00 a 10:00 b

8.4 Examine the trade-off between operating an articulated bus (AB) with 75 seats and operating 3 minibuses (MB) with 25 seats each. The following data and information are given: AB operating cost = \$25/h, MB operating cost = \$15/h, value of passenger-riding time is \$0.5/h, frequency of AB and MB services are the same, MB travel speed is below AB travel speed of some operational tactics for the MB service are utilized, and for $P > 30$, in which P is the average number of passengers on board AB, use of the MB service is preferred.

(a) Find the values of P for preferring the MB service if the MB travel speed (described above) is reduced by 10%.

(b) What are the upper and lower bounds on the travel speed of the AB service for positive P values?

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9 Chapter 9: Crew Scheduling

One of the most time-consuming and cumbersome scheduling tasks is assigning the crew (driv

ers) to vehicle blocks. The task requires the service of imaginative, experienced schedulers, and

usually, it is performed automatically. Consequently, it is not surprising to learn that most of

the commercially available transit-scheduling software packages concentrate primarily on crew

scheduling activities. After all, from the transit agency's perspective, the largest single-cost item

in the budget is the driver's wage and fringe benefits. Because of the important implications of

crew scheduling for providing good transit service, practitioners ought to comprehend the root

of the problem and be equipped with basic tools to be able to arrive at a solution. This chapter consists of four principal parts, following an introductory section. Section 9.2

uses the deficit-function (DF) properties for constructing vehicle chains (blocks) that take into

account maximum paid idle time (swing time that is an unpaid break in a split duty). The latter

consideration is aimed at helping to construct crew schedules with minimum cost. Section 9.3

presents the basic mathematical formulation used in solving crew-scheduling and crew-rostering

problems. In addition, this section accentuates an approximate solution to produce low-cost

duty pieces of a crew schedule. Section 9.4 describes a case study performed for the New Jersey

Transit Corporation that needed a tool for analyzing the marginal cost impacts of changes in

train schedules and for producing near-optimal crew

assignments (duties), given a schedule

and a set of work rules. Finally, Section 9.5 discusses and provides examples of crew rostering.

A roster is a pattern of duties to be fulfilled for a certain number of consecutive days; commonly,

the pattern repeats itself in cycles (of a week, a month, or any other period). The chapter ends

with a literature review and exercises. Practitioners are encouraged to visit all the sections while skipping only the mathematical

formulation. They should, though, pay special attention to the examples and their figures. A crew schedule is called a pick. Often the selection of picks involves conflicts. Some may

say that a selection without a conflict is almost as inconceivable as a nation without crises.

There are two short stories: First, about being told that the pick is excellent. A man bought a

newspaper for 50 cents. He found, however, that the price marked on the paper was 35 cents.

He returned to the kiosk and asked about it. The reply was, "Don't believe everything written

in newspapers." Second, two rabbits arrive at the planning department of a transit agency, and

one stopped to cry. "Why are you crying?" she was asked. And she answered, "Because here

they cut the fifth leg of every rabbit." "But you have only four legs, so there's nothing to worry

about," she was told. She replied, "Yeah, but here they cut first and then count."

9.1 INTRODUCTION

The functional diagram of a typical transit operations planning process, Figure 1.2, ends

its fourth and last planning activity with c

rew scheduling, the aim being to assign drivers according to the outcome of vehicle scheduling. This activity is often called driver run

cutting (splitting and recombining vehicle blocks into legal driver duties, shifts, runs, or

assignments). Part of a vehicle block is called a duty piece or a task. This crew assignment

process must comply with some constraints, which are usually dependent on a labor con

tract. The purpose of the assignment function is to determine a feasible set of driver duties

in an optimal manner. Usually, the objective is to minimize the cost of duties so that each

duty piece is included in one of the selected duties. It should be noted that vehicle-scheduling

activity for railways is not cumbersome; however, it is important to consider how to build

the work schedule of the train crews (drivers and conductors together) efficiently.

The criteria for crew scheduling are based on an efficient use of manpower resources while

maintaining the integrity of any work rule agreements. The construction of the selected crew

schedule is usually a result of the following subfunctions: (1) duty piece analysis, (2) work

rule coordination, (3) feasible duty construction, and (4) duty selection. The duty piece

analysis partitions each vehicle block at selected relief points into a set of duty p

pieces. These

duty pieces are assembled in a feasible duty construction function. Other required information

are travel times between relief points and a list of relief points designated as required

duty stops and start locations.

Theoretically, each relief point may be used to split the vehicle block into new duty pieces.

Usually, it is more efficient to use one or more of the following criteria to select the relief

points: (a) minimum duty piece length; (b) next relief point, selected as close as possible to

the maximum duty piece time (maximum time before having a break); (c) only a few (say,

two) relief points in each piece; and (d) operator decisions. In order to utilize any crew

scheduling method, a list of work rules to be used in the construction of feasible driver

duties is required. The work rules are the result of an agreement between the drivers (or their

unions) and the public transit agency (and/or public authorities).

The determination of different feasible sets of duties may be selected on the basis of, for

example, one or more of the following performance measures: (1) number of duties (drivers),

(2) number of split duties, (3) total number of changes, (4) total duty hours, (5) average duty

length, (6) total working hours, (7) average working time, (8) number of short duties, and

(9) costs. Sections 9.2 through 9.4 provide tools for handling both the preparation for and

analysis of the construction of efficient crew schedules (duties).

Once the set of duties are established, it is common to group them into rosters. A roster is

defined as a duty assignment over a certain amount of consecutive days, guaranteeing that

all the trips are covered for a certain (usually cyclic) period. Commonly, a roster contains a

subset of duties covering six consecutive days (called weeks). The length of a roster is typically

usually between 30 and 60 days (5-10 weeks). The usual rostering problem is to find a feasible

set of rosters to cover all the duties, using one or more of these four objectives: (a) minimum

number of crews required, (b) minimum sum of roster costs, (c) minimum of the maximum

roster duration, and (d) balancing (the equity of) workload and days-off. The rostering

problem is presented in Section 9.5.

9.2 VEHICLE-CHAIN CONSTRUCTION USING A MINIMUM CREW-COST APPROACH

There are two predominant characteristics in transit operations planning: (a) different

resource requirements between peak and off-peak periods and (b) working during irregular

hours. These characteristics result in split duties (shifts) with unpaid periods in between.

It

is often called swing time. The inconvenience accompanying split duties led driver (crew)

unions to negotiate for an extension of the maximum allowed driver's idle time for which

the driver can still get paid. It is common, therefore, to have a constraint in a labor union

agreement specifying this maximum paid idle time (swing time), to be termed $T_{\max}$.

The crew-scheduling problem (CSP) from the agency's perspective is known to be the

minimum crew-cost problem. With this minimum-cost orientation in mind, we can use the

deficit-function (DF) properties to construct vehicle chains (blocks) that take into account

$T_{\max}$, in other words, to maximize idle times (swing times) that are larger than $T_{\max}$ and

hence to reduce crew costs. What follows is based on the first edition of this book and on

Ceder (2011).

9.2.1 Arrival-departure joinings within hollows

The following description uses the notation and definitions associated with the DFs of

Section 6.5.1. Each hollow of a DF, $d(k, t)$ at terminal k , contains the same number of

departures and arrivals, except for the first and last hollow at the beginning and end of the

schedule horizon. This is due to the fact that each arrival reduces $d(k, t)$ by one and each

departure increases it by one, so that the hollow starts and ends at $D(k)$.

For a given hollow, H let $I_{m,k}$, be the set of all arrival epochs t in H and let $J_{e,m,k}$, be the set of

all departure epochs t in H s.t. $m \leq t \leq k$. The difference in time between departure and arrival is defined

as Δ_{ij} s.t. $i \leq j \leq i+m$ for $t \in I_{m,k}$. The joining (connection) between t and t_e is j in a vehicle block is

effectively the idle time between trips; hence, Δ_{ij} may represent this idle time. We may also define

local peak u, v within hollow $H_{m,k}$ as $d(k, t_{uv})$ between t and t in which $t \leq t_s \leq u \leq v \leq t_{m+1}$, $k \leq t \leq k$.

where $H_{m,k}$ starts and ends at e and $m+1, k$, respectively. Note that if the start and/or end of a

local peak, u, v , has more than one departure or arrival, then it suffices to refer to only one

of them (as u or v). Let d_{uv}, m_k be the number of departures in $H_{m,k}$ before and including t_s and

a_{uv}

m_k be the number of arrivals in $H_{m,k}$ before t_s .

Lemma 9.1:

The number of arrival-departure joinings in hollow $H_{m,k}$ before t_s must be d_{uv}, m_k .

Proof: If some departure epochs before a local peak, u, v , are left without a joining, it will

be impossible to connect them with arrival epochs after t_e . That is, each departure epoch

before and including t_s must have a joining to an

earlier arrival time within H_{mk} . This can be

seen in Figure 9.1a.

Lemma 9.2:

The number of arrival-departure joinings that can be constructed after $t_e v$ within H_{mk} is

$$(\sum_{u \in V} d_{uv})$$

$$, m_{uv}, m_k - .$$

Proof: Given hollow H_{mk} and local peak uv , then based on Lemma 9.1 and the characteristics

of local peaks in hollows, d_{uv}, m_k , departure epochs must and can be joined to earlier arrival

epochs in H_{mk} ; hence, the number of arrival epochs left over without joinings is $(\sum_{u \in V} d_{uv}, m_{uv}, m_k -$

for all local peaks. Figure 9.1b displays this explanation.

Lemma 9.3:

The sum of all idle times within any hollow is a fixed number and independent of any

procedure aimed at joining arrival and departure epochs, that is, $\sum_{i,j} \Delta_{ij}$ constant Δ .

Proof: Let H_{mk} have n arrivals and n departures. We noted previously that the number of

departures and arrivals are the same within each middle hollow (i.e., excluding the first and

last hollows). Let two different n -joining arrangements with idle times Δ_{ij} and Δ'_{ij} for all join

ings between $i \in I_m \in k$ and $j \in J_m \in k$ be expressed as follows: $\Delta_{ij} = \sum_{s=1}^j e_1(i, j) - \sum_{s=1}^j e_1(i, j) - t_1(i, j) + t_2(i, j) = \sum_{s=1}^j e_1(i, j) - \sum_{s=1}^j e_1(i, j) = 0$

and similarly $\Delta'_{ij} = \sum_{s=1}^j e_2(i, j) - \sum_{s=1}^j e_2(i, j) - t_2(i, j) + t_1(i, j) = \sum_{s=1}^j e_2(i, j) - \sum_{s=1}^j e_2(i, j) = 0$

We know that the sum of all departure or arrival times in a hollow is a fixed number; hence,

$\Sigma = \Sigma = i, j \mid i, j \mid \text{constant} \Delta \Delta 1 2$. Figure 9.1c further clarifies this argument. (b) (a) $d(k, t) d(k, t_{uv}) H k m e$
 $k m t u s t v s s k m+1 4 3 2 1 0$ (c) $d(k, t)$ Time 4 3 2 1
 $0 d(k, t) 6 5 4 3 2 1 0$ a k, m u 1 v 1 a k, m u 1 v 1 -d k, m
u 1 v 1 a k, m u 2 v 2 -d k, m u 2 v 2

Figure 9.1 Part (a) describes examples of arrival-departure joining to support Lemma 9.1, part (b) interprets Lemma 9.2, and part (c) shows two 4-joining examples of Lemma 9.3.

9.2.2 Objective function and formulation

In constructing the blocks at each DF, we wanted to maximize the number of times in which

$\Delta_{ij} \geq T_{\max}$, in other words, to reduce crew cost. At the same time, however, for cases in

which $\Delta_{ij} < T_{\max}$, we wanted from a crew's fairness perspective to attempt to have equitable

paid idle times. This was for a simple reason, to eliminate a situation in which some drivers

will have long and some short paid idle times. We will start with a formulation of the main

objective and continue with a secondary objective.

For a given hollow $H m k$ at terminal k , let x_{ij} be a 0-1 variable associated with a trip joining

between the arrival of the i th trip to $H m k$ and the departure from $H m k$ of the j th. The problem

of finding the maximum number of idle times greater than or equal to $T_{\max}$ in hollow $H m k$ is

as follows:

Problem P 4 $\text{Max} \sum_{i \in I, j \in J} x_{ij} = \epsilon \in \Sigma \sum_{m, k} k_{ij}$ (9.1)

Subject to $x_{ij} \mid i \in I, j \in J, m, k \leq \epsilon \in \Sigma 1, k$ (9.2) x_{jJ}

$i \in I, m \leq \epsilon \in \Sigma 1, k, k$ (9.3) $x_{, i \in I, j \in J} \mid m, m = \epsilon$

$\in \{, \} \mid 0 1 k, k$ (9.4)

The binary decision variables are determined by

$x_{ijt} \mid T$ otherwise $i, j \in i = - \geq 0 0 0 0 0 1 0, , \text{max}$

A solution with $x_{ij} = 1$ indicates that joining trips i

(arrival epoch) and j (departure epoch) results

in an idle time larger than or equal to $T_{\max}$. Constraints (9.2) and (9.3) ensure that each trip in

H_m

k may be joined with, at most, one successor trip and one predecessor trip, respectively.

Trips that were not joined in the solution of P_4 are subject to a secondary objective: equita

ble paid idle times. It is shown in Theorem 9.1 that joinings with this secondary objective are

based on the first-in, first-out (FIFO) rule. Balancing Δ_{ij} for $\Delta_{ij} < T_{\max}$ is the same as minimiz

ing the difference between each Δ_{ij} and its average Δ_{ij} either by absolute difference or by least

square difference. The FIFO rule used for this balancing is stated in the following theorem.

Theorem 9.1

Minimizing the least-square differences between Δ_{ij} and each Δ_{ij} for all $i \in I_m \in k$ and $j \in J_m \in k$ in

H_u

k is accomplished by constructing joinings using the FIFO rule.

Proof: It is sufficient to prove Theorem 9.1 on a simple but generalized example, as illus

trated in Figure 9.2, with a hollow containing two arrivals and two departures.

We may show that $(\Delta_{12} - \Delta)(\Delta_{23} - \Delta)(\Delta_{34} - \Delta)(\Delta_{41} - \Delta) \geq 0$ (9.5)

where Δ is the average arrival-departure joining length in the example. Using an algebraic

expression, then (9.5) becomes $\Delta^4 - 2\Delta^3 + \Delta^2 - 2\Delta + 1 \geq 0$ (9.6)

Lemma 9.3 states that $\Delta_1 + \Delta_2 = \Delta_3 + \Delta_4$, and hence the right-hand side of expres

sion (9.6) is zero. From Lemma 9.3, we can further obtain $(\Delta_1 + \Delta_2)^2 = (\Delta_3 + \Delta_4)^2$ or

$\Delta_1^2 + \Delta_2^2 + 2\Delta_1\Delta_2 = \Delta_3^2 + \Delta_4^2 + 2\Delta_3\Delta_4$. The latter is inserted into (9.6) to yield $\Delta_1\Delta_2 - \Delta_3\Delta_4 < 0$ (9.7)

Based, again, on Lemma 9.3, let $\Delta_3 - \Delta_1 = \Delta_2 - \Delta_4 = B$ or $\Delta_1 = \Delta_3 - B$ and $\Delta_4 = \Delta_2 - B$; these last

two equations are inserted into (9.7) to obtain $\Delta_3(\Delta_2 - B) < \Delta_2(\Delta_3 - B)$, which yields $\Delta_3 > \Delta_2$.

The last result must be correct from Figure 9.2, and therefore, it agrees with expression (9.5).

9.2.3 Procedure for determining the maximum number of unpaid idle times

The mathematical programming formulation in Equations 9.1 through 9.4 is aimed at

maximizing the number of idle times that are longer than or equal to $T_{\max}$. However, this is a FIFO scheduling problem. Non-FIFO scheduling problems are more complicated.

Figure 9.2 Example of a comparison of joinings based on FIFO and other rules.

formulation may involve a very large number of computations (nonpolynomial [NP])

complete, see Section 5.5), hence entailing the use of a more sophisticated procedure.

Such a procedure is described in a flow diagram in Figure 9.3 and contains both $T_{\max}$ and

FIFO rule considerations; the latter is for joining arrivals and departures with paid idle

times. Let us call this procedure a algorithm TMF.

The input for a algorithm TMF for each terminal k consists of two arrays, the arrival and

departure arrays, and a given $T_{\max}$. This input enables constructing $D(k)$ and obtaining

$D(k)$ following the insertion of deadheading (DH) trips and the shifting of departure times

for minimizing fleet size (see Chapters 6 and 7). Because $D(k)$ vehicles are required at k , we

assume their arrivals there to be at (or before) T_1 (the start of the schedule horizon). Algorithm

$T_m F$ moves by steps on $d(k, t)$, in which each step refers to a change in $d(k, t)$ or the detection

of a dot on $d(k, t)$; the dot means that a rival and departure epochs at k overlapped at t . $d(k, t) = d(k, T_2)$? A departure exit? Anymore disjointed departure array? Adding a disjointed arrival Adding departure array in a paid joining (list of departures) Erasing arrival array Adding unpaid joining array Constructing FIFO joining arrays Adding joining arrays Constructing FIFO joining arrays Adding joining arrays STOP STOP Initialization: arrival and departure times at k ; $T_{\max}$ Construct $d(k, t)$ Find $D(k)$ List $D(k)$ arrivals at T_1 each with trip number 0 Step move on $d(k, t)$ An arrival exists, too? Yes Yes Yes Yes Yes Yes No No No Being at $D(k)$, or is Lemma 9.2 applied? No No No $\Delta_{ij} \geq T_{\max}$?

Figure 9.3 Flow diagram of algorithm $T_m F$.

Algorithm $T_m F$ continues with a check of the end of the schedule horizon and detects the

nature of the change (or dot) in $d(k, t)$. For each departure epoch, Δ_{ij} is examined to deter

mine whether it is greater than or equal to $T_{\max}$; if greater, then an unpaid joining array is

added; otherwise a disjointed departure time array is added. Each arrival epoch (detected in

a step move in Figure 9.3) is added as a disjointed arrival array. If a departure epoch is identi

fied in a step move, the algorithm looks for a possible dot on $d(k, t)$, adding its arrival epoch

to the list of disjointed arrival arrays. At the end of the process, the algorithm constructs

joining arrays from the disjointed arrival and departure arrays, using the FIFO rule. The

complete process is shown in Figure 9.3.

An example of constructing vehicle chains (blocks), including the employment of algo

rithm T m F, is shown in Figures 9.4 through 9.6. The example, consisting of three terminals

and a 24-trip schedule, is exhibited in Figure 9.4, including DH travel time matrix, shift

ing tolerance, T_{max} , and schedule horizon. It should be noted, though, that DH travel time

between terminals b and c is considered in both directions although there is only a service

route between c and b. The fleet-reduction procedure, involving the shifting of departure

times and DH trip insertions, is shown in Figure 7.12; here it is applied to the example in

Figure 9.5. Two DH trips and two shifts are introduced into the process to reduce $D(a)$ and

$D(c)$ from 4 to 3, resulting in a fleet size of 11 vehicles. The shifts are shown in Figure 9.5 DH travel times (min) a b c a – 25 45 b 25 – 35 c 45 35 – Shifting tolerance= ± 3 min $T_{max}=40$ min $[T_1, T_2] = [6:00, 9:15]$ Trip number
Departure terminal Departure time Arrival terminal
Arrival time 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
19 20 21 22 23 24 a a a a a a a b b b b b b b b c c c
c c c 6:00 6:15 6:20 6:43 7:10 7:30 7:50 8:00 8:10 6:00
6:00 6:15 6:15 6:20 7:30 7:35 7:45 8:05 6:30 7:00 7:15 7:15
7:20 7:38 b c c c c b b b b a b b a a a a a b a b a b b
6:30 7:20 7:25 7:38 8:00 8:25 8:35 8:50 8:50 6:30 6:40 7:00
7:00 7:30 8:30 8:40 8:50 9:00 7:15 7:40 7:45 7:50 8:50 9:03
Routes a b c

Figure 9.4 Example consisting of 24 trips and 3 terminals for constructing vehicle chains with the T_{max} constraint. (6:43) 4

$d(b, t)$ 24 DH 1 DH 2 $D(b)=5$

$d(c, t)$ Time $D(c) = 4 \ 3 \ 24 \ 4$ Time

6:00 6:15 6:30 6:45 7:00 7:15 7:30 7:45 8:00 8:15 8:30 8:45
9:00

6:00 6:15 6:30 6:45 7:00 7:15 7:30 7:45 8:00 8:15 8:30 8:45
9:00

$d(a, t)$ Trip number $D(a) = 4 \ 3$

4

2

0

4

2

0

4

2

0

Figure 9.5 The 24-trip example (depicted by 3 DFs) undergoing a DH trip-insertion procedure, combined with shifting departure times. Joining based on T m F $d(b, t)$
6:00 6:30 7:00 7:30 8:00 8:30 9:00 10 11 12 13 14 1 11 12
19 15 16 21 17 18 6 7 8 9 23 24 DH 1 4 2 0 $D(k) = 5$ Trip
number (a) Time Joining based on FIFO 6:00 6:30 7:00 7:30
8:00 8:30 9:00 $d(b, t)$ 10 11 12 13 14 1 11 12 19 15 16 21
17 18 6 7 8 9 23 24 DH 1 4 2 0 $D(k) = 5$ (b)

Figure 9.6 Arrival-departure joinings for constructing vehicle chains (blocks) in the middle hollow of terminal b, utilizing in (a) the T m F algorithm and in (b) the FIFO rule.

by their shifting length and trip number. We can see from this figure that the only middle

hollow containing more than a single departure is the second hollow of $d(b, t)$; hence, only

this hollow is subject to the process of algorithm T m F.

Figure 9.6a describes the solution for a
algorithm T m F in comparison with a solution based

only on the FIFO rule in Figure 9.6b. The
trip numbers of the example, appearing in Figure

9.4, are added to Figure 9.6. A algorithm T m F
results in two unpaid joinings between the

arrivals of trips 11 and 12 and the d
epartures of trips 15 and 17, respectively. In
both cases,

$\Delta_{ij} > T_{\max} = 40$ min. The remaining joinings in Figure
9.6a are based on the FIFO rule. The

use of only the FIFO rule for the entire
process results in only one unpaid joining (that

between trips 11 and 15) as is shown in Figure 9.6b.

The final phase of the arrival-departure joining process is to construct vehicle blocks. This

will contain the joinings created and other FIFO-based joinings in order to make a complete

set of blocks. The 11 blocks of the 24-trip-schedule
example, based on algorithm T m F (at

terminal b), are given by their numbers in
the following list: [1-DH 1-22-7], [10-4-24],

[11-15], [12-23], [12-17], [13-5], [13-DH 2-9],
[14-6], [19-16], [20-8], [21-18]. The process

based only on the FIFO rule results in the
same blocks, except for the fifth and ninth
blocks,

which become [12-16] and [19-17], respectively.

9.3 MATHEMATICAL SOLUTIONS

Crew scheduling and rostering are extensively treated

mathematically in the books edited by

Wren (1981), Rousseau (1985), D aduna and Wren (1988), Desrochers and Rousseau (1992),

D aduna et al. (1995), Wilson (1999), and Voss and D aduna (2001) and in a forthcoming

volume by Hickman et al. (2008). Specific detailed studies are reviewed in the succeeding

text in Section 9.6.

The basic OR formulation of the CSP is a zero-one integer linear programming (ILP),

called a set-partitioning problem (SPP). SPP has as its objective the selecting of a minimum

cost set of feasible duties, such that each duty piece is included in exactly one of the duties.

CSP is often illustrated by rows and columns, in which the rows are the duty pieces and

the columns are the duties, each accompanied by a duty cost. In the latter, SPP is aimed at

minimizing the cost of a set of columns, such that each row is included exactly once in one

of the columns in the solution. The following is the SPP formulation:

Problem P5 $\min \sum_{q \in Q} c_q x_q$ (9.8) Subject to $\sum_{q \in Q(j)} x_q = 1, j \in J$ (9.9) $x_q \in \{0, 1\}, q \in Q$ (9.10)

where

Q is the set of all feasible duties

c_q is the cost of duty $q \in Q$ $Q(j) \in Q$ is the set of duties covering duty piece $j \in J$

A binary zero-one variable x_q is used to indicate whether duty q is selected in the solution

otherwise. Constraint (9.9) assures that each piece will be covered by exactly one duty. The

running time of P5 belongs to the class of NP complete (see Section 5.5).

An easier way to solve the CSP is to relax constraint (9.9): $x_{ij} \geq \epsilon \in \Sigma 1$ (9.11)

Equation 9.11, together with Equations 9.8 and 9.10, represents a new problem, called a

set-covering problem (SCP). SCP is usually solved first (before the SPP), and the solution

is changed to handle SPP by deleting overlapping trips. This experienced-based deletion

process involves changes in the duties, in the SCP solution, and in a crew member who is

so assigned and will make the trip as a passenger. Freling et al. (1999) explain that such a

change affects neither the feasibility nor the cost of the duties considered.

One way to speed up the running time of the SPP of CSP is by the use of the Lagrangian

relaxation method. In this method, the complicated set of constraints shown by Equation

9.9 is removed and put into the objective function (9.8). Each constraint removed is weighted

by a given Lagrange multiplier. Another useful and commonly used method for reducing

the SPP complexity is the column-generation (CG) technique. The number of columns (rep-

resenting duties) in the SPP is usually a very large number. The CG technique, instead of

examining all the columns over and over again, selects subsets of columns and either checks

each subset for optimality or solves several subproblems. Some of the articles reviewed in

Section 9.6 cover these methods.

A known heuristic approach to solve CSP within a framework of resolving both CSP and

the vehicle-scheduling problem (VSP) is discussed by Bodin et al. (1983) and presented by

Ball et al. (1981, 1983). This approach is chosen to illustrate the inherent combinatorial

difficulties that exist in CSP. The basic steps of the approach are as follows: (1) generate all

feasible pieces of work (to be derived from the vehicle blocks); (2) establish an internal piece

cost (based on piece characteristics and past experience); (3) create an acyclic network from

each block, in which nodes are the relief points and arcs represent the cost of each feasible

piece in the block; (4) solve the shortest-path problem in order to establish the best (minimum

cost) pieces; and (5) solve a matching problem while using at most three legal-piece

combination, and if the solution is not feasible, reiterate part of the process by updating the

piece combination cost and redo the shortest path (Step 4) until the solution is feasible and

satisfactory. Figure 9.7 presents the procedure schematically.

The example presented in Figure 9.4 is used for demonstrating the shortest-path and

matching (SPM) approach exhibited in Figure 9.7. The 11-block results of the TMF algorithm

of this example appear in Table 9.1; they are re-extracted from the analysis shown in Figures

9.5 and 9.6a. Given that all three terminals a , b , and c are relief points, each block can be

considered a driver's schedule (duty) or be partitioned into alternative pieces covering all

possible combinations. Figure 9.8 shows how to partition the blocks into combinations of

pieces. In addition, each piece is assigned an internal piece cost based on the piece characteristics

(e.g., time of day, arrival and departure locations, type of vehicle required) and past

experience (the cost of a similar piece in past crew schedules). Two sets of blocks have the

same pieces and costs (numbered 3 and 5, and 9 and 11). It should be noted that the rest set

includes unpaid idle time as a result of algorithm TMF.

The right-hand column of Figure 9.8 contains the a cyclic network of each block, representing

senting all the possible pieces and their internal costs. This a cyclic network has undergone a

shortest-path analysis, such as the known algorithm Dijkstra (1959). For the sake of clarity,

the Dijkstra algorithm is described with an example in Appendix 9.A; it will also be used in

Chapters 11, 12, 15, and 16. The results of the Dijkstra procedure are emphasized in Figure

9.8, a long with the minimum piece cost needed to cover the whole block. These results are

illustrated in Figure 9.9 for each block. Given that each piece is eligible to be covered by 1

driver, then Figure 9.9 shows that the 11 blocks require 17 drivers at a total cost of 82 (the

cost units are meaningless for the example). However, there are more possibilities for matching

some pieces if they are legal (from the labor-agreement perspective) and can reduce the

total cost. Initialization: vehicle blocks Generate all feasible pieces Establish internal piece cost Create acyclic network Solve shortest path Solve matching problems to create duties* Update piece cost STOP Yes No * Commonly one-, two-, or three-piece legal Feasible solution ?

Figure 9.7 Flow diagram of the SPM heuristic approach to creating duties.

Table 9.1 Solution for vehicle blocks and routing of the example appearing in Figures 9.4 through 9.6a, using the algorithm TMF

Block number Trips in block (see Figure 10.4) Block routing (see Figure 10.4) (# represents DH trip)

1 1-DH 1-22-7 a-b # c-a-b

2 10-4-24 b-a-c-b

3 11-25 b-b-a

4 2-23 a-c-b

5 12-17 b-b-a

6 13-5 b-a-c

7 3-DH 2 -9 a-c # a-b

8 14-6 b-a-b

9 19-16 c-b-a

10 20-8 c-a-b

11 21-18 c-b-a

Taking into account the DH travel times in Figure 9.4 as a measure of moving between

a, b, and c, there are five possible matchings that need to be examined for the first piece of

block 1 and one possible matching for the first piece of block 6. These possibilities are shown

by the dashed lines in Figure 9.9. Usually, if the best (minimum cost) feasible matching

results in a cost reduction (i.e., less than 82), then this matching is selected and the number

of drivers can be reduced by one or two (to 16 or 15). The more pieces examined for a single

duty, the more running time the matching process requires (Ball et al., 1983). Commonly,

the check is performed on two and three legal pieces. More details on possible optimal and

heuristic matching procedures can be found, for example, in Ahuja et al. (1993).

The simplified example using an SPM approach demonstrates some of the complexities

involved in handling CSP, hence justifying the use of OR software. Moreover, in addition to the

combinatorial problems inherent in the scheduling tasks, there are human dissatisfaction issues

concerning the crews that deserve attention and make this undertaking even more cumbersome. Block number Feasible pieces, (*) optimal solution Internal piece cost Shortest-path network and solution (best pieces emphasized) 1 a-b (*) a-c a-a a-b b-a (*) b-b c-a c-b a-b (*) 3 6 10 16 5 12 4 8 3 2 b-a b-c (*) b-b a-c a-b c-b (*) 3 6 12 4 9 4 3, 5 b-b b-a (*) b-a 3 5 4 4 a-c (*) a-b c-b (*) 5 11 4 6 b-a (*) b-c a-c (*) 4 10 5 7 a-c a-a a-b (*) c-b a-b 5 9 11 3 4 8 b-a (*) b-b a-b (*) 4 10 4 9, 11 c-b c-a (*) b-a 5 9 5 10 c-a c-b (*) a-b 6 10 5 a b c a b 3 2 4 3 6 10 16 8 5 12 11 DH trip Shortest-path's length 9 a 3 4 5 5 bb a c b 3 2 4 6 12 10 b b 5 4 11 9 ca c 4 5 10 9 a b b 4 4 10 8 ab b 6 5 10 10 ac a 5 5 9 9 b c a b 5 3 4 9 11 11 a c 7

Figure 9.8 Partitioning of blocks into minimum-cost pieces.

9.4 CASE STUDY: NJ COMMUTER RAIL

This section describes a case study that was performed for the New Jersey (NJ) Transit

Corporation. In this study (Tykulska et al., 1985), we will experience that all computers

wait at the same speed; what counts is the flexibility of the solution and the alternatives that

can be produced.

9.4.1 Background

NJ Transit provides a large commuter rail operation, carrying over 70,000 passengers and

running more than 400 trains daily. At the time of the study, NJ Transit had decided to take

under its wing both the service and labor force then maintained by Conrail. The collective

bargaining agreements then in effect between Conrail and the various bargaining units

operating in the labor force would then no longer remain in effect. Thus, NJ Transit was

placed in the position of negotiating new collective bargaining agreements.

With 1 labor-crew costs representing a major component of commuter rail costs, it was

critical that NJ Transit management understand the cost ramifications of different work

rules. Given the variations in these rules that already existed for the different NJ commuter

rail lines and the multitude of permutations, it was important that NJ Transit have some

mechanism for quickly analyzing options. Thus, the objective of the project was to provide

the NJ Transit management with a tool that would enable them to quickly analyze the

implications of work rule changes for labor costs. This tool would be used during the nego-

tiation process, since it would provide management with the ability to quantify the impacts

of various proposals and counterproposals. In addition, NJ Transit needed a tool that could

be used on an ongoing basis in order to (1) analyze the marginal cost impact of changes in

train schedules and (2) produce near-optimal crew assignments (duties), given a schedule

and a new set of work rules. 6:00 1 2 3 4 5 6 7 8 9 10 11
6:30 7:00 7:30 8:00 Time Time Block number c a a a a a a a
a a c c c c a a a b b b b b b b b c c c b b b 8:30 9:00
6:00 6:30 7:00 7:30 8:00 8:30 9:00

Figure 9.9 Result of partitioning the blocks into best pieces with feasible matching alternatives.

There were two primary bases of pay applicable to commuter rail. The first pay basis

was similar to a non-industrial pay basis: pay is based on hours worked, with a guarantee of

8 h/day. Any time in excess of 8 h is paid at time and a half. The second pay basis was called

the dual basis of pay. Employees were guaranteed pay for a standard day, defined as 8 h and

a prescribed number of miles (100 for engineers and 150 for trainmen, in the case of NJ

Transit); if either the time or the mileage limits were exceeded, additional payments were

made. Therefore, it was possible to receive both overtime and overtime payments, a double

penalty (for the company) for long assignments.

There were also two primary work rules, other than those incorporated in the basis of pay,

that had a large impact on labor costs. The more important of the two relates to interlining

restrictions. In many cases, crew members could only be assigned to a single line or group

of lines within their seniority district. This limited the ability to assign crews in the most

efficient manner available. The other work rules that affected labor costs involved manning

requirements. For example, men may not be needed for the operation of trains but may

be required, in certain seniority districts, because of prior collective bargaining agreements.

Also, the minimum number of trainmen (conductors, brakemen, and ticket collectors) is usu

ally specified by contract, but the maximum number is determined by a management decision

relating to the need to protect revenue, based on passenger volume and trip duration.

9.4.2 Crew assignment procedures

9.4.2.1 Overview of model

To address the issues described, a crew assignment/work rule model was developed. The

approach consisted of three steps: (1) defining train schedules (arrival and departure times

and locations, trip distances, and manpower requirements) and work rules (interlining

restrictions, DH rules, minimum layovers, hours of service limits, and the locations of the

beginning and ending of duties), (2) generating a range of alternative crew assignments, and

finally, (3) selecting the set of best assignments (duties).

Crew assignments were generated, starting with the earliest departing train, by finding

the first, second, third, etc., feasible departing connections at each location. From each of

these trains, additional connections were explored; thus, a tree of assignments was found

for each train. The size of the tree can be limited by several user-supplied parameters. If the

generation is limited, the entire procedure is repeated, starting with the latest arrivals and

using feasible connections. This is to ensure that a good set of morning and evening assign

ments are generated. Assignments were then costed by applying the appropriate bases of pay.

The costing procedure allowed for hourly or mileage-pay bases; daily, weekly, and monthly

guarantees; both overtime and overmile payments; spread premiums; and split duties.

The last step was to determine the actual crew assignments. Several heuristic techniques were

developed to reduce the magnitude of the computational problem. An optimization package

was used separately for weekdays, Saturdays, and Sundays. For each day, a relaxed (noninteger)

SCP was formulated and solved, assignments in the solution were set with overlapping trains,

additional assignments were made without the overlap, and this reduced SCP was solved for

an integer solution. Finally, the separate daily solutions were combined, producing assignments

with the required relief days. Figure 9.10 presents an overview of the model's components.

9.4.2.2 Model development

The first stage of the project was to identify an appropriate methodology. To accomplish the

assignment-selection procedure, careful consideration was given to various mathematical

programming and heuristic techniques. It is worth mentioning that in a comparison of bus-

and rail-crew scheduling, interlining and DH issues are more appropriate to bus-crew assign

ments. The combinatorial complexities of bus-crew scheduling, introduced by interlining and

DH, may dictate sequential optimization techniques. For the case study, use of the rest
gen

eral technique of eliminating assignments prior
to optimization was found to be sufficient.

We have already noticed that it is time consuming and
expensive to solve large-size prob

lems by exact integer-programming procedures. C
onsequently, it is common either to relax

the integer restriction (solving SCP) or to u
se heuristics. The SCP formulation was deemed

more appropriate for the rail CSP of the c
ase study, since crews can avoid deadhead o
n

trains. The SCP is formulated in P5 in E
quations 9.8, 9.10, and 9.11; in the case s
tudy, the

elements in P5 have the following interpretation: the set Q
is all feasible duties (assignments),

c_q is the crew cost of duty (assignment) $q \in Q$, $Q(j) \in Q$
is the set of duties (assignments)

covering a set of trains (or trips) $j \in J$, a
nd x_q is used for indicating whether duty q is to be

worked (selected in the solution) or not.

In commuter rail operations, crews occasionally p
erform DH trips while traveling on

service trains. In bus operations, crews may
deadhead via a route other than the service

route, with or without their vehicle. Usually,
bus-scheduling programs include a special DH

travel-time matrix so that the DH crew/vehicle can reach
the next departure point prior Assignment generator
Assignment limiter Manual assignment verifier Train
schedules Network description Basis of pay Assignment
costing Solve relaxed-set covering problem Resolve

overlapping situations Combine weekly, Saturday, and Sunday into weekly assignments Solve integer-set covering problem Represents data flow Represents control flow

Figure 9.10 Overview of the CSP process in the NJ case study.

to the arrival of a service trip (with which they overlap in the SCP solution). Although the

early availability of DH crews may create new opportunities for bus operations, this situa-

tion does not apply to commuter rail operations; the problematic issue with the latter is the

existence of DH (overlapping) loops in the solution.

A DH loop refers to a particular set of consecutive trains on a given assignment that fulfills

the two following conditions: (a) The departure station of the first train on the loop coincides

with the arrival station of the last train on the loop, and (b) each train on the loop is covered by

other assignments in the SCP solution. The loops are identified by first creating a train-coverage

table, which shows how many times each train is covered. The procedure is then iterated over

all the assignments in the solution set and over all trains in each assignment. When a train is

covered more than once, all candidate loops are reiterated (initially for each assignment, there

are no candidate loops). If the train completes a loop, the trains on the loop are eliminated and

the loopless assignment is found. This train

then starts a new candidate loop. Eliminating DH

loops will usually lower the value of the objective function. One should note, however, that it

is possible to obtain a minimum-cost solution that contains DH loops if these assignments are

already coded at the minimum daily pay (in other words, loop elimination cannot lower the

cost). From a operations and safety perspective, though, all DH loops should be eliminated.

To overcome the undesirable situation of DH loops, the process chosen used three main

steps as shown in Figure 9.10: (1) solving a relaxed (linear) SCP formulation, (2) re-

solving DH loops by forming new loopless assignments, and (3) solving an exact (integer) SPP

formulation with the incumbent solution plus the new loopless assignments.

9.4.2.3 Model implementation and results

The model was implemented as a set of a assignment filters: each program was separated and

acted to process or filter a file of assignments. This provided a structured modular package.

In addition, it was found necessary to build extensive checks for error and inconsistency into

this procedure, since the optimization procedure assumes that all assignments presented to it

are feasible. The existing assignments were then coded and used as seeds for the optimization

procedures. Figure 9.11 illustrates the computational process of the model in a flow diagram.

A major component of the model was the assignment generator. Assignments were gen

erated without regard to crew cost. This is due to the fact that the model was explicitly

designed to investigate alternative bases of pay. The generation was carried out in two

passes, forward and backward, which helped to ensure that each train was covered at least

once; also, this provides better-quality assignments at the start and finish of the day. Five

parameters were used in order to limit the number of assignments generated: (1) limit on the

connection-time window, (2) limit on the number of connections that need to be examined

for each train in the search tree, (3) maximum number of assignments to be formed from

each root tree, (4) minimum total time for any assignment, and (5) maximum total time for

any assignment.

Split duty and part-time assignments were generated by specifying three additional param

eters: the time at which a split can occur, the minimum length of the split, and the minimum

length of a part-time operation. In splitting a duty, the generator simply enforces a layover

of at least the minimum length at the appropriate time.

Once a set of regular, split, and part-time assignments was generated, a separate procedure

was employed to reduce the number of assignments. This is done so as not to overwhelm the

optimization procedure. This reduction first computes a productivity measure for each assign

ment. Several measures can be used: pay hours per total hour (effective hourly rate), pay dollars

per train hour, pay dollars per train, total hours per train hour, and so forth. It was decided to

use total hours per train hour because it was felt to be politically less sensitive to use a measure

divorced from pay. The assignments were then sorted by productivity measures. The reduction

was carried out by specifying a coverage limit—how many times each train must be covered

before an assignment is dropped. The assignments were processed in order of productivity

measure, updating a coverage count for each train on the assignment. If all the trains on a n

assignment are already covered to the limit, then that assignment is dropped. However, if a t

least one train is not at the limit, then the assignment is passed on. Thus, it is possible for some

trains to be covered more than the coverage parameter. Furthermore, a train could be covered

less than the coverage limit if the assignment generator did not produce that many assignments

for this train. In practice, a coverage limit of 20 times per train was found to work well.

The assignment-costing procedure was designed to be as flexible as possible. It must be very

simple in order to recost some or all of a set of assignments, using alternative bases of pay.

More important, it must be easy to modify the procedure to incorporate completely new con-

cepts in costing assignments. The mechanics of costing are as follows: For each assignment,

select the appropriate pay basis (from the line-basis-of-pay definition), apply the appropriate Input information (routes, stations, timetable, time elements) Existing work rules (1) Split-duty (2) Part-time (3) Interlining (4) DH of crew (5) Various time and distance allowances (6) Seniority district restrictions Trip chains (equipment schedule) Generate all feasible assignments Without DH With DH Optimization package for SCP (MPSX) Identify efficient pieces on trip chains (dual basis of pay) Heuristic CSP technique Estimation of cost components Feasibility check DH travel time matrix Extension of feasible joinings Form low-productivity assignments Form part-time assignments Parttime allowed ? Splitduty allowed ? Optimize split-duty from remaining pieces (matching) Establish assignments Yes Yes No No (4) (6) (3) Select a set of assignments Possible changes in work rules

Figure 9.11 Flow diagram listing the major computational elements included in the NJ case study.

rates to the miles and hours, and compute any guarantees. New pay bases are easily added by

inserting the necessary code and referring to a previously unassigned parameter.

After costing, the assignments were ready to be optimized. As described previously, the

process was to solve a relaxed SCP, resolve overlapping by forming new loopless assign

ments, recost, and then solve an SPP using both the new assignments and the relaxed SCP

solution. The SPP was much smaller than the SCP; typically, it limits the relaxed SCP to 20

assignments per train and solves the problem with an average of 4 trains per assignment.

Thus, the number of assignments for the SPP could be reduced by two orders of magnitude.

Another technique used to reduce the problem dimension is to partition the weekly crew

assignment problem into three components: weekday, Saturday, and Sunday. Three separate

solutions were generated and solved and then combined with the three solutions to produce

a full weekly set of assignments.

The optimization steps used the software package MPSX/370 with mixed-integer program

ming. The number of assignments (columns) ranged between 3 379 and 7 591 covering 2 50 trains

(rows). The central processing unit (CPU) times were in the order of 1 00 s, during which it

was found that the DH loop-elimination procedure was needed to produce good solutions: the

objective function reductions ranged from 0 % to 2 0 % but typically were around 2 % to 5 %. The

second linear optimization (for the relaxed solution plus the loopless assignments) usually terminated

the integer, eliminating the need for branch and bound. For cases requiring the employ

ment of branch and bound, the value of the

objective function usually increases by less than 1 %.

The model generally produced very good results; typical savings for manually generated

crew assignments ranged from 2 % to 20 %. However, the D H Loop-elimination procedure

occasionally produced assignments with very low productivity. For example, an assignment

with only two trains would uniquely cover only one train. The reason is that there are

no penalties for these poor assignments: the choice among competing assignments dur

ing optimization is done solely on a cost basis and not productivity. NJ Transit used the

model extensively during the transition period and during recent labor negotiation to evalu

ate alternative bases of pay and work rules. The Operations Department has exhaustively

reviewed the assignments produced, with positive reactions.

9.5 CREW ROSTERING

The CSP establishes duties in an optimal manner with the aim of minimizing crew assign

ments for a given set of constraints. Usually transit agencies then arrange the duties in a set of

patterns for a specified time horizon. Each pattern is defined as a roster, containing duties to

be fulfilled over a certain number of consecutive days. Commonly, the pattern repeats itself

in cycles (whether a week, a month, or any other period). The crew-rostering problem (CRP)

is usually to find a feasible set of rosters to cover all duties, using one or more of t

these four

objectives: (a) minimum number of crews required, (b) minimum sum of roster costs, (c) minimum

imum of the maximum roster duration, and (d) balancing (equity of) workload and days-off.

9.5.1 Literature review and problem definition

A broad literature survey on CRP by Ernst et al. (2004) reveals that the number of articles

on rostering in public transit is relatively small. The survey refers to a total of 193 articles on

staff scheduling and rostering, with a classification of applications, problems, and models.

Basically, the solution methods proffered for CRP in this survey are similar to those of CSP.

That is, math-programming (e.g., SPP, SCP), heuristic, and metaheuristic approaches. The

last is based on methods drawn from artificial intelligence, neural engineering, biological

evolution, and more (see, e.g., Lourenco et al., 2001). Among the few studies on CRP,

Carraresi and Gallo (1984) and Bianco et al. (1992) utilized network concepts for solving

CRP. Catanas and Paoxao (1995) used a SPP and SCP formulation combined with heuristic

rules for CRP. Caprara et al. (1998) determined rosters complying with the minimum number

of weeks in which each duty is carried out but once a day; mixed-integer programming

was used as a base to develop an efficient heuristic algorithm. Sodhi and Norris (2004)

decomposed the CRP at the London Underground into two stages: The first established a

roster pattern for each depot, followed by the insertion of duties into each of the patterns,

and the second was formulated as an assignment problem (a known problem in OR; a special

case of the minimum-cost flow problem on networks) with side constraints.

The CRP involves compliance with work and legal rules, institutional requirements, and

established in labor contracts necessitate constraints, as in CSP, that are difficult to treat

analytically, for instance, the minimum number of hours required between two consecutive

working days, exact (or between minimum and maximum) number of days-off per week,

minimum number of weekend days in days-off per month, maximum working hours per

day, equity of weekend working hours, and more.

The rosters are commonly arranged in cyclic patterns; the crew can have the same cyclic

pattern or rotate between patterns, depending on the arrangement between the transit

agency and their employees. The upper part of Figure 9.12 illustrates seven weekly roster

types covering all possible rosters complying with two consecutive weekly days-off. Each

duty is denoted by d_{jq} , in which q is the day in the week and j is the index of a specific duty.

Certainly some of the duties (among a total of 35) can be the same if, for example, t

here

are fewer than ve duties for a given day. Moreover, more rosters of the same type can be

used when more duties are involved. A single crew member may be tied to a single roster or

switched between rosters every month (or every few weeks). In the latter case, the number of

days-off in the transition period can be fewer or greater than the required 2 days.

The lower part of Figure 9.12 presents a sequence of quintuplets (five consecutive work

ing days) for a single crew member, which certainly results in one weekly day-off. The complete

cycle of these quintuplets covers 36 duties in 6 weeks, some of which can be the same.

Observation of the upper part of Figure 9.12 will show quintuplets if the same crew member

repeats the same roster, but with two weekly days-off and not one. Rosters generally established

in transit agencies may contain a combination of quintuplets, quadruplets, and triplets of work

ing days.

9.5.2 Heuristic approach

This section provides a heuristic approach to a simplified CRP; this is done for the sake of

illustrating the different planning elements involved and for illustrating the complexity of

the analysis. The heuristic method is p

(See Table 9.2; each roster in this table contains one duty per day for days marked with a "V"). Table 9.2 Input data for the example problem

Day 1	Day 2	Day 3	Day 4	Duties distributed
d 1 1	d 1 2	d 1 3	- d 2 1	- d 2 3 d 2 4 - d 3 2
d 3 3	d 3 4	Maximum roster hours	- - d 4 3	- Type of roster
R 1 V V V - 25	R 2 - V V - 18	R 3 - - V V 18	Duty for day	q (q = 1, 2, 3, 4) Start and end times Duty length (h), L _j
d 1 q 6:00-16:00 10	d q 2 12:00-20:00 8	d q 3 16:00-23:00 7	d 4 q 18:00-midnight 6	Mon Tue Wed Thur Fri Sat Sun Roster type
R 1 d M 1 d M 16 d M 21 d M 26 d M 31 d M 1 d M 7 d M 13 d M 19 d M 25 d	Tu d W d Th Off Off	R 2 Off Off	R 3 Off Off	R 4 Off Off
R 5 Off Off	R 6 Off Off	R 7 Off Off	Complete cycle of duties assigned to a single crew member, by week	1 Off 2 Off 3 Off 4 Off 5 Off 6 Off 2 3 4
d F 5 d Tu d W 6 d Tu 22 d Tu 27 d Tu 32 d Tu 2 d Tu 8 d Tu 14 d Tu 20 d Tu 21 d W 11 d W 28 d W 33 d W 3 d W 9 d W 15				

d W 16 d W 32 d Th 12 d F 13 d Sa 14 d Su 15 d Th 17 d Th
 34 d Th 4 d Th 10 d Th 21 d Th 33 d Th 27 d F 18 d Sa 19 d
 Su 20 d F 23 d F 23 d F 16 d F 22 d F 28 d F 34 d Sa 24 d
 Su 25 d Sa 29 d Sa 11 d Sa 17 d Sa 23 d Sa 29 d Sa 35 d Su
 30 d Su 30 d Su 6 d Su 12 d Su 18 d Su 24 d Su 30 d Su 36 7
 d Th 8 d F 9 d Sa 10

Figure 9.12 Roster types with two weekly consecutive days-off and possible duties assigned to a single crew member with one weekly day-off for a sequence of quintuplets (five consecutive working days).

The following heuristic procedure intends to find rosters with the maximum workload

if possible, thus fulfilling the objective of arranging the minimum number of crew members.

9.5.2.1 Procedure roster

Step 0: Initialization; determine index j of R_j by arranging all roster types in a list in decreasing order of their maximum possible lengths. For cases with the same length, assign a higher ranking in the list to roster types that contain more consecutive days (duties).

Step 1: Select the first nontreated roster type from the list in Step 0; construct a feasible network $G = \{N, A\}$ consisting of a set of nodes N and a set of arcs A per roster type, using a source node "s" and a target node "t"; let the arc's value be v_{ij} , $j \neq 0$, otherwise L_j , N_j , t_j , $j \neq 0$, $= \infty$

where L_j is the duty length. Label the (unlabeled) nodes (say, by "v") contained in List-1 and List-2 paths; if no more rosters, go to Step 4.

Step 2: Solve a longest-path problem by using the rearranging node number (RNN) algorithm in Appendix 9.A (by multiplying all v_{ij} , $(i, j) \in A$ by (-1) and applying the shortest-path procedure); if no s-t path exists, go to Step 1.

Step 3: Delete and store in List-1 paths (arcs and nodes) belonging to the solution attained, except nodes s and t; if the current solution contains unlabeled node(s), store their path and the sum W in List-2, where $W = L_j p_{j,s,t} = \sum_{j \neq 0} p_{j,s,t}$ and $p_{j,s,t}$

is a longest path determined, and delete only the incoming arcs to the labeled nodes (in p^*); go to Step 2.

Step 4: Determine the best solution using the paths (rosters) with the highest W values from List-1 and List-2, such that each duty is covered only once. List all nodes (duties) that remain outside any path solution for a manual rostering-assignment process.

The example problem in Table 9.2 undergoes a roster procedure. The index j assigned to the

three types in Table 9.2 conforms to Step 0 of the procedure. A preliminary stage to Step 1

in roster is a feasibility check. Table 9.3 lists by roster type all infeasible duty connections;

these infeasibilities affect Step 1. The three feasible networks constructed for R_1 , R_2 , and R_3

appear in Figure 9.13. The longest-path solution is attained in Step 2 by algorithm RNN, Table 9.3 Infeasible consecutive duty connections

Roster type	Infeasible connection ($q = 1, 2, 3, 4$)	Dominant constraint
R_1	$d d d 1 q 1 q 1 q - - d d d 1 q 1 q 2 q - -$	$d d d 1 q 1 q 3 q - - d d d 1 q 1 q 4 q - - d d d 2 q 1 q 1 q - - d d d 1 q 1 q 2 q - -$
R_2	$d d d 1 q 1 q 2 q - -$	Maximum roster hours
R_3	$d d 3 q 1 q -$	Minimum hours between consecutive working days

presented in Section 9.A.2. The RNN procedure is designed for finding the shortest path in

networks with any arc value (it can also be negative) and especially without cycles (in order

to avoid the possibility of an infeasible solution derived from negative cycles). The latter

characteristic is compatible with the networks constructed in Step 1. The longest path is

determined by the shortest path using RNN, in which $v(i, j)$ is replaced by $-v(i, j)$. $d 3 2 d 3 1 d 3 1 d 3 3 d 3 4 d$

1 1 d 1 2 d 1 2 d 2 3 t s R 1 W=25 h W=25 h 8 0 0 0 0 10 10
 7 7 7 6 6 8 7 s d 3 1 d 3 1 d 3 2 d 3 3 d 3 4 t R 2 v v v v
 10 7 d 2 3 d 2 1 8 8 7 6 7 6 0 0 0 0 W=18 h W=14 h d 3 2 d
 3 3 d 3 4 t s d 3 1 d 3 1 d 4 2 d 4 3 R 3 10 8 7 6 (a) (b)
 8 8 8 8 7 7 7 7 0 0 v v W=18 h W=15 h W=14 h W=13 h (a) (b)

Figure 9.13 Heuristic rostering process for the example problem, using a longest-path approach for each roster type (R 1 , R 2 , R 3).

The first feasible network for R 1 in Figure 9.13 results, with the use of roster, in a two

path solution, both with $W = 25$ h; these paths are inserted into List-1. The second network

for R 2 also results in two paths of pairs of duties (second-type rosters), but all four nodes—

besides s and t—in the solution are labeled; that is, they were included in a previous solution.

An analysis of R 2 provides the longest path, s d d t 1 2 2 3 – – – with $W = 18$ h. Then according to

Step 3 of roster, a rc s d- 1 2 is deleted and the second longest path is found. Both paths of R 2 are

inserted into List-2. The third and last roster-type network, R 3 , begins Step 2 with a nonla

beled path, s d d t 1 3 2 4 – – – with $W = 18$ h. The second solution, marked (a) in Figure 9.13, contains

a labeled node; thus, only a rc s d- 2 3 is deleted before continuing with the solution marked (b).

In the latter case, a rc s d- 3 3 is deleted, thereby allowing a fourth longest path with $W = 13$ h.

The first three steps of roster conclude with List-1 containing the solutions of R 1 and first

and last solutions of R 3 and List-2 containing the solutions of R 2 and the second and third

solutions of R 3 . If we continue now to Step 4 of roster, List-1 (usually having paths with

the highest W values) covers all 10 duties and, therefore, presents the ultimate solution.

Explicitly, four (as a minimum) crew members are required, two of roster type R_1 and two

of R_3 . The rosters are as follows: [] d d d 1 1 3 2 2 3 - - , [] d d d 2 1 1 2 3 3 - - , [] d d 1 3 2 4 - , and [] d d 4 3 3 4 - with $W = 25, 25,$

18, and 13, respectively. The total hours of all duties are a given number (81 hint the

example). It should be noted that (1) if d 4 3 , for instance, is removed from the input data, then

d 3

4 will be left for manual assignment, and (2) List-2, though it may seem unnecessary, could

open up opportunities for practical solutions utilizing a partial manual-assignment process.

9.6 LITERATURE REVIEW AND FURTHER READING

The CSP has been discussed abundantly both in and out of the transportation literature; rel

evant papers are found in journals related to mathematics, computing, operations research,

and specialized scheduling resources. We will focus here mainly on the latest developments

in this field. The literature for the CRP is sparse (see previous section).

CSP is often formulated as a nSCP, as is explicitly shown in Equations 9.8, 9.10, and

9.11. For this purpose, a large set of driver workdays is defined, and a subset is then chosen

that attempts to minimize costs, subject to constraints that make sure that all the necessary

driving duties are performed. Most CSP f

formulations also verify that the labor-agreement

rights of all drivers are maintained. A full classic CSP normally includes the generation of

the feasible workdays as a first separate stage and then the choice of a subset that best satisfies

all needs as a second stage. In most practical transit networks, there can be thousands

of driving duties to be carried out, hence the need for millions of possible workdays. It is,

therefore, common to construct in advance only a limited number of feasible workdays.

Most traditional CSP formulations concentrate on the work of bus drivers. Some of these

formulations are transferable to railway use. This fact is emphasized throughout the review

of papers that refer specifically to rail crews. The use of digital computers and heuristics

to solve the CSP has attracted the attention of researchers since the middle 1960s (Elias,

1964). Wren and Rousseau (1995) reviewed the bus driver-scheduling problem and clas-

sified the methods used to solve the problem. Mitra and Darby-Dowman (1985) proposed

a set-covering solution for CSP, using ILP. ILP was introduced to CSP-solution practice

mainly during the 1980s as a result of improved computer power. A special feature of the

formulation presented is its willingness to allow some uncovered duties for cases in which

an optimal covering solution is not found.

Smith and Wren (1988) introduce another CSP f

ormulation based on SCP, solved by

using ILP. The elements covered are duty pieces; the covering process uses slack and surplus

variables. Using pieces rather than entire duties helps to reduce the computational effort

involved. The approach presented by Desrochers and Soumis (1989) also uses SCP but with

a CG-solution procedure rather than ILP. The first stage of the CG process seeks a subset

that best covers all duties within the range of the feasible workdays, which are already

known. A second stage follows, in which new feasible workdays that improve the existing

solution are generated iteratively.

Paixao (1990) formulates another SCP, using dynamic programming. The search for

solutions employs a state-space relaxation method. This approach is extended in Parias and

Paixao (1993), who show that this method finds the lower-bound solution. Carraraesi et al.

(1995) propose another CG approach, one that starts with a feasible set of workdays and

iteratively replaces some workdays to obtain a better solution. The preconstructed workdays

are built of duty pieces, and the solution uses a Lagrangian-relaxation method. Another

somewhat similar CG method is proposed by Fores et al. (1999).

Clement and Wren (1995) introduce a solution for the CSP using a genetic algorithm

(GA): a group of chromosomes, each of which represents a feasible crew schedule, is subject

through repeated mutations, crossovers, and other actions, based on the idea that the search for

an optimal solution can follow rules similar to a genetic survival mechanism. Several greedy

algorithms are used for assigning duties to pieces of work. Another CSP-solution procedure

based on a GA is presented by Kwan et al. (1999).

The approach demonstrated by Beasley and Cao (1996) does not follow an SCP concept;

only a single type of workday is considered rather than attempting to search a broader

range. A Lagrangian-relaxation method provides a lower-bound solution, which is later

improved by using subgradient optimization. Next, a tree-search algorithm is used to obtain

the local optimum. Beasley and Cao (1998) again use a similar approach but, instead of the

Lagrangian-relaxation tool, seek the optimal lower bound by using a dynamic program

minimization algorithm.

Another method that does not rely on SCP is suggested by Mingozzi et al. (1999). The

authors describe two different duty-based heuristic solution procedures in which relaxed problems

are formulated; their solutions also solve the original CSP. A third proposed solution procedure

is based on SP. The dual concept of linear relaxation programming is used to obtain

a lower-bound solution. The number of variables in this problem is then reduced by using this

lower bound; Finally, the reduced-size problem is solved through a branch-and-bound tech

nique. Haase et al. (2001) present an approach for the simultaneous solution of vehicle sched

uling and CSPs in a single depot with a homogeneous vehicle fleet. The CSP is based on a n SP P

formulation that incorporates side constraints for the bus itineraries. Their proposed approach

consists of a CG process for the crew schedules integrated into a branch-and-bound scheme.

Lourenco et al. (2001) bring a multiobjective CSP, led by the concept that in practice there

is a need to consider several conflicting objectives when determining the crew schedule. The

multiobjective problem is tackled using metaheuristics, a Tabu-search technique, and GAs.

Banihashemi and Haghani (2001) formulate a CSP as a duty-based network-flow problem.

First, minimum cost is sought in a binary programming problem that is a relaxed version of

the original, omitting labor-rights constraints. Then an iterative CG procedure using two

sets of constraints is performed: hard constraints restricting the building of specific work

days (e.g., too long) and soft constraints penalizing inefficient workdays, but not strictly

forbidding their inclusion in the solution.

Freling et al. (2001a) and Huisman et al. (2005) present an integrated approach to solve

a VSP and a CSP on a single bus route (i.e., they assume that different routes do not share

the same drivers or vehicles). This combined methodology enables an examination of

such scenarios as and when drivers are allowed to change vehicles between any two runs.

First, CSP and VSP are defined separately. VSP is described as a network-flow problem, in

which each path represents a single feasible vehicle schedule and each trip a node. CSP is

an SCP that is solved after the VSP to form a sequential solution. An integrated approach

is then presented, in which the network formulation of the VSP is combined with an SSP

formulation for the CSP. The integrated model is one programming problem instead of

two, but it cannot use pregenerated workdays, since there are too many options; therefore,

dummy pieces are used.

Shen and Kwan (2001) introduce a process that involves partitioning a predetermined

vehicle schedule into a set of driver duties. The focus is on redefining an existing small set

of workdays; hence, the methodology does not include the common stage of generating

all feasible solutions. A Tabu search is used to improve the given crew schedule. Tabu

search is a class of metaheuristic that tries to avoid being trapped in a local optimum

solution by basing the solution choice in each iteration on a few-iterations-back analysis;

sometimes, this means that a solution is chosen even if it leads to a poorer performance

than the previous iteration. Forsee et al. (2001) describe a traditional ILP formulation

of the CSP, with some added flexibility. The formulation accepts different objective

functions (minimize the number of duties, minimize costs, or a combination), different

optimization techniques (primal CG or dual-steepest edge techniques), and different cri

teria for reducing the number of feasible workdays. The optimization technique chosen

is used to solve a relaxed noninteger problem; a branch-and-bound process then finds

an integer solution.

Freling et al. (2001b) discuss differences between bus and train CSPs and propose a

methodology for the scheduling of train crews. The problem, formulated as an SCP with

additional constraints, is solved using a heuristic branch-and-price algorithm. Branch

and-price is a special application of branch-and-bound, in which a CG technique is used to

solve linear programming relaxations with a huge number of variables. Feasible workdays

are generated in a network, in which each node corresponds to a duty piece and each path

through the network to a feasible duty; the optimal solution is sought through dynamic

programming, that of a resource-constraint, shortest-path algorithm. Finally, Kroon and

Fischetti (2001) present an SCP for railway crews, which allows some flexibility in specifying

penalties for undesirable types of workdays.

A dynamic CG procedure is used; hence,

duties are not generated a priori but in the course of the solution process. Regeneration

and reselection of workdays are carried out in each iteration. Generation is performed

in a network in which trips are represented by arcs. To solve the SCP, a Lagrangian

relaxation method and subgradient optimization are used instead of the common linear

programming.

Huisman et al. (2004) proposed a solution approach to the dynamic single-depot and

multiple-depot VSP. A cluster-reschedule heuristic is proposed to solve the multiple-depot

VSP. By using this heuristic, the authors first assign trips to depots by solving the static prob

lem and then solve dynamic single-depot problems. Sensitivity analysis about the deviations

of the actual travel times from estimated ones is conducted.

Yunes et al. (2005) proposed a way to combine pure-integer programming and declar

ative constraint logic programming techniques into hybrid CG algorithms that solved,

to optimality, huge instances of these real-world crew management problems. Hybrid

CG algorithms are designed to solve the problems. They reported that the hybrid algo

rithms always performed better, when obtaining optimal solutions, than the two previ

ously isolated approaches. In particular, the algorithms proved to be much faster for the

scheduling problem.

Desaulnier (2007) developed a new exact dynamic scheduling procedure compatible with

CG that starts with a relatively small scheduling cost value and increases it iteratively until

optimality is reached. This procedure can overcome some drawbacks of the traditional

CG method.

Portugal (2009) developed new alternative models that, from a practical point of view,

are better for the DSP-solving process. The models are based on set partitioning/covering

models and take into account the bus operator issues and the user's standpoint and environment.

Commercial ILP solver, Cplex, is used to solve the new mathematical models. It is

demonstrated that the solutions generated by these models can easily be implemented in real

situations.

Tian and Niu (2012) studied the modeling and algorithms of crew roster problem

with given cycle on high-speed railway lines. Two feasible compilation strategies to work

out the crew-rostering plan, with given cycle of high-speed railway lines, are discussed.

Subsequently, an integrated compilation method is proposed to obtain a plan with rel

atively higher regularity in execution and lower crew members arranged. The process

of plan making is divided into two subproblems, which are decomposition of crew legs

and a adjustment of nonmaximum crew roster scheme.
The decomposition subproblem is

transformed to finding a Hamiltonian chain with the
best objective function in network,

which was solved by an improved ant colony
algorithm, whereas the adjustment of non

maximum crew-rostering scheme is finally presented as an
SCP and solved by a two-stage

algorithm. Finally, the effectiveness of the pro-
posed models and algorithms are tested

through a numerical example. The authors con-
clude that although the compiling method

of crew-rostering plan presented in this study
has its advantages, it sacrifices both objec-

tives to some extent. However, this algorithm
can be used separately when repairing the

crew-rostering plan with a single requirement.

Chen and Shen (2013) present an improved CG
strategy to enhance the efficiency of the

proposed crew-scheduling approach. The main idea
presented is to precompile a reasonably

large set of good potential shifts (called a
shift pool) using problem-specific knowledge.

Significant time savings can be achieved by
imply selecting the negative reduced cost col-

umns from the shift pool to add to the restricted master
problem, which reduces the times of

solving the resource-constrained shortest-path
problem. Experiment results have been pre-

sented, which demonstrate that the improved CG
algorithm can outperform the traditional

CG algorithms. Future work has been proposed
in the area of designing more sophisticated

upper-bound strategies and more efficient branching rules that expedite the overall solution

process.

Shen et al. (2013) present an adaptive evolutionary approach incorporating a hybrid

GA for public transit CSPs. This GA is utilized to iteratively generate a near-optimal

schedule. Unlike the traditional GA-based methods for crew scheduling where the chromosome

length is fixed, a novel shift-based approach has been introduced where the

chromosome length varies adaptively during the iterative process, and its initial value

is laborately designated as the lower bound of the number of shifts to be used in an

unachievable optimal solution. Next, the hybrid GA with such a short chromosome

length is employed to find a feasible schedule. During the GA process, the adaptation of

chromosome lengths is achieved by genetic operations of crossover and mutation with

removal and replenishment strategies aided by a simple greedy algorithm. If a feasible

schedule cannot be found when the GA's termination condition is met, the GA will

restart with one more gene added. The aforementioned process is repeated until a feasible

solution is found. Computational experiments based on 11 real-world CSPs in China

show, compared to a fuzzy GA that is known to be well performed for crew scheduling,

that better solutions are found for all the testing problems. Moreover, the authors dem

onstrate that the algorithm works fast, has achieved results close to the lower bounds

obtained by a standard linear programming solver in terms of the number of shifts, and

has much potential for future developments.

The main features of the methodologies reviewed are summarized in Table 9.4.

Table 9.4 Summary of the characteristics of the methodologies reviewed

Source	Approach	Solution method	Comments
Elias (1964)	Set covering	Heuristics	
Mitra and Darby-Dowman (1985)	Set covering	ILP	Allows leaving some tasks uncovered
Smith and Wren (1988)	Set covering	ILP	
Desrochers and Soumis (1989)	Set covering	Column generation	
Paixao (1990); Paiao and Paixao (1993)	Set covering (dynamic)	ILP; state-space relaxation	
Carraraesi et al. (1995)	Set covering	Column generation; Lagrangian relaxation	
Clement and Wren (1995)	Genetic algorithm	Genetic algorithm	
Beasley and Cao (1996)	Single-type workdays	Lagrangian relaxation; subgradient optimization; tree search	
Beasley and Cao (1998)	Single-type workdays	Dynamic program	
Kwan et al. (1999)	Genetic algorithm	Genetic algorithm	
Mingozzi et al. (1999)	Set partitioning	Linear relaxation; branch and bound	
Haase et al. (2001)	Set partitioning	Column generation;	

branch-and-bound Simultaneous crew and vehicle scheduling

Lourenco et al. (2001) Multiobjective program Tabu search;
genetic algorithm

Banihashemi and Haghani

(2001) Network-flow problem Column generation

Freling et al. (2001a);

Huisman et al. (2005) Set covering; set partitioning
Column generation Simultaneous crew and vehicle scheduling

Huisman et al. (2004) Zero-one integer programming Dynamic
optimization

Shen and Kwan (2001) Set partitioning Tabu search Workday
generation; method not included

Fores et al. (2001) Set covering Primal column generation;
dual-steepest edge; branch and bound Varying objective
functions accepted

Freling et al. (2001b) Set covering; network-flow problem
Branch-and-price Rail crews

Kroon and Fischetti

(2001) Set covering (dynamic); network-flow problem Dynamic
column generation; Lagrangian relaxation and subgradient
optimization Rail crews

Yunes et al. (2005) Mathematical programming; constraint
logic programming Hybrid column generation (Continued)

EXERCISES

9.1 Given 29 arrivals epochs and 29 departure
epochs (a total of 58 trips) at terminal k
between 6:00 and 8:10 a.m. (see the following
table) and a maximum paid idle time of
 $T_{\max} = 35$ min

(i) Apply algorithm TMF and find $D(k)$ and the maximum
number of joinings at k in which $\Delta_{ij} \geq T_{\max}$; list all
joinings found.

(ii) Use the FIFO rule for constructing the j
oinings at k; list all joinings and compare

the results with those found in (i). Departures Arrivals
 Trip number Departure time Trip number Arrival time 1 6:00
 12 6:25 2 6:00 13 6:30 3 6:00 14 6:30 4 6:05 15 6:30 5 6:10
 16 6:35 6 6:10 17 6:35 7 6:15 21 6:50 8 6:15 22 6:55 9 6:20
 23 6:55 10 6:20 24 6:55 11 6:20 25 7:00 18 6:40 29 7:15 19
 6:45 30 7:15 20 6:45 34 7:30 26 7:05 35 7:35 27 7:05 36
 7:35 28 7:10 37 7:35 31 7:20 38 7:35 32 7:25 48 8:00 33
 7:25 49 8:00 39 7:40 50 8:05 40 7:45 51 8:05 41 7:45 52
 8:05 42 7:50 53 8:05

Table 9.4 (Continued) Summary of the characteristics of the methodologies reviewed

Source Approach Solution method Comments

Desaulnier (2007) Lexicographic bilevel objective formulation Column generation

Portugal (2009) Set covering/partitioningbased models ILP, Cplex

Tian and Niu (2012) Set covering Ant colony algorithm

Chen and Shen (2013) Set covering Column generation

Shen et al. (2013) Set covering Hybrid genetic algorithm
 Departures Arrivals Trip number Departure time Trip number
 Arrival time 43 7:50 54 8:05 44 7:50 55 8:10 45 7:55 56
 8:10 46 7:55 57 8:10 47 7:55 58 8:10

9.2 Given two vehicle blocks with their relief points, Block 1 starts at relief point a and con

tinues through points b, c, d, c, and b, in which the piece from b to c is a DH trip. Block 2 starts at relief point a and continues through points b, c, a, and d. The following table lists the segments of consecutive blocks with their associated internal piece cost (ignore cost unit). Find and list the best sets of duty pieces (minimum cost) for each block. Block 1 Block 2 Feasible pieces Internal piece cost
 Feasible pieces Internal piece cost a-b 3 a-b 3 a-c 5 a-c 7 a-d 9 a-a 12 a-c 14 a-d 24 a-b 20 b-c 3 b-d 7 b-a 8 b-c 8 b-d 16 b-b 12 c-a 7 c-d 5 c-d 12 c-c 5 a-d 8 c-b 9 d-c 2 d-b 4 c-b 4

9.3 Use procedure roster to determine the minimum number of drivers required to operate the

following given scheduling data: Two constraints are imposed—one is 10 has the minimum

minimum rest period required between two consecutive working days and the second concerns

the maximum allowed roster hours for each roster type (see these two constraints in

the tables of this exercise). Each roster can contain one duty period for days marked “V.” Mon Tue Wed Thur Fri Sat Sun Duties distributed d 1 1 d 1 2 d 1 3 – d 1 5 d 1 6 – d 2 1 d 2 2 d 2 3 d 2 4 d 2 5 – – – d 3 2 d 3 3 d 3 4 d 3 5 d 3 6 – Maximum roster hours d 4 1 – d 4 3 d 4 4 – – d 4 7 Type of roster R 1 V V V V V V – – 43 R 2 V V – V V – – 36 R 3 – – – – V V 16 Duty for day q (q = 1, 2, 3, 4) Start and end times Duty length (hours), L j d 1 q 6:00–16:00 10 d 2 q 12:00–20:00 8 d 3 q 16:00–23:00 7 d 4 q 18:00–midnight 6

APPENDIX 9.A: SHORTEST-PATH PROBLEM

The following description is based on Ceder (1978). Further reading can be found in almost

any OR book. This appendix supplements Sections 9.3 and 9.5, as well as material in

Chapters 11, 12, 15, and 16.

Consider a connected network $G = \{N, A\}$ with set of nodes N , set of directed arcs A , and

given distances (travel time, cost, etc.), $d(i, j) \forall (i, j) \in A$. The number of nodes and arcs are

$|N|$ and $|A|$, respectively. In a connected network, the shortest path may have the following

four categories:

(a) From one node to all other nodes with $d(i, j) \geq 0, \forall (i, j) \in A$

(b) From one node to all other nodes with $d(i, j)$ any

(c) From all nodes to all nodes with $d(i, j) \geq 0, \forall (i, j) \in A$

$d(i, j)$ is the length of the shortest path from node i to node j .

The shortest $u-v$ path, termed p , is defined as a $u-v$ path, $u, v \in N$, such that

$d(p) = d(i, j) = \min_{p \text{ is a } u-v \text{ path}} d(i, j)$.

9.A.1 Dijkstra algorithm (one to all, $d(i, j) \geq 0$)

The Dijkstra algorithm (Dijkstra, 1959) produces a spanning tree of G containing shortest

paths from a given node to all other nodes in G , with nonnegative arc lengths. It is described

as follows.

Step 0: Initially, assign the origin node u with a permanent label, $\pi^*(u) = 0$.

Let the set of nodes with permanent labels be $N(u) = \{u\}$. All other nodes $i, i \in N, i \neq u$, have temporary labels: $\pi(i) = \infty$, if $i \neq u$, $i(u, i) < \infty$, otherwise $\pi(i) = \infty$.

Note that a permanent label $\pi^*(i)$ represents the shortest distances from node u to node i . The symbol " $:=$ " means is replaced by in the following steps.

Step 1: Find the node with a temporary label, say, node k , such that $\pi(k) = \min_{i \in N(u)} \pi(i)$. Set $\pi^*(k) = \pi(k)$ and mark/store the arc that leads from $N(u)$ to k . Add k to $N(u)$.

Step 2: If $N(u)$ contains all nodes $i, i \in N$, terminate; otherwise, continue.

Step 3: For each arc (k, j) such that k is a node determined in Step 1 and $j \notin N(u)$, $\pi(j) := \min[\pi^*(k) + d(k, j), \pi(j)]$, go to Step 1.

The following example illustrates the Dijkstra algorithm with four iterations. The tree

emphasized (i.e., connected paths without cycles) after the fourth iteration is the solution.

with all nodes permanently labeled. The arcs emphasized represent the desired spanning

tree. The minimum spanning tree from node 1 to all other nodes is provided. $\pi(1) = 1$, $\pi(2) = 3$, $\pi(3) = 5$, $\pi(4) = 2$, $\pi(5) = 6$, $\pi(6) = 7$, $\pi(7) = 0$. $\pi(2) = 11$, $\pi(3) = 0$, $\pi(4) = 1$, $\pi(5) = 3$, $\pi(6) = 11$, $\pi(7) = 0$. 1st iteration (k=4) 2nd iteration (k=3) 3rd iteration (k=5) 4th iteration (k=2) 2 Rearranging node numbers (Step 0) 6 1 -9 2 4 5 7 3 2 -6 -2 -4 u 3 1 -9 2 4 7 6 3 2 -6 -2 -4 1 Given network u 3 1 5 1 3 4 5 6 7 0 -2 1 3 4 6 7 0 1 2 7= $\pi(2)$ 0 1st iteration (j = 2) 3rd iteration (j = 4) 4th iteration (j = 5) u 2 3 6 7 0 1 2 1 1 2 2nd iteration (j = 3)

Theorem 9.A.1

(a) Dijkstra algorithm terminates in, at most, $n-1$ iterations.

(b) Dijkstra algorithm produces a spanning tree whose arcs correspond to the shortest paths between u and all other nodes i , $i \in N$ (the π^* values are the lengths of these shortest paths).

Proof

(a) Almost trivial, because one node is permanently labeled per iteration, and the network considered is connected.

(b) By induction. The assumption is that at the k th iteration, the tree emphasized represents the shortest paths from u to the nodes belonging to that tree. From this point, it is not difficult to complete the proof, and therefore, it is left as an exercise.

Remarks

(1) If the interest is in the shortest u-v path (between only two nodes), then terminate once $\pi(v)$ becomes permanently labeled; that is, $\pi^*(v)$.

(2) If the interest is in finding the shortest paths between all nodes $i \in N$ and a node u (from all to one), simply reserve all arcs $(i, j) \in A$; that is, (i, j) will become a $\pi^*(j, i)$ arc; then apply the Dijkstra algorithm (from node u to all other nodes).

(3) For a complete graph (there is an arc between each two pairs of nodes), the Dijkstra algorithm requires a approximately $\frac{1}{2}N^2$ additions and $\frac{1}{2}N^2$ comparisons, or $3 \cdot \frac{1}{2}N^2$ operations. By using the complexity notation in Section 5.3 (Chapter 5), the algorithm runs in $O(N^2)$ time for dense (in arcs) networks. For sparse networks, other algorithms may improve the running time; see, for example, Ahuja et al. (1993).

9.A.2 Algorithm rearrange-node-numbers (one to all, $d(i, j)$ any, no cycles)

The major limitation of the Dijkstra algorithm is that it can run only on nonnegative arc

lengths. There are other algorithms that handle optimally any $d(i, j)$ that can be found, for

example, in Ahuja et al. (1993). This section introduces an algorithm aimed at special cases

in which $G = \{N, A\}$ contains no cycles. The algorithm, which is called rearrange-node

numbers (RNN) because of its first step, is described as follows:

Step 0: Initially, let the origin node be $u = 1$ and number all nodes $i, i \in N$, such that an arc $(i, j) \in A$ if and only if $i < j$. Remove all nodes with assigned numbers less than 1 (since they cannot be reached from u). If this step cannot be done, a cycle is found. Set $j := 2, \pi(1) := 0, ATx = \emptyset$.

Step 1: $\pi(i) = \min_{j \in A(i)} [d(i, j) + \pi(j)]$. Add (i, j) to AT . If there is no arc, connecting AT with j , assume $\pi(i) = \infty$, and continue.

Step 2: Let $j := j + 1$. If $j = N$ terminate. The tree $T = \{N, AT\}$ is the required spanning tree, and $\pi(i)$ are the shortest paths. For $\pi(i) = \infty$, there is no path between u and i in G . If $j < N$, go to Step 1.

Theorem 9.A.2

(a) Algorithm RNN terminates in $n - 1$ iterations.

(b) Algorithm RNN produces the required spanning tree; the $\pi(i)$ value is the shortest path from u to node i (for $\pi(i) = \infty$, there is no path from u to i).

Proof

(a) Self-evident, because one node is added per iteration.

(b) By induction. The assumption is that at the m th iteration, the tree emphasized represents shortest paths from u to the nodes on the tree. The rest is left as an exercise.

Remarks

(1) If the interest is in the shortest $u-v$ path (between only two nodes), then terminate once i equals the node number of v .

(2) If the interest is in finding the shortest paths between all nodes i and a node u (from all to one), simply reverse all arcs; that is, (i, j) will become (j, i) arc; then apply algorithm RNN.

(3) The RNN algorithm is basically applied to known project-scheduling methods, such as the program evaluation and review technique (PERT) and critical path method (CPM). Small changes in RNN enable one to introduce earliest-starting and latest-end times of each task (described by an arc).

The example in the succeeding text, in which some $d(i, j) < 0$, shows the RNN algorithm in

four iterations. The solution is represented by the last tree, with shortest paths from $u = 1$

indicated as $\pi(i)$ on the nodes.

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10 Chapter 10: Passenger Demand

One of the main measures of the improvement of transit services is the increase in passenger

demand. However, there is a trade-off between the cost of improvement and the extra benefit

gained by the additional demand. It is common to find transit managers who are puzzled about

the relationship (if any) between service changes and their impact on changes in demand. In fact,

the following humoristic aphorism readily applies: "The more unpredictable the transit demand

becomes, the more we rely on predictions." Although it is easier to try out service changes for

bus services (with conclusions drawn after a learning period), the rail industry requires more

assurances. The chapter contains four main parts following the introductory section. Section 10.2 presents

and discusses the basic attributes and tools used in passenger-demand analysis; among these attributes

are fares, travel time, service frequency, walking time, routing and transferring, and comfort

and inconvenience elements. Section 10.3 provides an example of a transit-demand forecast method

that is adapted to predict, for a given transit service, the future patronage of a specific

set of routes in certain required years. Section 10.4 exhibits a known share model that divides

passengers among various travel modes according to each mode's relative desirability; such a technique

has been commonly used to determine modal split, although its accuracy depends heavily on

the underlying mathematical model. The last main part, Section 10.5, presents a literature review

on estimating origin-destination matrices (ODMs), which constitute an essential input for most

transit planning and design procedures. The chapter ends with exercises. Practitioners are advised to read the entire chapter, rather than just Section 10.4. It should

be noted, however, that demand-prediction models, no matter how good they are, should be

treated carefully because of the vast number of assumptions (behavioral and others) that are

inherent in them. In other words, one should not believe that the models are reality (don't eat

the menu), but should properly weigh practical issues against modeling precision. The following

story may teach us about life and precision (squareness). During the French Revolution, three professionals were sentenced to be executed by guil-

lotine. The first person had been collecting practical data, the second analyzing practical data,

and the third modeling the precision of data. For the third professional, precision was above all

else (a square person). The first person was asked if he wanted his head facing up or down. He

requested down, and the guillotine dropped, stopping 2 cm before his neck; there was an old

rule that if the guillotine stopped, you're sent free, and indeed he was released. The second per-

son was asked about his head position; he asked face up, and the guillotine again dropped and,

unbelievable, for this had never happened before, stopped 1 cm before his throat, allowing him,

too, to go free. The third person, for whom precision was all important, replied to the same

question: "I don't want to go under that damn machine before you fix it!"

10.1 INTRODUCTION

Chapter 9 ends a main subject of the book, transit scheduling. Chapters 12 through 17

are more design and behavioral oriented, focusing on service, network, route, and shuttle

design and user behavior. Before any discussion of transit-design elements, however, it

would be only natural to present an overview of the tools and models used in analyzing

passenger demand (Chapter 10) and the related prediction of the way the demand will flow

on the transit network (Chapter 11). The first question confronting us in designing a tran-

sit service concerns the size, composition, and distribution of passenger demand; it is the

dominant input parameter for any new, improved, or redesigned undertaking. The sort of

sandwich position of this and the next chapter between the subjects of the book is displayed

in Figure 10.1.

The purpose of transit-demand studies is to estimate and evaluate passenger demand by

collecting and analyzing data and models pertaining to current and future transit needs.

Transit-demand studies and models form an essential part of any transit planning process.

The venerable transportation-planning practice is based on a four-step sequential process: trip

generation (number of trip ends associated with a zone of land), trip distribution (distribution

of trips among zones), modal split or mode choice (choice among travel modes for personal

trips), and traffic assignment (choice among available

origin-destination [O-D] routes and

the resulting accumulated traffic). A thorough overview of this four-step process and its

variants appears in de Dios Ortuzar and Willumsen (2011). This planning practice for trip

forecasting contains trips associated with transit modes, thereby calling for the utilization of

some of its features in analyzing passenger demand.

The four-step sequential planning process is discussed in Boyce and Daskin (1997) in

conjunction with solving optimal travel-choice models. Those researchers explained that

the four-step process had to be solved with feedback in order to forecast equilibrium traf-

fic conditions. In their iterative procedure involving the four-step process, the solution of

the trip-assignment (route choice) model is required to solve the trip-distribution (O-D

choice) model—that is, equilibrium travel times and costs. A variant of the four-step process

(de Dios Ortuzar and Willumsen, 2011) indeed considers route choice as the step preced-

ing the O-D choice step. Boyce and Daskin (1997) believe that the four-step process can at

best be seen as a heuristic for solving a traffic-network equilibrium problem because it lacks

the mechanism by which the process could converge to the desired equilibrium solution.

Following this explication of the conceptual issues and limitations surrounding the four

step process, the following sections will provide examples of analyses of passenger-demand

forecasting. For given data: balancing and optimization of

timetables and vehicle and crew scheduling (Chapters 3 through 9) Service, network, and route design: optimization encompassing scheduling and behavioral elements (Chapters 12 through 17) Passenger demand and assignment analyses (Chapters 10 and 11)

Figure 10.1 Position of Chapters 10 and 11 in the sequence of main subjects of the book.

10.2 TRANSIT DEMAND, ITS FACTORS, AND ELASTICITY

The analysis and estimating of transit demand are complex mechanisms mainly because

they involve the aggregate behavior of individuals. Thus, in estimating demand, it is better

to doubt what is true than accept what is not.

The known attributes affecting transit demand are changes in fare, travel time, service

frequency, walking time, routing and transferring, stop location, comfort and inconvenience

elements, information, socioeconomic factors (e.g., income), external factors (e.g., land use,

security), and competition from other transportation and transit services. Usually, passengers

(travelers) have a travel choice among alternatives; it is reasonable to assume that their

selection process is based on maximizing utility or finding the best way to realize their travel

objectives while complying with constraints. To some extent, this selection process varies

among individuals; thus, economic theories describing the behavior of a customer search

for services or goods can be helpful. A practical guide on transit demand appears in a

Transport Research Laboratory (TRL) report by Balcombe et al. (2004).

10.2.1 Factors affecting demand

The choice between public and private transportation is an individual decision that is influ

enced by government/community decisions. These various decisions often send mixed signals

to transit and potential transit passengers while failing to recognize their more system-wide

and integrated implications. Generally speaking, most localities have effectively encour

aged the use of private cars through planning (dispersed land use in the suburbs), infra

structure (available parking and circulation traffic flow), pricing, and financial decisions.

Consequently, there is growing confusion in many of these cities about what to do. It is

therefore important to identify the factors affecting transit demand and to use them for

improvements.

For the most part, increasing the quality of service (e.g., in terms of service coverage, ser

vice frequency, and fare reduction) has been found to have a significant positive outcome on

transit demand. Two practical guides that cover the factors affecting demand are Balcombe

et al. (2004) and TranSystems et al. (2006).

The greatest impacts from increasing the transit demand are extracted from a combina

tion of strategy-related actions and initiatives.

TranSystems et al. (2006), based on a large

survey of transit agencies in the United States, categorized these actions and initiatives in the

following list (in decreasing order of effectiveness):

1. Service adjustments or improvements (increased route coverage, route restructuring, improved schedule/route coordination, increased service frequency, increased span of service, improved reliability/on-time performance, improved travel speed and reduced stops, targeted services, passenger-facility improvements, new/improved vehicles, increased security and safety)

2. Partnerships and coordination (university and school passenger programs; travel demand management strategies; privately subsidized activity center service; consistent regional, and interagency, operating policies; coordination with other transportation agencies; promotion of transit-supportive design)

3. Marketing, promotional, and information initiatives (targeted marketing and promotions, general marketing and promotions, improved informational materials, improved customer information and assistance)

4. Fare collection and fare structure initiatives (improved payment convenience, regional payment integration, fare structure simplification, fare reduction)

In addition, both Balcombe et al. (2004) and TranSystems et al. (2006) indicated that

there are external factors influencing transit demand. The factors listed as significant by

TranSystems et al. (2006) are as follows:

1. Population characteristics and changes (general growth in the region, high and increasing immigration, high and increasing number of elderly, high and increasing tourism, high number of college students)

2. Economic conditions (employment/unemployment levels, per-capita income levels, household auto-ownership levels)

3. Cost and availability of alternative modes (fuel and toll pricing, parking pricing and availability, taxi fares, fuel taxes, auto purchase and ownership costs, availability of commuter-benefit programs for employers)

4. Land use and development patterns and policies (density of development; relative locations of major employers and residential areas; from, e.g., increasing suburbanization; land use and zoning controls and incentives)

5. Travel conditions (climate and weather patterns, traffic congestion levels and highway capacity, traffic disruptions owing to, e.g., major construction projects)

6. Public policy and funding initiatives (air quality mandates, auto emissions standards, federal and state operating capital and transit-funding levels)

Another indirect set of factors affecting transit demand is related to mode choice elements

elements extracted from various types of public policies. Examples of these elements given by

TranSystems et al. (2006) are as follows:

1. Price and availability of each mode (cost of auto use, cost of transit, parking cost)

2. Quality of service of each mode (travel time, convenience, comfort, service reliability, perceived personal security and safety, perceived overall image of each mode)

3. Trip characteristics for each particular trip (trip length and purpose, number of people to be making the trip, multiple destinations)

4. Sociodemographic characteristics of the traveler (income, origin and destination locations, status)

Retaining a high level of passenger satisfaction while fully maintaining the protection

of access to less-affluent citizens is another way to look at factors for improving transit

sit demand. For instance, TranSystems et al. (2006) reported on large-scale surveys conducted

for the Washington Metropolitan Area Transit Authority. The results of these

surveys, shown in Table 10.1, indicate significant differences between the most desired

transit-service improvements for riders and nonriders. It was concluded that targeting new

riders warrants a different set of actions and initiatives than those used to influence existing

riders.

Often, desired service improvement items can be interpreted by the willingness of riders

and potential riders to pay for an increase in their satisfaction (Ceder, 2004). In one study,

Kottenhoff (1998) employed the stated-preference method to ascertain Swedish train pas-

sengers' willingness to pay. The main question was how to increase the attractiveness of the

rail system (the willingness to pay) and, simultaneously, to decrease its costs.

To perform these quantifications of willingness to pay, train passengers were asked to

answer computer-assisted self-interviews. Table 10.2 presents the results by percentage of

fare levels for which passengers are willing to pay extra in order to improve the specified

item. These results were also checked (and agreed with in general) against the results from

other European countries.

One example of a train service for which the conditions were improved and the pay

ment was increased is the Danish IC/3, marketed under the name Kustpilen, which replaced

simple rail-bus services. The number of travelers rose by almost 200% in the next 4 years

as reported by Kottenhoff and Lindh (1996). The traveling time from Karlskrona to Malmö

was reduced by 15 min (7%), and passengers traveling all the way no longer had to make a

transfer. The frequency was increased from 6 to 11 trains per day in each direction. Although

the fare level was reduced by about 10%-30%, the enhanced comfort offered by the new

IC/3 was found to be the main attribute for the increase in patronage.

10.2.2 Demand function and elasticity

Theoretically, the (transit) demand can be mathematically expressed as a function of the

attributes (explanatory variables) mentioned earlier. That is,

$$D_f = f(y_1, y_2, \dots, y_m) \quad (10.1)$$

where D_f is transit demand (in passengers) y_i , $i = 1, 2, \dots, m$, are the attributes

Basically, no preference is given to any functional form of Equation 10.1; the form is determined

empirically for the type of application under question, using statistical techniques.

The responsiveness (sensitivity) of demand to changes in y_i attributes is known as elastic

ity. This is to say, demand elasticity concerning y_i is the ratio of the percentage change in

demand to the percentage change in y_i . Assuming the existence of a demand function D_i for

a given attribute y_i ,

$D_i = f(y_i)$, $i = 1, 2, \dots, m$ (10.2) Table 10.1

Transit-service improvements desired by bus riders and nonriders in the Washington, DC, region

Improvement item	Riders (%)	Nonriders (%)
Better reliability	49	6
More frequent service	31	16
More convenient stops	13	18
Better information	9	30
Better shelters	10	21
Faster service	10	16
Better vehicle condition	8	13
Service to destination	12	13
Longer hours	25	N/A
Less crowding	22	N/A
Lower fares	8	N/A
Better customer service	7	N/A

Source: TranSystems et al., Elements needed to create high ridership transit systems: Interim guidebook, TCRP Report 32, Transportation Research Board, Washington, DC, 2006. Note: Percentage opting for each item. a Not applicable for nonriders.

then there are two main direct elasticity definitions as explicated by Balcombe et al. (2004):

point elasticity and arc elasticity. The direct point elasticity of y_i is calculated as the slope of

the demand curve times the ratio of the values of y_i to D_i : $\epsilon_i^{\text{point}} = \frac{y_i}{D_i} \frac{\partial D_i}{\partial y_i}$ (10.3)

The direct arc elasticity is calculated between two points on the demand curve, using a logarithmic form extracted from the differential

$\frac{dD_i}{D_i} / \frac{dy_i}{y_i} = \frac{d(\ln D_i)}{d(\ln y_i)}: \epsilon_i^{\text{arc}}$

$\epsilon_i^{\text{arc}} = \frac{\ln D_2 - \ln D_1}{\ln y_2 - \ln y_1} = \frac{\log D_2 - \log D_1}{\log y_2 - \log y_1}$ (10.4) Table 10.2

Passengers' willingness to pay for train service in Sweden (by percentage of fare level to be paid as extra)

Improvement item	Fare level (%)
Timetable factors	
Change of train: one less transfer	19
Speed: 20% less travel time	15
Frequency: 2 h-1 h	5
Comfort	
Shaking and vibrations: a little less	11.5
Noise: a little less noisy	10
Climate: better ventilation	10
Seat adjustment: reclining seat backs	8
Seat orientation: face to face or face to back	8
Leg room: 10 cm less or more	6.5
Seat size: 5 cm wider	4
Onboard services	
Restaurant with hot food	13
Bistro with some food	10
Division: reading salon/quiet salons	9
Division: play areas	9
Coffee: free	

coffee and tea in the car 6 Entertainment: video or cinema on board 5 Entertainment: music/radio outlets 3 Office: service 2 Quality satisfaction On-time arrivals: from 80% to 90% 16 Lavatories: modern and roomy 14 Modernity: modern coach 9 Reservation: optional seat booking 9 Cleanliness: clean inside 5 Mode: going by double-decker 3.5 Source: Kottenhoff, K., Trans. Res. Record, 1623, 144, 1998.

where D_{i1} and D_{i2} exhibit the values of the demand before and after the change of the attribute

from y_{i1} to y_{i2} , respectively.

In a competitive environment, a change in an attribute (e.g., fare) for one transit service

may affect the demand for another transit service. The sensitivity of these cross effects is

indicated by the phenomenon of cross-elasticity, expressed mathematically as ϵ_{iuv} in y_i D_i $y_D = \partial \epsilon_{iuv} / \partial y_i$ (10.5)

where ϵ_{iuv} is the (point) cross-elasticity of demand for service u concerning the change in attribute y_i of service v D_{iu} is the value of the demand function for attribute y_i , using service u y_{iv} is the value of attribute y_i , using service v

In practice, it may be difficult to measure cross-elasticity (since direct effects frequently

outweigh cross effects). The following relationship shows how to calculate it from direct

elasticities: $\epsilon_{iuv} = \epsilon_{iu} / \epsilon_{iv} \cdot D_{iu} / D_{iv}$ (10.6)

where ϵ_{iu} (an absolute value) is the direct elasticity of demand for service u for the change in attribute y_i of (the same) service u D_{iu} / D_{iv} is the ratio between the competing volumes of services, u and v $\partial D / \partial y_{iv}$ is the proportion of the demand change in services v and u as a result of the change in y_{iv}

More on elasticity functions and approaches can be found in Balcombe et al. (2004).

10.2.3 Overview of elasticity results

Litman (2004) and Balcombe et al. (2004) summarized and compared numerous studies on

elasticity values for several attributes, with an emphasis on transit fares. Litman (2004) con

cluded the research with the recommendation to use ranges rather than point values because

of the inevitable uncertainty inherent in elasticity analysis.

Litman's (2004) recommendations are shown in Table 10.3 for short and long-run pre

dictions. The short run refers to periods of less than 2 years and the long run to periods

of 5-10 years. The transit demand considered is elastic for elasticity values approaching or

above 1.0 (or approaching or below -1.0). Measurable attributes influencing the quality

of service are included in Table 10.3 under attributes of service, that is, frequency of ser

vice, access and egress times, waiting time, in-vehicle time, and transfer time. According to

Litman, it is assumed in practice that demand is inelastic for the short run; a fare increase

and/or service reduction will have a marginal effect on the demand, thus increasing net

revenue.

Both Balcombe et al. (2004) and Litman (2004) indicated that fare elasticities were about

twice that for off-peak and leisure trips as for peak and commuting trips. In addition,

short-distance trips and trips made in small towns have higher fare elasticities than do lon

ger trips and trips made in large cities.

Balcombe et al. (2004) separated the attributes i

influencing the quality of service into those

that can be quantified and those that can be
only indirectly estimated. The latter include

attributes related to transit reliability, comfort, convenience, and safety. In the first group of

attributes, Balcombe et al. (2004) used both elasticity values and attribute weighing in terms

of equivalent in-vehicle time. They estimated that the elasticity of demand to vehicle-km for

buses was between 0.1 and 0.7 for short runs and between 0.2 and 1.0 for long runs; for

rail, it is roughly 0.6-0.9 for short run. The in-vehicle time elasticity for (local) bus ranges

is -0.4 to -0.6 and for rail -0.6 to -0.8.

The use of attribute weighing was applied to access/egress walking time, waiting time,

and service headway. Balcombe et al. (2004) estimated that access/egress walking time can

be on average 1.68 times the value of in-vehicle time (depending on trip length and walk

ing time); when transfers are involved, this average is around 1.81, which may also reflect a

transfer penalty. Waiting time was valued at 1.76 times that of in-vehicle time (varying by

trip length). Service headway was valued at 0.77 times in-vehicle time.

An additional attribute considered by Balcombe et al. (2004) was income, including car

ownership effects. Based on the national travel survey in the United Kingdom (1985-1997),

they recommended values of -1.08 and 0.34 for long-run elasticities, for this attribute, for

commuting bus and rail trips, respectively, and values of -0.33 and 0.42 for long-run elasticities

for leisure trips for bus and rail, respectively. They noted that the bus-income elasticity,

including car-ownership effects, was negative, compared to positive rail-income elasticity.

Hensher (2001), using mixtures of revealed-preference and stated-preference data, demonstrated

an example of direct elasticity and cross-elasticity in Sydney. Revealed-preference

applies to observed data (e.g., by agencies in regard to ticket sales, direct home surveys);

stated-preference applies to questionnaires inquiring the relative importance of factors that

are not easily appraised through revealed-preference methods. Table 10.4 presents Hensher's

results for commuters; each column has one direct (in *italic* and bold) and six cross

elasticities. For instance, concerning the change in the single-fare bus ticket (fourth column

in Table 10.4), an increase of 10% in this ticket's fare tends to reduce its sales by 3.57% and

to increase the sale of single-fare train tickets by 0.57% and to increase the number of car

users by 0.66%. Similarly, this fare change can apply to the other types of tickets.

Another study on fare (or trip cost) direct elasticity and cross-elasticity for commuting

trips in Sydney was presented by Taplin (1997). This study, whose findings are summarized

in Table 10.5, furnished adjusted fare (or cost) elasticities for trips performed by different

transit modes and cars. The direct elasticities are emphasized (in *italic and bold*) and have

Table 10.3 Recommended direct-transit elasticity values
Market segment Short run Long run

Transit demand in regard to the attribute of fares Overall
-0.5 to -0.2 -0.9 to -0.6 Peak hours -0.3 to -0.15 -0.6 to
-0.4 Off-peak hours -0.6 to -0.3 -1.0 to -0.8 Suburban
commuters -0.6 to -0.3 -1.0 to -0.8

Transit demand in regard to the attributes of

service quality Overall 0.5 to 0.7 0.7 to 1.1

Transit demand in regard to the attribute of car

operating cost Overall 0.05 to 0.15 0.2 to 0.4

Car users in regard to transit costs Overall 0.03 to 0.1
0.15 to 0.3

Source: Litman, T., J. Public Trans., 7(2), 37, 2004, Table 11.

negative values. For example, an increase of 10% in train tickets in Sydney tends to reduce

train-passenger demand (sale of this ticket) by 1.56% and to increase bus and ferry riders

and car users by 0.63%, 0.39%, and 0.16%, respectively.

Finally, elasticity may possibly be modeled through the functional relationship between

demand changes and fare (or trip cost) changes. For

example, Hong Kong transit agencies,

according to Zhou et al. (2005), are more concerned with the transit-fare structure than

with frequency of service in maximizing their profits. This concern derives from the rela

tively large sensitivity of passenger demand to fares. Zhou et al. (2005) mentioned that for a

1% reduction in transit fares, there is a consequent 1.33%-1.45% increase in transit patron

age in Hong Kong ($\epsilon < -1.0$), compared with only a 0.3% ($\epsilon = -0.3$) increase in Canada.

They illustrated the demand-elasticity effects on revenues through changing fares in the

following assumed exponential function: $D = B e^{-\alpha p}$ (10.7)

where D is the transit demand, p is the route fare, B and α are constant parameters (estimated from observed data)

The point elasticity of Equation 10.3 for fares is $\epsilon = \frac{p}{D} \frac{dD}{dp} = -\alpha$ (10.8)

Table 10.4 Direct elasticity and cross-elasticity for Sydney commuters

Mode Train Bus Car

Type of ticket Single fare Weekly Pass Single fare Ten fare Pass –

Single fare (train) -0.218 0.001 0.001 0.057 0.005 0.005 0.196

Weekly (train) 0.001 -0.093 0.001 0.001 0.001 0.006 0.092

Pass (train) 0.001 0.001 -0.196 0.001 0.012 0.001 0.335

Single fare (bus) 0.067 0.001 0.001 -0.357 0.001 0.001 0.116

Ten fare (bus) 0.020 0.004 0.002 0.001 -0.160 0.001 0.121

Pass (bus) 0.007 0.036 0.001 0.001 0.001 -0.098 0.020

Car 0.053 0.042 0.003 0.066 0.016 0.003 -0.197

Source: Hensher, D., Modal diversion, in: Hensher, D. and Button, K., eds., Handbooks of Transport Systems and Traffic Control,

Elsevier Science, Amsterdam, the Netherlands, 2001, pp. 107-123, Table 3.

Table 10.5 Adjusted fare (or trip cost) direct elasticity and cross-elasticity for commuting trips in Sydney

Mode of travel Train Bus Ferry Car

Train -0.156 0.032 0.003 0.037

Bus 0.063 -0.070 0.006 0.046

Ferry 0.039 0.037 -0.195 0.003

Car 0.016 0.011 0.000 -0.024

Source: Balcombe, R. et al., The demand for public transport: A practical guide, TRL Report, TRL593, TRL Limited,

Crowthorne, U.K., 2004, Table 9.22.

which is typically negative. Zhou et al. (2005) also referred to a denoted revenue R , in

which $R = D \cdot p$. By using the aforementioned relationships, the revenue has the following

derivative: $\frac{dR}{dp} = D + p \cdot \frac{dD}{dp} = D(1 + \epsilon)$ (10.9)

For elasticity ϵ values less than -1.0 (thus, $\epsilon < -1/p$), an increase in fare will result in a

decrease in revenue. Consequently, Hong Kong transit agencies can increase their revenue

by actually decreasing their fares.

10.3 EXAMPLE OF A DEMAND FORECASTING, METHOD AND PROCESS

This section presents and describes the rationale for a transit-demand forecast methodol

ogy adapted to predict, for a given transit service, which we will call Transit-A, the future

patronage of a specific set of routes in certain required years. This example of a forecasting

method also considers relevant survey results of passenger demand for competitor tran

sit services associated with the same underlying set of routes. Taking into account all the

parameters that affect transit-service demand is an exhaustive undertaking. Thus, simple

but useful models addressing the major attributes of the passengers' decision process will be

shown in this section. The method presented follows Ceder (2006) in a study of passenger

demand for Hong Kong ferries.

10.3.1 Framework

We assume that Transit-A patronage relies not only on certain attributes but also on the

changes in the levels of service of the competitive transit services. Therefore, whenever there

is a change in the level offered by these other services, an alternative model, such as modal

split, should be employed for a coarse estimate

In brief, the applied approach of the demand forecast covers the following four compo

nents: (1) identification of main attributes and their weights in affecting potential Transit-A

passengers; (2) demand prediction using a calibrated growth factor from past Transit-A

demand figures for any projected changes in the established Transit-A routes; (3) transit

modal split between specific O-D pairs on any new Transit-A

routes and/or changes in fares

and/or journey times in competitor routes; and (4)
development of Transit-A O-D demand

matrices for each alternative network in a required year.

The forecast approach principally relates to the changes in
Transit-A-relevant attributes

when there are changes in passenger demand; whenever
changes in competitive transit links

and/or new Transit-A routes arise, a modal split is
necessitated for establishing the O-D

demand. The relationship between change in attribute values
and the corresponding change

in Transit-A demand is estimated by conducting a survey
(see Chapter 2 for methods) to cap

ture the different behavior of existing and potential
Transit-A users. A demand forecast is

usually network-based. The existing Transit-A network
provides the base framework from

which other networks can be established. Specifically, each
network contains a definite set

of routes and fares, an operational strategy, and different
extents of competition (from other

transit services). In addition to the network scenario, a
route scenario can also be estab

lished to analyze individual routes when it is reasonable
to assume that they are independent

of one another. The latter was the case with
the ferry study (Ceder, 2006).

In order to integrate the demand forecast for Transit-A
into transportation-planning stud

ies of the local area being considered, the
ODMs for all transit services should be derived

directly from the local area studies, except for those of the existing Transit-A routes, which

are to be revealed by the survey. Combining all this information, the future demand for

existing routes can be predicted with regard to any change in the level of service. For new

routes and/or changes in other transit services, the common attributes of all relevant services

between specific O-D pairs need to be identified before a multinomial logit (MNL) model

(see de Dios Ortuzar and Willumsen, 2011) can be used for modal split purposes. The coefficient

values in the MNL model adopted the values of the local transportation-planning studies.

The following section explicates this MNL model.

The possible input/output lists for a transit-demand forecast methodology are shown in

Table 10.6. The framework for this forecasting methodology is presented in Figure 10.2,

which summarizes the flow of the forecasting tasks with the use of two models. It is worth

noting that the left-hand side of Figure 10.2 relates not only to changes in the existing

Transit-A routes but also to changes in the candidate routes following the establishment of

their O-D demand on the right-hand side of this figure.

Another important input in Figure 10.2 is the growth factor for a required year. This

factor should be updated continuously to allow for more accurate forecasting. The growth

factor is extracted from percentage changes in patronage for a given design year over the

past few years. It contains all the changes incurred, including changes in travel behavior.

Extrapolation (e.g., utilizing regression models) of the growth factor, which can be negative,

exhibits this ongoing patronage change.

10.3.2 Opinion survey (input) interpretation

This section explains how to apply the opinion survey information to demand forecasting.

Usually the main attributes affecting transit passenger demand are as follows: fare, trip

time, waiting time, walking time, and comfort level. Two sets of questionnaires can be

designed to obtain information on the aforementioned attributes, one set for the Transit-A

survey (possibly on board) and the other for the competitor(s). The first part of the ques

tionnaires can collect O-D information on commuters, which would help to develop the

base-year Transit-A demand matrices. The second half helps in establishing the commuter's

choice under different scenarios of changed services. Finally, basic questions on the trip pur

pose and personal details can be included for the sake of acquiring more information. For a

better understanding of people's behavior and, hence, for more accurate future passenger

demand predictions, a stated-preference type of questionnaire can be designed for fare.

Table 10.6 Input and output lists for forecasting models

Input Output

1. O-D survey 1. Base-year Transit-A patronage and potential increase in Transit-A patronage from competitors

2. Opinion survey 2. Attribute weights in linear regression models

3. Estimation of transit-passenger demand from

local transportation-planning studies 3. Expansion of the base-year Transit-A patronage to the design-year patronage at low, medium, and high levels

4. Local new-route service assumptions 4. Patronage estimate of new Transit-A routes

5. O-D matrices, fare matrices, travel-time matrices, and calibrated split-model coefficients taken from

local transportation-planning studies 5. O-D demand forecast by modes given by MNL modal split for a design year as a base for any further changes in new Transit-A routes

This will enable a direct deduction of the relationship between a fare level and a change

in demand. A similar arrangement and rationale holds for the item of comfort level, which

requires indirect transformation.

The attribute values can then be converted into equivalent monetary values in order

to refine the prediction mechanism by making use of the direct relationship between the

fare level and a change in demand. For developing a global model for all Transit-A routes,

regardless of their O-D characteristics, tailor-made questionnaires, addressing differences

in the fare and the other attributes, need to be constructed for individual routes. A list

of comfort-level items, from which a respondent can choose some but not all of the most

important ones, will be included in the questionnaires. By combining this information with

other fare-related information from other questions, an indirect deduction of the monetary

value of each comfort level can be achieved and consequently used in the prediction model.

Examples of stated-preference questions are as follows:

1. Will you still ride on a Transit-A vehicle if the fare level is increased to (1) \$7, (2) \$6.50, (3) \$6, (4) \$5.50, or (5) \$5.30?
2. If the Transit-A trip travel time is 5 min longer, how much do you think the fare should be reduced? (Mark an amount.)
3. If the Transit-A service waiting time is 3 min longer, how much do you think the fare should be reduced? (Mark an amount.)
4. Please mark up to three comfort items that are most important to you for improving the service? (Choose from a given list.)
5. Given improvements in the comfort items you marked in (4), are you willing to pay for an increase in the Transit-A fare to (1) \$10, (2) \$7.50, or (3) \$6? Opinion survey (questionnaire-based) O-D demand matrices for a required year Key: Input components Identify attribute values Attribute weighting (regression analysis): Changes in the demand derived from changes in each attribute For a given network/route scenario Growth factor for a required year O-D demand (survey board) Modal split (multinomial logit) derivation of new O-D proportions between the public transit modes Relevant O-D of competitive public transit links Main attributes: Fare Trip time Waiting time Comfort level Identify changes in all attributes from base year Consider new routes and changes in main attributes Consider existing routes with examination of changes in new routes having a derived O-D demand

Figure 10.2 Framework of a given forecasting method for a

given transit mode.

A linear regression model can be utilized as a tool to relate changes in the attribute value to

corresponding changes in demand. Because of different bases in most of the attributes associated

with different routes, a global method can be established by expressing the percentage

change for both the attribute values and the demand. For simplicity, each attribute can be

assumed to be independent of all other attributes in that they can be examined separately

while developing the model. For different levels of changes in fare, a corresponding percent

age change in Transit-A patronage can be deduced from the survey result. This information

can then be combined, using a linear regression, to determine the relative weight of the fare

attribute from the slope of the regression line. Similar procedures may be used for other

attributes after they are converted to monetary values, thereby making use of the fare-

demand relationship developed. A procedure to combine existing riders and potential riders

from a competitor within Transit-A is shown in Figure 10.3.

The regression lines are supposed to intersect the origin (0, 0), assuming that if there is no

change in the attribute, there will be no change in the demand. However, it is known that

the answers of survey respondents are biased,

reflecting riders' strong wishes to improve the

ride without additional cost and/or the natural resistance to increase Transit-A fares and/or

worsen its other attributes. As a result, the best fit for the regression line does not intersect

the origin; it shows that demand is dropped even if there is no change in the attribute. One

basic way to correct this bias, over and above its calibration with actual data, is to ignore

the free constant element in the linear regression and to allow the regression line to intersect

the origin.

10.3.3 Example of attribute derivation

A detailed example will now be provided as a useful device for understanding the underlying

procedures. This example explains how to calibrate the weights of attributes from the Slopes (linear regression) of a survey of Transit-A service Slopes (linear regression) of a survey of a relevant competitor service Adjustment through calibration of the regression line to account for actual observed demand changes owing to attribute changes Let the regression lines be divided into two parts, with two different slopes and their intersection at the origin point (0, 0). The part associated with reduced demand (an increase in fare, travel time, waiting time, and walking time, and a decrease in comfort level) is based on the Traffic-A survey and observed data, and the part associated with the increased demand (a decrease in fare, travel time, waiting time, and walking time, and an increase in comfort level) is based on the competitor's survey Conversion of the relevant amount of competitor patronage to the scale of the Transit-A service

Figure 10.3 Procedure for combining the survey results of the existing and potential users of Transit-A service.

raw survey data. It is assumed that 20 passengers were interviewed and their choices for

each question randomly assigned.

First, let us consider the question: Will you still ride on a Transit-A vehicle if the fare level

is increased to (1) \$ 7, (2) \$ 6.50, (3) \$ 6, (4) \$ 5.50, or (5) \$ 5.25? These five options correspond

to five levels of fare increase from \$ 5 (by 5 %, 10 %, 20 %, 30 %, and 40 %), with “ 1” being

assigned to the option selected. In case a passenger is unwilling to pay for any fare increase,

no option is assigned. The existing fares range between \$ 2.20 and \$ 5.30. The acceptable

increase in fare equals the existing fare times the increase option chosen by the passenger

interviewed. For example, the acceptable fare increase for one passenger was $\$ 5.00 \cdot 10\% =$

$\$ 0.50$; that is, for \$ 5.50, this passenger will still ride on Transit-A. The demand change for

the fare is calculated from the number of passengers still willing to ride Transit-A given a

fare increase. For instance, for a fare increase of 30 %, we (out of twenty) still will ride on

Transit-A; this means a 75 % patronage decrease. Plotting the percentage change in demand

versus the percentage change in fare for all routes and cases will construct a data set suitable

for a regression analysis.

Second, we will now present a full analysis of the question of an increase in travel time.

Similarly this analysis can be performed, for example, for waiting time, walking time/dis-

tance, and comfort items. Table 10.7 presents the raw survey data output for the question:

If the Transit-A trip travel time is x min longer, how much do you think the fare should be

reduced? (Interviewees need to mark an amount.)

The value of travel time is calculated by the ratio of the first two columns in Table 10.7.

A set of travel-time changes is tested, whether or not passengers still ride Transit-A. For

example, for the first passenger interviewed (first raw data in Table 10.7), an increase in

Table 10.7 Interpretation of survey results concerning a change in travel time

Value of x

more

minutes (\$) x (min) Value of travel time (\$/min)
Existing travel time (min) Change in travel time 40% 30%
20% 10% 5% 1%

1 3 0.33 7 0 0 0 1 0 0

0.5 3 0.17 7 1 0 0 0 0 0

1 3 0.33 8 0 0 0 1 0 0

0.5 3 0.17 8 0 0 0 0 1 0

1 5 0.20 13 0 0 0 1 0 0

2 5 0.40 13 0 0 0 1 0 0

2.5 3 0.83 7 0 0 1 0 0 0

1.5 3 0.50 7 0 0 0 0 0 0

3 3 1.00 5 0 1 0 0 0 0

3 3 1.00 5 0 0 0 0 0 1

2 3 0.67 7 0 0 0 0 1 0

3 3 1.00 7 0 0 1 0 0 0

2.5 5 0.50 11 0 0 0 0 0 1

1.5 5 0.30 11 0 0 1 0 0 0

2 5 0.40 12 0 0 0 0 1 0

2.5 5 0.50 12 0 0 0 0 1 0

1 5 0.20 15 1 0 0 0 0 0

2 5 0.40 15 0 0 0 0 0 0

2.5 5 0.50 10 0 0 1 0 0 0

3 5 0.60 10 0 0 0 0 0 1

Demand change for travel-time change -90% -85% -65% -45%
-25% -10%

travel time of 40% is equivalent to a fare increase of
 $7 \cdot 40\% \cdot 0.33$ (existing travel time times

travel-time change times the value of travel time) =
\$0.924. Because the acceptable increase

in fare for the first passenger was calculated (in the first
question) to be \$0.44, this passen

ger will not ride on Transit-A when the travel time
increases 40% or more, and thus "0" is

assigned. If, however, the travel time should increase by
only 10%, this would be equiva

lent to a fare increase of $7 \cdot 10\% \cdot 0.33 = \0.231 , which is
less than the maximum acceptable

increase, and therefore "1" is assigned in Table 10.7.
Thus, the demand change for the

travel-time change can be calculated with this conversion; this is plotted in Figure 10.4 with

the resultant fitted regression line. The R-squared value (0.816) is the fraction of the vari

ance in the data, which is explained by the regression.

10.4 MULTINOMIAL LOGIT MODEL

The MNL model (see, e.g., de Dios Ortuzar and Willumsen, 2011) is a share model that

divides the individuals among various travel modes according to each mode's relative desir

ability for any given trip. Such a technique has been commonly used to determine modal

split in a number of studies, though its accuracy depends heavily on the given data and

underlying mathematical format of the logit model.

This modal split model is usually used when there are some factors that the linear regres

sion model cannot fully address, for example, the introduction of new Transit-A routes and

changes in the degree of competition with other public transit modes (e.g., via improve

ments, fare changes). Four public transit modes are commonly included and examined in

the modal split process along with the private vehicle. These modes are train, bus, ferry,

and taxi.

The input attributes for the MNL model should be consistent and valid for all the avail

able transit modes and services. We will consider here the two main attributes affecting

modal split-fare and travel time. Because of the limited number of attributes employed in

the model, it may be necessary to re-tune Transit-A demand.

The MNL model takes the following mathematical form: $P_{em} = \frac{e^{u_m(i)}}{\sum_{n=1}^N e^{u_n(i)}}$ (10.10)
 0% -20% -40% -60% -80% -100% -120% 0% 5% 10% 15% 20% 25%
 30% 35% 40% 45% Change in demand (%) Change in travel time (%) $y = -2.655x$ $R^2 = 0.816$

Figure 10.4 Regression line of change in travel time against change in demand.

where P_m is the probability of an individual's choosing mode m or the proportion of trips using mode (or service) m , $u(i)$ is the utility of the i th mode, which includes the modal parameters and coefficients for a certain trip among the n available modes

For instance, the utility function with reference to buses can take this form:

$$u(i) = T(t_i - t_{bus}) + C(c_i - c_{bus}) + B_1 + B_2 \quad (10.11)$$

where t_i is the travel time incurred by mode i , c_i is the cost (fare) by mode i , T and C are the calibrated coefficients, B_1 and B_2 are biases for the mode and movement limitations, respectively, varying according to whether the mode is accessed by walking or by another transit mode and according to the degree of access difficulty

The following are typical MNL assumptions used in the forecast: (1) given zoning system;

(2) fares and travel times for railway, bus, ferry, and taxi taken from given matrices for peak

and nonpeak periods, as well as the cost and travel time of private vehicles; (3) total transit

patronage taken from given local transportation-planning studies (aimed at capturing the

right mixture of population and economy growth

affecting transit demand) and based on

self-zones selection; (4) given number of peak and off-peak hours for the local area considered

ered; and (5) specific zone selection depending on the proportion of transit users.

Following is an example of the use of the MNL model. Suppose a new railway line is

introduced in a transit network consisting of railway, bus, and passenger ferry. In order

to arrive at an estimate of daily patronage for this new line, the corresponding zones of

the line's end points should initially be identified; say, Z1 and Z2. The corresponding

transportation-planning study figures for this particular O-D pair for a designated year

are as follows:

- Total daily transit demand: 164,000 trips (Z1 to Z2) and 161,000 trips (Z2 to Z1)
- Travel times (includes the trip, waiting, walking, and transfer times) and fares for each of the three transit modes: Travel time (min) Mode Z1 to Z2 Z2 to Z1 Fare (\$) Railway 27.1 27.1 7.22 Bus 51.9 51.5 7.91 Ferry 49.2 48.2 5.37

The travel times and fares are inserted into the utility function (10.11) with the following coefficients: Mode T C B 1 B 2 Railway -0.00373 -0.00153 0.202 0.267 Ferry -0.00373 -0.00153 0.491 -1.1

The utility value of each of the transit modes is then calculated by Equation 10.11:

$u(\text{railway}) = 0.567409$, $u(\text{bus}) = 0$, $u(\text{ferry}) = -0.595789$

Utilizing the MNL model in Equation 10.10 yields the following modal split results: Modal split by railway $e e e e = + + = - 0.567409$ 0.595789 0.0

567409 53 2 % Modal split by bus e e e e = + + = -
 0 0 595789 0 0 567409 30 2 . . . % Modal split by ferry e e
 e e = + + = - - 0 595789 0 595789 0 0 567409 16 6 %

Applying these results to the new railway line between Z1
 and Z2 will determine the esti

mated p atronage l evel: (164,000 + 1 61,000) · 53.2%
 = 1 72,900 d aily t rips.

10.5 LITERATURE REVIEW AND FURTHER READING (ORIGIN-DESTINATION ESTIMATION)

ODMs constitute an essential input for most transit
 planning and design procedure. The lit

erature describes many methods of matrix estimation, but
 most studies refer to the genera

tion of car-trip matrices based on data from traf#c counts.
 This section will review methods

of e stimating t ransit-trip O DMs b ased o n p
 assenger c ounts o r s urveys.

Simon and Furth (1985) describe the method developed by
 Tsygalnitsky in 1977 for cre

ating a route O-D without O-D survey data. A good
 representation of transit demand can

usually be achieved if the trip matrix is based on a
 survey, in which passengers are asked

directly about their precise origin and destination;
 however, such surveys are expensive and

time-consuming; therefore, several methods deal with
 situations in which data from surveys

are not available. Tsygalnitsky's method uses only
 passenger on-off counts; the model's

passenger flow from one stop to another along the route is
 given in terms taken from fluid

mechanics. The matrix is generated using a recursive
 algorithm. Simon and Furth analyze

results statistically from the implementation of

Tsygalnitsky's method and compare them

to actual trip demand. They conclude that this method yields reasonable results in cases in

which the route structure is not too complicated.

Ben Akiva et al. (1985) review and test several techniques for generating route-level trip

matrices. The common practice of expanding onboard survey results by means of total

boardings is compared with expanding results by iterative proportional fitting, constrained

generalized least squares, and constrained maximum likelihood methods. An intervening

opportunity model, which does not use the onboard survey, is also tested. The more com

plex methods achieve better accuracy and reduced bias by combining the survey data with

passenger counts. An empirical case study demonstrates that under the assumption of error

free ride-check data, the proportional fitting technique is preferred because of its compu

tational ease without any loss of accuracy. Ben Akiva (1987), in a different study discusses

methods of combining transit data from different sources of partial information to create

a transit-trip matrix. The same matrix-estimation techniques described by Ben Akiva et al.

(1985) are examined. The use of alternative data sources, such as ticket-sales information,

is illustrated. Aspects concerning the level of accuracy of combined data, biases, and com

putational difficulties are discussed.

Nguyen et al. (1988) propose a method for the d

dynamic estimation of transit-trip matrices

ces, using passenger counts. The matrix period is divided into several time intervals, and

formulations that connect the link flow at a given time to link flows over time segments are

developed. The paper also presents an assignment model that is used for the estimation. The

optimization is based on maximum entropy and maximum likelihood approaches.

Furth and Navick (1992) analyze the relationship between Tsygalnitsky's recursive

method, which is commonly used for creating trip matrices from counts, and the biop

roportional method, which combines data from counts and from an OD survey. In the

bioproportional method, the survey data are used to create an initial seed matrix, which is

then adjusted iteratively by balancing row and columns to match on-off totals. The authors

show that the Tsygalnitsky method is, in fact, a special case of the bioproportional method.

As a seed matrix, it implicitly uses a null matrix, which contains information about trip

directionality and minimum length. The paper illustrates why the recursive method is inap

propriate when there is significant competition among routes. In addition, the authors offer

a correction for cases in which the on-off data have been aggregated to the segment level.

Navick and Furth (1994) discuss possible sources of the seed matrix, which is often

used as an initial basis for adjustments, and propose a model for generating this matrix.

According to this model, the number of trips in each matrix cell is influenced by two factors:

popularity, which is a feature of the destination zone and is not explained by models,

and propensity, which is a feature of each O-D pair and depends on the distance between

them. The researchers develop a propensity function that is the product of a power term

and an exponential term and is equivalent to a gamma distribution. The power-function

exponent is estimated by maximum likelihood, based on survey data. The gamma seed

combined with the bioproportional method to match O-D totals is shown to be effective in

generating ODMs.

Gur and Ben-Shabat (1997) develop another model that uses passenger counts for building

or improving trip matrices. Their model is formulated as a nonlinear programming

problem; for each pair $i-j$ of stations, the model calculates the probability that a passenger

who boards at i will alight at j . The problem takes advantage of information that is embedded

in counts of individual vehicles and in the differences between different counts. The

least-squares method is used for adjusting boarding totals to counts, and a minimum information

model (Fratton) then generates a detailed trip table.

Wong and Tong (1998, 2003) propose a methodology for creating transit-trip matrices

from passenger counts, using a maximum entropy programming formulation. The matrix

generation process depends on more detailed time data than do most previous methods; it

uses a schedule-based dynamic assignment model to determine least-cost paths between all

O-D pairs and clock-arrival times. Cases with and without a given seed matrix are examined

ined. The authors also present a solution algorithm in which the matrix reduces computer

memory requirements in an efficient way.

Friedrich et al. (2000) present a technique for the continuous updating of demand matrix

estimates based on fuzzy set theory. According to this theory, counts are treated as intervals

with lower and upper bounds, not as precise values. Matrix values are estimated such that

all totals fall within the proper bandwidths. The method, formulated as a programming

problem, may be implemented for estimating either car or transit-passenger matrices.

Nuzzolo and Crisalli (2001) and Nuzzolo et al. (2003) develop another method for esti

mating time-varying matrices from time-varying counts. The reference period is divided into

several subperiods, and detailed timetable information is considered. A dynamic schedule

based model of a stochastic type is used to predict passengers' path choices. The method is

formulated as a maximization problem without constraints. The match between matrix and

counts is achieved by minimizing generalized least squares.

Cui (2006) documents the development of an algorithm to estimate a bus passenger

trip ODM based on automatic data collection system (ADCS) archived data including

automated fare collection (AFC) data, automatic passenger count (APC) data, and auto

matic vehicle location (AVL) data. The proposed algorithm consists of three steps: data

preparation, estimation of single-route ODMs for all routes, and estimation of an network

level ODM using transfer flow information. The proposed single-route ODM estimation

utilizes AFC and APC data to develop the seed matrix and marginal values, respectively.

Using iterative proportional fitting and maximum likelihood estimation (MLE) tech

niques, the single-route ODMs based on seed matrices and marginal values are estimated

and results compared. The transfer flows for network level ODM estimation are derived

by considering the consecutive transactions from AFC data. The resulting network level

ODM is provided at a route segment level of detail. This ODM estimation algorithm has

been applied to a selected corridor of the Chicago Transit Authority bus network. Finally,

the author recommends that the MLE method be used to estimate the single-route ODMs

and the proportional distribution method be used to estimate the transfer flow ODM.

Farzin (2008) describes the process used to create an ODM in São Paulo, Brazil, with

data available from ADC systems. The approach used in this paper illustrates how these

large quantities of ADC systems data can be utilized to take advantage of the constant

improvements in computer processing speed to derive new planning and measurement

applications. The proposed methodology includes development of ODM using three data

sources: bus stops, AVL data, and AFC data. The results indicate reasonable patterns of

ODMs. Furthermore, using AFC system provides a larger sample set in comparison to tra

ditional survey data, thus allowing more comprehensive analysis and generating more spe

cific results. Future improvements suggested by the author include addition of transfer-level

information to the ODM and verification of available bus stop data.

Chapleau et al. (2008) argue that with a well-defined methodology, the transaction board

ing data, collected via smart card AFC system, can be used to monitor the supply and consumption

tion of the transit system, the mobility of its users, and the activity pattern at trip generators.

The authors provide descriptions of the validation and enrichment procedures as well as their

logic engaged, so it could be adopted for other similar systems. Furthermore, examples of

characterization of various objects in the transit system and a multiday analysis are presented

and discussed. The authors emphasize that more

transit data will become available as more

transit agencies adopt the smart card AFC technology, thus creating opportunities to develop

innovative ways to extract and analyze data for transit planning and modeling purposes.

Wang (2010) explores the application of archived data from A DCS to transportation

planning with a focus on bus passenger OD differences at the bus-route level and on travel

behavior, using London as an example. The feasibility and convenience of the trip-changing

methods' application, to obtain bus passengers' boarding and alighting locations, has been

demonstrated. These results have subsequently been validated by comparing against the

bus passenger OD survey data in London. The inferred ODs and the AVL data enable

estimation of alighting times for bus passengers, therefore allowing bus journey stages to

be conveniently linked to form complete journeys based on the difference between the sub

sequent trip's boarding time and the previous trip's alighting time for each bus passenger.

Comparison of interchange time and the connecting bus route's headway can provide a

methodology to evaluate the connecting bus services and bus passengers' interchange pat

terns. Finally, the author suggests that this research can be expanded to a full bus network

and other travel modes, thus enabling effective intermodal network planning through the

development of comprehensive databases.

Horváth (2012) investigated methods for obtaining reliable passenger data to construct

time-dependent ODMs, in order to facilitate effective planning of public transit systems.

A forecasting model, which consists of three stages, has been developed to derive time

dependent passenger data. First, full-scope cross-section data are collected with personnel

or with an automatic counting system. Then, boarding and alighting data are linked to

obtain the ODM for each run. Finally, ODMs of the run-through transfers are combined by

assuming that the probability of a transfer between two runs is a given stop is proportional

to the travel possibilities in this relation. The overall methodology has been tested through

an application in a Hungarian city (Dunaújváros). The author concludes that these results

were reliable, so they could be used in the planning process.

Table 10.8 Summary of characteristics of the origin-destination methods reviewed

Sources Method requires seed matrix? Mathematical approaches used Time segmentation (static/dynamic model)

Simon and Furth (1985)

(method by Tsygalnitsky) No Static

Ben-Akiva et al. (1985) Yes Iterative proportional fitting, constrained generalized least squares, constrained maximum likelihood Static

Ben-Akiva (1987) Yes Static

Nguyen et al. (1988) No Maximum entropy, maximum likelihood Dynamic

Furth and Navick (1992) Methods that do/don't require a seed matrix are compared Bioproportional method Static

Navick and Furth (1994) A method is presented for creating a seed matrix Bioproportional method, maximum entropy Static

Gur and Ben-Shabat

(1998) Optional Least squares, minimum information model (Fratar) Static

Wong and Tong

(1998, 2003) Cases with/without a seed matrix are examined Maximum entropy Dynamic

Friedrich et al. (2000) Yes Fuzzy set theory Static

Nuzzolo and Crisalli

(2001) and Nuzzolo

et al. (2003) No Generalized least squares Dynamic

Cui (2006) Yes Maximum likelihood estimation method is used to estimate the singleroute ODMs, and the proportional distribution method is used to estimate the transfer -ow ODM Static

Farzin (2008) No Static

Chapleau et al. (2008) No Static

Wang (2010) Yes Static

Horváth (2012) No Static

Tirachini et al. (2013) No Multinomial logit model Static

Tirachini et al. (2013) investigated the impact of crowding on the passenger-demand esti

mation. They argued that people's perception of crowding should be considered as a source

of disutility for passengers when estimating passenger demand for public transport. They

found that (1) a model that assumes users are indifferent to the occupancy levels of vehicles

overestimates the value of travel-time savings (VTTs) for low load factors and underesti-

mates VTTs for high load factors, and likewise (2) a model that is insensitive to crowd-

ing levels underestimates demand if vehicles are uncrowded and overestimates demand if

vehicles are crowded. These findings were not proven using empirical studies. However,

the authors analytically found that when crowding was omitted, there was a load factor

threshold that marks the underestimation or overestimation of passenger demand by using

an MNL model for mode choice.

The main characteristics of the methods reviewed previously are summarized in Table 10.8.

EXERCISES

10.1 Assuming that a rail company has the following linear demand relationship: $p = 20 - 0.04q$, where p is the route fare and q is the number of hourly sold tickets. Find the total revenue associated with each (p, q) pair and indicate when its maximum value is reached. In addition, specify the p - q zones under which the point elasticity is elastic and inelastic.

10.2 Given the following power demand function, $N = p^{-0.3} \cdot t^{-0.3} \cdot a^{0.2} \cdot c^{-0.3}$, where N is the number of transit trips, p is the fare (in dollars), t is the travel time (in hours), a is the cost of a automobile trip (in dollars), and c is the average income (in dollars). (a) Given 20,000 passengers per hour who currently use the transit system at a fare of \$1.20 per trip, what will be the change in N for a fare of \$0.80 and what will be

the transit agency gain? (b) Given the cost of an automobile trip (including parking fee) of \$ 4.00, what will be the change in N if the parking charges increase by \$0.60?

10.3 Given a calibrated utility function, $u = b - 0.04C - 0.02t$, where C is the cost of travel (in cents), t is the travel time (in minutes), and the MNL model is as described in the chapter. (a) What will be the modal split (percentage demand traveling by each mode) for the following data given? (b) Given rising gasoline prices so that C for an automobile increases by \$1.20, what impact will this change have on the modal split? Mode b C t Bus -0.30 85 30 Light rail -0.35 100 50 Automobile -0.25 110 35

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11 Chapter 11: Route Choice and Assignment

Once we have an idea of how many people want to travel between two points, by time of day

and transit mode, the next phase is to determine the routes (direct or via transfers) they should

take to reach their destination. This determination relies on passenger behavior in regard to

choosing a route if there are alternatives. What is the typical (average) choice strategy if there

is a slower or less direct route, but one that does not involve long waiting times? This and other

route choice issues are discussed in this chapter in order to prepare the ground for assigning

demand on the transit network. Utilizing an assignment procedure will allow for assessing/pre

dicting changes in transit service design at the network level. The following riddle may indirectly indicate the complexity of the analysis exhibited in this

chapter. A passenger arrives at an intersection with two bus stops in opposite directions of a

desired route. He doesn't know which direction to take, since there is no indication of any des

tinuation. The passenger meets two people who know the required direction, but one tells only

the truth and the other always lies (both these people, who are friends, know that). The pas

senger has no idea who is telling the truth and who is lying. How can the passenger, by receiving

an answer from one of these two people to but one question, obtain the information required?

(See the end of this practitioner's corner for the answer, but try first to think of the answer.) The chapter consists of five main parts, following an introductory section. Section 11.2,

using probabilities, analyzes the alternatives facing a transit passenger at a stop who is decid

ing whether to board an arrived vehicle or to wait for a later vehicle that will have a shorter

inv ehicle time. This route choice dilemma is viewed as a decision between two categories of

routes, slow and fast, in Section 11.3, from which the proportion of each category is derived.

Section 11.4 crystallizes the proportions between the two categories for the case of regular

vehicle arrivals. Section 11.5 presents some of the transit assignment features that are related

to route choice modeling; emphasis is placed on estimating the accumulated demand on each

segment of the network. The last part, Section 11.6, provides a literature review on transit

assignment studies, with guidance for further reading. Practitioners may find this chapter too technical, especially the mathematical handling of

probability formulae. However, it is recommended that the introduction to each section be read

in order to capture the essence of the subject. For instance, knowing the rationale behind pas

sengers' route choices may help overcome planning and operational problems. One humorous

example concerns passenger behavior in situations characterized by a high in-ation rate; then

it will be better to ride a taxi than a bus because in the bus you pay upfront and in the taxi at

the end of the trip. As for the question to be put to one of the two friends (it doesn't matter which one): If I'll tell

your friend that this direction (points to a direction) is the correct one, what will be his answer?

The passenger will then choose the opposite direction of the answer given.

11.1 INTRODUCTION

In constructing or revising a transit network of routes, it is not sufficient to know only the

demand, its elasticity and factors affected by it. What is needed in addition is a comprehensive

understanding and insight into passenger behavior when it reflects flows dispersed along the network

of routes. What, if any, are the passenger paths (use of more than one route with transfers)

through the network? What factors influence a passenger's choice of routes? The answers to

these questions represent route choice behavior (modeling), which is then used in assigning

passenger demand for prediction purposes (of traffic volume, capacity analysis, fleet size

required, design and control elements needed, and more). This chapter will review the subject

of passenger route choice and assignment, placing some emphasis on passenger waiting

time strategies.

One of the most crucial characteristics of a transit network is the existence of overlapping

routes that share the same transit segments while running on common segments.

Emerging from this characteristic is the fact that more than one transit route can serve

the demand between a certain origin-destination (O-D) pair. We will focus in this chapter

on the decision-making process of the individual passenger who has to select the most

efficient routes serving a given stop. This process is one of the main obstacles in developing

transit assignment procedures.

An additional complication of transit assignment methods in real-world problems relates

to the possibility that several transit modes (bus, train, light-rail, metro, etc.) and a walk

mode exist, each with its own characteristics. Moreover, each mode is perceived dif-

ferently by passengers. To simplify the presentation of transit assignment approaches, this

chapter includes the simple case of a single transit mode. The analysis follows Marguer

and Ceder (1984) and Israeli and Ceder (1996). The literature review of this theme appears

in Section 11.6. Summaries of the main contributions of this (transit assignment) theme

can be found in Bell (2003), Bell and Schmöcker (2004), Nuzzolo (2003a,b), and Nuzzolo

and Crisalli (2004).

11.2 ROUTE CHOICE USING WAITING-TIME STRATEGY

The decision problem facing a transit passenger at a stop with alternative choices is whether

to board an arrived vehicle or to wait for a later vehicle that will have less in-vehicle time.

This issue reflects real choice situations in which the passenger can distinguish between two

categories of overlapping routes: slow and fast. The objective is to minimize total travel

time

(waiting time and in-vehicle time), which would result from adopting an optimal strategy in

the choice process. This action depends on certain parameters: time between each O-D pair,

vehicle regularity, distribution of passenger arrivals at the stop, and the structure of possible

paths constituting the network.

The analysis shown in this section deals with the problem of passenger waiting strategies

by using mathematical formulations in a probabilistic fashion in order to derive

the proportion of passengers for each category (slow and fast). This calculation depends

on the evaluation of a probabilistic function that considers such affecting parameters as

vehicle-frequency share, in-vehicle time difference, passenger arrivals, and vehicle head

way distribution. In addition, theoretical evaluations are presented in order to achieve

correct implementation of (a) mean waiting-time assumptions and (b) the intuitive rule

circumstances (according to which, passengers select routes in proportion to their vehicle

frequencies). The outcome of the analysis can be integrated into transit assignment models

to approach realistic situations.

A practical passenger-assignment procedure on transit networks consists of the following

characteristics: (1) the exact structure of the network between each origin and destination

tion (direct routes, parallel routes, sequential routes, and transfer paths); (2) in-vehicle time

(direct routes and transfer-path times); (3) passenger waiting time at a stop; (4) passenger

waiting strategies at a stop served by overlapping routes; (5) the circular problem (because

frequencies are not always known at this stage) in which the passenger flows depend on

route frequencies and, conversely, the frequencies rely on the load profiles of the routes,

which are the outcome of the flows; (6) network size (treating each O-D pair separately and,

subsequently, considering the entire network simultaneously can help to combat computa

tional complexities of large networks); and (7) failure-to-board probabilities. This section

treats the second, third, and fourth characteristics and their relationship to the first, fifth,

and sixth; the seventh characteristic, as well as all the others, are discussed in the literature

review in Section 1.1.6.

11.2.1 Passenger waiting-time dilemma

The objective of this section is to study the problem of route choice encountered by a pas

senger who is able to select one of several routes (alternatives) in order to travel from stop

A to stop B. The common characteristic of all the transit routes considered is that they stop

at both points A and B. The central idea is that a passenger wishing to travel from A to B

may disregard some of the vehicles arriving at A because of their association with relatively

long travel times.

The analysis, following Manguier and Ceder (1984), is performed in a probabilistic

context, in which the passengers, based on their experience, have a fairly good sense of

the characteristics of each route. That is, they have some information about the headway

distribution (route frequency) and expected in-vehicle time from A to B. The passen

gers are also influenced in the route selection process by the amount of time they have

already waited since their arrival at the transit stop, which can be considered additional

information available to the passengers. Total travel times, consisting of waiting time

and in-vehicle travel time for each route, are compared in the analysis. The waiting

time, with its formulation shown in the following text and in Chapter 18, clearly plays

an important role in total travel time because of its relationship to and dependency on

vehicle-interval reliability and the passenger-arrival process at stop A. Before proceed

ing with the main thrust of the analysis, let us introduce some related formulations and

assumptions.

11.2.2 Basic relationships

Chapter 18 (on transit reliability) will show how to arrive to the following commonly

used formula for mean passenger waiting time under the assumption of random passenger

arrivals: $E(w) = E(H) - \frac{1}{2}E(H) = \frac{1}{2}E(H)$ $\text{Var}(H) = E(H^2) - [E(H)]^2 = \sigma_H^2$

$$w = E(H) - E(w) \quad (11.1)$$

where w is the waiting time $E(w)$ is the mean waiting time $E(H)$ and $\text{Var}H$ are, respectively, the mean and variance of the time headway H between vehicles

Underlying this relationship is the following assumption: the average waiting time of pas-

sengers arriving at an interval of length t is λt , and the average number of passengers

arriving during such an interval is proportional to t . In addition, from data on off-peak

bus intervals, the following relationship was found: $\text{Var}H = A + E(H)^2$ (11.2)

where A is a constant (with time squared dimension) between 0 and infinity.

$A = 0$ corresponds to the deterministic headway case (regular vehicle arrivals). $A \rightarrow \infty$

corresponds to the completely random case (Poisson vehicle arrivals, i.e., exponential head-

ways). Actual values found for A are between 1.5 and 3.5.

The transit vehicle interval irregularity is actually characterized by the headway distribu-

tion coefficient of variation C , which is defined as the ratio of the headway standard devia-

tion to the headway mean, that is, its square is $C^2 = \text{Var}H / E(H)^2$ (11.3)

For transit vehicle (especially buses) headway distributions, C^2 ranges between 0 and 1,

where $C^2 = 0$ corresponds to perfectly regular vehicle arrivals (deterministic headway) and

$C^2 = 1$ to the completely random case (exponential headway).

Combining Equation 11.2 and the intrinsic definition of C^2 given by Equation 11.3

yields $C^2 A E(H)^2 = 2 + (11.4)$

It may be noted that the parameter A lies between 0 (for which $C^2 = 0$) and infinity ($C^2 \rightarrow 1$).

Using, respectively, the notations $f_H(t)$ and $F_H(t)$ for the probability density function and the

cumulative distribution function of the headway H , defining $F(t) = 1 - F_H(t)$, and denoting

the vehicle frequency as $F = 1/E(H)$, yields the following:

1. For $A \rightarrow 0$ (deterministic) F and $F(t)$ are given by $F(t) = 0$ for $t < 0$ and $F(t) = 1$ for $t \geq 0$.

2. For $A \rightarrow \infty$ (exponential) $F(t) = F e^{-Ft}$ and $F(t) = 1 - e^{-Ft}$.

Equation 11.4 may also be rewritten as $A = C^2 / F^2 (1 - C^2)$, which shows how, for a given

frequency F , the parameter A and the coefficient of variation are uniquely related.

The idea, then, is that for a given A (or C^2 when using the aforementioned relation

ship), the headway distribution belongs to a family of functions that can approach the

two extremes, the deterministic and the exponential cases, whereas Equations 11.2 and

11.4 are imposed. Two such families are considered here. The first one, developed by

Marguier and Ceder (1984) and referred to as power distributions, verifies the following

relationships: $f(t) = \frac{2F C^2}{(1 - C^2)^2} \left(\frac{1 - C^2}{1 + C^2} \right)^{1/C^2} \left(\frac{1 - C^2}{1 + C^2} \right)^{1/C^2} F t^{1/C^2 - 1} e^{-F t^{1/C^2}}$ and $F(t) = 1 - \left(\frac{1 - C^2}{1 + C^2} \right)^{1/C^2} \left(\frac{1 - C^2}{1 + C^2} \right)^{1/C^2} e^{-F t^{1/C^2}}$ (11.5)

Underlying the definition of this family of distributions is the fact that for such distributions

at the limit for $A = 1$ (see the relationship between A and C^2 in the preceding text), the dis

tributions approach the exponential case in a fashion corresponding to the way the binomial

processes approach the Poisson process.

The second family is made up of gamma distributions and verifies the following

$$\text{relationship: } f(t) = \frac{t^{C^2-1} e^{-t/C}}{\Gamma(C^2) C^{C^2}} \quad (11.6)$$

where Γ is the gamma function, that is, $\Gamma(C^2) = \int_0^\infty x^{C^2-1} e^{-x} dx$.

The idea of selecting gamma distributions is that the exponential is a particular case

of gamma and the generic gamma distribution has two independent parameters related

to the mean and variance, which allows the constraint given by Equation 11.2 to be met.

Expressions (11.5) and (11.6) are used here in the route choice problem in order to illustrate

the generality of the results.

Certainly, both families give identical curves for $C^2 = 0$ and $C^2 = 1$, since these values

correspond to the deterministic and the exponential cases, respectively. Real-life data show

headway histograms with increasing irregularity (C^2) along a transit route; the histograms

have a maximum point, as observed for the gamma situation, and an intersection with the

ordinate axis (different from the origin), as observed for the power situation distributions.

Therefore, realistic situations fall somewhere between the power and the gamma situations.

The general assumptions of this analysis are the following:

1. The (between days) vehicle-arrival variability is high enough so that passengers cannot identify any minimum wait in order to time their arrival at the transit stop. This assumption is particularly appropriate for urban routes without published timetables in which the values of A and C_2 , or F , are relatively large. This assumption can be relaxed in accordance with the study by Jolliffe and Hutchinson (1975).
2. The proportion of passengers arriving coincidentally with a vehicle (see and rush) is fixed and independent of other parameters. Observations in urban areas (in London and Paris) found this proportion to be up to 16% (Manguier and Ceder, 1984). Together with assumption (1), we assume then that a major proportion (e.g., 84%-100%) of the passengers arrives randomly.

Assumptions related to the route choice problem are the following:

1. Passengers are assumed to know the routes and to have a general knowledge of the headways and waiting time from experience. Thus, they select a strategy of choice among routes by attempting to minimize the sum of expected waiting and in-vehicle times. This assumption could be extended by considering, in a similar fashion to Jolliffe and Hutchinson's (1975) methodology, that only a certain proportion of the passengers follow the principle of minimization of total waiting and in-vehicle times, whereas the others will take the first bus that arrives and then continues on to point B.
2. At point A, the routes are statistically independent of each other; this is true if the routes do not share another common section, upstream of point A.

11.3 PROPORTION OF PASSENGERS BOARDING EACH ROUTE

The route choice dilemma can be viewed as a decision between two categories of routes.

When there are more than two routes to travel from point A to point B, it is not irrational

to assume that passengers will not consider each route individually, but rather will tend to

group them into categories. Two likely categories are faster routes and slower ones. Passengers

will then not choose between routes, but only between categories, and thus take the rest

vehicle to arrive in the category selected. The routes belonging to one category cannot then be

viewed as forming one equivalent route. For simplicity's sake, we will refer in the following

text only to the faster route, Route 1, and to the slower route, Route 2. The following

analysis is suitable for individual Routes 1 and 2, but may be extended for the concept of

two-route categories.

Using the aforementioned formulations, we can then derive the estimated proportion of

passengers that travel from points A to B by Route 1 and the complementary proportion that

uses Route 2. This proportion represents the quantity of interest in the travel-assignment

process for different transit routes. The estimated proportion depends on two factors: the

frequency of each route and the headway coefficients of variation. The latter can be replaced

by Equation 11.4.

Following Marguier and Ceder (1984), use of the optimal strategy will result in Route 1

where RW_{1-1} is the reciprocal of the RW_1 function (RW_1 can be reciprocated, since it decreases monotonically)

Figure 11.1 Proportion of passengers boarding the slow Route 2 (complementary to the fast Route 1 proportion) as a function of the slow Route 2 frequency share of $(\tau_2 - \tau_1)/H_1 = 0.54$. $P_2 \approx 0.8, 0.6, 0.4, 0.8, 0.6, 0.4, 1$ for $C_{21} = 1$ for Powerdistributed headway Gammadistributed headway $0, 0.2, 0.4, 0.6, 0.8, 1.0, 0, 0.2, 0.4, 0.6, 0.8, 1.0, 0, 0.2, 0.4, 0.6, 0.8, 1.0, 0, 0, \infty, \infty, \infty$ $C_{22} = 0, C_{22} = 0, C_{22} = \infty, C_{22} = \infty, C_{21} = \infty, C_{21} = \infty, C_{21} = 1, H_1(\tau_2 - \tau_1) = TH_1 =$

Figure 11.2 Proportion of passengers boarding the slow Route 2 (complementary to the fast Route 1 proportion) as a function of $(\tau_2 - \tau_1)/H_1$ for $FS_2 = 0.5$.

Figure 11.1 represents the market share-frequency share curves, in which P_2 increases

with FS_2 . In Figure 11.1 for $C_{21} = 1$, it should be noted that the proportion P_2 for $C_{12} =$

remains fixed, equal to 0. This can be seen from Equation 11.9 because $TH_1 = 0.54 > 0.5$ for

C_{11}

$C_{12} = 0$. The curves in Figure 11.2 show that P_2 is either approximately or exactly constant

with respect to small values of $\tau_2 - \tau_1$. For higher values, however, the value of P_2 drops rap-

idly and reaches 0. This drop signals the possibility of a threshold for the passengers' route

choice decision.

All in all, this analysis shows that the general intuitive rule $P_2 = FS_2$ (i.e., passengers

board the routes proportionally to their frequencies) does not provide a good approxima-

tion in all cases. In addition, Manguier and Ceder (1984) present a revised formulation of

Equation 11.9 for a case in which the proportion of passengers whose arrival at the transit

stop coincides with the arrival of the first vehicle is not zero. The next section constructs a

full description of the proportion boarding each route, but only for regular vehicle arrivals.

11.4 PROPORTIONS DERIVED FOR REGULAR VEHICLE ARRIVALS

This section follows Israeli and Ceder (1996). Based on the random arrival of passengers

and regular transit-vehicle arrivals (deterministic headways), the waiting time at stops asso

ciated with $i = 1, 2$ (referring to Routes 1 and 2) is uniformly distributed: $f(t) = \frac{1}{\tau_i} \cdot \frac{1}{\tau_i} \cdot \frac{1}{\tau_i} \cdot \frac{1}{\tau_i} = \frac{1}{\tau_i^4}$ (11.11)

The relevant component of interest is the distribution of the difference between the waiting

times of Routes 1 and 2, $w_1 - w_2$. Israeli and Ceder (1996) showed that this distribution

could be simplified in the following uniform manner: $f(t) = \frac{1}{\tau_1 + \tau_2} \cdot \frac{1}{\tau_1 + \tau_2} \cdot \frac{1}{\tau_1 + \tau_2} \cdot \frac{1}{\tau_1 + \tau_2} = \frac{1}{(\tau_1 + \tau_2)^4}$ (11.12)

with $E(w_1 - w_2) = \frac{\tau_1 + \tau_2}{2}$ (11.13)

The cumulative distribution (choice probability), therefore, is given by $Pr(w_1 - w_2 \leq t) = \frac{t}{\tau_1 + \tau_2}$ (11.14)

to obtain $F(t) = \frac{t}{\tau_1 + \tau_2}$, $f(t) = \frac{1}{\tau_1 + \tau_2}$ (11.15)

The complementary cumulative distribution $F(t) = 1 - \frac{t}{\tau_1 + \tau_2}$ can be obtained by $1 - F(t)$.

The probability (proportion) P_1 of selecting Route 1 (fast) vehicle can be obtained by

assigning $t = \tau_2 - \tau_1$ to Equation 11.15. Because $\tau_2 > \tau_1$, then $t > 0$ and Equation 11.15 becomes $P_1 = \frac{\tau_2 - \tau_1}{\tau_1 + \tau_2} = \frac{\tau_2}{\tau_1 + \tau_2} - \frac{\tau_1}{\tau_1 + \tau_2}$ (11.16)

Based on Equation 11.16, the complementary probability P_2

(at the planning stage) changes in routes, l

ocation of stops, selection of operational

strategies, priority schemes for transit vehicles, traffic and parking arrangements, environmental

impact studies, and more. Section 11.6 provides a literature review of transit

assignment studies. The section presents some of the transit assignment features that are

related to route choice modeling.

Passenger path may involve transfers and, hence, the use of more than one transit route.

An arc in the transit network is a road segment that can have more than one crossing route,

thus creating the possibility of overlapping transit services. Figure 11.4 illustrates the difference

between a road network and a transit network. A passenger traveling from node A to

node D on the transit network in Figure 11.4 has two possible paths: Routes I and II with a

transfer at B, or routes I and II with a transfer at C. For instance, in the network in Figure

11.4, we can examine routes A-C-D and A-B-C, in which route A-C-D replaces route

A-B-C, thus making node C a single transfer point from B to D. This examination (as well

as other sets of alternative routes) can use passenger-assignment procedures for given fre-

quencies; the latter will be derived from the O-D demand. Based on given criteria or objec-

tives, different routing recommendations can be made by the assignment algorithms. This

simple description is further analyzed by Guan

et al. (2006) in combining transit assignment

and route configuration.

The proportion of passengers boarding each transit route (Figures 11.1 through 11.3)

can be integrated into an assignment model for an entire transit network. The use of such

a procedure can provide realistic considerations to some extent of passenger behavior in

the assignment procedure. Note that the previous section assumes that the routes operate

independently and do not share another common segment upstream of the transit stop with

a waiting-time dilemma. Here, this assumption is extended to all segments of the transit

network. A B D C Road network Transit network A B D C Route I Route II

Figure 11.4 Road and transit network illustrations of four locations.

Passenger assignment on the transit network is interpreted as intermitted flow of passen

gers: waiting to board or transferring at stops and riding on the vehicles. At node u , the total

flow of arriving passengers who want to obtain service (boarding or transferring) and the

frequency parameters of the service will determine the total waiting time W_u at that node.

The total expected waiting time, using Equation 11.1, can be formulated as $E[W_u] = \sum_{r \in R} \sum_{v \in V} p_{uvr} \frac{1}{f_r} + \sum_{r \in R} \sum_{v \in V} p_{uvr} \frac{1}{f_r} \sum_{H_r} \frac{1}{H_r} = + \dots$ (11.18)

where p_{uvr} is the number of arriving passengers at u from node v who seek to board/transfer to a route (or group of routes) r , H_r is the headway (or combined headway for a group) of r

One objective (e.g., Spiess and Florian, 1989) can be to

minimize the sum of the cost of pas

sengers and their expected waiting times, subject to constraints for all nodes in the network.

The constraints ensure conservation-of-flow conditions (what goes into a node equates with

what comes out) and that the flow on each arc is greater than or equal to 0 and less than or

equal to a function of the frequencies and waiting times on that arc.

Passenger-choice strategy was defined as being between fast and slow route category

ries, but in a network context. As the proportion boarding a vehicle in each category is

dependent on the frequency share and the time difference between the two categories,

it is necessary to define a combined frequency and equivalent in-vehicle time for each

category. The network structure contains different possible paths between each origin

and destination, that is, direct routes and transfer paths. The latter can consist of over

lapping (parallel) routes, successive routes, or both. Although calculating the equivalent

in-vehicle time for each category is simple enough (passengers tend to make an average of

all possible paths connecting the origin with the destination), the combined frequency is

more complicated, as will be explained. The complex network structure accounts for why

researchers have tried to avoid this problem by presenting nonrealistic simplifications,

such as using only the frequencies of routes that connect directly to the transit stop or

employing different frequencies for each arc of the network. The latter requires smooth

ing techniques in order to find the correct frequency even for direct routes, a process that

leads to unrealistic results.

At the outset of the process, as has been seen, passengers have to choose a category (slow

or fast routes); then, the route within this category needs to be chosen. These passengers

who do not distinguish among route lengths (i.e., n-vehicle times) within each category board

the first vehicle to arrive. That is, passengers are assigned to the routes in proportion to the

routes' frequencies.

Each category, which refers to a subnetwork between the corresponding origin and des-

tination, is part of a modified transit network: the nodes are transit stops with overlapping

routes (i.e., enabling transfers), and the arcs are direct route segments between these nodes.

The number of arcs in a possible path between an origin and a destination denotes the num-

ber of vehicle changes, minus one.

Figure 11.5 shows the combined frequency calculation and, consequently, the demand

assignment within a category, or the routes between i and j . The combined frequency for pair

allel routes is their sum, and for successive routes, it is their minimum. In such a manner, pas-

sengers between k and j in Figure 11.5a experience a combined frequency of $3 + 1 = 4$ veh/h,

while their effective frequency along $i-k-j$ is mi-

$n(6, 4) = 4$ veh/h. The combined frequency

of all categories between i and j is $6 + 4 = 10$ veh/h, which basically is the minimum cut

of the small network in Figure 11.5a (for the definition of minimum-cut frequency, see

Section 6.4). This result (which can be either F_1 or F_2 , depending on the type of category) is

then utilized in Equations 11.9, 11.16, and 11.17 in order to predict the proportion of passengers

between i and j for the two categories.

Further analysis of the network shown in Figure 11.5a is illustrated in Figure 11.5b

through an and called network synthesis. It is based on recursive calculations of the combined

frequencies in a subnetwork, until a single combined frequency arc is reached. Network

synthesis is analogous to an analysis made for capacitor (electric) networks, with parallel

and successive summations. A parallel summation of frequencies is equal to the sum of the

frequencies associated with the cut of the parallel arcs; a successive summation of frequen-

cies equates with the minimum among the frequencies considered. The calculation of the

divided demand can be viewed in a reversed process. The demand of $p_{ij} = 180$ passengers in

Figure 11.5e, which is a single-frequency arc, is divided (going backward) in Figure 11.5d on

the basis of the proportional frequency share rule, for example, see Equation 11.10, that is, Route number Frequency Path-based demand Traditional demand 108 [90] 54 [67.5] 18 [22.5] Min-cut i - j frequency

(=10) 72 [90] 2(6) 4(3) 3(1)k 1(6) Passenger demand i 180
 2(6) 72 3(1) 18 54 4(3) 108 1(6) k j i j i 180 2(6) 72
 3+4(4) 72 108 1(6) k j i 180 2+3+4(4) 72 108 1(6) j i 180
 180 1+2+3+4(10) j (b) (c) (d) (e) $P_{ij} = 180$ (a)

Figure 11.5 Passenger assignment on an hourly basis, in which part (a) shows that the traditional assignment is a path-based assignment and parts (b) to (e) describe in steps the network synthesis process.

$180 \cdot (6/10)$ and $180 \cdot (4/10)$. The latter demand of 72 is kept along i-k-j in Figure 11.5c and

a gain (going backward) is divided proportionally in Figure 11.5b to 54 and 18.

The separation into two route categories for each i-j pair (if passengers notice a time

difference between them) creates two separate sets (subnetworks) for each route; this

means that a route or path serving the demand between i and j (directly or via transfer)

will be included in the subnetwork of either the fast or slow category in this pair, but not

in both. Thus, assignment equations can be formulated in such a manner that combined

frequencies and demand shares will be formulated separately for the two route categories,

while the relation between variables in each category will be given in the frequency

calculation constraints.

Considering the interrelationship of all O-D pairs in the network enables more man

ageable handling of large, complex networks. This interrelationship is affected by the

mutual dependence of the unknown frequencies and demand flows (serving as vari

ables) in an assignment procedure. While a ctual

frequencies are based on the methods of

Chapter 3 (consisting, via ride and point-check counts, of a full O-D demands for a route),

the demand flows are usually based on the frequency share (with or without considering

route travel times). More on transit assignment studies and considerations appears in

Section 11.6.

11.6 LITERATURE REVIEW AND FURTHER READING

One of the vital ingredients in transit planning is the prediction of the paths that passengers choose for transit routes to take them from origin to destination. This prediction

relies on the use of transit assignment models that have appeared in the literature of professional papers on the subject in the past 40 years. Much progress was achieved in past

research in adding more realistic features of passenger behavior and operational planning

elements to these models. Among the added features are waiting times at stops, transfer

times between routes, preferred set of routes for a particular pair from a passenger's perspective,

preferred passengers' strategies, distributed travel and waiting times, crowding level

on vehicles, and failure-to-board probabilities. A summary of some of the papers can be

found in Bell (2003), Bell and Schmöcker (2004), Nuzzolo (2003a,b), and Nuzzolo and

Crisalli (2004). This section reviews briefly and chronologically the main contributions to

transit assignment modeling.

Early methods of transit route choice and assignment are works by Dial (1967) and

Le Clercq (1972), who used heuristic rules in considering both waiting and travel times

when computing the shortest paths on a network. Passengers take the first vehicle

to arrive, and a transfer penalty is equal to the expected waiting time. Chriqui and

Robillard (1975) introduced, for the first time, a choice-behavior feature by which pas-

sengers can select, from a set of alternative routes, a subset of routes from which they

will board the first vehicle to arrive. This subset of routes, called attractive routes,

represents routes that have a better chance to offer shorter travel times (based on prior

knowledge). The researchers presented their ideas with the use of a simple network and

a single origin and destination.

Chriqui and Robillard's direction was extended in a doctoral study by Spiess in 1984.

Nguyen and Pallotino (1988) introduced a graph interpretation for a strategy of select-

ing a set of attractive routes at a boarding point; this graph representation was denoted a

hyperpath. Part of Spiess's original study was incorporated into a work that became known

(Spiess and Florian, 1989). The latter developed a two-part algorithm to assign passenger

demand from the user's perspective. Spiess and Florian assumed that passenger behavior

reflected a minimization of the expected value of a cost, that is, of waiting and in-vehicle

times or a weighted sum of these time elements. The first part of their algorithm computes

the total expected time elements between origin and destination, including transfers; the

second part assigns demand according to the strategy of choosing a set of attractive routes.

Both Spiess and Florian (1989) and Nguyen and Pallotino (1988) considered the effect of

the inconvenience of crowded vehicles through discomfort functions in their equilibrium

assignment models.

De Cea and Fernández (1993), inspired by Spiess and Florian, introduced a transit

equilibrium assignment model in which waiting times on access links depended on pas-

senger flows, that is, they applied congestion functions to passengers requesting to board

the first-arrived vehicle. De Cea and Fernández (1993) incorporated heuristically the dis-

comfort effect of crowded vehicles and crowded stops. Their model was solved by Jacobi's

method of using a similar diagonalization algorithm to that used by Florian (1977).

Wu et al. (1994), using the concept of hyperpaths (strategies) proposed by Nguyen

and Pallotino (1988), introduced a network consisting of road-based and transit route-

based arcs. Their passenger-related arcs included the time elements of walking, waiting,

boarding, in vehicle, transferring, and alighting. Their assignment model considered

that the time required to board a vehicle i

increased with flow (of transit vehicles); the

distribution of flows across the set of attractive routes is proportional to the minimum

frequency share.

The foregoing articles may be grouped under frequency-based models, in which within

day dynamics (in transit vehicle headways) are not taken into account. In the mid-1990s,

some route choice and transit assignment models started explicitly to consider different

headways (timetable) as an input. These studies were grouped under schedule-based models.

In most such studies, passenger-choice behavior is interpreted by a utility function. That is,

passengers are assumed to assign a utility value to each alternative route from a given choice

set (of routes) and to select the one with maximum utility. Usually, the utility values of the

alternatives are treated as random variables, thus converting the choice process (among

routes) into a search for maximum (utility) probability. Each utility function is dependent on

a set of passenger attributes. Hickman and Bernstein (1997) used a sequential deterministic

path-choice approach with a utility function consisting of stochastic travel-time attributes

and passenger information. They studied high-frequency service, including congested net

works, using equilibrium and dynamic processes.

Tong and Wong (1999) showed the differences between frequency-based and schedule

based approaches and constructed a stochastic schedule-based dynamic model using

simulation. In their model, passengers either move on a walking segment, wait (queue)

on a network segment, or travel on a transit vehicle along a selected route. They employ

a time-dependent branch-and-bound method for the least-cost path (shortest path); the

path's costs contain the time elements of walking, waiting, in vehicle, and transferring

penalty.

Lam et al. (1999) introduced a frequency-based, stochastic user-equilibrium model for

the transit assignment process. Their model considers capacity constraints on each transit

route; the route on which the capacity constraints are not fulfilled is excluded from the set

of attractive routes. For a nonovercrowded service, some passengers (who could already be

on board) may choose alternative services, which in practice is the case only with boarding

passengers who see a loaded vehicle. Bell and Schmöcker (2004) concluded that the model

by Lam et al. (1999) could not simulate situations with spare capacity, such as seat reservation or high

fare systems.

Cominetti and Correa (2001) and Bouzaïene-Ayari et al. (2001) advanced the con-

sideration of limited vehicle capacities with more realistic waiting-time functions at

stops. Their frequency-based models used the e-

effect of changes in frequencies (because of

overcrowded vehicles) rather than minimum-frequency flows here. Their equilibrium

assignment models take into account queuing processes at transit stops to reflect more

realistic changes in establishing the set of attractive routes. However, no specific algo

rithm was provided for computing their models.

In a schedule-based study, Nuzzolo et al. (2001) used a random utility function with

passenger information for a transit assignment procedure. They included departure-time

choice and stop choice in their stochastic model. Stop choice was defined as the probability

of selecting the boarding point from a set of stops located within a given access distance.

They developed the model for high-frequency service with a sequential choice process for

both within-day and day-to-day dynamics.

Kurauchi et al. (2003) introduced a frequency-based model in which a cost is assigned

to the probability of failing to board the transit vehicle. They considered the separation

between passengers on board a crowded vehicle and passengers at stops seeking to board.

The latter have a reduced priority to flow on the network. Kurauchi et al. used Markov

chains, in which the boarding probability is dependent on the leftover capacity of the transit

vehicle. They also discussed the use of the Markovian approach in stochastic and determin

istic user-equilibrium route choice processes.

Cepeda et al. (2006) provided a frequency-based formulation built on the model by

Cominetti and Correa (2001). The former introduced a transit assignment algorithm

intended for large-scale networks and tested it on real-size networks. Cepeda et al. (2006)

discussed the assumptions made, the difficulties in reaching a satisfactory solution, and

some possible future improvement directions.

Cats et al. (2011) present a dynamic transit analysis and evaluation tool that represents

timetables, operations strategies, real-time information (RTI), adaptive passenger choices,

and traffic dynamics at the network level. A mesoscopic transit and traffic simulation

model has been developed to represent traffic dynamics, dwell times, timetables and vehi-

cle scheduling, control strategies, and passenger path-choice processes. This model incor-

porates passengers' progress within the transit network as a sequence of discrete choice

decisions on boarding, walking, and alighting. At each decision point, consideration has

been given to the time-dependent expectations of passengers about travel-time compo-

nents. This model has been applied to the metro network area of Stockholm, Sweden,

under various operating conditions and information provision scenarios, as a proof of

concept. Analysis of the results indicate substantial path-choice shifts and potential time

savings associated with more comprehensive RTI

provision and transfer coordination

improvements. The authors suggest that transit operators should provide RTI at station

levels in order to enable more informed travel decisions, which will ultimately result in

time savings. Future work on this model has been proposed to allow the development of a

mixed frequency and schedule-based passenger-arrival patterns to represent more realistic

scenarios.

Szeto et al. (2011) propose a nonlinear complementary problem (NCP) formulation for

the risk-averse stochastic transit assignment problem in which in-vehicle travel time,

waiting time, capacity, and the effect of congestion are considered as stochastic variables

simultaneously and both their means and variances are incorporated into the formulation.

This model has indicated a number of advantages compared to the frequency-based transit

assignment model. A new congestion model is developed and captured in the proposed

NCP formulation to account for different effects of onboard passengers and passengers

waiting at stops. Surveys and numerical studies have been performed to validate the risk

aversion behavior of passengers and to illustrate the properties of the problem and effect

tiveness of the proposed solution method. The results indicate that underestimating the

congestion effect and ignoring the risk-aversion behavior can overestimate the patronage

of transit service. As such, the proposed model can be incorporated in the transit network

design model to determine the optimal service frequency and fare structure, thus allowing

the transit operators to maximize profitability without compromising the service quality.

Future work would include extension of this model to account for multiple passenger

classes, and also, consideration can be given to the perception errors on travel time and

waiting time.

Wu (2013) aims to find an efficient solution algorithm for the reliable a priori shortest

path (RASP) problem. The subject problem is analyzed from two aspects: the convolution

method and the stochastic dominance (SD) scheme. An alternative to the convolution

method is developed on the basis of the adaptive discretization approach, which was originally

proposed to solve the RASP problem in static networks. In comparison, the higher

order SD rule can be engaged to reduce the number of nondominated paths, thus improving

the overall computational efficiency. The experiments undertaken illustrate that the second

order SD rule provides a approximation of reasonable quality while reducing the central processing

unit time by at least 50%.

Trozzi et al. (2013) present a dynamic user equilibrium (DUE) for bus networks where

recurrent overcrowding results in queues at transit stops. The route choice model embedded

in the dynamic assignment explicitly considers common lines and strategies with alternative

tive routes. As such, the shortest hyperpath problem is extended to a dynamic scenario with

capacity constraints where the diversion probabilities depend on the time the stop is reached

and on the expected congestion level at that time. The bottleneck-queue model with time

varying exit capacity is extended to reproduce congestion for all the lines sharing a stop.

This model is applied to separate queues for each line in order to satisfy the first-in, first-out

principle within a very attractive set while allowing overtaking among passengers having

different attractive sets but queuing single. The resulting DUE can reproduce the results

from the static model in the uncongested scenario with time-independent (constant) travel

variables; however, it clearly illustrates the changes in the hyperpath, total travel cost, and

flow pattern when these become dynamic. Future work has been proposed to extend this

approach to other transport modes. Furthermore, this new model can be tested on large

transit networks to verify the value of heuristic solution methods and algorithm speedup

techniques.

Tirachini et al. (2013) examine the multiple dimensions of passenger crowding related

to public transit demand, supply, and operations, including effects on operating speed,

waiting time, travel-time reliability, passengers' well-being, valuation of waiting and

in-vehicle time savings, route and bus choice, and optimal levels of frequency, vehicle

size, and fare. The authors illustrate the impact of crowding on the estimated value of

in-vehicle time savings and demand prediction by estimating the externalities for rail

and bus services in Sydney. Furthermore, using multinomial logit (MNL) and error com

ponent models, the authors show that alternative assumptions concerning the threshold

load factor that triggers a crowding externality effect do have an influence on the value

of travel time savings (VTS) for low-occupancy levels (all passengers sitting); how

ever, for high-occupancy levels, alternative crowding models estimate similar VTS.

Importantly, if demand for a public transport service is estimated without explicit con

sideration of crowding as a source of disutility for passengers, demand will be overes

timated if the service is designed to have a number of standees beyond a threshold, as

analytically shown using an MNL choice model. The authors conclude by stating that

more research is needed to explore if these findings hold with more complex choice

models and in other contexts. For instance, the negative impacts of crowding on the

reliability of public transport services can be analyzed along with engineering and

design factors in vehicles and stations that

may help to reduce the effects of crowding

on increasing travel time and worsening the experience of traveling in public transport.

Ceder et al. (2013) present a study with two main objectives. The first objective is to

determine the effects of uncertainty, in out-of-vehicle times during transfers, on transit

users' willingness to use transfer routes. The second objective is to determine the influence

of out-of-vehicle facilities, offered by public-transport operators, on transit users' perception

of trip attributes related to transfers. A user preference survey has been conducted at

two major transit terminals in Auckland, New Zealand. The survey data were modeled

using cumulative prospect theory (CPT) and fuzzy logic. The results indicate that for

all trip attributes, except for comfort, transit users' exhibited risk-averse behavior; users

revealed greater preference for the transfer route with less uncertainty in the out-of-vehicle

times. For comfort, transit users were found to display risk-taking characteristics when

the waiting time for a seat was less than 5 min. Such findings suggest that increasing the

consistency in out-of-vehicle times will increase attractiveness of transfer routes, thus

enabling a more efficient and integrated network of transit routes to result in enlargement

of ridership. The authors assert that the policymakers and transit planners must focus

on methods of reducing uncertainty in out-of-vehicle times during transfers. A analysis of

transit users' perception of trip attributes, given their current station, revealed statistical

evidence of differences for two trip attributes, transfer waiting time and vehicle delay.

Such findings indicate that transit users who are accustomed to better out-of-vehicle facilities

have a lower tolerance for uncertainty in transfer waiting times and delay times. To

the authors' knowledge, this study provides for the first time in literature a comparison

between the two cognitive models. The comparison revealed that CPT and fuzzy logic

models are both capable of representing transit users' decision-making process. However,

while CPT provides an indication of transit users' preference for a transfer route, fuzzy

logic is capable of providing a closer approximation of the proportion of transit users

preferring a transfer route.

Hamdouch et al. (2014) present a schedule-based transit assignment model considering

supply uncertainties and assuming that users adopt strategies to travel from their

origins to destinations. Unlike previous transit assignment studies with fixed timetables,

the proposed model is capturing the stochastic nature of the transit schedules and in

vehicle travel times due to road conditions, incidents, or adverse weather. The solution

method developed does not rely on any path enumeration or column generation tech

nique; the transit network loading procedure relies on the usage of Bellman's recursion

principle (dynamic program). The model developed ensures that onboard passengers,

continuing to the next stop, have priority, and waiting passengers are loaded on a first

come-first-serve basis considering vehicle capacities. The authors use a mean-variance

approach considering the covariance of travel time between links in a space-time graph;

the equilibrium conditions are stated as a variational inequality involving a vector

valued function of effective strategy costs. The equilibrium solution is found by method

of successive averages in which the optimal strategy of each iteration is generated by

solving a dynamic program. The numerical examples presented in this study produce a

few interesting results, namely, that for an increase of travel-time variance, passengers

may decide to leave later and that vehicle capacity and early/late arrival penalties have

no effect on departure time choices.

A summary of the articles reviewed appears in Table 11.1, in which separate attention is

given to frequency- and schedule-based models.

Table 11.1 Summary of features concerning the articles reviewed

Advance in transit

assignment research Frequency-based features Schedule (timetable)based features Route choice features

Dial (1967),

Le Clercq (1972) Heuristic consideration of waiting time (and travel time) in shortest-path approach. Boarding the first transit vehicle to arrive.

Chriqui and Robillard

(1975) Probabilistic selection of a subset of routes to minimize the expected sum of (wait + travel) time. Boarding the first vehicle from a set of attractive routes.

Nguyen and Pallotino

(1988) Origin to destination is interpreted on an acyclic directed graph (called hyperpath). Boarding the first vehicle (from a set of routes) with a planned strategy of path movement.

Spiess and Florian

(1989) Choosing a set of routes to minimize expected sum of (access + wait + travel) time; using equilibrium model with linear programming. Boarding the first vehicle using a strategy of choosing only among attractive routes.

De Cea and

Fernández (1993) Incorporating a limited capacity for each route (of an attractive set) at stops in which waiting time depends on passenger flow; using asymmetric equilibrium model with Jacobi method. Boarding the first vehicle (from a set of routes), given that passenger flows do not exceed a route's capacity.

Wu et al. (1994) Hyperpaths are used for walk, wait, board, in-vehicle, transfer, and alight time elements; boarding time increases with flow, using equilibrium model with Jacobi method. Boarding the first vehicle using a strategy of choosing only among attractive routes.

Hickman and

Bernstein (1997) Model for high-frequency service using sequential choice approach. Deterministic utility path-choice model with passenger information.

Tong and Wong

(1999) Simulation model consisting of a network of routes with a given number of departure times; using

shortest path of weighted (walk + wait + travel) time and route change penalty. Random utility path-choice model for frequent service. (Continued)

Table 11.1 (Continued) Summary of features concerning the articles reviewed

Advance in transit

assignment research Frequency-based features Schedule (timetable)-based features Route choice features

Lam et al. (1999) Stochastic user equilibrium with explicit route capacity constraints. Boarding the first vehicle; for overcrowded service, some passengers may choose alternative services.

Cominetti and

Correa (2001),

Bouzaïene-Ayari

et al. (2001) Congestion functions at stops obtained from queuing theory to increase waiting time and affect passenger-flow share. Boarding the first vehicle (from a set of routes) with available capacity.

Nuzzolo et al. (2001) Introducing a set of departure-time choices and a set of stop choices for a given access distance. Random utility path-choice model for frequent service with possible passenger information.

Kurauchi et al. (2003) Introducing failure-to-board probability in which the demand exceeding capacity remains on the platform; using Markov chains and user equilibrium approach. Boarding the first vehicle of a chosen single route in which route choice depends on risk of failing to board.

Cepeda et al. (2006) Congestion functions at stops with formulation for large-scale networks. Boarding the first vehicle (from a set of routes) with available capacity.

Cats et al. (2011) Represents timetables, operation strategies, real-time information, and traffic dynamics at the network level. Transit path choices are modeled as a sequence of boarding, walking, and alighting decisions that passengers undertake when carrying out their journey.

Szeto et al. (2011) A reliability-based user-equilibrium condition is defined based on the proposed generalized concept of travel-time budget referred to as effective travel cost in the formulation. A nonlinear complementarity problem formulation for the risk-averse stochastic transit assignment problem in which in-vehicle travel time, waiting time, capacity, and the effect of congestion are considered simultaneously.

Mu (2013) An alternative convolution method is developed on the basis of the adaptive discretization approach to solve the reliable a priori shortest path problem.
(Continued)

EXERCISES

11.1 Given the following four-route schematic transit network adapted from the EMMÉ/2 user's manual (reference: INRO Consultants Inc. EMMÉ/2 User's Manual, Release 9.6, May 2005), this network contains the following headways and average travel times. Route 1 (12 min headway) Route 2 (12 min headway) Route 3 (30 min headway) Route 4 (6 min headway) Node A X Y B 25 min 7 min 6 min 4 min 4 min 4 min

Table 11.1 (Continued) Summary of features concerning the articles reviewed

Advance in transit

assignment research Frequency-based features Schedule (timetable)-based features Route choice features

Trozzi et al. (2013) The shortest hyperpath problem is extended to a dynamic scenario with capacity constraints where the diversion probabilities depend on the time the stop is reached and on the expected congestion level.

Tirachini et al. (2013) Multinomial logit and error component models are used to test whether the threshold load factor that triggers a crowding externality effect does have an influence on the value of travel time.

Ceder et al. (2013) A user preference survey has been conducted at two major public transport terminals in Auckland, New Zealand. The survey data were modeled using cumulative prospect theory and fuzzy logic.

Hamdouch et al.

(2014) Both travel strategies and supply uncertainties are considered to capture the stochastic nature of the transit schedules and in-vehicle travel times; dynamic program is used. Onboard passengers, continuing to the next stop, have priority, and waiting passengers are loaded on a first-come-first-serve basis considering vehicle capacities.

Assume that walking times and transfer times are zero:

(a) Describe the possible travel options from A to B; include the possibility that a passenger may choose among different combinations of routes.

(b) Calculate expected travel times for each option.

(c) Which set of options gives the minimum expected travel time?

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12 Chapter 12: Service Design: Elements, Operational Parking, and Stop Location

After reviewing and investigating transit-scheduling components and passenger demand and

assignment, we open this chapter with a description of transit service and network and route

design components, subjects that will also be covered in the three following chapters. Service

design and stop placement issues, in the subjects of this chapter, facilitate an understanding of

the importance of transit planning in enhancing existing or new transit services. The planning

table is the mechanism with which to start changing the prevailing opinion that transit service

and problems are tied to each other as in the adage: urban areas with transit facilities are

exciting places where something is almost always happening, mostly unsolved. The chapter consists of four principle parts, following an introductory section. Section 12.2

outlines the predominant design elements, service strategies, and possible actions for both

existing and new transit services. In addition, this section provides standards and measures for

system and performance. Section 12.3 pinpoints and solves a specific design problem in which

transit vehicles need to park at route departure/controlled/holding stops located on street;

because of the lack of parking spaces, they block the traffic lane adjacent to the parking bay/lane.

The solution approach employs a surplus function model representing the surplus of parking

vehicles required at a particular departure point in a multiterminal transit system. Section 12.4

furnishes a framework and theoretical analysis for finding the smallest number of public transit

stops and their locations so that no passenger is further away from a stop than a preselected

distance. An optimal algorithm with an example is provided. Section 12.5 provides formulations

and framework of analyzing the location of public transit stops considering uneven topography

environment. This is to enable easy and convenience access and egress to and from the stops. Practitioners are encouraged to visit thoroughly all the sections besides Sections 12.3 through

12.5, except the introductions of these sections. They should, though, inspect the practical

problems portrayed in these three sections and the examples with their solution. Finally, among the important transit service and connectivity attributes are waiting time

and passenger information. When designing the service, it is worthwhile to put yourself into

the shoes of waiting passengers, who wish the wait had a fast-forward button. This may lead

the planner to recommend hanging the following sign at all stops: "Waiting times perceived are

larger than they are." In addition, useless information should be avoided, like that given to a pas

senger who asked where to get off the bus: "See where I get off, and get off two stops before."

12.1 INTRODUCTION

Profound service-design approaches, including well-coordinated service elements, comprise

the foundation of any successful public transit service. In fact, an adequate, well-designed

service is a present the transit agency gives to itself. The importance of service-design e

le

ments may be better comprehended after reading the following examples of two randomly

s elected reviews.

12.1.1 Transit service frustration

Examples of passengers' reviews are from the Washington Metropolitan Area Transit

Authority (WMATA, 2 014), w ebsite: <http://www.yelp.com/biz/washington-metropolitan-area>

transit authority washington-3 .

January 15, 2014 Being a n ative W ashingtonian a nd h aving e xperienced t he M etro s ince i ts o pening i n

1 976, i t p ains m e t o b e c ritical ... h owever, w ith t housands o f t rips o ver t he y ears, I f eel

I c an g ive a n h onest, u nvornished r eview. I 've l ived i n o r t raveled t o m any c ities w ith fa r

more dependable mass transit systems. Frequent s ervice d elays a nd m aintenance i ssues h ave b ecome t he n orm. H orrible s ound s ystem i n t he c ars m aking s tation a nnouncements e ither i n audible o r g arbled. S eating

a rrangement e ncourages " hogging" o f t wo s eats, c ompelling m any t o s tand, m ostly c rowding a round t he d oors a s t he a isles a re t oo n arrow f or t he o versized p assengers. T he t iered fare system causes confusion, particularly for tourists, newbies, and infrequent users. D on't g et m e s tarted o n p eople s tanding o n t he l eft s ide o f t he e scalators. T he o nly g ood t hing a bout b roken e scalators i s t hat i t f orces p eople t o a ctually m ove u p/down t he s teps. I g ive i t 2 -stars, b ut t here a re d ays w hen I c ompletely l ose fa ith i n M etro. N o s ystem i s p erfect, b ut t he M etro n eeds s ome s erious h elp t o g et i t u p t o a r easonable s tandard of reliability ...!

December 21, 2013 Welcome to the land of frustration! Where the escalators are always, ALWAYS broken, the train is ALWAYS delayed, and you never know where the hell you are because the driver doesn't know how to speak into a mike. Next stop ... j feiwroweir. What did you just say? L'Enfant? Perhaps I am spoiled. I mean, I did sit in my nice clean car for hours ... in traffic, but at least it was clean, rarely ever broke down and was on time! I imagine that? Even with the Metro D.C. areas horrible traffic ... why the hell can't metro be the same? Why is it so dirty? Why is it always late? Why is everything always broken? Where does all the money go? It's not like it's cheap ... riding the Metro is expensive. It cost me \$3.00 each way and to get my car out of the parking garage it's another \$4.75. I imagine the large crowds of people moving through the station like wild animals are probably spending about the same if not more. And what's with the weird smell? Why do the breaks on the train smell like a 10 day old shit hat has sat out in the sun? Why? I think metro is in desperate need of new management, repairs and updates. Metrorail needs to get it together! Let's start over.

12.2 SERVICE-DESIGN ELEMENTS

A discussion of service and evaluation standards and guidelines appears in Section 1.3. These

standards and guidelines establish service needs as seen by the local authority/government and

the transit agency, but they do not cover the entire spectrum of required service-design elements.

This section outlines the predominant design elements for both existing and new transit services.

The main service-design elements are as follows:

- Potential markets
- Network size and coverage

- Network structure, followed by route structure
- Route coordination (intra- and interagency)
- Route classification
- Span of service
- Service frequency, followed by public timetables
- Schedule coordination (intra- and interagency)
- Vehicle scheduling, followed by vehicle types and fleet size
- Crew scheduling, followed by rostering
- Fare policy
- Passenger amenities and information systems
- Data collection systems
- Determination of measures of performance
- Setting service and evaluation standards
- Ridership, cost, and revenue estimation

12.2.1 Description of service-design elements

The elements in the foregoing list will now be described briefly.

According to TranSystems et al. (2006), each potential market element requires an analysis

of demographics and travel patterns as well as market research. The analysis is used to identify

areas with the potential to support transit services and to locate current and projected travel

markets. The market research is intended to identify market segments, along with passengers'

the inclination to use/increase the use of the service, given specific improvements.

The network size and coverage element applies to the set of all routes—that is, the entire

system—and serves to × the recommended spacing (distance) between routes for varying

residential area densities. TranSystems et al. (2006) provide a few examples of a coverage

measure. For instance, it suggests the provision of transit service within walking distance

(defined as 400 m) of all residents living in areas with population densities greater than

2 000 people per square kilometer.

The network structure followed by route structure element can take one or a combi

nation of the four common forms: multimodal, radial, grid and time transfer, and pulse.

A multimodal network of routes coordinates short and long trips through different transit

modes, for example, short trips by bus and that feed long trips by rail. The network of radial

routes aims at providing a considerable amount of service to central points, for example, the

central business district (CBD). A grid structure network of routes, on the other hand, allows

easy access to the transit system but, on the other hand, requires many transfers; thus, the

transfers in this network are timed and preferably synchronized online. The last network

structure is based on a pulse system, in which routes are initiated at the same (central) point,

which becomes a transfer point; usually, this suits small urban areas.

The route coordination (intra- and interagency) element applies at the intra-agency front

to a route-design system with coordinated meeting points in terms of convenient passenger

transfer facilities. At the interagency front, coordination is manifested in terms of operating

policies and transit support promotion programs.

The route classification element exhibits various route-type needs for different geo-

graphical areas. Route classification schemes usually comprise one or a combination of the

following: line haul, local, express, feeder/distributor, branching, radial line haul, zonal,

commuter, circulator, crosstown, short turn, and limited stop.

The span of service element indicates the length of time that a service should be provided,

by time of day and day of week. For example, a span of service guideline can suggest that

the first trip should arrive no later than, and the last trip should depart no earlier than, the

(specified) times shown for weekdays, Saturdays, and Sundays.

The service frequency followed by public timetable element is described and discussed

explicitly in Chapters 3 and 4. Essentially, this element sets minimum service-frequency and

maximum vehicle-occupancy thresholds to guarantee a basic level of service for different

geographical areas. For the timetable, this design element sets the type of vehicle headway

to be utilized, for example, even headway, clock headway, even-load headway, and by time

of day and day of week.

The schedule coordination (intra- and interagency) element applies to the intra-agency

front to a design of timetables that will maximize simultaneous arrivals at transfer points

(see Chapter 5). At the interagency front, schedule coordination is expressed in terms of

jointly designed timetables by different agencies and the use of different transit modes to

allow for easy transfers between transit routes, with minimum waiting time.

The vehicle scheduling followed by vehicle type and fleet size element is described and

discussed in Chapters 6 through 8. Basically, this element ensures design efficiency in terms

of the fleet size required, balancing deadheading (DH) trips and shifts in departure times

and minimizing the cost involved in purchasing vehicles.

The crew scheduling followed by rostering element is described and discussed in Chapter 9.

Fundamentally, this element focuses on minimizing the crew wages involved and provides a

satisfactory working schedule from the crew's perspective.

The fare policy element aims at improving payment convenience and integration between

different transit services. For instance, a fare policy can be established with free t

transfers,

discount options, and the elimination of fare zones and special surcharges.

The passenger amenities and information systems element aims at improving passenger

facilities and vehicle amenities and improving passenger information for pretrip planning,

en route riding, and waiting at terminals and stops. This element is also concerned with

improving passenger safety and security.

The data collection system element is described and discussed in Chapters 2, 3, and 18.

This vital element constructs the foundation for effective and efficient transit operations

planning. The key components of this element are suitability, accuracy, and an adequate

amount of data.

The determination of measures of performance element aims at quantifying tools for

measuring the quality of service. The measures determined show whether the service is

appropriate, convenient, and reliable, especially from the passenger's perspective. Two basic

types of measures of performance are described later in this section: system performance

and connectivity performance.

The setting service and evaluation standards element is described and discussed in

Chapter 1. On one hand, standards maintain and improve the service; on the other hand,

they present a source of scalp pressure on the transit agencies. Evaluation standards aim at

improving the efficiency, effectiveness, and productivity of the transit service.

The last design element on the list is ridership, cost, and revenue estimation. This element

provides the tools for forecasting the three linked components of passenger demand, cost

of service, and revenue. Some of these tools are described in Chapter 10, which emphasizes

ridership prediction.

12.2.2 Service strategies and possible actions

A good summary of transit service strategies appears in the U.S. Transit Cooperative

Research Program (TCRP) H-32 report (TranSystems et al., 2006). The following lists

provide new design and design-adjustment examples by purpose and possible implementa

tion actions (in parenthesis):

1. New forms of service • Improve travel speed (introduction of express/zonal/rail/bus rapid transit service) • Attract new passengers (circulator shuttles, dial-a-ride service)
2. Area coverage service • Increased route coverage (service expansion; integrated circulator and line-haul services) • Increased span of service (late night, weekends, holidays)
3. New and adjusted routing • New routing (linking routes, splitting routes, feeder based, crosstown based) • Routing adjustment (route extension/shortening/realignment, express/zonal/local) • Coordination (intra- and interagency transfer centers/points)
4. New and adjusted scheduling • Introduce interlining (area based, trip based, vehicle type based) • Introduce/improve coordination (intra- and interagency synchronized/timed transfers) • Change frequency (increase/decrease, even headways, clock headways) • Change departure times (shifts within tolerances, even-load departure times) • Modify time elements (average travel

time, layover time, recovery time) • Improve reliability (see Chapter 18 for lists of possible actions)

5. Improved amenities • Introduce/improve passenger facilities (stops/station, transfer points, transit centers, park-and-ride amenities) • Introduce/improve vehicles (amenities, new vehicle type) • Increase safety (introduce/improve safety features on vehicles and at stops, increase awareness and preparedness) • Increase security (introduce/improve security agents and features on vehicles and at stops, increase awareness and preparedness)

12.2.3 Standards and measures of system and performance

This section presents service standards and measures of performance. The service stan

dards, accompanied by the required data, follow the list given in Table 12.3; these standards

are shown in Table 12.1. The measures are divided into measures of system performance

(MOSP) and measures of connectivity performance (MOCP).

The MOSP and MOCP, which should be selected at the design stage, quantify how well

transit routes are used. Table 12.2 lists 10 MOSP and 4 MOCP, along with their notation,

interpretation, and the data required. Measures of reliability of service are not included in

Table 12.2; they will be discussed in Chapter 18.

The measure of emissions E_{rt} in Table 12.2 can be extracted from real data. For instance,

Table 12.3 shows calculated data from the U.S. Environmental Protection Agency's (EPA)

MOBILE5 and PART5 software-based models. MOBILE5 is a vehicle-emission modeling

software, and PART5 is a software model for estimating particulate emissions from highway

vehicles. More information can be found in the EPA's national emission inventories air pollut

ant trend website:

www.epa.gov/ttn/chief/eiinformation.html. A more updated model by EPA

is MOBILE6, which can be found at www.epa.gov/otaq/m6.htm. Table 12.3 (using MOBILE5

and PART5 software) provides data by vehicle type, per single vehicle, and on a per-
er-passenger

basis; the last is based on an occupancy of 3.5 passengers/bus and 1.50 passengers/train.

12.3 SCHEDULING-BASED SOLUTION FOR OPERATIONAL PARKING CONFLICTS

One of the operational problems for urban transit, especially with buses, occurs in a situation

in which vehicles need to park at route departure/controlled/holding stops located on a street;

however, because of a lack of parking spaces, the buses block the traffic lane adjacent to the

parking bay/lane. Such operational scenarios are commonly observed at school dismissal times.

Although a special arrangement can be made for school/factory dismissal times—for example,

using side streets—the blocking of a traffic lane in other situations can result in severe traffic

congestion. One way to solve the problem is to construct more parking spaces at these

conflict points; however, this is a costly and time-consuming solution. This section presents a

scheduled-based design solution to this problem in which lane-blocked situations are caused at

route departure points, that is, eliminating or reducing the impact of lane-blocked situations

through the use of shifting departure times and/or inserting DH trips in timetables. This section

tion is based on Ceder and Israel (2009).

12.3.1 Surplus function model

The model constructed to solve the operational parking conflicts is basically a mirror image

of the deficit function (DF) model described extensively in Chapters 6 through 8. This model

is called a surplus function (SF) model because it represents the surplus of parking vehicles

required at a particular departure point in a multiterminal transit system. SF is a positive

step function that basically increases by one at the time of each trip arrival and decreases by

one at the time of each trip departure; its starting value depends on the starting and maxi

mum DF values. This definition shows a strong mutual dependency between DF and SF.

To construct the SF model, the only information needed is a timetable of required trips

and the number of available parking spaces at each departure point. The main advantage of

SF, as with DF, is its visual nature.

Let K be the set of all departure points, $p(k, t, S)$ denote the SF for departure point k for

a ll $k \in K$ a t t ime t f or s chedule S , a nd P_S
 k be the number of available parking spaces at k .

Table 12.1 Service standards and data required

Standard item Data required

Route length Average running time; distance

Stop spacing Population density; type of service (e.g.,
 local, express); number of passengers using a given stop

Route directness a Average running time; distance;
 passenger origin-destination count

Short turn a Average passenger counts by stop

Route coverage Population data; land-use data; public view

Route overlapping Scale maps of the network of routes

Route structure a Average passenger origin-destination
 counts; population data; public opinion

Route connectivity Maps of network of routes and feasible
 transfer points

Span of service Timetables by route and zone of operation

Load (crowding) level Average passenger counts by stop or
 at max load points

Standees a Passenger counts by stop

Headway upper limit a Average passenger counts; permits
 between operator and authority

Headway lower limit Average passenger counts; number of
 vehicles available

Transfers Average passenger counts; transfer counts;
 waiting time

Passenger shelters a Average passenger boardings by stop;
 number of elderly and handicapped

Schedule adherence a Onboard counts of departure and
 arrival times (manually or automatic device)

Timed transfer Passenger origin-destination counts by time

of day; trip timetables

Missed trips a Dispatcher log and maintenance records
(agency data)

Passenger safety a Accident reports, combined with average
passenger counts and km driven

Public complaints Ordinary mail, e-mails, fax, telephone,
visits (by category of complaint)

a Standards commonly in use.

The maximum value of $p(k, t, S)$ over the schedule horizon
 $[T_1, T_2]$ is designated $P(k, S)$.

For simplicity, S will be deleted when it becomes clear which underlying schedule is being

considered. The Definitions defined in Sections 6.5.1, 6.5.2, 7.3.1, and 7.3.3 will also be

utilized for the SF analyses.

SF, unlike DF, cannot be negative, because each departure must be preceded by an arrival,

thereby creating the need for a parking space between the arrival and the departure epochs.

DF is defined as the total number of departures minus the total number of trip arrivals at

terminal k , up to and including time t . SF is defined as the total number of arrivals minus

the total number of trip departures at terminal k , up to and including time t , in which the

Table 12.2 Notations and interpretation of transit measures of performance

Type of measure	Measure	Designated notation	Interpretation and data required
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Measure of			
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system

performance Number of passengers P_{rt} Number of passengers by route r and time of day t Revenue R_{rt} Agency revenue by route r and time of day t (\$) Vehicle trips V_{rt} Number of vehicle trips, timetable based, by route r and time of day t Vehicle-kilometers of travel VKT_{rt} Vehicle-kilometers of travel, based on V_{rt} , by route r and time of day t (veh-km) Passenger travel times PTT_{rt} Passenger travel times, based on trip times and load profiles, by route r and time of day t (h) Emissions E_{rt} Vector of average emission of four pollutants (CO, NO_x, VOC, PM₁₀), based on VKT_{rt} , by route r and time of day t (kg) Vehicle-hours of travel VHT_{rt} Vehicle-hours of travel, based on V_{rt} , by route r and time of day t (veh-hr) Passengers per veh-h PVH_{rt} Ratio of passengers per vehicle-hour, based on P_{rt} and VHT_{rt} , by route r and time of day t (pass/veh-h) Passengers per veh-km PVK_{rt} Ratio of passengers per vehicle-kilometer, based on P_{rt} and VKT_{rt} , by route r and time of day t (pass/veh-km) Revenue per passenger RP_{rt} Agency revenue per passenger, based on R_{rt} and P_{rt} , by route r and time of day t (\$)

Measure of

connectivity

performance Passenger waiting times PWT_{rt} Passenger waiting times, based on P_{rt} and service frequency, by route r and time of day t (h) Passenger transfers PTR_{rt} Number of passenger transfers, within and between modes of travel, by route r and time of day t Connectivityproduction cost CPC_{rt} Cost of passenger waiting time and transfer penalty plus the cost of vehicle-hour, based on VHT_{rt} , PWT_{rt} , and PTR_{rt} , by route r and time of day t (\$)

Table 12.3 Average emission of pollutant per vehicle (in parentheses, per passenger) in gram/mile CO NO_x VOC PM₁₀

Diesel bus 23.2 (0.66) 22.1 (0.63) 4.2 (0.12) 0.63 (0.02)

Automobile 23.0 (19.17) 3.9 (3.25) 3.7 (3.08) 0.09 (0.075)

Rail 0.03 (0.0002) 0.47 (0.003) 0.02 (0.0001) 0.009 (0.0001)

number of arrivals at T_1 is $D(k)$ minus the number of departures at T_1 . The next section

further describes these relationships and definitions in a more formal way.

Figure 12.1 depicts three operational parking situations with $P_S k = 2$, at t_1 , t_2 , and t_3 . Figure

12.1a shows a single parked bus; hence, $p(k, t_1) = 1$; because of the excess number of buses,

$p(k, t) - P_S k$ is zero (not -1). Figure 12.1b and c exhibit lane-blocked situations; in Figure

12.1c, $p(k, t_1) = P(k) = 4$, representing the maximum number of parking vehicles over $[T_1, T_2]$.

12.3.2 Minimum parking spaces required

Following the definition of SF, we can establish a few formulae and rules that will constitute

the basis of an algorithmic-based solution. The first of these is Theorem 12.1.

Theorem 12.1:

The minimum number of parking spaces required at k , $N_p(k)$ is equal to the maximum SF at k . $N_p(k) = \max_{t \in [T_1, T_2]} p(k, t) - P_S k$ (12.1)
 Bus 1 $p(k, t_1) = 1$ (Nonnegativity difference) $p(k, t_1) - P_S k = 0$ $P_S k = 2$ (a) Bus 1 Bus 2 Bus 3 $p(k, t_2) = 3$ $p(k, t_2) - P_S k = 1$ (b) Bus 2 Bus 1 Bus 3 Bus 4 $p(k, t_3) = 4$ $p(k, t_3) - P(k) - P_S k = 2$ (c)

Figure 12.1 Three situations of operational parking at $k =$ route departure stop (a) without lane-blocked, (b) lane-blocked by one bus, and (c) lane-blocked by two buses (the value of the surplus function and its difference from the two available spaces are shown).

Proof: Using the notation of Section 6.5.2, F_k is the number of vehicles present at k at the

start of the schedule horizon T_1 ; $s(k, t)$ and $e(k, t)$ yield the cumulative number of trips start

ing and ending at k from T_1 up to and including time t . The number of vehicles parking at

for all time $t \geq T_1$ is $F_k - s(k, t) + e(k, t)$, which by definition is $p(k, t)$. This expression must

be both nonnegative and less than or equal to $N_p(k)$; that is, $0 \leq p(k, t) \leq N_p(k)$, $T_1 \leq t \leq$

T_2 . The minimum number of parking spaces required at k , then, is equal to the maximum

$SFP(k)$.

Consequently, the objective of the operational parking conflict is to eliminate or minimize

cases in which $P(k) > P_{SK}$, in other words, to make sure that the minimum number of parking

spaces required at k is less than or equal to the number of spaces available. The following

Lemma 12.1 and Theorem 12.2 facilitate the interdependency of SF and DF through the

maximum $SFP(k)$.

Lemma 12.1:

Compliance with $p(k, t) \geq 0$ for all $k \in K$ is attained by shifting up the minimum image of $d(k, t)$

by $+D(k)$ to obtain $p(k, t)$.

Proof: The basic concept of the SF is that it is the minimum image of the DF . The minimum image

of $d(k, t)$ is simply $-d(k, t)$ for all $k \in K$; thus, the minimum value of the minimum image is

$-D(k)$. The compliance with $p(k, t) \geq 0$ is attained, therefore, by shifting up $-d(k, t)$ by $+D(k)$.

Theorem 12.2: $P(k) \geq D(\min_{t \in [T_1, T_2]} d(k, t)) \quad \forall k \in K$, (12.2)

Proof: Based on Lemma 12.1, $p(k, t) = -d(k, t) + D(k)$

$\} + D(k)$ for all $k \in K$; thus, the following holds:

$$\max p(k, t) = P(k) = D(k) - \min d(k, t) \text{ for all } k \in K.$$

Figure 12.2 illustrates an in-trip xed s schedule with two terminals/departure points. Both

$d(k)$ and $p(k)$ for $k = a, b$ are shown, with an emphasis on $PS_a = 2$ and $PS_b = 1$ available park

ing spaces. The value of $p(a, T_1) = 2$, $T_1 = 5:00$ is determined by shifting the mirror image of

$d(a, t)$ three units up because of $D(a) = 3$; thus, because the mirror image of the DF , $-d(a, t)$,

starts with $-d(a, T_1) = -1$, its shift will result in $p(a, T_1) = -1 + 3 = 2$. The same applies to

$$p(b, T_1) = 3.$$

Note that in order to have a sufficient number of parking spaces, $P(k)$ must be less than

or equal to PS_k . If it is not, attempts could be made to attain this constraint. The following

section utilizes the scheduled-based tools to reduce $P(k)$ for all $k \in K$, in terms of shifting

departure times and inserting DH trips.

12.3.3 Scheduled-based reduction of $P(k)$ procedure

The DF theory explicated in Chapters 6 and 7 is used for a heuristic procedure to reduce

required parking spaces for situations in which $P(k) > PS_k$. Ostensibly, this procedure involves Designated area (vehicle-hours) stretched over the time during which there is a lack of parking spaces Time a a a a b b b b a b a b a b a b 2 3 4 5 6 7 8 9 1 $d(a, t)$ $d(b, t)$ $p(a, t)$ $p(b, t)$ $PS_b = \text{Fixed schedule}$ $D(b) = 3$ $D(a) = 3$ 3 2 1 0 -1 3 2 1 0 -1 4 3 $PS_a = 2$ 1 0 4 3 2 1 0 Time 5:00 6:00 7:00 8:00 9:00 5:00 6:00 7:00 8:00 9:00 $P(b) = 3$ $P(a) = 4$

Figure 12.2 Construction of two surplus functions, dependent on their deficit functions (gray background), for a nine-trip example.

shifting trip-departure times and DH trip insertion so as to reduce $P(k)$, if necessary, with

out increasing $D(k)$ for all relevant $k \in K$.

The mirror-image configuration of the DF, including shifting up this configuration by

$D(k)$ to attain $p(k, t)$, has properties similar to the original DF. As a result, the two following

rules are applied for the construction of a heuristic procedure:

1. Shifting departure times (left, right, or in both directions), within their tolerances, for reducing $P(k)$ cannot increase $D(k)$, mainly because the maximum intervals of $P(k)$ and $D(k)$ do not overlap; this rule applies to both a single shift and a chain of shifts.

2. Inserting DH trip to reduce $P(k)$ is feasible only if the following two conditions are fulfilled: (a) the DH trip departs at or before the start of the first maximum interval of $P(k)$, and, at the same time, this DH departure must come at or after the last maximum interval of $D(k)$; (b) the DH trip must arrive, from k to k' , at or after the last maximum interval of k' , this rule applying both to a single DH trip and to a chain of DH trips.

Note that these two rules can also be formally stated and proved.

The schedule-based reduction of the $P(k)$ procedure appears in the flow diagram form

in Figure 12.3. This diagram contains similar features to the DF procedures described in

Figures 6.9a, b, 7.11, and 7.12. The procedure described is designed for an interactive per

son-computer system. The selection of a departure point/terminal can be made by the

scheduler by inspecting the SFs on a graphical display. The search for a reduction of $P(k)$ at

u (see Figure 12.3) can be performed manually by the scheduler or by requesting procedures

that are based on Theorems 12.1 and 12.2 and the two rules earlier. If a unit reduction

shifting chain of $P(k)$ is found, all the SF information is updated for this new schedule and a

new iteration initiated. In the following step, the procedure seeks to reduce $P(k)$ by DH trips

incorporated with shifting trip-departure times.

The modified URDHC (mixed with shifting trip-departure times [SDT]) subroutine,

described in Section 7.5.3, is then applied, but with changes to account for rule (2) previously

ously, that is, inserting a DH trip from (or before) the start of the maximum interval of $P(k)$,

compared to bringing a DH trip to the start (or before) of the maximum $D(k)$ interval in the

DF procedures. In this DH procedure, the feasibility requirement for DH trip insertion is

based on Equation 7.3. Another point to be mentioned is that DH trips that are added to the

schedule should include shifting tolerances for the next iterations of the SDT and modified

URDHC procedures. Finally, if a URDHC has been found, the DH chain cost involved is

compared with the saving of the cost of a single parking space. If the DH cost is higher than

the saving cost, the URDHC is cancelled. Otherwise, the set of DH trips and required shifts

(if any) { DH+SH} is added to the previous schedule; all SF information, including shifting,

is updated for a new schedule. The modified URDHC subroutine continues, with updated

S, until all terminals are examined.

A complete example of the required parking-space reduction procedure is illustrated in

Figure 12.4. This example consists of a nine-trip schedule with three terminals, a single DH

travel time of 75 min, and shifting tolerances of 15 min (both directions). The three DFs

of the example appear with a gray background after going through the minimum meet-size

reduction procedures; the optimum number of vehicles required is four. The SFs are illus

trated below the DFs, and the only complete minor-image configuration (multiplied by -1)

is that of departure point c, where the shifting up is not performed, because $D(c) = \emptyset$. The

procedure to reduce $P(k)$ first checks which departure points are characterized by $P(k) > P_S$
 k ;

in fact, this is the case for all $k = a, b, c$.

Departure point/terminal a is then selected in Figure 12.4 to search for possible shift

ing departure-time opportunities; indeed, for this search, two shifts of trips 6 and 8 are

found feasible to reduce $P(a)$ from 2 to 1, thus making it

equal to PS_a . These 15 min shifts

are shown in the upper part of Figure 1

2.4. The second step is the attempt to reduce $P(b)$

by shifting trips 1 and 5 (15 min each) in opposite directions, but this cannot be realized.

The reason is that by shifting trip 1 to the right, the value of $p(b, T_1)$ is changed from 1 to

2, which is the value of $D(b)$; the value reduces to 1 only at $p(b, 5:15)$. Thus, the two shifts

cannot reduce $P(b) = 2$. The next step examines the reduction of $P(c)$ by shifts, but here, too,

it cannot be performed. The procedure in Figure 1 2.3 proceeds to look for a DH trip inser

tion while fulfilling the two conditions of rule (2). Although it is impossible to insert a DH No Yes Yes No No Yes No No STOP Yes No Yes
Yes Initial input: Set of departure points K ; trip schedule S_0 ; set of tolerances for trip-departure times; DH time matrix; set of operational parking spaces PS_k .
Set $S = S_0$, $Q = \emptyset$ Computation of deficit and surplus functions information: $\{d(k, t, S)\}$, $D(S)$, $\{p(k, t, S)\}$, $P(S)$, PS_k Select next departure point u , with $P(u, S) > PS_u$, to examine for a possible reduction of $P(u, S)$ by shifting trip-departure times within their tolerances $P(u, S) > PS_u$? All terminals examined, $Q = K$? Update terminals examined. Set $Q = Q \cup u$ Reduce a parking space at u ; incorporate the new trip-departure times into S to form the schedule S_u ; set $S = S_u$ Select next terminal, q , to examine for a possible reduction of $P(q, S)$, using DH trip insertion incorporated with shifting trip-departure times $P(q, S) > PS_q$? Update terminals examined. Set $Q = Q \cup q$ Reduce a parking space at q ; add set of DH trips and shifts $\{DH + SH\}$ to the schedule: $S = S + \{DH + SH\}$. Set $Q = \emptyset$ Set $Q = \emptyset$ DH cost > parking space saving cost? Unit reduction, by shifts, of $P(u, t, S)$ found? All terminals examined, $Q = K$? Unit reduction of DH chain found at terminal q ?

Figure 12.3 Flow diagram of the required parking-space reduction procedure, involving the shifting of departure times and deadheading insertions (modified URDHC subroutine of Chapter 7).

$d(a, t)$ $d(b, t)$ $p(a, t)$ DH
 $D(b) = 2$ $D(a) = 2$ Time 2 1 0 2 1 0 1 0 -1 -2 2 1 0 2 1 0
 Time 5:00 6:00 7:00 8:00 9:00 10:00 5:00 6:00 7:00 8:00
 9:00 10:00 $d(c, t)$ $P(a) = 2$ 1 $P(c) = 2$ 1 $P(b) = 2$ DH
 $D(c) = 0$ PS $k=1$, $k=a, b, c$ $\Delta i(+) = \Delta i(-) = 15$ min Given:
 All DH travel times=75 min $p(b, t)$ $p(c, t)$ PS $a =$
 PS $b =$ PS $c =$ a b b c b a 5 1 Fixed schedule 7 9
 b b 4 a c b c 2 8 3 a c b a a c 6 Following DH trip
 insertion or shifting departure time

Figure 12.4 Nine-trip example of maximum reduction of the difference $P(k) - PS_k$, $k = a, b, c$ of the surplus functions, utilizing shifting and deadheading insertion (the impact of changes on the deficit functions [gray background] is also shown).

trip from departure points a and b, it is found that it is feasible to do so from c to b; this

results in reducing $P(c)$ from 2 to 1, making it equal to PS_c . The example ends up with only

one excess-parking situation in b, between 5:45 and 6:30, but with the elimination of this

undesirable circumstance in a and c.

12.4 OPTIMUM STOP LOCATION: NETWORK-BASED THEORETICAL APPROACH

Network coverage is among the service-design elements presented in Section 12.2, which

contains the criterion for providing transit service within a defined walking distance of all

residents living in a certain area. The present section furnishes a framework and theoretical

analysis for finding the smallest number and the locations of public transit stops so that no

passenger is further away from a stop than a preselected distance. An optimum algorithm

is described and discussed for a general road network in which the nodes are community

locations and the stops are to be located along the arcs (streets) or on nodes. The algorithm

follows the method described in Ceder et al. (1983).

In order to ensure a high level of service for public transit users, walking distances to stops

should be as short as possible. Farewell and Marx (1996) state that people consider walk

time to be much less convenient than in-vehicle travel time and proposed a maximum walk

ing distance of 400 m to a transit stop. Ceder et al. (1983) exhibit convenient ratios between

in-vehicle travel time and walking time: 2.0 in the Netherlands, 3.5 in Chicago, and 6.2 in

San Francisco. For instance, people in San Francisco consider 1 min walking to be 6.2 times

less convenient than 1 min of in-vehicle travel time; this high value may be attributed to the

difficult terrain in that city. Therefore, by reducing the length of the walking distance to

stops in that city, public transit agencies will make their service more attractive.

12.4.1 Framework of analysis

The problem under consideration is to find the smallest number and the location of transit

stops in a general network, so that no passenger is further away than a preselected d

istance,

assuming that demand is generated at specific locations/nodes along the arcs/streets/road

ways/segments. The stops could be located either on the nodes or on any point along each

arc. That is, the points lying on the arc are admissible sites each node. This problem, called

the "m-center" problem in the operations research (OR) field, will be reviewed briefly here.

Minieka (1970) first proposed the m-center problem as an optimal algorithm; this was

followed by an independent study by Christofides and Viola (1971), which also presented a

iterative optimal algorithm. Handler (1973) developed an improved algorithm for Minieka's

approach and showed that his method was preferable to that suggested by Christofides and

Viola, particularly as the problem size increases. Summaries of the algorithms developed

appear in Christofides (1975) and Minieka (1978). Several network examples serve as a

stimulus for further examination of this m-center problem, since both the Handler and the

Christofides and Viola approaches might require too many steps (computational time) for

the optimal solution. It should be noted, however, that the available procedures were con-

structed with the intention of finding the optimal locations of m-centers in a given network,

rather than locating centers according to a critical distance constraint. Both problems can be

e

solved by the same method, though more computational time is usually required for a given

m-center than for a given-distance problem.

Consider a connected network $G = \{N, A\}$ with a set of nodes N , a set of directed arcs

A , and given distances $d(i, j)$, $\forall (i, j) \in A$. The numbers of nodes and arcs are $|N|$ and $|A|$,

respectively. More notations are as follows:

$D(k, q)$ = shortest distance between k and q ; k, q are two points anywhere on G (shortest path algorithms appear in Appendix 9.A).

$SDM(i, j)$ = shortest-distance matrix for all pairs i and j ; $i, j \in N$.

$SDM_d = SDM(i, j)$ matrix information indicating the shortest paths by directions from node i to node j , $\forall i, j \in N$.

ℓ = critical distance (i.e., for each node i , $i \in N$ must be no further away than ℓ units from its closest stop).

$S_{u,v} = \{u, v \in N : D(u, v) \leq 2\ell\}$, that is, set of all node pairs u, v that have a value equal to or less than 2ℓ in $SDM(i, j)$.

$D_i(u, v) = \min[D(u, i), D(v, i)]$; $i \in N$; $u, v \in S_{u,v}$.

s = candidate stop.

$(i, j)_{u,v}$ = arc on which a candidate stop related to u, v can be located; $(i, j)_{u,v} \in A$.

$ST = \{s \in G : (i, j)_{u,v} \text{ exists} \mid u \{s = i, \forall i \in N\}$, that is, a set of all candidate stops, including each node as a candidate stop.

ST^* = set of stops in the optimal solution.

$d(i, s)$ = distance between node i and a candidate stop

points lying on $(i, i) u, v$.

$P_{u,v}$ = (potential active path: $(i, j) u, v$ exists, and the critical distance criterion is satisfied for a given pair u, v); that is, for the set of all possible paths in G between u and v , $u, v \in S_{u,v}$, so that (1) if the candidate stop is on either i or j , the shortest distance between u and v through i or j , respectively, is equal to or less than 2ℓ and (2) if the candidate stop lies on $(i, j) u, v$, excluding i and j , the shortest distance (simple path) between u and v through $(i, i) u, v$ is equal to or less than 2ℓ .

$P_a = \{u, v \in S_{u,v} \mid u, v \in P\}$, that is, the set of all potential active paths.

SCP = Set covering problem; the corresponding problem is related to a matrix in which each row represents a node (there are n rows) and each column a candidate stop. If the distance between node i and the candidate stop is less than or equal to ℓ , the entry of 1 is indicated. The set covering problem (SCP) is to find the least number of columns such that every row contains an entry 1 under at least one of the selected columns, that is, finding the minimum number of columns to cover all the rows.

The objective function is to find the minimum number of stops, $s \in S_{T*}$, so that $D(i, S_{T*}) \leq \ell$

for all $i \in N$, where $D(i, S_{T*}) =$ the shortest distance between node i and its closest stop, $s \in S_{T*}$.

$s \in S_{T*}$. Certainly, the SCP solution, where S_{T*} is represented by all the columns, is the solution

required for locating the transit stops. Note that there is always a feasible solution, since

each node is also a candidate stop; that is, for large ℓ values, each node will be its own stop.

12.4.2 Set-ST algorithm and explanation

The algorithm to find the set ST is constructed as follows:

Step 1: Construct $SDM(i, j)$ and its related matrix information, SDM_d .

Step 2: Identify $S_{u,v}$ from $SDM(i, j)$.

Step 3: Select $u, v \in S_{u,v}$ and search for $(i, j)_{u,v} \forall (i, j) \in A$ through the procedure to determine

ST (see the flow diagram in Figure 12.6); complete the procedure for ST for all $u, v \in S_{u,v}$.

Step 4: Store $SDM(i, j)$ and call SDM_d .

Step 5: Identify P_a (based on SDM_d).

Step 6: Construct SCP matrix for all $s \in ST$ (known from Step 3), based on P_a .

Step 7: Solve the SCP.

For Step 1 of the Set-ST algorithm, various methods can be implemented; one of them is

the Dijkstra's algorithm described in Appendix 9.A. More algorithms can be found, for

example, in Ahuja et al. (1993). Steps 2, 4, 5, and 6 in the algorithm are self-explanatory by

utilizing the aforementioned notations. Step 3 deserves clarification and Step 7 a comment.

In Step 3, the search for $(i, j)_{u,v}$ is performed on any $(i, j) \in A$ for each u, v pair. This search

enables one to check whether a candidate stop (i.e., whether the shortest distance to u and

to v is less than or equal to ℓ) can be located at any point on G . The interpretations of $(i, j)_{u,v}$

point to the existence of a simple path on G (with no repetitions of either nodes or arcs) a long

this path) between u and v , so that (1) its length is less than or equal to 2ℓ ; (2) its midlength

point, p^* , lies on $(i, j)_{u,v}$; and (3) the shortest distance between u and p^* and between v and

p^* is that along the simple path considered. In that last case, $p^* = s \in S \cap T$. The search for

$(i, j)_{u,v}$, is based on Lemma 12.2.

Before specifying Lemma 12.2, let us refer to a special diagram that was first demon-

strated by Hakimi (1964) and further discussed by Handler (1973) and Minieka (1978).

The x-axis in this special diagram represents the distance from node i to node j along (i, j) ,

and the y-axis represents the shortest distance from u and v to each point along (i, j) . The

distances are represented by 45° lines that can be monotonically decreased or increased or

can have a unique broken point of 90° along (i, j) .

Lemma 12.2:

If k is a point lying on (i, j) and $D(u, k)$ and $D(v, k)$ are described by the special diagram,

there are only four possible cases for $(i, j)_{u,v}$:

(a) If the lines for $D(u, k)$ and $D(v, k)$ intersect, a candidate stop is located on (i, j) .

(b) If the lines for $D(u, k)$ and $D(v, k)$ do not intersect, a candidate stop cannot be located on (i, j) .

(c) If the lines for $D(u, k)$ and $D(v, k)$ are the same at one node, say i , and split toward the other node j , then i becomes a candidate

stop.

(d) If the lines for $D(u, k)$ and $D(v, k)$ are the same along each point k , $k \in (i, j)$, then both i and j are candidate stops.

Proof: The proof of this lemma is almost straightforward with the use of the special diagram

configurations, such as the one illustrated in Figure 12.5. If we follow the line (triangle shape)

in Figure 12.5a for $D(u, k)$, it can be seen that the shortest path from u to a point between i

and k_1 is through i , whereas the shortest path from u to a point between j and k_1 is through j .

It is also clear that $D(u, i) + d(i, k_1) = D(u, j) + d(i, j) - d(i, k_1)$ because this is the peak point

(equal to the shortest distance from i and from j) of $D(u, k)$. Based on the interpretation for

$(i, j) u, v$, it becomes evident that point s satisfies all the requirements for a candidate stop. Figure

12.5b shows schematically that the solid line is the desirable simple path, whereas the dashed

lines exhibit the direction of the shortest distance from u and v to other points than s that lie

on (i, j) . The critical distance criterion can be checked by the length of the heavy solid line in

Figure 12.5a. It is postulated that the length of this line should be less than or equal to 2ℓ (note

that point s is the midpoint of the heavy solid line). The explanation for Figure 12.5 can be

applied to all the other special diagram configurations, that is, all the possible combinations

of the lines for $D(u, k)$ and $D(v, k)$ categorized by the cases of Lemma 12.2. For cases (c) and

(d), these configurations will include equal lengths of $D(u, i)$ and $d(v, i)$ or $d(u, j)$ and $d(v, j)$.

The basic rules concerning the search for candidate tops (Set ST) constitute Lemma 12.2.

These rules are integrated into a procedure to determine the set ST in Figure 12.6; this

procedure exhibits Step 3 of the Set-ST algorithm and examines each $(i, j), (i, j) \in A$ for all

u, v pairs. The set of arcs $(i, j) \in A$ should be arranged such that each node (except those

connected by a single arc) is considered at least once as node i of one arc (i, j) . Note that if a

node j is connected by a single arc (i, j) and can serve as a candidate top for i (and perhaps

for other nodes), then it can be replaced as a candidate top by node i . This efficient arrangement

allows the omission of a special check of node j as a potential candidate top, reduces

the computation time, and ensures a determination of all $s, s \in ST$, on G . Finally, in Figure

12.6, the critical distance is examined before the determination of $(i, j)_{u,v}$, an order of steps

that can be changed if the value of ℓ is relatively small; for example, $(i, j)_{u,v}$ can be determined

rst and then the critical distance examined.

Following a re r emarks p ertain t o S tep 7 o f t he
S et-ST a lgorithm. T he s olution t o l arge

scale S CPs o ften i nvolved a c onsiderable a mount o
f c omputation t ime (Cristo des, 1 975).

S CPs h ave a ttracted i ntensive r esearch a
ttention, p articularly b ecause o f t heir w ide a
pplica

bility, for example, for airline crew scheduling. Rubin
(1973) developed an effective heuris

tic a lgorithm f or S CPs, a nd B alas a nd P adberg
(1972) i denti ed s pecial p roperties o f a n S CP

m atrix t hat i mprove i ts s olution t hrough t he
s implex m ethod o f l inear p rogramming (by

a voiding d egeneracy d ifculties). T hus, S tep 7 c
an b e b ased o n k nown h euristics p rocedures

i n the OR eld. At this stage, it is possible to set the
following theorem:

Theorem 12.3:

The S et-ST a lgorithm a chieves a n o ptimal s
olution f or t he p roblem o f m inimum s tops f or
a

g iven critical distance in a nite number of steps. i j u
v i j Distance Distance k 1 k 2 s s $D(v, i) = D i (u, v)$
 $D(u, j) = D j (u, v) D(u, i) D(v, j) D(u, k) D(v, k)$ (a) (b)

Figure 12.5 Example of a con guration for case (a) of
Lemma 12.2 in order to search for (i, j) u,v and its
corresponding candidate stop where in (a) is the triangle
shape, and in (b) is schematic paths.

Proof:

1. Optimality i s a ttained b y c onstruction; t he S
CP s olution g uarantees t he s election o f t he m
inimum n umber o f s tops s atisfying t he d istance
c onstraint. T he p rocedure i n F igure 1 2.6, t
hrough L emma 1 2.2, e stablishes t hat a ll c
andidate s tops a re t o b e c onsidered for the

analysis of the SCP. Handler (1973) showed that the optimal solution for the problem combines only candidate sources. These simple arguments suffice.

2. Finite convergence is attained because a finite iteration of the Set-ST algorithm, which is performed only through the procedure in Figure 12.6, clearly indicates a finite operation (Su,v –see Figure 12.6–is based on a finite number |A|).

12.4.3 Numerical example

Let us consider a small network of roads for further explanations of the Set-ST algorithm.

This example is illustrated in Figure 12.7. The critical distance is $\ell = 500$ m. The shortest

distance matrix is given in Table 12.4, with the corresponding path (list of sequential nodes)

in parentheses; those distances forming Su,v are in bold.

The procedure (Step 3 of the algorithm) for Figure 12.6 is shown in Table 12.5. For

each u, v pair, there is a shaded symbol for those $(i, j) \neq (i, j)_{u,v}$; for each $(i, j)_{u,v}$, the Given: Su,v; ℓ ; $SDM(i, j)$ $(i, j) = (i, j)_{u,v}$; $ST d(i, s) = \lfloor [d(i, j) + D_j(u, v) + D_i(u, v)] D(v, j) > D(u, j) ? D(u, i) > D(v, i) ? D_i(u, v) > D_j(u, j) + d(i, j) \leq 2\ell ?$ Stop: ST Anymore (i, j) A? $D_i(u, v) \leq \ell ? D(u, i) = D(v, i) ? D(u, i) > D(v, i) ? D(u, j) > D(v, j) ? (i, j) = (i, j)_{u,v}$ $ST d(i, s) = 0$ Yes Yes Yes Yes Yes Yes Yes Yes No No No No No No No No No No Yes Anymore u, v Su,v?

Figure 12.6 Determination of a set of candidate-stop locations. Residential area 2 Residential area 4
Residential area 5 Residential area 3 Residential area 1
700 800 m 600 1200 900 300 250 400 100 200

Figure 12.7 Example of a network of roads consisting of five nodes (residential areas) and seven road segments, with lengths in meters; two stops (gray colored) are recommended at a specific location along a stretch of 100 m.

Table 12.4 Shortest-distance matrix (in meters) of the road network example in which the corresponding path (by

node number) appears in parentheses j

i 1 2 3 4 5

1 0 700 (1-2) 300 (1-3) 900 (1-3-4) 1200 (1-3-9)

2 0 800 (2-3) 300 (2-4) 1500 (2-4-5)

3 Symmetrical 0 600 (3-4) 900 (3-5)

4 0 1200 (4-5)

5 0

Table 12.5 Results of the procedure in Figure 12.6 for the road network example $(i, j) \in A$

$u, v \in S$ $u, v (i, j) \in A$ (1, 2) (1, 3) (2, 3) (3, 4) (3, 5) (4, 2) (5, 4)

1,2 700 (1-2) X X X X X

1,3 X 300 (1-3) X X X X X

1,4 1000 (1-2-4) a X X 900 (1-3-4) X X X

2,3 1000 (2-1-3) a X 800 (2-3) 900 (3-4-2) X X X

2,4 X X X X X 300 (4-2) X

3,4 X X X 600 (3-4) X X X

3,5 X X X X 900 (3-5) X X

a This $P_{u,v}$ is not the shortest-distance path between u and v .

value of $DL = D_i(u, v) + D_j(u, v) + d(i, j)$, which should be less than or equal to $2\ell = 1000$

m , is indicated in the upper part of each cell in Table 12.5. The path $P_{u,v}$ is shown

in the lower cell part for $(i, j) u, v$. In this example, none of the candidate stops is located

on a node.

The information and intermediate results related

those $S \in T$ are shown in Table 12.6, including

the SCP matrix. The optimal solution combines the two circle columns in Table 12.6, in

which $(i, j)_{u,v} = (3,5)_{3,5}$ and $(1,2)_{1,4}$. It is interesting, and important for planning purposes, to

note that the difference between 2.8 and the D.L. distance indicated in Table 12.6 allows for

flexibility of stop location. That is, the optimal stop lying on arc $(3,5)$ can be located 50 m

closer toward either node 3 or node 5 without violating the optimal solution. On the other

hand, the optimal stop on $(1,2)$ can be located only at one point (200 m away from node 2

on that arc).

In addition, it is possible to observe—or to obtain from the SCP matrix in Table 12.6—the

optimal solution for $\ell < 500$. For example, for $150 \leq \ell < 500$, three stops will be required.

Practically speaking, it makes sense to have the objective of maximum location flexibility;

such an objective yields the solution $(i, j)_{u,v} = (1,3)_{1,3}$, $(2,4)_{2,4}$, and $(5,5)_{5,5}$, where the last is

node 5.

This section approaches the problem of transit stop location in a network in which the

nodes are located in concentrated population areas. A more challenging problem distributes

the origins of passenger demand along the arcs (the continuous case) rather than in specific

nodes. Nonetheless, the Set-ST algorithm can also be used as a preliminary tool to assess the

location of transit stops in such a continuous case. For instance, artificial nodes (creating a

new set of arcs) could be added, such that the distance between two adjacent nodes on the

same arc would not be more than a given value.

12.5 STOP PLACEMENT CONSIDERING UNEVEN TOPOGRAPHY

Route coverage and stop spacing are among the service-design elements presented in

Section 12.2. This section provides a formulation and analysis for finding the location

of public transit stops to enable easy and convenient access and egress to and from the

stops. One of the major parameters considered in this section, to attain convenient access

and egress, is the effect of uneven topography on walking; this has barely been considered

in the literature, particularly due to a lack of data and the complexity involved with its

analysis.

Table 12.6 Information and results required for the set covering problem analysis of the road network example (first four rows) and the set covering problem matrix (last five rows)

u, v 1,2 1,3 1,4 1,4 2,3 2,3 2,3 2,4 3,4 3,5 1,1 2,2 3,3 4,4 5,5

DL a 700 300 1000 900 1000 800 900 300 600 900 0 0 0 0 0

(i, j) u, v (1,2) (1,3) (1,2) (3,4) (1,2) (2,3) (3,4) (2,4) (3,4) (3,5) 0 0 0 0 0

d(i, s) 350 150 500 150 200 400 450 150 300 450 0 0 0 0 0

1 b 1 1 1 1 1 1

2 1 1 1 1 1 1 1

3 1 1 1 1 1 1 1 1

4 1 1 1 1 1 1

5 1 1

a $DL = D_i(u, v) + D_j(u, v) + d(i, j)$, in meters; if s is on node I , then $DL = 2D_i(u, v)$.

b Node number for the rows of the SCP matrix.

It is estimated that the human heart beats approximately twice as fast while walking

u phill a t a g rade o f 1 5%, a s c ompared t o w alking o n l evel g round. T his m eans t hat a p erson

w alking u phill c onsumes t wice t he a mount o f o xygen a s a p erson w alking o n l evel g round

(Terrier a nd S chutz, 2 010). S lope n ot o nly a ffects w alking s peed b ut a lso r educes t he a ttrac

tiveness o f u sing t he t ransit s ervice a s sh own b y R odriguez a nd J oo (2004) i n t heir p aper

studying the factors of pedestrian and cyclist mode choice.

Locating b us s tops o n s teep s lopes i s a lso d isadvantageous t o b us o perators. T he

e ffect o f g ravity o n a lready w eak e ngines c an l ead t o a c onsiderable a dditional d elay i f

a b us h as t o a ccelerate f rom a s top l ocated a t t he b ottom o f a h ill. T he m arginal i mpact

o f g rade c auses a n a dditional s topping d elay r anging f rom 5 t o 1 1 s f or g rades g reater

than 3% (Furth and SanClemente, 2006). The bus stop infrastructure design guidelines

(Bus Stop, 2009) of the Auckland Regional Transport Authority include topography as

one of 10 primary factors to consider when locating new bus stops, noting that “in areas

where the topography is hilly or very steep, slower spacing of bus stops may be required.

Grade of roads should not impede accessibility.” In addition to this, a recent survey

published as part of the TCRP (2013) shows that 10 out of 42 agencies surveyed across

the United States consider steep hills to be a significant factor affecting the operation

of their fleet.

This section covers the methodology developed recently by Ceder et al. (2015) for optimal

stop placement considering uneven topography. This methodology was tested on the CBD

area of the city Auckland, New Zealand. The problem is formulated as follows: for a single

bus route r , select from a predetermined set of potential stop locations, the optimal set of

stops to serve demand accessing the route in both service directions considering the effects

of topography on access time and operational cost. The headway is assumed as a fixed input

to the model.

12.5.1 Framework and modeling

The underlying street network is represented by a digraph $G = \{N, A\}$ where A is a set of arcs

representing streets, roads, footpaths, etc., a

and N is a set of nodes representing intersections,

potential stop locations, demand points, origin points, and terminal points. G can represent

the street network at varying levels of complexity as appropriate for a particular application

of the model. For example, a street network exhibiting large changes in grade over a relatively

small area would be more appropriately modeled at a finer resolution. We define D as the set

of demand points, a set of theoretical points at which transit demand, generated by some facility

(e.g., building, land-use area, zone, or census block), enters or leaves the street network.

The total hourly demand between two demand points i and j is denoted d_{ij} , $\forall i, j \in D$ (pass/h).

A bus route, r , is assumed to be a simple path in G beginning at origin point o_r and terminating

at terminal point t_r . Potential stops are located along the route, on either side of the

carriageway, in order to serve the route in both service directions. $S = \{s_1, s_2, \dots, s_n\}$ is defined

as the set of potential stop locations along r , ordered by the distance from o_r . S is assumed as

an input to the model that in practice could be generated either manually by an experienced

planner or at fixed increments along the route as per Ibeas et al. (2010) and Ceder et al.

(2015). The optimal stop configuration will be a subset of S , selected to minimize operational

and accessibility costs. The decision variables $x_m(1)$ and $x_m(2)$ represent the selection of a poten-

tial stop $m \in S$ in the forward service direction (from o to r) and backward service direction

(from r to o), respectively. We define l_{mn} as the distance in kilometers between two stops

$m, n \in S$, l_r as the total length of the route in kilometers, h as the route headway in minutes,

and α_m as the slope (%) of the carriageway at stop $m \in S$, in the forward service direction.

The headway h is assumed to be a fixed input to the model, appropriately selected, to ensure

that the demand along the route is satisfied at a reasonable level of bus occupancy.

It is assumed that a trip from demand point i to j consists of three components: (1) walk-

ing from i to a stop m , (2) traveling in-vehicle from stop m to n , and (3) walking from stop

n to j . It is further assumed that passengers always select the paths from i to m and n to j

that have the least access cost denoted by ac_{im} and ac_{nj} , respectively. The cost of an access

path is equivalent to its total travel time in minutes (denoted by at_{im} and at_{nj}) weighted by an

attractiveness factor, ε . The factor ε represents the degree to which hilly terrain is perceived

as unattractive to a passenger (for reasons including increased energy consumption and

difficulty for the elderly and disabled). We note that in reality passenger route choice may

be influenced by a variety of other factors including bus stop proximity to an intersection,

safety issues, and land use along the access path. At this stage, such factors are not included

in the model for simplicity.

The attractiveness factor ϵ is based on an elasticity value relating pedestrian mode share to

estimated delay due to sloping terrain, derived by Rodriguez and Joo (2004) and defined as $E \frac{P}{P} \frac{X}{X} \frac{P}{P} \text{ped slope slope} = \delta \delta$ (12.3)

where

$\delta P \text{ ped} / P \text{ ed}$ represents a percentage change in pedestrian mode share

$\delta X \text{ slope} / X \text{ slope}$ represents a percentage change in delay due to slope

Rodriguez and Joo (2004) supply (in Table 3 of their paper) three different values for $E \frac{X}{X} \frac{P}{P} \text{slope ped}$,

relating to three modes of pedestrian, bicycle, and bus. We use the result of the pedestrian

mode for the one-level logit model with $E \frac{X}{X} \frac{P}{P} \text{slope ped} = -0.162$, significant at a 99 % level of

confidence. This value describes the percentage change in pedestrian mode share expected

for a 1% increase in delay due to sloping terrain. Thus, for example, a path with hills to topog-

raphy that causes a 20 % increase in travel time is 3.2% less attractive than a level path of

equal length. Conversely, the perceived cost of the hilly path is 3.2% greater than the level

path. This is based on the assumption that the relationship between access path hilliness and

demand, established by Rodriguez and Joo (2004), implies a similar, but inverse, relation

ship between route hilliness and access path cost. Defining $\varepsilon = -\frac{1}{E} \frac{dC}{dS}$ slope ped 0.162, we define

total access cost as $C_i = \frac{1}{w} l_i + \frac{1}{w} \sum_{j=1}^n l_{ij}$
 $C_i = \frac{1}{w} l_i + \frac{1}{w} \sum_{j=1}^n l_{ij} \quad (12.4)$

where

l_i is the total length of the access path between demand point i and stop m , $w(\alpha)$ is the walking speed on level terrain ($\alpha = 0\%$)

The access is similarly defined.

Let us first consider access time. The total access time at i between demand point i and

stop m is calculated based on a function of walking speed, $w(\alpha)$ in m/s that varies with

slope, α . The definition of $w(\alpha)$ is based on an approximation of data collected by Terrier

and Schutz (2010) and shown in Figure 12.8.

A linear piecewise function, obtained using least-squares regression with fixed inter

cept (i.e., fixing $w(0\%) = 1$), is used to approximate the data. This approximation is only

appropriate for grades between -15% and 15%. We assume that outside of this range,

the walking speed remains constant. This assumption is unlikely to have a considerable

impact on the analysis as road grades typically lie within the bounds shown in the follow

ing equation: $w(\alpha) = \frac{1}{w} \left(\frac{1}{1.058} - \frac{1}{1.058} \alpha \right) \quad \alpha \in [-0.15, 0.15]$
 $w(\alpha) = \frac{1}{w} \left(\frac{1}{1.058} - \frac{1}{1.058} \alpha \right) \quad \alpha \in [-0.15, 0.15]$
 $w(\alpha) = \frac{1}{w} \left(\frac{1}{1.058} - \frac{1}{1.058} \alpha \right) \quad \alpha \in [-0.15, 0.15]$ (12.5)

Slope is not constant, but varies along the length of the access path, producing a slope pro

file. The slope profile is calculated as the derivative of the elevation profile of the access path,

$E_{im}(x)$, where x is the distance along the access path from i . The total access time between

i and m is therefore calculated in Equation 12.6 as the integral of distance along the access

path divided by the walking speed (which is itself a function of distance along the access

path), over the length of the access path. That is, $at_{im} = \int_0^{l_{im}} \frac{dx}{w(E(x))}$ (12.6)

Walking speed (m/s)	Grade (%)
1.0	-10
1.1	-5
1.2	0
1.3	5
1.4	10
1.5	15

Approximation Preferred walking speed vs. grade Data

Figure 12.8 Relationship between slope and walking speed.

The at_{nj} access time between n and j is similarly defined. In reality, $E_{im}(x)$ is a smooth

continuous profile; however, in practice, it is likely represented as a discrete set of elevation

points $E_{im} = \{(x_1, e_1), (x_2, e_2), \dots, (x_n, e_n)\}$, where x_k is the distance along the access

path, from i to point k , e_k is the elevation at point k , and n is the total number of

elevation points. Assuming that gradient is constant between points in E_{im} , at_{im} can be

rewritten as follows: $at_{im} = \sum_{k=1}^{n-1} \frac{e_k - e_{k+1}}{w_k} \frac{x_{k+1} - x_k}{w_k}$ (12.7)

Equation 12.7 gives the correct total access time if $x_1 = 0$ and $x_n = l_{im}$, in other words, where

the first and last elevation points are located at the start and end points of the access path.

Where the elevation of start and end points is unknown, these may be calculated using

interpolation as follows: $e_1 = e_2 + \frac{x_1 - x_2}{x_3 - x_2} (e_3 - e_2)$ (12.8)

$e_n = e_{n-1} + \frac{x_n - x_{n-1}}{x_{n-2} - x_{n-1}} (e_{n-2} - e_{n-1})$ (12.9)

Furthermore, if the elevation points are determined at fixed intervals of elevation, δe ,

then interpolated start and end points should be constrained to within $\pm\delta e$ (in units of

m above sea level) of the nearest elevation point (i.e., $e_1 - \delta e \leq e \leq e_2 + \delta e$, and $e_2 - \delta e \leq$

$e \leq e_1 + \delta e$).

An example of an elevation profile from the case study of Ceder et al. (2015) is shown

in Figure 12.9. The figure shows the elevation profile of the access path and cumula

tive travel time in both directions, with and without accounting for the effects of slope

on walking speed. As can be seen, in the forward direction, the effect of slope causes

approximately 20 s of delay over a distance of approximately 650 m. In the backward

direction, the effect of slope is minimal. This is due to the fact that preferred walking

speed on downhill grades is not significantly faster than on level ground as shown in

Figure 12.8.

The last point of this section is the introduction of the concept of reachable stops and

reachable demand points. A stop m is reachable from a demand point i if $ac_{im} \leq \gamma$, where

γ is a predefined upper limit on access time in seconds. Conversely, i is reachable from m

if $ac_{mi} \leq \gamma$. Note that $ac_{im} \neq ac_{mi}$ since the elevation profiles E_{im} and E_{mi} are not equivalent

(and are, in fact, opposite). We thus define in Equations 12.10 and 12.1 the set of binary

variables β_{im} and β_{mi} where $\beta_{im} = 1$ if stop m is reachable from demand point i and $\beta_{mi} = 1$ if i

is reachable from m . $\beta \gamma i m$ is ac otherwise $i D m S =$
 $\leq \beta \gamma i m \forall \epsilon \in [1, \infty)$, (12.10) $\beta \gamma m i m$ is ac otherwise $i D m$
 $S = \leq \beta \gamma i m \forall \epsilon \in [1, \infty)$, (12.11)

12.5.2 Optimal selection of bus stop locations

The selection of bus stop location involves a hierarchical process in which at one level plan-

ners and operators make decisions in order to improve the overall performance of the route,

while at another level individuals make choices with regard to their selection of stops in

order to minimize their individual objectives, for example, travel time. Therefore, in order

to evaluate the cost of a certain bus stop configuration, user behavior in response to this

configuration must simultaneously be modeled. This property of the stop selection problem

naturally lends itself to formulation as a bi-level optimization problem.

A bi-level optimization problem is a class of optimization problem that has an optimization

problem embedded in its constraints. It has the following general formulation (Colson

et al., 2007): $\min_{x,y} F(x,y) \text{ s.t. } G(x,y) \leq 0, f(x,y) \leq 0, \epsilon \leq \epsilon \leq$ (12.12) Elevation profile Distance (m) 0 5 10 15 20 Elevation (m) 25 30 35 100 200 300 400 500 600 Travel time profile Distance (m) 0 100 200 300 400 500 600 0 100 200 300 400 500 Travel time (s) Equivalent travel time at level grade Equivalent travel time at level grade, Reverse direction Travel time accounting for grade Travel time accounting for grade, Reverse direction

(a)

(b)

Figure 12.9 Examples of (a) elevation and (b) travel time

problem.

The problem is divided into two levels, termed the upper and lower levels, where x and y

are the upper- and lower-level variables, $F(x, y)$ and $f(x, y)$ are the upper- and lower-level

objective functions, and $G(x, y)$ and $g(x, y)$ are the upper- and lower-level constraints. The

advantage of using a bi-level optimization formulation for this problem is to find a demand

distribution that is optimal according to some objective at the lower level, for a given s top

selection at the upper level, which is in turn evaluated according to a different objective.

In Ceder et al. (2015), a bi-level optimization problem has been developed in which the

upper level of the model minimizes a cost function that represents the perspectives of the

passenger, the agency, and the community (see more regarding these perspectives in Chapter

15). This function consists of a agency costs relating to the cost of operating the bus route and

user costs relating to accessing and using the route. The operational costs considered are

the total operating cost per kilometer of route covered (C_{km}), the total idling cost incurred

during dwell time at stops (C_{idle}), the total personnel costs over the duration of the analysis

period (C_{pers}), and the total fixed costs relating to the required fleet size (C_{fix}). The user costs

considered relate to the total access time to a and from stops (T_{access}) and the total

in-vehicle

time (T_{veh}). The upper-level variables represent the selection of stops along the route in for

ward and backward service directions and are denoted $x_{mS_m(1)}$, $\forall \epsilon$ and $x_{mS_m(1)-} \forall \epsilon$. The

upper-level constraints ensure that at least one reachable stop is supplied for each demand

point in both service directions and that each demand point is reachable from at least one

stop in both service directions.

The lower-level model serves to assign demand between demand points to the route via

selected stops. Two alternative lower-level objective functions are discussed by Ceder et al.

(2015), the first seeking to minimize the total passenger travel time and the second seeking

to minimize only the total passenger access time. Lower-level decision variables represent

the number of passengers traveling between any pair of demand points i, j , via a pair of stops

(m, n) at which they board and alight. The lower-level constraints ensure that all demand

d_{ij} is assigned to the route. In order to define the set of lower-level decision variables, the

concept of reachable stop pairs has been introduced. A pair of stops (m, n) is reachable for

a pair of demand points (i, j) , if m is reachable from i and j is reachable from n , or in other

words $\beta_{\beta i m m i} = 1$. For an arbitrary pair of demand points (i, j) , the set of reachable stop pairs,

S_{ij} , defined as $S_{m,n} \mid m \neq n \in S \times S, m \neq n, i, j \in D, D \subseteq I \times J, i \neq j, m \neq n, i \neq j$
 $= \forall i \in I, j \in J, i \neq j, \forall m \in S, n \in S, m \neq n, (i, j) \in D, (m, n) \in S, (i, j) \in S_{m,n}$
 (12.13)

The notation $(i, j) \in D \times D, i \neq j$, refers to all possible ordered pairs of demand points where

$i \neq j$, and $(m, n) \in S \times S, m \neq n$, refers to all possible ordered pairs of potential stops where

$m \neq n$. Then the set of lower-level decision variables can be defined as $\delta = \forall (i, j) \in D, (m, n) \in S, (i, j) \in S_{m,n}$ (12.14)

Where only passenger access time is considered in the lower-level objective function, the

model outlined previously can be classified as a mixed integer, nonlinear-linear, bi-level

programming problem where nonlinear-linear refers to the linearity of the upper- and

lower-level problems, respectively. Note that if in-vehicle time is included in the lower-level

objective function, then the problem becomes a mixed integer, nonlinear-nonlinear bi-level

problem. There is little treatment of such problems in literature, with most of the research

effort being invested in the study of linear bi-level problems (Li and Ji ao, 2008). Because

of their nonconvex, nondifferentiable nature, bi-level problems are inherently difficult to

solve, with even the simplest instance, the linear-linear bi-level problems, having been

shown (Colson et al., 2007) to be of NP-hard complexity (regarding complexity levels, see

Section 5.5). Thus, it is necessary to use a heuristic approach to approximate an optimal

bus stop configuration. The solution method used by Ceder et al. (2015) is a new evolutionary

algorithm based partially on the work of Li and Ji ao (2008) and Beyer and Schwefel (2002).

The approach outlined in this section is detailed by Ceder et al. (2015) with the complete

mathematical formulation of the bi-level optimization model, the solution algorithm and a

case study (in Auckland, New Zealand). The main benefits of this approach are as follows:

(1) the explicit modeling of access paths allows for a high degree of accuracy in relation to

route accessibility; (2) the technique developed for determining elevation profiles is reasonably

simple to implement using geographical information systems (GIS) software and enables

topography to be considered in bus network design; (3) the bi-level type model realistically

models the hierarchical nature of the transit planning process; (4) the GIS-based software

framework developed as part of the case study allows the street network to be represented

at a high level of detail, with the ability to include pedestrian-only paths; and (5) the model

also makes use of (mostly) freely available GIS data covering a very large geographic area;

thus, there is potential for the method to be applied to bus routes in almost every major city.

12.6 LITERATURE REVIEW AND FURTHER READING

This section reviews research works that discuss various indicators of the level of transit

service (LOS). Such indicators enable a measurement of the service quality of both existing

transit systems and proposed schemes and, therefore, serve as important planning tools.

Note that transit coordination/connectivity studies are reviewed in Chapter 5 and reliability

studies in Chapter 18.

Alter (1976) introduces a measure of transit LOS whose value is affected by accessibility,

travel time, reliability, service directness, service frequency, and onboard density criteria.

Agiven set of tables and conversion tables assigns grades, from A to F, to each criterion.

The accessibility grade is based on the walking time or distance to and from a stop. The

reliability grade is based solely on frequency. Travel time is evaluated according to the ratio

of transit travel time to car travel time on the same route. The grade for directness depends

on a combination of the number of required transfers and waiting time. Grades for peak

and off-peak frequencies are weighted by population density in the area served. Grades for

onboard density depend on the average area for a single passenger. All six measures are

aggregated into an overall LOS indicator. Some of the criteria use qualitative considerations,

but the whole process is generally quantitative. Although sensitive to various criteria, the

proposed evaluation methodology seems fairly simplistic.

Horowitz (1981) proposes an indicator for bus-service quality that incorporates the main

performance variables as perceived by transit riders: in-vehicle time, transfer time, walking

time, and waiting time. The need to wait and the need to transfer, which are known to have

asignificant effect on passengers' perception of LOS, are accounted for by both a numerical

variable and a dummy variable. To take equity considerations into account, LOS measures

are calculated separately for different ridership sectors and then weighted and summarized.

The weights used represent the equivalent travel time of a private car. The final LOS rating

depends on the number of people within each population sector who can travel within a

certain travel-time standard.

Polus and Shefer (1984) measure bus LOS by the ratio of average bus travel time to the

average car travel time on the shortest distance between the same origin and destination.

The model they propose for making forecasts of this LOS indicator is based on physical or

traffic-related variables: route length, stop spacing, number of intersections on route, etc.

It can, therefore, be used when detailed information about bus performance or demand is

unavailable, which makes their model useful for analyzing suggested route changes.

Madanat et al. (1994) use an ordered probit model to correlate discrete responses from

a bus-passenger survey with a quantitative variable representing onboard crowding levels.

The ordered probit technique assists in identifying the thresholds between successive ratings

presented to the respondents. The authors compare their analysis to the different onboard

crowding levels that were used in the highway capacity manual (HCM) manual at that time

to measure transit LOS. The results are very different from the HCM values.

Henk and Hubbard (1996) propose a measure for the availability of bus or rail services

that take into account service considerations: coverage, frequency, and capacity. Various

alternative measures of each of these considerations are discussed, and then preferred

indicators are chosen: coverage is calculated by the number of directional route kilometers

per square kilometer; frequency is indicated by the amount of vehicle-kilometers per

route-kilometer; and capacity is measured by seat-kilometers per capita. The values of the

three components in each zone of the network studied are normalized by subtracting the

mean and dividing by the standard deviation; an average of the three components is then

computed.

Murugesan and Rama Moorthy (1996) develop an index for bus LOS. Survey results

are processed using the theory of fuzzy sets, which enables a quantitative analysis based

on qualitative rating responses. Twenty LOS attributes are included in the analysis, but the

authors focus on a description of the fuzzy set methodology, not the LOS indicators.

Friman et al. (1998) study a database of transit riders' complaints and incidents in order

to determine which LOS attributes are perceived as the most important. The analysis is

qualitative, and the authors conclude that the service features most important to passengers

are a driver's behavior and service reliability. The simplicity of the transit network and the

provision of sufficient information are found to be important as well.

ATCRP report (1999) includes a detailed discussion of LOS and capacity issues for any

transit mode. The calculation of LOS takes account of passenger loads and the accessibility

of the service. In addition to the analysis of a system-wide LOS, there is a discussion of LOS

considerations for a specific stop or for a specific route segment.

Prioni and Hensher (2000) develop an LOS model based on stated-preference survey data

that depict the influence of various factors on the LOS as perceived by bus users. Factors

found to have the highest impact on LOS perception are the tariff, travel time, and access

time to the stop. Driver friendliness and the smoothness of the bus ride are also found to

be quite significant. The authors examine the differences in the LOS perception of buses

operated by different companies and find that there is some bias toward certain operators;

they show how the index developed could be used in the contracting process to monitor bus

operators and performance.

Ryus et al. (2000) introduce a measure for transit availability based on the concept that

transit routes serve a small group of potential users. The measure is defined as the percent

age of time in which transit service is available and accessible to an average person within

a reasonable walking distance. Computation of this indicator requires the use of a GIS.

Frequency, service coverage, stop location, and operation hours influence the value of the

indicator; population density affects it, too, but does not influence the demand for transit

services.

Guttenplan et al. (2001) propose a methodology for determining LOS in a combined bus,

cycling, and walking network. The bus LOS is influenced not only by common factors, such

as frequency, but also by intermodal factors that are unique to this method, such as d

if

culty in crossing the road or the connection between the bus stop and the sidewalk.

Yang and Wang (2001) evaluate LOS on the traf c-zone level, based on fuzzy c-means

method. Using GIS functions, the authors compute each zone's characteristics, such as its

route network density and population density. The fuzzy c-means method is used to identify

clustering patterns and to aggregate the detailed measurements into composite, zone-level

LOS indicators.

Tyrinopoulos and Aifadopoulou (2008) propose a methodology for the quality control

of passenger services in the public transit business. An overview of the methodology devel

oped by the Hellenic Institute of Transport to assess the levels of quality and performance

of public transport services is presented and discussed. Overall, a total of 39 indicators

are analyzed, classified in the following seven basic categories: safety, comfort, cleanliness;

ness; information, communication with the passengers; accessibility; terminals and stop

points' performance; lines performance; general elements of the public transport system;

and compound indicators based on the results of the indicators of the previous categories.

Among the compound indicators, a customer satisfaction measure is considered in order

to take into account customer perceptions. In fact, the authors suggest using factor analysis

sis and multinomial logistic regression for investigating the influence of the operational

performance indicators of the transportation system on customer satisfaction. This pro

posed methodology requires a adaptation to the local characteristics of the transportation

system under consideration. Furthermore, the indicator, the sample size, the surveys, and

other elements of the methodology must be customized based on the size of the network

under analysis. The adoption of the proposed methodology by a public transit operator

will enable a assessment of the performance of transportation services, development of

service improvement strategies, and monitoring the progress of service quality and more

importantly provide a better understanding of the needs and priorities of the passengers.

Lai and Chen (2011) explore the relationships between passenger behavior and the var

ious factors that affect them by constructing a comprehensive model considering public

transit involvement, service quality, perceived value, satisfaction, and behavioral intentions.

Notwithstanding the factors recognized by past studies, such as service quality, perceived

value, and satisfaction, this study addresses the importance of the involvement of public

transit services in passenger behavioral intentions. A real-life application has been presented

using passenger survey data from the Kaohsiung Mass Rapid Transit, a newly operating

public transit system in Taiwan. The s

structural equation modeling technique is applied to

analyze the conceptualized relationship model. The findings reveal that all causal relationships

are statistically significant. The relevant managerial implications arising from the findings

have subsequently been discussed.

Eboli and Mazzulla (2011) present a methodology for measuring transit service quality

by considering both passenger perceptions and transit agency performance measures. These

two types of service quality measurement, in conjunction, account for both perceived and

measured service quality, respectively. The basic formulation of this procedure entails that

when the variance of the objective/subjective indicator is close or equal to zero, the new indicator

tends to or coincides with the objective/subjective indicator. The proposed procedure is

subsequently applied to a real-life case study of a suburban bus line; a series of subjective and

objective indicators are calculated on the basis of users' perception about the service and mean

measurements provided by the transit agency. Variance of the objective indicators was found to

be lower than the variance of the subjective ones—thus leading to move the value of the new

indicator toward the value of the objective one. Therefore, it is imperative to carefully consider

the most appropriate objective indicators for m

measuring the quality of the service attributes.

Hadas (2013) presents a unified methodology for the extraction of transit data, data storage

age, and analysis, which enable relatively easy spatial analysis with GIS techniques based

solely on Google Transit feeds and any available road layer without need for additional data.

Four connectivity indicators are introduced for the analysis of transit systems: (1) road cov

erage level indicator, (2) intersection coverage level indicator, (3) stop transfer potential

indicator, and (4) routes overlap indicator. The model that has been developed is a simple-to

use tool enabling the decision makers to analyze the connectivity of transit networks and to

perform benchmarking. Automatic calculation of the connectivity indicators within the GIS

package enables the large public transit networks to be analyzed and compared effectively.

Finally, this study demonstrates the application of the proposed methodology with the anal

ysis of the transit systems in Auckland (New Zealand), Vancouver (Canada), and Portland

(Oregon, USA). The possibility of undertaking what-if analysis, such as altering the transit

network, changing frequencies, relocating stops, and merging stops for better accessibility,

has also been demonstrated. Furthermore, this tool allows checking the effect of different

stop transfer potential parameters in terms of policy making (maximum walking distance,

maximum transfer time, etc.). Subsequently, the analysis can be carried out based on the

dynamic properties of network, specifically based on the time of day and day of week.

The major features of the LOS indicators reviewed are presented in Table 12.7.

Table 12.7 Summary of the characteristics of the models reviewed

Source	Transit modes	Quantitative?	Factors influencing LOS
--------	---------------	---------------	-------------------------

Alter (1976)	All modes	No	Accessibility, travel time, reliability, service directness, service frequency, onboard density
--------------	-----------	----	---

Horowitz (1981)	Bus	Yes	In-vehicle time, transfer time, walking time, waiting time
-----------------	-----	-----	--

Polus and Shefer (1984)	Bus	Yes	External factors (route length, stop spacing, number of intersections on route, etc.)
-------------------------	-----	-----	---

Madanat et al. (1994)	Bus	Partially	Onboard density
-----------------------	-----	-----------	-----------------

Henk and Hubbard			
------------------	--	--	--

(1996)	All modes	Yes	Coverage, frequency, capacity
--------	-----------	-----	-------------------------------

Murugesan and Rama			
--------------------	--	--	--

Moorthy (1996)	Bus	Partially	(Twenty factors are discussed)
----------------	-----	-----------	--------------------------------

Friman et al. (1998)	All modes	No	Driver behavior, reliability, network simplicity, information
----------------------	-----------	----	---

TCRP 35 (1999)	All modes	Yes	Accessibility, passenger load
----------------	-----------	-----	-------------------------------

Prioni and Hensher			
--------------------	--	--	--

(2000)	Bus	Yes	Tariff, travel time, access time, driver friendliness, smoothness
--------	-----	-----	---

Ryus et al. (2000)	All modes	Yes	Frequency, service coverage, stop location, operation hour, population density
--------------------	-----------	-----	--

Guttenplan et al. (2001)	Bus	Yes	Factors related to the
--------------------------	-----	-----	------------------------

pedestrian environment (e.g. difficulty in crossing road, etc.)

Yang and Wang (2001) All modes Yes Network density, population density

Tyrinopoulos and

Aifadopoulou (2008) All modes Partially 39 LOS indicators presented

Lai and Chen (2011) All modes Yes Several core service and psychical environment factors discussed

Eboli and Mazzulla (2011) Bus Yes 11 service aspects and 26 service attributes discussed

Hadas (2013) Bus Yes Four connectivity indicators introduced for the analysis of public transit (PT) systems

EXERCISES

12.1 Given the following timetable with seven bus trips, three departure points, and shifting tolerances of ± 5 min and DH travel time of 2.5 min for all trips, the number of available parking spaces in each of the three departure points is one space. Note that the DH time is larger than the travel time (buses need to detour the service route when empty).

(a) Find the minimum fleet size required by using DF modeling.

(b) While maintaining the result of (a), find the best bus scheduling solution so as to minimize the number of departure points with maximum parking spaces required greater than the given number of available spaces; utilize SF modeling to arrive at the solution. Trip number Departure terminal
Departure time Arrival terminal Arrival time 1 a 6:00 b 6:15 2 b 6:25 a 6:40 3 b 5:40 c 5:55 4 c 5:45 a 6:00 5 a 6:30 b 6:50 6 b 6:35 c 6:50 7 a 6:20 c 6:42

12.2 Use the same technical data and queries as in Exercise 12.1, but with 10 trips, 4 departure points, and the following timetable: Trip number Departure terminal
Departure time Arrival terminal Arrival time 1 a 7:00 b 7:20 2 b 7:30 a 7:45 3 c 8:15 d 8:30 4 b 7:00 c 7:15 5 a 7:25 c 7:40 6 b 7:45 a 7:55 7 a 8:05 c 8:15 8 c 7:25 c

7:40 9 d 7:30 b 7:45 10 d 7:25 b 7:45

12.3 Employ the same technical data and queries as in Exercise 12.1, but with nine trips and this timetable:
Trip number Departure terminal Departure time Arrival terminal Arrival time
1 a 8:00 b 8:20 2 b 8:30 a 8:45 3 c 9:15 b 9:30 4 b 8:00 c 8:15 5 a 8:25 c 8:40 6 b 8:45 a 8:55 7 a 9:05 c 9:15 8 c 8:25 c 8:40 9 a 9:15 b 9:35

12.4 Given a network of 9 two-way street segments and 6 residential-area nodes in which the distances between nodes, in meters, appear in the following table and are symmetrical (same distances between nodes i and j and between j and i), Node 2 3 4 5 6 1 900 300 2 1100 1300 3 1000 400 4 700 500 5 600

(a) Find the minimum number of stops in the network, such that the critical distance between each node and its closet stop is less than or equal to 900 m.

(b) Find the optimal locations of four stops in the street network so as to minimize the maximum distance between a node and its closet stop; what is this distance?

(c) Find the optimal location of a single stop in the network using the same criterion as in (b); what is the minimax distance obtained, and from what node?

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13 Chapter 13: Coordination and Connectivity: Measures and Analysis

With changes in lifestyles, trip-making behavior has also changed. Commuters nowadays make

multiple stops between home and work. This has resulted in an increased demand for more

complex trip chains. In order to be competitive with private vehicles, transit operators need to

provide services that can handle such complex trip chains. Well-connected routes offer com

muters to access the complete network. Commuters are not confined to using direct routes

but are able to make transfers efficiently to go where they need to go. It allows them to take

control of their journey and their own destinations instead of having to choose predetermined

destinations by transit operators. This is similar to what we observe in transit-metro systems. To assess the connectivity of routes, we need a tool. This chapter outlines the measures that

can be used to assess the attractiveness of a route or multiple routes. The versatility of the

connectivity tool has been demonstrated by studies on comparison of routes and networks. The chapter consists of four main parts, following an introductory section. Section 13.2 shows how

to measure transit connectivity based on distinguished quantitative and qualitative attributes includ

ing the interpretation, significance, and application of each of the measures. Section 13.3 describes

directions and tools for detecting weak segments in interroute and intermodal transit chains (paths),

for possible revisions/changes. Section 13.4 provides a method, based on the max-flow concept in

networks, to determine weak routes in the transit networks

through an analysis to detect the bottle

necks existing in current transit networks. Section 13.5 presents an application of the transit connec

tivity measures in the area of social equity. The chapter ends with a literature review and exercises. Practitioners are basically encouraged to visit thoroughly all the sections. The writer James

Collins said: "an integrated life is one where you're able to fit the different pieces of your life

together in seamless fashion." Same will be with an integrated transit trip: just to move seam

lessly. This chapter interprets seamlessly.

13.1 INTRODUCTION

Service-design criteria always contain postulates to improve routing and scheduling coordination

(intra and interagency transfers/points and desynchronized/timed transfers). Obviously,

the lack of well-defined connectivity measures precludes the weighing and quantifying of the

result of any coordination effort. This section provides an initial methodological framework and

concepts for (1) quantifying transit connectivity measures and (2) directions and tools for detection

ing weak segments in interroute and intermodal chains (paths) for possible revisions/changes.

The importance of service-connectivity elements may be better comprehended after reading

ing the following selections of a newspaper article, written by its editorial writer. Although

the writer colors his personal experience with metaphors, it is hard to argue with the basic

facts of the article.

13.1.1 Transit connectivity frustration

From the Los Angeles Times, February 5, 2006 ("Taking the rapid out of transit" by Dan

Turner): Like many epic journeys of exploration, mine began not out of necessity but out of curiosity—the ancestral human urge to test the boundaries of endurance and knowledge. My quest: to get from my house in the Hollywood Hills to LAX, using only public transportation. “I had not anticipated that the work would present any great difficulties,” said Sir Ernest Shackleton after surviving his harrowing, failed attempt to reach the South Pole in 1915, his icebound ship by that time at the bottom of the sea ... Trains ferry passengers in and out of most big airports across the country, including Atlanta’s Hartsfield, Chicago’s O’Hare and even San Francisco International. But not at Los Angeles International Airport. It is the fifth-busiest airport in the world, with more than 60 million passengers a year, and more people start their flights there than anywhere else—yet it is not served by any rail line. Like reaching the Pole, getting to the airport using only public transit is a feat requiring courage, fortitude and very bad judgment ... Twenty-four minutes later the bus arrives. Knowingly, I put \$1.25 in the slot and take a seat. Leaving the bus at Hollywood Boulevard, I ask the driver for a transfer. He fixes me with a shy stare ... but does not issue transfers. You have to buy a day pass, which is \$3. I hold out a \$5 bill. The driver looks at it as if it’s a used tissue. He does not give change. So begins the 1.5-mile trek to the Hollywood and Highland Red Line station, with no assured dog or Sherpa to lead the way. Yes, I could take another bus, but I’m still steamed about the day-pass snub ... At least I didn’t have to contend with bus drivers anymore. Riding the escalator into the bowels of Hollywood, I enter the Mercedes of L.A. public transit, the \$4.5-billion Red Line subway. The 17.4-mile system is fast, semi-clean, quiet—a wonder of efficiency with nearly 120,000 boardings a day. It would attract many thousands more if only it went somewhere. Originally planned to run all the way down Wilshire B

oulevard, the city's densest corridor, it instead ends with a whimper at Wilshire and Western Avenue, its spine hacked off by community opposition and weak-kneed politicians ... From here it's 17 minutes to the 7th Street station in downtown L.A., where I transfer to the Blue Line. Eight minutes later, I'm flashing through downtown at 25 mph ... Several days later—or maybe it's 24 minutes—I'm at the Imperial/Wilmington station in Lynwood, prepared to transfer to the Green Line. Thirteen minutes later, I'm on the train heading toward LAX. At last I can see it up ahead—the LAX/Aviation Boulevard station. But my adventure isn't over. The Green Line from Norwalk was originally planned to end inside the airport, but in 1995, after the money ran short, so did the line. A \$11-billion plan to remodel LAX, approved in 2004, called for a people mover that would carry passengers to the terminals from a big transportation center connected to the Green Line, but when most of the plan was recently scrapped to settle a lawsuit with the airport neighbors, so was the people mover. Instead, there is a shuttle bus from the Aviation station ... My expedition from home to LAX takes two hours and 47 minutes, yet I am elated with the thrill of a accomplishment when the shuttle finally arrives. I am footloose and free, untied to a vehicle in a long-term parking lot. Records are sketchy, but I'm confident that no one else from the Hollywood Hills has ever attempted this journey. After all, they could drive or take a cab to LAX in about 40 minutes. Unlike Shackleton, I have reached my Pole.

Along the line of the article's metaphors, we can conclude that when it is dark enough,

one can see the stars; when the transit service is poor, the transit agency can see how

good it can be and proceed to attain this objective. The need to transfer between

routes generates a major cause of discomfort for transit users. Improving public transit

connectivity is one of the most vital tasks in transit operations planning. A poor connec

tion can cause some passengers to stop using the transit service.

13.2 CONNECTIVITY MEASURES

Before proceeding with the main themes of this section, let us review a perceived concept of

transit connectivity. One possible definition of a prudent, well-connected transit path is this:

a n advanced, attractive transit system that operates reliably and relatively rapidly, with

smooth (ease of) synchronized transfers, part of the door-to-door passenger chain.

Interpretation of each component is defined as follows:

Attractiveness: A available information (by telephone, Internet, newspaper, radio, TV, mail leaflets), simple communication (abbreviated telephone number, automatic storage of users' telephone number and address), clear service meeting-point characteristics (clear stop sign, vehicle color and logo), boarding/alighting/riding comfort (low floor, extra space next to driver, comfortable seats, features for physically challenged people, low noise), onboard service (newspapers, magazines, free coffee/tea, TV/video display of timetable, weather, etc.), simple payment (electronic ticketing, prepaid, transfers, smart-card ticketing).

Reliability: Small variance of measures of concern to passengers (total travel time, waiting time, in-vehicle time, seat availability), small variance of measures of concern to transit vehicles (schedule adherence, headways, on-time pullouts, missed trips, breakdowns, load counts, late reports), small variance of measures related to pretrip information using telephone communication (online timetable, travel time to caller); for more on reliability features, see Chapter 18.

Rapidly: Easy access/egress and comfortable stops (fixed stops with shelters and information, transit vehicle bays at time points, with a n

extra approach lane at signalized intersection), transit vehicle preference at unsignalized intersections (yield or stop according to traffic procedures, special bypass arrangements at strategic points), transit vehicle preference at signalized intersections passive priority by extending or shortening green, active priority using actuated transit vehicle signals (e.g., radio, inductive loop), purchase and validation of tickets (electronically, ordinary) on transit vehicles (one way, round trip, transfer, daily, weekly, monthly).

Smoothness (ease): Comfortable routing (maximum criterion for walking distance, round-trip deviation from designated route in bad weather, evolution of flexible routing and scheduling), special train-station entrance (transit vehicle special gate, passengers' special entrance door with comfortable stairs/escalator to/from the train platform), special train exit (exit door next to the train platform for transit vehicle ticket holders, transit vehicle waiting at the exit or under shelter with vehicle-arrival announcement on variable message signs [VMS]).

Synchronized: Online communication between all modes of transit vehicles (vehicle equipped with arrival information for the relevant stations and time difference, positive or negative, for synchronization), passenger subscriptions with serial numbers (adding a variable scheduling element to suit subscribers, planning the scheduling component with subscriber information), short-turn and shortcut routing strategies (computerized suggestions for the transit vehicle driver on short turn and shortcut, VMS onboard information on meeting time with another or the same transit mode); for more on routing strategies and timed transfers, refer to Chapters 17 and 20, respectively.

13.2.1 Developing connectivity measures

The common denominator for all transit services are the following quality-of-connectivity

attributes:

e 1 = average walking time (for a connection)

e 2 = variance of walking time

e 3 = average waiting time (for a connection)

e 4 = variance of waiting time

e 5 = average travel time (on a given transit mode and path)

e 6 = variance of travel time

e 7 = average scheduled headway

e 8 = variance of scheduled headway

These eight attributes, which can be measured, will be termed quantitative attributes.

However, there are other important attributes that cannot easily be quantified and mea-

sured. Three of these are

e 9 = smoothness (ease) of transfer (on a given discrete scale)

e 10 = availability of easy-to-observe and easy-to-use information channels (on a given discrete scale)

e 11 = overall intra- and interagency connectivity satisfaction (on a given discrete scale)

These hard-to-quantify attributes will be termed qualitative attributes. Nonetheless, it is true

that the value of all 11 attributes may be perceived differently by different passengers or even

by the same passenger in different situations. These different perceptions are captured in the

average weighting of each attribute. The weight of each attribute is surveyed based on and/or based

on the results of a mode (path)-choice model. The analysis framework proposed in this se-

ction

distinguishes between quantitative and qualitative attributes, though both can be combined

with some agreeable weighting factors. The reason for this separation is to make it easy for

decision makers to evaluate improvements and changes in the transit connectivity chains.

As noted earlier, measuring transit connectivity involves various parameters and com-

ponents. Therefore, the following notations are introduced in order to ease the explicit

construction of connectivity measures:

Notations

For a given time window (e.g., peak hour, average weekday),

$O = \{O_i\}$ (Koichi and Kambayashi) is the set of origins O_i

$D = \{D_u\}$ is the set of destinations D_u

$P_{Dk} = \{P\}$ is the set of interroute and intermodal paths to D_k

$P_{Ok} = \{P_i\}$ is the set of interroute and intermodal paths from O_k

$M_p = \{m\}$ is the set of transit routes and modes included in path p

$E_t = \{e_t\}$ is the set of quantitative attributes suitable for connectivity measures

$E_\ell = \{e_\ell\}$ is the set of qualitative attributes suitable for connectivity measures

e_{mp}

j is the value of attribute e_j , $j = t, \ell$, related to mode m on path p

α_e is the weight/coefficient for each attribute e_j , $j = t, \ell$

c_p

j is the quantitative and qualitative ($j = t, \ell$) connectivity measure of path p

F_p is the average number of passengers using path p

$c_p(i, j)$ is the capacity (flow of passengers) of arc (i, j) between route and mode i and between route and mode j ; each i can also be a non origin O_i or destination D_i ; (i, j) is contained in path p and is part of a network-flow model

Based on the notations earlier, the following equation-based notations are established: c_{ejt}, p_{je} $m_{j \in E} m_{j \in p} = \in \Sigma \alpha$, (13.1) C_{cjt}, D_{jpj} $p_{pk} D_k = \in \Sigma$, (13.2) $C_{cjt}, O_{jpj} p_{pk} D_k = \in \Sigma$, (13.3) $C_{cjt}, D_{jDj} D_k k = \in \Sigma$, (13.4) $C_{cjt}, O_{jOj} D_k k = \in \Sigma$, (13.5) $c_{cFjt} p_{jFp}$ $j_p = \cdot =$, (13.6) $C_{cjt} D_{jFp} jFp_{pk} D_k = \in \Sigma$, (13.7) $C_{cjt} O_{jFp} jFp_{pk} D_k = \in \Sigma$, (13.8) $C_{cjt} D_{jFD} D_k k = \in \Sigma$, (13.9) $C_{cjt} O_{jFO} jF D_k k = \in \Sigma$, (13.10)

Note that the required weight/coefficient α_e in Equation 13.1 for measuring the level/quality/

goodness of connectivity in Equations 13.1 through 13.10 can be estimated by the results

of both passenger surveys and the path-/mode-choice model (some of which may need to be

constructed, being site specific).

The interpretation, significance, and application of each of the proposed connectivity

measures appear in Figure 13.1. This figure also presents graphical examples of the relevant

intermodal path for each measure and, under the applications, the purpose of measure.

The purpose of the first connectivity measure in Figure 13.1, Equation 13.1, is to compare

paths, that is, chains of trips, each of

from a n o rigin t o a d estination, i ncluding t ransfers.

U sually, t his c omparison t akes p lace f ollowing a n i mprovement o r c hange i n o ne o r m ore

paths. Otherwise, this comparison, using the evaluation tool proposed, categorizes the dif

ferent p aths b y t heir a ccess/egress c onnectivity q uality. T he p urpose o f t he s econd c onnec

tivity m easure, E quation 13.2, i s t o c ompare d estinations. T he p urposes o f t he t hird a nd

fourth measures, Equations 13.6 and 13.7, are the same as for the f r st and second measures,

r espectively, b ut w ith t he c onsideration o f p assenger l ow, t hat i s, a d etermination o f t he

a verage n umber o f p assengers e xposed t o t he c alculated l evel o f c onnectivity. T he p urpose

o f t he f th m easure i n F igure 13.1, E quation 13.9, i s t o c ompare g roups o f d estinations i n

r egard to overall existing connectivity quality.

All connectivity measures that consider passenger l ows should be updated, following

r outing a nd/or s cheduling a nd/or s ervice i mprovements o r c hanges. W hen r eferring t o a

g roup o f d estinations (zonal b ased, p urpose b ased) i n t he l ast m easure i n F igure 13.1, p aths

c an h ave a s top a t o ne d estination a nd c ontinue t o o thers. F or e xample, i n t he l ast-row

g raphical r epresentation i n F igure 13.1, t he a ccess p ath w ith F 4 ' and F 4 " l ows has a stop at

D v for F 4 ' l ow between D j and D v and for F 4 " l ow between D v and D u .

Connectivity measure Interpretation Significance Graphical example Application

$c_{j,p}$ = Sum of all connectivity component measures along a given path k Overall connectivity value or quality for a given path k Evaluation of path k access-connectivity value Purpose: comparison among paths $c_{j,D,k} =$

$c_{jF,p} = c_{j,p} F_{j,p}$, $j=t, \ell$ $C_{j,p} = P_{D,k} P_{D,k} c_{jF,p}$, $j=t, \ell$ Sum of all connectivity component measures along all access paths to a given destination D_u Overall connectivity values for all paths related to D_u Evaluation of destination D_u access-connectivity value considering only existing paths (for new paths, C_{jFD} are applicable) Purpose: comparison among destinations Sum of all connectivity component measures along a given path k , weighted by the average amount of passengers using this path Overall exposure-connectivity measure (for all passengers) for a given path k Evaluation of path k people-access-connectivity value (considering amount of passengers exposed) Purpose: comparison among paths, considering passenger flow Sum of all connectivity component measures along all access paths to D_u , weighted by the average amount of passengers using these paths Overall exposure-connectivity measure for all paths related to terminal D_u Evaluation of terminal D_u people-access-connectivity value, considering existing and new paths Purpose: comparison among destinations, considering passenger flow Sum of all connectivity component measures along all access paths to a set of destinations, weighted by the average amount of passengers using these paths Overall exposure-connectivity measure for all paths in a given set of destinations Evaluation of a set of destinations, considering existing and new paths Purpose: comparison among destinations in a given set (e.g., zone-based, purpose-based) O_i Stop 1 Stop 2 D_u Bus Bus Rail $F_1 =$ Passenger Flow Passenger Ferry F_8 O_j Rail Bus Stop 1 D_u Bus Bus Stop 3 Shuttle F_1 O_i F_2 F_3 Stop 2 F_6 F_2 F_3 F_4 D_u O_j O_i D_v F_1 F_5 F_6 F_7 F_4 Stop 1 Stop 2 Bus Rail Bus Origin O_i Destination D_u Passenger Ferry Stop 1 Stop 2 D_u Bus Rail Bus Shuttle O_i Stop 3 Bus O_j

$m_{M,p} = e_{j,E} \alpha_{e,j} m_p$, $j=t, \ell$ *

$C_{jF,D,k} =$, $j=t, \ell$ $D_{k,D} C_{jF,D,k} C_{FD} =$, $j=t, \ell$

Figure 13.1 Interpretation, significance, and application of quantitative O-D connectivity measures* at different levels and for a given time period.

concepts for quantifying transit connectivity measures and uses these new measures in a

case study of Auckland's public transit network for possible revisions/changes. In February

2008, a busway was opened in the North Shore area of Auckland, New Zealand. This bus

way goes from Albany to Britomart in the central business district (CBD), through four new

stations in the North Shore area. Because of a dedicated road and comfortable and frequent

buses, it improves journey speeds and reliability and provides an attractive alternative to

private vehicle use.

For this busway project, local and express bus routes were designed to link into the bus

way through new stations, so that people can easily transfer and join the CBD from

everywhere in the North Shore. Figure 13.3 illustrates the North Shore bus network with

all the routes (A_{ij}), destinations (D_i), origins (O_i), and hubs (H_i) considered. Transferable

bus tickets were introduced, allowing passengers to change buses using only one ticket.

In addition, two park-and-ride stations were built at the Albany and Constellation sta-

tions, next to the motorway. People can park their car the whole day for free and travel

to and from their destination on the bus. All these improvements make transfers easier

and enhance people's willingness to use transit. In this section, the transit connectivity

measures refer to morning peak-hour period (7-9 a.m.) and to the flows of passengers to

the CBD (Britomart station) and Takapuna station—the two main employment areas of

Auckland. Northern Express line (busway) Feeder routes
 Point-to-point routes A 15 A 5 A 8 A 14 A 13 A 12 A 11 A 18
 A 4 A 17 A 3 A 6 A 2 A 7 A 23 A 22 A 24 A 25 A 26 A 9 A 10
 A 27 A 28 A 30 A 29 A 16 A 19 A 20 A 21 O 1 O 5 O 6 D 2 /O
 7 O 2 O 3 O 4 H 1 H 2 H 3 H 4 A 1 D 1

Figure 13.3 The selected bus network in Case study 1.

The computation of the attributes, of the network in Figure 13.3, was performed (Ceder

et al., 2009) for each transit-network arc (part of a path) as follows:

1. The average scheduled headway and its variance were derived by the bus timetables during the morning peak hour and actual observations.
2. The average waiting time was calculated by Equation 11.1, namely, $W = H \cdot \frac{\text{var}(H)}{H^2}$ where H : scheduled headway, $\text{var}(H)$: variance of headway.
3. The average travel time, its variance, and the average number of passengers per arc were estimated by the data collected of a survey and some data given by the bus companies.
4. The average walking time and its variance were estimated according to the configuration of the stations and the distances between stops.

The qualitative attributes were also included in this study. Qualitative attributes are more

of passenger-perception attributes; thus, it was decided to break down each of the attributes

and specify their related elements in the passenger survey as follows:

1. Smoothness (ease) of transfer (e.g.) was

estimated by

- a. Distance between the stops (meters)
 - b. Presence of stairs, sloping ground, etc. (scale of 1 -5)
 - c. Length of the transfer (minutes)
 - d. Timetable display (existence and clarity)
2. Availability and easy-to-use information channels (e 10) were derived by
- a. Presence of electronic signs indicating the expected waiting time
 - b. Presence of the station's directory/map
 - c. Timetable display (existence and clarity)
3. Overall intra and interagency connectivity satisfaction (e 11) was estimated by
- a. Presence of bus shelters
 - b. Presence of shops within the station

The results illustrated in Figure 13.4 clearly show the good performance of the busway

observed at its hubs (stations H 1 to H 4) compared with the local stations (origins O 1 to O 6); it

is to note that the lower the measure, the better. All arcs originating at one of these four hubs

have a connectivity measure of about 30% lower than the others because of a better transfer

arrangement within the stations, reduced travel times, and higher bus frequencies. The weak

arcs concerning connectivity are A 13 , A 28 , A 29 , and A 30 . The last three arcs are of bus route

858 with a low bus frequency and stations that have not been constructed for transfers.

In this first case study, it was concluded that the use of

the connectivity measures will

allow for pinpointing and prioritizing the required design elements for improvement, that

is, the elements of vehicle type and frequency, service-reliability issues, and quality of the

stations associated with access, egress, and transfer facilities. More analyses and results of

this study appear in Ceder et al. (2009).

13.2.3 Case study 2: Comparing transit connectivity measures of major cities

The second case study by Ceder and Teh (2010) presents a framework for determining and

comparing between the transit connectivity measures of the CBDs of Christchurch City,

Wellington City, and Auckland City in New Zealand. The morning time period 7–9 a.m.

was selected. According to statistics New Zealand's 2008 census population data, the popu

lation of Christchurch City CBD was 369,000 compared to Wellington CBD's population of

192,800 and Auckland CBD's population of 438,100. The study areas for these three CBDs

are defined based on the relative population of these cities. Auckland City's study area is the

largest among the three cities at 6.6 km², followed by 5.7 km² study area at Christchurch

City. Wellington City has the smallest study area of roughly 2.8 km² because of its smaller

population.

The network nodes selected are existing key timing points or key stops as indicated in the

public timetable. A directional arc was inserted into the bus network map where there is an

existing bus service operating. Among these network nodes, a list of origins/destinations,

which are the most frequently visited nodes, was determined. The arc connectivity values

were determined after establishing the bus network map. Because of not conducting actual

survey in the three study areas, the base data consist of only the existing bus frequencies and

timetables and of available data and information provided by the local transport authori

ties. Connectivity values of paths were calculated by summing up connectivity values of

individual arcs to form the path from a defined origin to a defined destination. A total of

115 paths have been selected for Auckland City, followed by 105 paths in Christchurch City.

Wellington City has the smallest sample size of 96 paths, which coincides with the relatively

smaller study area and population.

The comparison of path connectivity, shown in Figure 13.5, indicates that Auckland pro

vides the best path connectivity by having the lowest median and interquartile values and

the smallest range of data. Having the lowest interquartile value means that 50% of the

paths selected for Auckland has smaller variance and hence lesser uncertainties. It is further

noted that more than 75% of Auckland's selected path has a connectivity value of less than

Wellington's 25th percentile connectivity value. Further to the comparison of origins/desti

nations, it is desirable to compare the connectivity between central transit terminals because

of its importance to provide good connectivity for the public to enhance future patronage.

The qualitative attributes, of this second case study, were excluded from the calculation. 0

Dest.

Origin H 1 H 1 H 2 D 1 A 0 5 A 0 1 A 0 7 A 0 6 A 0 9 A 0 2
 A 0 8 A 1 0 A 2 6 A 0 3 A 2 7 A 0 4 A 1 1 A 1 2 A 1 3 A 1 5
 A 1 4 A 1 7 A 1 9 A 1 6 A 1 8 A 2 1 A 2 0 A 2 9 A 2 2 A 2 3
 A 2 4 A 3 0 A 2 8 A 2 5 D 1 D 2 D 2 D 2 D 2 D 2 D 1 D 1
 H 2 H 4 H 4 H 4 01 01 02 03 03 04 05 06 06 H 3 H 3 H 2 H 2
 H 3 H 1 H 2 H 3 H 3 H 3 H 4 H 4 H 3

0.1

0.2

0.3

0.4

0.5

0.6

0.7

0.8

0.9

Figure 13.4 Arc connectivity across Auckland's North Shore routes.

The central transit terminals in both Christchurch and Auckland are located on the main

street. However, Christchurch's terminal is located approximately 500 m south of the central

location, while Auckland's terminal is located at the northern end of the main street.

Wellington's central transit terminal is not located along the main street but at the northern

western end of the existing bus network. A
analysis of these three terminals shows that a
bus

terminal should be located at a more centralized location
along the main street to provide

the best connectivity. This should also be s
upplemented by providing frequent (lesser head

way) and more reliable (lesser variance in h
eadways and travel times) transit services a
long

the main street.

The conclusions of this second case study were
that Auckland City's transit outperforms

both Christchurch and Wellington cities with lower
connectivity measure overall. The loca

tion of the existing central city transit t
erminal in all three cities should be i
mproved to

provide better transit connectivity as both a n
origin and a destination. In general, more

centrally located transit terminal and more p
aths between each origin-destination (O-D)

pair could improve the transit connectivity. M
ore analyses and results of this study appear

in Ceder and Teh (2010).

13.3 DETECTING WEAKNESSES ON INTERROUTE AND INTERMODAL PATHS

An important aspect of a tool for transit c
onnectivity improvements is the ability to d
etect

current or anticipated weaknesses in interroute
and/or intermodal chains/paths. Figure 13.6

portrays a possible process identifying these w
eaknesses, based on the defined notations.

Whereas the first two lists (output in Figure 13.6) do not consider passenger flow, the third

and fourth lists do take this flow into account. The fourth list attempts to identify passenger

flow bottlenecks at the network level. The process of attaining the fourth list is explicated

in the next section. 1.925 2.093 1.732 2.471 1.728 1.180
2.726 1.162 1.151 1.447 1.699 1.793 1.432 1.046 1.064 1.017
0.000 Normalized connectivity value
0.500 1.000 1.500 2.000 2.500 3.000 Christchurch Wellington
Auckland Arc connectivity comparisons Mild outlier Extreme
outlier Average Upper quartile Median Lower quartile
Maximum Minimum 1.466

Figure 13.5 Arc connectivity ranges of three cities.

The first list of weak segments in Figure 13.6 is constructed by means of the value of each

attribute across all routes and modes on a given path. That is, the worst route (operated

by a certain mode) for each attribute on a given path is identified and can automatically be

considered a candidate for improvement. If improvement cannot be made, the second worst

route is examined, and so on. This process does not consider/weight the amount of passenger

flows on each path. It simply identifies weak segments (owing to their inherent features),

regardless of how many passengers use them. List of worst combination of routes and modes by values of attributes for a given path List of worst paths by sum of values of a given attribute Input Component Output (weak segments as candidates for improvement) List of worst paths by connectivity measures and type of measure (quantitative, qualitative) Lists of (1) bottleneck connections between routes and modes contained in destination Du or origin

(1) Flow network and (2) differences between existing flow and MaxFlow solution, in increasing order Estimated access and egress capacity (flow of passengers): Values of all actual and perceived connectivity measures: and p P O k , Finding the inter-route and inter-modal path with the worst measure of a given e j m p attribute; that is, Finding the worst (usually maximum) e j m p across all routes and modes in p , Finding (a) the passenger flow bottleneck (minimum cut) at each destination D u D and origin O i O using Max-Flow augmentation algorithm; and (b) differences between existing flow and max-flow solution p P D k , p P O k c j F p , $j = t$, ℓ c p (i, j) , p P D k i, j M p p P D k , p P O k ; that is, $\max (e$ j m p $), j = t, \ell$ Finding the inter-route and inter-modal path with worst connectivity measure for each destination and origin j ; that is, c j F p $\max = c$ j F p , $j = t$, ℓ p e j m p $= e$ j m p $*, j = t, \ell$ $\max p$ m M p m M p $*$ Value of all attributes p P D k , p P O k e l E l , e ℓ E ℓ e j m p , m M p , $j = t$, ℓ m

Figure 13.6 Analysis framework for detecting weak segments on interroute and intermodal paths.

The second list for detecting weak segments in Figure 13.6 considers the sum of the value

of each single attribute for all routes on a given path, for example, the sum of all waiting

times along a given path. The worst path, p $*,$ is identified and can serve as a candidate for

improvements or changes in the specific attribute. In practice, passenger flow could also

be considered in order to capture behavioral elements, though the inherent connectivity

features of each path are not dependent on this flow.

The third list in Figure 13.6 considers the exposure of passenger flow to all interroute and

intermodal connectivity features on each access/egress path. A list of worst paths can then be

established and treated accordingly. The next section explains and interprets this third list.

13.4 DETECTING BOTTLENECKS IN COORDINATED NETWORKS

The last process in Figure 13.6 intends to find weak connection links in the sense that those

links represent potential bottlenecks for passenger-access/passenger-egress flows. The basic

idea behind this approach is to review all intermodal activities and paths on a specially constructed

passenger-flow network. In any flow network (vehicles, people, or freight) on

a network, problems arise when the flow demand exceeds the capacity. These traffic problems,

well known in road traffic, are rooted in the bottlenecks of the system, some of which

are dynamic in nature (e.g., a slow truck in road traffic) and are not necessarily dependent

on physical characteristics. Our problem is to find the bottleneck obstructing passenger flow

in a coordinated transit network with required passenger transfers.

13.4.1 Finding bottlenecks in interroute and intermodal transit networks

A specially constructed passenger-flow network is illustrated in Figure 13.7. This network

contains two sets of elements: set N for nodes i , $i \in N$, and set A for arcs (i, j) , $(i, j) \in A$.

The network-flow model in Figure 13.7 consists of three subsets of nodes: r set, origin or C $k(i, j) = \text{capacity of arc } (i, j)$ contained in path k Origin or destination nodes Nodes representing transit routes and modes Single origin or

$C_k(O_n, j) C_k(j, D_n) C_k(j, D_u) C_k(O_i, j) C_k(i, j) \dots$ Artificial single origin (AD) or destination (AD) $O_i / D_u O_1 / D_1 O_2 / D_2 O_n / D_n$ Mode and route i Mode and route j $C_k(j, i)$

Figure 13.7 Conceptual O-D connectivity network-flow model.

destination nodes; second, nodes related to transit routes and modes; and third, a node

related to a single origin or destination. Each arc of this network has a capacity value for

either access or egress flow; the arc is associated with a certain interroute and intermodal

path. In the first subset of nodes, there is a source (origin) or sink (destination) node connected

artificially to all the O-D nodes. The process is to load this special network with

either access or egress passenger flows while having a capacity constraint on each arc.

Passenger-access flows originate in the artificial node AD in Figure 13.7 and move to

reach the destination DU; passenger-egress flows move from AD to O_i . Finding bottlenecks

requires maximizing the flow created at AD/AD and absorbed in D_u / O_i . In the maximum

possible-flow solution, those arcs having a flow equal to capacity naturally form the bottle

necks of the network.

A vital element of the network-flow model is the capacity assigned to each arc, in other

words, the maximum amount of passengers, for a given time period (peak hour, average

day), that can traverse each arc. Generally speaking, there are a finite number of arcs in

an actual transit network of routes and modes. For each path, there is a need to search the

capacity (passengers/hour) that can be offered by each route and mode, for instance, the

maximum feasible rail/bus/passenger ferry frequency times the vehicle's passenger capacity.

The solution for the maximum-flow (max-flow) problem corresponding to the network

flow model in Figure 13.7 appears in Appendix 6.A; it is based on the max-flow augmenta-

tion algorithm. An example of passenger-flow transit network is shown in Figure 13.8a.

This is an example of access intermodal paths from a given origin to two destinations,

D1 and D2. There are five transit modes, and the capacities (passengers/hour) are the num-

bers on the arcs. Based on the max-flow min-cut theorem in Appendix 6.A, the bottlenecks

of the network-flow model are the arcs along the minimum cut. That is, the solution is a

maximum produced flow of 1500 passengers/h. The max-flow augmentation algorithm is

based on the seven following paths (in each, more pass/h flow is added):

1. Origin-bus: D1-AD, Augmented (aug.) flow = 400
2. Origin-bus: D2-AD, Aug. flow = 200
3. Origin-bus: passenger ferry-D2-AD, Aug. flow = 100
4. Origin-shuttle bus-rail: D1-AD, Aug. flow = 300
5. Origin-shuttle bus-cable car-rail: D2-AD, Aug. flow = 250
6. Origin-shuttle bus-cable car-passenger ferry: D1-AD,

Aug. low = 150

7. Origin-shuttle bus-cable car-passenger ferry: D 2 -AD,
Aug. low = 100

Total low = 1500

The last augmented network low, based on the procedure in Appendix 6.A, appears in

Figure 13.8b. The sum of augmented lows on (i, j) in Figure 13.8b is represented by a

reverse arc (j, i) . The difference in low between the capacity on (i, j) and this sum of lows

is represented by a forward arc. Only those arcs with a low equal to the capacity do not

have forward arcs in Figure 13.8b. Those arcs that are emphasized are the bottleneck

segments.

The following is an additional procedure for the best exploitation of the results of the

max-low algorithm:

1. Solve the max-low algorithm for maximum capacity ranges.
2. Find differences between the max-low solution(s) and existing low (representing existing demand).
3. Check for any current bottleneck, in which existing demand is higher than the optimal max-low solution; if such exists, suggest immediate improvement.
4. Examine weak segments in the existing interroute and intermodal paths (see Figure 13.7).
5. Propose service modifications for actual and perceived attributes.
6. Update relevant attributes using path (mode)-choice model.
7. If no changes are made, continue; otherwise, go to 1.

8. Set final conclusions.

Practically speaking, it is recommended that the following data be collected from each relevant

transit agency. These data constitute the base for capacity calculation, estimation of attributes,

and the creation of interroute and intermodal connectivity proposals:

1. Frequencies—existing, minimum (estimated), maximum (estimated)

2. Vehicle sizes—existing (number of seats, allowed standees); which (and when, if not continuously) sizes are used Minimum cut (= maximum flow) Bus
 Rail Passenger ferry D 1 D 2 A 400 200 500 150 500 250 250
 Cable car 250 300 800 1500 600 ∞ ∞ Flow capacity
 (passengers/hour) 100 Shuttle bus Origin

(a) Bus D 1 D 2 400 200 200 250 200 250 300 700 ∞ ∞
 Passenger ferry 650 850 150 300 250 100 100 700 800 100 500
 50 Existing bottleneck obstructing passenger Origin A
 Cable car Shuttle bus Rail

(b)

Figure 13.8 Example of the network-flow model and its minimum cut in part (a) and of the last augmented network-flow in part (b).

3. Passenger load—by time of day

4. Average travel time—between each two nodes connected by time of day

5. Routing—existing; flexibility for routing adjustments

6. Routing strategies—existing; flexibility for adding strategies (e.g., skip stop, short turn, shortcut)

A good example of transit connectivity paths is a n a airport terminal. Table 13.1 demon

strates access and egress paths (k 1 to k 5) of transit service to an airport terminal.

13.4.2 Case study of two world sport events

This section is based on a study by Ceder and Perera (2014) who examined the transit network

works of the two world sport events from the coordination/connectivity perspective. It

is known that weak segments of the transit connectivity network will increase passenger's

congestion and will result in delays and also in passengers' frustration. The objective of the

two case studies was to use the aforementioned defined connectivity measures to analyze

and determine the weak routes in the transit networks.

The case studies present two major cities undergoing two major sporting events: the city

of London in the United Kingdom hosting the XXX Olympic Games in 2012 and the city of

Auckland in New Zealand that hosted the South Island Rugby World Cup in 2011. These cities

are used to offer a comparison of the bottlenecks in their transit networks.

The steps and their corresponding data requirements for identifying weak segments of a

transit network are outlined in Table 13.2.

In London, two origins were chosen, both from Central London, because most of the

tourists accommodated these locations during the Olympics season. One of the origin points

is the Charing Cross station and the other is the London Bridge station that is located on the

other side of the Thames River. The Wembley Stadium (with a seating capacity of 90,000

spectators) was considered as the destination because this is the venue staging the London

2012 Olympic Games Football competition. Frequencies and passenger capacities of each

of the transit modes selected were obtained from the information provided in Transport for

London website.

In Auckland, two origins were chosen, one from Central Auckland (Britomart Transport

Centre) and the other from the North Shore part of Auckland City (Devonport). These two

origins reasonably represent the majority of the spectators, both locals and tourists. The

Eden Park Stadium was chosen as the destination, because this was the venue for the 2011

Table 13.1 Access/egress transit modes and paths to airport terminal

Path	Access/egress mode	Path description	Notes
k 1	Train service	Auto-wait-train-wait-shuttle-	terminal
k 2	Public transit bus	Walk-wait a -public bus-	terminal
k 3	Scheduled airport bus	Auto-wait-airport bus-	terminal
k 4	Shared-ride door-to-door van	b Van-terminal	No waiting, but not a straight ride
k 5	Charter bus	Auto-wait-charter bus-	terminal Special ride

Note: Private vehicle access/egress to and from train, airport bus, and charter bus is assumed; passengers are assumed to walk to and from a public bus.

a There is an additional wait for each transfer between two buses.

b Additional distance and time for passengers using van, by time of day.

Rugby World Cup. The Eden Park Stadium is New Zealand's largest rugby stadium with a

seating capacity of 60,000 spectators. Frequencies and passenger capacities of each of the

transit modes selected were obtained from the information provided in M axx Auckland

Transport website.

The passenger-low transit network for London is shown in Figure 13.9a. This diagram

illustrates the access intermodal paths from two given origins (Charing Cross station

and London Bridge station) to a single destination (Wembley Stadium). There are four

transit modes, and the capacities (passengers/hour) are the numbers indicated on the

Table 13.2 Data requirements for identifying weak segments of a transit network

Step Data requirements for the case studies

Establish origin(s) and destination. There must be more than one feasible PT route between the origin(s) and destination. Create, first, origin nodes; second, nodes related to transit routes and modes; third, a node related to a single destination.

Consider all possible transit routes

from origin(s) to destination. Each arc in the network is associated with a certain interroute and intermodal path. All the routes considered entail a maximum walking distance of 1 km by the passengers, from origin to the destination including the walking involved during transfers.

Obtain the frequency of each transit

mode through published timetables. The hourly frequencies of each transit mode are obtained from the timetables by calculating the mean frequency between 8:00 a.m. and 8:00 p.m. on weekdays.

Establish the passenger capacities of

each transit vehicle. For simplicity, on each PT vehicle, the full passenger capacity (seated + standees) has been used.

Assign flow capacities (passengers/

hour) to each arc of the network. This is calculated as equal to the hourly frequency of transit vehicle times the passenger capacity of the vehicle.

Identify the minimum cut

(= maximum flow). The bottlenecks of the network-flow model are the arcs along the minimum cut. 11,580 1,800 10,980

Artificial origin Ferry (at Embankment Pier) London Bridge (origin 1) Tube (Jubilee line) Bus (24-29) Rail (London Euston) Wembley Stadium (destination) Bus (91) Charing Cross Station (origin 2) Tube (Northern) Bus (18) Tube (Bakerloo) 11,580 12,800 1,500 600 1,500 2,400 10,980 750 600 ∞ ∞ 750 2,400 12,800

(a)

Figure 13.9 London's network with capacities and min cut (dashed line) in part (a). (Continued)

arcs. Table 13.3 summarizes, as an example, the capacity calculations for each arc on

the London network.

The sum of augmented flows is shown in Figure 13.9b, and it is represented by a

reverse arc (j, i) . The difference in flow between the capacity on (i, j) and this sum of flows is

represented by a forward arc. Only those arcs with a flow equal to the capacity do not have

forward arcs in Figure 13.9b. Those arcs that are emphasized with a dashed arrow are the

bottleneck segments.

It is to note that Table 13.4 contains, as an example, all the possible paths from an origin

to the destination. However, the maximum augmentation algorithm is based on

only three of these paths as is shown in the succeeding text with an augmented flow greater

than zero.

Similar to the analysis explicated earlier for the London's case study, the passenger

flow transit network for Auckland's case study and its solution are shown in Figure

13.10a and b, respectively. This demonstrates the access intermodal paths from two given

origins (Britomart Transport Centre and Devonport) to a single destination (Eden Park

Stadium). There are three transit modes, and the capacities (passengers/hour) are the

numbers indicated on the arcs of Figure 13.10a. We note that in Figure 13.10b, like in

Figure 13.9b, only arcs that are emphasized with a dashed arrow are the bottleneck

segments.

It is evident from the network-flow diagrams of Figures 13.9 and 13.10 that the

Auckland network is not capable of handling as many passengers as the London network.

However, these two networks are similar wherein the bottlenecks of both networks lie

on the arcs adjacent to the destination. Therefore, a major passenger accumulation at

these specific stations can be expected, and indeed this was the case in Auckland in 2011.

That is, these bottlenecks, if not addressed properly, as was in the case in Auckland, will

cause major congestion problems at specific transit stations or will force the passengers

to seek alternative transportation modes and paths to the destination (including some

low walk ways). One possible solution would entail introducing a shuttle service 11,580 1800 9,480

Artificial origin Ferry (at Embankment Pier) London Bridge (origin 1) Tube (Jubilee line) Bus (24-29) Rail (London Euston) Wembley Stadium (destination) Bus (91) Charing Cross Station (origin 2) Tube (Northern) Bus (18) Tube (Bakerloo) 11,580 12,800 1,500 600 1,500 1,500 600 600 ∞ ∞ 750 1,500 12,800 900 1,500 900 9,480 750 0 0 0 0 0 0 1,500 0 14,300 1,500 0

(b)

Figure 13.9 (Continued) London's network with capacities and min cut (dashed line) its augmented flows in part (b).

from these points to the respective destinations to increase the capacity of the final arc

heading toward the destination.

In order to improve the capacity of the transit network (adapting the infrastructure to

the event), specific plans are made for the bus, train, metro, tram, and passenger ferry

services required for the event. These plans mainly consist of adapting the frequency of

the service, introducing larger vehicles (if available), extending the operating time, reduc

ing the crew rest period, closing the nearest overfilled stations, transferring stops and

rerouting lines, and issuing special day tickets. The planning process attempts to ascertain

the maximum capacity of the available transit networks (buses, railway lines, stations,

stairs, passageways). Such a review of capacity takes account of the expected access,

egress, and onboard demands. More analyses and results of this study appear in Ceder

and P erera (2014).

Table 13.3 Capacities of arcs in London transit network

Arc	Origin	Destination	Frequency (veh/h)	Capacity (pass/veh)	Total (= Frequency × Capacity)
1	Artificial origin	London Bridge	N/A	N/A	∞
2	Artificial origin	Charing Cross station	N/A	N/A	∞
3	London Bridge Tube (Jubilee Line)	Total egress flow	of 11,580	pass/h	
4	London Bridge Tube (Northern)	Total egress flow	of 10,980	pass/h	
5	London Bridge Ferry (London Bridge City Pier to Embankment Pier)	Total egress flow	of 1,800	pass/h	
6	Charing Cross station Tube (Bakerloo)	Total egress flow	of 12,800	pass/h	
7	Charing Cross station Bus (91)	Total egress flow	of 750	pass/h	
8	Charing Cross station Bus (24, 29)	Total egress flow	of 2,400	pass/h	
9	Tube (Jubilee line)	Tube (Bakerloo)	12 965	11,580	
10	Tube (Northern) Rail (London Euston)	12 915	10,980		
11	Ferry (London Bridge City Pier to Embankment Pier)	Tube (Bakerloo)	2.4 250	600	

12 Ferry (London Bridge City Pier to Embankment Pier) Bus
(24, 29) 2.4 250 600

13 Ferry (London Bridge City Pier to Embankment Pier) Bus
(91) 2.4 250 600

14 Bus (91) Rail (London Euston) 7.5 100 750

15 Bus (24, 29) Bus (18) 24 100 2,400

16 Tube (Bakerloo) Wembley Stadium 10 1280 12,800

17 Rail (London Euston) Wembley Stadium 3 500 1,500

18 Bus (18) Wembley Stadium 15 100 1,500

Table 13.4 Possible origin-destination paths and their
augmented flow

Route number Route Augmented flow (pass/h)

1 Artificial origin (AO)-London Bridge station-tube (Jubilee
line)-tube (Bakerloo)-Wembley Stadium 0

2 AO-London Bridge station-tube (Northern)-rail (London
Euston)-Wembley Stadium 1,500

3 AO-London Bridge station-ferry (London Bridge City Pier
to Embankment Pier)-tube (Bakerloo)-Wembley Stadium 0

4 AO-London Bridge station-ferry (London Bridge City Pier
to Embankment Pier)-bus (24-29)-bus (18)-Wembley Stadium 0

5 AO-London Bridge station-ferry (London Bridge City Pier
to Embankment Pier)-bus (91)-rail (London Euston)-Wembley
Stadium 0

6 AO-Charing Cross station-tube (Bakerloo)-Wembley Stadium
12,800

7 AO-Charing Cross station-bus (91)-rail (London
Euston)-Wembley Stadium 0

8 AO-Charing Cross station-bus (24-29)-bus (18)-Wembley
Stadium 1,500 Total flow 15,800 Artificial origin Devonport
(origin 2) Britomart (origin1) LINK bus Eden Park
Stadium (destination) Bus (015-163) Rail (Western line)
City Circuit (free bus) Bus (813) Bus (211-249) Ferry
(Downtown Ferry Terminal) Bus (839-875) 240 180 300 625
135 180 835 665 300 180 180 720 135 665 835 300 240 ∞ ∞

Artificial origin Devonport (origin 2) Britomart
 (origin1) LINK bus Eden Park Stadium (destination) Bus
 (015-163) Rail (Western line) City Circuit (free bus)
 Bus (813) Bus (211-249) Ferry (Downtown Ferry Terminal)
 Bus (839-875) 180 625 135 180 835 625 300 180 180 720 135
 625 835 300 240 0 0 0 0 0 0 0 1460 0 0 0 300 0 0 240 40
 40 ∞ ∞ (a) (b)

Figure 13.10 Auckland's network with capacities and min cut (dashed line) in part (a) and its augmented flows in part (b).

13.5 EQUITY IN TRANSIT PROVISION

Another application of the previously defined transit connectivity measures lands in the area

of social equity with a connotation somehow to the word public in public transit. This section

tion is based on a recent work by Kaplan et al. (2014) using data from Denmark.

The last decades are witnessing a slow but steady paradigm shift from planning mass

transit to considering equity and social inclusion as an integral part of the transit plan

ning process. While equity and social inclusion have been initially discussed with respect to

fare policies, concessionary fares, and transit subsidies, the perspective has been widened to

include population groups with mobility limitations. The interest in considering equity and

social inclusion was first manifested during the 1990s by discussing the need to integrate

equity as a policy goal in transport provision. Accessibility is broadly defined as the ability

and ease of reaching activities, opportunities, services, and goods, and accessibility gaps are

defined as the differences in accessibility across geographical areas, population groups, and

time. These accessibility gaps serve as indicators for identifying spatial, vertical, temporal,

and intergenerational inequities.

Kaplan et al. (2014) proposed to use transit connectivity as a comprehensive impedance

measure for the calculation of both location-based and potential-accessibility measures

that relate to equity assessment within transit planning and evaluation processes. While

previous accessibility measures focused on travel time, transit connectivity considers

travel time, passenger discomfort associated with waiting, transfer and access/egress times,

service reliability and attractiveness, frequency, and seamless transfers along multimodal

paths with specified travel demand as part of the door-to-door passenger chain. Thus,

transit connectivity is free of the aforementioned four limitations and offers a deep and

comprehensive understanding of accessibility gaps as equity indicators. In addition, for

each O-D pair, transit connectivity is calculated for a set of multiple and feasible transit

paths including the three shortest paths and the three most popular paths (i.e., the paths

with the maximum demand) in order to account for the probabilistic nature of transit

path choice.

This aspect is focused in Kaplan et al. (2014) on the multimodal transit system in the Greater

Copenhagen area (GCA), renowned for its transit-oriented urban plan for urban development.

The data consisted of the geographic information system (GIS) representation of the multi

modal network including metro, trains, and buses and detailing service lines, timetables, and

stations. Origin-destination matrices provided information about the current use of the network

work, and estimates regarding the importance of the travel time components were obtained

from the Danish national transport model called Landmark model (LTM). Zone-level data

were available regarding zone size, population, and socioeconomic characteristics including

employment status, age, and income. Transit connectivity was calculated for each origin-destination pair,

and distributions of origin and destination connectivity were compared to the distributions

of socioeconomic population characteristics to assess spatial and vertical equity. Notably, this

study is the first to apply the connectivity measure to a large-scale GIS-based metropolitan size

network.

13.5.1 Calculation of measures

The calculation of transit connectivity at the path, origin, and destination level allows com

puting location-based and potential-accessibility measures. Location-based measures relate

to the calculation of transit connectivity at the zone level. Consider the set $P_{O_i} = \{P_i\}$ of

multimodal paths p_{ij} from origin O_i and the set $P_{D_j} = \{P_j\}$ of multimodal paths p_{ij} to destina

tion D_j . Having computed the connectivity c_p of each path p , it is possible to calculate the

connectivity C_i for each zone as the origin of transit trips and the connectivity C_j for each

zone as the destination of transit trips: $C_i = \frac{1}{P_i} \sum_{p \in P_i} c_p$ (13.11) $C_j = \frac{1}{P_j} \sum_{p \in P_j} c_p$ (13.12)

The origin connectivity C_i measures the possibility of each zone i to reach every other

zone in the study area and hence the prospect for the population of zone i to reach

opportunities (e.g., jobs, universities, hospitals) in every other zone. The destination con

nectivity C_j measures the possibility of each zone j to be reached from every other zone

in the study area and hence the prospect of the opportunities in zone j to be accessible

by the population of every other zone. Notably, the connectivity C_{ij} for each $O_i - D_j$ pair

expresses the possibility of the population in zone i to reach the opportunities in zone j .

When comparing origin and destination connectivity across zones, a classification of the

zones according to connectivity distribution percentiles provides insight into the spatial

equity in the study area. When adding to the comparison of a layer of socioeconomic

characteristics of each zone, the comparison of the connectivity distribution percentiles

with the socioeconomic characteristic percentiles provides insight into the vertical equity

in a study area.

The transit network of the GCA consists of seven major modes: (1) metro, (2) local trains,

(3) suburban trains (S-trains), (4) regional and
intercity trains (Reg-trains and IC-trains),

(5) regular buses, (6) high-frequency buses (A-buses), and (7) suburban and express buses

(S-buses and E -buses). The metro serves central Copenhagen and the airport. The Reg-

and IC-trains lead north and west of Copenhagen, while the S -trains follow the radial

lines from central Copenhagen to the mentioned five cities. E-buses and S-buses

serve the S -train stations (primarily in rings), A -buses operate in central Copenhagen,

and the remaining buses run in Copenhagen, the suburbs, and the rural areas at a lower

frequency.

Figure 13.11 presents the distributions of origin and destination connectivity of the GCA

zones. The origin connectivity is presented also as a graph of connectivity values alongside

the corresponding map, in order to provide an example of the construction of the maps from

the calculated connectivity values. The values in the graph are presented at an aggregate

zone level because of the large number of zones in the study area, while the maps are pre-

sented at the traffic zone level.

From the origin connectivity perspective emerges the general advantage of the transit

oriented development in terms of spatial equity. Zones with higher population density in

the metropolitan core and along the train line

lines enjoy better connectivity. Nevertheless, the

regional is only partially associated with spatial equity because the same origin con

nectivity is not observed for all the regions. From the destination connectivity perspective,

Kaplan et al. (2014) found that while most jobs are located in the metropolitan core and

along the train lines of the regional, not all enjoy the same connectivity. In terms of

vertical equity, while the zones with the highest average income enjoy better connectivity,

the zones that are less better off in terms of connectivity are not much worse off in terms

of income. Nevertheless, there are exceptions in zones that are highly populated, have a

low-average income, and have a relatively low-connectivity level, albeit their proximity to

the metropolitan core. The connectivity activities associated with these zones were found in

all the investigated aspects of the equity assessment. More on this theme appears in Kaplan

et al. (2014).

13.6 LITERATURE REVIEW AND FURTHER READING

This section reviews research works that discuss various indicators of the level of transit

service (LOS). Such indicators enable a measurement of the service quality of both existing

transit systems and proposed schemes and, therefore, serve as important planning tools.

Note that transit coordination/connectivity studies are reviewed in Chapter 6 and r

eliability

s studies in Chapter 17.

Alter (1976) i ntroduces a m easure o f t ransit L OS
w hose v alue i s a ffected b y a ccessibility,

t ravel t ime, r eliability, s ervice d irectness, s
ervice f requency, a nd o nboard d ensity c riteria.

A given set of tables and conversion tables assigns grades,
from A to F, to each criterion.

T he a ccessibility g rade i s b ased o n t he w
alking t ime o r d istance t o a nd f rom a s top.
T he

r eliability g rade i s b ased s olely o n f
requency. T ravel t ime i s e valuated a ccording t o
t he r atio

o f t ransit t ravel t ime t o c ar t ravel t ime o
n t he s ame r oute. T he g rade f or d irectness d
epends

o n a c ombination o f t he n umber o f r equired t
ransfers a nd w aiting t ime. G rades f or p eak

and off-peak frequencies are weighted by population density
in the area served. Grades for

o nboard d ensity d epend o n t he a verage a rea f
or a s ingle p assenger. A ll s ix m easures a re

a ggregated i nto a n o verall L OS i ndicator. S ome
o f t he c riteria u se q ualitative c onsiderations,

Origin connectivity values Ballerup Rødovre Herlev Furesø
Gladsaxe Glostrup Brønshøj Kastrup Lufthavn Brøndby
Frederiksberg V Lyngby-Taarbæk Christianshavn Vallensbæk
Amager Nord Valby Tårnby Greve Hvidovre Frederiksberg Ø
Gentofte V Rudersdal Østerbro Ishøj Vesterbro Emdrup
Østerport Gentofte Ø København Centrum Nørrebro 0 10 20 30
Origin connectivity 40 50 Dragør Origin connectivity
Connectivity at Origins Best quantile Second best quantile
Second worst quantile Worst quantile Train lines Average
quantile

0 5 10 20 0 Destination connectivity Connectivity at
Destinations Best quantile Second best quantile Second

worst quantile Worst quantile Train lines Average quantile
5 10 20 kmkm

Figure 13.11 Connectivity to employment opportunities.

but the whole process is generally quantitative. Although sensitive to various criteria, the

proposed evaluation methodology seems fairly simplistic.

Horowitz (1981) proposes an indicator for bus-service quality that incorporates the main

performance variables as perceived by transit riders: in-vehicle time, transfer time, walking

time, and waiting time. The need to wait and the need to transfer, which are known to have

a significant effect on passengers' perception of LOS, are accounted for by both a numerical

variable and a dummy variable. To take equity considerations into account, LOS measures

are calculated separately for different ridership sectors and then weighted and summarized.

The weights used represent the equivalent travel time of a private car. The final LOS rating

depends on the number of people within each population sector who cannot travel within a

certain travel-time standard.

Polus and Shefer (1984) measure bus LOS by the ratio of average bus travel time to the

average car travel time on the shortest distance between the same origin and destination.

The model they propose for making forecasts of this LOS indicator is based on physical factors

traffic-related variables: route length, stops
spacing, number of intersections on route, etc.

It can, therefore, be used when detailed information about
bus performance or demand

is unavailable, which makes their model useful
for analyzing suggested route changes.

Madanat et al. (1994) use an ordered probit
model to correlate discrete responses from

a bus-passenger survey with a quantitative
variable representing onboard crowding levels.

The ordered probit technique assists in
identifying the thresholds between successive
ratings

presented to the respondents. The authors compare their
analysis to the different onboard

crowding levels that were used in the
highway capacity manual (HCM) manual at that
time

to measure transit LOS. The results are very different
from the HCM values.

Henk and Hubbard (1996) propose a measure for
the availability of bus or rail services

that takes into account service
considerations: coverage, frequency, and capacity.

Various alternative measures of each of these
considerations are discussed, and then

preferred indicators are chosen: Coverage is
calculated by the number of directional

route-kilometers per square kilometer, frequency
is indicated by the amount of vehicle

kilometers per route-kilometer, and capacity is
measured by seat-kilometers per capita.

The values of the three components in each
zone of the network studied are normalized

by subtracting the mean and dividing by the standard deviation; an average of the three

components is then computed.

Murugesan and Rama Moorthy (1998) develop an index for bus LOS. Survey results

are processed using the theory of fuzzy sets, which enables a quantitative analysis based

on qualitative rating responses. Twenty LOS attributes are included in the analysis, but the

authors focus on a description of the fuzzy-sets methodology, not the LOS indicators.

Friman et al. (1998) study a database of transit riders' complaints and incidents in order

to determine which LOS attributes are perceived as the most important. The analysis is

qualitative, and the authors conclude that the service features most important to passengers

are a driver's behavior and service reliability. The simplicity of the transit network and the

provision of sufficient information are found to be important as well.

Highlights of the Transit Capacity (1999) include a detailed discussion of LOS and

capacity issues for any transit mode. The calculation of LOS takes account of passenger

loads and the accessibility of the service. In addition to the analysis of a system-wide

LOS, there is a discussion of LOS considerations for a specific stop or for a specific route

segment.

Prioni and Hensher (2000) developed a new LOS model based on stated-preference survey data

that depict the influence of various factors on the LOS as perceived by bus users. Factors

found to have the highest impact on LOS perception are the tariff, travel time, and access

time to the stop. Driver friendliness and the smoothness of the bus ride are also found to

be quite significant. The authors examine the differences in the LOS perception of buses

operated by different companies and find that there is some bias toward certain operators;

they show how the index developed could be used in the contracting process to monitor bus

operators and performance.

Ryus et al. (2000) introduce a measure for transit availability based on the concept that

transit routes serve a small group of potential users. The measure is defined as the percentage

of time in which transit service is available and accessible to an average person within a rea-

sonable walking distance. Computation of this indicator requires the use of a GIS. Frequency,

service coverage, stop location, and operation hours influence the value of the indicator;

population density affects it, too, but does not influence the demand for transit services.

Guttenplan et al. (2001) propose a methodology for determining LOS in a combined

bus, cycling, and walking network. The bus L

LOS is influenced by common factors, such as

frequency, but also by intermodal factors that are unique to this method, such as difficulty

in crossing the road or the connection between the bus stop and the sidewalk.

Yang and Wang (2001) evaluate LOS on the traffic zone level, based on fuzzy-c means

method. Using GIS functions, the authors compute each zone's characteristics, such as its

route-network density and population density. The fuzzy-c means method is used to identify

clustering patterns and to aggregate the detailed measurements into composite, zone-level

LOS indicators.

Tyrinopoulos and Afadopoulou (2008) propose a methodology for the quality control of

passenger services in the transit business. Essentially, the work assesses the levels of quality

and performance of transit services. In this work, 39 indicators are analyzed, classified in the

following seven categories: safety-comfort-cleanliness, information-communication with the

passengers, accessibility, terminals and stop points' performance, lines performance, general

elements of the transit system, and compound indicators based on the results of the indicators

of the previous categories. Among the compound indicators, a customer satisfaction measure

is considered in order to take into account customer perceptions. In fact, the authors suggest

to use factor analysis and multinomial logistic regression for investigating the influence of the

operational performance indicators of the transportation system on customer satisfaction.

Lai and Chen (2011) study the passenger behavioral intentions and explore the relation

ships between passenger behavioral intentions and the various factors that affect them.

Apart from the factors recognized by past studies, such as service quality, perceived value,

and satisfaction, this study addresses the importance of the involvement of public transit

services in passenger behavioral intentions. By using passenger survey data from the

Kaohsiung Mass Rapid Transit (KMRT), a newly operating transit system in Taiwan, the

structural equation modeling technique is applied to analyze the conceptualized relationship

model. The findings reveal that all causal relationships are statistically significant. Finally,

the managerial implications are discussed.

Eboli and Mazzulla (2011) present a methodology for measuring transit service quality.

This methodology is based on the use of both passenger perceptions and transit agency

performance measures involving the main aspects characterizing a transit service. The com

bination of these two types of service quality measurement fulfills the need to provide a

reliable and possible measurement tool of the transit performance. The proposed procedure is

then applied to a real case study of a suburban bus line; a series of subjective and objective

indicators are calculated on the basis of users' perception about the service and measures

provided by the transit agency.

Mishra et al. (2012) propose a graph theoretical approach to determine connectivity mea-

sures for multimodal transit systems. The purpose of this work is to construct a methodology

enabling important information on transit system performance, especially multimodal

transit systems, with the lowest possible data requirements to support the decision-making

process of both transit agencies and users. Their methodology consists of performance indi-

cators of connectivity index from four different levels: node connectivity, line connectivity,

transfer center connectivity, and region (large area) connectivity. An example problem and

a real-life case study in Washington-Baltimore region are employed to illustrate how the

proposed methodology can be used to quantify and evaluate the network connectivity of

multimodal transit systems.

Hadas (2013) presents a unified methodology for extraction transit data, data storage,

and analysis, to enable spatial analysis with GIS techniques. The techniques are based solely

Table 13.5 Summary of the characteristics of the models reviewed

Source Modes Quantitative? Factors influencing LOS

Alter (1976) All transit modes No Accessibility, travel time, reliability, service directness, service frequency, onboard density.

Horowitz (1981) Bus Yes In-vehicle time, transfer time, walking time, waiting time.

Polus and Shefer (1984) Bus Yes External factors (route length, stop spacing, number of intersections on route, etc.).

Madanat et al. (1994) Bus Partially Onboard density.

Henk and Hubbard (1996) All transit modes Yes Coverage, frequency, capacity.

Murugesan and Rama

Moorthy (1998) Bus Partially (Twenty factors are discussed.)

Friman et al. (1998) All transit No Driver behavior, reliability, network simplicity, information.

TCRP 35 (1999) All transit modes Yes Accessibility, passenger load.

Prioni and Hensher (2000) Bus Yes Tariff, travel time, access time, driver friendliness, smoothness.

Ryus et al. (2000) All transit modes Yes Frequency, service coverage, stop location, operation hour, population density.

Guttenplan et al. (2001) Bus Yes Factors related to the pedestrian environment (difficulty in crossing road, etc.).

Yang and Wang (2001) All transit modes Yes Network density, population density.

Tyrinopoulos and

Aifadopoulou (2008) All transit modes Partially Thirty-nine LOS indicators are presented.

Lai and Chen (2011) All transit modes Yes Several core service and psychical environment factors are discussed.

Eboli and Mazzulla (2011) Bus Yes Eleven service aspects and twenty-six service attributes are discussed.

Mishra et al. (2012) Multimodal transit Yes Four connectivity indicators from the node, line, transfer center and region levels.

Hadas (2013) Bus Yes Four connectivity indicators are introduced for the analysis of PT systems.

on Google transit feeds and any available road layer with no need for additional data. Four

connectivity indicators are introduced for the analysis of transit systems: (a) road coverage

level indicator, (b) intersection coverage level indicator, (c) stop transfer potential indicator,

and (d) routes overlap indicator. Finally, this study demonstrates the proposed methodology

with the analysis of the transit systems in Auckland (New Zealand), Vancouver (Canada),

and Portland (Oregon, United States).

The major features of the LOS indicators reviewed are presented in Table 13.5.

EXERCISES

13.1 Given the following network (the numbers on the arcs represent capacities in thousand passengers/hour),

(a) Find the maximum value 1-8 (s-t or O-D) flow, its distribution and min cut.

(b) Find the maximum value 4-5 (s-t or O-D) flow, its distribution and min cut.

Use for both (a) and (b) the augmentation algorithm shown in Appendix 6.A. Flow capacity (in thousand passengers/hour) 1 4 5 3 8 7 4 2 2 1 2 3 3 6 1 1 1 3 1 3 6 2 4

13.2 Given the following network (the numbers on the arcs

represent capacities in thousand passengers/hour),

(a) Find all the minimum cuts between nodes 1 (origin) and 6 (destination).

(b) Assume that the cost to increase the capacity of each arc by one unit (each unit is in terms of 1,000 passengers per hour) is \$1,000 (e.g., increase frequency of service). What is the minimum-cost procedure to increase the maximum value of the 1-6 (O-D) flow by one unit? 1 3 6 5 4 2 2 3 1 3 1 4 2 1 Flow capacity (in thousand passengers/hour)

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Prioni, P . and Hensher , D . (2000). Measuring service quality in scheduled bus services. J ournal of Public T ransportation, 3(2), 51-74.

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Tyrinopoulos, Y . and Aifadopoulou, G. (2008). A complete methodology for the quality control of passenger services in the public transport business. European T ransport, 38, 1-16.

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14 Chapter 14: Coordination and Connectivity: Behavioral Aspects

There are always two sides to a story. The same can be said about perception of public transit

services. While operators are focused on minimizing cost of service, passengers are concerned

about maximizing ease of use. Travelers demand an efficient and reliable door-to-door service

from transit systems to consider it a viable alternative to private vehicles. Transport agencies

have responded by implementing integrated multimodal systems with effective interconnectivity

as a strategy for attracting and retaining patronage. Integration supports service providers

in reducing operational costs while increasing network coverage. It also means more routes

involving transfers for users. Many studies have shown that passengers consider transfers to be

an inconvenience. This chapter discusses strategies to increase public users' willingness to make

transfers. The goal is for passengers to perceive transfer as a beneficial component of their trip

rather than a disturbance. This chapter consists of five principle parts, following an introductory section. Section

14.2 discusses the psychological factors that create intentions to make transfers. Section 14.3

outlines the two cognitive models that can be used to assess transfer behavior. Section 14.4

describes the results of a study that adopted Weber's law just noticeable difference to determine

the travel-time and cost savings desired by public transit users. Section 14.5 provides a guideline

for policymakers and planners for designing well-connected transfers in an integrated transport

network. Each principle section ends with recommendations for practitioners. The chapter

ends with a literature review and exercises. Methods and solutions presented in this chapter are expected to assist network planners and

operators to include seamless transfers in an integrated multimodal transit system as a service

to passengers. This inclusion will offer travelers with a viable alternative to private cars and

thus increase patronage for service providers. Three-part notes for practitioners appear at the

end of Sections 14.2.2, 14.4.4, and 14.5.2. Finally, the complications, though solvable, involved

with understanding transit users can, to some extent, be described by the saying of Friedrich

Nietzsche: "You have your way. I have my way. As for the right way, the correct way, and the

only way, it does not exist."

14.1 INTRODUCTION

In today's society, for reasons ranging from the increase in demand for employees' flex-

ibility in the labor market to the decline of traditional household travel patterns, private

vehicles play a dominant role in travel behavior. In 2004, 23% of the world's energy-related

greenhouse gas emissions were from the transport sector and private vehicles contributed

44% to that share. The need for user-friendly transit systems has become crucial with road

transport contributing significantly toward climate change. The Intergovernmental Panel

on climate change recommended modal shift from private vehicles to sustainable transport

to be a key climate change mitigation strategy. In March 14, 2008, the Transportation

Research Board released a research problem statement titled What Do Passengers Want,

What Do Passengers Need? The statement discussed that little is known regarding what

passengers want, need, or expect from intermodal facilities. Both the motivation to change

and the means to facilitate such change, by the individual overcoming the perceived barriers,

are required for altering travel behavior. The following article discusses sustainable travel

behavior as a new era for passenger transport.

From The Economist, September 22, 2012 (The Road Less Travelled).

Modern life is unimaginable without the car. The automobile has powered the growth of cities and steered their sprawl. Its manufacture has created millions of jobs and eased the development of many millions more. In rich countries, 70% of journeys are now by car. More than a billion cars now roll on the world's roads. Measured globally, car use will go on rising, for as people in emerging markets get rich, they want the mobility and status that car-ownership offers. But in the rich world the decades-long link between rising incomes and car use has been severed, and miles driven per person have been falling. That is partly because of recession and high oil prices, but the trend pre-dates 2007. Other, longer-term, factors are at work. One is generational: car-ownership is reaching saturation. Congested roads, smog and fears about global warming have led many cities to try to change

the way people move around. Tokyo has shown that mass-transit systems need not be poor

or dirty. Portland, which grew with the car, has since the 1980s developed its light rail.

London has devoted more space exclusively to buses and cycles; cars pay to enter the centre.

Singapore has congestion-pricing too. For the past 30 years Copenhagen has cut the number

of parking bays by 3% a year. By contrast, in places where petrol is under-taxed so the

motorists are shielded from the costs of the pollution (America) or where urban design has

included public transport as an afterthought (Los Angeles), policy has supported the car.

Governments in emerging markets, where hundreds of cities are taking shape, should learn

from mistakes and successes elsewhere. Policies that encourage people to drive into urban

centres—by, for instance, requiring businesses to offer parking spaces for employees and

customers—condemn metropolitan areas to heavy car use and congestion. Planning that

provides mass transit systems and good pedestrian and cycle ways can make them more

efficient. That is happening in some places. China is building rail networks in more than

80 cities. Eighteen Indian cities are developing metro systems. Yet many cities continue to

drive themselves round the same bend as in the developed world. Bangkok, Dhaka and

Jakarta are building more freeways in response to already clogged ones.

The car will bring freedom and fun to millions in emerging

markets, just as it has done

in the rich world. But if technology and policy are enabling people to do cheaper and

cleaner ways to work and enjoy themselves, that is all to the good.

To make public transit a viable alternative to private vehicles, globally, transport agencies have

been developing strategies to produce an effective integrated multimodal system. The objective

of such integrated systems has been to provide travelers with a wide spectrum of destination

choices through a convenient, accessible, comfortable, safe, speedy, and affordable transport systems

tem. This can be achieved by strategically positioning transfer points in the network to optimize

resources. However, travel behavior specialists have established that passengers find transfers to

be an inconvenience to their journey. As such, transport modelers give routes involving transfers a

penalty termed transfer penalty. With this in mind, this chapter attempts to answer the question:

What can be done to increase passengers' willingness to use routes that involve transfers?

Leaving one's comfort zone is never easy. Change is always undertaken with caution by

every individual in matters of concern. The idea of sustainability is no longer new, but the

concept of sustainable transport is. As such, transport practitioners must assist in the devel

development of creating more attractive integrated transit systems, and the rest step in doing so

is to understand what is desired by the passengers.

14.2 PSYCHOLOGICAL FACTORS INFLUENCING PASSENGERS' INTENTIONS TO MAKE TRANSFERS

Many researchers have confirmed that travel behavior does not depend directly on the

objective service level. Instead, it depends on psychological factors, such as beliefs, that can

be influenced by the service level of the transport system. The theory of planned behavior

(TPB), a well-established social psychological model, has been commonly adopted in travel

behavior studies to gain such knowledge.

14.2.1 Theory of planned behavior

The TPB was first proposed by Ajzen (1985). As shown in Figure 14.1, one central assumption

of the theory is that intention is the main direct determinant of behavior. Intention

is viewed as the resultant of three antecedents: attitude, social norm, and perceived

behavioral control (PBC). The strength of intention provides an indication of how hard

people are willing to try, of how much effort they are willing to exert, in order to conduct

the behavior. In other words, intention captures the motivational factors that influence

behavior. According to the TPB, intention is viewed as the resultant of three anteced

ents: attitude, social norm, and PBC. These intention antecedents are based on behavioral

beliefs, normative beliefs, and control beliefs, respectively. Ajzen (2005) claims that these

beliefs are the foundation of behavior and changes in these beliefs should lead to change

in behavior. The rationale for a direct link between PBC and behavior is that given a suf

ficient degree of actual control over the behavior, people are expected to carry out their

intentions when requisite opportunities and resources (e.g., time, money, skills) are avail

able (Ajzen, 2002).

Individuals create beliefs about behavior by associating it with certain attributes. These

beliefs are termed behavioral beliefs and form the basis of attitude. Attitude is defined as the

individual's positive or negative evaluation of performing the intended action. For example,

Iseki and Taylor (2009) have shown that passengers perceive transfers to be burdensome

due to the extra effort required, that is, generally, there is a negative attitude toward making

transfers. Normative beliefs are created from approval or disapproval of important reference
Behavioral beliefs Normative beliefs Control beliefs
Attitude Social norm Perceived behavioral control
Intention Behavior

Figure 14.1 Theory of planned behavior. (From Ajzen, I., Attitudes, Personality and Behavior, McGraw-Hill, New York, 2005.)

group about performing behavior. These beliefs form the basis of social norm defined as the

individual's perception of social obligation to perform or not perform the intended action.

For example, Bamberg et al. (2007) revealed that proenvironmental reasons, through social

norm, have a greater effect on the intention to use public transit if travelers are in a society

that is aware of the negative environmental consequences caused by cars and supports tran

sit use.

Control beliefs are created based on barriers that are perceived by the individual to under

take action. When individuals perceive a higher level of control, the control factor is no

longer important and the emphasis is placed on whether the individual has the intention to

perform the behavior, thus strengthening the intention-behavior relationship. When control

beliefs are weak, the individual is unlikely to have the intention to perform the behavior,

thus weakening the intention-behavior relationship. PBC refers to an individual's perception

of ease or difficulty in performing the intended behavior. For example, Bamberg and

Schmidt (2001) have shown that PBC is more positive when ease of using public transport

has been improved through better service quality.

14.2.2 Effects of PBC on intention and behavior

Two recent studies were undertaken to investigate the effects of PBC on passengers' inten

tions to use routes that involve making transfers.

14.2.2.1 Effects of improved trip attributes on passengers' intention

The first study by Chowdhury and Ceder (2013a) investigated the effects of improvements in

trip transfer-based attributes on passengers' intention to use routes with transfers. The trip

attributes selected are reliability of the connection, personal safety, transfer time (walking

and waiting), journey time, and information on making transfers. Figure 14.2 shows the

conceptual framework investigated by this study.

A survey was undertaken in two major transport hubs: New Lynn Transport Center and

Northern Busway, in Auckland, New Zealand. A five-point Likert scale (strongly disagree

to strongly agree) was used in the questionnaire. A neutral point (no opinion) was included

in the scale to acquire responses for all items in the questionnaire and to keep participants

interested in completing the questionnaire. The questionnaire was designed to be completed

within 10 min. To determine statistical difference in the responses for each of the items, the

data were fitted into generalized linear models (GLMs) of the Poisson family. Table 14.1

gives the results of the GLM models.

Findings of this study support the proposed theoretical framework, illustrated in

Figure 14.2. Statistical analysis showed that as transfer walking and waiting times increased,

passengers were less willing to use transfers. For delay in connecting vehicle, users were

negatively disposed to any delay greater than 5 min, even if occurrence was rare. During the

survey, participants verbally expressed that they are more comfortable with the direct route

service because it is inconvenient to make transfers due to past experience with unreliable

services. Reliability of the transfer duration and reduction in travel time were shown to have

the most influence on users' intention to use transfer routes. This result is in line with And

ings of Beirao and Sarsfeld-Cabral (2007).

Findings confirm the importance of the selected trip attributes. Results suggest that the

service provided by operators must match or exceed passengers' expectation for users to

have the intention to make transfers. According to TPB, users are required to possess strong

underlying control beliefs to produce a positive PBC, which, in turn, creates a higher inten

tion to use transfer routes. More details are in Chowdhury and Ceder (2013a).

14.2.2.2 Effect of PBC on passengers' intention

The second study by Chowdhury and Ceder (2013b) was conducted to further investigate

the effect of PBC on passengers' intention to make transfers. PBC was broken down to its

constituents, self-efficacy, and perceived controllability (PC) as shown in Figure 14.3.

Self-efficacy is defined as confidence in one's ability to perform the behavior. PC refers to

one's control over performance or nonperformance of the behavior. There exists sufficient

clear and consistent evidence for the distinction between self-efficacy and P.C. Findings

by Armitage and Conner (2001) have shown that the control beliefs used to measure self

efficacy and P.C. overlap. A certain level of correlation also exists between the two components

(Kraft et al., 2005). Ajzen (2002) discussed that past studies commonly used different

control beliefs to measure self-efficacy and controllability.

PC has been suggested to be predominantly dependent on external factors such as a

available resources and opportunities. Similarly, for the study of Chowdhury and Ceder (2013b),

control beliefs of PC were measured using operational trip attributes: reliability of connection,

transfer walking time, and transfer waiting time. Self-efficacy has been suggested to

be predominantly dependent on internal factors such as ability, perceived inconvenience, Service quality does not meet users' preference. Service quality exceeds or meets users' preference. Weak PBC-intention relationship Public transport users' preference Significant improvement in transfer quality is required to strengthen control beliefs. Strong PBC-intention relationship Travelers are willing to use routes with transfers, given improvements. Behavior Positive control beliefs are created. Negative control beliefs are created.

Figure 14.2 Effect of public transit users' preference for quality of interchange attributes on their perceived behavioral control-intention relationship.

Table 14.1 Results (Chowdhury and Ceder, 2013a) from generalized linear models

Likert item (variable) 1 (Strongly disagree) Intercept
 2 (Disagree) 3 (Neutral) 4 (Agree) 5 (Strongly agree)

Delay in connecting vehicle arrival

5 min 0.14 0.17 0.22 0.25 0.22 Reference (0.400) (0.058)
 (0.008) (0.036)

5-10 min 0.26 0.26 0.23 0.16 0.08 Reference (1.000) (0.446)
 (0.019) (0.000)

10-15 min 0.45 0.19 0.18 0.12 0.06 Reference (0.000)
 (0.000) (0.000) (0.000)

Transfer walking time

5 min 0.04 0.05 0.14 0.24 0.53 Reference (0.827) (0.002)
 (0.000) (0.000)

5-10 min 0.17 0.12 0.32 0.25 0.14 Reference (0.213) (0.001)
 (0.050) (0.548)

10-15 min 0.39 0.19 0.22 0.11 0.09 Reference (0.000)
 (0.000) (0.000) (0.000)

Transfer waiting time

5 min 0.06 0.04 0.13 0.25 0.51 Reference (0.534) (0.001)
 (0.000) (0.000)

5-10 min 0.14 0.15 0.30 0.30 0.11 Reference (0.901) (0.000)
 (0.000) (0.287)

10-15 min 0.40 0.22 0.22 0.12 0.04 Reference (0.001)
 (0.000) (0.000) (0.000)

Willingness to pay for security

Better lighting 0.25 0.09 0.25 0.14 0.27 Reference (0.000)
 (0.924) (0.012) (0.710)

CCTV 0.23 0.13 0.27 0.21 0.15 Reference (0.016) (0.398)
 (0.545) (0.054)

Security personnel 0.34 0.17 0.25 0.10 0.13 Reference
 (0.000) (0.0829) (0.000) (0.000)

Transfer information

Real-time display 0.04 0.03 0.10 0.17 0.66 Reference

(0.796) (0.014) (0.000) (0.000)

Advice on the Internet 0.05 0.06 0.15 0.24 0.50 Reference
(0.549) (0.002) (0.000) (0.000)

More information about tickets and routes 0.07 0.03 0.17
0.30 0.43 Reference (0.041) (0.004) (0.000) (0.000)

Intention to use transfer given improvement to service

Reliability 0.06 0.06 0.23 0.31 0.35 Reference (1.000)
(0.000) (0.000) (0.000)

Connection 0.04 0.07 0.32 0.29 0.28 Reference (0.151)
(0.000) (0.000) (0.000) (Continued)

and willpower. For the study, the control beliefs of self-efficacy were measured using trip

attributes that are commonly evaluated by user perception: personal safety at terminals

(Atkins, 1990) and need for information to make transfers (Molina and Chorus, 2009).

Three hypotheses were formed and tested. Figure 14.4 shows the hypothetical relationship

among the variables assessed for significance in passengers' willingness.

Hypothesis 1: The stronger is passengers' PC and self-efficacy, the greater will be their intention to use transfer routes.

Hypothesis 2: The stronger is the passengers' intention of use, the greater will be the probability of actually using a transfer route.

Hypothesis 3: Passengers' sociodemographics and trip characteristics have a direct effect on their self-efficacy and PC and an indirect effect on their intention.

A user preference survey was undertaken at two key transport hubs: New Lynn Transport

Centre and Newmarket Train Station in Auckland, New Zealand. These two, chosen from

the Auckland public transit network as survey locations, support a high percentage of

passengers who make either an intermodal or an intramodal transfer. The questionnaire

consisted of two sections: Section A and B. Section A contained sociodemographic and

trip characteristic questions. Section B consisted of measurement items for PBC, intention,

Table 14.1 (Continued) Results (Chowdhury and Ceder, 2013a) from generalized linear models

Likert item (variable) 1 (Strongly disagree) Intercept 2 (Disagree) 3 (Neutral) 4 (Agree) 5 (Strongly agree)

Information 0.06 0.13 0.41 0.19 0.20 Reference (0.0252) (0.000) (0.000) (0.000)

Travel-time savings 0.05 0.05 0.27 0.30 0.33 Reference (0.67) (0.000) (0.000) (0.000)

Choosing to use transfer route if all four improvements were made

New route with transfer 0.15 0.07 0.16 0.30 0.32 Reference (0.018) (0.718) (0.002) (0.000)

Old direct route 0.22 0.13 0.21 0.18 0.26 Reference (0.028) (0.739) (0.359) (0.527) Perceived behavioral control
Self-efficacy Perceived controllability Control beliefs
Control beliefs

Figure 14.3 Perceived behavioral control.

and behavior. The questionnaire was designed to be completed within 10 min Table 14.2

gives the measurement items and their notations. The data were analyzed using structural

equation modeling (SEM). As the TPB contains a chain of mediating causal variables, SEM

provides an appropriate statistical tool to determine the effects and relationships among the

variables. Figure 14.5 gives the final SEM model based on the definitions of Table 14.2. Fit

indices—such as the goodness-of-fit-index (GFI) value, 0.96; adjusted GFI, 0.90; normed

index, 0.91; comparative index, 0.97; and the root mean square error of approximation,

0.04—have indicated the model to be of good fit.

Taken together, results from the final model support the adoption of the TPB to explain

passengers' intention to use routes requiring transfers. In line with past findings, the two

constructs of self-efficacy were seen to have discriminant validity and similarity. Similar

to the findings by Ajzen (2002) and others studies (Armitage, 1999; Armitage and Conner,

2001), it is evident from the analysis that self-efficacy is more closely associated with intention

and behavior than PC. Only two observed variables of PC had a significant effect on

intention: next vehicle in 10 min plus a 5–10 min walk (PC4) and next vehicle in 15 min plus

a 5–10 min walk (PC5). The next vehicle in 10 min plus a 5–10 min walk (PC4) variable had

a positive effect, whereas other significant variable (PC5) had a negative effect. This finding

is in accordance with the design of the measurement items. The next vehicle in 15 min plus

a 5–10 min walk (PC5) variable was designed to be the worst connection scenario of the

four items related to transfer waiting and walking time. Overall, the results seem to

upport

Hypothesis 1: the stronger the public transit users' PC and self-efficacy, the greater will be

their intention to use transfer routes. The significant positive coefficient (0.29), concerning

the path from intention to decision, supports Hypothesis 2 (Chowdhury and Ceder, 2013b).

Contrary to Hypothesis 3, it was found that sociodemographics and trip characteristics

have direct effects on intention. Only gender, frequency of transit use, and current use of

transfer routes had significant effects on intention. The positive effect of frequency and cur

rent use of transfers on intention indicates that positive control beliefs are related to regular

use of public transit. Such finding is consistent with past studies that have shown that the

transfer penalties given by infrequent passengers tend to be higher than those given by regular

passengers (Guo and Wilson, 2007; Blainey et al., 2012). Regular passengers tend to be more

familiar with the system and therefore are less likely to face uncertainties when transferring. Self-efficacy
Perceived controllability

Sociodemographics and trip characteristics Control beliefs
Control beliefs Intention Behavior Perceived behavioral
control

Figure 14.4 Hypothetical model explaining the causal relationship among the latent and manifest variables.

Male participants were seen to be less willing to use transfer routes than female participants.

A probable reason for the gender effect is that the different travel patterns between men and

women, due to household responsibilities, childcare, and employment status.

14.2.2.3 Notes for practitioners

The importance of well-connected public transit routes has been highlighted by the findings

of the two psychological studies. Passengers must possess strong and positive underlying

controlling beliefs, particularly those associated with confidence in their ability to make

transfers, for them to have the intention to use routes involving transfers. The perceived

benefits of using routes involving transfers need to be greater than their perceived barriers

of making the connections. This section recommends that improvements in service quality

of transfers need to be targeted at supporting the users' ability to make transfers successfully

Table 14.2 Measurement items

Measurement items Statements

Self-efficacy

Need for information (CON1) I will feel more confident to use transfer routes with high-quality information services.

Perception of security (CON2) I will feel more confident to use transfer routes with good security facilities at stations/stops.

Need for information (PD1) With high-quality information services, how easy or difficult will it be for you to use transfer routes?

Perception of security (PD2) With good security facilities, how comfortable or uncomfortable will it be for you to use transfer routes?

Perceived controllability

Reliability

(PC1) I will feel more in control (less anxious) about using transfer routes with highly reliable connections.

Transfer walking and waiting time Personally, I will feel more in control (less anxious) about making the transfer if

(PC2) The next vehicle is scheduled to arrive in 5 min and walking to the next vehicle takes 2-5 min.

(PC3) The next vehicle is scheduled to arrive in 10 min and walking to the next vehicle takes 2-5 min.

(PC4) The next vehicle is scheduled to arrive in 10 min and walking to the next vehicle takes 5-10 min.

(PC5) The next vehicle is scheduled to arrive in 15 min and walking to the next vehicle takes 5-10 min.

Intention

(Inten1) Given this opportunity, I am willing to use transfer routes for my daily trips (e.g., work, recreation).

(Inten2) I intend to use transfer routes to combine most of my daily trips (e.g., work, recreation).

Behavior

(Decision) I will choose to use transfer routes for most of my daily trips (e.g., work, recreation).

Source: Chowdhury, S. and Ceder, A., Int. J. Trans., 1(1), 3, 2013b.

and with ease. Therefore, authorities need to focus on developing attractive transfer routes

(travel-time s avings) w ith c omfortable t ransfers
(easy s witch b etween v ehicles), f rom a u ser

perspective, to encourage ridership and increase patronage of public transit.

14.3 EFFECTS OF VARIATION IN OUT-OF-VEHICLE TIMES ON PASSENGERS' DECISIONS TO MAKE TRANSFERS

Out-of-vehicle times are perceived as being more onerous than in-vehicle times by passen-

gers when making transfers. In this section, two cognitive models will be discussed to assess

the effects of variation in out-of-vehicle times on passengers' decisions to make transfers.

Out-of-vehicle times are defined as the duration of time when beginning a trip (waiting

times) and when making a transfer (walking and waiting times). Iseki and Taylor (2009)

show that perceived waiting time is more onerous than the actual waiting time and is dependent on

waiting conditions such as personal safety, reliability of connection, and comfort.

These authors further show that reducing perceived walking and waiting times, for trans-

fers, substantially increases the attractiveness of public transit.

14.3.1 Route choice problem

When modeling route choice in travel behavior studies (see Chapter 11), it has been a com-

mon practice to assume that passengers have perfect knowledge about their choices and

make rational decisions based on utility maximization. Discrete choice models derived

using expected utility theory (EUT) and random utility theory (RUT) have been com-

monly adopted to analyze choice in travel behavior studies. An increasing number of route

choice studies have been challenging the assumption of absolute rationality of passengers by Decision PC5
 Intention z1 Inten1 Inten2 e8 0.287*** 0.623*** 0.565***
 Self-efficacy 0.422*** Gender Frequent Transfer PC4
 -0.149** 0.208** 0.399** -0.266** 0.504***

e4

e5 CON1 CON2 PD1 PD2

e6

e7 Perceived difficulty z2 1.0 0.754*** Confidence z3
 0.853*** 1.0 1.0 0.726*** 0.30 e9

Figure 14.5 Final structural equation modeling model (**p ≤ 0.01, ***p ≤ 0.001).

showing evidence of violations of EUT (Avineri, 2004; Xu et al., 2011). Statistical models

using the assumption of utility maximization overlook the fact that human decision-making

process is approximate rather than precise (Ceder et al., 2013).

14.3.2 Cumulative prospect theory: A brief introduction

In 1992, Amos Tversky and Daniel Kahneman introduced the cumulative prospect theory

(CPT), an extension of the prospect theory (Tversky and Kahneman, 1992). The theory

conceives human choice behavior to be predominantly a n intuitive process rather than a

conscious deliberate one. The central feature of CPT is that it is a ble to model diminishing

sensitivity in human decision-making process (Brandstatter et al., 2002). Pesendorfer (2006)

discussed that one of the main shortcomings of CPT is the dependency of the outcome frame

ing on the reference point. The reference point is often obtained by researchers from experience

mental settings and thus essentially unobservable outside the experiment. Researchers have

often dealt with this issue by treating the reference point as a free variable chosen to match

the observed behavior. Others have proposed the use of multiple reference points (Senbil

and Kitamura, 2004). Van de Kaa (2010) suggested that, for the moment, the estimates of

the CPT parameters obtained by Tversky and Kahneman (1992) offer the best functional

description of choice under uncertainty in a wide variety of travel behavior context.

14.3.3 Fuzzy logic: A brief introduction

Fuzzy logic, first introduced by Zadeh (1965), enables formation of logical statements to

compute vagueness. Using the concept of approximate reasoning, fuzzy logic makes it

possible to model imprecision in human reasoning and thus decision making. Each fuzzy

logic system can be divided into three stages: fuzzyification, fuzzy inference, and defuzzi-

cation. Figure 14.6 shows the link among the stages and the input and output of each

stage. A disadvantage of fuzzy logic is the task of fine-tuning the membership functions and

adjusting the fuzzy rules. Construction of the membership functions and the fuzzy rules

is a trial-and-error process until an appropriate set of the input-output set is achieved. It is

evident from a number of route and mode choice studies (Murugesan and Rama Moorthy,

1998; Teodorovic, 1999) that fuzzy logic is capable of modeling ambiguity in perception and

appraisal of trip attributes. Input numerical values Fuzzy inference Output linguistic variables Input linguistic variables Linguistic terms Numerical values Output numerical values Fuzzification Defuzzification

Figure 14.6 Fuzzy logic.

14.3.4 Assessment of modeling

Ceder et al. (2013) conducted a study that utilized the two models for assessing the effects

of variation in out-of-vehicle times. A user preference survey was undertaken at two major

terminals: Britomart Transport Centre and Newmarket Train Station, in Auckland, New

Zealand. Respondents of the survey were presented with a total of 15 hypothetical cases,

three cases (A, B , C) for each of the out-of-vehicle time variables. The variables chosen

are transfer walking time, transfer waiting time, delay in connection, security at stops,

and seating availability for comfort. Each case consisted of a direct route (Scenario 3) and

two transfer routes (Scenarios 1 and 2) with an equal travel-time savings of 15-20 min in

comparison to the direct route. For the transfer routes, one of the scenarios was designed to

be perceived as being less conservative (risky route) and the other to be more conservative

(less risky route). Three cases with varying probabilities of uncertainty in trip attributes

we re p resented to b etter u nderstand t he v ariations i n u sers' r isk-taking b ehavior w ith v ari

ations i n r isk p robabilities. P articipants we re i nstructed to s elect o ne o f t he t hree s cenarios

fo r e ach c ase. F igure 1 4.7 s hows o ne o f t he c ases (Case A) fo r t ransfer w aiting t ime.

14.3.4.1 Results o f a s s e s s m e n t

14.3.4.1.1 C u m u l a t i v e p r o s p e c t t h e o r y m o d e l i n g

According t o T versky a nd K ahneman (1992), t he m a i n f o r m u l a e u s e d i n c a l c u l a t i n g t h e

c u m u l a t i v e p r o s p e c t v a l u e s a r e $V(f) = V(f) + \sum_{i=1}^n \pi_i \nu(x_i)$ (14.1) $V(f) = \sum_{i=1}^n \pi_i \nu(x_i)$ (14.2) $V(f) = \sum_{i=1}^n \pi_i \nu(x_i)$ (14.3)

where $V(f)$ is the cumulative prospect value $V(f +)$ is the prospect gain value

$V(f -)$ is the prospect loss value 20% Scenario 1 15 min of waiting 9 min of waiting 80% Scenario 2 20 min of waiting 5 min of waiting 30% 70% Tick one -Scenario 1 (more conservative) -Scenario 2 (less conservative) -Scenario 3 (direct route)

Figure 14.7 Case A for transfer waiting time.

The decision weights for gains π_i^+ and losses π_i^- are given by $\pi_i^+ = \omega_i^+ p_i^+$ (14.4) $\pi_i^- = \omega_i^- p_i^-$ (14.5) $\pi_i^+ = \omega_i^+ p_i^+$ (14.6) $\pi_i^- = \omega_i^- p_i^-$ (14.7)

The functional form is used to calculate value functions for gains $(x \geq 0)$ and losses $(x < 0)$

as f o l l o w s : $\nu(x) = \lambda x^\alpha$ (14.8)

The w e i g h t e d f u n c t i o n s f o r g a i n s a n d l o s s e s a

-5.67 -7.02 -7.22 -5.27 6.5 0.09 0.14 -0.21 0.58 -1.18 1.71
 11.5 3.72 4.28 4.36 4.55 4.55 5.47 Proportion of
 respondents from survey data (%) 37.0 55.0 32.0 60.0 22.7
 69.3 Delay time Case A Case B Case C S1 S2 S1 S2 S1 S2
 Reference point (min) Cumulative prospect value 2.5 -5.10
 -6.69 -6.25 -10.33 -6.17 -7.15 7.5 0.11 0.80 -1.12 -1.30
 -0.76 -0.55 12.5 4.62 4.90 3.11 3.64 4.12 4.49 Proportion
 of respondents from survey data (%) 22.0 70.0 31.7 59.7
 30.7 61.0 Comfort (seating while waiting) Case A Case B
 Case C S1 S2 S1 S2 S1 S2 Reference point (min) Cumulative
 prospect value 2.5 -14.02 -12.54 -6.25 -10.88 -17.02 -9.46
 4.5 -10.86 -9.04 -3.71 -5.93 -14.83 -5.93 6.5 -7.55 -14.48
 -1.97 -3.27 -8.23 -3.27 Proportion of respondents from
 survey data (%) 53.3 38.3 51.0 40.3 21.0 71.0 Personal
 security (at terminal) Case A Case B Case C S1 S2 S1 S2 S1
 S2 Reference point (min) Cumulative prospect value 1 -7.57
 -9.07 -7.40 -8.08 -9.03 -8.48 3 -4.54 -5.60 -4.54 -3.63
 -6.03 -5.00 5 -2.79 -2.50 -2.79 -1.45 -4.19 -1.84
 Proportion of respondents from survey data (%) 37.7 54.3
 24.7 67.3 21.0 71.0 Source: Ceder, A. et al., Trans.
 Policy, 27, 112, 2013.

Step 2: Fuzzy inference

Fuzzy inference contains a set of if-then logic statements.
 The fuzzy rules were in the following form.

If [trip attribute] is [X TA] and [ΔWT] is [X
 WT], then ridership is [Y R].

Based on Figure 14.8, the input data are fuzzified, and then
 based on the fuzzified input data, the corresponding fuzzy
 rules were used. The max-min composition method is applied
 for making the fuzzy inference. This procedure
 is shown in Table 14.4.

Step 3: Defuzzification

Defuzzification is the final stage of the fuzzy system. A
 common approach is the center of gravity method (Zhang and
 Prevedouros, 2011). The process involves converting the
 fuzzy inference outputs into an aggregated (crisp) value.
 The aggregated value, denoted y^* , will change continuously
 with continuous change in the input values, and its ex-
 pression is shown in Equation 14.7. $y^* = \frac{\int y \mu(y) dy}{\int \mu(y) dy}$ (14.11)

An example from Ceder et al. (2013) shows the resulting
 ridership categories from the fuzzy

inference; these results were F (0.38) and G (0.62). The center of gravity method of this example

is illustrated in Figure 14.9. The shaded area represents the ridership categories (F and G) for

which the y^* calculation was undertaken (Zhang and Prevedouros, 2011). Based on the result

shown in Figure 14.9, 24% of transit users who were willing to use transfer routes preferred

Scenario 1 and 76% preferred Scenario 2 (see Figure 14.7 for the two scenarios of Case A).

Table 14.5 presents the comparison between the fuzzy model output and the survey data.

Analysis of the survey data revealed that for all trip attributes, except for comfort, users

exhibited greater preference for the scenario that was perceived to be more conservative despite

a higher probability of shorter out-of-vehicle time in the less conservative scenario. For comfort,

users displayed risk-taking characteristics when the waiting time for an available seat was less

than 5 min. Results in Tables 14.3 and 14.5 from the CPT and fuzzy logic model, respectively,

illustrate that the two models are capable of representing users' out-of-vehicle behavior when

making transfers. The cumulative prospect values were able to accurately reflect public transit

between the cumulative prospect values for the two scenarios increased, and the difference in

the proportion of participants' preference also increased, with participants favoring the higher

cumulative prospect value scenario. Analysis also revealed no statistical evidence of significant

difference between outputs from the fuzzy system developed and the survey data. Outputs of

the fuzzy system were within 5% of the actual response. Therefore, while the cumulative pros

pect value is able to provide an indication of users' preference, fuzzy logic is capable of providing

the proportion of users preferring a transfer route. More on this study is in Ceder et al. (2013).

14.4 JUST NOTICEABLE DIFFERENCE OF TRAVEL AND COST SAVINGS FOR ROUTE CHOICE

Two trip attributes commonly considered by passengers during route choice are travel time

and travel cost. A recent study was undertaken by Chowdhury et al. (2015) to determine the

minimum travel-time and cost savings required from passengers' perception to use routes

with t ransfers. T he s tudy a dopted W eber's l aw j ust n oticeable d ifference (JND). 0 0.2 0.4 0.6 0.8 1 5 10 15 20 25 μ Transfer waiting time (min) Low Moderate High 0 0.4 0.8 1.2 1.6 2 2.4 0 0.2 0.4 0.6 0.8 1 μ WT of waiting time (min) Low Moderate High 0 0.2 0.4 0.6 0.8 1 0 2 4 6 8 10 12 14 μ Transfer walking time (min) Low Moderate High 0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 WT of walking time (min) Low Moderate High 0 0.2 0.4 0.6 0.8 1 4 6 8 10 12 14 16 18 20 μ Transfer delay (min) Low Moderate High 0 0.4 0.8 1.2 1.6 2 2.4 0 0.2 0.4 0.6 0.8 1 μ 0 0.2 0.4 0.6 0.8 1 μ WT of delay (min) Low Moderate High 0 0.2 0.4 0.6 0.8 1 0 5 10 15 20 μ Personal safety (min) Low Moderate High 0 0.2 0.4 0.6 0.8 1 1.2 1.4 WT of personal safety (min) Low Moderate High 0 0.2 0.4 0.6 0.8 1 5 10 15 20 25 μ Seating availability (min) Low Moderate High 1 1.4 1.8 2.2 2.6 3 3.4 3.8 0 0.2 0.4 0.6 0.8 1 μ 0 0.2 0.4 0.6 0.8 1 μ WT of seating (min) Low Moderate High 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1 0 10 20 30 40 50 60 70 80 90 100 μ Pro porti on of users willing to use transfer route (%) AB C D E FGH

Figure 14.8 Fuzzy sets for input and output variable form. 0 0.1 M e m b e r s h i p v a l u e (μ) 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1 0 10 20 30 40 Proportion of transit users willing to use transfer route (Scenario 1) 24%

Figure 14.9 Defuzziication for ridership G and F using center of gravity method for transfer delay time of Case

A, Scenario 1. Table 14.4 Max-min composition method
 Fuzzification of input data for Scenario 1 Input variable
 Input data Fuzzified category Membership grade DT 3 min Low
 1.0 12 min Moderate 0.88 High 0.12 WT 1.1 Low 0.38 Moderate
 0.62 Fuzzy inference for Scenario 1 Rule no. Input data
 Ridership Max-min compositionDT WT 9 Low Low F Min (1.0,
 0.38) = 0.38 (1.0) (0.38) 3 Low Moderate G Min (1.0, 0.62)
 = 0.62 (1.0) (0.62) 1 Moderate Moderate G Min (0.88, 0.62)
 = 0.62 (0.88) (0.62) 2 High Moderate G Min (0.12, 0.62) =
 0.12 (0.12) (0.62) 8 High Low F Min (0.12, 0.38) = 0.12
 (0.12) (0.38) 7 Moderate Low F Min (0.88, 0.38) = 0.38
 (0.88) (0.38) Ridership F: Max (0.38, 0.12, 0.38) = 0.38
 Ridership G: Max (0.62, 0.12, 0.62) = 0.62 Source: Ceder,
 A. et al., Trans. Policy, 27, 112, 2013.

14.4.1 Just noticeable difference

JND, also known as the difference threshold, originates in the field of experimental psychol

ogy (psychophysics). The concept of JND was first discovered by the experimental investi

gations conducted by Ernst Weber (1795–1878), an anatomist and psychologist (Baird and

Noma, 1978). Weber's law states that when two stimuli are compared, rather than simply per

ceiving the difference between the stimuli being compared, human beings perceive the ratio of

difference to the magnitude of the stimuli. Weber's law has been mathematically formulated as $\Delta I / I = k$ (14.12)

where

ΔI is the change required for a JND in the stimulus magnitude

I is the stimulus magnitude

k is the constant for the particular sense

Table 14.5 Comparison between fuzzy model and survey data

Trip attribute Case Scenario Fuzzy output (set B) Survey data (set B) Chi-squared/p-value

Transfer waiting time A S1 97 107 1.773/0.183 S2 42 32 B S1

56 54 S2 84 86 C S1 56 49 S2 83 90

Transfer walking time A S1 56 59 0.191/0.662 S2 84 81 B S1
56 54 S2 84 86 C S1 28 33 S2 112 107

Transfer delay time A S1 28 37 1.041/0.307 S2 112 103 B S1
42 47 S2 97 92 C S1 42 41 S2 98 99

Comfort A S1 82 83 0.175/0.675 S2 55 54 B S1 68 73 S2 68 63
C S1 27 29 S2 110 108

Safety A S1 55 50 0.071/0.789 S2 82 87 B S1 27 34 S2 110
103 C S1 27 29 S2 110 108

Source: Ceder, A. et al., Trans. Policy, 27, 112, 2013.

The values of k have been seen to vary over low, medium,
and high magnitude ranges

of the stimuli. As the k value decreases, the perceptual
sensitivity improves (Baird and

Noma, 1978).

14.4.2 Survey as a case study

A user preference survey was conducted (Chowdhury et al.,
2015) at the University of

Auckland's city campus to test the JND concept. The study
location is shown in Figure 14.10.

The variables in Equation 14.12 were defined as I
representing the current total travel time

or travel cost experienced by the individual and ΔI defined
as the change in travel-time or

cost savings that motivate the passenger to switch to the
alternative route with a transfer.

Two surveys were conducted, with the main focus of one
survey being travel-time savings

(Survey I) and the other being travel-cost savings (Survey
II). The influence of comfort at Study area Auckland domain
Grafton Rd Stanley St St Pauls e University
of Auckland Grafton Rd Symonds St e
University of Auckland Churchill St Alton Rd
e University of Auckland Alfred St St Paul St Wake

University of Auckland
Victoria St E

Vulcan Lane Bacon Ln High St

Q

Queen St Albert park Kitcheners St Lorne St
Queen St Bowen Ave Chancery St Water
loo Quadrant High court Princess St Wyny
ard St Symonds St University of Auckland City
Campus

Key Bus stop within study area Mayor's Dr Court
house Ln Kitcheners St Mount St

Figure 14.10 The study area of the survey conducted by
Chowdhury et al. (2015).

the connection point was included in both surveys to
determine its effect on the JND con

stants for travel-time and cost saving.

14.4.3 Results of the case study

To determine the average k values, the trimmed means of the
two survey data were calculated.

Wilcox (2009) recommended a 20% trimming of the data to
account for outliers. The 20%

Winsorized mean (W) represents the average k for users'
travel-time and cost savings. The

ranges of the average k were determined by calculating the
95% Winsorized confidence

interval (C.I.). Table 14.6 includes the 20% Winsorized
variance (s_w^2) and W of the two data

set with the range for both comfort levels at
the interchange.

A curve fitting regression analysis was undertaken using the
Statistical Package for

the Social Sciences (SPSS) with 300 data points for each
comfort level. Gender of the

participants did not have a statistically significant effect on the k values. To deter

mine the relationship between the k values and participants' current travel time, the

best fitted curve model was chosen through observation and the R² value. The selected

curve illustrated an inverse relationship between the k values and participants' current

travel time. The following equations give the relationship for each comfort level at the

connection point: k current travel time (mi basic
interchange comfort() . . = +0.249 3.08 n) (14.13) k
0.172 3.77 current travel high quality interchange
comfort() = + time (min) (14.14)

Distinct patterns were identified among the k values. It is shown in Figure 14.11 that 94%

of the participants selected a ΔI following inverse curves at intervals of 5 min. The diagram

illustrates that majority (70%) of the participants desired a travel-time savings between 10

and 20 min for basic comfort at the interchange. The change from a basic to a high-quality

comfort at the connection point increased the number of participants following the ΔI curve

of 5 min (41-77). The number of participants on the ΔI curve of 10 min also increased

(87-103). Figure 14.12 illustrates that majority (77%) of the participants desired a time sav

ing between 5 and 15 min for a better-quality connection point.

For travel-cost savings, gender of the participants did not have a statistically significant

effect on the k values. The k values were grouped in accordance with participants' current

travel cost. Table 14.7 gives the 20% Winsorized mean ($\bar{W}$) and variance (s_w^2) of the k values

for each group.

Table 14.6 JND for travel-time and cost savings

Trip attribute Basic interchange facilities High-quality interchange facilities

Travel-time savings $\bar{W}$ 0.33 ± 0.014 0.25 ± 0.012 s_w^2
 0.0057 0.0043

Travel-cost savings $\bar{W}$ 0.16 ± 0.011 0.10 ± 0.010 s_w^2
 0.0029 0.0025

Source: Chowdhury, S. et al., J. Trans. Geogr. 43, 151, 2015. $\Delta I=5$ min $\Delta I=10$ min $\Delta I=15$ min $\Delta I=20$ min $\Delta I=25$ min $\Delta I=30$ min

0.00

k

v a

l u

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r c

h a

n g

e

0.10

0.20

0.30

0.40

0.50

0.60

0.70

0.80 0 10 20 30 40 50 60 70 80 90 Current travel time (min)
Data points following the ΔI inverse curves at 5 min
interval Data points that are in between the ΔI inverse
curves at 5 min interval

Figure 14.11 Inverse relationship between k values and
transit users' current travel time for basic quality
comfort at interchange. $\Delta I=5$ min $\Delta I=10$ min $\Delta I=15$ min $\Delta I=20$
min $\Delta I=25$ min $\Delta I=30$ min

0.00

0.10

0.20

0.30

0.40

0.50

0.60

0.70

0.80 0 10 20 30 40 50 60 70 80 90 Current travel time (min)

k

v a

l u

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h

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q u

a l i

t y

c o

m

f o

r t

a t

i n

t e

r c

h a

n g

e Data points following the ΔI inverse curves at 5 min interval Data points which are in-between the ΔI inverse curves at 5 min interval

Figure 14.12 Inverse relationship between k values and transit users' current travel time for high-quality

comfort at interchange.

14.4.4 Discussion of the JND analysis

Results of the analysis, given in Table 14.6, have shown that the average JND constant for

travel-cost savings is less than the average JND constant for travel-time savings in both

interchange comfort scenarios. An explanation for this finding is that the majority of the

survey participants were university students. Comfort at the connection point was seen to

have a positive effect in reducing the average JND constant for both travel-time and travel

cost savings. On average, passengers desired at least a 33% reduction in their current travel

time and at least a 16% reduction in their current travel cost given basic comfort at the

interchange. For an interchange with more comfort, on average, passengers desired at least

a 25% reduction in their current travel time and at least a 10% reduction in their current

travel cost. This finding suggests that high-quality interchanges are more likely to encourage

the use of more efficient routes involving transfers.

The statistically significant inverse relationship between passengers' current travel time, in

minutes, and the k values suggests that travelers with a longer journey time have more will

ingness to use routes involving transfers given some travel-time savings. For travelers with a

shorter journey time, the intention to make transfers is lower possibly because of the reason

that these travelers are satisfied with their current travel time. Figures 14.11 and 14.12 illus

trated that majority of the participants followed Δt (minimum travel-time savings required

to use route with transfer) inverse curves at intervals of 5 min. This finding supports prior

research (Van der Henst et al., 2002) that discussed the psychological disposition of round

ing during verbal communications related to time in multiples of 5 min. Analysis of the travel

cost-savings data, given in Table 14.7, has shown that the average k was of a similar value for

each of the current travel cost groups. The finding suggests that fare discounts need to be given

in proportion to users' current travel costs to increase their willingness to make transfers.

14.4.4.1 Notes for practitioners

The importance of high-quality services for public transit users' satisfaction is no secret.

It has been claimed by researchers that users are unwilling to make transfers. The studies

discussed in Sections 14.3 and 14.4 have shown that users are willing to make transfers

given that the connection is reliable and saves time and cost. Such attractive connections can

be achieved through integration among transit services.

Rachel Kyte, World Bank's vice president of sustainable development, stated the impor

tance of integrated transport system to insure quality. The following extract is from the

article presented in the magazine Global, "An international briefing" (Kyte, 2011):

Having a large number of small operators allows for low-cost services, but the qual

ity is poor due to severe competition. Other disadvantages include dangerous driving

practices, pollution and a tendency to have too much service on profitable routes and

virtually no service on non-profitable routes. Meanwhile, single publicly owned entities Table 14.7 Average JND k constant for current travel cost Current travel cost (NZ\$) \$21 \$28 \$35 \$42 k (basic interchange comfort) W 0.160 0.178 0.149 0.162 s w 2 0.003 0.003 0.002 0.003 k (high-quality interchange comfort) W 0.102 0.121 0.089 0.106 s w 2 0.003 0.004 0.002 0.003 may offer higher quality of service but costs tend to be high and the quantity of service

is often inadequate. There is increasing recognition that the best industry structure falls

somewhere between the two. Having a single public entity that plans the network and

determines the quality of service, with a small number of private operators providing

services under structured contracts, allows a balancing of public good needs with the

operational efficiency of the private sector.

14.5 PLANNED AND UNPLANNED TRANSFERS: A GUIDELINE FOR POLICYMAKERS

14.5.1 Definition of planned and unplanned transfers

Chowdhury and Ceder (2013c) proposed a definition of planned and unplanned transfers.

A planned transfer is a connection that has been intentionally designed by policymakers and

network planners in the planning stage of the multimodal transport network to improve

service efficiency and convenience for passengers. An unplanned transfer is defined as a

connection that has been created by the p

passengers from a available services without any additional

tional guidance on how to make the connection. As given in Table 14.8, planned transfer

consists of five attributes: network integration, integrated timed transfer, integrated physical

connection of transfers, information integration, and fare and ticketing integration.

14.5.2 Effects of planned and unplanned transfers on passengers' perception

This section covers two studies focused on the effects of planned and unplanned transfers

on passengers' perception of transfers.

14.5.2.1 Effects of attributes

The first study by Chowdhury and Ceder (2013c) is based on a survey conducted in

Auckland, New Zealand. The Northern Busway and Britomart terminals were chosen as

the survey locations. Local feeder routes are connected to ve designated stations along the

Northern Expressway. Transfers occur between the local routes and the main line. Majority

of the buses entering and leaving Auckland central begin and end their trip at Britomart,

thus creating the opportunity to make transfers. A assessment of the facilities at each station

revealed that the transfer occurring on Northern Expressway is more closely aligned to

being planned transfer than those in Britomart.

The questionnaire was composed of general sociodemographics, trip characteristics, and

hypothetical scenario questions. The bandwidth for the

sociodemographic questions, age

and income, were adopted from the New Zealand census questionnaires. Top participants

who are currently using a transfer route, questions on details of the transfer connection and

usage satisfaction were asked. Top participants who are not using a transfer route, a multi

choice question was asked on improvements to which trip attributes will increase their will

ingness. A 5-point Likert scale (very poor/very good) was used as the response measure.

Of the participants from Northern Bussway, 61% rated their transfer route to be 4 (good).

Of the participants from Britomart, 54% rated their transfer route to be 3 (neutral) and none

rated their routes to be greater than 3. The average transfer waiting time and walking time for

users of Northern Bussway was found to be 12 and 8 min, respectively. Similarly, for Britomart,

the average transfer waiting and walking time was found to be 6 and 5 min, respectively.

Figure 14.13 shows the proportion of passengers willing to use transfer routes given

improvements made to the trip attributes. Users of Britomart were willing to use transfer

routes given more connected routes, more information on transfers, and the total transfer

time being less than 20 min. Network integration was shown to have the most influence

on willingness. Northern Busway passengers' willingness depended on better seating area,

sheltered walking area, and integrated ticketing system, which offers no additional cost for

transfers. Cost of transfer was shown to have a greater influence than the trip attributes

related to comfort for users from both locations.

Table 14.8 Attributes defining a planned transfer

Attributes Definition Characteristics

Network

integration Routes are required to be connected from a network perspective to allow passengers to access a wider range of destinations. Proper integration of multimodal transport system will reduce wasteful duplication of route services, thus improving the utilization of resources. • Physical overlap of service lines. • Combination of high-frequency routes and low-frequency routes (feeder services). • Network coverage. • Easy accessibility to public transit network.

Integrated

timed

transfer The aim of integrated timed transfers is to interconnect the multimodal transport network such that the transfer times are minimized. This is achieved by operators synchronizing their scheduled routes to develop a pulsed-hub network. Since the performance of timed transfer is dependent on schedule reliability, implementation improves the overall reliability of the transport system. Routes and scheduling are required to be designed simultaneously. • Minimize transfer waiting time. • Synchronize scheduled routes.

Integrated

physical

connection

for transfers Terminals are required to be physically connected for the transfers among them to be considered as

being planned. Integration between terminals has been defined as being sheltered walkways between terminals, security measures at connected walkways, and information provisions such as signage providing guidance between the connected terminals, map of the local street area, and location of connected walkways.

- Sheltered walkways.
- Security measures to protect users between separate stations/ stops.
- Types of information, such as directional signage and maps, to link stations/stops at separate locations.

Information

integration An integrated information system is essential to facilitate urban and interurban multimodal trip planning. With many advanced information systems available, real-time information can be made accessible directly to the user en route. Such an information system assists users to preplan transfers and then provides guidance en route, thereby reducing the chances of missed connection and providing travel support (Grotenhuis et al., 2007; Zhang et al., 2011).

- Journey planner to assist users in planning their transfers among all public transit services.
- Real-time information (arrival/ departure/DTs) at stations/stops.
- En route guidance providing real-time information.
- Maps and timetables for all public transit services at stations/stops.

Fare and

ticketing

integration Fare system integration of a multimodal transport network has been shown to facilitate seamless transfers, thus encouraging the use of transfer routes (Sharaby and Shiftan, 2012). A simple user-friendly integration, such as smart cards, can improve the efficiency of boarding and egressing.

- Smart cards used for all services.
- No additional cost for transfers.

Source: Chowdhury, S. and Ceder, A., J. Public Trans., 16(2), 1, 2013c.

14.5.2.2 Effects of explanatory variables

The second study by Chowdhury et al. (2014) was undertaken to determine the effects of

planned and unplanned transfers on captive users' (passengers who do not currently make

transfers) decision to use transfer routes. The survey locations in Auckland, New Zealand, were selected—Northern Busway and Britomart. All participants were asked a question on whether they would choose an alternative hypothetical transfer route when comparing the route to a direct route with travel time of 40 min. The hypothetical transfer routes scenarios had travel-time savings of 10, 15, and 20 min with varying types of connection. This question was designed to determine passengers' perception of planned and unplanned transfers.

Table 14.9 gives one of the scenarios. The survey was conducted for 10 weekdays (5 days for each location) during commuter morning peak period (7–10 a.m.). Target participants were limited to only commuters because this group represents the highest proportion of users.

For the hypothetical scenarios, as the response is dichotomous, a binary logistic regression model was developed to predict the dichotomous dependent variable as a function of the predictor variables. Equation 14.15 gives its common form. A statistical package, SPSS, was used to fit the two data sets into binary logistic regression models. $\text{logit}P(X) = \ln \frac{P(X)}{1-P(X)} = \alpha + \sum \beta_j X_j$ (14.15)

Table 14.10 gives the model for the Northern Busway data. The Hosmer-Lemeshow test is a goodness-of-fit statistic used to determine whether the developed model reasonably approximates the data (Kleinbaum and Klein, 2010). Results of the analysis indicated that

the model adequately fit the data because the
 significance value for Hosmer-Lemeshow test

was 0.27.

Compared with the travel-time savings of 10 min,
 only travel-time savings of 20 min was

significant. Holding all other factors constant,
 current nontransfer route users were 2.92

times more likely to use a transfer route with
 travel-time savings of 20 min savings (odds

ratio = 2.92) when compared to a travel-time
 savings of 10 min. In addition, users were 0%
 10% 20% Proportion of participants 30%
 40% 50% 60% 70% 80% Northern Busway Britomart Better
 seating and sheltered waiting areas
 Sheltered walking area Transfer time
 is less than 20 min More information
 about transfers One ticket with no
 additional cost More security More
 connected routes

Figure 14.13 Proportion of participants willing to use
 transfer routes given improvements in trip attributes.

12.46 times more likely to use a transfer route with the
 presence of shelters when compared

to a transfer connection without shelters.

The following gives the logit for the fitted model for the
 Northern Busway data set: $\text{logit}(\text{Transfer}) = \beta_0 + \beta_1 \text{Travel min} + \beta_2 \text{Travel min}^2 + \beta_3 \text{Shelter} + \beta_4 \text{Information} + \beta_5 \text{Gender} + \beta_6 \text{Frequency}$
 $\beta_0 = -1.51, \beta_1 = 0.32, \beta_2 = 0.07, \beta_3 = 1.15, \beta_4 = 2.20, \beta_5 = 0.52, \beta_6 = 0.69$
 Frequency + + + (14.16)

Table 14.11 gives the model for Britomart. Holding all other factors constant, current

nontransfer route users were 1.52 times more
 likely to use a transfer route with
 travel-time

savings of 15 min savings (odds ratio = 1.52)
 when compared to a travel-time savings of

10 min. Public transit users were 2.94 times more likely to use a transfer route with high

quality information. When comparing connections with and without shelter, users were

5.19 times more likely to make a transfer with shelter (odds ratio = 5.19). Table 14.9 Hypothetical transfer scenarios Ten minutes travel-time savings, given the following service Option 1 • 5 min waiting time for the next vehicle • 5 min walking time to the next vehicle • Sheltered waiting and walking area • High-quality information about transfer Yes No Option 2 • 5 min waiting time for the next vehicle • 5 min walking time to the next vehicle • Sheltered waiting and walking area • Have to search and find your own transfer point Yes No Option 3 • 5 min waiting time for the next vehicle • 5 min walking time to the next vehicle • No sheltered waiting and walking area • Have to search and find your own transfer point Yes No Source: Chowdhury, S. et al., Trans. Plan. Technol., 37(2), 154, 2014. Table 14.10 Explanatory variables for Northern Busway Explanatory variables Coefficient Wald df p-value Odds ratio 95% C.I. of odds ratio Lower Upper Northern Busway (planned transfer) Travel 21.350 2 0.000 Travel (15) 0.322 2.082 1 0.149 1.380 0.891 2.137 Travel (20) 1.072 20.674 1 0.000 2.922 1.840 4.638 Shelter (yes) 2.522 112.250 1 0.000 12.458 7.813 19.865 Information (yes) 0.693 9.068 1 0.003 1.999 1.274 3.138 Gender (female) 1.770 63.216 1 0.000 5.869 3.794 9.079 Frequent (yes) 1.203 26.119 1 0.000 3.331 2.100 5.284 Constant -3.511 100.875 1 0.000 0.030 Source: Chowdhury, S. et al., Trans. Plan. Technol., 37(2), 154, 2014. Test of model fit (Hosmer-Lemeshow test) chi-squared = 9.9, df = 8. Nagelkerke R² = 0.46, log likelihood (final) = 710.066. Omnibus test (intercept public transport only and final model) chi-squared = 314.81, df = 6.

The following equation gives the logit for the final logistic regression model for the

Britomart data: $\text{logit}(\text{Transfer}) = -1.48 + 0.42 \text{ Travel} + 1.65 \text{ Shelter} + 0.15 \text{ Information} + 0.15 \text{ Travel} + 0.15 \text{ Shelter} + 0.15 \text{ Information}$ (14.17)

14.5.2.3 Discussion of both studies

Overall, results from these two studies suggest that users of public transit network with

planned connections are more likely to use transfer routes than users of network with

unplanned connections. Physical integration was seen to have a greater effect than infor-

mation integration on passengers' willingness to make transfers. Participants from both

transport terminals were willing to use the transfer routes given shelter provided during

transfer walking and waiting times (Northern B usway odds ratio = 1.246 and Britomart

odds ratio = 5.20) and high-quality information services (Northern B usway odds ratio =

2.0 and Britomart odds ratio = 2.94) while holding transfer walking and waiting times to

be constant at 5 min each. Therefore, it can be recommended that for transfers between

physically separated terminals, planning must focus on design measures to create good con-

nections. At the very least, connections between terminals need to be weather-protected and

sufficient way-finding signage must be present. Findings from the two studies have high-

lighted the importance of including transfers as a service component in the main planning

stage of an integrated multimodal transport network development to provide users with

seamless transfers.

14.5.2.4 Notes for practitioners

The idea behind providing planned transfers is to make public transit services more efficient

and attractive by reducing travel times and increasing reliability and connectivity. During

the development of an integrated transit n

network, planners are encouraged to address all
the

attributes of planned transfers for the
connections to be perceived as being seamless
by passengers.

The various attributes that need to be
achieved for transfers to be categorized as
being

planned are (a) network integration, (b) integrated timed
transfer, (c) integrated physical

connection of transfers, (d) information
integration, and (e) fare and ticketing
integration.

It should be noted that if all the
attributes are not achieved, then the
connection cannot be

categorized as being a planned transfer. Table 14.11
Explanatory variables for Britomart Explanatory variables
Coefficient Wald df p-value Odds ratio 95% C.I. of odds
ratio Lower Upper Britomart (unplanned transfer) Travel
(15) 0.420 4.770 1 0.029 1.522 1.044 2.218 Shelter (yes)
1.648 52.315 1 0.000 5.196 3.325 8.121 Information (yes)
1.079 21.075 1 0.000 2.943 1.856 4.665 Constant -1.478
52.496 1 0.000 0.228 Source: Chowdhury, S. et al., Trans.
Plan. Technol., 37(2), 154, 2014. Test of model fit
(Hosmer-Lemeshow test) chi-squared = 1.634, df = 4.
Nagelkerke R² = 0.306, log likelihood (final) = 651.72.
Omnibus test (intercept/public transport only and final model)
chi-squared = 151.14, df = 3.

14.6 LITERATURE REVIEW AND FURTHER READING

The literature review of transit behavioral-based
studies is quite extensive. In what follows

are three themes with a review linked directly
and indirectly with user's behavior related to
the

making transfers on transit routes.

14.6.1 Mode choice between public transport and private vehicles

Literature analyzing the general attitude of travelers toward public transit use has shown

that although improvement in service quality is likely to increase ridership, the level of

increase can be limited if travelers hold prejudices toward the image of transit service and of

becoming users (Murray et al., 2010). One of the strategies used by a government to amend

travelers' misconception of transit is mass media information campaigns on the benefits of

use (Tertoolen et al., 1998). The aim of the campaigns is to induce positive attitudes about

transit use in society. Beale and Bonsall (2007) revealed that persuasive marketing messages

on the benefits of using transit increased the negative attitude of car users. On the other

hand, for habitual users, the marketing messages encouraged more use. Guo and Wilson

(2007) stated that frequent users are more familiar with the network and services and therefore

have fewer negative perceptions than infrequent users and nonusers. Pedersen et al.

(2012) discussed that car users frequently underestimate their potential satisfaction with

transit use and inaccurately recall their satisfaction associated with any previous use, which

leads to persistent negative attitude toward modes with. Taniguchi and Fujii (2007) inves-

tigated the psychological effect of an intervention, which consisted of newsletters (a market

ing technique) and free bus tickets, on travelers' decisions to use transit. The results showed

that including marketing techniques in interventions is an effective method of increasing

ridership of transit services.

Several other studies have focused on the proenvironmental aspect of travel behavior.

Nilsson and Kuller (2000) concluded that the intention to behave in a proenvironment way

depends considerably on the individual's attitude related to environmental issues. Factual

knowledge was seen to play a subordinate role. In terms of the attitudinal factors, envi-

ronmental concern and attachment to one's car was suggested to be most influential in

determining proenvironmental behaviors. Individuals living in a heavy traffic area showed

more interest in proenvironmental behaviors than individuals living in a low-pollution

area. Nordlund and Garvill (2003) stated that values and problem awareness significantly

affect personal norms, which in turn influences the individual's willingness to incorporate

proenvironmental behaviors. Collins and Chambers (2005) discussed that individual's pro-

environmental behavior is a result of the interaction among their social beliefs, biospheric

beliefs, egoistic beliefs, and situational factors such as travel by public transport cost, travel

time, and access.

14.6.2 Trip attributes

Redman et al. (2013) stated that supply-oriented quality of transit is only part of the whole

picture. It is also important to consider the quality of service as perceived by passengers to

meet a consistent satisfactory level over time. This section focuses on the trip attributes that

have been identified to be significant in passengers' perceived ease of making transfers. The

importance of personal safety at terminals has been echoed in several travel behavior studies

(Atkins, 1990; Volinski and Page, 2006). Currie and Delbosc (2013) discussed that users are

concerned about traveling with unknown passengers. Transit users feel unsafe particularly

during night travels (Blainey et al., 2012; Currie and Delbosc, 2013). A study in the United

Kingdom has shown that an additional 10.5% of rail trips would be generated if users' fears

were addressed (Currie and Delbosc, 2013). Reliability is one of the most important operational

attributes of transit services. A number of studies on timetable scheduling have been

done to determine methods of improving reliability. Muller and Furth (2009) examined how

better planning can minimize the inconvenience to users who are making transfers. Results

emphasized the importance of optimal offset in schedule planning to minimize transfer

waiting times as well as to reduce missed connections. Methods such as holding have been

suggested to improve reliability by reducing additional journey times from schedule deviation

ance (Van Dordt and van Nes, 2009; Van Dordt et al., 2012).

Other studies have shown that better information systems are required to reduce users'

perceived inconvenience of making multimodal trips (Zografos et al., 2008). Integration

between various operators is required for an information system to facilitate urban and

interurban multimodal trip planning (Zografos et al., 2008). Molin and Chorus (2009)

showed that transferring to a lower-frequency service induces a need for more information

tion and thus a higher willingness to pay for information. Pricing is another key attribute

that attracts ridership of transit (Redman et al., 2013). Integration of fare systems across

operators has a positive effect on the perceived ease of making transfers (Buehler, 2011).

Scharaby and Shifan (2012) demonstrated that a well-integrated fare system, which allows

no additional cost for transfers, has a positive impact on ridership by improving travelers'

willingness to use routes with transfers.

14.6.3 Policy and infrastructure

Two distinctions can be made among the different policy measures: push and pull. The

objective of push-type measures is to reduce the attractiveness of private vehicles, while

pull-type measures aim to increase the attractiveness of sustainable transport. There are

four types of policies: legal policies, information and educational policies, economic policies, and physical changes (e.g., infrastructure) (Cools et al., 2009). Some examples of legal

policy measures are speed restrictions, parking restrictions, and prohibiting cars in city centers (Cools et al., 2009). Types of parking restrictions include availability of parking spaces,

parking costs, parking time limits, provision of free employee parking, and residential parking

permits. As for economic policy, some of the measures developed are congestion charges

(already implemented in cities like London, Stockholm, and Durham), toll roads, and taxes

on cars and fuel (Stopher, 2004; Cools et al., 2009).

In regard to information and educational policy measures, primarily in Europe and

Australia, governments have identified the need to encourage voluntary travel behavior

change toward sustainable modes of transport (Pramberg, 2004). Personalized travel planning

initiatives are targeted marketing approaches that empower individuals to change

their travel behaviors for their personal benefit (Ampt, 2004). Physical policies aim at

increasing the attractiveness of transit services by improving service quality, providing

infrastructures such as separate bus lanes and better land-use planning for shorter travel

times (Cools et al., 2009). Such pull-type measures are seen to be more easily accepted

by the public than push-type measures as they are considered to more effective and fairer

(Eriksson et al., 2008). Integration of multimodal transport systems has received particular

interest in recent years to promote mode switch. Hu ll (2008) emphasized the importance of

maintaining interoperability and interconnectivity across service providers to optimize an

intermodal transport system.

Table 14.12 provides a summary of the factors that influence transit users' willingness to

make transfers.

EXERCISES

14.1 The TPB has been used as a common psychological model to determine travel behavior.

(1) Define the three antecedents: attitude, social norm, and PBC.

(2) Describe the two links, from intention and PBC to behavior.

(3) State the three most influential trip attributes that will improve public transit users' control beliefs to make transfers.

14.2 Given as follows is the choice of two alternative routes that involve making a transfer. The case shows the variation in transfer waiting time for Scenarios 1 and 2. The assumption is that the scheduled travel time of both alternatives is equal.

(1) Determine what alternative users are most likely to choose using the CPT. Assume that the reference point is equal to 9 min.

(2) Describe the steps that will be undertaken to determine users' route choice using fuzzy logic. 20% Scenario 1 15 min of waiting 3 min of waiting 80% Scenario 2 12 min of waiting 5 min of waiting 30% 70%

-Scenario 1 (less conservative) -Scenario 2 (more conservative)

Table 14.12 Summary of factors influencing public transport users' willingness

Factors influencing willingness

to use transit Source Users Quantitative?

Education Beale and Bonsall (2007) Bus users No Murray et al. (2010) Car users Partially Tertoolen et al. (1998) Car users Partially

Frequency of use Guo and Wilson (2007) Train users Yes Pedersen et al. (2012) Car users No

Proenvironmental behavior Nilsson and Kuller (2000) Car users Partially Nordlund and Garvill (2003) Car users Partially Collins and Chambers (2005) Students Partially

Integrated information systems Zografos et al. (2008) Train users No Molin and Chorus (2009) Train users Yes

Integrated ticketing system Sharaby and Shiftan (2012) Bus users Yes Buehler (2011) Car and transit users No

Safety Currie and Delbosc (2013) Students Partially Blainey et al. (2012) Car users No Atkins (1990) Transit users No

Reliability Muller and Furth (2009) N/A Yes Van Oort et al. (2012) N/A Yes

14.3 A government transport agency is developing a new guideline for designing their transit network. The current network mainly comprises train and bus services for the high- and low-frequency routes, respectively. Describe a planned transfer between a high-frequency train service (every 20 min) and a low-frequency bus service (every 60 min).

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15 Chapter 15: Network (Routes) Design

This chapter focuses on the first activity of the operational planning process described

in Chapter 1: public transit route design and evaluation at the network level. The problem

addressed is that of how to design a new transit network or to redesign an existing network.

Naturally, this activity would be appropriate only very infrequently because of the disruption

that would be imposed on passengers if wholesale changes were made to the transit network. In

addition, many transit agencies have not gone through such a reappraisal, mainly because of the

lack of clear-cut, practical, and measurable design criteria for evaluating the goodness of transit

routes and comparing sets of routes. Theodore Roosevelt said: "Nine tenths of wisdom is being

wise in time." Thus, a design and evaluation tool may provide a timely boost for reconstruct

ing transit routes, though one should bear in mind the required inter-connectivity of a transit

network. Following the introductory section of this chapter, Section 15.2 proposes a framework for

the construction of the operational objective functions of the transit network design problem.

This framework takes into account the passengers as well as agency and community/ government

interests—the three perspectives emanating from the broad spectrum of transit activities.

Section 15.3 utilizes the objective functions described for facilitating the construction of a method

ology for the practical design of an efficient transit network of routes. This methodology

combines the philosophy of mathematical programming approaches with decision-making tech

niques; it allows the transit planner to select a set of routes from a number of alternatives.

Section 15.4 introduces a model and a solution algorithm to take into account seasonal demand

variation, which is a somewhat neglected aspect of transit demand. The solution algorithm

developed is a hybrid method that optimizes the design of a bus network at the route and net

work levels. Section 15.5 introduces a method for solving bus network design problem using

genetic algorithm (GA), and Section 15.6 shows how to consider connected exclusive lanes

within optimal solution of bus networks. Practitioners are encouraged to visit all the sections except Section 15.5. In particular, it is

recommended that they look at the introductory parts and examples illustrated in all of the

sections. It is worth noting that a new design or a redesign of a transit network needs time to be fully

comprehended; this is the same time defined as nature's way of keeping everything from happen

ing at once. Last, this Danish proverb seems most apt: "Better to ask twice than to lose your

way once." In our case, it is better to present the new routes in at least two clear ways than to

have the user misinterpret them even once.

15.1 INTRODUCTION

There are two main approaches to restructuring transit routes: (1) at the route level or

for a small group of routes and (2) at the network level. For the first approach, Pratt and

E vans (2004), i n T CRP R eport 9 5, s uggested t hat r estructuring w as t o s implify r outes,

accommodate new travel patterns, ease or eliminate transfers, reduce route circuitry,

o r o therwise a lter r oute c onfiguration. T his a pproach i s d ealt w ith t o s ome e xtent

i n Chapters 1 2, 1 6, a nd 17. T he p resent c hapter p roffers p ractical s olutions u tilizing t he

s econd a p proach.

Only a few researchers have studied the interrelationship of the scheduling and the

n etwork-design p lanning a ctivities sh own i n F igure 1 .2. T he i nterrelationship h as t wo c at

egories: (a) e ach s et o f r outes, b ased o n d emand, y ields a d ifferent s et o f f requencies a nd

t imetables a nd, u ltimately, t he r equired l eet s ize; (b) t he o perational c ost d erived f rom t he

s cheduling a ctivities a nd p assenger l evel o f s ervice a ffects t he s earch f or t he o ptimal r oute

design while relying on a compromise solution between the agency and the passengers.

For m any p ublic t ransit a gencies w hose n etwork o f r outes h as n ot b een r eappraised f or

a nywhere f rom 2 0 t o 5 0 years, i t i s h igh t ime t o c onsider s uch a n u ndertaking. T his sh ould

p rovide s ufficient m otivation t o s eek a n ef ficient n etwork r oute-design m ethod, b ased o n

c ertain o bjective f unctions a nd a s et o f c onstraints. T he m ain p urpose o f t he m ethodologies

presented in this chapter is to transport a given origin-destination (O-D) demand through

the transit network in the most cost-effective way.

The special characteristics of route-design problems are as follows: (1) passenger demand

is spread throughout the entire network, where it is generated and terminated at many

points along the network's links and can be grouped in terms of an O-D matrix; (2) the

demand must be transported simultaneously (usually during peak hours); and (3) over a

given planning horizon, it is impossible to reconstruct the routes (i.e., once the route network

is designed, it will remain as is over an entire planning period).

Prior approaches to the public transportation network-design problem can be grouped into

those that simulate passenger flows, those which deal with ideal networks, and those based

on mathematical programming. Simulation models are presented in Dial and Bunyan (1968),

Hearn et al. (1968), and Vandeboina and Richardson (1985). These models require a

considerable amount of data, and their proximity to optimality is uncertain. Ideal network

methods are based on a broad range of design parameters and a choice of objectives reflecting

user and agency interest. Such methods appear in Kocur and Henrickson (1982), Tsao

and Shonfeld (1984), and Kuah and Perl (1988). These methods are adequate for screening

or policy analyses, in which approximate design parameters rather than a complete design

are determined; thus, these methods cannot represent real situations. Mathematical pro-

gramming models are divided into generalized network-design models and transit-specific

network models. Known generalized network models are well summarized and reviewed in

Kim and Barnhart (1999); also, see as an example the heuristics developed by Farvolden and

Powell (1994). The transit-specific network-design models are inevitably heuristic because

of the extremely high computational effort required. These partial optimization approaches

appear in Lampkin and Salmans (1967), Silman et al. (1974), Dubois et al. (1979), and

(1979), and Kuehl (1988). None of them models and methods mentioned in this section

has actually been applied. However, two survey papers by Guihaire and Hao (2008) and

Kepaptsoglou and Karlaftis (2009) review methods to solve the problem of bus network

design. The latter performs a three-part systematic review of the problem: design objectives,

operating environment parameters, and solution approaches. This three-part classification

can be used in defining further research. Section 15.7 provides a further literature review.

An overview of prior work done on this theme calls for a method that, given the availabil-

ity of typical data, will be more practical and less complex than other methods and models.

Such a method, described in Section 15.3, may increase the chances of its acceptance by

most transit agencies. The method developed and its related methods, appearing in this

chapter, seek to have transit routes as close as possible to the shortest paths.

Finally, it is interesting to observe Figure 15.1, which is currently a distributed info-page

at the taxi stand at the Washington (DC) National Airport. This page includes the following

note: Fares are based on shortest route; however, the shortest route may not be the quickest

route. (Which is also a nice way of reminding us that laughter is the shortest distance

between two people.) In our case, the shortest path intends to be the quickest one because

of its time units; in real-time, however, this shortest path may not hold because of traffic

congestion (e.g., caused by road accidents or road repairs, etc.).

15.2 OBJECTIVE FUNCTIONS

This section proposes a framework for the construction of the operational objective func

tions of the public transit network-design problem. This framework takes into account the

passengers, the agency, and community interests. It follows the studies by Ceder and Israeli

(1992), Israeli and Ceder (1995), Ceder (2001), Ceder et al. (2002), and Yin et al. (2005).

15.2.1 Three perspectives and four criteria

From the literature review presented earlier and that at the end of the chapter, there are no

clear-cut, practical, or measurable criteria for evaluating the goodness of transit routes and

comparing sets of routes at the network level. The only comprehensible matter is that the

design of transit routes should be looked at simultaneously from three perspectives: that of

the passenger, the agency, and the community/government. These three perspectives ema

nate from the broad spectrum of transit activities.

Figure 15.1 Current use of the term shortest route by the taxi service at the Washington (DC) National Airport (see the marked section under APPROXIMATE FARES ...).

Four criteria can be considered when measuring the quality of a transit route: (1) mini

mum passenger waiting time; (2) minimum empty seat/space time; (3) minimum time differ

ence from shortest path; and (4) minimum fleet size. The first three criteria are measured in

passenger-hours, and the last one in number of vehicles. Clearly, criterion (1) represents the

passengers' perspective; criteria (2) and (4), the agency's perspective; and criterion (3), both

the passengers' and the community's perspectives.

When the purpose of measuring is to compare sets of transit routes or different transit

modes, monetary weights can be introduced to the four criteria. Optionally, criterion (3)

can be replaced by the total monetary loss (or saving, if it is negative) if all the transit

passengers are switched to the shortest path.
For instance, when comparing a set of bus

routes and a set of metro lines, in which the latter is the
shortest path (in time), criterion

(3) will provide the total monetary loss/saving if all the bus passengers switch to the

metro lines.

The next section constructs a quantitative framework for the four criteria in order to

devise tools for the design of an optimal set of transit routes. This quantitative framework

furnishes a clearer picture of the four criteria, with two optional versions of criterion (3).

15.2.2 Formulation

The following established notations will be used throughout the analyses of this chapter.

15.2.2.1 Notations

Consider a connected network composed of a directed graph $G = \{N, A\}$ with a finite number

of nodes $|N|$ connected by $|A|$ arcs. Define:

Route Progressive path initiated at a given transit terminal and terminated at a certain node while traversing given arcs in sequence.

Transfer path Progressive path that uses more than one route.

$R = \{r\}$ Set of transit routes.

$TR = \{tr\}$ Set of all transfer paths.

$S = \{sp\}$ Set of all shortest paths (minimum average travel times).

N_r Set of nodes located on route r .

N_{tr} Set of nodes located on transfer path tr .

N_{sp} Set of nodes located on the shortest path sp .

d_{ij}

r Passenger demand between i and j , $i, j \in N$, riding on route r .

d_{ij}

tr Passenger demand between i and j along the transfer path t_r .

d_{ij}

sp Passenger demand between i and j along its shortest path.

F_r Vehicle frequency associated with route r .

F_{min} Minimum frequency (reciprocal of policy headway) required.

t_{ij}

r Average travel time between i and j on route r .

t_{ij}

tr Average travel time between i and j on transfer path t_r (can include transfer penalties).

t_{ij}

sp Average travel time between i and j on its shortest path.

t_r Overall travel time on route r between its start and end.

L_r Maximum passenger load on route r .

w_r Passenger waiting time on route r .

d_o Desired occupancy on each vehicle (load standard).

$a_{tr} = 1$ if i, j transfer through route r otherwise. 0

α Maximum allowed deviation from shortest path for any OD pair on a transit path (including transfers).

k Maximum degree of transfer path t_r (number of vehicle changing).

15.2.2.2 Two principal objective functions

The transit network-design problem is based on two principal objective functions, minimum

Z_1 and minimum Z_2 , across the different sets of transit routes:

$Z_1 = \sum_{i,j \in N} \sum_{r \in R} \alpha_{ijr} (WT_{ijr} + \epsilon_{ijr} (PH_{ijr} - DPH_{ijr}))$, for single set $\alpha_{ijr} = 1$ if $i, j \in N$ and $r \in R$, $\alpha_{ijr} = 0$ otherwise, $\epsilon_{ijr} = 1$ if $i, j \in N$ and $r \in R$, $\epsilon_{ijr} = 0$ otherwise, WT_{ijr} is the waiting time in a transit vehicle on a hourly basis; PH_{ijr} is the passenger hours between nodes i and j , $i, j \in N$ (defined as passengers' riding time in a transit vehicle on a hourly basis; it measures the time spent by passengers in vehicles between the two nodes) DPH_{ijr} is the difference in passenger hours between PH_{ijr} and total passenger hours from i to j when using only the shortest path, $i, j \in N$ (15.1)

$Z_2 = FS$ (15.2)

where PH_{ijr} is the passenger hours between nodes i and j , $i, j \in N$ (defined as passengers' riding time in a transit vehicle on a hourly basis; it measures the time spent by passengers in vehicles between the two nodes)

DPH_{ijr} is the difference in passenger hours between PH_{ijr} and total passenger hours from i to j when using only the shortest path, $i, j \in N$

WT_{ijr} is the waiting time between nodes i and j , $i, j \in N$ (defined as the amount of time passengers spend at the transit stops between the two nodes)

EH_r is the empty seat/space-hours on route r (defined as the unused seats/spaces in a transit vehicle on a hourly basis; empty seat/space-hours measures the unused capacity on vehicles) FS is the fleet size (number of transit vehicles needed to provide all trips along a chosen set of routes)

α_k is a monetary or other weights, $k = 1, 2, 3, 4$

Equation 15.1 contains the two options for the Z_1 objective function, which can be interpreted as minimum waiting time and maximum u

tilization; for given weights of 1 or with

units, this equation results in units of passenger hours. Equation 15.2 is simply the

required minimum fleet size.

15.2.2.3 Objective function components

Equations 15.1 and 15.2 essentially combine the objective function components. The first

straightforward objective is to minimize passengers' total waiting time. This is strictly the

perspective of the transit user. The formulation of this objective takes the following form: $\sum_{i,j \in N} a_{ij} W_{ij}$ (15.3)

where a_{ij} is the monetary value of i 's waiting time.

The second objective is to minimize the total unused seat capacity so as to allow more

viable transit service. This is strictly the perspective of the agency, which wishes to see more

occupation of the available seats. The following is the formulation of this objective: $\sum_{r \in R} a_{r2} E_{r2}$ (15.4)

where a_{r2} is the equivalent of i 's average monetary revenue divided by the average number

of hourly boarding passengers. This objective is to minimize the total monetary value of

unused seat capacity.

The third and fourth objectives are two versions of the same objective: (a) to minimize

the total time loss (in monetary value) between riding the transit vehicle and traveling by

automobile (assumed to be the shortest path) and (b) to minimize the total loss (in monetary

value) if all the passengers are switched to the shortest path. Versions (a) and (b) take the

following forms, respectively: $\text{Min } a \text{ DPH}(i, j) + 4 \sum_{i,j \in N} (15.5)$
 $\text{Min } a \text{ PH}(i, j) + a \text{ DPH}(i, j) + 3 + 4 \sum_{i,j \in N} - [\] \in \Sigma, (15.6)$

where $a + 3$ is the equivalent of 1 h's difference in average cost/fare between riding the shortest path (automobile or a transit competitor) and the transit route $a + 4$ is the monetary value of 1 h's in-vehicle time

The value of Equation 15.6 is the total monetary loss (or saving, if it is negative) if all the

passengers are switched to the shortest path, where $a + 3 \text{ PH} =$ total monetary loss, with respect

to cost/fare only, if all the passengers are switched to the shortest path; and $a + 4 \text{ DPH} =$ total

monetary value of the time saved if all the passengers are switched to the shortest path. The

latter is Equation 15.5. These objectives represent the perspectives of the community/gov

ernment and the passengers.

The fifth objective is to minimize the number of vehicles required for a given set of routes

and frequencies (timetables). This is strictly the agency perspective, which wishes to perform

all transit trips using a minimal number of vehicles. This objective takes the form:

$\text{Min FS} (15.7)$

Objectives (15.3) through (15.6) are all in terms of passenger-hour cost; for the sake of sim

plicity, therefore, this can be summed up to Min Z 1 as shown in Equation 15.1. Objective

(15.7) stands alone to some extent and is termed $\text{Min } Z_2$ in Equation 15.2.

15.2.2.4 Calculation of Z_1 functions

The objective function Z_1 is based on the passenger-load profile. Figure 15.2a presents the

example of a small transit network. The input data during a given time period (usually peak

hours) consist of average travel times, an estimated O-D demand in Figure 15.2b, and a $k = 1$

for all $k = 1, 2, 3, 4$. Figure 15.2c displays the feasible routes complying with the following

practical constraints: (i) maximum route length of 30 min; (ii) maximum allowed

deviation of 40% from the shortest path; (iii) maximum of one transfer for all O-D pairs;

(iv) a chosen route should not be included in another feasible route; and (v) no circular routes

considered.

Because routes can start only at terminals (Node 1 in the example), a mapping process is

applied, with constraint (i) considered first. This results in six routes: [1-2], [1-2-3], [1-2-

3-4], [1-3], [1-3-2], and [1-3-4], all of which comply with the maximum of 30 min average

travel time. In parallel, a shortest-path algorithm (see Appendix 9.A) is applied. Constraint

(ii), maximum deviation from shortest path, can then be checked using the shortest-path

information. Table 15.1 provides this check while describing the procedure for a determina

tion of feasible routes, that is, making sure that a feasible route will not be included in an

another feasible route in order to avoid overlapping.

Once the feasible routes are established, their load profiles are constructed, based

on the input data. The load profile of route [1-3-4] is presented in Figure 15.2d. The

load on 1-3 is the maximum load ($L_{1-3-4} = 680$ passengers) of the O-D demand: 1-3,

1-4, 2-3, and 2-4 (see Figure 15.2b). The load on 3-4 is 300, consisting of the O-D

demand: 1-4, 2-4, 3-4. According to the 680 maximum load, the frequency of route

[1-3-4] would be $(680/50)$, $F_{1-3-4} = 13.6$ veh/h. The shaded area of the load profile

between 680 and 300 passengers along the 16 min segment represents 101.3 empty

seat-hours for $d_o = 50$ (number of seats on the vehicle) and $F_{min} = 4$ during the associated

time period of the example. In addition, the load profile of the other feasible route, 1-2,

is simply rectangular with a maximum load of $L_{1-2} = 200 + 150 + 80 = 430$ passengers

and $F_{1-2} = 8.6$.

The first function of Z_1 in Equation 15.1 is the total wait-time hours both at the tran

sist stops and during transfers. Different formulations of the expected passenger waiting

time appear in Sections 11.2 and 18.4. Utilizing Equation 11.1, in which passengers arrive 1243 Terminal 25 1610 5 (a) j i 2 3 4 1 200 350 100 2 150 80 3 120 (b) Emptyseat hours Passenger load Average travel time (min) 680 80 50 00 350 120 80 100 300

100 (d) 2 1 1 3 4 Route 1-2 Route 1-3-4 Transfer (c)

Figure 15.2 Example network, its given data, and the construction of a load profile: (a) sample network with average travel time in minutes (given); (b) symmetric O-D matrix in passenger/h; (c) feasible routes in both directions, complying with the four criteria; and (d) the load profile of route 1-3-4 (derived).

randomly at the transit stop and headways are deterministically distributed, we obtain the

expected waiting time on route r , which is half the headway: $w_r = \frac{1}{2} \text{ for all } r \in R$ (15.8)

Thus, $WT(i, j) = \sum_{r \in R} d_{ij} \cdot \frac{1}{2} \cdot \frac{1}{N_{ij}} = \frac{1}{2} \sum_{r \in R} \frac{d_{ij}}{N_{ij}}$ (15.9)

Applying Equation 15.9 to the example of Figure 15.2 yields (for both directions of routes): $WT(i, j) = \frac{1}{2} \sum_{r \in R} \frac{d_{ij}}{N_{ij}} = \frac{1}{2} \left(\frac{1}{1} + \frac{1}{1} + \frac{1}{1} + \frac{1}{1} \right) = 2$ (15.9)

where $d_{ij} = 50$. Note that any demand between i and j , $i, j \in N$, can split (e.g., some via

a direct route and the remaining via transfers); in this case, an assignment procedure

(see Chapter 11) can be applied. However, this is not the case of the example.

The second function of Z_1 in Equation 15.1 describes the total empty-space hours or

empty-seat hours (when d_{ij} equals the number of seats on the vehicle). This function repre-

sents an unproductive measure for the agency (e.g., unused seat capacity). Its formulation is $EH_r = \sum_{i,j \in N} (L_r - d_{ij}) \cdot \frac{1}{N_{ij}}$ (15.10)

In the example of Figure 15.2, this equation yields $EH_r = 2(101.3 + 0) = 202.6$ passenger

hours for both directions of the feasible routes.

Table 15.1 Determination of feasible routes in the example

problem

Route end points 1 → 2 1 → 3 1 → 4

Shortest-path (min) 5 10 26

Shortest-path route 1-2 1-3 1-3-4

Symmetric (both directions), O-D served optimally 1-2 1-3
1-3, 1-4

Other routes not complying with the maximum length

criterion (40%) 1-3-2 1-2-3 1-2-3-4

Symmetric (both directions), O-D served by other routes,

and their deviation (%) from the shortest path 1-2 (600%)
1-2 (0%) 1-2 (0%) 1-3 (0%) 1-3 (200%) 1-3 (200%) 2-3 (67%)
2-3 (67%) 1-4 (77%) 2-3 (67%) 2-4 (32%) 3-4 (0%)

Feasible route 1-2 1-3-4

Symmetric (both directions), O-D served optimally via

one transfer 2-3, 2-4

It is to note that the empty-seat hour calculation is based
on the provided frequency usu

ally max load based; however, if this frequency is the
minimum required, the empty-seat

hour would be calculated based only on the frequency. For
instance, if in route [1-3-4] we

want to have a clock-headway of 4 min (frequency of 15
veh/h) then the load profile is as

shown in Figure 15.3, and $EH_{par} = - + - \dots$
 $\dots = \dots 2\ 750\ 680\ 750\ 300\ 16\ 60\ 240\ 4$
 $10\ 60\ ()\ () . ss.-h$

The first version of the third function in Equation 15.1,
which appears in Equation 15.5, is

the total passenger-hour difference between $PH(i, j)$ on r
and $PH(i, j)$ on the shortest path sp : $DPH_{ij} = PH(i, j)_{dt}$
 $ij_{sp} - ij_{sp\ i,j} N_{sp} S_{ij} N_{ij} N_{sp} (,) , , = - \dots \Sigma \Sigma \Sigma$
(15.11)

in which, for the example in Figure 15.2, $\Sigma DPH = 0$; the reason for this is that routes [1-2]

and [1-3-4] include only the shortest paths.

The second version of the third function in Equation 15.1, which appears in Equation

15.6, has two parts, the second part being Equation 15.11. The first part is total passenger

hours in the routing system: $PH(i, j) d_{ij} t_{ij} r_{ij} r_{ij} N_{ij} tr_{ij} tr_{ij} N_{tr} TR_{tr} R_{ij} N_{tr} tr = \epsilon \epsilon \epsilon \epsilon \epsilon \Sigma \Sigma \Sigma \Sigma + , , (15.12)$

For the example in Figure 15.2, Equation 15.12 yields $PH = 2458.3 \text{ pass.-h}$ $i, j, N = \times + \times + \times [] \in \Sigma () () () , 680 \ 10 \ 300 \ 16 \ 430 \ 5 \ 60$

in which the transfers (both directions of 2-3, 2-4) do not include penalties (estimated

extra-effort cost). Passenger load Average travel time (min) 680 80 1501 1001 350 120 80 100 300 100 2626 750 (provided capacity)

Figure 15.3 The load profile of route [1-3-4] in case of minimum frequency provided.

15.2.2.5 Estimation of Z₂

Estimation of the minimum fleet size can utilize the simple calculation of Section 6.2 or the

definition function modeling described in Chapters 6 and 7. Note that it may be sufficient to use

the procedure to determine the stronger fleet size lower bound (see Section 7.2.2) for this

estimation. Practically speaking, the design of optimal transit routes involves a vast amount

of computations of sets of routes. Thus, the lower-bound-based Z₂ calculation can ease the

computation effort for each route considered.

15.2.3 Applications

The framework of the objective functions described is believed to be a useful tool set for

more than designing a new transit network of routes. For instance, this framework can

apply to

- Optimal design for expansion or curtailment of an existing transit network of routes
- Assessment of the performance of an existing transit network from the aspects of: (1) agency efficiency (fleet size, empty-seat hours, length of routes, number of transfers) and (2) passenger level of service (average waiting time, deviation from shortest path, crowding level)
- Sensitivity analysis of transit network performance for a variety of system parameters (such as different fleet sizes, different levels of service, changes in passenger demand, changes in frequencies, changes in travel time, and more)

One application that utilized the objective functions described was presented by Yin

et al. (2005). This application proposes a deployment-planning framework that pro-

vides, in a sequence of steps, a general structure for the optimal deployment of buses in

a rapid transit (BRT) system. The following BRT elements were considered for system

enhancement:

1. Bus signal priority (extended over existing deployment)
2. Exclusive lanes
3. Articulated buses
4. Multiple door boarding and alighting

5. Stop enhancements

6. Electronic fare payment

7. Precision docking

Given these BRT elements, the deployment-planning framework was used to determine cost

efficient combinations for system enhancement. Equations 15.1 through 15.6 were then

utilized to calculate performance measures for each combination of BRT elements. From

these calculations, a non-optimal combination was selected and recommended for deployment.

Table 15.2 presents seven alternative combinations of BRT elements considered in the study,

together with their estimated cost (in millions of \$US).

In the Yin et al. (2005) study, the budget limit for implementing the BRT elements was

considered as given in the amount of \$90 million. If the number was high enough, there

would be no trade-off between elements. Therefore, the financially feasible alternatives

are C, D, and G. Moreover, by refining Alternative E through excluding element e, stop

enhancement, and making it financially feasible, a new Alternative H is created that includes

the elements a, c, d, and f. The total cost is \$83.9 million.

The results of the Yin et al. analysis are shown in Table 15.3, in which the objective is to

find the best alternative with minimum Z1 and minimum Z2. It can be seen from Table 15.3

that Alternative H is dominated by Alternative G and that both G and D are non-dominated.

Although the capital cost of Alternative D is much less than that of Alternative G, the former

requires a much larger fleet size, which may lead to higher operating and maintenance costs.

Therefore, transit agencies should look carefully at the trade-off between these two objectives

the two recommended alternatives (G and D) should be deployed.

15.3 METHODOLOGY AND EXAMPLE

The objective functions described facilitate the construction of a methodology for the practical

design of an efficient transit network of routes. This methodology, outlined in this

section together with an example, combines the philosophy of mathematical programming

approaches with decision-making techniques; it allows the transit planner to select a set of

routes from a number of alternatives. The methodology and the example follow Ceder and

Israeli (1992) and Ceder (2003).

15.3.1 Six-element methodology

The transit route-design methodology consists of six elements, shown in Figure 15.3. The

first element generates every feasible route and transfer (throughout the entire network)

from all terminals, including the shortest-path computation. Initially, the network contains

average travel times covering a time window, which is usually the peak period. These mea-

sured average travel times are then used as input for the calculation of the shortest path

between each origin-destination (O-D) pair. Each candidate route determined meets the

route length-factor constraint; that is, one procedure of this element screens out routes

according to given boundaries of a route length. In addition, there is a limit on the route's Table 15.2 Cost estimates for combinations of BRT elements Alternative BRT elements Cost (\$M) A a, b, c, d, e, f 99.9 B a, b, c, d, e, f, g 101.4 C a, b, e, f 19.9 D a, b, e, f, g 21.4 E a, c, d, e, f 93.9 F a, c, d, e, f, g 95.4 G a, b, c, d 86.6 Source: Yin, Y. et al., Transport. Res. Rec., 1903, 11, 2005. Table 15.3 Evaluation results for combinations of BRT elements Passenger travel time (pass.-h) Passenger waiting times (pass.-h) Empty seat-hours (pass.-h) Z1 Z2 Alternative D 1999 94 1614 23,966 138 Alternative G 2025 139 1639 28,648 84 Alternative H 2075 139 1665 29,065 86 Source: Yin, Y. et al., Transport. Res. Rec., 1903, 11, 2005.

average travel time between each O-D pair. That is to say, a given passenger demand, usu

ally during peak hours, cannot be assigned to a candidate route if its average travel time

exceeds the shortest-path travel time by more than a given percentage. Feasible transfers

(between O -D pairs without direct routes) are based on establishing additional direct routes

between O -D pairs characterized by high O -D demands (predetermined O -D); feasibility is

determined by the travel-time limit in comparison with the shortest path. These extra direct

routes are initiated and/or terminated at non-terminal nodes, and consequently deadhead

trips are responsible for their connection to the terminals. The feasible transfers are

created using a mapping algorithm (branching of routing possibilities along with a check of

constraints). Finally, low O -D demand, without a direct route, may not be considered for

implementing a transit service. The following example further explains this element.

The second element in Figure 15.4 creates a minimum set of routes and related transfers,

such that connectivity between nodes is maintained and their total deviation from the shortest

est path is minimized. This problem is defined as a set-covering problem (SCP) similar to the

one in Section 12.4.3. The SCP determines the minimum set of routes from the matrix of

feasible routes, in which each column represents either a feasible route or a feasible transfer. Network nodes and arcs; average travel time on each arc; public transit origin-destination (O-D) demand; set of terminals Element 1 Construction of public transit routes and determination of transfers Element 2 Construction of complete sets of routes and transfers (set-covering problem) Element 3 Demand assignment procedure for Element 4 Calculation of the optimization criteria Z_1 and Z_2 (in terms of total passenger hours, difference in passenger hours from shortest or alternative path, waiting time, empty-seat hours, minimum fleet size) frequency calculation Element 5 Generation of more suboptimal solutions of (Z_1, Z_2) Element 6 Multiobjective user decision Maximum percentage increase from shortest or alternative path length; maximum route length Additional routes and transfers given by the agency Maximum required frequency (inverse of policy headway); desired vehicle occupancy; set of routes and transfers given by the agency (e.g., existing); weights of passenger waiting, transfer, and in-vehicle times Monetary weights for the optimization criteria elements Constraints on the effectiveness of each set of solutions Key: Input data

Figure 15.4 A methodology for designing public transit routes.

The procedures included in this element are described in the first edition of this book, in

Section 14.4.

The third element assigns the entire O-D d

emand to the chosen set of routes. The assignment

ment algorithm follows the procedure described in Section 1.5 and includes steps that are

related to a route choice decision investigation. That is, the algorithm includes a probability

listic function for passengers who are able to select the transit vehicle that arrives first or,

alternatively, who wait for a faster vehicle. The passengers' strategy is to minimize the total

weight of wait, transfer, and in-vehicle times.

The fourth element represents the optimization criteria from the passengers', agencies',

and community/governments' perspectives. It is detailed in the previous section and based

on Equations 15.1 through 15.6.

The fifth element is responsible for constructing alternative sets of routes in order to search

for additional (Z_1 , Z_2) values in the vicinity of their optimal setting. The procedure for this

search is based on incremental changes in the set of routes, much like the known reduced

gradient methods. Given the set of routes associated with the minimum Z_1 value, the single

route that is the worst contributor to Z_1 is deleted and then the SCP is resolved, followed by

the execution of the third and fourth elements. This process could continue, but there is no

guarantee that a previous alternative will not be repeated. In order to overcome this problem,

a new matrix is constructed with the idea of finding the minimum and worst set of candidate

routes for possible deletion in each iteration, that is, a new SCP matrix is constructed in

which the candidate routes are the columns and each row represents a previous set of routes

that was already identified in the vicinity of the optimal (Z_1, Z_2) setting. The solution to this

new SCP matrix is a set of rejected routes so as not to repeat a previous alternative solution.

During this process, a number of unique collections of routes are termed prohibited columns,

as they are the only ones that can fulfill a certain demand. These prohibited columns are

assigned an artificially high cost value so as not to be included in the solution. This process

also involves some bounds on the number of (Z_1, Z_2) solutions and number of iterations.

The sixth element, and the final one in Figure 15.4, involves multi-objective programming

of the two objective functions, Z_1 and Z_2 . Given the alternative sets of routes derived in

the fifth element, the purpose is to investigate the various alternatives for the most efficient

(Z_1, Z_2) solution. The method selected in this element is called the compromise-set method.

The outcome of the compromise-set method is the theoretical point at which (Z_1, Z_2) attains

its relatively minimum value. The results can be presented in a table or a 2D graph showing

the trade-off between Z_1 and Z_2 . These results also indicate the optimal zone or the

so-called Pareto front. The decision maker can then decide whether or not to accept the

proposed solution; for example, the decision maker can see how much Z_1 is increased by

decreasing Z_2 to a certain value and vice versa.

15.3.2 Example

A simple eight-node network for a bus service will be used as an example to demonstrate

some of the procedures discussed; it is depicted in Figure 15.5 with two terminals (from

which trips can be initiated). The input of passenger demand appears in Table 15.4; in addition

it is given that $a_k = 1$ for all $k = 1, 2, 3, 4$. The aim is to find the best bus routes in the

network while complying with given constraints.

The outcome of the first element in Figure 15.4 is presented in Table 15.5, while using the

maximum deviation from the shortest path as $\alpha = 0.4$, that is, no route length or portion

of it can exceed its associated shortest travel time by more than 40%. Construction of the

nine feasible routes emanating from terminal 1 is shown in Figure 15.6. The first element is

based on an algorithm that mainly produces feasible transfers throughout the entire network.

The transfers that are created using a mapping algorithm are shown in Table 15.6 for the

example problem, the numbers in parentheses being the route numbers of Table 15.5 that

comprise the transfers. In the transfer-path description, the numbers outside the parentheses

represent nodes while those inside the parentheses represent routes.

The second element of the methodology creates a minimum set(s) of routes and related

transfers, defined as a nSCP matrix. Each row in this matrix represents either a feasible

route or a transfer. The “1” and “2” in the matrix are inserted whenever a nOD demand

can be, respectively, feasibly and optimally (shortest path) transported by the route or transfers

or “0” otherwise. The word covering in the SCP refers to at least one column with “1”

in each row. Section 14.4 of the first edition of this book provides further description of the

SCP analysis. In the example problem, about 100 sets of routes were generated. An example

of one set is {4, 6, 9, 11, 25, 28, 32, 46}, in which the numbers refer to those in Tables 15.5

and 15.6; ve routes and two transfer paths cover all the OD pairs in the defined feasible

manner. The results of the SCP analysis for this set of routes appear in Table 15.7. 3 2 5 6 8 7 1 4 15 25 35 25 5 25 5 30 30 20 20 10 10 20 Terminal Terminal Average travel time (min) Input : Maximum increase (from shortest/alternative path) factor, $\alpha=0.4$ Maximum number of transfers allowed, $k_{tr} = 2$ Desired vehicle occupancy, $d_o = 50$ passengers

Figure 15.5 Example of an eight-node network with its basic input.

Table 15.4 Passenger demand between nodes (assumed to be symmetrical) for the example problem

Nodes 2 3 4 5 6 7 8

1 80 70 160 50 200 120 60

2 120 90 100 70 250 70

3 180 150 120 30 250

4 80 210 170 230

5 Symmetrical 250 40 130

6 130 120

7 70

The third element assigns the entire O-D demand to the chosen set of routes. Vehicle

frequency in this element is derived from the passenger-load profile of each route; the

load on each route is determined by the demand and assignment method. The fourth

element calculates the optimization parameters P_H , DPH , WH , EH on a selected

route basis for computing Z_1 , and determines the minimum fleet size Z_2 required to

meet the passenger demand. For instance, the calculated values of these optimization

parameters of the set of routes, of Table 15.7, are shown in Figure 15.7. The step-by

step derivation of these parameters is shown in Figure 15.7 for Route 4 (the first route

in the set); that is, given the assigned passenger demand for each O-D pair (the result of

the third element), a load profile is constructed. Then, the max load, frequency (from

which the waiting time is derived), and empty-space hours can be calculated. The values

of P_H and DPH depend on the average travel time for each O-D pair as well as on its

assigned demand.

The 4th element is responsible for constructing alternative sets of routes in order to

search for additional (Z1, Z2) values in the vicinity of their optimal setting. In the example

problem, nine sets were produced by this element; these sets are as shown in Table 15.8,

including the resultant Z1 and Z2 values. Table 15.5 All feasible routes of the example problem generated by Element 1

Route no.	Description
1	1 - 2 - 2 - 1 - 2 - 4 - 3 - 1
2	1 - 2 - 5 - 4 - 1 - 2 - 5 - 6 - 5 - 1 - 2 - 5 - 7 - 6 - 1 - 2 - 5 - 7
3	1 - 8 - 7 - 1 - 2 - 5 - 7 - 8 - 6 - 8 - 1 - 3 - 9 - 1 - 3 - 6 - 10 - 4 - 2
4	11 - 4 - 2 - 1 - 3 - 12 - 4 - 2 - 1 - 3 - 6 - 13 - 4 - 2 - 3 - 14 - 4 - 2
5	15 - 4 - 2 - 5 - 16 - 4 - 2 - 5 - 6 - 17 - 4 - 5 - 18 - 4 - 5 - 3
6	19 - 4 - 5 - 6 - 20 - 4 - 5 - 7 - 21 - 4 - 5 - 7 - 8 - 22 - 4 - 5 - 7 - 8
7	23 - 4 - 7 - 24 - 4 - 7 - 5 - 25 - 4 - 7 - 5 - 3 - 26 - 4 - 7 - 5 - 6
8	27 - 4 - 7 - 8 - 28 - 4 - 7 - 8 - 6 - 8 - 5 - 6 - 5 - 2 - 3 - 3 - 2 - 4 - 5 - 7 - 3 - 4 - 5 - 6
9	3 - 8 - 7 - 8 - 6 - 3 - 4 - 1

Key: Cannot be connected because of the maximum increase factor (over the shortest/alternative path)

Figure 15.6 Example solution steps in generating all (nine) feasible routes from Terminal 1.

Table 15.6 All transfers for the example problem connecting nodes 3 and 8 during their 35 min shortest path

Transfer no.	Description
29(5, 18, 27)	3(18) - 5(5) - 7(27) - 8
30(5, 18, 28)	3(18) - 5(5) - 7(28) - 8
31(6, 18)	3(18) - 5(6) - 7(6) - 8
32(6, 25)	3(25) - 5(25, 6) - 7(6) - 8
33(7, 18)	3(18) - 5(7) - 7(7) - 8
34(7, 25)	3(25) - 5(25, 7) - 7(7) - 8
35(18, 20, 27)	3(18) - 5(20) - 7(27) - 8
36(18, 20, 27)	3(18) - 5(20) - 7(28) - 8
37(18, 21)	3(18) - 5(21) - 7(21) - 8

38(18, 22) $3(18) - 5(22) - 7(22) - 8$

39(18, 24, 27) $3(18) - 5(24) - 7(27) - 8$

40(18, 24, 28) $3(18) - 5(24) - 7(28) - 8$

41(18, 26, 27) $3(18) - 5(26) - 7(27) - 8$

42(18, 26, 28) $3(18) - 5(26) - 7(28) - 8$

43(21, 25) $3(25) - 5(25, 21) - 7(21) - 8$

44(22, 25) $3(25) - 5(25, 22) - 7(22) - 8$

45(25, 27) $3(25) - 5(25) - 7(27) - 8$

46(25, 28) $3(25) - 5(25) - 7(28) - 8$

Note: Route numbers in parenthesis.

a When there is a possibility of more than one transfer, the node in which the transfer is considered indicates the two

routes in the parentheses.

The sixth element of the methodology involves multiobjective programming of the two

objective functions, Z_1 and Z_2 . Given the alternative sets of routes derived in the 5th ele

ment, the purpose here is to investigate which set provides the more efficient solution. The

trade-off situation regarding the example problem is depicted in Figure 15.8. The lower

left corner of the envelope contour of the nine solutions represents the best sets; for a bi

objective problem, this boundary is called the Pareto front (Coello et al., 2002). The user is

then able to choose a desired solution with this information in hand. In the example case,

the choice is between $[Z_1 = 787, Z_2 = 106]$, $[Z_1 = 866, Z_2 = 102]$, and $[Z_1 = 997, Z_2 = 101]$

for sets 1, 4, and 6. More details of the multi-objective analysis appear in the next section

of this book. Table 15.7 SCP matrix of one set of routes and transfers

O-D pair	Routes	Transfer path	4	6	9	11
25 28 32 46	1, 2 2 2 - 2 - - - -	1, 3 - - 2 2 - - - -	1, 4 - - - 2 - - - -	1, 5 2 2 - - - - -	1, 6 - - 2 - - - - -	1, 7 - 2 - - - - -
1, 8 - 2 - - - - -	2, 3 - - - 2 - - -	- - 2, 4 - - - 2 - - -	2, 5 2 2 - - - - -	2, 6 2 - - - - 2, 7 - 2 - - - - -	2, 8 - 2 - - - - -	3, 4 - - 2 1 - - - 3, 5 - - - - 2 - - - 3, 6 - - 2 - - - - 3, 7 - - - - 2 - - - 3, 8 - - - - - 2 a 2 a 4, 5 - - - - 1 - - - 4, 6 - - - - - 2 - - 4, 7 - - - - 2 2 - - 4, 8 - - - - 2 - - 5, 6 2 - - - - - 5, 7 - 2 - - 2 - - - 5, 8 - 2 - - - - 6, 7 - - - - - 2 - - 6, 8 - - - - - 2 - - 7, 8 - 2 - - - 2 - - -

The O-D pair is not covered by the route. 1, Covered but not optimally. 2, Optimally covered. a Optimum path: contains a single transfer. Load profile computation for Route 4

Arc on Route 4	Load on arc (passengers)	O-D pair	Assigned demand (passengers)
(1, 2)	186	1-2	1-5
1-6	145	(2, 5)	271
1-5	1-6	2-5	2-6
(5, 6)	465	1-6	2-6
5-6	22	19	19
145	37	70	145
70	250	Routes in set	
Emptyspace hours			
Waiting time (pass.-h)	DPH	Max load (passenger)	Frequency (veh/h)
4	6	9	11
25	28	55.6	105.5
25.3	50.2	128.8	210.2
29.2	30.3	27.7	32.3
34.1	32.8	12.1	0
0	0	13.3	0
465	781	175	482
430	729	9.3	15.6
3.5	9.6	8.6	14.6
Total	575.6	186.4	25.4
Empty-space hours			
Load (passengers)	600	400	200
186	271	465	Max load Time (min)
1	2	5	6
Node	0	10	20
30	40	50	

Figure 15.7 Computation of the load profile of Route 4 and the analysis required for Set 1 of routes (#4, 6, 9, 11, 25, 28) and transfers (#32, 46). Table 15.8 All selected sets of the example problem and their objective function values

Set	Description	Z_1	Z_2	Z_3	Z_4
1	{4, 6, 9, 11, 25, 28, 32, 46}	787	106	2	{7, 9, 11, 19, 25, 27, 34, 45}
2	{7, 9, 11, 19, 25, 27, 34, 45}	900	109	3	{7, 9, 11, 25, 28, 34, 46}
3	{7, 9, 11, 25, 28, 34, 46}	1105	117	4	{6, 9, 11, 16, 25, 28, 32, 46}
4	{6, 9, 11, 16, 25, 28, 32, 46}	866	102	5	{6, 12, 19, 25, 28, 32, 46}
5	{6, 12, 19, 25, 28, 32, 46}	937	103	6	{7, 12, 25, 27, 34, 45}
6	{7, 12, 25, 27, 34, 45}	997	101	7	{7, 12, 25, 28, 34, 46}
7	{7, 12, 25, 28, 34, 46}	1213	113	8	{4, 6, 12, 25, 28, 32, 46}
8	{4, 6, 12, 25, 28, 32, 46}	869	103	9	{6, 12, 16, 25, 28, 32, 46}
9	{6, 12, 16, 25, 28, 32, 46}	961	105		

15.4 BUS NETWORK DESIGN WITH DEMAND VARIATIONS

Chapter 10 lists and describes the main a

tributes to affect the transit demand. That is,

changes in fares, travel time, service frequency, walking time, routing and transferring,

stop location, comfort and inconvenience elements, information, weather changes, tour

ism flows, and more. This section portrays the different demand-related characteristics and

focuses on the design of transit network for cases in which the demand varies in a certain

pattern during the year.

15.4.1 Transit demand characteristic in bus network design problem

Three types of demand variations attribute-related can be identified as follows.

15.4.1.1 Service dependency

Transit demand is an output of the transportation-demand analysis process. Modal-split

procedure defines the share of transit demand, out of the total demand in a city, by using

the utility of transportations modes. The utility of a transit mode is derived from current

transit system attributes, such as travel time, number of transfers, and convenience. It can

be easily seen that transit network design would result in a new set of utilities and consequence

quently with a new transit demand curve. This variation of transit demand is called service

dependency. Lee and Vuchic (2005), Fan and Machemehl (2006), and Van Nes et al. (1988)

have all considered service dependency of demand in their attempt to solve a transit network

design problem. Figure 15.9 shows the concept of service dependency with respect to the bus

network design problem (BNDP). 600 700 800 900 1000 1100
1200 1300 120 2 2 (fleet size) 2 1 (minimum
waiting time and maximum utilization) 1 2 3 7 6 9 5 8 4
Compromise solution 110 100 90 Min 2 1 Min 2 2 Set #

Figure 15.8 Trade-off between 2 1 and the minimum fleet size 2 2 in the example problem.

15.4.1.2 Temporal variation

Passenger demand constantly changes during the days of a week and the weeks of a month

according to people's activities and seasonal requirements. These temporal variations can

produce a continuous demand function with the possibility to characterize it by peak

and off-peak demand. One important aspect of temporal variations is related to seasonal

demand with its annual variation pattern. For instance, summer tourism can significantly

increase the transit demand. Generally speaking, in many cities recreational trips increase

the demand during summer with a reduction of educational and work trips; it results in a

complete different demand pattern than in other seasons. This multi-period demand consid

eration has been studied on one occasion by Chang and Schonfeld (1991). They developed

an analytic model for optimizing bus service, considering the time dependency and elasticity

in demand characteristics. Their analytic model has been developed for feeder bus systems

and cannot be applied for the entire network. Although their method was a step forward

in this area, the simplifications made of network description and decision variables do not

allow for utilizing it with real-life transit networks. For instance, dividing the city into trian

gular zones and optimizing line spacing for the feeder network is not a realistic assumption,

especially in irregular city networks.

15.4.1.3 Randomness

Random demand variation is another aspect of changes in transit demand. This aspect

was studied by Yan et al. (2006) in their investigation of intercity bus routing and sched

uling. However, the random fluctuations of demand are negligible compared with the

significant changes attribute based. For example, the peak (or off-peak) hours demand

fluctuations between two days are significantly smaller than the differences between the

observed peak and off-peak demands for a given day. Accordingly, and also because of

the complexity involved with the consideration of demand distribution functions, this

type of variation is assumed to be insignificant in the analysis of the network (routes)

design problem.

A good choice related to transit-demand characteristics can make the problem closer to

reality. There are a number of cities all over the world that experience demand variations

as a result of tourist arrivals and other demand-related

activities. Currently, these situations

are the main reasons for the change in schedule of the existing transit service. Naturally,

there is a need to investigate the extent to which changes in the transit routing can assist in

attaining a better network-design solution when considering the variation of demand. Transportation demand analysis Transit demand Bus network design Transportation demand analysis Transit demand Bus network design New utility parameters

(a) (b)

Figure 15.9 Service dependency (fixed vs. elastic demand) related to the BNDP. (a) Fixed demand consideration for bus network design and (b) elastic demand consideration for bus network design.

The idea of this section is to construct a base-robust network (BRN) considering the

variation of demand throughout the year, that is, different demand matrices for different

seasons rather than considering only one demand matrix for the whole year. This consid

eration would complicate the problem, making it difficult to solve. Multiple-demand sce

narios mean multiple calculations of the objective function in a problem that is already with

NP-hard complexity (see Chapter 5). In what follows are two methods proposed to solve

this complicated problem based on Amiripour et al. (2014a,b).

15.4.2 Problem statement

In cities that experience different demand conditions during the year, the first strategy that

transit agencies choose is to reschedule the routes and provide extra services where needed.

If this strategy does not satisfy the raised demand, then some new routes are introduced to

cover new origin-destination demand. The need to introduce new routes implies that the

regular network cannot provide the required service when a new demand arises. In other

words, the base network that has been designed for one demand scenario is not optimum

in a new demand scenario. Thus, the question is whether it is possible to have a base-robust

network (BRN) that keeps its optimality in all demand scenarios occurring during the year

and remains intact.

For clarification, a four-node network is illustrated in Figure 15.10 with two demand

matrices, D1 and D2, representing the first and the second half year (6-month) demand,

respectively. The D1 and D2 represent two seasons. The idea of the BRN is illustrated on

this network. There are 12 possible routes (R1 to R12) covering the four nodes of the net

work of Figure 15.10, given that routes can start and end at any of the four nodes. These

routes are shown in Figure 15.10. The initial objective is to minimize the total travel time

(TTT) in passenger-minutes such that all the node-to-node demand is satisfied.

Basically, there exist two possible strategies to handle the variation of the demand. The

first is to use/implement the same network of routes for the entire year, and the second is

to use a different (best suited) network for each season. However, the second strategy is not

convenient and confusing from the passenger's perspective, thus it is not really applicable.

The main question, therefore, is which set of routes to apply for the entire year so as to best 2 3 1 4 9 2 5 5:
 Travel time (min) R1: 1-2-3-4 R2: 1-3-2-4 R3: 1-4-2-3 R4:
 1-4-3-2 R5: 1-3-4-2 R6: 1-2-4-3 Demand matrix 1 of first
 season (D1) Nodes 1 0 10 12 15 0 SYM SYM 0 15 0 10 30 12 1
 2 3 4 20 400 10 30 0 0 15 0 2 3 4 Nodes 1 1 2 3 4 2 3 4
 Demand matrix 2 of second season (D2) R7: 2-1-3-4 R8:
 2-1-4-3 R9: 2-3-1-4 R10: 2-4-1-3 R11: 3-1-2-4 R12: 3-2-1-4
 5 5

Figure 15.10 Example of four-node network with 12 possible routes.

accommodate the variable demand. An optimization approach is to select a network of

routes that minimizes the total travel time with the fulfillment of all pairs of the demand.

Table 15.9 shows the TTT for satisfying D1, D2, and the sum of both representing the TTT

for the entire year.

Fundamentally, there are four different scenarios to consider. That is, Scenario 1, best case

(applying each half a year a different bus network); Scenario 2, applying the best network

of the first half year throughout the entire year; Scenario 3, applying the best network of

the second half year throughout the entire year; and Scenario 4, applying the network with

the minimum TTT for the entire year (BRN). The solution for Scenario 1 is to apply Route

10 and Route 9 for the first and second half year, respectively, with a TTT of 1349 (650 +

699) passenger minutes. The solution for Scenario 2 is to

apply only Route 10 throughout

the year with a TTT of 1 529 (650 + 8 79) passenger minutes. The solution for Scenario 3 is

to apply only Route 9 throughout the year with a TTT of 1 414 (699 + 7 15) passenger min

utes. Finally, the BRN solution is to apply Route 4 throughout the year with a TTT of 1 388

(659 + 729) passenger minutes.

Figure 15.11 presents a comparison between the highlighted results of Table 15.9 for

the yearly TTT results; this is the last column of Table 15.9 except for Scenario 1 in which

the best results are combined of each half year demand (650 + 6 99). Although Scenario 1

is the best result, as told, it is not convenient and will confuse the user who will have to

deal with different routes in the first and second half year for the same origin-destination

(OD) pair. Thus, the BRN (Route 4 and Scenario 4) is the best to represent both the first

and second half year demands using a single route.

In this simple example, also shown in Amiripour et al. (2014b), a single route covers the

demand. However, in the real problem of finding the BRN the complicated configurations of

routes should be considered and the objective function would not be the simple TTT. That

is, the objective function is usually a multi-objective function covering the three perspectives

of the users, operators, and community.

15.4.2.1 Problem formulation

All possible transit networks of all of the demand scenarios must be examined under an

optimization framework. This is in contrast to cases where the network has been designed

for a single demand scenario. The general formulation of the problem is described as follows. Table 15.9 Total travel time of all possible routes in the sample network

Route no.	Path	TTT-D1 (passenger-min)	TTT-D2 (passenger-min)	Sum of TTT -D1 and TTT-D2 (passenger-min)
1	1-2-3-4	968	1113	2081
2	1-3-2-4	860	830	1690
3	1-4-2-3	719	854	1573
4	1-4-3-2	659	729	1388
5	1-3-4-2	760	945	1705
6	1-2-4-3	928	1353	2281
7	2-1-3-4	1040	1185	2225
8	2-1-4-3	814	1149	1963
9	2-3-1-4	715	699	1414
10	2-4-1-3	650	879	1529
11	3-1-2-4	1240	1303	2543
12	3-2-1-4	1063	1084	2147

Assume n demand scenarios (D_q), $q = 1, 2, \dots, n$, $D_q \in DS$, where DS is a demand-scenario

set, and each D_q occurs at a specific time window during the year. In addition, θ_q indicates

the percentage of year that D_q occurs for each q , thus $\sum_{q \in DS} \theta_q = 1$. Let $Z_{BN D p q}$ be the objective

function of transit network $BN p$ on demand scenario D_q . $BN p \in BNS$, where BNS is the set

of different networks designed for different demand scenarios. Thus, the objective in a year

implementing $BN p$ would be $\sum_{q \in DS} \theta_q Z_{BN D p q}$.

For each demand scenario D_q one optimum transit network $BN q$ can be designed. One pos

sibility is to have n transit networks each fitting best to its corresponding D_q with an objective

function value of $\sum_{q \in DS} \theta_q Z_{BN D q q}$. Another possibility is to consider a single network $BN p$ for all

demand scenarios D_q , and attain $\sum_{q \in DS} \theta_q Z_{BN D p q} \leq Z_{BN BNS p q \theta}$. Between these two possibilities, it

is therefore reasonable to construct the following formulation of the BRN to be termed BN r with

the objective function $\sum_{q \in Q} DS_q BN D Z r q \theta$ for demand scenario $D q$. This yields the following problem: $\text{Min } \sum_{q \in Q} BN D q DS r q \theta \in \Sigma$ (15.13)

subject to $\theta \theta q BN D q DS q BN D q DS p Z Z BN BNS r q p$
 $q \in \Sigma \Sigma < \forall \in$ (15.14)

The goal is to minimize $\sum_{q \in Q} DS_q BN D Z r q \theta$ with the result being a base-robust network for all

demand changes.

Constraint (15.14) implies that the robust network will give better results than cases where

only one (most frequently used) demand matrix is used for the transit network design problem; in

these cases, the variation of demand is considered only by changes of schedules, not in the routing.

Equation 15.15 shows that certainly the objective function value of the sum of all best net

works for each scenario is less than or equal to the one searched for (BRN). The left-hand

side of Equation 15.15, therefore, represents the best-case scenario. However, in practice,

this is unrealistic because transit agencies and the authorities would not agree to frequent 1250 Scenario 1
 Scenario 2 Scenario 3 Four scenarios of networks Scenario 4
 1300 1350 Total travel time (passenger
 minutes) 1400 1450 1500 1550

Figure 15.11 The TTT yearly results of the four scenarios related to the sample four-node.

routing changes (a few times per year) of the entire network system; this would obviously

create confusion among the passengers: $\theta \theta q BN D q DS q$
 $BN D q DS Z Z q q r q \in \Sigma \Sigma \leq$ (15.15)

The objective function used is the summation of

passengers.

a_3 : Monetary value of 1 h in-vehicle time.

a_4 : Monetary value of 1 h's operation of one transit vehicle.

a_5 : The time penalty associated with an unsatisfied transit user.

f_k : Corresponding headway of route k .

$f_{\min}$: Its minimum allowable value (policy-headway based).

L_k : Length of route k , $L_{\min}$, and $L_{\max}$ are its minimum and maximum allowable values.

T_{ijk} : Travel time between i and j on route k .

T_{ijsp} : Travel time between i and j using shortest path.

α : Maximum allowable deviation of one route from shortest path.

N_k : Set of nodes in route k .

$Tr_{ij \text{ covered}}$: Number of transfers for a trip between i and j if the OD pair is covered.

$Tr_{\max}$: Maximum allowable number of transfers.

$UD_{\max}$: Maximum allowable unsatisfied demand.

$FS_{\max}$: Maximum allowable fleet size.

Constraint (15.17) guarantees that the robust network would be better compared with other

networks designed for a single demand scenario.

Constraint (15.18) limits the model to

define frequencies for all routes within an acceptable range. Constraint (15.19) is the upper

and lower bounds of the route length, and Constraint (15.20) assures that the route length is

below an acceptable deviation from the shortest path.

Constraint (15.21) limits the number

of transfers below a given maximum number. Constraint (15.22) is the maximum allowable

unsatisfied demand, and Constraint (15.23) represents budget limitation associated with the

operator's perspective; it keeps the fleet size below a given acceptable level.

Solving this problem would take a great deal of computational effort (NP-hard) because

of the calculations involved with the transit assignment process and of the objective function

for (D S) times. The next section, therefore, proposes two methodologies enabling solu

tions for large-scale transit network sizes.

15.4.3 Two solution methodologies

Based on Newell (1979), the transit network design problem is a problem of astronomical

proportions. This quote is another way of showing the NP-hard nature of the transit network

design problem. Among the three categories of the problem solution methods discussed

earlier (math solutions, heuristic methods, and meta-heuristic methods), the math methods

cannot be used for a real-life transit network design problem because of the level of sophistication

it requires for the problem. Developing a heuristic application to different large-scale

transit networks is also a complicated procedure that requires different types of evaluations

and definitions within the algorithm. Thus, there are not many heuristics available for the

transit network design problem in contrast to meta-heuristic methods.

Considering seasonal demand variations in the transit network design problem would

make it more complicated; the required time depends on how many demand scenarios exist.

The more demand scenarios the more required computational effort to solve the problem.

The use of meta-heuristic methods, such as genetic algorithm, is one way to attain a satisfactory

solution. Let us call this solution method DMH (direct meta-heuristic). In a DMH method,

the fitness function would be the weighted objective function for different demand scenarios.

Although DMH is a straightforward solution, it cannot be the ultimate procedure because

of the required computational time. Hence, two-solution methodologies, enabling to solve

the problem in a reasonable amount of time, are presented in the following text.

15.4.3.1 Solution methodology 1(DMH-F)

Following Amiripour et al. (2014a), a hybrid method, called DMH-F, is constructed. This method

is developed in two major steps: route-level optimization and network-level optimization. The

route-level optimization limits the set of feasible routes to routes that have greater potential

to enter BRN. This heuristic process ignores routes that do not satisfy the BRN requirements

and creates a smaller feasible set for the next step. The network-level optimization is a meta-

heuristic method where the possible BRNs are created to

check whether or not they can be con

sidered as robust using genetic algorithm. Afterward, a heuristic algorithm checks the created

BRNs for possible improvements. Figure 15.12 illustrates this procedure schematically.

In order to solve the transit network design problem with seasonal variable demand amid

the proposed algorithm, the sample network with eight nodes (see Figure 15.5) is used as

an explanatory device to describe the new method developed. The method is based on the

following four steps.

15.4.3.1.1 Step 1: Data collection

In this step, demand scenarios and their corresponding demand matrices are identified for

each time period during the year. Other data related to policies and levels of service are also

inserted for creating the constraints.

Sample-related: Four demand scenarios are considered for the sample network of Figure

15.5; it is depicted in Table 15.10 (one for each season) with SYM, meaning symmetrical.

15.4.3.1.2 Step 2: Route-level optimization

15.4.3.1.2.1 STEP 2.1: CREATING AN INITIAL SET OF FEASIBLE ROUTES

In this step, all possible routes are identified using the shortest path or the K-shortest path

algorithm (Yen, 1971) between all nodes or terminal points. This requires the input of the Inputs (step 1) Demand matrix 1 Heuristic method at route level Genetic algorithm at network level Heuristic method on each route for possible improvement Demand matrix n Route-level optimization (step 2) Network-level optimization (steps 3 and 4)

Figure 15.12 Procedure of DMH-F for transit network design with seasonal variable demand.

maximum allowable deviation from the shortest path (α) and the parameter K . That is,

the second shortest path will be based on the shortest path (see Appendix 9.A), the third

shortest path will be based on the second shortest path, and so on. Yen (1971) provided an

algorithm to solve it efficiently.

Sample-related: For the sample network, of Figure 15.5, $K = 10$ and the allowable deviation

from shortest path (α) is 50%; this results in the creation of 113 feasible routes.

15.4.3.1.2.2 STEP 2.2: ELIMINATING SHORT OR LONG ROUTES

Excessively short or long routes are not economic and operable. Thus, a range of accept

able route length (in terms of time or distance) is defined such that routes outside this range

are not acceptable. This range is defined by using the mean and standard deviation of all

the routes created in the set of feasible routes. Equation 15.24 represents a route length

criterion where routes that do not comply with it will be deleted from the feasible set. The

criterion is: $l_k - \leq \cdot \mu \leq \sigma$ (15.24)

where l_k is the route length μ and σ are mean and standard deviations of all route lengths of the created set of feasible routes

Parameter ξ is the ratio that defines the range and it is determined by performing a sensitiv

ity analysis on each network considered. If a large number is chosen for ξ on ly few routes

would be eliminated; if choosing a small number, most of the routes would be eliminated.

Hence, a starting number of $\xi = 2$ is proposed. It is to note that a sensitivity analysis can be

useful for determining the amount of ξ for different transit networks.

Table 15.10 Four demand matrices of the sample network in Figure 15.5

0	0	80	70	160	50	200	120	60	0	80	70	40	50	120	120	60
1	0	120	90	100	70	250	70	0	120	40	100	70	250	70		
2	0	180	150	120	30	250	0	40	150	120	30	250				
3	0	80	210	170	230	0	80	210	170	230						
4	0	250	40	130	0	250	120	130								
5	SYM	0	130	120	SYM	0	65	120								
6	0	70	0	70												
7	0	0														
0	0	80	120	160	90	200	120	60	0	20	70	160	50	200	120	60
1	0	120	90	40	120	20	120	0	120	90	250	70	250	70		
2	0	180	150	120	30	20	0	120	250	120	30	250				
3	0	80	210	1120	10	0	20	210	170	230						
4	0	20	40	130	0	250	40	130								
5	SYM	0	130	120	SYM	0	130	120								
6	0	120	0	70												
7	0	0														

Sample-related: In the sample network, the mean and variance length of the 113 routes,

initially selected, are 46.9 and 19.3 (in min) with $\xi = 2$; it results with an acceptable route

length within the time (in min) range [8.2-85.6]. Four routes were not within this range and

thus removed to make 109 routes in the feasible set.

15.4.3.1.2.3 STEP 2.3: IMPORTANT NODE COVERAGE

Covering important demand-related attraction/production (A/P) points within a transit

network is a target to guarantee reaching meaningful demand points. This step identifies

important A/P points and the routes are ranked by the number of important nodes they

cover; then routes with the lowest number of important nodes covered are eliminated from

the set of feasible routes. That is, using the following procedure:

1. For all nodes ($i \in N$) in the network:

Set $ODI_i = d_i + \sum_{j \in N} d_{ij}$.

2. Sort all the nodes by their ODI_i .

3. Mark top $\lambda\%$ of the nodes as important A/P points.

4. For all the routes in the set of feasible routes, count the number of important A/P points.

5. Sort all the routes by their important A/P points count.

6. Eliminate $\beta\%$ of the routes with the lowest important A/P point count.

where ODI_i is the total amount of demand that node i attracts or produces; d_{ij} is the amount

of demand between nodes i and j .

Determination of λ and β is a decision to take place after a sensitivity analysis.

Sample-related: On the sample network, the ODI s

of nodes 4 and 6 (see Figure 15.5) are

larger than others, thus chosen to be important A/P points. Routes that do not cover nodes

4 or 6 are removed. The procedure, with $\lambda = 0.3$ and $\beta = 0.2$, results with a removal of

21 routes, bringing it to 88 routes in the feasible set.

15.4.3.1.2.4 STEP 2.4: DEMAND VARIABILITY INDEX

A good route of the process portrayed is perceived also in terms of having sufficient passen-

ger loads. In other words, a good route will attract passengers continually; this type of route

is expected to be part of the BRN. Thus, in this step, the covered demand by each route

in different demand scenarios is calculated including the mean and standard deviation of

the demand covered. Then the remaining routes are sorted by their coefficient of variation

$(CoV) = \sigma/\mu$. The $\gamma\%$ of the routes having the largest CoV are dropped (γ is selected based

on sensitivity analysis). This filter assures that all routes of the set of feasible routes do not

have large variations of covered demand for the different demand scenarios.

Sample-related: In the sample network, the CoV of each of the 88 routes is considered using

the four demand scenarios. An initial value of $\gamma = 0.3$ is considered. By doing so, 26 more

routes were deleted by this filter bringing it to 62 routes in the set of feasible r

outes, from

which the best solution will be found.

15.4.3.1.2.5 STEP 2.5: PRACTITIONERS' ROUTES

It is to note that following the three filtering steps, there is another step allowing practitio

ners to insert existing routes chosen on top of the routes in the feasible set. This will form

the final set of feasible routes. This additional step is important in practice, but was not

considered for the sample network.

15.4.3.1.3 STEP 3: GENETIC ALGORITHM OPTIMIZATION

Step 3 employs the approach of genetic algorithm (GA) to optimize the specified objective

function. GA is a stochastic optimization algorithm, inspired by body-evolution process,

searching for a good solution of problems in which the feasible set is excessively large.

This known algorithm (Amiripour, 2014b) has been widely used for complex problems.

Operators of crossover, mutation, and elitism are the major components of GA. First, a

random initial population of solutions, called chromosomes, is made; in this section, chro

mosomes are transit networks where the routes come from a set of feasible routes. Then

the fitness function or the objective function value of each chromosome is calculated as

is shown schematically in Figure 15.12. By using a weighted random selection, two chro

mosomes are chosen to enter the mating pool for the crossover operator, called parents.

Crossover operator combines two parents and makes two new children (termed offspring)

as is depicted in Figure 15.13. If the offspring have a better fitness value than their parents,

then the parents are replaced by the offspring; otherwise the parents stay in the popula

tion. This process, described in Figure 15.12, continues until the new generation is born.

To avoid stagnation in terms of local optima, the mutation operator randomly changes

some solutions in each population as is shown in Figure 15.13. Elitism operator gets the Initial population
Fitness evaluation of each individual in population Take the best chromosome to the next generation (elitism)
Randomly choose two individuals (weighted) Do these two steps until the required chromosomes for new generation are created
Crossover operator—take parents or children to the next generation
Mutation operator Build the next generation
Termination: number of generations > max generation? Yes End
No

Figure 15.13 Genetic algorithm (GA) procedure.

best chromosome in each generation and passes it to the next. This ensures that the itera

tive process of GA always gets better or at least remains the same.

There exist many variations of GA in the literature applicable to a variety of problems.

The classic form of GA is illustrated in Figure 15.13. Crossover operator, shown in

Figure 15.14, combines two parent chromosomes (P1, P2) and produces two children chro

mosomes (C1, C2); the mutation operator changes one random gene in one chromosome of

P1 (Figure 15.14). Narrowing the set of feasible routes affects the convergence speed of the

algorithm thus reducing the computational effort for

Figure 15.14 Crossover and mutation operators in GA.

can be applied in this step using weights for each demand scenario. The modified procedure is described as follows:

1. For each route (not examined) create the set of nodes that can be added into the route such that each added node has a direct link with one of the two end points of the route; add the nodes to the set NA.

2. Calculate for all nodes of NA the total weighted demand ΔD_w (weighted across all demand scenarios) covered by the route with the added node as defined in Equation 15.25: $\Delta D_w = \sum_{D \in \Sigma} (CD_{D,q} - CD_{D,q-1}) w_{D,q}$ (15.25) where $CD_{D,q}$ is the covered demand by the corresponding network in demand scenario D before adding one node to the specified route in the bus network $CD_{D,q-1}$ is the same parameter after adding the node to the network

3. The node from NA with the largest ΔD_w will be added to the corresponding route if the characteristics of the new route comply with the constraints of max capacity, max length, max deviation from shortest path, and available fleet size; if not complying with all the constraints, the added node is deleted from NA and the next node from NA with the largest ΔD_w is examined; otherwise, add the checked node to the route, delete it from NA, and create a new route (with the added node); go to (a) with the new route; if no more nodes exist in NA then STOP.

The ΔD_w is the total demand covered, by the route, with the consideration of the added node(s). This total demand is the weighted sum of passenger demand across all scenarios.

Figure 15.15 illustrates the elements of Step 4 using two routes [1 to 5] and [6 to 9]. That is,

the set NA comprises nodes 10-15 of the [1 to 5] route and of nodes 14-19 of the [6 to 9] route. The ΔD_w is then calculated for each possible added node by figuring out the total new demand (weighted across all scenarios) of its corresponding route. In this example, there will be 12 such calculations from which the node associated

with max ΔD w will be selected

to be added. Then the process continues with additional nodes to be added to the set NA

coming from the new added node that will become a new end point. This process follows

major parts of the route expansion procedure of Baaj and Mahmassani (1995). 10 1 11 12 13 5 15 14 6 16 7 8 19 17 9 18 4 3 2

Figure 15.15 Interpretation of Step 4.

Sample-related: In the sample network, shown in Figure 15.5, no node could be added

because of the constraints given.

Step 4 allows taking into account the maximization of the annual, not season-based,

passenger demand. The total weighted demand ΔD w can be calculated at the network level

considering also trips with transfers. At the end of this four-step procedure, the demand

considered for the BRN is increased and ready for processing.

15.4.4 Analysis of the sample network

Table 15.11 presents a summary of results and processes of Steps 1 and 2 of the developed

procedure for the sample network (Figure 15.5).

In Step 3, the objective is to minimize the weighted summation of waiting time (in

passenger-hours), deviation from the shortest path (in passenger-hours), empty-seat hours,

the minimum fleet size, and unsatisfied demand as presented in Equation 15.16.

The population size for the GA is 1000 chromosomes for each generation; the chromosome

structure is a binary vector with the length of the number of feasible routes (62: the reduced

number for the 8-node network). Crossover rate is set to 0.6 and mutation is set to 0.003 in

a population. In addition and in order to provide more diversity of solutions and avoid stagnation

at a local optima, a new population is randomly generated once every 50 generations. The

next generation will be formed by selecting the top chromosomes from both the current and

the new population. The elitism operator takes the best chromosomes to the next generation.

Figure 15.16 refers to the sample network and shows the performance of the robust network

by comparing its objective function value (Equation 15.16) with other possible networks.

The best objective function values belong to a set of networks termed Best Networks that

consider the best possible network for each demand scenario—in other words, to set a differ

ent transit network for each season, thus being unacceptable in practice. This is used as the

best-case scenario for comparison purpose. The regular networks in Figure 15.16 are those

that have been designed using only one demand matrix. Figure 15.16 shows that the BRN is

closer to the best-case scenario than for the regular (usually existing) network results, that

is, the BRN has the potential to significantly improve

existing networks.

The advantage of the B-RN is one aspect of the problem studied. The other aspect of the problem is the solution methodology. In order to solve the previously stated problem, either a DMH or the DMH-F algorithm can be used.

Figure 15.17 shows the results, of the sample network, using the DMH-F and DMH methods.

Both methods are iterative as is seen in Figure 15.17; the convergence to a better

solution is an important factor which shows that the DMH-F converges much faster than

the common DMH method. The initial listing of the set of feasible routes narrowed the

Table 15.11 Summary of implementing the route-level process on the sample network in Figure 15.5

Set of feasible routes 113 routes Running K-shortest path algorithm between all node pairs with $K = 10$ and allowable deviation from shortest path = 50%.

Eliminating short or

long routes 4 routes deleted Mean and variance of routes are 46.9 and 19.3 min. For $\xi = 2$ the acceptable route length is in the range [8.2, 85.6] in min.

Important node

coverage 21 routes deleted Values of $\lambda = 0.3$ and $\beta = 0.2$ are used. Two nodes identified as important nodes; 21 routes are removed because of zero coverage of A/P points.

Demand variability

index 26 routes deleted Initial value of $\gamma = 0.3$ considered, and leading to the removal of 26 more routes.

Remaining routes 62 routes

area and consequently the GA may start from a better solution. It is to note that according

to the stochastic nature of the GA the last statement is not always true, but it is expected that

the initial population will be better in terms of objective function.

As represented in Figure 15.17, the DMH-F converges to a network with an objective

function value of 281,976 (in monetary units) by choosing routes out of the set with

62 routes. The DMH chose routes among 113 routes and resulted in a network with the

same objective function value of 281,976. This means that the DMH results are the same as

DMH-F, but the GA reaches this result in iteration 222 (DMH) compared with iteration 36

for the DMH-F method. In other words, the DMH-F converges to the sample-network solu

tion six times faster. In addition, the chromosome length is shorter in the DMH-F method

than in the DMH to reduce the runtime for each generation. It is to note that in small net

works the risk of narrowing the set of feasible routes is much higher than for large networks. 0 Season 1 Season 2 Season 3 Free demand scenarios Objective function value for each network (in Ks of monetary units) Best networks Robust network (BRN) Regular network Season 4 50 100 150 200 250 300 350 400 450 500

Figure 15.16 Objective function values using three scenario networks of the sample network. 1 265 Objective function value (in Ks of monetary units) 270 275 280 285 290 295 300 305 310 51 101 151 201 GA iterations DMH DMH-F 251 301 351

Figure 15.17 Comparing the proposed algorithm and direct GA method for the sample network.

That is, by narrowing the set of feasible routes some of the good solutions are removed and

it would affect the final results. More of this methodology is in Amiripour et al. (2014a).

15.4.5 Solution methodology 2 (HBRD-I)

The second solution methodology is based on Amiripour et al. (2014b) in which a hybrid

method, called HBRD-I, for solving the transit network design with seasonal demand varia

tion (finding BRN) is preferred. The objective function and constraints are those shown in

Equations 15.16 through 15.23. This HBRD-I procedure utilizes GA efficiently in creating

the BRN. Figure 15.18 illustrates the HBRD-I procedure in general terms, based on eight

specific steps described as follows:

Step 0: Network set of BRN starts with no routes.

Step 1: Design one corresponding network ($BN_p \in BNS$) for each demand scenario considering the routes of the network set of BRN as fixed. These networks would be the best possible networks for the corresponding demand scenarios (using GA).

Step 2: Calculate the overlapping score (OS) for all the routes using the following process: For all the routes ($k \in BN_p$) in all networks ($BN_p \in BNS$) Calculate overlapping index of route k with network BN_q (OI_{k, BN_q}) Compare k with all routes (k) of the other networks ($BN_q | BN_q \in BNS, q \neq p$) using $OI_{k, BN_q} = \frac{L_{k, k'} \cap L_{k, k'}}{L_{k, k'} \cup L_{k, k'}}$ (15.26) with $L_{k, k'}$ common k, k' (15.27) End Design the rest of the network using genetic algorithm directly If the CoV of the most overlapped route is acceptable the continue, otherwise try the next route Calculate the CoV of passenger-km traveled in the most overlapped route Find the most overlapped routes using overlapping score Yes No Start Design one network for each demand scenario Consider the selected route(s) fixed, remove the most overlapped routes with recently chosen routes from the potential-route set, and continue Eliminate routes, in the feasible set, that are overlapped with the designed network Is $\tau\%$ of all demand scenarios satisfied?

Figure 15.18 Flowchart of the HBRD-I method. The overlapping score of route k^* would be $OS_{OI(k,k^*)}$, BN_{BN} $BS, q_p q_q^* = \epsilon \neq \Sigma$ (15.28) where L common is the common length of route k^* and k . L_{k^*} and L_k are length of routes k^* and k . A route with a higher OS_{k^*} is more likely to enter the network set of BRN because it will be selected in different demand scenarios.

Step 3: Sort all the routes in all the networks according to their OS_{k^*} .

Step 4: Calculate the total passengers-km of the route with the largest OS_{k^*} in all demand scenarios within the designed network. Calculate the coefficient of variation (CoV) (σ/μ) of passenger-km for the chosen route.

Step 5: If the coefficient of variation of passenger-km for the chosen route is within the acceptable range, this route would enter the designed-network (BRN) set and the procedure proceeds from Step 6. Otherwise, the next route with largest OS_{k^*} with the desired condition is chosen to enter the designed network and the process goes back to Step 4.

This step results with routes having an acceptable amount of passengers in all of the

demand scenarios. The coefficient of variation helps to find routes that are more robust in

terms of demand, thus it leads to a network with less frequency changes required during

the year.

Note: The acceptable range of CoV should be determined by performing a sensitivity analysis of

the network of feasible routes before running the HBRD-I.

Step 6: Remove all the routes that have overlapping index (OS_{k,k^*}) more than 8% with the recently chosen route(s) from the set of feasible routes.

Step 7: If the total covered demand by the network set of BRN, in all demand scenarios, is less than $\tau\%$ go back to Step 1. Otherwise, go to Step 8.

Step 8: Consider the network set of BRN as
x and run a DM algorithm to complete the
network, that is, to solve an optimization problem for
finding the rest of the routes in the final network
considering all demand matrices.

At Step 0 of the HBRD-I, a network set of B
RN is opened. Then in Step 1, one optimal
tran

sition network is designed for each of the a
ssumed demand scenarios using GA; the GA goes

for a limited number of generations in this s
tep. In Step 2, the overlapping score p
arameter

(OS) is calculated in order to find the most
overlapped route in all of the designed n
etworks.

It is assumed that the routes appearing in
the transit networks of all demand scenarios
can

be considered as the base skeleton of the B
RN. Thus, by using the overlapping score t
hese

routes are identified. In Steps 3 and 4, the
most overlapped routes are checked for c
overing

the demand in all of the demand scenarios b
ased on the COV of passenger-km. If the r
oute is

within an acceptable range of COV, it will
be part of the BRN, otherwise the next o
verlapped

route is checked (Step 5). In Step 6 all o
f the routes that are similar to the chosen
routes

(based on the overlapping index OI and the
predefined parameter of δ) are removed from

the set of feasible routes. This step n

arrows down the set of feasible routes and reduces the

chromosome length and feasible solution area for a better convergence of GA. At this stage

if $\tau\%$ of the demand, for each demand scenario, is covered by the BRN the process proceeds

with Step 8, which is running DMH to find the other routes of the BRN. Otherwise, the

process goes back to Step 1 to design a new transit network for each demand scenario by

considering the routes in the BRN as pre-chosen routes.

The HBRD-I procedure attempts to find routes to help construct the BRN incrementally.

At each iteration, one or more routes are added to the BRN and all the similar routes

are removed from the set of feasible routes. Then another iteration starts with previously

selected routes considered. After some iterations, the BRN covers a certain portion of

the demand, so that a DMH method can complete the process and finds the rest of the routes

to increase the demand coverage. The decision of the time to switch to a DMH method is

determined by the τ parameter; this parameter should be determined by performing a sensi-

tivity analysis of different sets within the network.

As mentioned earlier, the HBRD-I is a hybrid method using GA efficiently. Because of the

stochastic nature of the GA, the complexity of this method is unknown; however, search

ing for a solution in a smaller set of feasible routes is much faster than in a larger space

(it converges to the solution faster). Thus, the HBRD-I method is more efficient than the

DMH method because it makes the set of feasible routes smaller at each iteration. Besides,

searching for a smaller amount of routes at each iteration makes this procedure faster than

the DMH method. More of this methodology is in Amiripour et al. (2014b).

15.5 METHODOLOGY FOR ACTUAL-SIZE TRANSIT NETWORK DESIGN

Designing transit network for actual-size road networks is described by Bagloee and Ceder

(2011). This study focuses on providing systematic transit-planning tools for actual-size

networks considering the following issues collectively:

Practicality—being able to tackle actual-size problems.

Integrated planning—synchronizing routes and frequency setting.

Multiclass—design of a system containing different transit modes (e.g., metro, bus, ferry) and different route hierarchies (e.g., collector, feeder, mass (or dominant)-route) in a systematic manner.

Transit assignment—a transit assignment method is required for a given solution; it is noted that in transit network design most researchers, because of computational complexities, use simplifications.

Stop positioning—despite the importance of positioning the stops on the routes, only a handful of research papers consider it and these are mostly concerned with the issue of optimal stop spacing by the use of rules-of-thumb.

This method of Bagloee and Ceder (2011) undertakes actual-size cases with a process of

transit network design synchronized with f

frequency setting. The algorithm developed con-

siders the concept of clustering to identify the best set of candidate stops and of different

transit modes and hierarchies. Two transit assignment methods have been employed in the

planning process: capacity-free and capacity-constrained transit assignment. The principals

of Bagloee and Ceder (2011) methodology to design optimal transit network are as follows.

Given a road network, an exogenous fixed transit demand, and a limited budget, the prob-

lem is to derive a comprehensive transit plan (including routes and assigned vehicles) so that

the total passengers' discomfort is minimized. A route comprises two end points (terminals),

a sequence of links, a timetable based on frequency setting, and is 2-way (bidirectional).

The algorithm allows different and even distant paths (depending upon demand concen-

tration) for either direction, thus loop routes can also be considered. In any transit plan

(scenario), the discomfort index of passengers from origin i to destination j is designated as

generalized time $GT_{i,j}$: $GT_{i,j} = InVehTim_{i,j} + WalkTim_{i,j} + WaitTim_{i,j} + Transfer_{i,j} \cdot \omega_1 + \dots + \omega_n$ (15.29)

where $InVehTim_{i,j}$, $WalkTim_{i,j}$, $WaitTim_{i,j}$, $Transfer_{i,j}$ are in-vehicle time, access/egress walking time, waiting time, and number of transfers or boarding, respectively, to get from i to j

$\omega_1, \omega_2, \omega_3 > 0$ are weights for waiting time, walking time, and transfer/boarding penalty while in-vehicle's weight is assumed to be 1

The objective function (1) is established as the total saved generalized time with respect to

no-transit-plan scenario: $\max \sum_{i,j} \theta_{i,j} \times i,j \times i,j$
 $GT = - \sum \sum (15.30)$

where θ and x denote no-transit-plan and scenario number, respectively. This formulation

makes the definition of the objective function clearer in terms of discomfort index inter

preted as generalized times, and emphasizes the catering to captive users who have no other

travel choice than transit.

The solution algorithm consists of three components illustrated in Figure 15.19. In

Component 1, the location of stops is identified based on closeness to highly concentrated

transit demand places and clustering factor. Component 2 is inspired by the Newton grav

ity theory, to generate a set of candidate routes. In Component 3, a search algorithm is run

through the generated candidate routes to seek a good solution; this search procedure is

encoded by means of a genetic algorithm (GA) and equipped with a ant-system (AS) collective

points as the search engine.

The algorithm was tested on the actual-size transit network of the city of Winnipeg; the

results show that under the same conditions (budget and constraints) the set of routes resulted

in a reduction of 14% of total travel time compared to the existing transit network. In addition

tion, the methodology developed is compared favorably with other studies using the transit

network of M and I benchmark. The generality of the methodology was tested on the real

dataset of the larger city of Chicago, in which a more efficient and optimized scheme is proposed

for the existing rail system. More on this study appears in Bagloee and Ceder (2011).

15.6 CONNECTED PRIORITY-LANE CONSIDERATION IN BUS NETWORKS

This section is somewhat different from the previous sections of this chapter. It is about

the general problem of transit network design; it is about a special transit network that

consists of only exclusive or dedicated or priority bus lanes. One of the major drawbacks in

constructing priority bus lanes is the lack of a well-connected network of these lanes so as

to maintain the prioritized movement of its patronage.

Bus priority schemes and techniques on urban roads and highways have proven effective

for almost a half century. Many bus-priority strategies have been demonstrated worldwide.

Traditionally, priority is granted for bus operation at stops, at intersections, and by preferen-

tial/exclusive lanes. It is known that bus travel times, reliability of service, and vehicle pro-

ductivity are improved when buses are able to use higher-speed, uncongested lanes. These

improvements make the bus systems more attractive and thus increase the potential to gain

new riders.

Eight preferential treatments to buses on street lanes are known as follows: exclusive

curb lane, semi-exclusive curb lane (shared only with cars about to turn), exclusive median

lane (with stop island), exclusive lane in the center of a street, bus malls (limited to pedes

trians and buses), exclusive freeway/highway lanes, ramp bypass (for entering a freeway/

highway during traf c congestion), and congestion bypass (exclusive lanes to bypass traf c

bottlenecks).

Network-level consideration of priority/exclusive lanes is one way to improve transit per

formance. Mees (2010) introduced the term network effect where the transit network has

short waiting times, easy transfers, good coverage, and high reliability to allow it to compete

with the private cars. Thus, planning system-wide priority lanes share the same philosophy.

The  rst to introduce a system-wide approach for designing priority lanes were Mesbah

et al. (2008, 2010, 2011a,b). In these research studies, a bi-level optimization model is

proposed to combine priority lanes selection with traf c assignment. The model assesses Route selection filter (headway filter) Sophisticated generation selection upon collective point Scenario evaluation (capacity-constrained assignment) Scenario evaluation (capacity-free assignment) Stop category clustering concept 1. Stop location and assigning demand to stops Demand at the stop level Route generation module -Selecting terminals -Laying down the routes on manipulated shortest path -Hierarchies: (mass-feeder-local) Stochastic generation random selection

subject to budget

3. Route selection and frequency setting

2. Route generation Route detouring iter

Figure 15.19 Three components of the Bagloee and Ceder (2011) method.

the impact of exclusive lanes on private cars' travel time and optimize the overall weighted

travel times and distances. Because of the complexity of the problems, the authors used

meta-heuristics algorithms in their solution approach. However, few points are worth men

tioning in these four good studies: (i) handling of only two priority alternatives (exclusive

or mixed) compared with the eight preferential treatments mentioned earlier; (ii) resulted

priority lanes are not necessarily connected (or continuous); and (iii) consideration of only a

reduction of travel time may not result in the most efficient network coverage.

A recent study by Hadas and Ceder (2014) attempts to deal with the three points (i), (ii),

and (iii) mentioned in the studies of Mesbah et al. (2008, 2010, 2011a,b). In what follows is

a brief description of the modeling of Hadas and Ceder (2014).

Each arc in the sample network presented in Figure 15.20 (between two numbered nodes)

is a road section that can be constructed as part of a possible priority lane (exclusive or semi

exclusive). Each priority-lane alternative will be examined in terms of its cost and benefits

(time saving). All circled nodes are a set of possible origins and destinations for the priority

lanes. The goal is to construct a set of priority lanes that connects transit stations, transfer

hubs, routes' start/end stops, and link one priority lane to other priority lanes. By doing so,

the transit network will be characterized by uninterrupted routes (such as 15-14-32-13-

12-11-33 in Figure 15.20), as opposed to the construction of isolated priority lanes, which

often experience traffic bottlenecks in the form of non-prioritized sections. Balancing origin

and destination nodes is crucial for a well-connected network. For instance, optimizing time

saving alone can result with two North-South priority lanes compared with a balanced

result of one North-South priority lane and one East-West priority lane; the latter slightly

decrease the time saving, but maintain a balanced priority lanes network.

15.6.1 Model formulation

Let $G(N, A)$ be a directed network that comprises all road sections traversed by transit

routes. Let $x_{i,j,k}$ be a binary decision variable, such as "1" represents the selection of priority

lane alternative k for road section (i, j) , and "0" otherwise. Furthermore, let $c_{i,j,k}$, be the con

struction cost, $v_{i,j,k}$, the travel time saving per passenger, and $f_{i,j}$ the total passengers' flow of all

1 6 2 7 8 5 3 15 30 17 14 32 18 34 20 19 22 23 26 25 24 33
12 11 28 27 Potential transit lanes Transit nodes Terminal
nodes 30 10 29 9 13 31

Figure 15.20 Sample network.

routes passing through road section (i,j) . Let $I \subseteq N$ be a set of all nodes from which a priority

lane starts or ends. For constructing paths, let p_{ijmst} , , , be an indicator whether road section

(i, j) is part of path m that starts from node $s \in I$ and terminates at node $t \in I$. For clarity the

index m will be omitted henceforth. Let $x_{m,s,t}$ be a decision variable, such as “1” represents

the selection of path m that starts from node $s \in I$ and terminates at node $t \in I$. Let matrix P

represent the paths; P can be easily calculated, as is described in the following text. A gain,

for clarity, the index m will be omitted. Furthermore, let B be the budget available, and D_l ,

D_u be the lower and upper bounds for nodes' degree. The optimization formulation is $\max \sum_{i,j,k} x_{i,j,k} c_{i,j,k}$ (15.31) $\min \sum_{i,j,k} x_{i,j,k} t_{i,j,k}$ (15.32)

subject to $x_{i,j,k} \leq B_{i,j,k}$ (15.33) $x_{i,j,k} \leq 1$ (15.34) $x_{i,j,k} \geq 0$ (15.35) $x_{i,j,k} \in \{0,1\}$ (15.36) $x_{i,j,k} \leq 1$ (15.37)

Equation 15.31 maximizes total time saving resulted from using the selected transit prior

ity lanes. Equation 15.32 maintains a balanced connectivity between the selected terminal

nodes. This balance is maintained by maximizing the minimal in-degrees and out-degrees

(the number of nodes directly connected to/from a given node) of all terminal nodes among

all feasible solutions (SL). An unbalanced priority lane set will impact the overall reliability

of the transportation network and reduce the level of service. Constraint (15.33) enforces budget

availability and constraint (15.34) maintains the selection of only one alternative. Constraint

(15.35) enforces the following: if at least one path $\sum_{s,t} s_{i,j} s_{s,t} p_{px} \neq 0$ () $\geq \sum_{s,t} s_{i,j} s_{s,t} p_{px} \neq 0$ () if from s to t is

selected ($p_{x,s,t} = 1$), then one alternative ($x_{i,j,k}$) for road section (i,j) must be selected given that

the road section is part of the path from s to t ($p_{i,j,s,t} = 1$). This constraint also maintains the

continuity of each selected priority lane.

The complex multi-objective problem can be converted to a single-objective problem.

That is, Equation 15.32 can be substituted by the following constraints: $p_{x,Dl} s_{I, s,t} s_{t,t} \sum \geq \forall e \neq$ (15.38) $p_{x,Du} s_{I, s,t} s_{t,t} \sum \leq \forall e \neq$, (15.39) $p_{x,Dl} t_{I, s,t} s_{t,t} s_{\sum} \geq \forall e \neq$ (15.40) $p_{x,Du} t_{I, s,t} s_{t,t} s_{\sum} \leq \forall e \neq$, (15.41)

Equations 15.38 and 15.39 enforce that the out-degree is between upper and lower bounds.

Equations 15.40 and 15.41 enforce the in-degree bounds. Hence, it is possible to explore

different solutions with the adjustments of D_u and D_l .

The input to this model is based on matrix P ; this matrix is a set of all possible paths

(shortest, quickest, etc.) starting at nodes and terminating at nodes. Pre-calculating P will

result with an efficient optimal algorithm, because it is not required to calculate on-the-fly

paths from G . Constructing matrix P is a straightforward process of implementing an all

pairs shortest path algorithm (see Appendix 9.A), K-shortest path (Yen, 1971), and multiple

reasonable routes (Park et al., 2002). Finally, a case study is presented in Hadass and Ceder

(2014) to demonstrate this approach.

15.7 LITERATURE REVIEW AND FURTHER READING

This section contains a review of papers that propose methods of optimizing the configura-

tion of transit-route systems. The output of such methods is a route itinerary that usually

includes headways or frequencies. Papers that focus on headway determination, without a

route-itinerary design, are discussed in Chapter 3. Papers in which special attention is given to

unique objectives, such as the minimization of transfer times, are concentrated in Chapter 5.

Hobeika and Cho (1979) present a method for determining the structure of a bus-route

system. A heuristic algorithm partitions the existing bus stops in an urban area into sectors

and seeks a way to link the stops while trying to minimize the total distance traveled by all

buses. In the optimization process, each bus is subject to capacity and distance constraints.

The routes developed are improved iteratively; in each iteration, a disaggregate choice model

is used to examine passengers' behavior. E

equilibrium between supply and demand is reached

when the proportion of passengers using buses cannot be increased by improving the bus

network.

Marwah et al. (1984) develop a method for the simultaneous design of routes and fre-

quencies. First, passenger flows are assigned on the road network. Then, a large set of pos-

sible bus routes that satisfy certain constraints is generated. Finally, routes that minimize

the number of system-wide transfers are selected. Heuristics are used for the concentration

of flows on the road network and for the initial generation of routes. Linear programming

is used for the selection of optimal routes and for assigning frequencies.

Van Nessel et al. (1988) formulate a programming problem that sets route itineraries and

frequencies with the objective of maximizing the number of trips in which a transfer is not

needed. The constraints include a given fleet size and budget limitations. The authors dis-

cuss the advantages of formulating the route-design problem as a programming problem,

such as the ability to add further constraints. With additional constraints, it is possible to

use the model in systems in which some existing routes may not be changed or there is a

given limit to the number of routes, etc.

List (1990) describes a methodology for preparing optimal sketch-level service plans. The

sketch-level plan does not include precise routes, but it does determine passenger-flow values

and frequencies on the road network. The model includes constraints for demand satisfaction,

fleet size, minimal frequency, load factor, junction capacity, train length, and crew

requirements. Several sub-models are developed that help to formulate those parts of the

model that stem from limited resources, such as energy consumption. The model makes sure

that network flows are balanced and takes into account multi-period passenger demand.

Baaj and Mahmassani (1991, 1992, 1995) develop transit network design methods based

on artificial intelligence (AI). The methods discussed are developed by a typical formulation

of the network-design problem as a programming problem with minimum frequency, load

factor, and fleet size constraints. The first paper (1991) uses flowcharts to present a quantitative

description of a three-stage design process for a route network. In the first stage, a large

set of routes is generated; the second stage involves network analysis and determination

of frequencies; the third stage is network improvement. The second paper (1992) focuses on

a method of representing the transportation network by using lists and arrays to make the

solution procedure efficient. The third paper (1995) concentrates on the stage of creating

the initial set of routes, which are supposed to be modified now and improved later on. In

order to generate this initial route set, a set of basic skeletons is created a long the shortest

paths between nodes with high passenger demand; the skeletons are expanded, using a set

of node-insertion manipulations.

Spasovic and Schonfeld (1993) introduce a method for determining optimal route lengths,

route spacing, headways, and stop spacing in a radial network with one central business

district (CBD). Cost functions are developed for both the operators and the users by mini-

mizing their sum; equations are derived for the optimal values of all decision variables.

A many-to-one demand pattern is assumed, and passenger density can be either be uniform or

decrease linearly with distance from the CBD. In addition, a solution algorithm is developed

that incorporates realistic vehicle-capacity constraints.

Spasovic et al. (1994) extend the model described in the previous paragraph. The deci-

sion variables in this version include route lengths, route spacing, headways, and fares. Two

alternative optimality criteria are examined—operator profit and social welfare, the latter

being the sum of user and operator surpluses. Social welfare is optimized with both uncon-

strained subsidies and break-even constraints. Analytical solutions are sought for a rectan-

gular transit corridor with elastic demand, uniformly distributed passenger-trip density, and

many-to-one travel patterns.

Ramirez and Seneviratne (1996), using GIS, propose two methods for route-network

design with multiple objectives. Both methods involve ascribing an impedance factor to

each possible route and then choosing those routes that have the minimum impedance. In

the first method, the impedance factor depends on passenger flow and on the road length

traveled. This method requires the use of an assignment model. In the second method, the

impedance factor depends on the number of employees who have a reasonable walking distance

from the route.

Pattanik et al. (1998) present a methodology for determining route configuration and

associated frequencies, using a genetic algorithm. Solutions are chosen in an iterative process

from a large set of possibilities in which the chances of a solution's surviving through

the iterations are higher if it yields a high value for a given fitness function. The method

presented here adopts the typical programming formulation of the route-network-design

problem with the objective of minimizing a weighted combination of passenger-time costs

and operator-time costs; the objective function provides the basis for the calculation of

the fitness-function values. A methodology is also presented for the coding of variables as

strings with a fixed or variable length.

Soehodo and Koshi (1999) formulate a programming problem for designing transit routes

and frequencies. Similar to other models, the problem is solved by first creating a list of feasible

routes and then choosing an optimal subset. In addition to some traditional components, such

as minimum frequency and fleet size constraints, the problem adds some unique elements,

such as the inclusion of private car user costs, transit-passenger crowding costs, and transfer

costs, to the minimized objective function. A sub-model is developed for each of these cost

types. Equilibrium of network flows is another constraint. The model assumes that demand

is elastic, and therefore the shift of passengers between different modes of transport plays a

major role. Both transit and non-transit demand-assignment models are used.

Bielli et al. (2002) describe another method for designing a bus network, using a genetic

algorithm. As in other genetic algorithms, each population of solutions goes through repro-

duction, crossover, and mutation manipulations, whose output is a new generation of solu-

tions. In the proposed model, each iteration involves demand assignment on each network of

the current set of solutions and a calculation of performance indicators based on the assign-

ment results. These indicators supply input to a multi-criteria analysis of each network,

leading to the calculation of its fitness-function value.

Van and Chen (2002) present a method for designing routes and timetables that aims at

optimizing the correlation between bus-service supply and passenger demand. The method is

based on the construction of two time-space networks: a fleet-low network and a passenger

low network. Both networks are depicted in bidimensional diagrams in which the hori-

zontal dimension represents bus stops and the vertical dimension represents time. While the

fleet-low network shows the potential activities of the bus fleet, the passenger-low network

illustrates trip demand. The objective of the model is to feed buses and passengers at minimum

cost in both networks simultaneously. A mixed-integer, multiple-commodity, network

flow problem and a solution algorithm based on Lagrangian relaxation are presented.

Van Nes and Bovy (2002) and Van Nes (2003) investigate the influence of the definition of

different traveler groups, respectively. Van Nes and Bovy develop an analytical model, in which

the objective functions are defined for the traveler, the agency, and the authorities. Two

alternative objectives for the traveler and six objectives for the authorities are examined, and

the results of the interactions of different combinations are discussed. In addition, various

alternative assumptions regarding demand elasticity are accounted for, and the difference

in outcomes between different city sizes is analyzed. Van Nes (2003) investigates the same

model, but with different weights for different traveler groups. Van Nes found that the opti

mal result of the network of routes was similar to the traditional single-user-class approach,

thus concluding that a more realistic description of passenger groups was not necessary for

the network-design theme.

Tom and Mohan (2003) continue the development of genetic methods for route-network

design. In the current model, frequency is the variable; thus, it differs from earlier models in

terms of the coding scheme adopted. Whereas fixed-string length coding and variable-string

length coding were used in previous models, a combined route and frequency-coding model

is proposed here.

Zhao and Ukkari (2004) present a mathematical methodology for transit route network

optimization. The goal of this study is to provide an effective computational tool for the

optimization of large-scale transit route networks. The objectives are to minimize transfers

and optimize route directness while maximizing service coverage. The formulation of the

methodology consists of three parts: (1) representation of transit route network solution

space; (2) representation of transit route and network constraints; and (3) solution search

schemes. The methodology has been implemented as a computer program and has been

tested using previously published results. Finally, the results of these tests and results from

the application of the methodology to a large-scale realistic network optimization problem

in Miami-Dade County, Florida, are presented.

Zhao and Zeng (2008) present a meta-heuristic method for optimizing transit network

works, including route network design, vehicle headway, and timetable assignment.

Given information on transit demand, the street network of the transit service area, and

total fleet size, the goal is to identify a transit network that minimizes a passenger cost

function. The methodology described in this paper includes a representation of the

transit network variables search spaces (route network, headway, and timetable); a user cost

function based on passenger random arrival times, route network, vehicle headways, and

timetables; and a meta-heuristic search scheme that combines simulated annealing, tabu,

and greedy search methods. Finally, this methodology has been tested with problems

reported in the existing literature, and applied to a large-scale realistic network optimization

problem. The results show that the methodology is capable of producing improved

solutions to large-scale transit network design problems in reasonable amounts of time

and computing resources.

Mauttone and Urquhart (2009a) proposed a heuristic method, called pair insertion algo-

rithm (PIA), for designing bus network exhibiting the viewpoints of both users and operators.

They modified the heuristic route generation algorithm of Baaj and Mahmassani

(1995), where its original expansion of routes is replaced by a strategy of insertion of pairs

of vertices. Using a case of the city of Rivera, Uruguay, they show that their model has an

advantage over the Baaj and Mahmassani model from the operator perspective in terms of

number of routes, and total travel time.

Mauttone and Urquhart (2009b) developed a method to solve the multi-objective trans-

it network design problem using greedy randomized adaptive search procedure (GRASP)

meta-heuristic. They implemented their method on Mandl's benchmark network and in a

case study of the city of Rivera, Uruguay. Their results show that the proposed algorithm

produces more non-dominated solutions than the weighted-sum method with the same com-

putational effort.

Wan and Lo (2009) developed a framework to design transit services taking into account

financial viability requirement. Inter-route and inter-modal transfers are modeled through

the state-augmented multimodal (SAM) network a

approach. They developed a two-phase

methodology; the first phase seeks to maximize the social welfare by providing the largest

set of bus service connections, subject to the financial viability constraints, and the second

phase seeks to minimize the operation cost by detailing the bus routings and frequencies to

fulfill the service connections determined in the first phase. They implemented their method

on an illustrative sample network.

Fan and Mumford (2010) presented a framework for solving the urban transit routing

problem, dealing with the following components: (1) a suitable representation of the

problem, (2) an initialization procedure to construct randomly feasible sets of routes, and

(3) a make-small-change routine to generate neighborhood moves. They implemented hill

climbing and simulated annealing for testing their technique and showed that on Mandl's

network their method gives better results than those of previously published papers. In addition

their method was examined for use on large-scale networks.

Feng et al. (2010) employ fuzzy programming and use four indicators representing sustain

able transportation, such as accessibility, operator profit, user cost, and reduction of exter

nal cost, to optimize urban bus route allocation. The proposed model is applied to Taoyuan

County, the most rapidly growing city in Taiwan, and managerial implication are identified

to help decision-makers align bus routes based on sustainability criteria. Consideration is

also given to extending the model to more comprehensive evaluation functions.

Szeto and Wu (2011) designed the bus network for Tin Shui Wai, a suburban residential

area in Hong Kong, to improve the existing bus services by reducing the number of transfers

and the total passenger travel time. They proposed an integrated solution method to solve

the route design and frequency setting problems simultaneously. In their proposed solution

method, a genetic algorithm for the route design problem is hybridized with a neighborhood

search heuristic for the frequency setting problem. In order to illustrate the robustness and

quality of their solutions, computational experiments are performed based on 1000 per

turbed demand matrices. The test results showed that the design obtained by the proposed

solution method is robust under demand uncertainty.

Poorzahedy and Safari (2011) utilized an ant colony optimization algorithm for bus network

design problem and minimized total passenger travel time on the transit network taking into

account fleet requirements. They applied and calibrated the model based on the network of

Sioux Falls. They also implemented their method on the case study of the city of Mashhad,

Iran. Finally, this study compared the developed method with a genetic-algorithm method

on two case studies and showed that their results are superior to those obtained by the genetic algorithm.

Yu et al. (2012) created a traveler density model with the maximization of demand density

of routes considering resource constraints. Their transit network design problem is divided

into three stages: (1) skeleton route design, (2) main route design, and (3) branch route

design. They used an ant colony optimization technique for solving the model on a case

study from Dalian city in China. Their results showed that the approach can improve the

solution quality if the transfer coefficient of their objective function is reasonably set.

Cipriani et al. (2012) describe a procedure for solving the bus network design problem

and its application in a large urban area (the city of Rome), characterized by (a) a complex

road network topology, (b) a multimodal transit system (rapid rail transit system, buses, and

tramways lines), and (c) a many-to-many transit demand. The solving procedure consists of

a set of heuristics, which includes a first routine for the route generation based on the low

concentration process and a parallel genetic algorithm for finding a suboptimal set of routes

with the associated frequencies. The final goal of the research is to develop an operative tool

to support the mobility agency of Rome for the bus network design phase.

Nikolic and Teodorovic (2013) developed a model for solving the transit network design

problem based on the swarm intelligence (SI) concepts of artificial intelligence. The objective

tives were to maximize the number of served passengers, minimize the total in-vehicle time

of all served passengers, and to minimize the total number of transfers in the network.

Their proposed bee colony optimization (BCO) algorithm can generate high-quality solutions

within reasonable CPU times. They used Mandl's benchmark network and a larger

test network to examine their development.

Nayeem et al. (2014) developed a population-based model for the transit network design

problem to maximize the number of satisfied passengers, the total number of transfers,

and to minimize the total travel time of all served passengers. They used genetic algorithm

with elitism (GAWE) for solving the problem. The study claims to have the best results on

Mandl's network compared with all previous methods.

Szeto and Jiang (2014) developed a bi-level transit network design problem, where the

transit routes and frequency settings are determined simultaneously. They also developed

a hybrid artificial bee colony (ABC) algorithm to solve the bi-level problem. The upper

level problem is formulated as a mixed integer nonlinear program with the objective of

minimizing the number of passenger transfers, and the lower-level problem is the transit

a ssignment problem with capacity constraints.

Table 15.12 summarizes the main features reviewed and discussed in this section.

Table 15.12 Summary of the characteristics of the methods reviewed

Source	Special features	Unique objectives/constraints	Intermodal considerations
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Hobeika and Cho			
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(1979)	Stops are divided into sectors.	Yes	
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Marwah and Umrigar			
--------------------	--	--	--

(1984)	Minimizing the system-wide number of transfers.	No	
--------	---	----	--

Van Nes et al. (1988)	Maximizing the number of direct trips.	No	
-----------------------	--	----	--

List (1990)	Sketch-level design only; multi-period demand. Junction capacity, train length, and crew-requirement constraints.	Yes	
-------------	---	-----	--

Baaaj and Mahmassani			
----------------------	--	--	--

(1991, 1992, 1995)	Artificial intelligence.	No	
--------------------	--------------------------	----	--

Spasovic and			
--------------	--	--	--

Schonfeld (1993)	Minimizing the sum of user and operator costs; capacity constraint.	No	
------------------	---	----	--

Spasovic et al. (1994)	Maximizing either operator profit or social welfare, with unconstrained subsidies and break-even constraints.	No	
------------------------	---	----	--

Ramirez and			
-------------	--	--	--

Seneviratne (1996)	GIS-based; the second proposed model takes into account potential demand (based on employment density), and not just existing demand.	No	
--------------------	---	----	--

Pattanik et al. (1998)	Genetic algorithm.	No	
------------------------	--------------------	----	--

Soehodo and Koshi			
-------------------	--	--	--

(1999) Elastic demand. Private car user costs, transit passenger crowding costs, and transfer costs are included in the minimized objective function. Yes

Bielli et al. (2002) Genetic algorithm. The fitness function is based on multi-criteria analysis. No

Yan and Chen (2002) Network-flow problem formulation. No

Van Nes and Bovy

(2002) Elastic and fixed demand examined. Multiple objectives of the traveler, the agency, and the authorities are defined and compared. No

Van Nes (2003) Different traveler groups. Multiple objectives in which travel time and relationship between supply and demand depend on traveler groups. No

Tom and Mohan

(2003) Genetic algorithm. No

Zhao and Ubaka

(2004) Minimizing transfers and optimizing route directness while maximizing service coverage. Yes
(Continued)

Table 15.12 (Continued) Summary of the characteristics of the methods reviewed

Source	Special features	Unique objectives/constraints	Intermodal considerations
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Zhao and Zeng

(2008)	The formulation includes definitions of solution search spaces (route network, vehicle headway, and timetable) and a stochastic solution search scheme. Minimizing the total user cost of the transit service area subject to the operator budgetary constraint and a set of route, headway, and timetables feasibility constraints.		No
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Mauttone and

Urquhart (2009a)	Heuristic algorithm. The interests of users (total in-vehicle time) and operators (total route mileage).		No
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Mauttone and

Urquhart (2009b) Multi-objective consideration in bus network design problem. Minimizing total travel time as the first objective and fleet size as the second. No

Fan and Mumford

(2010) Heuristic method. Minimizing travel time and number of transfers. No

Feng et al. (2010) Employ fuzzy programming, and use four indicators representing sustainable transportation. Four objectives, including accessibility, operator profit, user cost and the reduction of external cost, referring to criteria of sustainable transportation are formulated. No

Szeto and Wu (2011) Solving the route design and frequency setting problems simultaneously. Sum of the number of transfers and total passengers' travel time. No

Poorzahedy and

Safari (2011) Use of ant colony optimization for bus network design. Minimize the total travel time of the users of the network, while being feasible in fleet requirements. No

Yu et al. (2012) Used ant colony optimization for bus network design. Maximize the total demand density of the route (the number of transit demands divided by the length of the route). No

Cipriani et al. (2012) Genetic algorithm. The objective function is defined as the sum of operator's costs and users' costs plus an additional penalty related to the level of unsatisfied demand. Yes

Nikolić and

Teodorovic (2013) Based on the bee colony optimization (BCO) meta-heuristics. Maximize the number of satisfied passengers, minimize the total number of transfers, and the total travel time of all served passengers. No
(Continued)

EXERCISES

15.1 Given: (1) A bus service network consisting

of four nodes (one terminal) and vehicle arcs with average travel times (minutes) for both directions of travel. Average travel time (min) 3 20 40 10 4 2 30 20 1
Terminal

(2) An O-D symmetrical demand matrix for a 2 h peak period, the arrival pattern of passengers is assumed to be homogeneously distributed for each cell. From To 2 3 4 1
960 380 160 2 220 200 3 Symmetrical 240

(3) The desired occupancy is 60 passengers per bus.

(4) Bus trips are generated (start and end) only at the terminal (Node 1) and deadheading trips are not allowed.

(5) Two routes (round trips) serve the given network demand, Route A: 1 → 2 → 3 and backward to Node 1; and Route B: 1 → 2 → 4 and backward to Node 1.

(6) Boarding, alighting, and transfer times are neglected.

(a) For the two existing routes, derive the required headways based on the maximum load section method; if more than one alternative exists, select the most appropriate one and explain your decision.

(b) What is the minimum fleet size required for Routes A and B, based on the derived schedule (headways)?

Table 15.12 (Continued) Summary of the characteristics of the methods reviewed

Source Special features Unique objectives/constraints
Intermodal considerations

Nayeem et al. (2014) Genetic algorithm with elitism. Minimizing number of satisfied passengers, the total number of transfers, and the total travel time of all served passengers. No

Szeto and Jiang

(2014) Artificial bee colony (ABC) algorithm. Minimizing the number of passenger transfers. No

(c) Your task is to evaluate all possible single routes (round trips—same route forward and backward from t

he terminal) in the given network, visiting all nodes during the 2 h peak period. Note that the round trips may end after the peak hours.

(c-1) Define all possible combinations of single routes (visiting all nodes).

(c-2) Set an appropriate criterion (criteria) for your evaluation.

(c-3) Construct the load profile with respect to travel time for each possible single route.

(c-4) What is the minimum fleet size required for each single route?

(c-5) Suggest a single route and compare it with the existing two routes.

15.2 Given: (1) The following bus-service network: 18 7 2
12 20 15 4 18 15 6 20 5 10 12 14 16 10 10 8 3 Average
travel time (min) 1 10

(2) In accordance with their respective sequence of nodes, the following three routes exist: the first employs a articulated bus, 1-2-4-6, with a desired occupancy (do) of 75 passengers; the second, 4-5-3, uses a standard bus with do = 50; and the third, 1-8-7, employs a minibus with do = 25.

(3) Symmetrical O-D matrix of a peak period, by number of passengers. From To 2 3 4 5 6 7 8 1 80 100 120 100 180 30
40 2 100 100 140 200 50 30 3 100 80 120 20 30 4 60 140 30
20 5 Symmetrical 100 0 40 6 10 20 7 50

(a) For each route r (one direction), calculate: difference in passenger-hours, PHr; empty-space hours, E Hr; difference between passenger-hours and shortest path, D PHr (without transfers); and passenger waiting time, wr.

(b) Calculate total DPH for all three routes, including transfers consideration.

15.3 One of two independent transit modes is considered for implementation, either fast ferry or fast train. The route examined is $A \rightarrow B \rightarrow C$. The following data are given:
Data item Mode Transit segment $A \rightarrow B$ $A \rightarrow C$ $B \rightarrow C$ Average
travel time (min) Fast train 40 50 10 Fast ferry 70 90 20

Expected fare (\$) Fast train 35 45 12 Fast ferry 30 35 9
O-D demand (passengers) Both modes 280 1100 700

Additional given data and information: (1) desired occupancy, in terms of number of seats, on the fast train and fast ferry is 600 and 450 passengers, respectively; (2) value of passenger waiting time is \$9/h; (3) value of passenger travel time is \$3/h; (4) passengers arrive randomly at stops A and B; and (5) timetables in both modes are based on even headways and Method 2 (see Section 3.2).

(a) Calculate the hourly waiting-time cost of each alternative mode.

(b) Calculate the loss cost of hourly empty seats and travel times.

(c) Calculate the hourly income and profit of each alternative mode.

(d) Based only on the given data, suggest the preferred mode.

(e) List briefly actual cost elements that were neglected in this exercise.

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16 Chapter 16: Designing Short-Turn Trips

Subsequent to the overview and methods for designing routes at the network level, the next

two chapters deal with specific design features at the route level. This present chapter presents

a set of procedures to design transit timetables efficiently with trips that are initiated beyond

the route departure point and/or terminated before the route arrival point. Such trips are called

short-turn trips. In practice (see Chapter 4), transit frequency is determined at the heaviest

load-route segment, whereas the operation at other segments may be inefficient because of

situations, characterized by empty seats. Transit planners attempt to overcome this problem

by manually constructing short-turn trips with the objective of reducing the number of vehicles

required to carry out the transit timetable. The purpose of this chapter is to improve and auto

mate this task. The following riddle may serve as a stimulus for the need to handle a design element

only after comprehending the process in which this element appears. The riddle: given a

microorganism-biological process in which a microbe that is put into a glass splits into two after

1 s, the two then split into four after another second, the four into eight after another second,

and so on; after 1 min, the glass is full of microbes. How many seconds are required to fill half

of the same glass? (Answer is given at the end of this corner.) The chapter contains four main parts, following an introductory section. Section 16.2 outlines

the framework of the methodology for the efficient design of short-turn trips. Section 16.3

identifies a minimum number of candidate short-turn points and presents the example that is

subsequently used throughout the other sections of the chapter. Section 16.4 provides a procedure

to adjust the number of departures at each short-turn point to that required by the load

data, provided that the maximum headway (associated with passenger waiting time) attained is

minimized. Section 16.5 constructs another procedure to minimize the number of short-turn

trips while ensuring that the minimum fleet size is preserved. The chapter ends with a literature

review and exercises. Practitioners are encouraged to visit Sections 16.1 through 16.3. In addition, they should

follow all the figures, tables, and corresponding paragraphs of the chapter that are related to

the example problem. It is known that an error in the premise will appear in the conclusion. Often, transit agen-

cies avoid a profound design because of their premise that it would be costly, time consuming,

and unrewarding. It is analogous to a common saying, and perhaps even belief, that good things

in life are illegal, immoral, or fattening. It will be shown in this chapter that a profound design

(of short-turn trips) can, on the contrary, reduce cost, reduce time, and be rewarding. Last,

the answer to the riddle is 59 s (in the 60th second, the half will multiply itself); obviously this

answer is strongly dependent on the process described.

16.1 INTRODUCTION

Chapter 15 discussed the attainment of an efficient network of transit routes. This chapter

focuses on improving the cost-effectiveness of each single route. Transit planners certainly

understand the need to accommodate the observed passenger demand as well as possible.

At the same time, however, their efforts are also directed at the minimization of vehicle and

driver costs. The trade-off between increasing the passenger's comfort and reducing the cost

of the service makes the planner's task cumbersome and complex. The design of short-turn

trips is one such trade-off. A short-turn trip is initiated beyond the route's departure terminal

and/or terminated before its arrival terminal. The possibility of generating short lines

opens up an opportunity to further save on vehicles while ensuring that the passenger load

on each route segment does not exceed the desired occupancy (load factor).

Planners in most transit agencies usually include a short-turn operating strategy in their

attempts to reduce the cost of the service. The procedures commonly used are based only on

the visual observation of the load profile, that is, a potential turn point is determined at the

time point (major stop) nearest to a stop at which a sharp decrease or increase of passengers

is observed. Although this procedure is intuitively correct, planners do not know whether

all the short-turn trips are actually needed to reduce the

fleet size. Unfortunately, each short

turn trip limits the service and, hence, tends to reduce the passenger level of service.

The major objectives of a short-turn operating strategy, and therefore of this chapter, are

as follows:

- To identify minimum candidate short-turn points based on passenger load profile data
- To adjust the number of departures at each short-turn point to that required by the load data, provided that the maximum headway obtained is minimized; this objective results in the maximum possible short-turn trips and the minimum required fleet size (including shifting departure times and deadheading (DH) trips)
- To minimize the number of short-turn trips, provided that the minimum fleet size attained (with short-turn trips) is maintained; for a given timetable, this objective results in increasing the level of service as seen by the passengers

Several methods will be presented to meet these objectives. These methods are based on pro-

cedures and algorithms that use data commonly inventoried or collected by most transit

agencies. The chapter is based on the methodology and modeling presented by Ceder (1990,

1991), with some improved and corrected elements.

16.2 METHODOLOGY

The proposed methodology relies on the following input data: (1) a complete timetable of

all route time points, (2) passenger load profiles, by load and distance, for each time period,

(3) minimum frequency or policy headway, and (4) a set of candidate short-turn points.

These data are given for both directions of the route (each direction with its own data).

Candidate short-turn points are usually all the major route stops (time points) at which the

public timetable exists. In some cases, it may be limited to only those time points at which

vehicles can actually turn back.

The comprehensive tasks needed to accomplish the objectives of the methodology are

described in flow-diagram form in Figure 16.1; this methodology also appeared in Ceder

(2003). It starts with a procedure to determine the set of feasible short-turn points R_j among

the candidate points. Then the deficit function (DF) theory (Chapters 6 and 7) is used to

derive the minimum number of vehicles required to carry out all the trips in the complete,

two-direction timetable, N_{min} . The required number of departures is determined at each of

the feasible short-turn points, and then the so-called Minimax H algorithm is applied. The

basis of the algorithm is the elimination of some departures from the complete timetable in

order to obtain the number of departures required. In this procedure, the algorithm mini

mizes the maximum difference between two adjacent departure times (headway). At this

stage, as shown in Figure 16.1, the DF method derives the minimum fleet size required with

short turns, $\leq N_{min}$. If this minimum is less than the size required without short turns, then

another procedure is applied. This second procedure inserts (back) the maximum possible

departures among those previously eliminated, provided that the minimum fleet size, $\leq N_{min}$,

is maintained. The final step of the overall program is to create vehicle blocks to cover all

the trips that appear in the last version of the two-direction timetable. Yes No Initialization: Timetable (set of departure times at all time points) Passenger loads (for each time period across all time points) Minimum frequency (policy headway) Set of candidate short turn points Creation of $R = \{R_j\}$ = set of minimum candidate short turn points for each period j Construction of deficit functions at the end points A procedure to shift departure times and/or insert deadheading trips to minimize the route's fleet size Determination of the required number of departures (Max load/desired occ.) for each time point $t \in R$ Result: Minimum fleet size without short turns, N_{min} Deficit function theory and algorithms Networkbased minimax H theory and algorithm Result: A new timetable with the maximum possible short turn trips Construction of deficit functions at each time point A procedure to shift departure times and/or insert deadheading trips to minimize the sum of all the maximum deficit function values Theory and algorithms to maximize the extensions of short turns Procedures to construct vehicle schedules Construction of vehicle schedules Result: Final optimal timetable with short turns Result: A timetable without any short turns Apply the procedure to extend the short turns toward their original trips without increasing N_{min} $N_{min} \leq N_{min}$? Result: Minimum fleet size with short turns, N_{min} $j \in R$ A procedure to exclude departure times from each time point j except for Max load points; it is carried out to minimize the maximum obtained headway while reducing the number of departures to that required $\in R$

Figure 16.1 Flow diagram describing the design of efficient short-turn trips.

16.3 CANDIDATE POINTS AND EXAMPLE

The short-turn points are usually route time points at which the vehicle can turn back without

interfering with the traffic flow. It is therefore anticipated that for each route, the initial

set of candidate short-turn points will be given by the transit planner.

16.3.1 Minimum candidate short-turn points

Let the set of candidate short-turn points be designated as set R_1 for one direction and R_2

for the opposite route direction. Note that R_1 does not necessarily coincide with R_2 . More

specifically, $R_1 = \{r_{11}, r_{12}, \dots, r_{1n}\}$ and $R_2 = \{r_{21}, r_{22}, \dots, r_{2q}\}$ (16.1)

where r_{ij} is the j th candidate short-turn point in the i th direction ($i = 1, 2$) and n and q are such points for directions 1 and 2, respectively

For a given time period, the fluctuation of a passenger load along the entire route (load profile)

may reveal that some short-turn points are actually redundant; thus, we can establish

a set of a minimum number of candidate points. For example, say a load profile consists of

13 stops and 5 candidate short-turn points as shown in Figure 16.2. Theoretically, each seg-

ment between two adjacent short-turn points can be treated independently with respect to

its required frequency. This frequency is determined by the maximum observed load on the

segment, which is marked by the hatched area in Figure 16.2. In short-turn strategy, how-

ever, all the trips must serve the heaviest load segment

of the route (in the example, all trips must cross the $r_4 - r_5$ segment). Another observation is that fewer trips are required between r_3 and r_4 than between r_2 and r_3 while both groups of trips must cross the $r_4 - r_5$ segment. Consequently, the point r_3 is redundant.

The exclusion of the redundant points at each time period j results in a set of minimum candidate short-turn points, R_j ; this analysis is important from the viewpoint of computational time. The formal description of the algorithm to determine the set R_j includes an additional analysis of the difference between the required frequencies at the short-turn point.

Passenger Candidate short-turn point r_1 r_2 r_3 r_4 r_5
 Needed? Distance

Figure 16.2 Construction of a set R of feasible short-turn points.

associated with the max load segment and a considered short-turn point. If this difference is small, the considered short-turn point can be deleted. Note that the difference in the frequencies is equivalent to the difference in the load, and there are always stochastic variations of that load. Hence, if this difference is small, it is not reasonable to consider short-turn trips from the associated short-turn point. This is actually similar to the manual procedure performed in current practice in which the planner selects short-turn points only on the basis of an observed sharp increase or decrease in the load profile. For subsequent analyses, the union of all R_j for all (time periods) j is denoted as the set R .

16.3.2 Example

A simple example is used as an expository device to illustrate the DF approach and the

procedures developed. This example appears in Figure 16.3 for a 2 h schedule of departure

times at the max load points. The route includes three time points (a, b, c), which is the

set R. The average travel times for service and DH trips are given in the following timetable

in Figure 16.3; no shifting in departure times is allowed.

Construction of $d(a,t)$ and $d(c,t)$ can then take place, and the minimum number of vehicles

required without DH trips is $D(a) + D(c) = 11$. However, a DH trip can be inserted from a

to c, departing after the last maximum interval of $d(a, t)$ and arriving just before the start

of the first maximum interval of $d(c, t)$. Both $d(a, t)$ and $d(c, t)$ are then changed, as seen by

the dashed line in Figure 16.3; thus, $D(c)$ is reduced from 6 to 5, and the overall fleet size

is reduced from 11 to 10. After that, it is impossible to further reduce the fleet size through

DH trip insertion; hence, $N_{\min} = 10$. This condition can also be detected automatically by

the lower-bound test. That is, the maximum of the combined DFs is 10, and therefore, $N_{\min}$

reaches its lower bound.

16.4 EXCLUDING DEPARTURE TIMES

The basic information required to consider short turns is the route's load profile. Based on

this load-profile information, each route segment between

two adjacent short-turn points

can be treated separately. That is, the required number of trips between the $(k - 1)$ th and k th

short-turn points for a given direction and given time periods is similar to Equation 3.2 in

Chapter 3: $F_k \geq \frac{P_k}{d_o} \geq F_{\min}$ (16.2)

where P_k is the maximum load observed between the two adjacent short-turn points d_o is the desired occupancy $F_{\min}$ is the minimum required frequency

F_r is the route frequency, determined as in Chapter 15.

16.4.1 Level-of-service criterion

The manual procedure usually undertaken by the planner to create short-turn trips simply

involves the exclusion of departure times in order to set the frequency at each short-turn

point k to F_k instead of to F_r . This exclusion of departure times is performed without any

systematic instructions in the (scheduler) belief that it is possible thereby to reduce the num

ber of vehicles required to carry out the timetable.

The result of excluding certain departure times is that some passengers will have to extend

their wait at the short-turn points. To minimize this adverse effect, it is possible to set the

following (Minimax H) criterion: Delete $F_r - F_k$ departure times at k with the objective of

minimizing the maximum headway obtained.

The Minimax H criterion attempts to achieve the minimization of maximum passenger

waiting time; the result may represent a nadequate passenger level of service whenever short

turn strategy is employed. Hours at the max load point
Travel times (min) DH times (min) a-b b-c c-b b-a a-c a-b
b-c 7:00 -8:00 8:00 -9:00 15 20 25 25 20 25 15 15 25 15 5
Time D(a) = 5 d (a , t) 6 4 2 0 d (c , t) DH trip 7:00
7:30 8:00 8:30 9:00 9:30 D(c) = 6 5 7:00 7:30 8:00 8:30
9:00 9:30 6 4 2 0 Direction a c c a Time period a b c c b a
Time period 7:00-8:00 (at b) 7:00 7:10 7:25 7:35 7:40 7:15
7:25 7:40 7:50 7:55 7:40 7:50 8:05 8:15 8:20 7:00 7:15 7:20
7:25 7:30 7:40 8:05 8:15 8:20 8:25 7:20 7:35 7:40 7:45 7:50
8:00 8:30 8:40 8:45 8:50 7:35 7:50 7:55 8:00 8:05 8:15 8:45
8:55 9:00 9:05 7:00- 8:00 (at c) 8:00- 9:00 (at c)
8:00-9:00 (at b) 7:45 7:50 8:00 8:15 8:25 8:40 8:05 8:10
8:20 8:35 8:45 9:00 8:30 8:35 8:45 9:00 9:10 9:25 a b c
Route: Max load point Stop 7:50 8:10 8:25

Figure 16.3 Example of a 2-way route with a 2 h schedule for which 10 vehicles are required (based on the graphical deCit function method).

16.4.2 Minimax H algorithm

To solve the optimization problem with the Minimax H criterion, a theory based on three

stages was developed by Ceder (1991): (1) representation of the problem on a directed network

with a special pattern; (2) application of a modified shortest-path algorithm on the network to

determine the Minimax headway; and (3) application of an algorithm to ensure that the exact

number of required departures is included in the optimal solution. The Minimax H algorithm

will now be outlined and then applied to the example problem presented in Figure 16.3.

Let $G_m = \{N_m, A_m\}$ be the special network consisting of a finite set of nodes N_m and a finite

set A_m of directed arcs. Figure 16.4 presents a general illustration of the special network,

accompanied by an example. There are given

departures from the complete timetable, and

the requirement is that only $m < n$ need remain to satisfy the Maximax Heuristic. The construction

of G_m is based on $m - 2$ equally spaced departure times between the first and last

given departures, t_1 , and t_n , respectively. These equally spaced departure times are denoted

by $t'_j = t_1 + (t_n - t_1) \cdot \frac{j-1}{m-2}$, $j = 2, 3, \dots, m-1$, and have an equal headway of $\Delta t = (t_n - t_1) / (m - 1)$.

The G_m network has the following six characteristics:

1. G_m consists of m rows; the first and last rows are nodes t_1 and t_n , respectively, and there is a row for each t_j , $j = 2, 3, \dots, m - 1$.

2. Each node in N_m represents a departure time in the given set of departures; however, it is not necessary that all given departures be included in N_m (see 7:10, 8:50 in the example of Figure 16.4), and also the same departures may be represented by several nodes (see 7:45, 8:00, 8:20 in Figure 16.4).

3. The nodes in each row are organized from left to right in increasing time order with respect to their associated t'_j . That is, all the given nodes t_i , such that $t'_j \leq t_i < t'_{j+1}$, are positioned twice, once to the right of t'_j and once to the left of t'_{j+1} , where t'_j , t'_{j+1} are two adjacent, equally spaced departure times. An exception occurs in the second and the $(m - 1)$ th rows, where only one node is positioned to the left of t'_2 and one to the right of t'_{m-1} , respectively. These single nodes, t_3 and t_{n-2} in Figure 16.4, are selected such that t_3 is the closest node to t'_2 , provided that $t_3 - t'_2 < t'_3 - t_3$, and t_{n-2} is the closest node to t'_{m-1} , provided that $t'_{m-1} - t_{n-2} < t_{n-2} - t'_{m-2}$.

4. The directed arcs in A_m connect only nodes from the k th row to the $(k + 1)$ th row, $k = 1, 2, \dots, m - 1$.

5. A directed arc from t_i to t_j is included in A_m if $t_j > t_i$, and from t_i to t_i (in subsequent rows) if and only if G_m is disconnected without this arc.

6. The length of an arc from t_i to t_j is exactly $t_j - t_i$.

After constructing G_m , a modified shortest-path algorithm is applied as the second stage of

the Minimax Hop procedure. This is a modified version of the Dijkstra algorithm described

in Appendix 9.A. The Dijkstra algorithm is based on assigning temporary labels to nodes,

the label being an upper bound on the path length from the origin node to each node. These

labels are then updated (reduced) by an iterative procedure. At each iteration, exactly one of

the temporary labels becomes permanent, implying that it is no longer the upper bound but

rather the exact length of the shortest path from the origin to the considered node.

The modification of the Dijkstra method takes place in the computation step, in which the

labels are updated. It is modified from $\pi(t_i) = \text{Min}[\pi^*(t_k) + (t_i - t_k), \pi(t_i)]$ (see Step 3 of Dijkstra

algorithm in Appendix 9.A), to

$$\pi(t_i) = \text{Min}\{\max[\pi^*(t_k), (t_i - t_k)], \pi(t_i)\} \quad (16.3)$$

where $\pi(t_i)$, $\pi^*(t_k)$ are temporary and permanent labels of nodes t_i and t_k , respectively.

This modified-Dijkstra algorithm, instead of minimizing the sum of arc values, searches

for the minimum of the maximum arc value. The algorithm is applied to G_m when the origin

node is t_i ; the algorithm terminates when the temporary label on node t_n becomes permanent.

for the second largest headway, is introduced
 to find the difference of m-p departure times;
 Departure times Network representation of the Minimax H
 procedure Given : n Required: m Departures 7:00 7:10 7:20
 7:30 7:45 8:00 8:00 8:20 8:30 8:40 8:50 9:00 Given: 9
 Required: 5 Departures 7:30 8:00 8:30 7:00 7:20 7:45 8:20
 8:00 7:45 8:20 8:00 8:40 9:00 Example $t_1 t_2 t_3 t_4$
 $t_4 t_4 t'_2 t'_2 t'_3 t'_3 t'_{m-2} t'_{m-2} t'_{m-1} t'_{m-1} t_5$
 $t_5 t_5 t_{n-4} t_{n-4} t_{n-4} t_{n-3} t_{n-3} t_{n-3} t_{n-2} t_{n-2}$
 $t_{n-1} t_n t_n$

Figure 16.4 General network representation of the Minimax H procedure at one terminal, accompanied by an example.

this is, basically the third stage. The third stage of the Minimax H procedure ensures that

the optimal result includes exactly the required number of departures (m). Although the

modified shortest-path algorithm for G_m determines the value of the Minimax headway, it

does not ensure that the result will include all the m required departures.

The third stage is interpreted utilizing the 10 different network configurations illustrated

Figure 16.6. Each example in Figure 16.6 is based on a different initial DT set of ve

departures, from which one of the middle three is unnecessary. Cases (e) and (f) consist of

a single path with three departures, whereas four departures are required. The additional

departure required will be selected between t_2 and t_3 for case (e) and between t_3 and t_4 for

case (f). The Minimax $H = 45$ min in both c

4 t 2 t 2 t 2 t 4 t 5 t 5 t 3 t 3 t 3 t 3 t 3 t 3 t 3 t
 3 t 3 t 3 t 2 t 2 t 2 t 2 t 2 t 2 t 2 t 2 t 4 t 4 t 4 t
 4 t 4 t 4 t 4 t 4 t 4

Figure 16.6 Different networks (based on a given five-departure timetable at one stop) for a required schedule of four departures from 7:00 to 8:00.

The Minimax H algorithm is now applied to the example problem presented in

Figure 16.3. The given and required numbers of departures for each hour and direction of

travel are determined by the load profiles and $\alpha = 50$ in Figure 16.7. These required numbers

(values of m) then undergo the Minimax H procedure in Figure 16.8. Four Gm networks

are constructed to derive the Minimax headway. This derivation appears in the figure with

a emphasized line indicating the optimal path of Gm and the labels of the shortest-path

algorithm according to Equation 16.3. A dashed line and a rowhead indicates the direc-

tion of another optimal solution. Also note that between 7:00 and 8:00 t_n becomes t_1 for

8:00-9:00 in order to preserve the continuity of the analysis. The other parts of Figure 16.8

are self-explanatory. In all four cases, there is no need to proceed to the third stage of the

Minimax H algorithm, because all the required departures are determined by the modified

shortest-path algorithm.

The results of the Minimax H algorithm are then applied to the example problem in

Figure 16.3. After the deletion of departures

at time points a and b in directions $a \rightarrow c$ and

$c \rightarrow a$, it is possible to construct the new timetable, a long with the DFs. This time, how

ever, all three time points (a, b, c) are involved. That is, in the modified timetable, some

trips are initiated at b and some terminate at b in directions $a \rightarrow c$ and $c \rightarrow a$, respectively.

Thus, point b also becomes an end/start point, and the DF description can be applied to

it. The new timetable and DFs are presented in Figure 16.9; the resultant timetable (with

maximum short turns) appears in the upper part of this figure. The corresponding DFs show

	250	150	300	200	350	200	200	150	Passenger load	Direction
A										
C										
Direction	C	A	a	b	c	c	b	a		
	7:00-8:00	8:00-9:00	7:00-8:00	8:00-9:00						
Distance	300	200	100	0	300	200	100	0	300	200
	0	300	200	100	0					

Figure 16.7 Load profiles of the example problem with a desired occupancy of 50 passengers per vehicle.

that 10 vehicles are required to carry out the timetable without DH trips and that 9 vehicles

($\lceil N_{\min} \rceil$) are required with a single DH trip from $d(c, t)$ to arrive at $d(b, t)$ before or at 8:35.

Following the condition of Figure 16.1 for $\lceil \cdot \rceil < N_{\min}$, where $N_{\min} = 10$ vehicles, the next

step is to attempt to reduce the number and impact of short-turn trips, provided that $\lceil \cdot \rceil < N_{\min}$

is maintained.

16.5 MAXIMUM EXTENSIONS OF SHORT-TURN TRIPS

There are two similar undertakings for possible extensions of short-turn trips: (1) convert

trips to points $r_i \in R$ from DH to service trips and (2) extend arrival and departure trips to

points $r_i \in R$ toward their original schedule. Direction a; c; max load point = b; determined departures at a 7:00-8:00 8:00-9:00 Given (n=6) : 7:00, 7:10, 7:25, 7:35, 7:40, 7:45 Result (m=4): 7:00, 7:10, 7:25, 7:45 Given (n=6): 7:45, 7:50, 8:00, 8:15, 8:25, 8:40 Result (m=4): 7:45, 8:00, 8:25, 8:40 Direction c; a; max load point = c; determined departures at b 7:00-8:00 8:00-9:00 Given (n=8): 7:20, 7:35, 7:40, 7:45, 7:50, 8:00, 8:10, 8:30 Result (m=5): 7:20, 7:40, 8:00, 8:10, 8:30 Given (n=4): 8:30, 8:40, 8:45, 8:50 Result (m=3): 8:30, 8:40, 8:50 20*= Minimax H 7:15 7:30 10* 15* 25 0* 7:00 7:10 7:25 7:25 7:35 7:45 8:03.3 15* 15* 30 0* 25*= Minimax H 25* 8:21.7 8:15 8:15 8:40 Alternative solution 7:45 8:00 8:25 7:37.5 7:35 7:50 7:45 8:30 8:10 7:50 7:45 7:40 8:00 8:12.5 7:55 20*=Minimax H 20* 15* 30 35 20* 25 20 15* 15* 15* 15* 0* 25 30 20* 7:20 8:00 8:10 7:40 10*=Minimax H 10* 0* 8:50 8:40 8:40 8:30

Figure 16.8 Minimax H algorithm and results for the example problem where * refer to Equation 16.3.

16.5.1 Converting DH to service trips

To convert DH to service trips, start by denoting the modified timetable with maximum

short turns by DT' ; the route's end points by e_q , $q = 1, 2$; and the intermediate short-turn

points by $r_i \in R$, $i = 1, 2, \dots, V$, where there are V short-turn points. To attain $N_{\min}$, the

overall schedule carrying out DT' might also include DH trips. This overall schedule is des

ignated S . In this section, the DF properties will be exploited to check whether a DH trip can

be interpreted as an extension of a short-turn trip in DT' .

By using DF theory, a DH trip can be inserted into a certain time window in order to

reduce the fleet size by one. To simplify this possibility, a DH trip is inserted from one ter

minial to terminal k so that its arrival time always coincides with the first time that $d(k, t)$ Direction a; c; a

Stop a b c c b a 7:00 7:10 7:25 - - 7:45 - 8:00 - 8:25 8:40
 7:15 7:25 7:40 *7:50 *7:55 8:05 *8:10 8:20 *8:35 8:45 9:00
 7:40 7:50 8:05 8:15 8:20 8:30 8:35 8:45 9:00 9:10 9:25 7:00
 7:15 7:20 7:25 7:30 7:40 7:50 8:05 8:15 8:20 8:25 7:20
 7:35* 7:40 7:45* 7:50* 8:00 8:10 8:30 8:40 8:45* 8:50 7:35
 - 7:55 - - 8:15 8:25 8:45 8:55 - 9:05 * Departures and
 arrivals at b (creating short-turn trips) Time d (a , t)
 D(a) = 3 d (b , t) D(b) = 1 0 d (c , t) D(c) = 6 DH Max
 load point 2 0 -2 7:00 7:30 8:00 8:30 9:00 9:30 4 2 0

Figure 16.9 New timetable with maximum excluded departure times (following the Minimax H procedure) and associated deficit functions.

attains its maximum. The following steps attempt to describe the procedure to convert DH

trips in S into service trips, used in the original timetable:

Step 1: Select a DH trip in S and call it DH; if there is no DH trip in S , stop.

Step 2: If the DH is from r_i to $e_{q'}$ ($q' = 1$ or 2 , $r_i \in R$), go to Step 3; if the DH is from $e_{q'}$ to r_i , go to Step 4; and if the DH is from r_i to r_k ($r_i, r_k \in R$), go to Step 5.

Step 3: Examine the arrival in $d(r_i, t)$ to the left of the departure time of DH (start with the one closest to that departure time) to see whether it can be extended to $e_{q'}$ (by replacing DH). If the arrival considered is associated with trip p_1 , the extension can be executed but if and only if the following three conditions are met: (a) the arrival time of p_1 at r_i is within the hollow that contains the DH departure time; (b) p_1 was originally planned to continue toward $e_{q'}$; and (c) the originally planned arrival time of p_1 at $e_{q'}$ is equal to or less than the arrival time of DH. If all the three conditions are fulfilled, delete DH from S , update DT' , $d(r_i, t)$, and $d(e_{q'}, t)$, and go to Step 1; otherwise, DH remains in S , go to Step 1.

Step 4: Examine the departure in $d(r_i, t)$ to the right of the arrival time of DH (start with the one closest to that arrival time) to see whether it can be extended to $e_{q'}$ (by replacing DH). If the departure considered is associated with trip p_2 , the extension can be executed but if and only if the following three conditions are met: (a) the departure time of p_2 at r_i is less than or

equal to the arrival time of DH ; (b) p_2 was originally planned to start at $e_{q'}$; and (c) the originally planned departure time of p_2 at $e_{q'}$ is within the hollow that contains the DH departure time. If all three conditions are fulfilled, delete DH from S, update DT' , $d(e_{q'}, t)$, and $d(r_i, t)$, and go to Step 1; otherwise, DH remains in S, go to Step 1.

Step 5: Set $r_k = e_{q'}$ and use the procedure in Step 3; if it is terminated successfully (DH is converted to a service trip), execute Adjustment A and go to Step 1. Otherwise, set $r_i = e_{q'}$ and use the procedure in Step 4; if it is terminated successfully, execute Adjustment A and go to Step 1. Otherwise, DH remains in S, go to Step 1. Adjustment A: delete DH from S, update DT' , $d(r_i, t)$, and $d(r_k, t)$.

16.5.2 Extensions at short-turn points

Following the use of DH trips from/to short-turn points, this section will now describe the

principles and procedures for possible extensions of short-turn trips toward their original

schedule without increasing N_{min} . To this end, let the updated timetable DT' be denoted by

DT_1' , including the conversion of DH trips to service trips by extending their associated short

turn trips. An extension of a short-turn trip can be viewed as stretching the trip toward the

route's end points, $e_{q'}$. An extension does not necessarily mean that the short-turn trip is

converted to a full trip along the entire route, because it can be only partially extended; that

is, an extension can be performed only from r_i to r_k ($r_i, r_k \in R$). The three stages at which the

extensions at $r_i \in R$ can be analyzed and executed are as follows: (a) zeroing the maximum

DF; (b) s tretching t he m aximum i nterval; a nd (c) t reating t he D F h ollows.

16.5.2.1 Zeroing the maximum deficit function

On the basis of the DF properties, it is possible to prove that while $\sum N_{\min}$ is preserved,

the number of extensions in each $r_i \in R$ from r_i to $e_{q'}$ is greater than or equal to $D(r_i)$.

This rule is based on the observation that in each $r_i \in R$, exactly $D(r_i)$ departures can

be extended to their original departure point without increasing $\sum N_{\min}$. This procedure

will eventually lead to $D(r_i) = 0$ for all $r_i \in R$. These extensions are obtained through the

following basic steps. Note, for subsequent steps, as well, that t_s and t_e denote the begin

ning and the end of the DF maximum interval.

Step 1 (Initialization): Set $R = R_0$.

Step 2: Select $r_i \in R$; if $R = \emptyset$ (empty), stop.

Step 3: Check to see whether $D(r_i) = 0$; if so, delete r_i from R and go to Step 2; otherwise continue.

Step 4: Identify a trip (there might be more than one) whose departure time is at t_s of $d(r_i, t)$ and extend this departure to its original time at $e_{q'}$; update DT_1 , $d(r_i, t)$, and $d(e_{q'}, t)$ and go to Step 3.

Figure 16.10 illustrates an example of three extensions to $d(r_i, t)$. The first two (numbered 1

and 2) induce $D(r_i)$ to decrease from two to zero. Each extension in Figure 16.10 is followed

by an update of $d(r_i, t)$. After the three extensions, only two (of five) short-turn trips remain

at r_i . Note that these two trips could also be extended if both are associated with the same

terminal k and that their arrival time in the original schedule is at or before the departure

time $a_{t_k} - 1$. $D(r_i) = 1$ $D(r_i) = 2$ $D(r_i) = 0$ $D(r_i) = 2$ $d(r_i, t) = 1$ * Can be extended only if both arrival and departure

activities are at the same terminal and if the original arrival time is at or before the departure time $D(r_i) = 0$
 $3 \ 0 \ -1 \ -2 \ D(r_i) = 0 \ * \ * \ 0 \ -1$

Figure 16.10 Schematic example of updated deficit functions at an intermediate short-turn point after each of the three indicated extensions.

16.5.2.2 Stretching the maximum interval

After reducing all $D(r_i)$ to zero, it is possible to prove the following rule: while preserving

$\tau \leq N \min$, further extensions can be performed from $r_i \in R$ to $e_{q'}$, up to the point at which the

maximum interval is stretched over the whole span of the schedule horizon. This rule is

based on the observation that certain arrivals can be extended without increasing $D(r_i)$

above zero. The span of the schedule horizon is determined by the earliest departure and

latest arrival in the original timetable.

These additional extensions to the route's end points are executed using the following

steps, where DT_2' denotes the updated timetable after the stage described previously (zeroing

the $D(r_i)$):

Step 1 (Initialization): Set $R = R'$.

Step 2: Select $r_i \in R'$; if $R' = \emptyset$ (empty), stop.

Step 3: Check whether the r_i maximum interval (from t_s to t_e) coincides with the span of the schedule horizon; if so, delete r_i from R' and go to Step 2; otherwise, continue.

Step 4: Identify a trip (there might be more than one) whose arrival is at t_e of $d(r_i, t)$ and extend this arrival to its original time at $e_{q'}$; update DT_2' , $d(r_i, t)$, and $d(e_{q'}, t)$ and go to Step 3.

The preceding procedure is demonstrated by E

extension 3 in Figure 16.10.

16.5.2.3 Treating the deficit function hollows

At this third stage, a search is made to determine more extensions at $r_i \in R$ regarding departures and arrivals in hollows. Each hollow in $d(r_i, t)$ contains the same number of arrivals as

departures. The procedure developed does not treat hollows consisting of only one point. In

Figure 16.10, for example, the third DF with Extension 3 has one hollow that consists of two

arrivals followed by one departure. DF theory, as outlined in Chapters 6 and 7, permits the

construction of the following extension search procedure, in which DT_3' denotes the updated

timetable after the stages just discussed:

Step 1 (Initialization): set $R = \emptyset$.

Step 2: Select $r_i \in R$; if $R = \emptyset$ (empty), stop.

Step 3: Check the next (with respect to time) trip in $d(r_i, t)$; if it is the last departure, go to Step 2; if it is an arrival, go to Step 5; otherwise, continue.

Step 4: Examine this departure by extending it to its original time at $e_{q'}$; execute this extension if $D(e_{q'})$ is unchanged or if $D(e_{q'})$ is increased, but it can also be reduced (back) through the unit reduction DH chain (URDHC) procedure; update DT_3' and all the DFs involved. Then, if the end of $d(r_i, t)$ does not coincide with the right boundary of the schedule horizon, go to the extension procedure described in the previous stage (stretching the maximum interval); otherwise go to Step 3. If the extension cannot be made, repeat this extension examination toward a different short-turn point r_k (instead of $e_{q'}$) each time, selecting them backward from $e_{q'}$ to r_i .

Step 5: Examine this arrival by extending it to its original time at $e_{q'}$ and use the URDHC procedure to check whether $D(r_i)$ can remain the same; if so, execute

the extension; update DT 3' and all the DFs involved; otherwise, repeat this extension examination the same way as in Step 4.

Finally, if this procedure leads to the introduction of a new DH trip, the procedure for extensions of DH trips needs to be repeated.

16.5.3 Extensions of the example problem

The example problem, after the Minimax H procedure, is shown in Figure 16.9; this result

will now be subjected to possible extensions related to the arrivals and departures at $d(b, t)$.

First, the extensions of DH trips procedure will be used to determine whether or not the

DH trip can be converted into a service trip. In the example it is found that it cannot be

converted, and hence the single DH trip remains in the schedule.

Second, the procedures described for extensions of short-turn points are applied; their

execution is shown in Figures 16.11 and 16.12. Because $D(b) = 0$ (after inserting the DH

trip), the algorithm in the first stage cannot be utilized. However, because of the algorithm Direction a c c a Stop a b c c b a Timetable with maximum short turns,* including DH trip from c to b (8:30-8:35) 7:00 7:10 7:25 - - 7:45 - 8:00 - 8:25 8:40 7:15 7:25 7:40 *7:50 *7:55 8:05 *8:10 8:20 *8:35 8:45 9:00 7:40 7:50 8:05 8:15 8:20 8:30 8:35 8:45 9:00 9:10 9:25 7:00 7:15 7:20 7:25 7:30 7:40 7:50 8:05 8:15 8:20 8:25 7:20 7:35* 7:40 7:45* 7:50* 8:00 8:10 8:30 8:40 8:45* 8:50 7:35 - 7:55 - - 8:15 8:25 8:45 8:55 - 9:05 * Departures and arrivals at b. Time $d(a, t)$ $D(a) = 3$ $d(b, t)$ $d(c, t)$ $D(c) = 6$ DH 2 4 2 7:00 7:30 8:00 8:30 9:00 9:30 6 4 2 0 Max load point 2 0 -2 3 DH e 1 $D(b)=1$ 0

Figure 16.11 Modified timetable and deficit functions following the Minimax H procedure, along with an indication of three short-turn trip extensions.

in the second stage, Extension 1 can be performed,

increasing $d(b, t)$ at 8:45 from -1 to 0.

Then, the algorithm in the third stage is used. It can be observed that Extension 2 alone

affects $D(b)$, increasing it by one at 8:10; the URDHC procedure, therefore, searches for

a DH trip that can arrive at b at 8:10. Such a DH trip, designated DH e in Figure 16.11, is

inserted from a . Note that the insertion of DH e is possible because $d(a, t)$ drops to 2 at 7:50

after resuming the original arrival time associated with Extension 2; thus, $D(a)$ remains 3.

The final step in the third stage is to check DH e with the procedure of extensions of DH

trips. This permits Extension 3 to be performed. Consequently, three of the eight short-turn

trips in the timetable of Figure 16.10 were extended to their original schedule, while $N_{\min}$

remains 9. In other words, the procedures developed identify only the minimum (crucial)

short-turn trips that are required to reduce fleet size. Figure 16.12 illustrates the updated

DFs after the three extensions. It can be observed that no more extensions can be made.

The timetable in Figure 16.12 achieved, for the example problem, both the minimum fleet

size required (of nine vehicles) and the minimum required five short-turn trips to reach this min

imum fleet size. The next step is to construct the nine blocks using either the FIFO method or Directional method. Stop at each block a Final timetable with minimum number of short turns, but with the same minimum number of vehicles 7:00 7:10 7:25 -- 7:45 (7:50)* 8:00 -- 8:25 8:40 7:15 7:25 7:40 7:50 7:55 8:05 8:10 8:20 8:35 8:45 9:00 7:40 7:50 8:05 8:15 8:20 8:30 8:35 8:45 9:00 9:10 9:25 7:00 7:15 7:20 7:25 7:30 7:40 7:50 8:05 8:15 8:20 8:25 7:20 7:35 7:40

7:45 7:50 8:00 8:10 8:30 8:40 8:45 8:50 7:35 (7:50)* 7:55 –
 – 8:15 8:25 8:45 8:55 (9:00)* 9:05 (...) * is an extension
 (see the following). Time $d(a, t)$ $D(a) = 3$ $d(b, t)$
 $D(b) = 0$ $d(c, t)$ $D(c) = 6$ 2 3 1 4 2 0 7:00 7:30 8:00
 8:30 9:00 9:30 6 4 2 0 2 0 -2

Figure 16.12 Optimal timetable for the problem and associated deficit functions and extensions.

the within hollow method described in Section 6.5.5.
 Finally, a summary of performance mea-

sures of the nine blocks can be calculated for
 administrative and further evaluation purposes.

For instance, Table 16.1 lists the final schedule
 of Figure 16.12 in order of increasing

departure times, when for the same departure
 times those at a come before those at c and

the last departures are at b. The 23 trips
 listed are subjected to the FIFO procedure and

nine blocks are constructed, grouped by trip number, as
 follows: [1,9,20], [2,10,21,22],

[3,11,23], [4,13], [5,15], [6,16], [7,12,17], [8,14,18], [19]. A summary of the performance

measures of each block is shown in Table 16.2. Table 16.1
 List of trips of the example problem in order of increasing
 departure times Trip number Departure time Departure
 location Arrival time Arrival location 1 7:00 a 7:40 c 2
 7:00 c 7:35 a 3 7:10 a 7:50 c 4 7:15 c 7:50 a 5 7:20 c
 7:55 a 6 7:25 a 8:05 c 7 7:25 c 7:45 b 8 7:30 c 7:50 b
 9 7:40 c 8:15 a 10 7:45 a 8:30 c 11 7:50 c 8:25 a 12 7:50 b
 8:15 c 13 7:50 a 8:35 c 14 7:55 b 8:20 c 15 8:00 a 8:45 c
 16 8:05 c 8:45 a 17 8:15 c 8:55 a 18 8:20 c 9:00 a 19 8:25
 c 9:05 a 20 8:25 a 9:10 c 21(DH) 8:30 c 8:35 b 22 8:35 b
 9:00 c 23 8:40 a 9:25 c

Table 16.2 Performance measures of the nine-block result of
 the example problem

Block

number Service time (min) DH time (min) Idle time (min)

	Service (km)	DH (km)	Block time (min)
--	--------------	---------	------------------

1	120	0 10 72	0 130
---	-----	---------	-------

2	100	10 10 76	14 120
---	-----	----------	--------

3	120	0 15 72	0 135
---	-----	---------	-------

4	75	0 0 48	0 75
---	----	--------	------

5	80	0 5 48	0 85
---	----	--------	------

6	80	0 0 48	0 80
---	----	--------	------

7	80	0 5 52	0 90
---	----	--------	------

8	85	0 0 52	0 85
---	----	--------	------

9	40	0 0 24	0 40
---	----	--------	------

Total	785	10 45 492	14 840
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16.6 LITERATURE REVIEW AND FURTHER READING

This section reviews primarily the research on the design of short-turn trips. It ends with

several studies on using short turning as a real-time control strategy.

Furth (1987) presents a methodology for schedule coordination between short-turn and

full-length trips in a system consisting of several transit modes. The objective is to find an

optimal trade-off between the minimum possible fleet size and the minimum passenger

waiting time. The schedule offset between full and short-service patterns is determined as

are the turning points of the short-turn service. Optimal service headways and vehicle size

are also calculated. The frequency of the short-turn route is set to a multiple of that of the

full-length route, and the ratio of frequencies is called scheduling mode; different models

are developed for different scheduling modes. Vehicle capacity is given significant attention.

The researcher shows that a problem of overcrowding may occur even when overall capacity

exceeds the volume on every road link. The timetable-design problem includes an integer

minute constraint.

Miller and Bunt (1987) describe a computer program that simulates light-rail operations

on a specific route in Toronto. This program is designed to analyze the impact of a range of

operating policies on the regularity of the streetcar service and to compare alternate means

of improving regularity. One of the strategies compared is the introduction of short-turn

service. The model assists in determining where and when a vehicle will be directed to turn

around before the end of the route. This determination is based on a set of decision rules.

The basic rules recommend turning a vehicle when a gap is found that exceeds a given

threshold between successive vehicles traveling in the opposite direction of the vehicle to be

turned. Other rules make sure, for example, that the gap in the original direction, which

occurs as a result of the turning, does not exceed a given threshold.

Vijayaraghavan and Anantharamaiah (1995) examine the possibility of reducing the

required number of bus uses operated on a single route by using two strategies: partial service

(i.e., short turning) and express service. Fleet-assignment options and their effect on the

efficiency of fleet use are analyzed graphically.

Dell S. and Filippi (1998) present a model for bus-service optimization under an elastic

demand. The model is formulated as a programming problem in which decision variables

are the locations of turning points, the time offsets between departures of the full-length

and the short-turn routes, frequencies, and fares. Vehicle size is also considered as a variable

but is represented indirectly. The model enables taking into account different demand

patterns at different periods. A different frequency and offset are ascribed to each period;

for other variables, a single optimal value is determined. A numerical procedure is presented

to solve the problem.

Ulusoy et al. (2010) develop a model for cost-efficient operation that optimizes a full-stop,

short-turn, and express transit services and the associated frequencies by considering hetero-

geneous demand. For a transit line with given stations and origin-destination demand, the

objective function of total cost is optimized by considering a set of constraints that ensure

frequency conservation and sufficient capacity subject to operable fleet size. A numerical

example is designed that is based on a real-world rail transit line to demonstrate the applicability

and effectiveness of the developed model. Results show that optimized integrated

service patterns and associated service frequencies significantly reduce total cost.

Cortés et al. (2011) develop a model that combines short turning and deadheading in an

integrated strategy for a single transit line, where the optimization variables are both of a

continuous and discrete nature: frequencies within and outside the high demand zone, vehicle

capacities, and those stations where the strategy begins and ends. The authors show that

closed solutions can be obtained for frequencies in some cases, which resembles the classical

square root rule. Unlike the existing literature that compares different strategies with a

given normal operation (no strategy–single frequency), the authors use an optimized base

case, in order to assess the potential benefits of the integrated strategy on a fair basis. It was

revealed that the integrated strategy can be justified in many cases with mixed load patterns,

where unbalances within and between directions are observed. In general, the short turning

strategy may yield large benefits in terms of total cost reductions, while low benefits are asso-

ciated with deadheading, due to the extra cost of running empty vehicles in some s-

ections.

The main characteristics of the models reviewed to this point are summarized in Table 16.3.

The aforementioned papers relate to the design of short-turn trips. Some other papers also

discuss short turning as a real-time control strategy.

Huddart (1973) discusses short-turn trips as a strategy to avoid bus bunching. The author

explains that in order to use this strategy efficiently, a bus should be taken out of a bunch

in one direction and be entered into a schedule gap in the opposite direction. In addition,

difficulties in real-time decision-making regarding short-turn trips are described.

Strathman et al. (2001) mention that a bus chosen to turn around should ideally be one

with a small number of passengers, a small gap from the preceding bus, and a small gap

from the following bus. The authors mention that damage is caused for passengers whose

destination is further from the turning point, since they are forced to transfer from a short

turn trip to a full-length trip.

Shen and Wilson (2001) develop a disruption-control model for rail transit systems. The

model, formulated as a deterministic mixed-integer program, is used to examine the intro

duction of several real-time control strategies, including short turning. The main conclusion

is that the best system performance is a

chieved when holding a nd short-turn s strategies a re

combined. The paper does not include a methodology for short-turn strategy design.

Table 16.3 Summary of features of the models reviewed

Source	Decision variables	Required demand input	Operation period
Furth (1987)	Offset between full and short service, turning points, headways, and vehicle size	O-D matrix	Constant Single
Miller and Bunt (1987)	Turning points and times	None	Not considered Single (afternoon 5 h peak)
Vijayaraghavan and Anantharamaiah (1995)	Turning points, number of stops	Total demand	Constant Single
Dell Site and Filippi (1998)	Offset between full and short service, turning points, headways, fares, and vehicle size	O-D matrix and elasticities	Elastic Multiperiod
Ulusoy et al. (2010)	The decision variables of the objective function of total cost include integrated service patterns and associated service frequencies	O-D matrix	Elastic Multiperiod
Cortés et al. (2011)	The impact on waiting time cost, in-vehicle time cost and operator cost	O-D matrix	Constant Single

EXERCISES

16.1 A bus route is given between terminals a and c (both directions of travel) with a single short-turn point b along the route. The input data appear in the following table for the period 6:00-8:00.

Direction a → c c → a a → b b → c c → b b → a

Original departure times 6:00; 6:12; 6:17; 6:25; 6:32; 6:40; 6:46; 6:53; 7:00; 7:10; 7:18; 7:25; 7:35; 7:40; 7:48

– 6:00; 6:10; 6:18; 6:25; 6:32; 6:40; 6:50; 6:55; 7:10;
7:18; 7:25; 7:32; 7:40; 7:48; 7:53 –

Service travel time (min) 15 20 25 10

DH travel time (min) 10 17 20 8

Passenger load 6:00–7:00 400 250 150 200 7:00–8:00 400 300
450 550 The desired occupancy of a bus is 50 passengers;
there is a minimum frequency of 3 veh/h, and no shifting
in departure time is allowed.

(a) Apply the method for designing short-turn trips so as to minimize both the number of vehicles required and the number of short-turns; provide the final timetable.

(b) Establish schedules/blocks for all vehicles using the FIFO procedure.

16.2 The following two-route network is given, with stop *a* as a single candidate short-turn point for both routes in both directions of travel. *c* *a* *e* *b* *k* The following table contains the data required.

Route and direction *a* → *c* *c* → *a* *e* → *k* *k* → *e* *a* → *b* *b* → *c* *c*
→ *b* *b* → *a* *e* → *b* *b* → *k* *k* → *b* *b* → *e*

Service travel time (min) 20 30 25 15 10 15 20 15

DH travel time (min) 18 25 20 12 7 10 15 10

$H_{min\ i}$ (min) *a* 5 10 8 13

$H_{max\ i}$ (min) *a* 6 15 7 20

Passenger load 6:00–7:00 150 250 250 400 7:00–8:00 250 350
150 200

$H_{min\ i}$ and $H_{max\ i}$ are the minimum and maximum headways permitted, respectively, between two adjacent departures on route *i* (see Chapter 5 for more detailed description and examples). In addition, it is given that the determined frequency for the 2 h period considered (departures between 6:00 and 8:00) is 12 vehicles/period for each route and direction of travel; no shifting in departure time is allowed, and the derived timetable starts at 6:00 (the first fixed departure).

(a) Construct the timetable for only routes $a \rightarrow c$ and $c \rightarrow a$, using the method of synchronization described in Chapter 5.

(b) Apply the method to create short-turn trips for routes $a \rightarrow c$ and $c \rightarrow a$ so as to minimize both the number of vehicles required and the number of short-turns; provide a final timetable.

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17 Chapter 17: Smart Shuttle and Feeder Service

The choice between public and private transportation is an individual decision that is in-

fluenced by government/community decisions. These decisions often send mixed signals (e.g., “use

transit”; “we’re trying to reduce traffic congestion to ease your automobile driving”) to transit

and potential transit passengers while failing to recognize their more system-wide, integrated

implications. Chapter 13 facilitates an understanding of the importance of service-connectivity

elements in enhancing existing or new transit services. One such element is the design of an

integrated, smart feeder/shuttle service. Such a service may stem, for instance, from the need

to overcome the problem of an excessive number of automobiles arriving and parking at a train

station, resulting in high parking demand around the station. The purpose of this chapter is to

examine an innovative feeder/shuttle system that will comply with (1) passengers’ needs and

desires, (2) intelligent transportation technologies, and (3) an agency’s viability. After all, a solu-

tion for being transported by a door-to-door service is the mother of attraction. This chapter contains eight main parts, following an introductory section. Section 17.2 con-

tinues the analysis of the minimum fleet size required that was the subject of Chapter 6, but

for a radial/shuttle route having a single departure and arrival point. Section 17.3 proposes 10

different feeder/shuttle routing strategies with various combinations of fixed/flexible routes,

fixed/flexible schedules, a uni- or bidirectional concept,

and shortcut (shortest path) and/or

short-turn (turn-around) concepts. Section 17.4 investigates these strategies by employing

a simulation model specifically developed and constructed for this purpose. This simulation

model is used in Section 17.5 for a case study of Castro Valley in California, where the feeder/

shuttle service is coordinated with the metro (BART) service; the 10 routing strategies are

compared for 4 -foot size scenarios. Section 17.6 reports on a survey conducted in the case

study area concerning (a) willingness to use the smart-shuttle service, (b) willingness to pay

for the service, and (c) the attributes that would enhance the shuttle service. Section 17.7

introduces a framework for the optimal design of circular feeder/shuttle fixed routes, including

a method for estimating potential passenger demand. Section 17.8 shows the modeling of an

optimal circular route that can handle any size road network. The chapter ends with a literature

review and exercises. Practitioners are advised to visit all sections of the chapter, except perhaps Section 17.8,

which is more of a mathematical nature. They are especially encouraged to follow the examples

and the case study. Finally, an idea that has been introduced is that passengers who want to take the shuttle

service should be able to use an online, intelligent information system through a telephone/

cellular call. The system will announce the arrival time to the point closest to the caller unless

this estimated time is not reliable enough. In the latter case, the system will call back once an

estimation becomes reliable. It may even have a special ring for easy recognition. Does the name

Pavlov ring a bell?

17.1 INTRODUCTION

When exploring the design possibility of a transit feeder/shuttle service, we can refer to

Section 13.2, in which a detailed coordinated service is portrayed. For simplicity, we will

include the transit-feeder service within a shuttle service in the remaining parts of the chapter.

Ideally, a smart transit shuttle system will provide an advanced, attractive service that

operates reliably and relatively rapidly, and has smooth, synchronized transfers as part of

the door-to-door transit-passenger chain. In order to achieve all the design of such a system,

new integration and routing concepts had to be developed. This chapter integrates work

done by Ceder and Yim (2002), Yim and Ceder (2006), Jerby and Ceder (2006), Ceder

(2009), and Ceder (2013).

A growing concern for public transit is its inability to encourage people to switch their

mode of transportation from solo to shared driving. The majority of large cities have

encouraged private car use through planning (dispersed land-use in the suburbs), infra

structure (available parking and circulation infrastructure), pricing, and financial decisions.

Consequently, there is growing confusion among drivers in many of those cities about what

to do. One way to handle the known decline in transit use is to retain a high level of service

for all transit users while fully maintaining the protection of accessibility for less

affluent travelers.

What follows is a description of the motivation that was employed by research reported in

this chapter. Although overall transit ridership is declining in cities, an encouraging trend is

increased ridership in long-haul express bus or rail transit. When long-haul express transit

systems were built in the 1970s and 1980s, parking facilities were also provided for the rider

ers, under the rubric of park and ride. The concept was readily accepted by the public, and a

large number of commuters preferred to take a long-haul bus or train to avoid rush-hour traffic

and high parking costs. However, most of the arrivals at the train stations are made by

private cars, hence creating traffic congestion and parking overloading in the station area.

Such a case was observed in the Bay Area around San Francisco in connection with the use

of the local metro system, called BART (Bay Area Rapid Transit). This observation, which

served as the motivation for this chapter, appeared on the front page of the San Francisco

Chronicle on three consecutive days in 2001. Figure 17.1 shows the relevant sections of the

newspaper, providing news about the problem and commentary in the form of suggested

parking solutions (bottom right of figure might be helpful to read). The main parking solu

tion considered by BART was this: "Building reserved parking lots, where users would be

charged a fee for a guaranteed space." An obvious question arises: Why not employ a smart

bus shuttle service instead?

Mark T. Main said: "You cannot depend on your eyes when your imagination is out of focus."

Our eyes see what transit services are currently providing for high and low density communities.

Our eyes can read reports covering urban transportation characteristics, the influence of

transportation investment, ground transportation strategy, and passenger transportation action

plans. However, we cannot depend on our eyes alone to trigger our imagination. As Einstein

said, "Imagination is more important than knowledge." Consequently, in order to design a n

imaginative, ideal shuttle service (see definition in the first paragraph of this section), it will be

worthwhile to start almost from scratch in order to attain such a smart service.

The purpose of this chapter is to describe the conceptual construction of an innovative

shuttle system that will (1) meet the needs and desires of end users, (2) utilize intelligent

transportation technologies, and (3) increase operational efficiency. A simulation model,

a passenger survey, and a case study will help
us take a practical approach to the optimal

design of the shuttle route(s).

17.2 MINIMUM FLEET SIZE REQUIRED FOR A CIRCULAR (SHUTTLE) ROUTE

Section 6.2 discussed the case in which interlinings are
not allowed and each route is operated

separately. Let T_r be the average round-trip time,
including layover and turnaround times,

of a circular route r (whose departure and arrival points
are the same). The minimum fleet

size is then equal to the largest number of vehicles that
depart within T_r (Salzborn, 1972).

Although Salzborn's modeling provides the basis for fleet
size calculation, it relies on two

assumptions that, in practice, do not hold for a circular
(shuttle) route: (1) vehicle departure

rate is a continuous function of time and (2) T_r is the
same throughout the period under

consideration. In practice, departure times are discrete
(see Chapters 4 through 6), and aver

age trip time is usually dependent on time-of-day. The
analysis of Section 6.2 is used in the

present section for the case of a circular (shuttle) route.

Let route r start and end at b . Let T_{rj} be the average
trip time on route r for a vehicle

departing at t_j from b , including its layover time at b .
Also, let n_j be the number of departures

from b including departure j at t_j until $t_{j'}$, but
excluding departure j' . We further define that j

arrives at b and continues with departure j' at $t_{j'}$,
which is the first feasible departure from b

Figure 17.1 Motivation for the study effectively appeared on the front page of the San Francisco Chronicle on 3 consecutive days.

at a time greater than or equal to $t_j + T_{rj}$. Theorem 6.1 (see Chapter 6) without n_{ia} but with

the setting $n_{jb} = n_j$ yields the following: $N_{rj} \geq \lceil \frac{n_j}{T_{rj}} \rceil$ (17.1)

where N_{rj} is the minimum fleet size required for the circular route r .

The timetable for both in the example shown in Figure 6.3 is used for the shuttle case in

Figure 17.2. Both a single T_{rj} value of 30 min throughout the timetable and a case in which

T_{rj} varies are used in Figure 17.2. On the left-hand side of Figure 17.2, the time windows are

all with the same length (30 min); on the right-hand side, they are T_{rj} dependent. The maxi-

mum number of departures within T_{rj} is emphasized for each case, leading in both instances

to a determination of the minimum fleet size, N_{rj} , according to Equation 17.1.

Following the determination of N_{rj} , the construction of vehicle chains (blocks) can be car-

ried out by using the FIFO (First-in, first-out) rule of first-feasible connection. For example, the case with $T_{rj} = 30$ minutes for all trips j

Timetable	5:00	5:30	6:00	6:30	6:50	7:05	7:10	7:15	7:20	7:30	7:40	8:00
T_{rj} (min)	30	30	30	30	30	30	30	30	30	30	30	30
N_{rj}	3	2	4	3	1	1	1	2	5	4	2	2
Timetable	5:00	5:30	6:00	6:30	6:50	7:05	7:10	7:15	7:20	7:30	7:40	8:00
T_{rj} (min)	30	25	25	40	20	20	20	20	20	20	20	20
N_{rj}	3	3	3	2	3	3	3	3	3	3	3	3

Timetable $N_{rj} = 5$ $N_{rj} = 6$ e c a s e w i t h d i e r e n t T_{rj} f o r e a c h t r i p j

Figure 17.2 Example of the derivation of radial-route fleet size when the round-trip time, including layover time, can either be the same (left side) or vary (right side) for each departure time.

for the case with varied T_{rj} in Figure 17.2, the first

block is [5:00-5:30-6:00-6:30-7:05-

8:00], in which the 7:05 departure will be ready for another departure at 7:45 (Tr6 = 40 min).

The second block, using the FIFO rule, is [6:50-7:20]. Because there is need for a minimum

of six vehicles, some trips of the first block can certainly be performed by the second block,

thereby opening some idle time for the vehicles associated with these blocks. For instance,

the first two blocks can be [5:00-6:00-6:50-7:20] and [5:30-6:30-7:05-8:00].

Finally, two points are worth noting: (1) When shifting in departure times is allowed, a

further reduction in the fleet size can be achieved. (2) When more than one circular (shuttle)

route is operated from the same point, it is recommended that the deficit-function method

(of Chapters 6 and 7) be used to arrive at the minimum fleet size required for all routes.

17.3 ROUTING STRATEGIES

The previous section dealt with a fixed route and a fixed shuttle-service schedule; however,

other routing strategies can be imagined and should be explored. These strategies represent

the flexibility and part of the attractiveness of the transit system. Before embarking on a

study of these other strategies, we will first present an overview of various known and relevant

concepts of transit-shuttle operation.

In order to alleviate the problems encountered

in traditional transit services, several

flexible services were studied. Dial-a-ride and door-to-door paratransit have played a

vital role in providing equitable transportation service to elderly and handicapped per-

sons who have difficulty in accessing regular public transit systems (Cervero, 1998). Such

a demand-responsive transit (DRT) system, which was investigated by Joachim et al.

(1995) and Brondorfer et al. (1999), does not, though, fulfill the needs of the entire tran-

sit population. An interesting study by Melucelli et al. (2001) distinguishes between two

classes of users, so-called passive users and active users. The passive users make use of

traditional transit, that is, boarding and alighting at compulsory stops. No reservation

is necessary, since vehicles are guaranteed to serve each compulsory stop within a given

time window. The active users, who board or alight at a non optional stop, must issue a

service request and specify pickup and drop-off stops, as well as the earliest departure and

latest arrival times. In the Melucelli et al. study, transit vehicles had to be rerouted and re-

scheduled in order to satisfy as many requests as possible, complying with passage-time

constraints at compulsory stops while activating optional stops between two compulsory

stops on demand.

Several studies have made use of simulation as a tool to arrive at satisfactory routing and

scheduling DRT solutions. Two waves of simulation studies can be traced in the literature.

The first wave consists of the research conducted by Wilson et al. (1970), who evaluated

various heuristic routing rules and algorithms used in a computer-aided routing system.

These rules and algorithms, developed for mainframe computers, have limitations in their

handling of large-size road networks with different routing strategies. The second wave of

research was conducted by Fu (1999, 2001) and Fu and Xu (2001), who considered the use

of advanced technologies. Their studies present a simulation model, Sim-Paratransit, which

was developed for evaluating advanced paratransit systems, such as AVL (automatic vehicle

location) and CAD (computer-aided dispatch) systems. The simulation model is described

by Fu (1999, 2001), and the evaluation of AVL and CAD systems by Fu and Xu (2001). The

ability to track the location of a transit vehicle continuously allows for the introduction of

intelligent paratransit systems, which will naturally lead to operating such systems at a significant

improved level of productivity and reliability.

This chapter, among other themes, investigates 10 routing strategies (Ceder, 2013);

1. Fixed route with a fixed schedule (timetable) and fixed direction (of travel)
2. Fixed route; flexible (demand-driven) schedule; fixed direction

3. Fixed route; flexible schedule; bidirectional
4. Fixed route; flexible schedule; fixed direction; possible short-turns
5. Fixed route; flexible schedule; bidirectional; possible short-turns
6. Fixed route; flexible schedule; fixed direction; possible shortcuts
7. Fixed route; flexible schedule; bidirectional; possible shortcuts
8. Fixed route; flexible schedule; fixed direction; possible short-turns and shortcuts
9. Fixed route; flexible schedule; bidirectional; possible short-turns and shortcuts
10. Flexible (demand-responsive) route with a flexible schedule

Fixed direction means that the shuttle will always maintain the same direction of travel

(same sequence of stops), whereas bidirectional allows for flexibility in selecting a direction

based on real-time demand information. The term shortcut means that, based on certain

loading-threshold and synchronization criteria, the shuttle will not continue its fixed route

and, instead, will use the shortest path (minimum travel time) to arrive at the required

point, for example, the train station. For convenience, we will use the shuttle terminal as

a train station throughout this chapter, though this connection point can also serve other

transit modes.

The loading threshold is a given (input) number of passengers on board the shuttle. The

synchronization criterion means matching the shuttle's new (via shortcut) arrival time

with a new earlier train than that originally planned if the entire route were completed. The

term short-turn means that, based on certain loading-threshold and synchronization criteria,

the shuttle will not continue on its fixed route. Instead, it will turn around and arrive

at the train station in the opposite direction in order to try to pick up passengers who were

too late to catch this shuttle when it first passed their stops. The loading-threshold and

synchronization criteria for the short-turn strategy (including the consideration of more

pickups) and the shortcut strategy are similar. Each strategy allows for the flexibility of

the other. In other words, the loading threshold of the shortcut strategy is initially higher

than the loading threshold of the short-turn strategy. If the latter is reached and there is a

possibility of picking up x passengers (after turning a round), where x is equal to or greater

than the difference between the two loading thresholds, then the short-turn strategy is

recommended.

Figure 17.3 represents the 10 strategies on a small network with two shuttle routes, one

marked with a dashed line and the other with a dotted line. Strategies 2-10 are characterized

ized by a flexible schedule, and it is emphasized by not timetable in Strategy 2. Arrows in

both directions of the route means a bidirectional situation. The shortcut strategy has lines

with arrows in Figure 17.3 that indicated deviations from the fixed route. The short-turn

strategy has arrows indicating a turnaround at a certain point in the network. Both repre

sentations appear in Strategies 8 and 9, involving a possible combination of the two. The last

strategy is for a DRT-type of service, allowing for the creation of a new route every time,

based on trip bookings.

The idea of covering almost every possible practical routing strategy stems from the need

to satisfy user desires and understandings. Certainly, there is no intention that all strate

gies be used at the same time; rather, it is to examine which strategy is best for a given

demand pattern and magnitude while taking into consideration the real-time traffic situation

in the area of the shuttle's trips. A simulation model was devised for that purpose. Two routes Fixed route, fixed schedule (timetable-based), fixed direction Base network Train station Fixed route, flexible schedule, fixed direction, short turn Fixed route, flexible schedule, bidirectional, short-turn Fixed route, flexible schedule, fixed direction, shortcut Fixed route, flexible schedule, fixed direction, short-turn, shortcut Fixed route, flexible schedule, bidirectional, short-turn, shortcut Demand-responsive (DRT) Current demand Fixed route, flexible schedule, bidirectional 1 2 3 4 5 6 7 9 10 8 Fixed route, flexible schedule, bidirectional, shortcut Fixed route, flexible schedule, fixed direction

Figure 17.3 Ten routing strategies.

This simulation tool, explained in the next section, enables a comparison of the various

strategies, based on the following measures:

- Sum of total time (in passenger-hours) from passenger pickup to train-departure time
- Sum of total time (in passenger-hours) riding the shuttle vehicle
- Sum of total waiting time (in passenger-hours) for the train
- Sum of total waiting time (in passenger-hours) for the shuttle vehicle
- Total number of transit vehicles (by number of seats) required to meet the demand

These measures of travel and waiting times and number of vehicles characterize the effec

tiveness and efficiency of each strategy. The strategy selected for a given passenger demand

is the one with the minimum weighted travel and waiting times (user perspective) and the

minimum number of vehicles (agency perspective).

17.4 SIMULATION

The purpose of the simulation model (Ceder, 2013) is to examine the 10 different routing

strategies. Passenger demand can either be inserted as part of the input or be generated ran

domly on the network.

17.4.1 Simulation input variables

Following are the input variables of the simulation program, which was written in C++

language. Each variable is presented by its simulation name, as well as a name explanation and

interpretation of its substance. What is referred to here as bus can be applied to any shuttle

vehicle.

Bus2Train = Time in seconds that a bus must be at the station before a train arrives in order to ensure an efficient meeting

Train2Bus = Time (seconds) that a bus must wait after a train arrival to ensure pickup

SizeType = Number of seats in this bus type

Quantity = Number of vehicles of this SizeType

FixPick = Fixed time (seconds) for one passenger pickup, including bus slow down

FixDrop = Fixed time (seconds) for one passenger drop-off, including bus slow down

FixBoard = Fixed (additional) boarding time (seconds) per passenger

FixAlight = Fixed (additional) alighting time (seconds) per passenger

NodeNo = Node index at section end points

SectionNo = Section number between two nodes

StopNo = Stop-number starts with SectionNo, representing an intersection, not a node

MeanDemand = Mean number of potential travel requests per given hour and SectionNo, to the train station

MeanDestin = Mean number of potential travel requests per given hour and SectionNo, from the train station

MeanTime = Mean section travel time (seconds)

StDevTime = Standard deviation of section travel time (seconds)

Min4Turn = Minimum number of on board passengers to allow a short-turn

Min4Cut = Minimum number of on board passengers to allow a shortcut

Min4Trip = Minimum number of travel requests by calls to allow a non-scheduled trip

Min4Dep = Minimum of number of waiting passengers to allow a non-scheduled trip

RouteNo = Unique route index

RouteDir = Direction of RouteNo (start westbound or eastbound)

TTimeTable = Fixed train timetable in hhmmss form, hh is from 00 to 24

BTimeTable = Fixed bus timetable in hhmmss form, hh is from 00 to 24

Layover = Fixed time for driver rest at the end of each trip

All these variables interact in each simulation iteration as part of the different strategies and

other internal features.

17.4.2 Simulation procedures

The simulation model is based on events. The simulation starts with reading the input data

and proceeds by arranging train-arrival events and passenger-arrival events. Figure 17.4

presents the basic event-oriented simulation logic.

There are eight main events, classified in Figure 17.5. Event 1 represents passengers walking

into the stop to wait for the next shuttle in order to arrive at the train

tation. Event 2 represents

passengers who have arrived on the train and are now waiting for the next shuttle.

Event 3 occurs when a vehicle becomes available for the next trip. Event 4 is when the shuttle

arrives at a node (intersection) on the road network being considered. Event 5 represents the

arrival of passengers who want to ride the shuttle from its stop to the train station. Event 6

represents passengers who are about to arrive at the train station and will seek to ride the

shuttle. Event 7 is the arrival of the train at the station, including the time for the passengers

to arrive at Event 2. Last, Event 8 is the time when the shuttle departs, in accordance with

the timetable. End simulation Print results No Yes Start simulation Read input data Schedule train-arrival events and passenger-arrival events Process next event (see Figure 17.5) More events ?

Figure 17.4 Basic simulation logic.

The actions taken for Events 1, 2, and 3 appear in Figure 17.6. They start with enquiring

whether the number of passengers who want the service has reached the minimum required

for dispatching a vehicle. For the DRT strategy (no fixed route), the procedure in Figure

17.6 identifies the section with current (booked) demand and the application of a shortest

path algorithm (e.g., see Appendix 9.A). The dynamic-routing procedure involves moving

the shuttle from the train station across all demand points to the first demand point that is

within the shortest path from the station. From the last point, the shortest-path algorithm is

used again for all the other demand points that were not visited until all the points have been

included in the dynamic route. This DRT routing procedure has been found to be effective

and convenient to use in the simulation.

Once a vehicle is available (see Figure 17.6), the next event is of type 4, described in Figure

17.7. Thus Event 4, in Figure 17.7, starts either with a station node (train station), whether

at the end or at the beginning of the shuttle ride, or at an intermediate node. For any inter

mediate node, the procedure checks whether the minimum number of passengers on board

the shuttle has reached the threshold for either a short-turn or a shortcut strategy. The procedure

then checks to see whether creating a short-turn or a shortcut will enable passengers

to arrive at an earlier train than that which would be met by completing the entire route. A vehicle becomes available for a ride (see Figure 17.6) Passengers arriving on train, wait for a shuttle, and are ready to be distributed (see Figure 17.6) Passenger arrives at a shuttle stop and waits to be picked up (see Figure 17.6) Shuttle arrives at a node on the route (see Figure 17.7) Arrival of a passenger requesting a ride from a stop to the station Arrival of a passenger who boarded the train at a remote station; will arrive on the next train and will request a ride from the station to a stop; will arrive with train at Event 7

Arrival of a train: all passengers alight; after a given time, Event 2 occurs Shuttle-departure time is in accordance with the timetable (see Figure 17.7) Which event type (number)? Event 1 2 3 4 5 6 7 8

Figure 17.5 Event classification.

Finally, there are the actions taken in the simulation for Event 5 following the process

of informing the passenger through the available online service. There are two possible

actions: (a) the passenger will be able to reach the shuttle stop on time after learning of the

expected arrival time and (b) the passenger will not be able to arrive on time and will ask

to be notified of the next arrival time by callback. It is assumed that passengers will either

call an automated system or look at the shuttle website to ascertain the arrival time. The

passenger will then be asked to click, for instance, “1” (for wanting to use the service) or “2”

(for not wanting to use it following the announcement of the expected arrival time). Only

those who click “1” (OK) are taken into account in the simulation process. The simulation

model can either consider a given demand curve or be used to generate a random demand

based on the residential density of each section of the network. In the fixed-route case, pas-

sengers reach their closest stop on the network. The travel time is a random variable with a

normal distribution, and the simulation model calculates the probability of being on time.

If this probability is below 90%, the user is notified to wait for a callback. In this way, the

system uses the philosophy of advanced technologies and maintains a highly reliable service. Continue to next event Next is Event 4 Append r to R, then append u to R; remove u from S, and let X be W
 threshold for dispatching a shuttle reached ? Vehicle available? Fixed route? Denote: S= set of sections with positive demand X= train station R= route to be constructed; empty for the time being Find shortest route r from X to the train station and append it to R Find shortest route r from X to Y, where Y is a node on a section in S; denote this section by u = (Y, W), where the end points (nodes) of the section are Y and W S empty? Yes Yes Yes No No No No yet

Figure 17.6 Actions taken for Events 1, 2, and 3.

17.5 CASE STUDY

In order to test a real-life situation, the area of Castro Valley in California was selected

for data collection and simulation runs (Ceder, 2013). The BART station in Castro Valley

is on the blue line, Dublin/Pleasanton to Daly City. A site observation was conducted in

the Castro Valley area, from which the base network and stops were created. The optimal

routing procedures given in Sections 17.7 and 17.8 were not considered in this case study;

instead, two routing scenarios starting at the BART station were introduced, one with a Yes Yes Yes No No No Which node ? Station node at the end of ride: all passengers on the shuttle alight; vehicle will be available (Event 3) after a layover time threshold for shortturn/cut reached? No more passengers on existing route? Station node at the start of ride: all passengers from Event 2 that a vehicle can accommodate will board Intermediate node: all passengers in this section from Event 2 alight; all passengers from Event 1 on this

section board (given seat availability) Next event for this shuttle is the next node on the route after the time spent traversing the section considered (including the time for pickups and drop-offs) Type of route's cut Try short-turn (turn around on same route back to station) Try shortcut (shortest route back to station) Alter the route An earlier train caught?

Figure 17.7 Actions for Events 4 and 8.

single route and one with a two-route shuttle system. The routing decided on followed a site

visit. Figure 17.8 presents the base network (described in Section 17.7.1), both the single

route case (solid line) and the two-route case (dashed lines), and shows the location of the

BART station at which the routes start and end. The two circular routes (dashed lines) are

extended from the station—one to the left and one to the right.

A large number of simulation runs were executed across the 10 routing strategies described

for four different groups of (available) shuttle buses: 1, 2, 3, and 4 buses. The estimated

demand was 400 passengers daily, generated randomly. Table 17.1 summarizes the results

obtained for the waiting time per passenger in 40 cases related to the single-route scenario.

The minimum (best) passenger-waiting-time results are indicated in this table by an asterisk

for each group of available buses; the second-best results are marked by “✓.” The waiting

time in Table 17.1 is the average time per passenger, in minutes, that elapses from the time

of a phone call to the shuttle bus-information center until pickup occurs. This time period

Figure 17.8 Routing map of the Castro Valley case study:

base network is defined by gray lines, single-route case by a solid line, and the two-route case by dashed lines.

includes the walking time from the place of the call to the bus stop (assuming the shortest

walking distance) and the waiting time until the bus arrives.

The results presented in Table 17.1 show that the fixed route-fixed schedule strategy (#1)

results in the highest (longest) waiting times. At the same time, the flexible route-flexible

schedule (demand-responsive) strategy (#10) does not always provide the best results; hence,

it cannot a priori be superior to the other strategies. In fact, the best routing strategies

observed in this real-life test are those with two asterisks and two "✓": fixed route-flexible

schedule and bidirectional, with possible shortcut-turn (#5); fixed route-flexible schedule

and bidirectional, with possible shortcut (#7); and fixed route-flexible schedule and

bidirectional, with possible shortcut-turn and shortcut (#9). The shortcut-turn, shortcut, and bidirectional

directional-based routing strategies indeed prove worthwhile to consider. These three uncommon

strategies reflect the current availability of offline information and communication.

In addition, six more simulation runs were performed for the two-route scenario, using

strategies #1, 8, and 10 with a group of four buses for both picking up passengers for the train

station and distributing them from the station. An additional criterion, attention to w

high

had to be paid at the phone location, was established for maximum waiting time on these six

runs: 20 min (could be changed in the simulation runs). This criterion reflected the fact that

the caller would not actually wait if the announced waiting time for the shuttle bus was more

than 20 min; in such a case, the caller would cancel the request. Table 17.2 summarizes the

Table 17.1 Simulation results for waiting time per passenger (in min), using different combinations of strategies and number of buses for the Castro Valley case study

Strategy Number of buses 1 Bus 2 Buses 3 Buses 4 Buses

1 51 22 20 20

2 25 22 17 15

3 24 ✓ 23 15 * 14 ✓

4 25 17 * 16 ✓ 15

5 24 ✓ 18 ✓ 15 * 12 *

6 24 ✓ 17 * 16 ✓ 15

7 24 ✓ 18 ✓ 15 * 12 *

8 24 ✓ 23 16 ✓ 15

9 24 ✓ 18 ✓ 15 * 1 *

10 22 * 18 ✓ 15 * 15

Note: Given demand = 400 buses daily.

* = Best result.

✓ = Second-best result.

Table 17.2 Simulation analysis for the two-route, four-bus case, with a 20 min criterion (maximum waiting after call)

Strategy → Fixed route, Fixed schedule (#1) Fixed route, Flex schedule (#8) Flex route, Flex schedule (#10)

Average wait from phone call to bus arrival (min) 13.3 8.9 6.8

Standard deviation (min) 3.0 3.7 2.5

Average wait at train station (min) 13.6 1.6 1.4

Standard deviation (min) 15.3 8.5 5.0

average waiting time per passenger for these two pickup and drop-off cases, including the

standard deviation determined for each simulation run.

The cumulative curves for the waiting times in these six situations appear in Figures 17.9

through 17.11. Table 17.2 shows that the average waiting time for distributing passengers in

the fixed schedule case is much higher than for the flexible schedule cases. Furthermore, the

standard deviations are lower in the pickup cases than in the drop-off cases. More precise

configurations for these results are shown in Figures 17.9 through 17.11.

In these figures, the upper cumulative curve refers to the pickup case, and the lower

curve to the drop-off case. Obviously the waiting time in the drop-off case depends on the

bus-departure time from the train station. This is the reason that the cumulative curves

for waiting at the train station have the shape of a large step function. It should also be

mentioned that the x-axis scale is not the same in all cases; it simply reflects the resultant

waiting-time range. A comparison between Strategies 1 and 8 (Figures 17.9 and 17.10) for 400 350 300 250 200 150 100 50 0 No. of passengers (waiting the value of X or less) Waiting time (min at home/work/stop) 0 5 10 15 20 25 Strategy 1: Fixed route, fixed schedule, waiting from phone call to vehicle arrival No. of passenger s (waiting the value of X or less) 350 300 250 200 150 100 50 0 Waiting time (min at the train station) 0 10 20 30 40 50 Strategy 1: Fixed route, fixed schedule, waiting at the train station

Figure 17.9 Waiting time (phone call-to-bus arrival in upper curve, and at the train station in lower curve), in minutes, for Strategy 1 (Fixed route-Fixed schedule); note that waiting-time scales are not the same.

the pickup case reveals that whereas the wait ranges from 5 to 20 min in the Fixed schedule,

it ranges from 3 to 18 min in the flexible schedule. In the demand-responsive case (Strategy

10, Figure 17.11), the waiting time for the pickup case ranges from 3 to 13 min.

These simulation runs are only preliminary steps toward the examination of a smart

shuttle operation. More simulation runs are required for different numbers of fixed routes

and various demand levels, along with further sensitivity analyses of the input parameters.

17.6 CUSTOMER SURVEY

To obtain the needed consumer information, the case study used the survey-research method

described by Yin and Ceder (2006). The test market was identified as being within a 2-mile

radius of the Castro Valley BART station. Four hundred telephone interviews were conducted 0 20 40 60 80 100 120 140 400 350 300 250 200 150 100 50 0

No. of passengers (waiting the value of X or less) No. of passengers (waiting the value of X or less) Waiting time (min at home/work/stop) Strategy 8: Fixed route, flex-schedule, short-turn and shortcut, waiting from phone call to bus arrival 0 2 4 6 8 10 12 14 16 18 20 350 300 250 200 150 100 50 0 Waiting time (min at the train station) Strategy 8: Fixed route, flex-schedule, short-turn and shortcut, waiting at the train station

Figure 17.10 Waiting time (phone call-to-bus arrival in upper curve, and at the train station in lower curve), in minutes, for Strategy 8 (Fixed route-flex schedule, with short-turn and shortcut); note that waiting-time scales are not the same.

in this market area, using a random-digit-dial sample and the computer-aided telephone

interview (CATI) technique. The following criteria were used for screening survey participants:

- Eighteen years old or older
- Permanent resident of the household surveyed
- Said that BART was a possible means of transportation for them
- Commute or make their most frequent trip a way from home by some means other than walking or bicycling

The margin of error for a 400-responder sample is $\pm 5.0\%$ at the 95% level of confidence.

The survey questions covered the following topics: (1) trip characteristics, (2) mode of

access to transportation to BART, (3) willingness to use a smart shuttle, (4) willingness to pay 400 350 300 250 200 150 100 50 0 No. of passengers (waiting the value of X or less) Waiting time (min at home/work/stop) 0 2 4 6 8 10 12 14 Strategy 10: Flex-route, flex-schedule (demand-responsive), wait from phone call to bus arrival No. of passengers (waiting the value of X or less) 400 350 300 250 200 150 100 50 0 Waiting time (min at the train station) Strategy 10:

Flex-route, flex-schedule (demand-responsive), waiting at the train station 0 5 10 15 20 25 30 35

Figure 17.11 Waiting time (phone call-to-bus arrival in upper curve, and at the train station in lower curve), in minutes, for Strategy 10 (demand-responsive); note that waiting-time scales are not the same.

for the service, (5) desired attributes of the shuttle service, and (6) demographic characteristics

of the survey respondents. The study identified the features that would attract consumers

in terms of routing characteristics (i.e., intermediate stop options, express service), travel

time, waiting time, number of stops, and willingness to pay for the shuttle service.

17.6.1 Survey results

Several attributes were investigated with respect to the design of the shuttle service. Among

them were the number of pickups and drop-offs, the size of a shuttle vehicle, acceptable

number of riders, travel time, and waiting time. The results for these attributes and for other

questions are as follows.

17.6.1.1 Important attributes for shuttle design

The respondents said that the most important attribute was the cost of the shuttle service.

The second most important was overall travel time, including waiting time for the shuttle

either at BART or at the pickup location. The third most important was the on-time reliability

of the service at the pickup location or at the BART station.

17.6.1.2 Pickups and drop-offs

Most people expected four to five pickups (median of five pickups) on the way to the BART

station. Similarly, they expected four to five drop-offs on the way home (median of five

drop-offs).

17.6.1.3 Travel time

The question was, if the average travel time to the BART station was slower than it currently

is, do you think you might use the BART shuttle service? Nearly one-third (29%) said they

would take the shuttle, whereas half (53%) said they would take the shuttle if it took about

the same time. Only 12% said they would take it only if it were faster. The survey suggests

that people are willing to accept a longer travel time when using a shuttle because of what

benefits they perceive it affords.

17.6.1.4 Arrival time and schedule information

One of the reasons that people are hesitant to go by public transit is the uncertainty associ-

ated with vehicle-arrival times. Advanced transit information system (ATIS) can disseminate

real-time vehicle-schedule information to those who are regular transit users, as well as

to the occasional transit rider. A TIS vehicle schedules can also attract those who seldom or

never used public transit in the past.

17.6.1.5 Cost

The cost question for riding a shuttle was constructed to ask about the higher price first and

then lower prices. Suppose the cost of the shuttle service were \$1 for a one-way trip, how

likely would you be to use this service? According to the survey, half the respondents said

they would be interested if the cost were \$1; furthermore, 41% said that they would still be

interested in taking a shuttle at the price of \$2.00 for a one-way trip. As expected, consumer

interest in using the shuttle service is highly elastic with respect to the cost as the results in

Figure 17.12 illustrate. However, the survey found that price elasticity was not directly pro

portional to the cost of the shuttle service. The willingness to use the service is significantly

(95% level of confidence) different between the low and the high cost of the shuttle service.

17.6.1.6 Waiting time

Questions about waiting time were also asked at the same time as the cost of the shuttle ser

vice. If the waiting time was 20, 15, 10, and 5 min, how likely would respondents be to use

the shuttle service? The results are shown in Figure 17.12. The survey showed that the longer

the waiting time, the less willing people are to take the shuttle; however, there was not a

significant difference between a 10 min waiting time and a 5 min waiting time. This sug

gests that half of the shuttle users are willing to wait 5 min and up to 10 min for a vehicle.

17.6.1.7 Frequency of using the shuttle service

Most respondents said that they would use the service two to three times a week.

17.6.1.7.1 Payment method

Over half (55.4%) of the respondents were interested in paying for the service on a per-use

basis. Only 11.2% responded favorably to a weekly fee basis, and 31.2% said they could

work with a monthly subscription arrangement.

17.6.1.7.2 Preferred means of receiving information about the shuttle

The survey showed that 62% would like to receive their information from a pamphlet.

Approximately one-third (30.9%) preferred to retrieve it through the Internet, and only

5.9% would like to receive it by telephone.

10 0 0 1.- 2.- 3.- Fare (\$US) Waiting time (min) 0 5 10 15 20 30 40 50 60
Percent age of respondents

Figure 17.12 Willingness to pay and wait for the shuttle service.

17.6.1.7.3 Benefits of the shuttle

A question pertained to the biggest benefits derived personally from using the shuttle service.

It was an open-ended question and accepted up to three responses; therefore, the per

centages shown in this section are not mutually exclusive. Respondents mentioned a variety

of personal benefits. Among them were these: (1) convenience, including no need to park

(25%), avoid walking in bad weather (2%), avoid wear on vehicle (21%), and others (30%);

(2) safety, including reduced stress and anxiety (8%), less chance of an accident (2%), and

the avoidance of traffic lights (18%); (3) travel time savings (14%); (4) less cost (18%); (5)

reduced pollution (7%); and (6) the chance to meet people and to socialize (2%).

17.7 OPTIMAL ROUTING DESIGN: BASE NETWORK

This and the following sections present, respectively, a methodology and a procedure for

designing optimal shuttle (circular) routes, following Ceder (2009) and Jerby and Ceder

(2006). The objective set forth for an urban road network is maximal coverage of potential

passenger demand to a transportation center (e.g., train station), with travel time along the

entire route not exceeding a certain threshold value. Generally, an objective function that

aspires for maximum coverage is supposed to create a long, winding route, which in this

case is blocked by the route-time constraint to ensure an adequate level of service. The problem

input consists of an urban network with trip-generation nodes and a single destination

(main) node, average travel time between each segment on the network to the main node,

and a constraint defining maximum travel time along the route.

The methodology described is based on a modular approach, enabling the partitioning of

a complex problem into a series of sub-problems. Hence, each sub-problem can be referred

to as a single independent component. The methodology consists of eight main stages as shown

in Figure 17.13. The first stage is the characterization of urban network attributes for deriving

an urban network for route design. The second stage deals with the insertion of a average

travel times into the links of the road network, while the third stage defines the service area.

The fourth stage focuses on a method to estimate potential demand for trips on a designed

route. This method includes the development of a potential demand measure for

each link on the road network, based on urban and spatial criteria and using density and

walking-distance parameters. This measure is used as a single input to an optimization model

developed in the fifth stage. The model enables the automated design of an optimal circular

route complying with a given total travel-time constraint. Test runs using the optimization

model are performed in the sixth stage, showing that its complexity is high and cannot be

efficient for medium and large-size networks. Accordingly in the seventh stage, an alternative

tive heuristic algorithm is proposed and developed. The algorithm, which enables the auto

matic design of circular routes, ensures good, mostly but not always optimal, results. The

eighth stage examines the heuristic algorithm on different networks.

17.7.1 Base road network

Theoretically there could be a situation in which all city streets are used for the passage

of transit vehicles; in actuality, this is not the case. For various reasons, some of the net

work links will never be used for the passage of transit vehicles; therefore, these links can

a priori be excluded from the planning process. Such an action will define a more reduced

transit-road network (the base network). Working with a base network allows the planner

to reduce the complexity of the problem and enables the transit agency and the passengers to

rely on a simpler network that is easier to understand and operate. Determination of a base

road network consists of the following elements: street characteristics (width, slope, park

ing arrangements); spacing between parallel streets; and safety considerations or any other

criterion set by the transit agency's constraints on a case-by-case basis.

According to the stages in Figure 17.13, the travel time in each network link is determined

after creating the base network. This value can be either

measured directly or calculated

according to the length of the link and the average travel speed. The next stage in Figure

17.13 is to determine the service area that can be served reasonably by a single circular route.

The size of the service area depends on the route's travel-time constraint, that is, links that

cannot be covered in the framework of the travel time allocated should be eliminated from the network. OutputComponentInput Result: A computerized method of planning a circular public transit route Analytical framework Urban planning framework Base network Base network containing link to travel times Defined service area for the base network Potential passenger demand on every link of the base network Optimal model Model's strengths and weaknesses Algorithm's strengths and weaknesses Stage one—reduce to basic network Urban street network Average travel speeds Sample networks (different sizes) Constraint on maximum circular travel time Assumptions on passenger demand and walking distances Problem definition Sample networks (different sizes) Heuristic rules Stage eight—test algorithm Stage seven—develop a heuristic algorithm Stage six—test model Stage five—formulate an optimal model Stage four—determine demand indexes Stage three—define service area Stage two—insert travel times Heuristic algorithm

Figure 17.13 Urban planning and analytical frameworks for a circular route design.

the base network. The proposed process includes testing of each of the base network links

in order to determine whether a circular route can be created from the link to the main node

and back while complying with the time constraint. For convenience, the main node will

continue to be referred to as the train station. Should the duration of the shortest route (in

travel-time values) from the link to the train station and

back exceed the time constraint,

such a link should be eliminated from the network. Applying this procedure on all links of

the base network will result in a further reduction of the network and the elimination of

distant links that should not have been taken into consideration in the first place.

17.7.2 Potential passenger demand

The demand potential is assessed on the basis of spatial data from the urban network,

and is subject to a number of assumptions. Methods for assessing transit demand, which

were presented in Chapter 10, include the counting of passengers, conducting surveys, using

known databases of residential addresses and work places, and weighting of socioeconomic

and demographic data of the region. In addition to these methods, following is a two-phase

alternative method that uses spatial and demographic data.

17.7.2.1 Phase A: Calculation of average walking distance per link

This phase considers the maximum acceptable walking distance to the shuttle route for poten

tial passengers. This variable is known to be part of a catchment area. It is customary to assume

that it encompasses an area of up to 400 m from every side of the road link although some pas

sengers will be willing to walk a longer distance under certain conditions. Figure 17.14 illustrates Bahai Gardens Central railway station Subway station CBD
0.....200.....400 m

Figure 17.14 Catchment areas for the base network of the case study (City of Haifa).

the catchment areas for pedestrians in a case study area of downtown Haifa (Israel) as polygons

on the base network. In this case, the catchment area amounts to 300 m from every side of the

route. In Figure 17.14, one can identify areas where overlapping is created between two or more

polygons; the assumptions discussed in the following text are used for such overlapping cases.

The first assumption is that passengers choose the road link nearest to the point of origin

as the waiting point for the shuttle vehicle. The second assumption is that passengers will

walk to the nearest possible waiting point by taking the shortest bee line distance (a rea

sonable assumption, since the passenger in many cases can walk to the stop in an almost

straight line through parks or alleys). Therefore, in cases in which catchment areas overlap,

every point in the overlapping area can be attributed to the nearest bee-line link. A value

assessing the average walking distance in the area will now be defined.

The average walking distance $wd_{i,j}$ on link (i, j) is influenced by land uses within the catch

ment area, and is calculated as follows: $wd_{pop_{i,j}} = \frac{1}{m} \sum_{b=1}^m wd_{b,i,j}$ (17.2)

where

m represents the total number of buildings in the catchment area

b is a certain building in the catchment area

pop_b represents the amount of population (residents, employees) in building b

wd_b is the walking distance from building b to the

nearest road link

i and j represent network nodes $pop_{i,j}$ is the amount of population in the catchment area between nodes i and j

17.7.2.2 Phase B: Assessment of the demand potential as dependent on walking distances

In this phase, the demand potential from each of the catchment areas can be assessed, based

on two more assumptions. The first of these assumptions is that the entire population in the

pedestrian catchment area can be related to as potential users, regardless of their motoriza

tion and socioeconomic levels; this means that there is a correlation between building den

sity and demand potential. The second assumption is that the demand for trips also depends

on the walking distances from the land uses around the road to the shuttle stop, that is, the

further the land use from the road link, the less the demand potential.

Based on these assumptions, a demand potential index is proffered for every network link

(i, j) as follows: $pd_{i,j} = \frac{pop_{i,j}}{wd_{i,j}}$ (17.3)

where $pd_{i,j}$ expresses the ratio between the amount of population at each link and the aver

age walking distance per link. In other words, the demand potential measure (index) in

Equation 17.3 is in direct proportion to the population density and in inverse proportion to

the average walking distance.

The two-phase method described for estimating demand requires quality geographic infor

mation. This method was implemented in the case study shown

in Figure 17.14, with bands

around each network link. When no exact data existed on the number of residents in each

block, assessments were carried out based on a multiplication of the number of building floors

(using the number of apartments per floor) and the approximate size of households. The output

comes from the case study in Figure 17.14 is presented in the schematic network of Figure 17.15.

17.8 OPTIMAL ROUTING DESIGN: ALGORITHM

This section contains two main parts. First, an optimization model formulation is described,

using operations research (OR) terms (see Section 5.5) and following Ceder (2009). Second,

a heuristic algorithm, following Ceder (2009) and Jerby and Ceder (2006), is presented for

handling large-size networks.

17.8.1 Optimization model

Let $G(N, A)$ be defined as the graph representing a network of streets in an urban area,

where network nodes are represented by the set $N = \{ 1, 2, \dots, n \}$ and network streets (graph

links) as $A = \{ (i, j) : i, j \in N ; i \neq j \}$. For each of the links $(i, j) \in A$ an average travel time

of $t(i,j)$ minutes by time-of-day is defined, and also a demand potential index $pd(i,j)$. The

potential demand value is assumed to be homogeneously distributed at the link. The model

described produces a circular route r , starting and ending at node n_1 . The model maximizes

the demand potential of passengers while complying with the constraints of maximum cir

culating travel time T . The case study of Figure 17.15 includes the optimal circular solution

for $T = 25$ min and shows the values of $t(i,j)$ and $pd(i,j)$ for each link.

Model formulation is as follows: $\text{Max } \sum_{i,j \in A} y_{i,j} A_{i,j} t(i,j)$ (17.4) Haifa central railway station Bahai Gardens Hameginim Ave. To Haifa Bay To Tel Aviv 4, 3 3, 6 Hagefen St. Allenby St. Subway i,j Optimal solutions 3, 6 6, 3 6, 3 5, 9 7, 2 7, 2 7, 3 7, 3 2, 4 2, 4 5, 3 5, 3 2, 5 5, 3 $t_{ij} = 7, 4 = pd_{ij} \quad t_{ji} = 5, 4 = pd_{ji}$

Figure 17.15 Schematic description of the case-study network and its optimal circular route solution for $T = 25$ min.

Subject to: $t(i,j) \leq T$ (17.5) $y_{i,j} \in \{0,1\}$ (17.6) $\sum_{i,j \in A} y_{i,j} = 1$ (17.7) $\sum_{i,j \in A} y_{i,j} t(i,j) \leq T$ (17.8) $y_{i,j} \in \{0,1\}$ (17.9) $y_{i,j} \in \{0,1\}$ (17.10)

The objective function (17.4) contains $\{0,1\}$ variables $y(i,j)$ in which $y(i,j) = 1$ means that the

link (i,j) is part of the circular route r , or zero otherwise. Equation 17.5 represents the maxi

mum travel-time constraint. Equation 17.6 implements a condition for creating a closed cir

cular route in the network; it follows the known Euler theorem (e.g., see Eiselt et al., 1995),

in which the number of incoming and outgoing links of every node is equal.

Equation 17.7 ensures that only circular routes originated at node n_1 will be considered,

where S is a set of all the network-node combinations, which include at least two nodes with

out node $n-1$. Equation 17.7 ensures that the number of route links connecting the nodes of any

sub-set s_m of S will be smaller than the number of nodes in the set. In other words, Equation

17.7 requires that all chosen links must be connected to each other, thus avoiding the forma

tion of more than one continuous circular route. Equations 17.8 and 17.9 ensure that node $n-1$

is included in any solution of circular routes, and finally constraint (17.10) defines $y(i,j)$.

Note that the preceding model formulation cannot create circular routes in which two (or

more) routes overlap in some of the nodes (e.g., eight-shaped routes). To attain such routes,

it is possible to identify potential overlapped nodes and to split them each to two nodes with

the same incoming and outgoing links (artificially), that is, a potential node i will be split

into i' and i'' . Another important note is that only Equation 17.7 increases in a manner that

cannot be described by a polynomial equation, hence results in increasing the complexity

of the problem with the increase of the network size. The conclusion, therefore, is that solv

ing the problem of real-life road networks through the integer-programming formulation

described is impractical. This complexity in the number of calculations warrants the exami

nation of heuristic solutions that is given in the next section.

17.8.2 Heuristic algorithm

The algorithm described is based on an efficiency criterion. For that purpose, an impedance

ratio, z_{ij} , is defined as follows: $z_{ij} = \frac{t_{ij}}{\omega + \lambda d_{ij}}$ (17.11)

where ω , λ are coefficients.

As mentioned, the potential demand is a value that is in direct proportion to inhabitant

density and in inverse proportion to walking distances. The efficiency criterion utilized in

the algorithm is that of the shortest path between each two links of a selected group of links

in accordance with Equation 17.11.

The heuristic algorithm is shown in a flow diagram in Figure 17.16. The algorithm

constructs an efficient circular route from an initial group of links ($p \leq P$) characterized

by relatively high demand potential; it utilizes Equation 17.11 and the shortest path between Yes Yes No No No Yes Yes Have all possible routes for p been tested? Create the group of main-demand links according to the defined demand-potential index Does $p=P$? Input: Basic urban network Most efficient routes matrix between each two links Initial size (p) of a mandatory group of links Maximal size (P) of the mandatory group of links Does the route tested comply with the time constraint? Choose a combination of p links from the group created (order of links considered) and construct an efficient circular route that passes the p links and the main node Is the potential demand on the route the highest found so far? Define the route as optimal among those already tested STOP $p=p+1$ No

Figure 17.16 Flow diagram of the heuristic procedure.

the links in the selected group p . A small sub-group of links is chosen in every iteration

of the algorithm. The algorithm examines the possibility of a circular route that will pass

through the sub-group links and meet the travel-time constraint. Then the algorithm com

pares the potential demand of the candidate route to each of the routes examined so far.

Lastly, the best circular route found is chosen.

Several test runs were made enabling a comparison of the heuristic algorithm with the

optimal model. The results are shown in Table 17.3. The rest network is that of Figure 17.15;

these even networks that follow a resmall, randomly selected networks; last is a larger-size

network. One important feature of the heuristic algorithm is group p , which is the number

of initial mandatory links chosen for route r . Figure 17.6 defines P as the maximum number

of such mandatory links. Table 17.3 shows the ratio between P and the number links in the

network, $A \cdot P = 4$ was found to provide reasonable computer running time, including the

42-link last tested network. Finally, the results support the heuristic algorithm, as they are

the same as the optimal across all small networks.

17.9 IMPLEMENTATION STAGES

In order to secure the potential success of a new shuttle service, certain steps should be

undertaken gradually and carefully. Essentially, these steps, shown in Ceder (2013), involve

an initial quantitative analysis and then a pilot plan.

17.9.1 Initial analysis

Figure 17.17 schematically outlines five components that are necessary to complete the ini

tial analysis: (1) constructing a base street n

etwork; (2) creating groups of fixed routes; (3)

constructing short-turn, shortcut, and bidirectional strategies; (4) creating a DRT-type of

service; and (5) comparing different strategies with a given passenger demand.

Component 1 of Figure 17.17 uses site observations and measurements for a base road

network configuration, including traffic-light locations and an 85 percentile of time to and

from the train-station platform. This 85 percentile of the different times observed ensures

a adequate walking time for the majority of people. The determination of the base road network

work considers the following elements: approximate (low, average, high) density of residential

area, street characteristics (width, slope, parking arrangements), spacing between

parallel streets, and the road-network shape of each zone in the area. Table 17.3 Summary of the heuristic algorithm runs, given different networks

Network tested	No. of nodes	P/A ratio	Objective function value a (optimal)	Objective function value a (heuristic)	
From Figure 17.15					
7 4/9	26.6	26.6	Random 1 7 4/9	191	
191	191	Random 2 7 4/7	208.8	208.8	
Random 3 7 4/7	206.4	206.4	Random 4 7 4/8	79.2	
79.2	Random 5 7 4/9	338	338	Random 6 7 4/9	243.6
243.6	Random 7 8 4/9	204	204	vRandom 8 (large)	25 4/42
Too long	running time	142.2	a	In potential units of demand,	following Equations 17.2 and 17.3.

Component 2 of the initial analysis creates the fixed routes to be considered. These circu

lar routes can be constructed by the heuristic algorithm described in Section 17.8, includ

ing a measure of its demand potential. The maximum length of the routes to be selected is

influenced by the number of shuttle vehicles available and,

if given, the minimum frequency

required.

Component 3 constructs the operational strategies that can continuously ensure an ade

quate level of service. These strategies, outlined and explained in Section 17.3, basically Input Component Outcome Nodes and arcs of base network, with indication of traffic-light locations and time to and from train-station platform Conclusions Input 1 2 3 4 Creating shortcut, short-turn, and bidirectional strategies for groups of fixed routes, using objective functions and constraints Tool: Simulation Creating groups of fixed routes (one fixed route, two fixed routes, etc.) using specific objective function(s) and constraints Tool: Algorithmic approach Construction of base street network Tool: Site observation and measurements with simple analytical calculations 5 Online demand-responsive transit (DRT) Shortcut, short-turn, and bidirectional strategies on fixed routes (fixed routes, flex-schedule) Creating demand-responsive strategies with objective function(s) and constraints Tool: Algorithm+ Simulation Comparison of strategies Set of fixed routes, depending on the number of vehicles employed and average frequency required (fixed route, fixed schedule) Street network configuration Street network characteristics (width, slope, parking arrangements) Spacing criterion between parallel streets Residential area density Location of train (hub) station 85 percentile of time to and from the train-station platform Number of vehicles available Frequency and headways required Train schedule 85 percentile of walking time to and from the trainstation platform Average time required to pickup and drop-off (including vehicle slow down and merging with traffic times) Minimum number of on-board passengers required for shortcut (to deviate from fixed route) Minimum number of on-board passengers required for short-turn (conditional on min. no. of reserve pickups when turning back) Min. level of pickups Train schedule 85 percentile of walking time to and from trainstation platform Average time required for pickup and drop-off (including vehicle slow down and merging with traffic times)

Figure 17.17 Practical methodology for constructing a feeder/shuttle service.

cover short-turn, shortcut, and bidirectional possibilities. They can be analyzed by a simula

tion tool similar to that explicated in Sections 17.4 and 17.5.

Component 4 in Figure 17.17 creates DRT strategies, given certain input elements. That is,

the input is based on the minimum number of passengers for whom a vehicle may be sent for

a pickup; the train schedule, in order to match the expected arrival time of the DRT vehicle

with the train's arrival; the 85 percentile of walking time to and from the train-station plat

form; and average times required to pickup and drop-off passengers. The tool to create this

type of dynamic, online routing is based on an algorithm (e.g., shortest path from point to

point), using the simulation tool discussed in Sections 17.4 and 17.5.

Finally, component 5, which consists of the outcome of components 2, 3, and 4, compares

the different strategies. This comparison can cover different demand levels, different num

bers of available vehicles, and different input parameters (travel times, threshold values for

dispatching a trip, short-turn, shortcut, etc.). The comparison will lead to a strategy that can

better fit a given situation (by time-of-day and demand level).

17.9.2 Pilot plan

Once the analysis of the shuttle service is completed, the application of a pilot study is rec

ommended. The implementation of such a pilot in the area being considered can follow, for

example, the 12 steps shown in Figure 17.18. These 12 steps can serve as a framework for

a master plan, with each outcome of a step becoming an additional input for the next step,

except for Step 6.

The pilot master plan starts with a demand analysis by time-of-day and day-of-week in

order to find the origin-destination pattern and consumer-oriented features.

The second step is to establish (if done previously in Sections 17.7 and 17.8) or design the

fixed-routing and stop system. The third step is to determine the base frequencies and time

tables for each route or, alternatively, to set the operational strategies in accordance with

Section 17.3. The fourth step is to determine the number and size of the shuttle vehicles and

to create the chain of trips (vehicle schedules) that will serve the fifth step, constructing the

crew schedules.

The pilot plan continues in Step 6 with the establishment of effective information channels

and instruments (e.g., call center, Internet, newspapers, radio, TV, and mail leaflets)

that will lead to the development of user-friendly communication procedures between the

users and a selected operating agency in the

next step (Step 7). Step 8 constructs the D
RT

operational strategies without the use of the
xed routing/stop/schedule system. Step 9

determines the testing scenarios of the pilot
while Step 10 presents the process for s
elect

ing an adequate operator. Step 11 uses proper
advertising tools to approach an operable

pilot study. Finally, the last step of the plan (Step 12)
is aimed at improving the instru

ments, procedures, and strategies with the use
of innovative intelligent transportation

systems (ITS).

17.10 LITERATURE REVIEW AND FURTHER READING

This section describes models for the design o
f shuttle or feeder routes. Only models cor

responding with a fixed route fall within the scope of this
review.

Mirasinghe (1980) formulates an analytical model
for a system in which bus routes

feed a railway line during peak hours. It is assumed that
the street network is a perfect

rectangular grid and that there is no need f
or synchronization between the feeders and the

rail line. Two cases are presented. In the first case, the
locations of railway stations are

given, and the optimal density and headways o
f the buses determined. In the second case,

the spacing of feed points is not given, a
nd therefore all three parameters (station s
pacing,

route spacing, and headways) are optimized.

Kuah and Perl (1987) present an optimization methodology for the design of an integrated

bus-rail system. A mathematical programming problem is formulated in which the itinerary

and frequency of each route are determined under a fleet size constraint. In addition, a heuristic

solving method is developed. For the basic problem, a many-to-one demand pattern is assumed. Input Survey Road map Standards, travel time Vehicle sizes, travel times Operator work rules Survey, routes, timetables Travel times, call back Short-turns, shortcut; and bidirectional options Survey, candidate scenarios Operations process Determined procedures Innovative ITS (1) Demand analysis (2) Routing + stops (3) Frequency + timetables (4) Vehicles (No., sizes) + scheduling (5) Crew scheduling (6) Information (7) Communication (8) Operational strategies (9) Testing scenarios (10) Bidding (11) Advertisement + execution (12) Update O-D, user features Routes, stops Timetables Vehicle schedules Crew schedules Information channels Caller information Deviation elements Operational scenarios Operator Pilot Pilot changes Input Step Outcome

Figure 17.18 Overview of a possible feeder/shuttle pilot master plan.

This demand pattern, which is assumed in most feeder-route models, means that passen-

gers from various origins travel to either one destination or various destinations, but they

must pass one transfer hub that is viewed as their destination when the feeder system is

designed. The authors show, however, that the model can be generalized to represent many

to-many demands. It enables simultaneous network-wide optimization, meaning that the

bus system designed may serve more than one rail line. The model is not based on binding

assumptions regarding the street network; s

streets may form any structure, and zones do

not have to be rectangular. In addition, the feeder routes are not necessarily perpendicular

to the railway line.

The same researchers (Kuah and Perl, 1988) describe an analytical model for the design

of buses feeding an existing rail line. The feeder routes in this case are perpendicular to the

rail line in a grid-shaped network. The variables optimized are headways, route spacing, and

stop spacing. The paper examines three different stop-spacing policies: uniform network

wide stop-spacing, uniform line-specific stop-spacing that may change from one route to

another, and stop-spacing that can vary along the route. They conclude that headways and

route spacing are not sensitive to system characteristics.

Chang and Schonfeld (1991) present a model for designing feeder routes to one transfer

station in a simple rectangular service area. The many-to-one demand is elastic to service

quality and fare. The model allows the designed service to change, a long period of the day,

according to demand characteristic, operating costs, and vehicle speeds, which are given

separately for each period. The suggested model makes it possible, therefore, to determine

the optimal route spacing that does not change during the day, with varying route headways

within each period. Models are formulated and compared for four types of conditions:

steady demand, cyclical demand, steady equilibrium demand, and cyclical equilibrium

demand. The results for the four cases are compared. All demand patterns (with

minor exceptions) are found to have an optimal constant ratio between the headway and

route spacing.

Chang and Lee (1993) and Chang and Yu (1996) develop mathematical models for optimizing

fixed-route and flexible-route feeder services in a simple rectangular area. Separate

models are formulated for each of the two alternatives with the objective of maximizing

welfare under a subsidy constraint; the optimal results of both types of services are com-

pared. Decision variables are route spacing, headways, and fares; their optimal values are

examined under several possible demand functions.

Chang and Schonfeld (1993) propose an analytical model for the design of parallel

routes that feed a central terminal. Demand is assumed to vary a long the day, and there-

fore different headways are assigned to different day periods. Other decision variables—

zone size, route length, and route spacing—are determined without differences between

periods.

The model developed by Chien and Schonfeld (1998) strives for a joint optimization of a

rail route and its feeder buses. The variables optimized are rail-line length, feeding-station

spacing, bus headways, bus stop-spacing, and bus route-spacing. Street network is assumed

to be a simple grid, with the feeders perpendicular to a single rail line. On the other hand,

the model is also based on the realistic assumption of many-to-many demand.

Chien (2000) presents a model for defining bus-feeder routes and their headways. One

central route is fed into one major station, and the demand is many-to-one. The service area

is shaped as a partial grid, that is, a grid with some of the links missing. The form of each

zone must be rectangle-like in a simple grid, but the zone size may be heterogeneous. The

model enables the user to decide the number of feeder routes to be defined.

Chien et al. (2001a) introduce a methodology for the design of a feeder-bus route feed

ing a major intermodal transfer station in a suburban area. The model is formulated as a

linear programming problem with geographic, capacity, and budget constraints. Decision

variables are route locations and headways. The street network has a partial grid pattern,

and the demand pattern is many-to-one. Two solution algorithms are suggested. The

first is an exhaustive search algorithm that promises, in principle, an optimal solution;

however, it is very time-consuming in most realistic networks. The second suggested algo

rithm is a genetic algorithm (GA), that is, an algorithm in which the search for optimal

solutions is based on the natural evolution process. In each iteration of a GA, a set of

solutions is selected based on the previous set; solutions that show good performance in

each iteration have a higher probability of being selected for the next iteration. Genetic

algorithms are efficient in terms of calculation time, but may result in only a near-optimal

solution.

Chien et al. (2001b) compared a nondispatchable route systems serving as feeders for a

transportation center close to a rectangular service area. Optimal vehicle size, route spacing,

and headways are determined for both types of service. The models assume a probabilistic

demand, which varies over periods of the day. Passengers' value of time is assumed to be

non-additive, that is, the value that a passenger ascribes to a 2 min period is higher than

twice the value ascribed to a 1 min period. The researchers find it best to operate conventional

bus routes at peak hours and dispatchable routes in off-peak periods. The threshold value of

demand at which one system is more cost effective is calculated.

Shrivastava and Dhingra (2001) present a heuristic algorithm for the development

of bus routes feeding railway stations. The algorithm uses different node-selection and

insertion strategies to determine route itineraries under many-to-many demand. Routes

are developed for two types of time criteria. The first is a maximum-demand-deviated,

shorter, time-path criterion; this means that when nodes are inserted into the middle of

a route, the length of the deviated route should not exceed a given acceptable limit. The

second criterion is a path-extension time criterion, meaning that if nodes are inserted

at the end of a route, then the maximum route length in terms of travel time should

not exceed a certain maximum value. Use of the suggested method is possible in any

network, no matter the street-network structure. A case study is presented, in which

the use of this method enabled serving the same demand as previously with a smaller

number of routes, a smaller fleet, and better schedule adherence because of the shorter

routes.

Another paper by Shrivastava and Dhingra (2002) presents a methodology for design

ing a coordinated schedule for existing bus routes feeding a railway station. The model

presented aims at minimizing total operating costs and total transfer times under the

constraint that each transfer time should be executed within given boundaries. The model

is formulated as a nonlinear, non-convex programming problem with many variables,

and it is therefore difficult to solve. A GA is suggested to provide near-optimal results.

The results show that the designed schedule is most sensitive to the parameters (such as

the transfer penalty). The conclusion is that the value of this penalty should be carefully

considered.

Zhao and Dessouky (2008) study the service capacity design problem for a mobility

allowance shuttle transit system—a transportation concept that merges the flexibility of

DRT systems with the low-cost operability of fixed-route bus systems. The authors analyze

the relationship between service cycle time, and the length and width of the service area

under a desired service level constraint. The analysis can be used as a guide for designing

service area parameters. Given that a shuttle travels in a rectilinear pattern and utilizes

a non-backtracking nearest-insertion algorithm, closed-form approximate solutions are

derived for the service capacity design problem. Finally, the authors demonstrate that setting

the length of the service area to half the travel speed of the shuttle multiplied by the cycle

time is an effective approximation.

Feeder lines often switch operations between a demand-responsive and a fixed-route

policy. In designing and running such systems, the identification of the condition justify

ing the operating switch is often hard to properly evaluate. In their study, Quadrioglio

and Li (2009) propose an analytical model and solution of the problem to assist deci

sion makers and operators in their choice. By employing continuous approximations,

the authors derive handy but powerful closed-form expressions to estimate the critical

demand densities, representing the switching point between the competing operating poli-

cies. Based on the results of one-vehicle and two-vehicle operations for various scenarios,

in comparison to values generated from simulation, the authors verify the validity of the

analytical modeling approach developed.

Ciaffaglia et al. (2012) investigate the transit network design problem related to the feeder

routes defined as transit services for the connection of local areas, where the demand has to

be gathered, with the stops of the main transit network, usually railway or underground sta-

tion. The objective of this study is the development of a procedure that simultaneously gen-

erates routes and frequencies of the feeder bus network in a real-size large urban area. The

solving procedure is articulated in two phases: in the first one, a heuristic algorithm generates

two different and complementary sets of feasible routes, in order to provide a good balance

between maximization of the service coverage area and minimization of the overall travel

time. The set of all feasible routes, generated taking into account only the main skeleton of

the road network, is then the input data for the second phase where a GA is utilized for find-

ing a sub-optimal set of routes with the associated frequencies. In the first phase, the first set

of feasible routes generated is composed by circular routes using the solution of the travel

ling salesman problem (TSP), and the second feasible set aims at developing feeder routes

more direct than the others using the k-shortest path algorithm. The proposed procedure has

been implemented on two real-life size networks, Winnipeg and Rome, in order to compare

its effectiveness with the performances of the existing transit networks. The results of the

applications of the design procedure show that the feeder routes imply a more integrated

transit network with a reduction of the total travel time, despite an increase of the number

of transfers, in a more efficient way as demonstrated by the reduction of the operating costs

and the increase of the average load factor.

Kim and Schonfeld (2014) present formulation and numerical examples of integration

between flexible- and fixed-route services. Their decision variables are service type, vehicle

size, number of zones per region, headway, and fleet size. The authors propose a probabilis

tic optimization model for integrating fixed and flexible bus services with timed transfers.

For the coordination of passenger transfers, same or multiple headways are used including

optimization of the fleet size. Kim and Schonfeld (2014) show that the integration of fixed

and flexible services is especially desirable when demand densities vary considerably among

the served regions. The proposed models quantify the delay and cost savings achievable

from optimized coordination. Their results confirm that limited transfers are desirable for

increasing the probability of vehicle connections at transfer terminals, and thus minimizing

passenger wait times compared to uncoordinated operations. A GA is used in their study as

a solution method.

Dikas and Minis (2014) studied the possibility to change the optimal routing of a lex

ible transit service so as to pick up passengers with limited mobility and drop them off

at their destination. The analytical problem is to maximize the number of served para

transit (of people with limited mobility) requests and reducing the total riding time of

the paratransit clients with the compliance of covering the optimal routing. The solution

of this problem uses an exact branch and price method including a labeling algorithm

with appropriate dominance rules in order to reach the optimal solution within reason

able computational times. The authors reported that their proposed algorithm improves

the computational time by up to 50% compared with existing approaches for similar

problems.

Braekers et al. (2014) are extending previous dial-a-ride transit studies by introduc

ing multi-depot heterogeneous dial-a-ride problem (MD-H-DARP). In other words,

their formulation considers different types of

users and vehicles. Two algorithms are

proposed to solve the MD-H-DARP: an exact branch-and-cut algorithm and a deterministic

annealing meta-heuristic. Their branch-and-cut algorithm is based on a similar

algorithm for the standard dial-a-ride problem (DARP) and applied on a compact

two-index formulation using binary variables. Results on several sets of benchmark

instances indicate that the algorithm is characterized by a small computational time.

The authors concluded that based on their findings the branch-and-cut algorithm may

be applied to solve small-size problems and the meta-heuristic algorithm is an efficient

tool for solving large-size problems.

Pan et al. (2014) develop modeling to handle simultaneously the decision of both the

service area of flexible feeder transit service and best routing. They propose a mixed

integer linear programming (MILP) to optimize both the service area and routing.

The transit service is of the type of demand-responsive service dealing with passengers

located in typical Chinese residential areas. The proposed model is based on a bi-level

math programming with an upper level to maximize the number of served passengers

by the feeder transit system and a lower level to minimize the operational cost for

transit operators. The authors use a case study to examine their modeling and selected

the network of the hi-tech district in Jinan

in China. The analyses conducted use the

tools of the CPLEX software and then a meta-heuristic method to show that the total

cost of the objective-function differs by 2 % between the optimization and heuristic

treatments.

Finally Yu et al. (2014) in their recent report attempted to optimize flexible feeder dedi

cated transit from bus stops to rail stations and vice versa. They call their designed system

circular service. That is, the authors formulated an optimization problem to determine the

feeder -service routing and the stop sequence to collect passengers, based on real-time pas

senger demand. The objective function is to minimize the total tour cost incurred by passen

gers and operators including the minimization of the walk time of each passenger . A bi-level

nonlinear mixed integer programming model was constructed and a tabu search method was

implemented with different local search strategies and neighborhood evaluation methods.

They introduced a sort of case study in Austin, Texas, comparing the cost performance of

three different situations: utilizing two vehicles, three vehicles, and considering fixed stops. It

is concluded that a penalty value should be set high for passengers who may miss the train,

and that forcing the route to go via more bus stops is linked directly with the budget avail

able to run the service.

Characteristics of the models reviewed are summarized in Table 17.4.

Table 17.4 Summary of models reviewed, with emphasis on fixed-route models

Sources	Model formulation	Decision variables	Street network structure	Feeding system	System being fed	Special demand features
Wirasinghe						
(1980)	Analytical model	Feeder spacing and headways.				
	Locations of feeding points—optional	Simplistic	Parallel routes	One rail route		
Kuah and						
Perl (1987)	Programming problem	Feeder itineraries and headways	Realistic	Any routes	Several rail routes	Many-to-one and many-to-many
Kuah and						
Perl (1988)	Analytical model	Feeder spacing, headways, and stop spacing	Simplistic	Parallel routes	One rail route	
Chang and						
Schonfeld						
(1991)	Analytical model	Feeder spacing and day-period-specific headways	Simplistic	Parallel routes	One transfer station	Multipleperiod elastic demand
Chang and						
Lee (1993);						
Chang and						
Yu (1996)	Analytical model	Feeder spacing, headways, and fares	Simplistic	Parallel routes	One rail route	Elastic demand
Chang and						
Schonfeld						
(1993)	Analytical model	Feeder spacing, length, and day-period-specific headways	Simplistic	Parallel routes		

One transfer station Multipleperiod

Chien and

Schonfeld

(1998) Iterative algorithm Rail line length, location of feeding points, feeder spacing and headways, feeder stop spacing Simplistic Parallel routes One rail route Many-to-many

Chien

(2000) Iterative algorithm Feeder itineraries and headways (the user determines the number of routes) Semirealistic (partial grid) Any routes One transfer station

Chien et al.

(2001a) Programming problem Feeder itinerary and headway Semirealistic (partial grid) One feeder route One transfer station

Chien et al.

(2001b) Analytical model Feeder spacing, headways, and vehicle size Simplistic Parallel routes One transfer station Multipleperiod probabilistic demand

Shrivastava

and

Dhingra

(2001) Iterative algorithm Feeder itineraries Realistic Any routes Several transfer stations Many-to-many (Continued)

EXERCISE

17.1 The following street network connected to a train station is given. Each link direction (there are two directions of travel per link, marked by arrows) is characterized by a pair of numbers: average travel time (in minutes, on the left) and demand potential ratio (in units of Equation 16.3, on the right). The maximum circulating travel time is given as $T = 30$ min;

in addition, $\omega = 1.0$, $\lambda = 1.0$, $p = 2$, and $P = 2$.

Table 17.4 (Continued) Summary of models reviewed, with emphasis on fixed-route models

Sources	Model formulation	Decision variables	Street network structure	Feeding system	System being fed	Special demand features
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Shrivastava

and

Dhingra

(2002) Programming problem Railway and feeder schedule
Realistic Any routes One/ several transfer stations

Zhao and

Dessouky

(2008) Optimization problem Maximizing potential demand
and minimizing walking distances Realistic Any routes
One transfer station

Quadrioglio

and Li

(2009) Analytical model Demand responsive policy that
might be converted to a traditional fixed-route policy
for higher demand Realistic Any routes One transfer
station Critical demand based on utility function

Ciaffaglia et al.

(2012) Heuristic and genetic algorithms Transit routes
Realistic Any routes Several transfer stations Fixed
demand

Kim and

Schonfeld

(2014) Analytical model Service type, vehicle size,
number of zones per region, headway, and fleet size
Realistic Any routes One transfer station Timed
transfer

Dikas and

Minis

(2014) Optimization problem Riding requests, travel time, routing Realistic Any routes One transfer station

Braekers

et al.

(2014) Optimization problem Routing, response time, type of vehicle Realistic Any routes Several transfer stations

Pan et al.

(2014) Optimization and metaheuristic Service area, feeder-transit routing Typical Chinese residential areas Any routes One transfer station

Yu et al.

(2014) Optimization and metaheuristic Locations of rail stations, Fixed bus stops, Fixed cost for routing Realistic Any routes One transfer rail station

(a) Create the matrix of the impedance ratio z_{ij} .

(b) Assume that the initial group of the highest potential demand units contains 5 links (out of 22 links, both directions) of the given network. Find the best single route that will maximize potential demand while complying with the constraint $T = 5, 43, 23, 16, 75, 34, 28, 21, 6, 42, 17, 14, 34, 24, 24, 64, 25, 34, 25, 24, 33, 34, 45, 54, 35, 29, 24, 5$ Demand potential units Average travel time (min)
Train station

Borndorfer, R., Grotschel, M., Klostemeier, F., and Kuttner, C. (1999). Telebus Berlin: Vehicle scheduling in a dial-a-ride system. In Computer-Aided Transit Scheduling. Lectures Notes in Economics and Mathematical Systems, vol. 471 (N.H.W. Wilson, ed.), pp. 391-422, Springer-Verlag, Berlin, Germany.

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18 Chapter 18: Service Reliability

Transit service reliability problems are very real and have had significant impact on transit efficiency

and productivity. Transit reliability can be defined as dependability in terms of time (waiting and riding), passenger load, vehicle quality, safety, amenities, and information. There is never

a good uncertainty or a bad certainty in transit operations. The question arises as to whether

remedies exist to service reliability problems and, if so, whether they are implementable and

exhaustive. This chapter will attempt to answer this question through a disaggregate perspective

of individual elements causing unreliable service. The chapter contains six main parts, following an introductory section with examples of

passenger complaints as reported in the media. Section 18.2 discusses measures of reliability,

sources, and indicators of unreliable service and the bus-bunching phenomenon. Section 18.3

provides the variables affecting service reliability and models for estimating travel time and dwell

time. Section 18.4 describes and analyzes statistical models for passenger waiting time at transit

stops. Section 18.5 focuses on introducing innovations into an advanced public transit system—

for example, automatic vehicle locators and automatic passenger counters—their objective being

to collect and control reliability-based data. Section 18.6 furnishes an overview of techniques to

improve reliability problems; it covers the subjects of better planning, online control strategies,

and vehicle-priority schemes. Section 18.7 reviews work done on three themes: measures of reliability

ability, passenger waiting time, and control strategies. The chapter ends with exercises. Practitioners may skip Section 18.4 as they read through the chapter. We argue that transit

agencies usually assess measures of system performance that do not reflect what individual pas

senger perceives, thus warranting measurements at the disaggregate level in order to capture

passengers' perceptions. We make the following suggestion to passengers; the story is illustrative: Never let a bus driver know you're in a hurry; it will work against you. This brings us to the

story. A priest and a bus driver arrive in heaven. The angel Gabriel approaches them and asks

them to follow him so he can show them their new houses. They come to a nice neighborhood

with villas. Gabriel pulls out a key and gives it to the bus driver: "This is your home," the angel

says, pointing to one of the villas. Gabriel continues to walk with the priest and, after a while,

they arrive at a poor neighborhood, with small houses that are almost tottering. The angel pulls

out a key and gives it to the priest: "This is your home," he says, pointing to one of the small

houses. The priest, looking completely puzzled, asks: "How come that is for me? After serving

God all my life, you give me such a house and you give the bus driver a villa?" Gabriel replies:

"Because when you're praying, everyone falls asleep, when he is driving, everyone prays."

18.1 INTRODUCTION

Service reliability in transit operations has been receiving increasing attention as agencies

become faced more and more with the i

immediate problem of providing credible service

while attempting to reduce operating costs. Unreliable service has been cited as the major

deterrent to existing and potential passengers. For example, Balcombe et al. (2004), in a

U.K. practical-transit guide, report that passengers' perception of local bus services is inter

preted in the following ranking of importance in weights (given here in parentheses) that

add up to 100: reliability (34), frequency (17), vehicles (14), driver behavior (12), routes (11),

fares (7), and information (5). In other words, for instance, it is twice as important from a

passenger's perspective to improve reliability as to increase the frequency of service.

There are many humorous sayings about transit reliability that represent part of the pas

sengers' accumulated frustration on this subject. Some of these sayings are as follows:

1. Passengers waiting long give a look you could have poured on a waffle.
2. Bus and men are alike—both aren't there when you need them.
3. To go nowhere, follow the crowd or stay on the bus (stuck in a traffic jam).
4. Ad on a bus: "Love is like a bus—if you missed one, another will come along, " to which one can add: like in love, you don't know when it will come and whether it will have room for you.
5. When the bus you are on is late, the bus you want to transfer to is on time (one of Murphy's laws).

Along the line of comprehending the magnitude of service reliability problems in regard

to transit demand, it is also interesting to read passengers' complaints. The following are

typical and have appeared in the media; they remind us what Alfred North Whitehead said:

"We think in generalities, we live in details."

Misleading frequency claim

From the New Zealand Herald, May 16, 2006, in an article by Claire Trevett ("Link

bus adverts 'misleading'). Stagecoach's claim that its Link bus service runs every 10 min has been ruled misleading after a passenger complained of sometimes waiting for up to 40 min ... Of the claim that the buses run "every 10 min," the board said: "There was a passenger expectation that a bus would be available every 10 min on the route and, as that had not been the case in the consumer's experience, the Complaints Board ruled that the website advertisement was misleading. ..." Jean McGeorge, a law student in her 30s: "They are really erratic. You'll get two buses at the same time and then nothing for 20 min and that's incredibly frustrating."

Missed connections

From the San Diego Union-Tribune, March 2, 2006, in a Letter to the Editor ("Transit

schedules are a nightmare"). In addition to the common complaints about public transit not being on time and going

where people want to go, the biggest problem I have had is in making connections.

Example 1: I am in class at Cajamarca College until 8:50 p.m., but the No. 858 bus

departs at 8:44 p.m., leaving me to wait until 10:02 p.m. As a result, I don't get home

until 11:30 p.m., then have to leave again

at 6:30 a.m. to beat my office in time
for work.

Example 2: I often have meetings in University Heights
and transfer from the Green

Line trolley to the No. 11 bus at San Diego State University. The No. 11 leaves at

about the same minute that the trolley arrives,
with no time to get upstairs to catch it.

Example 3: I can't begin to count how many times
I've seen the No. 855 bus leaving

Grossmont Transit Center right as the trolley
arrives. The point is that the schedules

don't seem to take each other into consideration.
And just for fun, try to get from

Santee Town Center to Scripps La Jolla. Scott Weselis,
Santee

Lack of amenities

From BBC.co.uk on October 28, 2005, by a participant Gerry
Chatham. Train announcements are rare. Customer
services are a joke. Trains always packed -

lucky to get a seat on some of the newer
trains and if you do - you're at risk of
catching DVT because of lack of leg room. Three out of
five morning trains this week were either
delayed or cancelled. Some (not all) staff are
rude/unhelpful; lack of staff at train
stations. Inability to purchase a ticket on a
Monday morning as tickets are now like
hot tuns. No visible security presence at
stations late at night. Dirty stations; ex-
pensive fares. ... It sounds like I am being
very negative - I have good reason. Kent is a
lovely place to live. It would be excellent if
we, the customers, were offered value for
money, clean trains that run on time, and
overall a train company that puts the customers
first.

their list of priorities. Just look at the comments left by everyone else here, then make

up your own mind.

Bus bunching

From the book by Larson and Odoni (1981), taken from a Sunday edition of the

Washington Post.

I have long been trying to discover why Metro buses on Wisconsin Avenue are

regularly bunched during rush hours instead of being evenly spaced. It is a common

experience to wait for 10 or 15 minutes in rain or snow, and then find three or four

buses coming along nose to tail.

As Metro refuses to reply to letters on this subject, I can only assume that it has

something to hide.

From the passenger's point of view, the advantages of even spacing are obvious -

shorter waits and buses that are evenly filled, instead of being packed up at the front

and half-empty behind. Traffic congestion is also eased.

Evidently, either the company or its drivers must prefer bunching. I continue to wonder

why. E. Peter Wright

The aforementioned citations cover four service reliability problems from a broader list:

misleading timetables, missed connections, lack of amenities, and bus bunching. In a general

sense, a service reliability problem can be defined as a degradation to system performance

because of (1) uncertainties in the operating environment, (2) lack of suitable data for ef

cient operations planning, (3) improper service design, (4) improper service monitoring and

control, and (5) failures in executing designed schedules. This chapter attempts to highlight

the main ingredients of transit service reliability and provide some insight into possible rem

edies to alleviate problems.

18.2 MEASURES OF RELIABILITY AND SOURCES OF UNRELIABLE SERVICE

Prior to analyzing the elements of transit reliability, we ought to view acceptable measures of

both reliability and perceived reliability and the sources of unreliable service. Abkowitz et al.

(1978) defined service reliability as the invariability of service attributes that influence the

decisions of travelers and transit providers. This section, too, focuses its examination on the

variability of attributes that are of concern to passengers and agencies.

18.2.1 Attributes and measures of reliability

Transit-related attributes that vary by time or space may be distributed. Therefore, the (statis

tical) characteristics of the distributions form the base for constructing measures of reliability.

Measures such as mean (average) value and variance, coefficient of variation (ratio of standard

deviation to mean), and percentage of observations for a value greater than the mean

can

represent the compactness and skewness measures of an attribute's distribution. The following

are three lists of attributes: from the passenger's and agency's perspectives and exogenous

attributes, all of which tend to vary by time of day, day of week, week of season, and space.

Reliability attributes of concern to passengers

- Waiting time
- Boarding time
- Seat availability
- In-vehicle time
- Alighting time
- Total travel time
- Transfer time
- Missed connections
- Pretrip information time
- Pretrip time required for changes in access path

Reliability attributes of concern to the agency

- Dispatching according to schedule adherence
- On-route schedule adherence
- Headway distribution
- Individual-vehicle headway
- Load-count distribution
- Individual-vehicle load count
- On-time pullout

- Missed trips
- Breakdowns
- Late (crew) report (arrival)
- Driver proficiency
- Dispatcher and street-inspector proficiency

Reliability exogenous attributes

- Traffic delays
- Road and other accidents
- Weather

Figures 18.1 and 18.2 illustrate the main aforementioned attributes, the exogenous attri

butes being associated with the agency. Figure 18.1 shows the process in which the attributes

of concern to passengers take place; if each attribute's probability of complying with passen

gers' expectation is high, then the service is perceived as reliable. Similarly, if the probabil

ity that a given run will comply with the agency's expectation is high for each attribute in Pre-trip time required for changes in access path Alighting time Transfer time and comfort Ride comfort Successful transfer Boarding time Successful boarding Seat availability In-vehicle time Total travel time Waiting time (on-time arrival at stop) Pre-trip information time Transit stop

Figure 18.1 Flow diagram of reliability attributes of concern to the passenger; sources of unreliable service are characterized by high variance. On-time pullout On-route schedule adherence Passenger load Headway distribution Dwell time Driver proficiency Successful connection Administrative attributes: Transit vehicle Missed trips Breakdown Exogenous attributes: Traffic delays Accidents (road and other) Weather

Figure 18.2 Flow diagram of reliability attributes of concern to the agency; sources of unreliable service are characterized by high variance.

Figure 18.2, then the agency may perceive the service as reliable. High probability coincides

with small variance for each attribute.

Conceptually, the importance of attributes being negatively or positively perceived by individuals

on reliable pretrip information time than will infrequent users. In addition, there are quality

based attributes that are difficult to quantify. These attributes especially concern passengers' sat

isfaction. Figure 18.1 presents ride comfort as one of these attributes; there are also expectations

in regard to the existence and proper functioning of on and off-vehicle amenities.

Concerning reliability-related standard measures, Figure 1.4 contains several standard

items, their common criteria, and corresponding remarks as follows.

Standard item	Typical criterion range	Remarks
---------------	-------------------------	---------

Schedule adherence	Min 80% on time (0-5 min behind schedule) at peak period, 90% otherwise	This guideline is usually relaxed for short headways.
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Timed transfer	Max of 3-8 min vehicle wait at transfer point	Used more in smaller agencies.
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Missed trips	Min 90%-95% of scheduled trips are OK	Missed trips may also not comply with trip-reliability criterion.
--------------	---------------------------------------	---

Passenger safety	Max 6-10 passenger accidents per 10 ⁶ passengers; max 4-8 accidents per 1.6 • 10 ⁵ vehicle-km	Depends on updated safety data.
------------------	---	---------------------------------

Public complaints	Limits on # of complaints per driver/pass./time period	Public comments and complaints always received.
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a Reflects data mainly from the United States.

b Standards commonly in use.

These five standards are only examples, and more can be established from the lists of attributes;

their criteria may vary by time, space, and agency.

Another theme affecting the reliability of system performance is maintenance reliability.

Examples of maintenance-based reliability attributes are the following:

- Missed trips
- Kilometers and hours of service lost because of road calls
- Number of vehicles not available for service
- Number of late starts
- Number of vehicles overdue for inspection
- Number of operator-trouble reports
- Absenteeism

It is important to analyze these attributes historically and to disaggregate information by

type of service and equipment.

18.2.2 Sources of unreliable service

Transit reliability problems can stem from a number of service factors. Abkowitz et al.

(1978) divided these factors into two groups: environmental and inherent. The first group

includes such factors as traffic signals, changes in traffic conditions, variation in demand,

and availability of crew and vehicles on any given route and day. The second group includes

setting and distribution of waiting times, headways, travel times, boarding and alighting

times, and transfer times on any given route and day.

Some causes of unreliable service are chronic and known in advance; suitable planning and

adjustments can address these causes. Other causes are unpredictable in nature and require

real-time responses, preferably taken from a library of options. The sources of unreliable

service may be found in the following lists of indicators of developing reliability problems.

Planning indicators of developing reliability problems

The items in this list, unlike those in the other lists, do not always/necessarily create reliability

problems, but can serve as diagnostic indicators:

- Long urban route
- Network of routes requiring many individual origin-destination (O-D) transfers
- Lack of feeder services
- Short spacing between stops
- Problematic stop location
- Single daily average running (travel) time
- Sticking, in principle, to even headways
- Different forms of fare payment
- Poor in- and off-vehicle amenities
- Poor security

Operational indicators of developing reliability problems

- Missed trips

- Late pullouts
- Poor on-time dispatching
- Significantly late/early trips
- Bunching
- Uneven loads
- Overloaded vehicles
- Unpredictable passenger demand
- Missed transfers
- High variance of scheduled headway and/or running (travel) time
- Insufficient/extended layover time
- Absenteeism
- Road calls
- Breakdowns
- Passenger complaints
- Dispatcher complaints
- Driver complaints
- Bad press
- Road and other accidents

Maintenance indicators of developing reliability problems

- Lack of vehicles available for service
- Long hours/kilometers between preventive maintenance activities
- Large number of vehicles overdue for inspection
- Large percentage of old vehicles
- Inadequate replacement policies and contingency plans

- Poor-quality level of spare parts
- Intensive vehicle use
- Poor vehicle design
- Inadequacy of maintenance facilities
- Lack of data on vehicle histories and maintenance effectiveness
- Improper maintenance-engineering staff

Troubleshooting in all the earlier mentioned three themes in terms of suitable recurrent

analysis or even in the form of a checklist can eliminate the need for diagnoses and improve

service reliability.

Finally, we will describe one of the most irritating phenomena in urban bus operations,

called bunching or pairing (see passenger complaints in Section 18.1). Undoubtedly, bus

bunching is a major cause of unreliable service. Figure 18.3 describes three possible sources

of the bunching phenomenon: (1) delayed vehicle, (2) early-dispatched and/or speeding vehicle. Scheduled trajectory (second vehicle) Actual trajectory
 Stop 2 Stop 1 Departure point Departure point Space Stop
 2 Stop 1 Stop 2 Stop 1 Departure point Delayed vehicle
 Bunching (pairing) 1 2 3 Vehicles (at departure times)
 Time From passenger overboard Early-dispatched speeding
 vehicle (driver behavior of third vehicle) 1 2 3 1 2 3
 Average speed Time Time (a) (b) (c)

Figure 18.3 Three typical processes creating a pair of vehicles (at stop 2), notwithstanding scheduled even headways and random passenger arrivals: (a) with second vehicle delayed because of traffic, (b) with third vehicle dispatched early and speeding up because of driver behavior, and (c) with extra demand at stop 1 for the second vehicle.

vehicle, and (3) unexpected passenger overflow. Each of these sources or some combination

of them can lead to the creation of two or more buses arriving nose to tail.

The base of Figure 18.3 consists of an even-headway timetable and random passenger

arrivals, from which three vehicle trajectories are shown on a time-space diagram; therefore,

the slope of each trajectory between two adjacent stops constitutes the average vehicle's

speed. In all three cases described, bunching occurs between the second and third vehicles at

stop 2. Figure 18.3a illustrates the case in which traffic conditions cause the second vehicle

to slow down before stop 1. This slowing down becomes more pronounced at each successive

stop, as more and more passengers accumulate after the departure of the first vehicle.

Meanwhile, the third vehicle finds fewer and fewer passengers waiting (shortened headways

between the second and third vehicles), and eventually, it comes together with the second

vehicle, in our case at stop 2. Figure 18.3b presents a scenario in which the driver of the third

vehicle is in a rush, departing early and speeding up. In Figure 18.3c, the second vehicle con

fronts passenger overflow at stop 1, which forces it to extend its dwell time at that stop. Thus,

this vehicle departs stop 1 late as is the case in Figure 18.3a. These three cases, in addition to

creating bunching, result in imbalanced loading on the second and third vehicles.

18.3 MODELING OF RELIABILITY VARIABLES

Following the diagnosis of the sources and causes of the development of reliability problems, this

and the next sections will present models and methods covering key aspects of service reliability.

These key aspects are time-related issues that result from the operating environment. This sec

tion provides the basic notation of the variables, the common formula for calculating dwell time,

and an analysis of travel time. The next section presents an analysis of passenger waiting time at

the stops. Table 18.1 lists the main variables and parameters of the analysis to follow.

The prediction of travel time, thus the arrival time at stops, consists of this information:

the running time R_{ik} and the dwell time D_{ik} . The variation measure of the running time can

be defined as
$$r_{kk} = \frac{\text{Var } R_{ik}}{R_{ik}^2} = \frac{\text{Var } R_{ik}}{R_{ik}^2} = \frac{\text{Cov } R_{ik} R_{ik}}{R_{ik}^2} = \frac{1}{R_{ik}^2} \left[\sum_{i=1}^k \left(R_{ik} - \bar{R}_{ik} \right)^2 \right] \quad (18.1)$$

Table 18.1 Notation and definition of reliability-based variables and parameters

Notation Definition

P_{ik} Passenger load onboard trip i , upon departure from stop k

λ_{ik} Average passenger-arrival rate for trip i , at stop k

H_{ik} Headway between trips i and $i - 1$, upon departure from stop k

R_{ik} Running (travel) time of the vehicle serving trip i between departure time from stop $k - 1$ and arrival time at stop k

r_{ik} Variance-based variation measure of R_{ik}

D_{ik} Dwell time of the vehicle serving trip i , at stop k , including the time required for acceleration and deceleration ($D_{ik} = 0$ if it does not stop at k)

b Dead time portion of the dwell time, including the time required for acceleration and deceleration ($b = 0$ if no stop at k)

B_{ik} Number of passengers boarding the vehicle serving trip i , at stop k

A_{ik} Number of passengers alighting from the vehicle serving trip i , at stop k

ΔB Marginal dwell time per boarding passenger

ΔA Marginal dwell time per alighting passenger

where $\text{Var}[\dots]$ and $\text{Cov}[\dots]$ designate the variance and covariance of the random variable in

brackets. If R_{ik} and $R_{(i-1)k}$ are uncorrelated, r_{ik} is just the variance of the running time; r_{ik} is

also the covariance between $(R_{(i+1)k} - R_{ik})$ and $(R_{ik} - R_{(i-1)k})$.

The correlation expressed in Equation 18.1, which exists in real life, between the

running times of two consecutive vehicles can be considered in an analytical or simulation

models.

18.3.1 Dwell time

The dwell time depends on the number of boarding and alighting passengers. Usually, it can

be expressed by the following linear model: $D_{ik} = b + B_{ik} \Delta B + A_{ik} \Delta A$,
 single-door vehicle, if B or $A \neq 0$, since $B_{ik} \Delta B + A_{ik} \Delta A$
 $k = + \cdot + \cdot > \Delta \Delta$ single-door vehicle, if $B = A = 0$,
 double-door vehicle, if $B \neq 0$ and $A \neq 0$,
 $k = + \cdot + \cdot \max(\Delta B, \Delta A)$ double-door vehicle,
 if $B \neq 0$ and $A = 0$,
 $k = + \cdot + \cdot \max(\Delta B, \Delta A)$ double-door vehicle, if $B = 0$ and $A \neq 0$,
 $k = + \cdot + \cdot \max(\Delta B, \Delta A)$ (18.2)

where the dead time b is made up of the time to open and close the door(s), the time taken

between sequential alighting and boarding events on single-door vehicles, and the time

taken by the driver to check the traffic: it includes, by convention, the penalty for stopping

because of the consequent deceleration and acceleration.

Equation 18.2 assumes that boarding and alighting passenger flows are distinct for

double-door vehicles. For a given number of boarding and alighting passengers, we can

assume that there is no intrinsic randomness. If there is (e.g., different groups of passengers

carrying different size of luggage), its variance can be assumed to be fixed and thus added to

the running-time variation in Equation 18.1.

The following factors influence dwell time:

- Form of payment
- Entrance characteristics
- Conflicts between boarding and alighting passengers
- Available space near drivers
- Traffic-flow characteristics
- Passenger characteristics

The following are practical (rounded) values of the marginal dwell time per boarding and

alighting passenger; these and the dead-time component of Equation 18.2 were extracted

from a number of studies, especially in the United Kingdom (Balcombe et al., 2004).

Boarding time per passenger

$\Delta B = 1.5$ s—pay conductor

$\Delta B = 2.5$ s—lat fare, with change

$\Delta B = 3.0$ s—lat fare, with change

$\Delta B = 5.0$ s—with change

$\Delta B = 6.5$ s—automated fare box

Alighting times per passenger

$\Delta A = 1.5$ s—without hand baggage and parcel

$\Delta A = 3.0$ s—moderate hand baggage

$\Delta A = 5.0$ s—considerable baggage

Dead time per stop

$b = 2.0$ s—pay conductor

$b = 5.5$ s—otherwise, but with extra space near driver

$b = 7.0$ s—without extra space

18.3.2 Analysis of travel time

From a reliability standpoint, one of the more crucial input elements in the planning process

is vehicle travel (running) time. Often the term travel time is broken down into the following

string: walk from origin, wait for vehicle, ride in vehicle, and walk to destination. Here, we

will apply this term only to the riding or running time and exclude layover (recovery) time.

The majority of the agencies do not build slack time into the scheduled travel time to

compensate for the variability of vehicle travel time; average travel time is then utilized for all

scheduling tasks. For convenience, let R_i be the travel time of the vehicle serving trip i across

the entire transit route. This travel time depends on the trip time (hour, day, week, season),

number of passengers, as well as on the habits of each individual driver.

Jeong and Rilett (2005) reviewed prediction models for travel time, including historical

cal data-based models, regression models, time series models, and neural network models.

They selected artificial neural network models to predict R_i , given real-time information

on traffic congestion and transit-vehicle dwell times at stops. The researchers found that,

given the availability of real-time data, their model outperformed both the historical data

based model and the regression model in terms of prediction accuracy. The increasing use of

advanced public transit/transportation systems (APTS) (see Section 18.5) by transit agencies

allows for better predictions of both R_i and R_{ik} in real time; this undoubtedly can be used

for reducing passenger uncertainties at stops and on the vehicles and eventually for improving

service reliability.

While Sections 18.5 and 18.6 focus on new technologies and techniques to improve

real-time operation, the present section provides a practical, statistical-based analysis for

deploying R_i at the planning level. It is based on Ceder (1981), which utilized four major

procedures: (1) exclusion of outliers, (2) division of the day into intervals, (3) union of inter

vals, and (4) union of days for weekly and seasonal cross sections.

18.3.2.1 Description of general method

The main objective of data processing is to create a database on vehicle travel times for different

cross sections of the day, week, and season (d-w-s). From a statistical viewpoint, the

existence of a large data bank that can be systematically updated means that an analysis of

variance (concerned with differences between means of groups) can be carried out (based

on the normality distribution assumption) in order to estimate the effects of d-w-s (the

independent variables) on the dependent variable, R_i . Afterward, the independent variables

that were found to affect the dependent variable (at a desired level of significance) can be

analyzed with the use of contrasts (comparison of the differences between pairs or combinations

of means). Finally, it is possible to apply a multiple variable regression analysis.

In order to effectively carry out an analysis of variance on d-w-s, a data bank must be

accumulated over a yearly period, on the assumption that there will be no physical changes

along the transit route or in the location or in quantity of stops. Because it is well known

that transit routes are liable to physical changes in a dynamic fashion, it is questionable

whether the data bank for R_i values can rely on a yearly base. Moreover, it is desirable that

the process that determines the R_i relationship allows for the intermediate involvement of

those responsible for data collection. This involvement can be expressed by the identity

tion of outliers, whose cause is known, and by the practical decisions relating to changes in

statistical criteria, which are combined in the process, for different transit routes.

As a result, a number of criteria can be utilized for characterizing outliers; dividing the

day into time used; dividing the week into homogeneous days; and dividing the year into

seasons. If no physical changes occur for a given transit route over a yearly period and a

data bank for R_i values exists, there is a possibility of carrying out an analysis of variance

and contrasts for d-w-s. Once again, it should be noted that the objective is not to build a

statistical model for simulation or control, but to construct a value system for R_i that can be

of greater aid in realistic planning than can a single mean value over a specified period of

time (e.g., year as some transit agencies do).

A possible method of analyzing R_i consists of four main components: (1) exclusion of out

liers (OUTLIER procedure), (2) division of day into intervals (INTERVAL procedure), (3)

union of intervals (UNIT procedure), and (4) union of days for weekly and seasonal cross

sections (UWEEKDAY-USEAS procedure).

The first three components relate to the data on a daily basis, and the fourth component

serves as a tool for determining the division of weeks and seasons. The procedure starts with

assembling data into two sets: (1) general data, which include the number of vehicles in the

set, route number, origin, destination, the day (or days) of the week, and the season, and (2)

specific data, which include vehicle departure time, travel time, and type of vehicle.

The outliers are deleted from these data, and then a number of methods are tested for

dividing the day into intervals (division for each hour, each 2 h, each 3 h, and one daily

average). Then statistical tests are conducted to determine the possibility of unifying the

intervals. The procedure continues for the series of days, and then the possibility of unify

ing the days is examined. The next step checks the possibility of unifying a number of days

from different seasons; this is possible if the division of days in each season (or at least those

chosen for unification) is similar. More details are shown in Ceder (1981).

18.3.2.2 Example of bus data set

For a given bus route (in Haifa, Israel) departing at a frequency of either every 15 or 30 min,

data were collected for 2 days (on the same day of the week for 2 weeks) at a total of 126

data points. The data points are shown in Figure 18.4, as well as the results obtained after

the INTERVAL and UNIT components. In the OUTLIER procedure, five outliers were

found and are marked on the upper portion of Figure 18.4; they also appear at the start of

Table 18.2. The latter exhibits a computer record.

Following a run of the INTERVAL procedure, the method designated C in Table 18.2

was chosen, that is, a division of the day into 3 h intervals, when the first hour is considered

separately as shown in the upper portion of Figure 18.4 and in Table 18.2. The record in

Table 18.2 is accompanied by F values, where F Found variation of the group averages Expected variation of the group averages . Outlier 60 50 40 30 20 10 0 After INTERVAL Time 60 50 40 30 20 10 0 After UNIT 5 00 7 00 9 00 11 00 13 00 15 00 17 00 19 00 21 00 23 00 Travel time (min) Travel time (min) (b) (a)

Figure 18.4 An example of travel time data points for one bus line with mean values determined by the INTERVAL procedure in (a) and by the UNIT procedure in (b). Table 18.2 Example of data analysis using outlier, interval, and unit for a given bus route; summary of computer record Stage Description Input 126 data points (see Figure 17.4) Outlier 1. Departure 7:45, travel time = 62 min 2. Departure 8:30, travel time = 55 min 3. Departure 15:15, travel time = 53 min 4. Departure 21:00, travel time = 44 min 5. Departure 23:00, travel time = 49 min Interval Method A: division by 1 h Method B: division by 2 h periods Method C: division by 3 h periods Method D: single average for the whole day Steps of comparisons: (A vs. D) → (A vs. C) → (A vs. B), until significant F difference found at the 0.05 level 1. A vs. D: F-calculated = 1.92, F-table = 1.35; conclusion: continue 2. A vs. D: F-calculated = 1.15, F-table = 1.35; conclusion: accept Unit Same-means hypothesis is based on t-test at the 0.05 level Integer interval Actual interval Mean value (min) Standard deviation (min) 5:00 < R i ≤ 6:00 5:00-6:00 29.4 3.5 6:00 < R i ≤ 9:00 6:15-9:00 38.2 8.1 9:00 < R i ≤ 15:00 9:30-15:00 30.1 6.1 15:00 < R i ≤ 18:00 15:15-18:00 44.8 4.5 18:00 < R i ≤ 24:00 5:00-6:00 36.9 7.0

The data then proceed to the UNIT procedure. Of the seven intervals in method C, two are

united and only five intervals remain as shown in the lower portion of Figure 18.4 and in

Table 18.2. This example does not include the

U WEEKDAY-USEAS procedure.

The last five intervals (each with a mean and standard deviation) are transmitted to the

bus company's scheduling department. This information can be transmitted either auto

matically or manually. Determination of the R_i value for planning purposes will depend on

the degree of certainty that the bus will arrive at its destination on time. Table 18.2 contains

both the mean value of R_i and its standard deviation for two possible objectives: (1) to utilize

the standard deviation for introducing slack time to account for the variability of R_i and (2)

to allow the scheduler more flexibility in creating/adjusting blocks, which in certain cases

are already performed by a specific driver.

18.4 PASSENGER WAITING TIME AT A STOP

This section attempts to clarify some of the probabilistic issues of passenger behavior at

transit stops. The section consists of two parts. First, the distribution of waiting times is

analyzed for the case of random passenger arrivals. This distribution is described by the

two families of headway distributions, the deterministic and the exponential, discussed

in Section 11.2.2, which can approach the two extremes. Second, the mean waiting-time

formulation is interpreted using observed data from two studies; in addition, two explicit

expressions for the mean waiting time are derived for the case of a suburban rail station. The

majority of this section follows Ceder and Manguier (1985).

18.4.1 Waiting-time distributions

Equation 11.1 presented the basic relationships between the expected (mean) waiting time

$E(w)$ and the mean $E(H)$ and variance $\text{Var}H$ of the time headway. This known equation

for $E(w)$ is based on two assumptions: (1) passengers can always board the first vehicle to

depart (no overloading situations) and (2) the passenger random-arrival rate at the stop is

independent of the vehicle-departure process and constant over the period. The following is

the derivation of Equation 11.1 on a per-route basis, in which $f(H)$ is the probability density

function (PDF) for headway H (from a passenger's perspective, compared to $f_H(t)$, which is

from the system's perspective); $w(H)$ is the mean waiting time per passenger for H , which is

$H/2$; and λ is the mean arrival rate of passengers, from which λH is the number of passen

gers arriving in the course of H minutes: () Total number of passengers arriving at the transit stop H
 $f(H) = \lambda \cdot H \cdot f(H) \quad dH = \infty \int$ Mean waiting time for passengers
 arriving randomly $E(w) = \int_0^\infty w(H) \cdot \lambda \cdot H \cdot f(H) \cdot dH = \int_0^\infty \frac{H}{2} \cdot \lambda \cdot H \cdot f(H) \cdot dH = \frac{\lambda}{2} \int_0^\infty H^2 \cdot f(H) \cdot dH = \frac{\lambda}{2} E(H^2)$
 $E(H^2) = \int_0^\infty H^2 \cdot f(H) \cdot dH = \int_0^\infty H \cdot H \cdot f(H) \cdot dH = \int_0^\infty H \cdot dH = \int_0^\infty H \cdot \lambda \cdot H \cdot f(H) \cdot dH = \lambda \int_0^\infty H^2 \cdot f(H) \cdot dH = \lambda E(H^2)$

The use of $\text{Var}(H) = E(H^2) - E^2(H)$ results in the known mean waiting-time formula shown

in Equation 11.1: $E(w) = E(H) + \frac{\text{Var}(H)}{2E(H)}$

Thus, the mean passenger waiting time is minimized at half the headway when the variance

of the headway is zero.

In practice, assumptions (1) and (2) may not always hold,

which would then argue in

favor of even-load headways, as was analyzed and proposed in Chapter 4, rather than even

headways. For instance, there are surges in the passenger-arrival rate at the start and end of

a factory shift and a school day.

Under the assumption of random passenger arrivals at the stop, the following quality

relates the waiting-time distribution to the headway distribution: let $f_w(t)$ be the waiting

time PDF: $f_w(t) = \int_0^t f_H(u) du + F_H(t) E(H) - F_H(t) E(H) F_H(t) w_H(t) H_H(t)$ (18.3)

where F_H is the frequency of service (vehicle-arrival rate) $F_H(t) = 1 - F_H(t)$ and $F_H(t)$ is the cumulative distribution function of the headway H (system's perspective as in Chapter 11)

The derivation of this relationship, which is shown in Larson and Odoni (1981), is related to

the phenomenon commonly referred to as random incidence.

Particular examples of the application of Equation 18.1 are provided by the deterministic

headway and exponential-headway cases. The following is obtained for the deterministic

headway in which $H = 1/F$: $F_H(t) = 1, t < 0, t > 1/F, H = 1/F$

hence, $f_w(t) = 1/F, t < 0, t > 1/F$ and $E_w = 1/F E(H) = 1/F$

In this case, the waiting time is uniformly distributed between 0 and the fixed headway $1/F$.

For the exponential-headway case, the following is obtained: $f_H(t) = F e^{-Ft}, F_H(t) = 1 - e^{-Ft}, F_H(t) = 1 - e^{-Ft}$

hence $f_w(t) = F e^{-Ft}$

ations, $C^2 = 0$ and $C^2 = 1$, both families give identical curves because these values must

correspond, respectively, to the deterministic-headway and exponential-headway cases.

For $C^2 = 0$, the uniformity of the waiting-time distribution (between 0 and 10) can be

noticed. For $C^2 = 1$, the curve is exponential. For both families, the waiting-time probability

densities are always decreasing. This is a general property, implied by Equation 18.3.

18.4.2 Average waiting time

Consider a transit route in which the vehicle departure times follow the timetable and the

traffic flow is not congested. In such situations, which are characterized by relatively low

(between days) variability, some passengers will attempt to follow the arrival pattern of

vehicles in order to reduce their waiting time.

Jolliffe and Hutchinson (1975) observed buses in suburban London with published time

tables and examined between-day variations. They found no correlations between the

between-day variability of bus-departure times and headways. When they considered the

(between-day) expected (mean, average) waiting time for passengers arriving at the bus stop

at time t , they found that the waiting-time function had a minimum wait, $w_{\min}$, before the

average departure time of each bus, the corresponding t (to $w_{\min}$) being the optimal arrival

time. The authors showed how, based on departure-time variability, it is possible to com

pute $w_{\min}$ theoretically. The observed average waiting time w was found to be higher than 102030 Time (min) 0 0.05 1.0 $C^2 = 0$ $C^2 = 0$ $C^2 = 0$ $C^2 = 0$ $C^2 = 1$ $C^2 = 1$ $C^2 = 0.05$ 1.0 $f_w(t)$ 10 20 30 Time $f_w(t)$ (a) (b)

Figure 18.5 Various waiting-time distributions showing (a) the power-distributed headways and (b) the gamma-distributed headways for a mean headway of 10 min and different values of C^2 .

$w_{\min}$ and lower than the expected value $E(w)$ given by Equation 11.1, indicating that some

passengers do time their arrival. It was also concluded that some passengers never wait—if

they see a bus coming, they rush to board it. Jolliffe and Hutchinson (1975) proposed to

represent the passenger-arrival pattern by considering three proportions: (1) passengers who

arrive coincidentally with a bus (see and rush); (2) a proportion p for passengers who, based

on the timetable and experience, arrive at the optimal time and wait only for $w_{\min}$; and (3)

passengers who arrive at random and wait on average for $E(w)$.

Jolliffe and Hutchinson found the proportion of coincidental arrivals to be relatively low

(16%), similar to what other studies have observed (Ceder and Marguier, 1985). The mean

waiting time for non-coincidental arrivals could then be expressed as follows: $w_p(w) = E(w) - (1-p)w_{\min}$ (18.6)

Jolliffe and Hutchinson used $w_{\min}$ to determine the proportion p and proposed this

relationship: $p e^g = 1 - \alpha$ (18.7)

where $g = E(w) - w_{\min}$ α is a constant (found by them to be 0.131 for the peak and 0.015 for the off-peak period data)

Let us consider a suburban rail station. Assume that the service is very reliable, say $C^2 = 0$,

with posted timetables, and that passengers cannot see coming trains in advance. Using the

earlier approach, which should be appropriate in this case, and inserting Equation 18.7 into

Equation 18.6, one obtains $w_g = \frac{1}{\alpha} E(w) g(1 - e^{-\alpha})$ (18.8)

For this deterministic headway, the expected waiting time (of randomly arriving passengers)

is $E(w) = (E(H))/2$; and Equation 18.8 becomes $w_g = \frac{1}{\alpha} E(H) g(1 - e^{-\alpha})$ (18.9)

Huddart (1973) in London observed, as expected, that for transit services with small head

ways, passengers arrive at random. However, if the service is sufficiently infrequent, pas

sengers may take advantage of this information to arrive at the boarding point just before

the vehicle is due to arrive. Huddart considers this situation to be rare for urban bus services

but fairly common at a suburban railway station.

If we consider the passenger-arrival rate $\lambda(t)$ as monotonically and continuously varying

with time, as shown by the typical passenger-arrival pattern given by Huddart (1973), the

mean waiting time can be expressed as $w = \frac{1}{E(H)} \int_0^\infty \lambda(t) dt$ (18.10)

For example, the data in Huddart (1973) can be fitted into a regression analysis for an

exponential function of the type $\lambda(t) = a \cdot e^{-bt}$, where a, b are fitted parameters. For different

passenger-arrival rates between 3 and 15 min before a train departure, the following model

agencies; they were asked about existing and planned APTS. In addition, the U.S. Federal

Transit Administration issued a report by Hwang et al. (2006) on the state of the art of

APTS, which emphasized the need for integrating APTS with emerging intelligent trans

portation systems (ITS) and ITS trends. Hwang et al. (2006), from lessons learned, stressed

the need for the provision of better data, standards, and voice communication in order to

(1) improve transit planning, maintenance, operations, and incident management and (2)

facilitate coordination, integration, and interoperability with transportation providers and

public safety organizations.

The main features contained in the U.S. survey and reports, some of which appear in

Khattak and Hickman (1998), are assembled according system category in Table 17.3.

Common APTS terms used in this table are AVL (automatic vehicle location), APC (auto

matic passenger counter), CAD (computer-aided dispatch), AFP (automatic fare payment),

and ATIS (advanced traveler information system). Other known terms appearing in the

literature, but not used here, are AVM (automatic vehicle monitoring), AVLC (automatic

vehicle location and communication), and a different CAD (computer-assisted design).

Current practices in transit agencies show that sufficient data seldom exist for both service

operations planning and improving reliability problems. Manual data collection is costly

and, consequently, must be used sparingly. As a result, detailed information on passenger

demand and service characteristics is generally not available at the route level. Without

this information, the efficient deployment of transit service commensurate with demand is

impossible. Thus, a major reason for transit agencies to be interested in the use of APTS data

is the hope of gaining badly needed information at a greatly reduced unit cost. The result

ing data are expected to improve utilization of vehicles and crew, as well as to resolve some

reliability problems.

Useful, automatically collected data are extracted from AVL, APC, CAD, and AFP systems.

Their features appear in Table 18.3. These data sources have the potential to enhance

schedule efficiency and to improve service reliability.

A general program for developing operations planning applications is depicted in Figure 18.6.

The program as a whole intends to serve transit routes with or without AVL/APC/CAD/

AFP data. Its major objectives are (1) to improve management and operations by developing,

improving, validating, and testing models and procedures for transit-operation planning; (2)

to improve levels of service through increased reliability resulting from better control and

response; (3) to improve productivity and efficiency by better matching supply and demand; (4)

to reduce data gathering, processing, and reporting costs; and (5) to develop vital

omponents

for a management information system pertaining to operations and passenger behavior.

Table 18.3 Advanced public transit systems applications

APTS applications System name Features

Monitoring systems AVL Automatic position determination by means of dead reckoning (using the vehicle's odometer and compass), GPS, signposts (transmitted signals picked up by vehicle), ground-based radio, and real-time reporting APC Automatic counts of boarding and alighting passengers (e.g., use of treadle mats or infrared beams placed by the door) Advanced communication Digital radio (binary information) and/or trunked radio (computer selection of frequency) CAD Dispatch software as a support tool Automated operations of software Software that displays vehicle positions, vehicle and agency data, and communications information Silent alarm Emergency signal triggered by the driver; possible hidden microphone for dispatcher/other listeners, onboard surveillance camera Vehicle component monitoring Automatic remote measurement of engine oil pressure, engine temperature, electrical system, tire pressure, etc.

Fare payment

systems AFP Payment by smart card, magnetic stripe card, credit/debit card, etc. Multicarrier fare integration Fare scheme that covers more than one transit service provider

Traveler information

systems ATIS Pretrip, in-terminal, and in-vehicle real-time passenger information Multimodal traveler information Available information covering multiple modes (i.e., transit and traffic or different transit modes)

Multipurpose

information systems Vehicle probe Automatic data from transit vehicles for estimating traffic travel times and speeds and road conditions Mobile data terminal Wireless device that can send and receive information Information sharing Sharing of information on traffic and incidents among agencies Mobility manager Coordination of travel requests and vehicle dispatching (multiple agencies)

Traffic signal control Manual priority Extended
green-activated by driver Automatic priority Automatic
extension of green phase

The applications for AVL/APC/CAD/AFP data in Figure 18.6
are rooted in two streams

of data flows: passenger-load data and vehicle
running-time data. Some of the methods of

handling the running-time data are discussed in
Section 18.3. The use of load data and its

appraisal appeared in Chapters 3 and 4. Moreover,
AVL/APC/CAD/AFP data can improve

route design, schedule adjustments, and vehicle
scheduling. The outcome of best exploitation

of the data is better operational strategies, which are
required to reduce reliability problems.

In an implementation study utilizing AVL and APC
systems, Kimpel et al. (2006) utilized

operations data to improve schedules and reduce
service reliability problems. The data used

were recovered by the TriMet transit agency (Portland, Oregon, United States), using a Load
data Running (travel)-time data Dynamic patterns:
variability of passenger loads and running times
Simulation of routes to improve scheduling procedures
Construction of alternative timetables, including
(overlapping) route-design elements Evaluation and
refinement of strategies for operation Schedule adjustments
(interactive mode) Development of AVL/APC/CAD/AFP monitor
Improving route-level management: evaluation of possible
short-term scheduling changes to substitute for
operational controls Vehicle-scheduling optimization
(generation of blocks): graphic and interactive
techniques AVL/APC/CAD/AFP collected data Automatic
passenger counters: test and evaluation Development and
appraisal of point and ride check methods for setting
frequency Algorithms of setting vehicles departure times
based on balancing passenger loads and smoothing headways
Arrangement of data concerning route characteristics and

passenger behavior Impact of run times on practical
schedule development methods Organization and aggregation
of data Key: Flow of information Interaction between
components

Figure 18.6 Program to develop applications for
AVL/APC/CAD/AFP data.

automated CAD system. At TriMet, 100% of the bus fleet is
equipped with AVL technology.

Today, while approximately 72% of the vehicles are
equipped with APC, TriMet obtains bus

location information at regular time intervals and
transmits it to two dispatch centers in

real time. The data collection component of the
AVL and APC systems includes arrival and

departure times, dwell times, door openings, lift
operations, and maximum speeds since the

previous stop, as well as the number of boarding and
alighting passengers on APC-equipped

vehicles.

The findings of Kimpel et al. (2006) show that the
on-time performance measure (per

centage arriving on time at stops) for 15
frequent-service bus routes rose from 73%-74%

to 75.5%, which is slightly above the agency
standard of 75%. In addition, they claim that

TriMet's scheduling and operations management has benefited
from the analysis of the AVL

and APC data, especially in reports generating
capabilities. The study concluded by recom

mending that efforts to improve schedule
efficiency should center on reducing excess runs

and layover times, as well as employing
supervisory actions to reduce operator
variability.

18.6 TECHNIQUES TO RESOLVE RELIABILITY PROBLEMS

Essentially, there are three basic mechanisms for improving transit reliability: (1) improve

planning and scheduling, (2) improve priority techniques for transit vehicles, and (3) improve

operations control. Abkowitz et al. (1978) classified reliability-improvement strategies as

preventive and corrective/restorative. Preventive strategies are fundamentally associated

with mechanisms (1) and (2) and corrective/restorative strategies, with mechanism (3).

TranSystems et al. (2006), in a study on attracting and retaining transit passengers, suggested

that these mechanisms should address one or more of the following parameters: travel time,

convenience, comfort, perceived personal security/safety, and perceived image of the transit

system. What follows is a description of tactics, strategies, and possible actions in conjunction

with each of the three mechanisms and an extended example of a holding strategy.

18.6.1 Improved planning and scheduling

Basically, all the methods presented in this book are intended to improve transit planning

and scheduling; inherently better reliability measures are incorporated into the resultant

improvement. Examples of specific actions, some of which can be found in TranSystems

et al. (2006), are presented in the following lists, by category.

Area coverage actions

- Increased route coverage; new routes
- Old set of routes replaced by a new set
- Service expansion; new local circulators/shuttles
- New/improved feeder services
- New/improved timed transfers; improved route coordination
- New/improved transit centers

Route restructuring actions

- Interlinings introduced
- Route shortening, extension, realignment, and removal
- Revised operating strategies
- Route splitting
- Joining routes to form a single route
- New/improved zonal, express, and local services
- Reduced number of stops

Scheduling amendment actions

- Increased span of service; longer late-night and weekend service hours
- Increased service frequency
- Average even-load headways introduced
- Adjusted departure times to suit new connections
- Modified running (travel) times by time of day
- Increased layover time
- Deadheading (DH) trips introduced for necessary reinforcement
- Shifts in departure times introduced as a corrective strategy

General system's organization actions

- New/improved geographic, O-D, route, stop, and transfer-point database
- New/improved frequency, timetable, and transfer-time database
- New/improved stop, station, transit center, and park-and-ride amenities
- New/improved passenger amenities
- New/improved real-time service information
- New/improved vehicles
- Increased security and safety

These types of actions are typically implemented in an effort to seek a better, more reliable

transit service.

TranSystems et al. (2006) described examples of successful actions taken in the United

States. Here are two typical examples: (1) Washington Metropolitan Area Transit Authority

extended late-night weekend hours on Metrorail from 02:00 to 03:00 (a.m.) on Saturday

and Sunday mornings. The trial program, which ran for 18 months, was projected to

attract an additional 3,000 passengers; actually performance exceeded these expectations

by 20%, and it was decided to make the extension permanent. (2) Bangor (Maine)

Area Comprehensive Transportation System introduced a new fleet of low-floor buses for

quicker, easier boarding and alighting; this action resulted in increased passenger demand,

which rose by 8% between 2003 and 2004. It is customary in

the United States to have an

approximately 2 00 m distance between bus stops, compared with about 3 20 m in Europe.

In Europe, moreover, one can usually find the name of the stop and updated timetable

information at each bus stop.

18.6.2 Improved priority techniques for transit vehicles

Transit-vehicle priority techniques are extensive, and many bus-priority strategies have

been demonstrated worldwide. Traditionally, priority is granted for transit-vehicle operation

at stops, at intersections, and by preferential/exclusive lanes. Usually there is a trade

off between granting priority to buses and improving traffic flow for other vehicles.

Local authorities (especially those in elected positions) are often reluctant to provide this

priority. In other words, with a drop of a humor, in order to convert a street lane into

strictly a bus lane (thereby increasing bus reliability and reducing traffic congestion, by

helping to switch from automobile to buses), one must first prove that there is no traffic

congestion on that street; this is analogous to the common experience that in order to

secure a loan, one must first show that he/she doesn't need it.

Priority at stops: Transit-vehicle stop locations at intersections should normally be designed

to prefer the far-side location (after crossing the intersection). In this way, the bus will not

block traffic intending to turn; it will have easier pull-in and pull-out maneuvers and will

experience fewer conflicts with pedestrians. However, a near-side stop is more appropriate

at locations where transit vehicles make a turn and where the crossing street is one way.

Other useful strategies are to prohibit parking near stops, to grant pull-out priority (reduc

ing the merging time with the traffic), and to extend the sidewalk at the stop location (elim

inating a pull-out maneuver and easing the boarding process).

Priority at intersections: At intersections, transit priority can be divided into passive and

active schemes. Traffic engineers usually employ passive priority at intersections in order to

utilize four measures: (1) exempt transit vehicles from turning prohibitions so as to facilitate

transit routes, (2) extend the green interval at signalized intersections for nonstopping tran

sit vehicles, (3) divide the green interval into two parts within the same cycle, and (4) provide

preference to streets carrying transit routes through YIELD and STOP signs. Active priority

permits transit vehicles to preempt traffic signals, using in general one or a combination of

the three following procedures: (1) immediate priority upon the arrival of the transit vehicle,

(2) priority dependent on the crossing-street t

traffic queue, and (3) priority granted only to

transit vehicles with late arrivals.

Priority through preferential/exclusive lanes: Preferential treatment to transit vehicles on

street lanes can be categorized according to these three lists:

Type of preferential lane

- Exclusive curb lane
- Semi-exclusive curb lane (shared only with cars about to turn)
- Exclusive median lane (with stop island)
- Exclusive lane in the center of a street
- Transit-vehicle malls (known as bus malls; limited to pedestrians and buses)
- Exclusive freeway/highway lanes
- Ramp bypass (for entering a freeway/highway during traffic congestion)
- Congestion bypass (exclusive lanes to bypass traffic bottlenecks)

Integration with traffic flow

- With-flow lane (by pavement markings and signs; enforcement problems)
- Contra-flow lane (easy enforcement)
- Exclusive lanes

Period of operation

- Single-peak operation
- Two-peak operation
- Permanent operation

Some exclusive lanes for transit vehicles are

shared with high-occupancy vehicles (taxis,

certain minimum number of people in a car, for encouraging carpools).

Another known type of priority involves bus rapid transit (BRT): a flexible, rubber

tire form of rapid transit that combines stations, vehicles, services, priority lanes, and

intelligent-system elements into an integrated system with a unique identity. TransSystems

(2006) described BRT as a system of buses with such features as signal priority, dedicated

right of way, automated and off-vehicle fare collection, automated information systems,

level boarding, modern vehicles, bus shelters with enhanced amenities, and unique graphics

identity (painted on the bus). Two known BRT systems in South America are found in Brazil

and Ecuador:

(a) Curitiba, Brazil: Curitiba runs one of the early BRT systems; the system features multiapplication smart cards (used for transit as well as other applications).

(b) Quito, Ecuador: This BRT system uses controlled-access stations adjacent to the exclusive lanes.

An example of a successful BRT implementation is that run by the Los Angeles County

Metropolitan Transportation Authority, which in 2000 launched the service in several

of Los Angeles' heavy corridors. TransSystem reported that bus ridership in the Wilshire

Whittier and Ventura Blvd. corridors has increased by 20% and 50%, respectively, since the

implementation of BRT; furthermore, up to one-third of BRT passengers had previously not

been transit users.

In Europe, numerous transit-priority projects have been executed; for example, in Athens,

Dublin, Munich, Turin, Vienna, and Zurich. Based on Ceder (2004), lessons can be learned

from these six case studies: Table 18.4 summarizes the benefits gained from the implementa

tion of transit-priority schemes in these cities.

18.6.3 Improved operations control

Fundamentally, there are two distinctive real-time transit-performance disruptions: (1) devi

ations from the schedule (timetable), but not necessarily creating an imbalance between supply

and demand and (2) creation of an imbalance between supply and demand (overloaded

and almost empty vehicles), but not necessarily deviating from the schedule. Given that these

disruptions are known in real time (e.g., by an APTS), corrective and restorative control

strategies can take place.

Table 18.4 Examples of transit priority results in six European cities

City (population)	Benefits gained
-------------------	-----------------

Athens (4 million)	• Reduction of travel time and its variance on bus lanes • 10% increase in patronage on some bus routes
--------------------	---

Vienna (1.6 million)	• Reduction in travel time on bus lanes • Possible (not clear) increase in patronage
----------------------	--

Munich (1.2 million)	• Reduction of 19% in travel time for tram priority at one traffic signal
----------------------	---

Dublin (1 million) • No reduction in travel time, but a reduction in its variance on bus lanes • Reduction in boarding times • Increased patronage with new fare scale • Increase in revenue • Reduction in the variance of travel time and headways using AVL system

Turin (1 million) • Reduction in tram travel time and more reliable tram reliability using AVL (called SIS) system

Zurich (0.36 million) • Increase of 19 km/h in average vehicle speed because of priority at traffic signals, AVL, and passenger-information systems

The main real-time control strategies are shown in the following list:

- Holding the vehicle (at terminal or at mid-route point)
- Skip-stop operation
- Adding a reserve vehicle
- Changes in speed (not above the lawful speed limit)
- Interlining operation
- DH operation
- Short-turn operation
- Shortcut operation
- Leapfrogging operation with the vehicle ahead

The first strategy on the list can be used for improving on-time performance (scheduled

based), eliminating bunching (headway based), and responding to unexpected demand;

however, it has a negative effect on onboard passengers. This holding strategy is further

discussed and analyzed in the next section. The skip-stop operation is meant to pass stops at

which no passenger wishes to alight; however, it has a negative effect on waiting passengers

at those stops. Adding a reserve vehicle as a reinforcement action is suitable in locations

where unexpected demand may appear. Changes in speeds (slowing down or speeding) can

improve on-time performance and eliminate bunching; in the slowing-down cases, it may

be better valued by the onboard passengers than is the holding strategy. Interlining (if com

monly not in the schedule), DH, and short-turning operations can be used as corrective

actions for possible no-show/lateness situations. A real-time shortcut decision to convert

a local service into an express (or semi-express) service between two points is feasible if it

targets the destinations of all onboard passengers. Finally, the leapfrogging operation will serve

to correct a load-imbalance scenario between two following vehicles, as well as ease the

bunching phenomenon.

The main drawback of possible real-time control actions is the lack of prudent model

ing and software that can activate these actions by a automatic/semiautomatic/manual mode.

Figure 18.7 illustrates schematically a computer-aided, real-time control system; such a system

can be employed in a transit-control center to allow for best exploitation of the real-time

information.

Another means of improving transit-operations control can be approached from the pas

senger side. That is, the match between supply and demand can be improved by utilizing

intelligent real-time passenger-information systems. Pretrip information has the potential

to change demand by time of day and to direct the passengers to the best service. En route

information can reduce passenger uncertainty, thus improving service reliability.

Current communication technology offers the opportunity to convey traveler information

via a variety of media: cell phones; in-terminal systems; variable message signs at bus stops,

train stations, and platforms; and in-vehicle systems, by signs or voice. Lessons may also be

learned from the following four major European cities:

(a) London: London Transport offers several types of traveler information service, for example, the Journey Planner, bus stop-specific timetables and maps, and the Countdown bus-arrival information system and Dyster, a regional smart card system.

(b) Paris: Régie Autonome des Transports Parisiens (RATP) has deployed Modeus, a regional smart card payment system, with multiple-application agreements for retail, telephone calls, and more; ALTAIR, a real-time information system that provides information on board and at bus stops; and SIEL, which provides information on regional rail service.

(c) Turin, Italy: A public-private partnership provides transit information, including itinerary planning, variable message signs, and at-stop displays.

(d) Helsinki, Finland: Advanced traveler information services include the Journey Planner, the transit agency's real-time system Helsinki Electronic Library Multimedia Information (HELMI), personal mobile traveler and traffic-information

ervice, use of smart cards, and real-time information via the Internet.

18.6.4 Holding strategy: Approximate method

Vehicle holding strategy is considered a possible remedy for reliability problems for three

main reasons: (1) prevents bunches from forming, (2) ensures that scheduled connections

(transfers) are made, and (3) minimizes total passenger waiting time. Holding strategies,

therefore, take place at dispatching points, connection points, and major stops (e.g., maximum

load point).

Traditionally, there are two classes of holding strategies: scheduled based and headway

based. The first strategy aims at holding the vehicle until its scheduled departure time; it is

especially useful for schedules with large headways. The second strategy, which usually is AVL/APC/CAD/AFP system Off-line data Tactics embodied in simulation and statistical models Evaluation (optimization) model Control decision to be transmitted Predicted delays and disruption Display Selected strategy (display) Set of possible control strategies Feasible strategies (display) Real-time data Semiautomatic Manual Actual delays and disruption Selection of decision mode Automatic

Figure 18.7 Schematic diagram of a computer-aided real-time control system.

schedules with short headways, has the objective of minimizing the total passenger delay

at stops. This optimization framework is the result of reducing headway variability (saving

delay) and creating extra delay (because of the holding strategy).

Undoubtedly, the use of a holding strategy is

complicated in practice because of the

uncertainty involved (in reaching its objective), its adverse effect on the onboard passengers,

and its impact on real-time operations at the network level. This situation may

justify the use of an approximation (isolated) method to describe the development of

basic holding rules. In general, Mirasinghe (2003) describes a cceptable approximate

methods in transit services, including dispatching policies, headway setting, and scheduled

travel times.

An early foundation for approximately analyzing dispatching and holding strategies in

transit operation was set down by Newell (1971, 1977, 1982). Section 18.7 reviews chronologically the essence of control strategies to improve transit reliability. Newell (1971) and

Mirasinghe (2003), in which Newell's work is discussed, present the best transit-vehicle

dispatching policy for minimizing total passenger waiting time, subject to a fixed number

of departures (dispatches). Newell found that the best dispatching rate was proportional

to (1) the square root of the random passenger-arrival rate for large (not constrained) vehicle

capacities and to (2) the random passenger-arrival rate, otherwise. Osuna and Newell

(1972) presented control strategies for dispatching immediately or holding a vehicle for a

hypothetical bus-route loop with only one service and control point. They used dynamic

programming and queuing techniques for distributed travel times and uniform passenger

arrival rates. The following is an approximate analysis for a holding strategy inspired by

Newell's work.

Figure 18.8 illustrates the process of passengers arriving at a maximum load and then

departing. Figure 18.8a refers to an even-load timetable and Figure 18.8b to an even-

headway timetable. In each part, the upper curve describes the accumulated number of

passengers according to the scheduled departure times; the lower curve exhibits a real-life

situation in which one early and one late arrival are introduced. The notations used in

Figure 18.8, some of which appear in Chapters 3, 4, and 6, are these: P_m is passenger load at

the maximum load point; d_0 is the desired occupancy on a single vehicle; t_i and t'_i are the i th sched

uled and real departure times at the maximum load point, respectively; λ_{im} is the average uniform

random passenger-arrival rate between the $(i-1)$ th and the i th departures at the maximum load

stop; H , H_i , and H'_i are the scheduled even headway, scheduled even-load headway, and real

headway, respectively; and Δ_1 , Δ_2 are the possible holding times for an early arrival and the

extra-late arrival time, respectively.

As expected, the accumulated number of p

assengers in the upper curve in Figure 18.8a

reaches d_0 when a departure takes place. This is not the case in Figure 18.8b, assuming a dif

ferent λ_{im} for the even-headway case. Improving reliability usually coincides with reducing

the total waiting time at transit stops. In the lower curves in Figure 18.8a and b, the shaded

area represents an additional passenger-minutes delay compared to the scheduled (expected)

situation. Although it is impossible to reduce the additional delay incurred by the late vehicle

arrival (by Δ_2 minutes), it is conceivable that the total passenger delay can be reduced by

holding the early arrival vehicle (by Δ_1 minutes). It should be noted that in real time, the situ

ation of late arrivals may also be ameliorated or avoided by the use of skip-stop or shortcut

strategies.

In an average and approximated sense, the dashed area of Δ_1 in Figure 18.8a is $\lambda_{H1} \cdot \lambda_{1m} \cdot \Delta_1$,

being the additional passenger minutes incurred if the t_1 departure is held until t_1 . Otherwise,

the additional approximated delay for passengers expected to board the t_2 departure is

$\lambda_{1m} \cdot \Delta_1 \cdot H_2$. Consequently, a simple strategy is this: hold the vehicle only if $\lambda_{H1} \cdot \lambda_{1m} \cdot \Delta_1 < \lambda_{1m} \cdot \Delta_1 \cdot H_2$.

However, because of the uncertainty about H_2 , its expected value is considered, that is, $E(H_2)$.

In general, the strategy for even-load timetables is to hold the vehicle if $\lambda_{H1} < \lambda_{1m} E(H_2)$ (18.12)

and to dispatch immediately, otherwise. The expected (mean) value $E(H_{i+1})$ requires data for

constructing the PDF (distribution) of H_{i+1} for all departures i , $i \neq 1$. $P_m d_0 P_m t_0 t_0 \lambda_1 m \lambda_4 m t_1 t_1 t_1' t_3' H_1 H_1 H_2 H_3 H_3 H_4 H_5 H_6 t_2 t_2 t_3 t_3 t_4 t_5 t_6 t_4 t_5 t_6$ Early arrival Time Late arrival $\Delta_1 \Delta_2 d_0 \lambda_1 m \lambda_5 m t_1 H H H H H t_0 t_2 t_3 t_4 t_5 t_6 t_1 t_0 t_1' H_1' H_3' t_2 t_3 t_4 t_5 t_6 \Delta_1$ Early arrival $\Delta_2 P_m d_0 P_m d_0$ (a) (b)

Figure 18.8 Cumulative number of passengers at the max load stop for an even-load timetable in (a) and an even-headway timetable in (b), in which the anticipated, scheduled situation in each part is in the upper curve and the real-life scenario is in the lower curve.

The simple rule of Equation 18.12 also applies to Figure 18.8b, the case of even headways.

That is, the vehicle is held from t_1 departure until t_1 if $\tau < H E(H) - 1$ (18.13)

and dispatched immediately, otherwise.

18.7 LITERATURE REVIEW AND FURTHER READING

The reliability of a public transit service is considered one of the main factors influencing

the level of passengers' satisfaction. There is a very abundant literature on various aspects of

transit reliability. This review describes, in chronological order, selected papers that focus

only on the following subjects:

- Development of quantitative measures of service reliability
- Models of passengers' waiting time at stops as a function of service reliability
- Suggested control strategies for improving reliability

18.7.1 Measures of service reliability

Service reliability is an abstract term. In order to include reliability considerations in a

detailed engineering design and to evaluate the differences between existing and suggested

service alternatives, it is necessary to describe reliability in mathematical terms. This section

includes a review of quantitative measures of service reliability.

One of the earliest discussions of transit service reliability measures was made by Polus

(1978), who focused on bus services. Reliability was defined as the amount of consistency

from day to day associated with any operational performance measure, the proposed reliability

indicator being the inverse of the standard deviation of travel times.

Silcock (1981) mentions some traditional reliability measures:

- Number of buses taking x minutes longer than scheduled
- Percentage of buses that depart from 1 minute early to 4 minutes late
- Average waiting time of passengers
- Excess waiting time (EWT) of passengers
- The difference between the actual average waiting time and the calculated waiting time

Turner and White (1990) formulated reliability measures in order to compare services provided by different bus sizes. They suggest the following

formula for EWT as a measure of

reliability: $EWT = \frac{1}{N} \sum_{i=1}^N (T_i - \bar{T})^2$

where T_i is the time duration of the reliability survey N is the number of observed buses during the reliability survey S is the

number of scheduled buses during the reliability survey
VarH is the variance of observed headways

Henderson et al. (1991) analyzed the advantages of using
odd ratios of on-time perfor-

mance as a measure of service reliability. An odd
ratio is the percentage of trains or buses

that arrive on time, divided by the per-
centage of those that were not on time. It
should

be emphasized that an odd ratio is not the
ratio of on-time arrivals to total arrivals, since

the denominator is not that of total arrivals.
Because of this formulation of the ratio, the

manner in which its value changes as the number of
deviations from the schedule increases

is not linear; the authors claim that this
formulation gives a better representation of pas-

sengers' perception of reliability than do most
other measures. For example, the reliability

of a transit service with a 70% probability of a-
rriving on time ($70/30 = 2.33$) is eight times

worse than the reliability of a service with 9
5% on-time arrivals ($95/5 = 19$). Hence, odd

ratios as reliability measures show passengers' high
sensitivity to the consistency of service

punctuality.

Rudnicki (1997) defined several unreliable p-
erformance measures. A suggested unreli-

ability measure is the chance that a passenger
will have to wait for a later departure be-
cause

of too-early departure of the expected v-
ehicle. An inconvenience measure is the aver-

age EWT. A degree-of-punctuality factor is the average value of a predetermined function,

which receives a value between 0 (worse) and 1 (best) according to the deviation from the

scheduled time. Another unreliability factor is the difference between the 95th percentile of

actual waiting time and the planned waiting time. The paper explains which of the measures

represent the interests of the passengers and which are more suitable to represent the operator's

point of view.

Hallowell and Harker (1998) present a method of predicting the schedule reliability (SR)

of a partially double rail line on which delays are caused by the need for one train to wait

for the passing train. Using a simulation model for the design of a reliable timetable, the

researchers describe measures of departure-time inaccuracy: at origin, arrival at destination

time, delay time from origin to destination, and slack time from origin to destination (slack

time is defined as planned ride time minus free running time).

Carey (1999) developed measures of estimating service reliability in advance. The

first proposed measure depends on the probability of delay for a specific train, result

ing either from schedule failure of this specific train or from a delay caused by previous

trains. Another measure depends not only on the chance of such delay but also on the

magnitude of the delay. Both of these measures require a PDF of route delay as the input.

Additionally, the author suggests two measures whose calculation does not require a

given PDF. One is based on the standard deviation and the mean of headways. The other

derives from the idea that a small initial delay grows larger if the delayed train, on the

way to its intended platform, has to cross paths used by another train; the fewer paths

crossed, the more reliable the service. The measures developed by Carey are normalized,

such that they all receive values between 0 and 1, with 1 representing the highest reliability

level.

Yin et al. (2002) develop a simulation-based approach to assess transit reliability, taking

into account the interaction between network performance and passengers' route choice

behavior. Measures are developed for three types of reliability: travel-time reliability (TTR),

SR, and waiting-time reliability (WTR). TTR is the probability that the average actual

travel time is less than a given threshold. SR is the probability that an actual line is frequency

larger than a given threshold. WTR is the probability that passengers' average waiting time

is less than a given threshold, taking into account the possibility of passengers' being unable

to board the first vehicle that arrives owing

to insufficient vehicle capacity. A Monte Carlo

simulation approach is used, incorporating a stochastic user-equilibrium transit-assignment

model with a capacity constraint and elastic frequencies. Analyzing simulation results using

the measures mentioned is very effective in evaluating the trade-off among alternatives for

service improvement.

Characteristics of the reliability measures reviewed are summarized in Table 18.5.

18.7.2 Passenger waiting time as a function of reliability

It is often claimed that passengers are more sensitive to changes in their waiting time than to

the other periods comprising the trip. Naturally, waiting time depends on the motion of the

anticipated vehicle, and service is considered more reliable from this point of view if that motion

is always the same. Prediction of vehicle-time variability is a complex task in itself, since it is

influenced by numerous factors, such as traffic conditions and passenger flow while boarding

and alighting. Waiting time also depends on other factors, some of which have to do with the

passengers' behavior. Several models for estimating waiting time will now be reviewed.

Table 18.5 Characteristics of works dealing with passenger waiting time as a function of reliability measures

Source Proposed measures Source of input data

Polus (1978) Inverse of standard deviation of travel times
Travel-time measurements for different days

Silcock (1981) Number of late buses, percentage of buses that depart on time, average waiting time, excess waiting time, difference between actual and calculated wait
Vehicle performance survey/simulation

Turner and White (1990) Excess waiting time Vehicle performance survey

Henderson et al. (1991) Odd ratio = percentage of on-time arrivals divided by percentage of not-on-time arrivals
Vehicle performance survey/simulation

Rudnicki (1997) Factor of unreliability on account of unpunctuality, factor of inconvenience owing to unpunctuality, degree of punctuality as a factor of operation regularity, factor of unreliability on account of irregularity Vehicle performance survey

Hallowell and Harker (1998) Departure-time error at origin, arrival-time error at destination, delay time from origin to destination, slack time from origin to destination
Simulation

Carey (1999) A measure depending on the chance of delay and a measure depending on the magnitude of delay A measure based on standard deviation and mean of headways A measure of the chance of delay owing to the need to wait for a crossing train PDF of route delay Vehicle performance survey/simulation Information about number of rail paths crossed

Yin et al. (2004) Probability that the average actual travel time is less than a given threshold, probability that an actual line frequency is larger than a given threshold, probability that the passengers' average waiting time is less than a given threshold Simulation

The traditional formula for average passenger time, depending on headway variability,

is shown in Equation 11.1 and discussed in Section 18.4.1. This formula was used by

Osuna and Newell (1972), as well as by other authors. For the sake of clarity, it is pre

sented again: $E(w) = H^2 / (2C) + H^2 / 2$ ()

where $E(w)$ is the average waiting time

H is the mean headway C_H is the coefficient of variation in headways (standard deviation/mean).

This formula is based on the assumption that passengers arrive at a stop randomly. The

average waiting time is greater than half the headway because more passengers arrive during

long headway intervals (before the arrival of a delayed vehicle) than during short intervals

(after the arrival of a delayed vehicle). The average wait, therefore, is longer when headways

are less uniform.

Turnquist (1978) developed a model of bus and passenger arrivals at a bus stop. He

investigated the impact of reliability on the average waiting time under the assumptions

that the bus arrival time behaves according to a lognormal distribution and that its mean

and variance are known. The bus-arrival-variance variable plays the role of the reliabil

ity indicator. Calculation of waiting time takes into account both passengers who arrive

at the station randomly and passengers who plan their arrival time. If the proportion of

random arrivals is known, then this model makes it easy to calculate waiting time for

planning purposes.

Bowman and Turnquist (1981) developed another model for evaluating average wait

ing time as a function of service reliability and frequency. This model is based on more

complex assumptions regarding passenger behavior: it takes into account the decision

making process that passengers go through when they choose

a time to arrive at the

bus stop. The model predicts the distribution of passenger arrivals when the mean and

average of bus headways are given. It enables operators to make a realistic prediction of

how a change in service reliability will alter the overall waiting time. This prediction is

sensitive to the differences in passenger behavior among lines with different frequencies.

The authors conclude that waiting time is much more sensitive to SR than to service

frequency.

Guenther and Sinha (1983) described another model for estimating waiting times. Their

model is unique in that it assumes that SR is influenced by the level of vehicle maintenance:

when there are not enough vehicles available to meet service demand, reliability is severely

affected. The tool presented consists of three models: a maintenance model, a reliability

model, and a performance model. The input for the maintenance model is the number of

spare buses and the number of mechanics: its output is a dependability factor that expresses

the point at which maintenance problems will lead to service failure. This factor is calcu

lated by using simulation. The dependability value is an input for the reliability model,

which determines the average waiting times both of passengers who arrive randomly and

of those who come at a preplanned time. Waiting-time values are transferred to the per

formance model, which yields predictions of ridership and

an evaluation of general system

performance. Implementing the proposed set of models requires some computer tools as

well as calibrated values of demand elasticities, which may not always be available.

Xuan et al. (2011) acknowledge that dynamic holding strategies based on headways

a lone cannot help buses adhere to a schedule. As such, the authors investigate a family of

dynamic holding strategies that use bus arrival deviations from a virtual schedule at the

control points. The virtual schedule is introduced whether the system is run with a published

schedule or not. It is shown that with this approach, buses can both closely adhere

to a published schedule and maintain regular headways without too much slack. However,

the modeling is based on a long list of assumptions such as unlimited capacity, unlimited

number of departures and stops, holding strategy being applied after boarding, and more.

The authors show that the only data needed in real time are the arrival times of the current

bus and of the preceding bus at the control point relative to the virtual schedule. The

simple method was found to require about 40% less slack than the conventional schedule

based method.

Li et al. (2010) review and critique the modeling frameworks and empirical measure

ment paradigms used to obtain willingness to pay (WTP) for improved TTR, suggesting

new directions for ongoing research. The authors review TTR studies in the context of

passenger transport (car, rail, and bus). Three types of empirical models to capture TTR

(i.e., the mean-variance model, the scheduling model, and the mean lateness model) were

presented, including associated theoretical issues, experimental design, and practical applica

tions. The authors estimated models to derive values of reliability, scheduling costs, and

reliability ratios in the context of Australian toll roads. They used the new evidence to

highlight the important influence of the way that trip time variability is included in stated

preference studies in deriving WTP estimates of reliability in absolute terms and relative to

the value of travel time savings. The study found that a choice experiment with multiple

travel time scenarios to represent early, late, and on-time trip activity produced significantly

higher reliability ratios, compared to a choice experiment with a single time component

(i.e., variability).

Table 18.6 summarizes the models reviewed.

Table 18.6 Characteristics of works on passenger waiting time as a function of reliability

Source	Required input	Assumptions regarding passenger arrival at station	Assumptions regarding vehicle arrival at station
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Osuna and

Newell (1972) Standard deviation and mean of headway
Random. None

Turnquist (1978) Standard deviation and mean of headway
Some passengers arrive randomly; others plan their
arrival time. Lognormal distribution

Bowman and

Turnquist (1981) Standard deviation and mean of headway,
probability distribution function of vehicle arrival A
utility function is ascribed to each possible arrival
time. None

Guenther and

Sinha (1983) Number of spare buses and mechanics Some
passengers arrive randomly; others plan their arrival
time. Lognormal distribution

Xuan et al. (2011) Scheduled and actual arrival times and
headways Stationary passenger arrivals, but station
dependent. Trip times are random

Li et al. (2010) Mean travel time, travel cost, and five
equiprobable arrival scenarios Different studies are
compared with. Different studies are compared with

18.7.3 Control strategies for improving reliability

Transit services are considered unreliable if they suffer
from significant travel-time variability.

Improvement in service reliability is,
therefore, likely to occur if actions are taken

to reduce time variability. Such actions are
usually aimed either at having control over

vehicle adherence to its schedule or at
introducing route patterns with low time
variability.

This section presents several strategies for
improving reliability. It should be noted that

short-turn strategies, although mentioned in some papers in

this context, are referred to in

Chapter 16.

Osuna and Newell (1972) presented the conflict between the desire for minimal route

time and the need to slow down system performance in order to increase its regularity.

They formulated a control problem that aims at increasing punctuality, thereby achiev

ing minimal waiting time. Dispatching and holding strategies are introduced: dispatch

ing strategies help in deciding what the actual headways would be between successive

departures from the starting terminal, while holding strategies mean that departure time

is also controlled at other time points along the route. The performance of controlled

and uncontrolled systems is analyzed, based on a PDF of travel time, which is required as

input. Other basic assumptions, declaratively simplistic, refer to an idealized network. For

example, only single-stop routes with one or two buses are discussed. Still, this chapter

represents a significant discussion of schedule control, since it led to many other research

works in this area.

Lesley (1975) suggested a procedure for designing a realistic schedule on a route along

which bus progress is monitored. A reliability index is calculated for each stop on the

basis of the level of headway variance. Stops at which the index is at least twice the average

are chosen as time points for holding c

ontrol. The required slack time at each point

is calculated.

Newell (1977) developed a method for determining the slack time allocated to each

time point when holding control is applied. With the aim of reducing vehicle time while

keeping schedule regularity, the lower boundary of slack time is calculated. Some of the

basic assumptions are simplistic, such as ignoring a lighting times (which is still a realistic

assumption) and supposing no correlation between adjacent buses. Passengers are assumed

to arrive at stops according to a Poisson distribution, and bus delay is assumed to behave

according to Fokker-Planck. Slack time is calculated for each stop, not just selected ones.

Koffman (1978) compared several real-time strategies for improving headway reliability

on a single-direction bus route with a single control point. Employing a simulation model,

the author examined the following strategies: holding, skipping stops, introducing bus-

priority signalization, and reducing dispatch uncertainty. All control strategies are analyzed

under different headway thresholds. The resulting conclusion is that the last two strategies

seem the most promising.

Jordan and Turnquist (1979) discussed reliability improvement through the introduction

of a special pattern for urban routes. According to this pattern, the whole path is

divided

into several zones, and each bus makes stops only in a specific zone. A dynamic programming

problem is formulated in order to determine the optimal number of zones, points

where zones switch, and the number of buses serving each zone. The objective is to find a

zone structure that minimizes trip-time average and variance. The model developed includes

submodels for predicting dwell time at stops, distribution of headways, mean and variance

of waiting time, and mean and variance of travel time; intensive calibration is therefore

required. The dynamic programming model is applied in a case study, which yields the

conclusion that even a very simple zone structure can lead to a substantial improvement in

reliability and a decrease in bus-fleet size.

Turnquist and Bowman (1980), using a set of simulation experiments, investigated the

effect of network structure on service reliability. Transit networks on which routes paths

form a radial pattern are compared to grid-shaped networks. The researchers found that

even though passengers were obliged to transfer more in grid networks, the uncertainty in

regard to the length of the transfer delay was smaller than in radial networks. The com

bined travel-time uncertainty is also smaller in grid networks. The authors conclude that

a concentration of transfers at the center node

of radial networks has a more disruptive

effect on reliability than does the dispersed distribution of transfers that occurs in grid networks.

Furthermore Nash (1985) suggested a method for improving adherence to a replanned

timetable, for which all routes started at a specific terminal. According to this method, the

pool of vehicles that belongs to the terminal serves all the round trips leaving that terminal

in a first-in-first-out sequence, instead of being assigned to specific trips in advance. Features

of this strategy are presented, and the reliability of the resulting service is estimated. A PDF

of bus travel times is needed as input.

Abkowitz et al. (1986) determined which single point was optimal for holding control.

They found that the location of the control point was sensitive to the passenger-boarding

profile. The optimal control point is usually located just prior to a group of stops having a

large number of boarding passengers, that is, toward the peak segment of the route (max

load point). Many other papers published in the 1970s and 1980s describe methods of

designing optimal holding strategies. Most of these methods use simulation techniques to

optimize the value of a waiting-time threshold, above which a stop is declared a time point

for controlling schedule adherence.

Seneviratne (1990) calibrated an expression for headway standard deviation as a function

of the number of time points. This function is a second-degree polynomial, meaning that

there is an optimal number of points and that increasing the points beyond that number will

lead only to diminished reliability.

Li et al. (1993) presented various real-time dispatching strategies to be used at the terminal

of a single route: instructing bus drivers to skip stops, directing them to cut out parts

of their route, or simply adjusting layover times. The design of these strategies is expressed

as a nonlinear program and as a stochastic binary integer program. Simulation is used for

testing this approach, but it can also be used to evaluate preestablished schedules or decisions

regarding stop-skipping strategies. Special attention is given to assessing the impact of

bus-location information on the dispatching decision. The researchers do not need for a

automatic vehicle-location system that covers the entire network for the purpose of real-time

dispatching. Location information from a limited number of points seems to be sufficient.

Wirasinghe and Liu (1995) formed a dynamic programming model for the design of hold

ing strategies for an isolated bus with fixed demand. The number and location of time points

are determined, as is the amount of slack time at each

point.

Carey (1998) formulated a problem of optimizing slack time between successive activities

in a vehicle schedule. Increasing slack time reduces the risk that a delay in one activity will

affect the next activity; on the other hand, allocating more spare time may cause the activity

to take longer because of behavioral reasons. The author claims that some percentage

of scheduled rides will always depart a few minutes late, independently of the length of the

slack time before the scheduled departure. A PDF of a activity time is a required input. Slack

times that give minimum total operating costs are calculated. It should be noted that many

other papers by this author deal with delay control and various other aspects of transit

reliability.

Hallowell and Harker (1998) proposed a simulation method mentioned earlier in this

section. Their method can be used for determining slack times allocated to stations

along a railway line in order to increase SR. The authors compare features of real-time

scheduling and master scheduling. They also discuss the difficulty in using simulation

methods for improving performance; it is due to the complexity of the computation

tasks involved.

Adamski and Turnau (1998) developed a tool for real-time schedule-control deci

sion making. The input for the model is real-time information about deviations from

scheduled times or from regular headways along a transit route; the output is a set of

recommended decisions regarding vehicle dispatching. The tool developed includes a

submodel that calculates the time spent at stops. Many other papers by Adamski deal

with dispatching control.

Eberlein et al. (1998) discuss the real-time strategy of DH, that is, running empty from the

terminal to the first station at which passengers are allowed to board while skipping several

stations en route. A programming model is formulated in order to determine which vehicles

to deadhead and how many stations to skip.

O'Dell and Wilson (1999) developed a real-time decision-support system for rail-transit

operations. The system includes a deterministic model of a rail system and mixed-integer

programming formulations for the choice of an optimal holding strategy. Several holding

strategies are discussed: hold at all stations, hold only at the first station after the disruption

occurs, and hold at a selected station. These are investigated with and without train

capacity constraints. The paper also includes a dwell-time submodel. Attention is given to

assumptions on the order in which trains from different branches enter junctions; this is

claimed to have a major effect on system performance. The results show that holding s

trate

gies at one station (whether the first after the disruption or any station) are almost as effective

tive as holding at all stations. They recommend holding control at the first station after the

disruption, since it is the easiest to implement.

Eberlein et al. (1999) considered several control strategies: two station-skipping strate

gies, holding at a given station, and combinations of the two. They formulated mathematical

cal models for all control problems under the assumption that real-time vehicle-location

information was available and investigated the advantages and disadvantages of each con

trol type. The models are deterministic but are also examined under various stochastic

conditions. The results show that a combination of station-skipping and holding strategies

can have very effective results on system performance. It is also shown that combined

strategies are more efficient than any single strategy although a holding strategy is slightly

preferable. The authors find that the optimal holding strategy depends mainly on the

route-headway pattern at the holding station and is independent of passenger demand

along the route.

Shen and Wilson (2001) developed a disruption control model for rail-transit systems.

The model, a deterministic, mixed-integer program, includes a dwell-time submodel. Several

control strategies are compared. The main conclusion is that best system performance is

achieved when holding and short-turn strategies are combined.

Liu and Mirasinghe (2001) presented another simulation model for the design of a

holding control. The simulation model consists of several submodels that predict perfor-

mance during various journey stages: linked travel time, passenger arrival, passenger

alighting demand, dwell time, and dispatching time at the starting terminal. The output

of the overall model is the choice of time points and the slack times allocated to each

of them.

Fu and Yang (2002) developed two simulation models for choosing time points for cases

in which holding control should be implemented. In both models, time points are allocated at

every stop where headways exceed some threshold. In the first method, only the headway

between a bus and the preceding bus is checked; the second method checks, in addition, the

headway between each bus and the following bus. The researchers find that control is most

useful when the control stops chosen are those with the highest demand and close to the

middle of the route. The number of control points is also investigated—scenarios with one

or two or all stops with control are examined. Control at all stops reduces waiting

time, but

it also results in a significant increase in vehicle time. Control at two stops, one of which

is the starting terminal, is found to be the optimal solution, since it enables the operator

to achieve a reduction in waiting time with no increase in in-vehicle time and only a slight

increase in bus-travel time.

Dessouky et al. (2003) examined simulated systems in which holding and dispatching

strategies are used; they also investigated the dependence of system performance on available

technologies. Systems in which communication, vehicle locating, and counting technologies

are available or unavailable are compared. The analysis addresses even different holding

strategy combinations, each of which differs in its definition of holding conditions and in

its approach to the use of holding for connecting buses. The results show that a advanced

technology is most advantageous when there are many connecting buses; the schedule slack

is then close to zero and the headway is large.

Chen et al. (2009) present an in-depth analysis of service reliability based on bus opera-

tional characteristics in Beijing. Three performance parameters are proposed in the study

for the evaluation of bus service reliability: punctuality index based on routes (PIR), deviation

index based on stops (DIS), and evenness index based on stops (EIS). The relation

ship among the three parameters is discussed using a numerical example. Subsequently,

through a sampling survey of bus lines in Beijing, service reliability at the stop, route, and

network levels is estimated. The authors claim that the three performance parameters

allow for reliability estimation from macro to microscale. In general, this study found

that $P_{IR} > D_{IS} > E_{IS}$. In terms of Beijing, the results imply that its transit system is

operated at a relatively low reliability level. In addition the study analyzed the effects of route

length, headway, the distance from the stop to the origin terminal, and the use of exclusive

bus lanes.

Yu et al. (2012) present a DH strategy for transit operations to improve transit service

of the peak directions of transit routes. The DH strategy takes place at different stops by

letting the vehicle to turn around and continues as a DH trip; it is called partway DH.

This strategy consists of two phases: reliability assessment of further transit service and

optimization of partway DH operation. The reliability assessment of further transit ser

vice, which is based on the current and recent service reliability, is used to justify whether

or not to implement a partway DH operation. The objective of the second phase is to

determine the beginning stop for a new service for the DH vehicle by maximizing the

benefit of the transit system. A heuristic algorithm is also defined and implemented to estimate

reliability of further transit service and to optimize partway DH operation. Then,

the partway DH strategy proposed is tested with the data from a transit route in Dalian

City of China. The authors analyzed the length of rolling horizon and the threshold value

of service reliability and validated the results. All in all, the authors illustrated that their

partway DH strategy yields the best performance measures compared with other partway

DH strategies and the no-control scenario.

The major characteristics of the models reviewed are summarized in Table 18.7. For

practical use, the main difference between analytical methods and methods based on simulation

should be mentioned. Analytical methods require less detailed information about

all network components; however, they usually rely on many assumptions. The problem

is then to simplify the representation of the system performance. Methods that use simulation

provide much more realistic results, but the input that they require is much more

detailed.

Table 18.7 Characteristics of works on control strategies for improving reliability

Source	Control type	Method	Place of control	Suggested solution
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Osuna and				
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Newell (1972) Dispatching, holding Analytical Single control point.

Lesley (1975) Holding Simulation Number and location of points are optimized.

Newell (1977) Holding Analytical All stops are controlled.

Koffman (1978) Real-time holding, stop-skipping, bus-priority signalization, dispatching Simulation Single control point (terminal). Bus-priority signalization and reducing dispatching uncertainty.

Jordan and

Turnquist (1979) Zone scheduling Dynamic programming

Turnquist and

Bowman (1980) Network structure Simulation Grid network.

Furth and Nash

(1985) Real-time dispatching Analytical Terminal.

Abkowitz et al.

(1986) Holding Combination of simulation and analytical Location of control point is optimized. Optimal control point is toward the peak segment of the route.

Seneviratne

(1990) Holding Simulation Number of control points is optimized.

Li et al. (1993) Real-time dispatching/ stop-skipping Nonlinear program/binary integer program (simulation is used for testing the method) Terminal of a single route.

Wirasinghe and

Liu (1995) Holding Dynamic programming Number and location of points are optimized.

Carey (1998) Dispatching Analytical Terminal.

Hallowell and

Harker (1998) Holding (real time/ master) Simulation All stops are controlled.

Adamski and

Turnau (1998) Real-time dispatching Simulation All stops are controlled

Eberlein et al.

(1998) Real-time deadheading (stop-skipping) Programming model Terminal.

O'Dell and

Wilson (1999) Real-time holding Analytical/ mixed-integer programming All stations/First station after disruption/Fixed strategies are investigated. Holding at First station after disruption. (Continued)

EXERCISES

18.1 Given a bus route on which passengers' mean waiting time, $E(w)$, can be estimated by $E(w) = E(H^2)/2E(H)$, where H is the headway variable,

(a) Explain and show (mathematically) how $E(w)$ can be reduced, using the waiting time formula. Consider headway distributions ranging from regular (deterministic) to exponential (random).

(b) Outline feasible (practical) ways to reduce $E(w)$, given fixed passenger demand.

18.2 With high volumes of bus services on major arterial roads, it becomes very expensive and often ineffective for bus agencies to use curbside inspectors and mobile supervisors. The inspector's or supervisor's knowledge of situations is limited to his or her immediate vicinity, with no indication of conditions elsewhere. Accordingly, bus drivers may be instructed in some cases to act in a fashion that is counterproductive from the system point of view. This represents only one group of problems that can be resolved by introducing an AVL system. In general, the real-time control provided by an AVL system enables the operator to increase both the utilization of each bus and the service level as seen by each passenger. F

Furthermore, the substantial amount of off-line AVL data can be used to improve the entire bus-planning process.

(a) List all possible strategies that supervisors can accommodate in the control room while using an advanced AVL system. Note that these strategies are carried out by instructing a driver or a group of drivers to follow a certain action.

(b) Outline the major potential benefits of (adequately used) AVL off-line data.

Table 18.7 (Continued) Characteristics of works on control strategies for improving reliability

Source	Control type	Method	Place of control	Suggested solution
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Eberlein et al.				
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(1999)	Real-time stopskipping, holding, combinations			Analytical Combination of strategies.
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Shen and Wilson				
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(2001)	Holding, short turn	Mixed integer program		Combination of strategies.
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Liu and				
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Wirasinghe				
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(2001)	Holding Simulation	Number and location of points are optimized.		
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Fu and Yang				
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(2002)	Holding Simulation	1, 2, and all strategies are investigated. Holding at two points.		
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Dessouky et al.				
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(2003)	Holding, dispatching Simulation	Seven alternatives are compared.		
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Chen et al. (2009)	Ordinary operations	Statistical analyses	Three performance indices are proposed.	
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Yu et al. (2012)	Deadheading (DH) strategy	on route		
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segments Optimization and heuristics At transit stops. New DH strategy is proposed.

Abkowitz, M., Eiger, A., and Engelstein, I. (1986). Optimal control of headway variation on transit routes. *Journal of Advanced Transportation*, 20, 73-88.

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19 Chapter 19: Operational Strategies and Tactics

The following riddle may help spread some knowledge on the substance of transit real-time

control actions, especially from a sensitivity perspective, and what we need and what we don't

need to pay attention to. Assume that a tight rope surrounds the earth along the Equator. The rope is very tight.

The question is: "If we add one meter to the length of the rope, can a cat then go underneath

this rope, or not?" (The answer appears at the end of this practitioner's corner.) This chapter consists of five main parts. Section 19.2 focuses on a multiagent transit sys

tem composed of the following agents: transit vehicles, passengers, road segments, and local

authority (or government). This section defines each agent and their interrelationship as pos

sible tools for real-time transit control actions. Section 19.3 analyzes the probability of the

s simultaneous arrival of two or more transit vehicles at a given road segment; it calculates the

road-segment encounter, with any point along the segment constituting a possible encoun

ter point, and discusses possible online tactical deployment measures. Section 19.4 develops a

dynamic programming model for minimizing the total travel time, resulting in a set of preferred

tactics to be deployed. An optimization model based on dynamic programming and a transit

simulation that validates the benefits of such a model are described. Section 19.5 investigates

how to use selected operational tactics in transit networks for increasing the actual occurrence

of scheduled transfers. The model determines the impact of instructing vehicles to either hold

at or skip certain stops or segments (one or more consecutive stops), on the total passenger

travel time (TPTT) and the number of simultaneous transfers. This section explains how to

create simulation and optimization frameworks for optimal use of the different scenarios and

compare between the scenarios using a case study. In Section 19.6, an intervehicle communica

tion (IVC)-based scheme is proposed to optimize the synchronization of planned transfers in

transit networks. By using this scheme, transit drivers can share their real-time information,

such as vehicle location, vehicle speed, and the number of passengers, with each other, through

a central communication coordinator. The chapter ends with a literature review. Fundamentally, practitioners are encouraged to visit Sections 19.1 and 19.4 through 19.7,

except Sections 19.5.1 and 19.6.3, and also to read the introduction and conclusions of Sections

19.2 and 19.3. The chapter offers challenges for devising alternative transit solutions that have

yet (if at all) to be introduced or that may need to negotiate the barrier of implementation.

Until the future actually reveals itself, practitioners can, and should, improve their readiness for

future transit operations under new conditions. We can now return to the riddle. The answer is: "Yes, indeed." The reason is that the length

of the rope is not related to R , the radius of earth. The length of the original rope is known to be

around $40,075,000 \text{ m} = 2\pi R$. Adding 1 m will create a new radius, say $R^* = R + \Delta$; thus, $2\pi R + 1 =$

$2\pi(R + \Delta)$, and $\Delta = 100 \text{ cm}/2\pi = 15.9 \text{ cm}$, which is enough room for a cat to go underneath.

19.1 INTRODUCTION

Following Chapter 18, transit-service reliability is receiving increased attention as agencies

are faced with immediate problems of providing credible service while attempting to reduce

operating cost. Unreliable service has also been cited as the major deterrent to existing and

potential passengers. Due to the fact that most of the transit attributes are stochastic, travel

time, dwell time, demand, etc., the passenger is likely to experience unplanned waiting

times and ride times. One of the main components of service reliability is the use of transfers

as is clearly indicated in Chapters 13 and 14. Transfers have the advantages of reducing

operational costs and introducing more flexible and efficient route planning. However, its

main drawback is the inconvenience of traveling multilegged trips. Some works introduce

synchronized (timed) timetables to diminish the waiting time caused by transfers. Their use,

however, suffers from uncertainty about the simultaneous arrival of two (or more) vehicles

at an existing stop. In order to alleviate the uncertainty of simultaneous arrivals, operational

tactics such as hold, skip-stop, and short turn can be employed considering the positive and

negative effects, of each tactic, on the total travel time

(TTT).

The arguments of Chapters 5, 12, and 15 demonstrate that the use of transfers in public

transit has the advantages of reducing operation costs and introducing more flexible, ef

ficient route planning. In contrast, the main drawback, from the passengers' point of view,

is the inconvenience of traveling multilegged trips. In the attempt to diminish waiting time

caused by transfers, the following sections will elaborate the new concepts.

As we noticed in Chapter 15, designing a public transit system produces conflicts among

the parties involved. The agency will usually prefer a small number of routes with transfer

centers in order to reduce fleet size. The passenger will prefer more routes and fewer transfers

because transfers naturally decrease comfort and reliability. This conflict may intensify

in areas with variable and sparse demand. The solution is to combine the advantages of a

transfer-based system (lower costs) with models that increase the comfort of transfers and

reduce waiting and travel time. Table 19.1 summarizes these options.

Last, using a metaphor, the only way to eat and enjoy a baguette sandwich (to arrive on

time) is sequentially (make control actions consecutively), and not across (planning all the

control actions concurrently), even if you are hungry (in a rush).

19.2 MULTIAGENT TRANSIT SYSTEM

In this section, we refer to what is known as multiagent systems (MAS) and start with a brief

background. Van Dijke Parunak (1997) defined MAS as collections of autonomous agents

within an environment that interact with each other for achieving internal and/or global

objectives. Minsky (1986) argued that an intelligent system could emerge from nonintelligent

parts. His definition of the "Society of Mind" makes use of small, simple processes, called

agents, each of which performs some simple action, and the combination of all these agents

forms an intelligent society (system). Bradshaw (1997) classified agents by three attributes:

autonomy, cooperation, and learning. He defined four agent types from these attributes: col-

laborative, collaborative learning, interface, and smart. The most interesting types so far are

the public transit system is concerned is the collaborative agent. Such an agent is simple and Table 19.1 Perception of transit options, by party Option Party Passenger Agency Authority

More routes	Better service	More expensive	Better coverage
More transfers	Reduced comfort and reliability	Less expensive	More user complaints
Smart transfers	Reduced waiting and travel time	Less expensive and increased ridership	Increased use of public transit

can perform tasks independently, but can collaborate with other agents if necessary in order

to achieve a better solution. Zhao et al. (2003) developed MAS for a bus-holding algorithm.

The authors treat each bus stop as an agent; the agents negotiate with each other, based on

marginal cost calculations, to devise minimal passenger waiting-time costs.

The multiagent transportation system (MATS) proposed here is based on Hadas and

Ceder (2008b). The MATS will be composed of the following agents: public transit vehicles,

passengers, road segments, transit agencies, and transit authorities. In order to construct the

system as a whole, it is necessary to define each agent and the interrelationship of agents.

Since the system is complex and cumbersome, each agent will be explored separately; this

will also make it easier to define the system's elements. Figure 19.1 presents the main activi

ties and interrelationships of the proposed system.

based on the real-time information available from road-segment agents (travel time) and

vehicle agents (routes and dwell times). Passengers will input to their cell phones or personal

digital assistant (PDA) every trip desired from point A to point B. The agent, referring to

Figure 19.1 Agents' activities and relationships.

available at that time. The passenger will choose among the possible options and book a

trip in the system. Then the passenger will be notified by SMS or a different means of com

munication about path changes owing to, for example, traffic congestion or a system-wide

optimization deploying tactics that change the planned schedule of route legs. In response

to the route changes, the passenger agent will try to find a better route, if possible changing

the existing planned path. The agent itself can be a small software program running on a

cell phone or PDA.

Road-segment agent: The road-segment agent can reside physically, as part of the road

infrastructure (e.g., a traffic light controller), or virtually, as part of a multiagent software

system. The agent is responsible for the following activities: estimating travel time, estimating

encounter probability, improving the system's objective function, and instructing vehicle

control deployment. Each road-segment agent continuously collects local traffic flow

information and estimates the travel time. The agent evaluates the encounter probability (for

vehicles transferring passengers), based on the adjacent road-segment travel-time estimation

and vehicle locations. This probability is described in the next section. Using dynamic

programming (DP), each agent or group of agents calculates the optimal tactical deployment

that will optimize the system's objective function, which is the total expected travel time.

Vehicle agent: The vehicle agent can be part of the onboard automatic vehicle location

(AVL) system or, virtually, part of a multiagent software system. The agent estimates the

vehicle's dwell time according to booked trips and demand forecasts, using travel-time estimation

from the road-segment agents and estimated deviations from the planned time

table schedule. The vehicle agent receives instructions for the deployment of tactics from the

road-segment agent in order to improve the system's objective function.

Agency agent: This agent is responsible for designing and managing the transit network, for

updating timetables, and for configuring the possible operational tactics available on each

road segment for each route.

Authority agent: The authority (local authority or federal government) is responsible for

monitoring system performance according to the determined/decided indicators.

The MATS offers the following benefits, which are inherent in the multiagent approach:

Extensibility: Easily allows the system's growth and adding of new resources. Each vehicle has computation power to contribute to the entire network. Adding a new vehicle is similar to plugging in a new computer to a local area network.

Fault tolerance: The proposed system will handle failures. Critical operations that are heavily dependent on computation and that are built on a standard central computing architecture must have a redundancy system in order to maintain a certain level of service. Redundant systems are expensive and cumbersome; however, MAS agents are distributed. Consequently, if some agents are down, the others will continue to perform because of their autonomous capability. Table 19.2 presents the mode of operation and outcome for different communication scenarios in case of communications disturbances.

Scalability: Distributed systems can theoretically grow without limit (e.g., the Internet).

Adaptability: Changing rules and transmitting data to the agent is quick and simple, similar to the spreading of a virus.

Efficiency: Negotiations between agents can reach a non-optimal or a near-optimal solution efficiently (e.g., see Raiffa, 1982).

Distributed problem solving: Cooperating agents can distribute subtasks to other agents that are idle and can contribute their computer time to solve these subtasks (e.g., see Smith [1980] and Davis and Smith [1983]).

Stability: The use of a closed set of operations tactics for each road segment in order to eliminate solution sets that are not stable.

The following is a simple example of the benefits of MATS. On board a shuttle heading

for the train station are 30 passengers, of whom 20 plan to take the 12:20 train. The

train headway is 30 min, and trains are assumed to depart on time. A long the shuttle

route, four more passengers are waiting (say, for the system's callback; see Chapter 17) to

take this shuttle service. According to the estimated arrival time, there is a probability of

0.60 of reaching the train station at 12:15 and a probability of 0.40 of reaching the sta-

tion at 12:25. In this case, the expected total waiting time for the train will be ($0.6 \cdot 5 +$

$0.4 \cdot 25) \cdot 24 = 312$ min. If the shuttle deploys a shortcut tactic, the estimated arrival time

probability will be increased to 0.95 for 12:15, in which case the four waiting passengers

will miss the 12:20 train and will have to board the next shuttle (the average headway being

20 min). The total new waiting time for the 20 passengers will be $(0.95 \cdot 5 + 0.05 \cdot 25) \cdot 20 =$

120 min. For the existing four waiting passengers, and assuming that the next shuttle

for the 12:50 train does not execute any operational tactic, the total waiting time will be

$(0.6 \cdot 5 + 0.4 \cdot 25 + 20) \cdot 4 = 132$ min. In terms of system optimization, the shortcut tactic

results in $120 + 132 = 252$ min total waiting time, compared with 312 min otherwise; thus,

the tactic will be preferred.

19.3 VEHICLE ENCOUNTERS ON ROAD SEGMENTS

MATS encapsulates activities and processes that the agents continuously perform in parallel

with one another. Some of these activities have been described in this book and can easily

be integrated into MATS: estimating travel time (see Section 18.3.2), estimating dwell time

(see Section 18.3.1), and finding the best route (see Chapters 15 and 17). However, this sec

tion presents a step-further method, in which MATS incorporates new modeling concepts

rooted in the determination of encounter probabilities. The section is based on Hadas and

Ceder (2008c).

Figure 19.2 presents the main activities of a road-segment agent. This agent is responsible,

among other things, to the public transit-system optimization. Hadas and Ceder (2008b)

show that modeling encounter probability, c

constructing a set of operational tactics, and

modeling a new operational objective function of the total public transit-system travel time

(travel times, transfer times, and waiting times) are essential for developing a DP model that

can optimize the system. Table 19.2 Communication scenarios Scenario Mode of operation Outcome Full communication Online collaboration Synchronized transfers Total travel-time reduction Partial communication No communication Autonomous Ordinary transit system with reliability problems According to timetables

19.3.1 Defining an encounter probability

Transit-vehicle behavior can be illustrated by a time-space (T-S) trajectory, an example of

which is shown in Figure 19.3. The notations S_0 and S_3 represent the entrance and exit of

the road segment, respectively; t_1 is the entry time of the vehicle to the road segment; t_2 and

t_4 are the arrival times at stops S_1 and S_2 , respectively; and t_3 and t_5 are the departure times

from these stops. The slopes from t_1 to t_2 and from t_4 to t_5 show the average travel time

(based on average speed) between two consecutive stops. The average travel speed depends

on traffic-flow characteristics, speed limits, geometric design, and driver behavior.

The dwell time depends on the number of passengers boarding and alighting, vehicle

configuration (number of doors, entry and exit doors, and payment methods), and driver

behavior.

The probability of an encounter between two transit vehicles, especially buses, is a

function of road-segment characteristics and vehicle travel characteristics. Road-segment Traffic deployment Dynamic programming model System optimum Current encounter probability Encounter probability upper bound Encounter probability estimation Maximal gap model Defining encounter probability Objective function modeling

Figure 19.2 Road-segment agent's main activities. Time (T)
Space (S) t₁ t₂ t₅ t₃ t₄ Stop 1 S₁ Stop 2 S₂ S₃ S₀

Figure 19.3 Time-space diagram illustrating transit-vehicle trajectory behavior on a road segment.

characteristics include segment length, number of stops, distance between adjacent stops,

and travel-time estimation. Vehicle characteristics include road segment, entry time, and the

number of passengers boarding and alighting at each stop. The following assumptions are

made: (1) a planned encounter (based on a given timetable) exists for two vehicles on a road

segment and (2) segment travel time for an arrival vehicle is dependent on the travel time of

the preceding vehicle. The first assumption is essential because transfers are planned on pre-

defined road segments (where two transit-vehicle lines intersect) rather than on a dynamic transfer

area. The second assumption, more realistic in urban bus systems, assumes the dependency

of travel times between sequential vehicles (experiencing similar traffic characteristics) as

long as their headway is not too large.

Transit-vehicle behavior is taken into account when constructing an estimation model

for the probability of an encounter. Such a model considers that (1) randomness exists

for travel time and dwell time and (2) dependency exists between the travel-time distribution

distributions of the vehicles examined. Figure 19.4 illustrates, through a T-S diagram, an

example of possible encounter situations between two vehicles and a possible miss of

encountering.

Figure 19.4a shows two vehicles that encounter each other three times (designated e 1, e 2,

and e 3) along the road segment between stops 1 and 2 and at stops 1 and 2. In contrast,

Figure 19.4b shows the case in which, because of a large headway, the two vehicles missed

their planned encounter; as a result, the passengers who wanted to transfer missed their

connection and will experience a longer wait. Stop 2 Stop 1 T S t 1 t 2 t 5 t 3 t 4 S 1 S 2 S 3 S 0 t p S p e 1 e 2 e 3 Stop 2 Stop 1 T S t 1 t 2 S 1 S 2 S 3 S 0 Vehicle #1 Vehicle #2 (b) (a) Vehicle #1 Vehicle #2

Figure 19.4 (a) Public transit vehicles making an encounter; (b) public transit vehicles missing an encounter.

T-S trajectories are constructed for each simulation run. With the use of line-intersection

techniques, the number of encounters (point and trajectory intersections as in Figure 19.4a)

is accumulated as is the number of missed encounters. Using this information, the following

formula may be established: $EP - SR - ME - SR = -$ (19.1)

where

EP is the encounter probability estimator

SR is the number of simulation runs

ME is the number of missed encounters

This simulation model can be integrated into the road-segment agent. The calculation of

EP will take place continuously for each group of transit vehicles planned for transferring

passengers.

19.3.2 Estimating encounter probability

A JAVA-based simulation was developed for estimating encounter probabilities. The simu

lation emulates the behavior of two vehicles along a road segment, given road-entry time

distributions, stop locations on the road segment, the number of passengers (boarding and

alighting at each stop), and travel-time distribution parameters between stops. Each vehicle

acts as an independent entity, using the threading capabilities of JAVA (the ability to emulate

the parallel processing of the simulation entities).

19.3.2.1 Postsimulation analysis

The postsimulation analysis includes the following: (1) finding the T-S overlap of each simu

lation run, (2) computing the encounter's probability estimator, and (3) computing the prob

ability of Vehicle #1 arriving before Vehicle #2 given that the vehicles did not meet along

the road segment.

Let $T_S v r v r$, be the time and space vectors for vehicle v at simulation run r , respectively,

representing the T-S trajectories constructed for each vehicle and simulation run, let $v r$ be the

entry time of vehicle v to the road segment at simulation run r , and E_r be the encounter coun

ter for simulation run r , ME_r be the missed encounter indicator for simulation run r , and Pv_r

be an indicator whether the vehicles did not meet and vehicle v_1 arrived before vehicle v_2 .

Using Equations 19.2 and 19.3 as described by Shaffer and Feustel (1991), it is possible to

enumerate all the encounter points and intervals (illustrated in Figure 19.4 as e_1 , e_2 , and e_3)

for each set of lines based on vectors $T S v$
 $r v r, : x x y x y x x x y x y x x x y y x x y y = - - -$
 $- - - 1 1 2 2 1 2 3 3 4 4 3 4 1 2 1 2 3 4 3 4$ (19.2) $y x y$
 $x y y y x y x y y y x x y y x x y y 3 4 = - - - - - 1 1 2$
 $2 1 2 3 3 4 4 3 4 1 2 1 2 4 3$ (19.3)

Moreover, it is possible to calculate the following: $ME = 1$
 $\text{sign}(E_r - r) = 1$ if $et_r < et_{r'}$ and $E_r = 0$ otherwise $r v_1 r v_2$
 $r r 1 1 0 0 = \leq = 0 0 0 0$ (19.5)

as well as construct the following encounter probability estimator (where SR is the number

of simulation runs): $EP_{SR} = \frac{ME_{SR}}{SR} = \frac{1}{SR} \sum (19.6)$

and these two conditional probabilities: $Pr_{v_1} = Pr_{v_1} | v_1 \text{ arrives before } v_2$
 $Pr_{v_2} = Pr_{v_2} | v_2 \text{ arrives before } v_1$
 $Pr_{v_1} = \frac{ME_{12}}{ME_{12} + ME_{21}} = \frac{1}{2} \sum (19.7)$
 $Pr_{v_2} = \frac{ME_{21}}{ME_{12} + ME_{21}} = \frac{1}{2} \sum (19.8)$

Equations 19.6 through 19.8 are the main constructors of the public transit-system objec

tive function of minimum TT (Hadas and Ceder, 2008b).

19.3.3 Deployment of tactics

The stochastic nature of transportation networks enforces a reduction in encounter probabili

ties. Hence, in order to improve encounter probabilities, a change of tactics must be imposed

on the vehicles. A list and description of possible online tactics in transit operations appear

in Section 18.6.3. The main real-time control tactics for the encountering- coordination

problem are as follows:

- Holding the vehicle (at terminal, midroute point)
- Skip-stop operation
- Changes in speed (but not above the lawful speed limit)
- Short-turn operation
- Shortcut operation

A set of tactics will be available for deployment for each road segment and vehicle. The

particular tactic or tactics deployed will be decided by the transit agency. For example, the

agency may decide to employ a shortcut tactic if, according to demand forecasts, passengers

seldom wait to embark a long the section(s) skipped.

19.3.4 Improving encounter probabilities

Chapter 18 indicates that an important aspect of operating a public transit system is the use

of online information to change vehicles routes and/or timetables dynamically. The changes

are made in order to increase the system's objective functions (TTT and encounter probability

for transferring passengers). Estimating an encounter probability is crucial to deciding

whether or not an encounter is plausible. Moreover, information of a maximum encounter

probability is important for deciding whether it is

worthwhile to change tactics (which

might cause increased travel time for waiting passengers).

In order to achieve maximum overlap of T-S trajectories, a time shift must be placed

on a vehicle's road-segment entry time. Figure 19.5 illustrates an overlapping situation

between two vehicles that depart at the same time from stop 1. Note that their dwell

times at stops 1 and 2 are not the same. This S-shaped overlap has a time length of

$(T_b - T_a)$ and a space length of $(S_2 - S_1)$. Two areas of Figure 19.5 are defined by A_1 and

A_2 in units of space multiplied by time; A_1 is between S_1 and S_0 , and A_2 is between S_3

and S_2 . The simulation program calculates these areas for an average speed between

each two adjacent stops for each vehicle. These areas are used to calculate an average

weighted headway, H_w , in time units between the two vehicles along the road segment

as follows: $H_w = \frac{A_1 + A_2}{S_3 - S_0}$ (19.9)

The rationale for Equation 19.9 is that the smaller the H_w , the larger is the overlapping area.

For two parallel trajectories, H_w will be the same as the constant headway between the two

trajectories. An algorithm was constructed, first, to calculate H_w for each S-shaped overlap

that exists between two adjacent stops and, second, to determine the overlapping case in

which H_w is the minimum. S_3 Stop 2 Stop 1 S_1 S_0
 t_1 t_2 T_a T_b T_A T_B

Figure 19.5 Time-space diagram illustrating an overlapping

situation.

19.3.5 Finding the time gap for maximum time-space diagram overlap

The model described (Hadas and Ceder, 2008c) finds the tactics that will minimize the

objective function of TTT based on a set of encounters that can be improved after calculat

ing the current encounter probability estimator and the maximum encounter probability.

The remaining-gap vector (for all encounters) is the state of the variable vector because each

combination of tactics, at each step, will reduce the remaining gap. Because the effect on the

objective function is calculated for each tactic, the optimal decision is known for each state.

Given road-segment characteristics, estimated time of road entry for two vehicles, and the

amount of passengers to board or alight, the following algorithm will produce the time gap

for maximum T-S diagram overlap (G). The algorithm calculates for each S-shaped overlap

that exists within the road segment (for each adjacent bus stop) the weighted average time

gap (H_w) and finds the overlap that produces $\text{Min}(H_w)$.

Let b denote the bus stop index, with $b = 0$ representing the segment entry point, and if

n bus stops are spaced along the road segment, $b = n + 1$ represents the exit point. Let d_{ib}

denotes the dwell time of bus i at bus stop b , and let $d_{i,i+1}$ denotes the average speed of

vehicle i between bus stops b and $b + 1$; L_b is the distance between bus stop b and bus stop

$b + 1$; L denotes the total segment length; t_i is the

entry time of vehicle i ; and A_b denotes the

areas boxed by the two vehicle T-S diagrams between bus stop b and bus stop $b + 1$. The

algorithm is as follows:

Initialization: $b = 0$, $T_i = t_i$, $G = 0$ Loop $G \neq T$,
 $H_w A_L$, $b \leq 1$ $T \neq L$ $V_d b \leq j \leq i$ $b \leq i \leq b$: $= - = + =$
 $+ \cdot + \sum 1^2$

If $b > n$ Finish, else return to Loop

Finish: $G \leq b$ $H_w b \leq b = \{ \} | : \min()$

G will be the maximum gap that must be overcome in order to achieve maximum encounter

probability.

19.3.5.1 Example

The following example is constructed for illustrating the simulation model. A road segment

has a total length of 3500 m and three bus stops serving two bus routes. Bus 2 is estimated

to enter the road segment $N(90,50)$ s behind Bus 1. The bus stops are spaced, 500, 1500,

and 3000 m from the road segment entrance. dwell time is 5 s per passenger with 12, 25,

and 20 passengers for Bus 1 and 10, 6, and 15 passengers for Bus 2. The travel speed has

lognormal (LN) distribution of x , based on Law and Kelton (2000), and is referred to as

$x \sim LN(\mu_x, \sigma_x)$. In this example, the travel speed from the entrance to bus stop 1 is in m/s

and has $LN(8, 10)$, from bus stop 1 to bus stop 2 is $LN(6, 20)$ m/s, and from bus stop 2 to

bus stop 3 is $LN(10, 13)$ m/s. The travel speed from bus stop 3 to the road segment exit is

$LN(8, 5)$ m/s. A general speed limit of 17 m/s

was enforced.

Four scenarios were constructed, each with a different S-shaped overlap and time gap, in

order to provide a simulation validation for the aforementioned algorithm. For each scenario,

the T-S diagram is shown in Figures 19.6 through 19.9, accompanied by 2D histograms of

a T-S encounter. These histograms show where the encounters will most likely take place. 4000 3500 3000 2500 2000 1500 1000 500 0

Space

ac

e (

m

) Time (s) Bus #1 105 s gap Bus #2 0 100 200 300 400 500
600 700 800 900 Encounters 40 35 30 25 20 15 10 5
0 Time (s) Space (m) 88 700 1313 1925 2538 3150 108
158 209 260 310 361 411 462 513 563

Figure 19.8 Time gap of 105 s.

4000

3000

2500

3500

2000

1500

1000

500 0 Time (s) Bus #1 130 s gap Bus #2 0 100 200 300 400
500 600 700 800 900 Encounters

Space

ac

e (

m

) 40 35 30 25 20 15 10 5 0 108 158 209 260 310 361 Time (s)

411 462 513 563 88 700 1313 1925 2538 3150 S p a c e (m)

Figure 19.7 Time gap of 130 s.

S p

a c

e (

m

)

4000

3500

3000

2500

2000

1500

1000

500 0 0 220 s gap 200 400 Time (s) Bus #1 Bus #2 600 800

1000 E n c o u n t e r s 40 35 30 25 20 15 10 5 0 108 158

209 260 310 361 411 Time (s) 462 513 563 88 700 1313 1925 S

p a c e (m) 2538 3150

Figure 19.6 Time gap of 220 s.

For each scenario, H_w was calculated, and the estimate of the encounter probability was

extracted from 100 s simulation runs. Table 19.3 summarizes the results. It can be seen that

Scenario (c) has the smallest H_w and the maximum encounter probability. These histograms

show where the encounters are most likely to occur. Such a histogram, aside from being a

tool for visualizing the simulation, can help the transit planner compare the most probable

encounter locations. This information, combined with the geometric features of the road

segment and the existing location of the transit stops, may help in relocating the stops so as

to expedite the transfer process.

19.3.6 Objective function improvement

Estimating encounter probability adds a new operational objective function, namely, the

total transit-system travel time (waiting, on board, and transfer times). The establishment of

this function will now be described. Let $V(i, j)$ be the vehicle number on the i th route leg for

the j th passenger in constructing the following equation:

$$PTT_{i,n} = ETR_{i,n} + \sum_{j=1}^{n-1} V(i,j) - V(i,j+1) + V(i,n) \quad (19.10)$$

where $PTT_{i,n}$ is the TTT for passenger i , the trip having n legs

$ETR_{i,j}$ is the estimated travel time on leg j
 $V(i,j), V(i,j+1)$ is the expected transfer time between leg j and leg $j+1$
 $0 \leq j \leq 10 \leq 15 \leq 20 \leq 25 \leq 30 \leq 35 \leq 40$ Encounter
 s

Sp

ac

e

(m

) 108 158 209 260 310 361 411 462 513 563 88 700 1313 1925
 2538 3150 Time (s) Space (m) 0

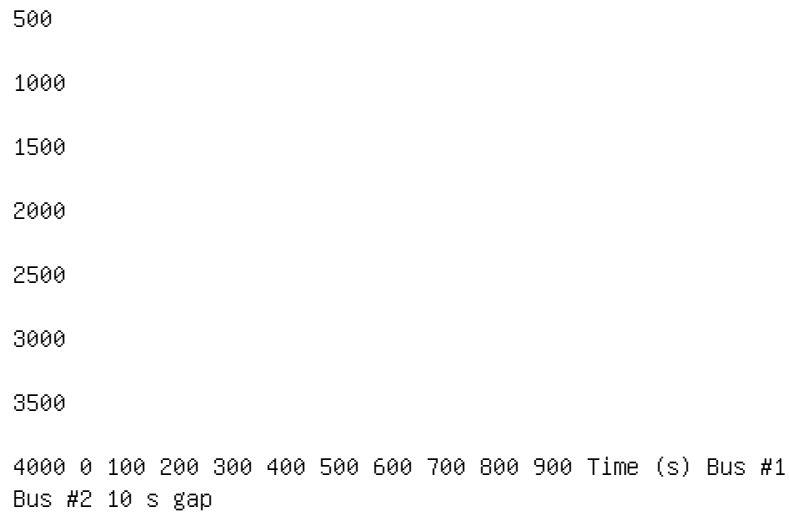


Figure 19.9 Time gap of 10 s.

Table 19.3 Simulation results

Scenario Headway at entry point (s) Weighted average headway (s) Encounters Missed encounters Encounter probability

(a) 220 153.6 90 58 0.42

(b) 130 51.4 263 7 0.93

(c) 105 45.7 400 0 1.00

(d) 10 59.3 144 25 0.75

The expected transfer time is

$$ETR_{V(i,j),V(i,j+1)} = EP_{V(i,j),V(i,j+1)} \cdot TR_{V(i,j),V(i,j+1)} + (1 - EP_{V(i,j),V(i,j+1)}) \cdot [Pr_{V(i,j),V(i,j+1)} \cdot W_{V(i,j),V(i,j+1)} + (1 - Pr_{V(i,j),V(i,j+1)}) \cdot W_{V(i,j)+1,V(i,j+1)}] \quad (19.11)$$

where $EP_{V(i,j),V(i,j+1)}$ is the encounter probability

$TR_{V(i,j),V(i,j+1)}$ is the direct transfer time $Pr_{V(i,j),V(i,j+1)}$ is the conditional probability that vehicle V_1 will arrive ahead of vehicle V_2 , given that they will not meet along the road segment $W_{V(i,j),V(i,j+1)}$ is the headway between two vehicles

The headway is an input to the model and is based on real-time information and estimated

time of arrival of the vehicles. The TTT of the system can be expressed by $TTT \min PTT_i = \Sigma$ (19.12)

Minimizing both travel time and waiting time prevents the possibilities of reducing

waiting time at the cost of much longer travel time, because of the operational tactics

deployment.

The objective function in Equation 19.11 can be optimized by a DP method (see Section

5.5 and Appendix 9.A for an example of DP formulation). Following the calculation of

both current and maximum encounter probability estimators, operational tactics (discrete

decision variables) are examined for minimizing the new objective function. Combining the

distributed computing capabilities of MAS with the DP approach can produce an efficient

algorithm for online use in optimizing real-world public transit systems.

19.4 SIMULATION MODEL FOR SYNCHRONIZED TRANSFERS

A brief review is introduced for describing synchronized transfers. The availability of

real-time information on bus locations, estimated arrival times, number of passengers,

and their destinations enabled Dessouky et al. (1999) to develop an algorithm of the bus

dispatching process at timed-transfer points. Such an algorithm can intelligently decide

whether or not a bus should be held in order to achieve a transfer with a late bus. In

another study, based on AVL technology, Dessouky et al. (2003) present a method to

forecast accurately the buses' estimated arrival times and to use bus-holding strategies

to coordinate transfers. The use of advanced public transit systems on a fixed route

and with paratransit operations was found to be important for improving departure

times and transfers by Levine et al. (2000). In order to overcome the complexity of trip

planning, Horn (2004) suggests an algorithm for the planning of a multilegged trip.

Its objective is to construct a journey while minimizing the travel time, subject to time

window constraints.

This section is based on Hadass and Ceder (2008a). Figure 19.10 illustrates the process

of improving the TTT, which is the sum of the trip legs and waiting time of all

passengers of a transit system. Use of operational tactics can help in performing transfers

at point X by changing the arrival time of a vehicle (or vehicles) to the transfer

point; this will decrease TTT (because of a shorter waiting time). On the other hand,

the deployment of tactics may increase the travel time of passengers on board the vehicle

(i.e., using hold tactics) or increase the waiting time of passengers at the bus stops

(i.e., hold and skip-stop tactics).

The illustration of Figure 19.10 shows that the time change can be reached by deploying

tactics along the upstream road segments of X (X - 1, X - 2, etc.) with the effect of

he

change on all road segments downstream to the furthest road segment in which tactics are

deployed ($X - 2$, $X - 1$, X , $X + 1$, $X + 2$, etc.). The optimization process needs to balance

travel-time changes in order to minimize TT. Clearly, this process fits into the description

of a DP problem that is formulated as an allocation problem with G (the estimated time gap

of arrival to the transfer point between the two vehicles) as a resource, TT as the objective

function, and N road segments as stages.

Because of incorporating stages in the model (each state is a road segment, as described

in Figure 19.10), multiple tactics (each tactic is an optional tactic that can be deployed at a road

segment), and transfer points (state variables representing the scheduled transfers based on

the synchronized timetables), its complexity will rise if the number of simultaneous transfers

is high. In order to reduce the level of complexity, a distributed DP model was developed

by Hadass and Ceder (2008a) breaking down the global problem into subproblems, each

with a single state variable. Such a model can be executed in a reasonable time because the

complexity of each subproblem is $O(n \cdot m)$; see Section 5.5 for comprehension of the measure

of complexity. This optimization model integrates the public transit network (timetables,

synchronized transfer points, operational tactics) with online data (such as public transit

vehicle locations, travel-time estimation, and demand forecasts) for the reduction of TTT

and increase of travel comfort. For example, if the estimated arrival time of a bus to the

transfer point is delayed (because of traffic congestion or increased dwell time) and as a

result the estimated encounter probability decreased significantly, the system optimization

model will recommend the deployment of tactics so as to increase the encounter probability

and at the same time to reduce the TTT.

19.4.1 Network simulation

The simulation model is constructed for (1) validating, evaluating, and analyzing the

benefits of the optimization and (2) comparing the performance of a highly complex global

optimization problem to a low-complexity suboptimal local optimization. This simulation

model is composed of two components: (1) simulation model of a public transit network and

(2) DP optimization model. X-2 X-1 Tactics deployment Tactic effect X X+1 X+2

Figure 19.10 Routes of two vehicles scheduled to transfer passengers at road segment X.

19.4.1.1 Simulation principles

The simulation model is discrete. In each step, buses are moved to the next road segment in

which the arrival time for each bus to each road segment is known, but all activities within

the road segment are performed in a single simulation step. The simulation model can be

executed in three modes:

- Not optimized, in which the operation of the bus is not altered during the simulation run; this is the basis for evaluating the optimization.
- Global optimization, in which the optimization is carried out for the whole network.
- Local optimization, in which the optimization is performed locally for each road segment.

For the three optimization modes, two parameters are in use: time horizon and space horizon.

The time horizon parameter determines the forecast range of optimization. Each time

optimization is carried out, all future arrivals of buses to the road segment within the time

horizon are treated. This parameter is relevant for global and local optimization processes.

The space horizon parameter is relevant for local optimization only. This parameter affects

the collaboration of neighbor road segments. If two or more road segments are within the

range of the space horizon parameter, then the optimization process is performed jointly;

otherwise, the optimization is carried locally.

Figure 19.11 illustrates the behavior of the space horizon parameter on a grid-shaped

transit network. The straight lines are the transit networks, the black dots are two road segments

with possible transfers between transit routes, and the circles are the space horizons.

In Figure 19.11a, calculations are done jointly as a single global optimization because of the intersection

section of the space horizons; in Figure 19.11b, there are two separate calculations, without

interactions, because both space horizons do not intersect.

The TTT is composed of riding time and waiting time according to the total travel-time

objective function. Riding time is estimated from a riding time estimation matrix, and wait

ing time is estimated as follows: (1) calculate the estimated arrival time of both buses to

the transfer road segment, (2) compute the direct transfer probability, and (3) compute the

waiting time.

An example, shown in Figure 19.12, represents a public transit network with the main

line connected to the railway and two feeding lines. It is difficult to solve such a network in a

real-life situation because of the three transfer locations associated with the main line shown

in all transfer areas. The example network of Figure 19.12 illustrates 14 road segments,

3 bus routes, and a train line; this network is used for evaluation and validation of the

(a) (b)

Figure 19.11 Space horizon parameters in a grid-shaped transit network: (a) joint calculation and (b) two separate calculations.

model. The bus routes and the train line characteristics are described in Table 19.4. Planned

transfers are along S 2 (for routes 1 and 2)
and along S 4 (for routes 1 and 3). The
transferring

passengers to the train line is from route
1 at a critical road segment (S13). The
transfer

road segments are marked with circles in F
igure 19.12. The demand between each pair of

road segments appears in Table 19.5. Passenger
arrival at each road segment is shown in

Table 19.6. The arrival time distribution reflects the case
in which most of the passengers are

aware of the published timetables and arrive before the
planned departure.

Riding time along each segment is distributed L
N with the parameters $L N(5, 2)$ in

minutes, except for the train road segment, w
hich is a constant of 30 min. Because the s
imu

lation model is only at the road segment l
evel, not dealing with intersegment c
haracteristics Route 1 S 1 S 2 S 3 S 4 S 5 S 6 S 7 S 8 S 9
S 10 S 11 S 12 S 13 S 14 Train line Route 3 Route 2

Figure 19.12 Network example. Table 19.5 Origin-destination
matrix in passengers per hour From To S 9 S 12 S 14 S 1 300
300 100 S 2 100 100 50 S 3 0 100 100 S 4 0 100 50 S 5 0 0
10 S 6 200 100 100 S 7 150 150 50 S 8 150 150 100 S 9 0 0 0
S 10 0 150 50 S 11 0 150 50 S 12 0 0 0 S 13 0 0 0 S 14 0 0
0 Table 19.4 Route characteristics Route no. First
departure Headway (min) Route layout 1 6:00 20 S 1 -S 2 -S
3 -S 4 -S 5 -S 13 2 6:25 20 S 6 -S 7 -S 8 -S 2 -S 9 3 6:40
15 S 10 -S 11 -S 4 -S 12 Train line 7:00 20 S 13 -S 14

(such as bus stops), a probability table for a direct
transfer was constructed based on the

time gap between two buses entering a road s
egment; that is, for a time gap of 1 min or
less,

the encounter probability is 0.95, whereas for

2, 3, 4, 5, and above the probabilities are 0.80,

0.60, 0.30, and 0.00, respectively.

For each pair of buses (or bus and train) according to the estimated time gap, an

encounter probability will be selected directly from that table. A number between 0 and 1

will be drawn from a uniform distribution, and if the result is less or equal to the encounter

probability, then a direct encounter occurred. In the case of a direct encounter, transferred

passengers will not experience any delay; otherwise, the transferred passengers will be

dropped at the road segment and wait for the next bus or train. For comparison of different

optimization types, the seed for the pseudorandom number generator can be selected, hence

generating the same set of data. The following two discrete tactics are available: (1) hold for

1, 2, 3, 4, or 5 min and (2) skip a segment that results with travel-time reduction of 4 min.

Output: For each passenger group departing the same origin, heading toward the same

destination, and arriving at the bus stop at the same time, the information of group number,

origin, destination, number of passengers, and start and end of trip time will be accumulated

and saved. For each trip leg, leg number, route number, bus number, transfer point,

transfer type (1 = direct, 2 = arrive before second bus, 3 = arrive after second bus), and

boarding and alighting time will be collected.

Scenarios: Eleven scenarios were constructed for the simulation on the basis of the program

parameters summarized in Table 19.7. These scenarios present the combined effect of major

attributes of public transit networks on the model: (1) headway, routes with long versus

short headways; (2) riding time variance, low variance (fewer disruptions to the traffic; low)

Table 19.6 Arrival time distribution
Arrival time
Percentage
Planned departure - headway 10
Planned departure - headway/2 20
Planned departure 55
Planned departure + headway/2 10
Planned departure + headway 5

Table 19.7 List of scenarios
Scenario Synchronized timetables
Time horizon (min) Headway Variance
1 Yes 10 Short Base 2 Yes 15
Short Base 3 No 10 Short Base 4 Yes
Yes + high load to train 10
Short Base 5 Yes 10 Long Base 6
Yes 20 Long Base 7 Yes 30
Long Base 8 No 10 Long Base 9
Yes 10 Short High 10 Yes 15
Short High 11 Yes 10 Short Low

versus high variance; and (3) synchronized transfers, whether or not the transit network

incorporates timed transfers (planned synchronization). The time horizon attribute is relevant

for the performance of the model and was included for comparison purposes.

Scenarios 1 and 2 present synchronized timetables. Scenario 3, for comparison, does

not have synchronized timetables; the purpose of the simulation, in this case, is to provide

insights on the effect of the model in synchronized and nonsynchronized networks. Scenario

4 provides a higher load to the train (which has fixed timetables, thus cannot deploy opera-

tional tactics). Scenarios 5 through 8 are characterized by long headways, and Scenarios

9 through 11 are characterized by high and low travel-time variance. These scenarios represent

typical examples of transit networks. One of the main assumptions for the model

is that the transit network of routes is synchronized (at the planning stage); the objective

of the optimization process is to approach better synchronization considering the changes

of travel times that may be the result of traffic congestion or change in passenger demand.

Hence, the simulation model was executed with synchronized timetables (Table 19.4) and

nonsynchronized timetables (Table 19.8).

Few options of increased time horizon were checked. The larger the time horizon, the

higher the complexity; however, it increases the ability to react earlier to more events. The

space horizon for the example was set to 1 in order to force local optimization.

Results: Each scenario was executed for a time period of 6 h, once per optimization type.

The full list of results appears in Hadass and Ceder (2008a). The main conclusions reached

in this study were as follows: (1) by using tactics, it is possible to attain a reduction in the

TT between 3% and 17%; (2) global optimization tends to yield better results than local

optimization; (3) because local optimization has only one

state variable and the execution of

each optimization problem is carried out in parallel, the methodology is suitable for online

optimization problems; (4) aside from the reduction in travel time, the number of direct

(without wait) transfers increased significantly, and this result has a significant effect on

the comfort of the ride and ease of transfer; and (5) the simulation results, which depend

on the scenario's characteristics as well as the optimization parameters, call for the devel

opment of a full-scale simulation system for the analysis of real-world networks and the

calibration of these parameters.

19.5 HOLDING AND SKIPPED/ SEGMENT TACTICS FOR TRANSFER SYNCHRONIZATION

The list of desirable features in designing efficient transit network appears in Chapters 12

through 15. We can then realize that in any public transit network, it is impractical to have

routes between every conceivable trip origin and every conceivable trip destination. There are

too many possible routes, so that service cannot be economically provided. Transfers in the

network enable fewer routes to provide the same coverage, and therefore, higher-frequency
Table 19.8 Headways for nonsynchronized scenarios
Route First departure Scenario 3 Scenario 8 Headway (min)
Headway (min) 1 6:00 21 32 2 6:25 18 42 3 6:40 12 27
Train line 7:00 20 20

and higher-quality services and any origin-destination (OD) demand can be accounted for

(Mees, 2000). Having fewer routes is also easier to remember, and this, along with increased

frequencies, is crucial for increasing the network attractiveness for passengers wishing to

travel without having to consult timetables or travel websites.

According to a study by Barton and Tsourou (2000) and Jakob et al. (2006), transfers

generally allow more flexibility in route planning and effective use of services, thus result

ing in more efficient public transit and the associated environmental, economic, health, and

social benefits. Knoppers and Muller (1995) argued that transfers are cited as a key rea

son for public transit being less attractive than cars. The stochastic nature of travel times,

del times, and passenger demands in transit networks leads to a noncounter probability

of two vehicles, termed a direct transfer. A direct transfer is defined as a no-wait transfer

between two vehicles meeting simultaneously at the same stop. When such a direct transfer

fails (missed transfer), it can cause frustration and longer waiting times for passengers who

miss connections, and a less efficient system as a whole. Transfer synchronization aims to

increase the number and probability of bus encounters and thus subsequent connections.

Domschke (1989), Ceder et al. (2001), and more recently Shafahi and Khani (2010) used

mathematical relationships to generate timetables

with the maximum opportunity for direct

transfers. Dessouky et al. (1999) show that it is also possible to control the time of dispatch

for a better synchronization, taking into account real-time information,

Another way of improving synchronization of transfers is by using specific operational

tactics in real time (Eberlein et al., 2001; Hickman, 2001; Ceder, 2007). The specific

tactics evaluated by Hadass and Ceder (2008a) in Section 19.4 were holding buses at stops

in anticipation of connecting buses and instructing skip-stops and shortcuts of routes to

meet subsequent connections. A recent study by Ceder et al. (2013) investigates how to

use selected operational tactics in transit networks for increasing the actual occurrence

of scheduled transfers. The model presented determines the impact of instructing vehicles

to either hold at or skip certain stops, on the TPT and the number of simultaneous

transfers.

This section is based on Nesheli and Ceder (2014), and it is a new extension of Ceder et al.

(2013). The section introduces the possibility of skip segment in addition to only skip a

individual stop, and for real-time operational control; the refinement takes place in the opti

mization formulation. That is, this section refers to three tactical scenarios: no-tactical case,

holding and skip individual stops, and holding and skip segments (H-SSs). The objectives

are to create simulation and optimization frameworks for optimal use of the three scenarios

and compare between the scenarios using a case study.

19.5.1 Formulation and modeling framework

The methodology of work commences by the use of TransModeler simulation tool (Caliper,

2013) to represent a real-life example and to generate random input data for the proposed

optimization model. Then a standard optimization software, ILOG (IBM, 2012), was used

to solve optimally a range of different scenarios determined by the simulation runs. Finally,

more simulation runs are made, containing the tactics determined by the optimization pro-

gram, so as to validate the results attained by the model.

19.5.1.1 Model description

The model developed (Nesheli and Ceder, 2014) considers transit networks consisting of

main and feeder routes. The transfers occur at separate transfer points for each route.

The formulation contains all the implemented tactics using a deterministic modeling.

Analytically, the model seeks to attain minimum TPTT and to increase, in this way, the

total number of direct transfers. The model formulates the tactics of holding vehicles, skip-

ping individual stops, and skipping segments, as well as indication of missing or making a

direct transfer. Thus, the components of the model are (1) the effect on TPT of holding a

vehicle, (2) the effect on TPT of skipping a stop/segment, and (3) the effect on TPT of

vehicle's being late, or not, to a transfer.

19.5.1.1.1 State variables

N Set of all bus stops, in which $n \in N$

R Set of all bus routes in which $\{r, r'\} \in R$

TF Set of all transfer points, in which $tr \in TF$

Q_r

max Passenger capacity of bus of route r

l_r

n Passengers' load of route r at stop n

b_r

n The number of boarding passengers of route r at stop n

a_r

n The number of alighting passengers of route r at stop n

$p_{rr'}$

n The number of transferring passengers of route r to route r' at stop n

d_r

n Bus dwell time of route r at stop n (in seconds)

h_r Bus headway of route r

c_r

n Bus running time of route r at stop n from the previous stop

A_r

n Bus arrival time of route r at stop n

D_r

n Bus departure time of route r at stop n

$\Omega(t)_r n$ Time penalty function of route r at stop n

$\Gamma_r n$ Time to reach a desired stop skipped of route r at stop n

T_r Bus schedule deviation of route r

TP_r

tr Transfer stop of route r at transfer point of tr

E_r Bus elapsed time of route r from the previous stop to the current position

m_r Maximum total number of stops of route r

k_r Positional stop of route r for a snapshot

ω Ratio between the average speed of a bus and the average walking speed of pedestrian (same ratio for all routes and stops)

19.5.1.1.2 Parameters

$\theta_r n$ The number of passengers of route r for a bus departing stop n

$\beta_r n$ The number of passengers waiting at stops further along the routes with respect to route r and stop n (future passengers)

$\gamma_r n$ The number of passengers who wish to have transfers at transfer points with respect to route r and stop n

$\lambda_r n$ The waiting time per passenger at previous stops due to applied tactics

19.5.1.1.3 Decision variables

$H0_r$

n Bus-holding time of route r at stop n

S_r

n Bus skipping stop of route r at stop n ; if stop, skipped
= 1, otherwise = 0

$Y_{rr'}$

n

γ Possible transferring from route r to route r' at
transfer stop n , pretactics; if a possible transfer
occurs = 0, otherwise = 1

$Z_{rr'}$

n

γ Possible transferring from route r to route r' at
transfer stop n , posttactics; if a possible transfer occurs
= 0, otherwise = 1

19.5.1.1.4 Assumptions

The model is designed deterministically. Therefore, the
following assumptions are made:

1. There is full knowledge of the route information, including average travel times, average passenger demand, average number of transferring passengers, and average dwell times.
2. Passenger demand is independent of bus arrival time.
3. Vehicles are operated in FIFO manner with an evenly scheduled headway.
4. Passengers will wait at their stop until a bus arrives (none leaves the system without taking the first arrived bus).
5. The bus arriving subsequently to a bus that skipped stop cannot use any of the tactics considered.
6. Passengers on board a bus that will skip segment will be informed on this action at the time of the decision so that they can alight before or after the skipped segment; it is to note that the formulation of optimization minimizes

these type of passengers and in most cases tested it is nil.

7. Stops where passengers want to transfer cannot be skipped.

19.5.1.2 Formulation and properties of holding tactic

Holding a vehicle is a tactic considered operationally for regulating undesired scheduled

deviations, reducing bunching and approaching a direct transfer at transfer points. However,

u sing t he h olding t actic h as s ome d rawbacks o
n t he T T T o f t he p assengers. T hat i s, t he
h old

ing t actic w ould a ffect t hree g roups o f p
assengers: (1) t hose o n b oard t he b us d efin
a s e r n ,

(2) those waiting for the bus further a long the route defined a section, and (3) those who wish

to have transfers defined as $r_n \gamma$. The following formulation can now take place. It is to note

t hat t he t erm " route" d escribes a t ransit s
ervice t hat s erves a s eries o f s tops. A r oute
i s m ade

u p o f a c o l l e c t i o n o f “ t r i p s”; e a c h t r i p r e p r e s e n t s a s i n g l e r u n, b a s e d o n a c e r t a i n d e p a r t u r e

t ime, a long t he s eries o f s tops o f t he r
oute. F or i nstance, a r n describes the number of
alighting

[illegible]

The effect of holding tactics on the change in total passenger travel time (ΔTPT_i) with

respect to route r and stop n is $\Delta TPTT$ Holding $H_0(r, n, r, n)$

$$r_n r_{n-1} r_{n-2} \dots r_1 () = + + \theta \beta \gamma \quad (19.16)$$

19.5.1.3 Formulation and properties of skip-stop and skip-segment tactics

Skip-stop: When a major disruption occurs, holding a vehicle, as the only tactic available,

cannot guarantee to obviate the headway variation from the schedule. This is true even with

holding the following vehicles because these actions may lead to greater schedule deviations

(Sun and Hickman, 2005). Skip-stop is a rather tactic that can be implemented to decrease

the irregularity of service and increase the number of direct transfers. The advantage of

skip-stop is for passengers who already are on board the bus and those to board down

stream. On the other hand, it has an adverse effect on passengers who want to alight or

board at the skipped stop.

Skip segment: The skip of individual stop tactic has the limit that no more than one stop can

be skipped in a row. However, if, for instance, some stops are close to each other and only

a very few passengers will be impacted from skipping those stops, a rather tactic can take

place as is illustrated by Figure 19.13. That is, consideration of a skip-segment tactic, where

a segment is defined as a group (one or more) of adjacent stops.

One of the assumptions made is that passengers on board a bus that will skip segment

will be informed immediately on this action at the time of the decision so that they can

Synchronized transfers mean that two or more buses, or other transit vehicles, meet at the

same time at a transfer point. As told, this can be improved by the use of real-time opera

tional tactics. For missed transfers, passengers have to tolerate a waiting time of $(h_r - T_r)$

where h_r is the headway at route r . Therefore, a direct transfer occurs if the following holds: $\forall r, r' \in N, r \neq r' : \tau_{rr'} = 0$ (19.24) $\forall r, r' \in N, r \neq r' : \tau_{rr'} = 0$ (19.25)

In this model, a possible transfer $\forall r, r' \in N$ (before implementing tactics) occurs if bus depar

ture time on route of r' is after the bus arrival time on route r and vice versa for $\forall r, r' \in N$.

Same arguments apply for $\forall r, r' \in N$ and $\forall r, r' \in N$ after utilizing tactics. If the following conditions

hold, direct transfers will be possible and none of the buses will be late at a transfer point: $\forall r, r' \in N, r \neq r' : \tau_{rr'} = 0$ (19.26) $\forall r, r' \in N, r \neq r' : \tau_{rr'} = 0$ (19.27) if $D \neq \emptyset$ then $\forall p, r, r' \in N, r \neq r' : \tau_{rr'} = 0$ (19.28)

Subsequently, the effect of transfers on the change of the TTT for route r and stop n is $\Delta TTT_{transfer} = \sum_{r' \in N, r' \neq r} (\tau_{rr'} - \tau_{rr'})$ (19.29)

19.5.1.5 Objective function

According to the formulations derived of the TPTT, an objective function for the proposed

model can be written as $\min \sum_{r \in N} \sum_{n \in N_r} (TPTT_{holding} + TPTT_{transfer})$ (19.30)

This objective function, of minimum TTT, is subject to the following constraints.

19.5.1.6 Constraints

tactics are applied: if $TP \cap TP \cap i \cap tr \cap TF \cap rr \cap tr \cap r \cap tr \cap r \cap tr \cap 1 \cap r$
 $r \cap tr \cap r \cap tr \cap S \cap * \cap Z \cap TP \cap TP \cap k = \epsilon' - - \sum \Pi < \forall > > () 1 1$
 (19.44) if $k \cap TP \cap r \cap i \cap tr \cap TF \cap rr \cap tr \cap r \cap tr \cap r \cap r \cap tr \cap r \cap r \cap tr \cap S \cap * \cap Z \cap TP$
 $k \cap TP = \epsilon' - \sum \Pi < \forall > > () 1 1$ (19.45) if $k \cap TP \cap r \cap i \cap rr \cap 1 \cap r$
 $1 \cap r \cap r \cap 1 \cap S \cap * \cap Z \cap TP \cap k = \epsilon' \sum < \forall > () 1$ (19.46)

Constraints (19.44 through 19.46) can be simplified. Let us denote I_m as $I_m \cap Z \cap tr \cap TF \cap rr \cap tr = \epsilon' \sum$ (19.47)

The maximum value of I_m implies that the use of the tactic does not result in a transfer and

$Z = 1$ for all routes at all transfer points. Thus, the following constraint ensures the use of

tactics when a transfer occurs: if I_m size of Z , then $S \cap TP \cap TP \cap k \cap rr \cap tr \cap i \cap TP \cap TP \cap r \cap i \cap r \cap tr \cap r \cap tr \cap 1 \cap r \cap tr \cap r \cap tr = \epsilon' > > \epsilon' - - \sum 1 \cap 0 \cap r ()$ (19.48)

The same applies for Constraints (19.44) and (19.45). For using skip-stop (not segment)

tactic, the following constraint is for skipping only one stop in a row: $S \cap r \cap n \cap r \cap n+1 \cap + \leq 1$ (19.49)

To restrict the number of passengers on board when the bus departs the stop, the following

constraint applies: $\theta \cap r \cap n \cap r \cap \max Q \leq$ (19.50)

19.5.1.7 Model optimization

The formulation of the problem with the three scenarios (no tactics, holding and skip-stop

[H-S], H-SS) can be solved using constrained programming (CP) technique. The CP is an efficient

approach to solving and optimizing problems that are too irregular for mathematical

optimization. The reasons for these irregularities as per IBM (2012) are as follows: (1) the constraints are nonlinear in nature, (2) an uncon-
 vexe solution space exists with many local

local-optimal

solutions, and (3) multiple disjunctions exist resulting in poor returned information by a linear

relaxation of the problem. A CP engine makes decisions on variables and values and, after each

decision, performs a set of logical inferences to reduce the available options for the remaining

variables' domains. This is in comparison with a mathematical programming engine that uses

a combination of relaxations (strengthened by cutting planes) and "branch and bound."

19.5.2 Case study of real-time tactics implementation

Examination of the model developed (Nesheli and Ceder, 2014) took place with a real-life

case study. The data of the case study were collected in Auckland, New Zealand. It is to

note that all buses of the Auckland transport network are equipped with AVL systems. The

primary benefits of the AVL system are in the communication and processing of data for

service monitoring, fleet management, and traveler information. In addition, AVL data are

used to analyze bus travel times and schedule adherence (to implement holding tactics). This

means that real-time information on bus locations can be used to predict bus arrivals at the

transfer points and to determine bus schedule deviations.

The transit network of the case study consists of three bus routes and two transfer points.

The first route is known as Northern Express with a dedicated lane that runs from the sub

urb across the Auckland CBD area and has quite a high number of passengers during peak

hours. The second route, route 858, runs north-south (east of the first route), and the third

route, route 880, is a loop that serves as a feeder route. Figure 19.14 illustrates the three

routes and the two transfer points used.

19.5.2.1 Data

The data were recorded and available are route characteristics based, that is, stop ID, lon

gitude, latitude, stop sequence, stop lag, stop code, stop name, route ID, user ID, point ID,

and route number. A bus capacity of 60 passengers is assumed with 40 seated passengers

and 20 standing. The simulation package used, TransModeler 3.0 (Caliper, 2013), considers

passenger crowding for the calculation of the dwell time. That is, it takes longer to board or

alight a bus if there are more standing passengers. In addition, bus dwell times were taken

from Dueker et al. (2004) for simulation use. It affects whether or not the stop be skipped.

The dead time (time spent at the stop without boarding or alighting) is set to 4 s as the Transfer point 1 Transfer point 2 Route Northern Express Route 858 Route 880 Transfer point

Figure 19.14 Bus system and the study routes, Auckland, New Zealand.

default value of TransModeler. Passenger demand in terms of OD matrix is being estimated

based on the average number of passengers boarding and alighting at each stop, from the

first stop of the route to a transfer point. T

that is, O D is estimated based on proportions of

the number of boarding, alighting, and onboard passengers. Average headways and dispatch

times were assumed to be known at both transfer point 1 and transfer point 2. In addition,

vehicle headways considered are 5, 10, 15, and 20 min (same for all routes).

The case study was simulated for analyzing and validating the performance of the model

before and after implementation of the tactics. Moreover, the transit network has been

simulated for handling the concept of synchronized transfers. Overall, the analysis includes

simulation (Caliper, 2013) and optimization (IBM, 2012).

19.5.2.2 Results

Three scenarios were designated “no tactics,” H-S, and H-SS. Note the difference between

skip-stop and skip segment. Figure 19.15 demonstrates the significant better results for the

H-SS combined tactic than for the H-S combined tactic especially for short headway cases.

It illustrates the effect of the model on the TPTT for transfer point 1.

In Figure 19.15a, the results of short headways are completely different from the results

of the long headways, in Figure 19.15b, in terms of the shape of the trend before and after

the no-tactic zone. Figure 19.15 shows that by using the combined tactics, compared with

the no-tactic scenario, a considerable reduction of TTT is attained. Figure 19.15a and b

shows that when the scheduled deviation ends too early, the maximum saving of TTT occurs

without the use of any tactics; this maximum travel-time saving coincides with maximum numbers of

direct transfers. It is also observed, from Figure 19.15, that the scheduled deviation interval,

in which no tactic is used, is larger for long than short headways.

The different shapes of trend in Figure 19.15a and b deserve explication. Figure 19.15a

illustrates that a larger reduction of TPTT is possible when the bus is behind schedule. This

suggests that passengers waiting for a late bus prefer, in the short headway cases, to con-

tinue to wait than to find an alternative solution (walking or using another travel mode). In

this case, it is worth applying the combined tactics. Figure 19.15b demonstrates a completely

different pattern; it shows that, in the long headway case, travel time is not saved much for

large scheduled deviations. That is, unreliable service for long headway cases cannot be com-

pensated by the use of tactics and passengers will tend to find, for such a service, an alternative

solutions. However, if the deviations are reasonable, the use of tactics can save travel time

and increase the number of direct transfers.

It is to note that the deviations (in seconds) of

Figure 19.15 are based on a few assumptions, thus cannot be translated to exact figures

of being behind or a head of the planned schedule. Instead, their trends provide a new insight

of when to use the combined tactics proposed.

More on this concept of evaluation and decision of best operational tactics in real-time

scenarios appears in Ceder et al. (2013) and Nesheli and Ceder (2014).

19.6 TRANSFER SYNCHRONIZATION USING IVC

This section considers another aspect related to transfer synchronization, that is, the consideration

of an intervehicle communication (IVC)-based scheme to optimize the synchronization

of planned transfers in transit networks. By using this scheme, transit drivers can share their

real-time information, such as vehicle location, vehicle speed, and the number of passengers,

with each other through a central communication coordinator. This scheme is especially useful

for transit drivers from different routes and driving toward the same transfer point because

they can communicate and collaborate with each other, and thus dynamically adjust their run

ning speed on transit-route segments or holding time at transit stops. Through this cooperative

driving approach, the frequency of simultaneous arrivals of transit vehicles at a transfer point

can be improved. The section follows the work of Liu et al. (2014).

19.6.1 Overview of IVC

IVC enables drivers to communicate with other drivers that locate outside the range of

line of sight and share real-time traffic information with each other. IVC has the ability to

Headway (s)	5 min (H-SS)	5 min (H-S)	10 min (H-SS)	10 min (H-S)
Direct transfers occur without tactic (Y=Z)	-9000	-8000	-7000	-6000
Change of total passenger travel time (s)	-5000	-4000	-3000	-2000
Direct transfers occur without tactic (Y=Z)	-9000	-8000	-7000	-6000
Change of total passenger travel time (s)	-4000	-3000	-2000	-1000
Headway (s)	-245	-206	-168	-128
Headway (s)	-92	-69	-47	-23
Headway (s)	0	27	60	100
Headway (s)	140	177	Schedule deviation (s)	15 min (H-SS)
Headway (s)	15 min (H-S)	20 min (H-SS)	20 min (H-S)	Direct transfers occur without tactic (Y=Z)

(a)

(b)

Figure 19.15 Total passenger travel time with the two combined tactics of H-S/segment at transfer point 1. (a) Δ TPTT for short headway (5 and 10 min) and (b) Δ TPTT for long headway (15 and 20 min).

extend the horizon of drivers and onboard devices (OBDs) and thus has the potential to

significantly improve road traffic safety, efficiency, and comfort (Luo and Hubaux, 2006).

It is recognized as an important component of intelligent transportation systems (ITS) and

has attracted significant research attention around the world. Since the 1980s, a number

of research projects focusing on IVC have been set up, such as CarTalk 2000, Fleetnet,

PATH, Car-to-Car Communication Consortium, Network on Wheels, and COMeSafety

(Reichardt et al., 2002; Sichitiu and Kihl, 2008). Recent research focuses mainly on vehicu

lar ad hoc networks, which are a kind of mobile ad hoc

networks offering direct communi

cation between vehicles (Hartenstein and Laberteaux, 2008).

The main application areas of IVC are collision warning, traffic monitoring, traffic coord

ination, traveler information support, traffic light scheduling, targeted vehicular com

munications, and car-to-land communication (Sichitiu and Kihl, 2008). These appealing

applications have attracted many researches. However, the majority of current research is

focused on cars. Not so much research has been done about transit vehicles.

19.6.2 Methodology Using IVC

We will describe in this section a semidecentralized control strategy for transit-vehicle trans

fer synchronization.

Consider a directed transit network $G = (N, R)$ with a set N of stops and a set R of

routes. The transit routes are divided into a few route segments by transfer points, as is

shown in the upper of Figure 19.16a. A communication control center is assigned to an $C_1 C_2 C_3 C_4 C_6$ Route 1 Route 2 Route 3 Route 4 $C_5 S_1 S_2 S_3 S_4 S_5 S_6 S_7 S_8 S_9 S_{10} S_{11} S_{13} S_{14} S_8 S_{16} S_{18} S_{15} S_{17} S_{12}$ (a) Segment C Section 1 Section 2 Section N Stop 1 Stop 2 Stop N-1 Stop N (b)

Figure 19.16 Public-transit network example illustrating network-level semidecentralized control strategy: (a) a basic transit network with communication centers and (b) an illustration of the concept of route sections.

intersection point or a shared route segment where passengers make a transfer. The com

munication control center is responsible for the communication coordination of vehicles

on the route segments leading to it. A route segment is divided into a few route sections

determined by transit stops along it, as shown in the lower of Figure 19.16b. The route

section is further divided into a few smaller intervals.

An interval is a basic unit controlled by the communication control center that delivers

advisory speed information to a transit vehicle when it arrives to each interval. New infor

mation is sent to the driver when moving to the next interval. A simple example is shown in

Figure 19.16a. In this example network, there are 4 transit routes R_i ($i = 1, 2, \dots, 4$), and the

whole network is divided into 18 route segments S_j ($j = 1, 2, \dots, 18$). The segments S_{15} and

S_{17} are shared road-segment encounter transfers, while the other segments direct to a fixed

single-point encounter transfer. Six communication control centers C_k ($k = 1, 2, \dots, 6$) are

assigned to the corresponding transfers and are responsible for their own route segments.

For example, C_1 is in charge of the coordination of vehicles on route segments S_1 and S_9 ,

while C_5 is in charge of S_6 , S_8 , and S_{15} .

Transit vehicles controlled by the same communication control center are in the same

group and can communicate with their peers to share traffic information. The communi

cation coordinator in the control center delivers advice on the real-time optimal running

speed. Drivers then can adjust their running speed in a cooperative manner in order to

achieve a simultaneous arrival. Once a transit vehicle passes the communication control

center, it automatically joins another group of vehicles. The communication group is a

dynamic and self-organized. It is to note that, from the entire transit-network perspective,

different communication groups can be performed at the same time. That is, the process

is a decentralized and parallel process. However, from a single-route segment perspective,

the communication activity is severely governed by the communication control center, and

thus, it is a centralized process. Thus, this control strategy is termed semidecentralized. By

doing so, drivers in the same communication group drive in a cooperative manner, and the

opportunity of simultaneous arrivals improves.

19.6.2.1 Conceptual illustration

A simple example is used as an explanatory device to show the main components of the proposed

IVC systems and to outline its potential benefits. The example is shown schematically

in Figure 19.17, in which two transit vehicles are moving toward the same single-point

transfer. Both vehicles belong to the same communication group and are controlled by the

same communication control center. The center comprises mainly a central server and a

communication coordinator.

An OBD is installed on the transit vehicle to receive signals from global positioning sys-

tem (GPS) satellites. The OBD can record information about vehicle ID, vehicle location,

vehicle speed, time, route direction, and route ID. A SIM card is embedded in the OBD.

The recorded data then can be transmitted to the database in the communication control

center through GSM/GPRS networks. These data are visualized in geographic information

system (GIS) maps. The communication coordinator has knowledge about the relatively

accurate location and speed information of the vehicles of the same group. Based on the

knowledge, a advisory optimal speed information is disseminated to the drivers in the group

so as to meet simultaneously or within a given time window at the same transfer point. The

advisory information can be displayed online to the driver on the onboard variable mes

sage sign installed in the vehicle. This will allow for a peer-cooperative communication.

We note that the basic assumption of the IVC system is that drivers will comply with the

recommended (feasible) speed and holding times so as to materialize the direct transfers of

passengers without waiting time. The control center will have a record of this compliance to

help in minimizing issues associated with driver behavior.

19.6.2.2 Main features

The main features of the CBVC-B are related to the whole transit-network communications

comprised of different decentralized and parallel groups.

The communication-based con

trol process can be performed at the same time between different communication groups.

However, technically, bus drivers of the same communication group are not exactly com

municated in a direct peer-to-peer manner; it is more of a client-server way. Therefore, the

whole control process is termed semidecentralized group communication. In a communication

group, bus drivers leading to the same transfer point serve as clients, and the communication

control center serves as the central server. Bus drivers through the central server share vehicle

and passenger information with their peers. The central communication center is responsible

for the communication coordination of vehicles on the route segments and delivers advice

on the real-time vehicle control tactics to bus drivers. Once a bus passes the communication

control center, it automatically joins another group of vehicles. The communication group is

self-organized.

A small transit network is used in Figure 19.19a to illustrate the basic concepts. This

network is from Ceder et al. (2001). The network is divided into four communication

groups by four transfer points shown in Figure 19.18b. Each group has two routes lead

ing to the transfer point, and a central server is assigned to be responsible for the com

munication between bus drivers. GPS satellite GSM/GPRS

Route 2 VMS VMS Vehicle 1 Route 1 Central server

Communication coordinator Database Transfer 2B* 40 min 1C*

25 min Vehicle 2 2A 20 min 1A

10 min 1B 10 min

Figure 19.17 Graphical illustration of transfer synchronization of two public transit routes based on the intervehicle communication systems.

19.6.3 Model

The model developed by Liu et al. (2014) assumes given GPS information of transit-vehicle

speed and location. It is a distance-based dynamic speed-adjustment model based on IVC

systems for vehicle running speed update under both scheduled single-point encounter and flexible

road-segment encounter scenarios. In this model, drivers of different routes can share

their real-time travel speed and location information through the IVC communication con-

trol center. Drivers can adjust their travel speed or holding time at transfer points dynami-

cally and cooperatively in order to achieve a preplanned direct transfer. The main objective

of the model is to increase the number of simultaneous arrivals at all transfer points. This

approach analyzes the impact of the changes on the TPTT, in-vehicle travel time, out-of-

vehicle waiting time at stops, and transfer waiting time. These impacts are studied on a net-

work-wide scale using the optimization model presented in Sections 19.6.3.1 and 19.6.3.2.

19.6.3.1 Notations

R Set of all PT routes $r \in R$.

S Set of all segments $s \in S$ of all routes $r \in R$.

N_{sr} Set of all stops on segment s of route r , $n \in N_{sr}$.

$|N_{sr}|$ Total number of stops on segment s of route r .

PI_n

sr Positional index (k) of a stop n on segment s of route r .

q_k

sr Vector listing all stops on segment s of route r in order defined by positional index (k).

M_{sr} Set of intervals on segment s of route r , $m \in M_{sr}$.

$|M_{sr}|$ Total number of intervals on segment s of route r .

p_{in}

sr Number of passengers assumed to be arriving in vehicle i at stop n on segment s of route r ; this is derived from a previous O D matrix; the number of the first stop on route r is the initial number of passengers on the vehicle.

e_{in}

sr Number of passengers entering vehicle i at stop n on segment s of route r .

l_{in}

sr Number of passengers leaving vehicle i at stop n on segment s of route r .

d_i

$s_{r,i}$ Length of segment s of route r on which vehicle i is moving.

$d_{i,m}$

$s_{r,i}$ Distance of vehicle i on segment s of route r in interval m to the nearest transfer point. 1 4 2 3 Route 3 Route 4 Route 1 Route 2 (a) (b) Group 1 Group 2 Group 4 Group 3 Bus of Route 1 Bus of Route 3 Bus of Route 2 Central server Bus of Route 4

Figure 19.18 Cooperative group communication between bus

drivers in a transit network: (a) an example of bus transit network and (b) semidecentralized group communication between bus drivers of the example.

d_s

$s_{r_i r_j}$ Length of shared road segment s between segment s_i of route r_i and segment s_j of route r_j .

t_{in}

sr Average running time of vehicle i on section between stop $n - 1$ and stop n of segment s of route r , posttactics.

t_{in}

sr

Average running time of vehicle i on section between stop $n - 1$ and stop n of segment s of route r , pretactics.

$\Delta t_{in sr}$ Average running-time difference of vehicle i on section between stop $n - 1$ and stop n of segment s of route r between posttactics and pretactics, that is, $\Delta = -t_{in sr} - t_{in sr}$.

A_t

i_q

$s_{r_i k} s_{i r_i i i}()$ Arrival time of vehicle i of segment s_i of route r_i at transfer stop $q k s_{i r_i}$, posttactics.

D_t

i_q

$s_{r_i k} s_{i r_i i i}()$ Departure time of vehicle i of segment s_i of route r_i at transfer stop $q k s_{i r_i}$, posttactics.

ΔD_q

m

k

$s_{ir i}$ Dwell time of vehicle i of segment s_i of route r_i within the interval m , that is, $\Delta D(t) A(t) q_m i q s_{r i q s r k s_{ir i k s_{ir i i k s_{ir i i i} = - .$

h_i

q

$N s_{ir i}$

$s_{ir i}$ Holding time of vehicle i of segment s_i of route r_i at transfer stop $q N s_{ir i s_{ir i} .$

$A(t)$

$i q$

$s_{r k s_{ir i i i}$ Arrival time of vehicle i of segment s_i of route r_i at transfer stop $q k s_{ir i}$, pretactics.

$D(t)$

$i q$

$s_{r k s_{ir i i i}$ Departure time vehicle i of segment s_i of route r_i at transfer stop $q k s_{ir i}$, pretactics.

v_{im}

s_r Average speed of vehicle i in interval m on segment s of route r .

$V_{min i, m s_{r i i}$ Minimum average speed of vehicle i in interval m on segment s of route r .

$V_{max i, m s_{r i i}$ Maximal average speed of vehicle i in interval m on segment s of route r .

p

$i q j q$

$s_r s_r$

$N s_{ir i}$

$s_{ir i N s_{jr j s_{jr j}$

$i i j j$ Number of passengers wishing to transfer from

vehicle i of segment s_i of route r_i at transfer stop $q_{N s_i r_i s_i r_i}$ to vehicle j of segment s_j of route r_j at transfer stop $q_{N s_j r_j s_j r_j}$.

twt

iq_{jq}

$s_r s_r N s_i r_i s_i r_i N s_j r_j s_j r_j$

$i i j j$ Transfer walking time required from vehicle i of segment s_i of route r_i at transfer stop $q_{N s_i r_i s_i r_i}$ to vehicle j of segment s_j of route r_j at transfer stop $q_{N s_j r_j s_j r_j}$. It follows that if transfer stops $q_{N s_i r_i s_i r_i}$ and $q_{N s_j r_j s_j r_j}$ are a shared transfer stop, then $twt_{iq_{jq} s_r s_r N s_i r_i s_i r_i N s_j r_j s_j r_j i i j j} = 0$.

19.6.3.2 Distance-based dynamic speed-adjustment model

Consider a transfer point of two transit routes. The time-distance diagram of vehicles for

fixed single-point encounter scenario and flexible road-segment encounter scenario is shown

in Figure 19.19a and b, respectively. The time horizon of each route segment is divided into

$|M_{sr}|$ intervals of equal size. The length of each interval is τ . Drivers will receive real-time

advisory speed-adjustment information by a communication coordinator for each interval

and holding time information for the last interval based on their distance to the transfer

point.

Fixed single-point encounter scenario: The update rules for average running speed adjust

ment, for the fixed single-point encounter scenario, are given as follows:

if $d_{dv} > a; v_{vi, m s_r j, m s_r i, m+1 s_r i, m s_r j, m+1 s_r j, m s_r i i j j i i i i j j j} \geq + =, j b-$.

else if $d_{dv} > b v_{vi, m s_r j, m s_r i, m+1 s_r i, m s_r$

$j, m+1$ s r j, m s i i j j i i i i j j j $\leq - \prime =, ; r j a + \prime$
 $.$

else d d i, m s r j, m s r i i j j = . if v v v v v i, m s r
j, m s r i, m+1 s r i, m s r i, m s r j, m s i i j j i i i i i
j $\geq - -$, max 1 2 r i, m+1 s r j i i Vmin () $\square \square \square \square \square \square$,
, v v v v Vmin i, m+1 s r i, m s r i, m s r j, m s r i, m+1 s r
i i i i i j j i i = - - () max , 1 2 $\square \square \square \square \square \square$; v v v
v Vmax j, m+1 s r j, m s r i, m s r j, m s r j, m+1 s r j j j j
i i j j j j = + () min , 1 2 - $\square \square \square \square \square \square$. d s r d s i r
i i d s j r j j Direct transfer point Route 1 Route 2 Time
D i s t a n c e 0 t (a) d s r d s i r i s j r j d s i r i D
i s t a n c e i d s j r j j Feasible direct transfer points
Shared route sections Time 0 s t Route 1 Route 2 (b)

Figure 19.19 Time-distance diagrams of two vehicles moving to the same transfer point: (a) Fixed singlepoint encounter transfer and (b) flexible road-segment encounter transfer.

else if v v i, m s r j, m s r i i j j < , v v v v Vmax i, m+1
s r i, m s r j, m s r i, m s r i, m+1 s r i i i i j j i i i i =
+ - () min , 1 2 $\square \square \square \square \square \square$; v v v v Vmin j, m+1 s r j, m
s r j, m s r i, m s r j, m+1 s r j j j j j j i i j j = - - ()
max , 1 2 $\square \square \square \square \square \square$.

else v v v v v i, m s r j, m s r i, m+1 s r i, m s r j, m+1 s
r j, m s r i i j j i i i i j j j j = = , ; . end.

where

a a nd a' a re a ccelerations

b a nd b' a re d ecelerations

The average running speed of vehicle i in interval m on
segment s of route r, $v_{im sr}$ is defined

as f ollows: v d d A(t) A(t) D im sr i, m+1 s r i, m s r i
m+1 i m q m i i i i k s i r i = - - - $\Sigma \Delta$ (19.50)

For distance d i, m s r i i larger than distance d j, m s r
j j , vehicle i will be accelerated by a and vehicle j will

be decelerated by b. If distance d i, m s r i i is less
than distance d j, m s r j j , then vehicle i will
decelerate by

b' and vehicle j will accelerate by a'. For distances d i, m
s r i i and d j, m s r j j equal and speeds v i, m s r i
i and v j, m s r j j

not equal, the next speeds $v_{i,m+1}$ and $v_{j,m+1}$ are to be adjusted to attain equal speeds. If both

the distance and speed are equal, the speeds remain unchanged. The values of a , b , a' , and

b' for each case are determined as follows:

Case I : $d_{i,m} s_{r i i} > d_{j,m} s_{r j j}$: Let $\lambda \tau =$
 $(d_{i,m} v_{i,m} s_{r i i} - d_{j,m} v_{j,m} s_{r j j}) / (v_{i,m} s_{r i i} - v_{j,m} s_{r j j})$,

where τ is the length of interval. The values of a , b are obtained by solving the following

minimization problem:

$$\min a + b - \lambda \quad (19.51)$$

subject to $v_{i,m} s_{r i i} \leq V_{\max}$ (19.52)

$v_{j,m} s_{r j j} \geq V_{\min}$ (19.53)

$$a \geq 0 \quad (19.54)$$

$$b \geq 0 \quad (19.55)$$

The objective function of the model is to minimize the distance gap between $d_{i,m+1}$ and $d_{j,m+1}$

in interval $m + 1$, where $d_{i,m+1} = d_{i,m} + v_{i,m} \tau$ and $d_{j,m+1} = d_{j,m} + v_{j,m} \tau$, in order to achieve a direct

transfer. The constraints are of minimum and maximum speed limits.

Let $p = (d_{i,m} v_{i,m} s_{r i i} - d_{j,m} v_{j,m} s_{r j j}) / (v_{i,m} s_{r i i} - v_{j,m} s_{r j j})$. Certainly, the optimal solution for this

problem is $a + b = \lambda$. The rules for determining the values of a and b , to minimize TPTT,

are as follows:

if $p \leq \lambda$, $a = \lambda$; $b = 0$;
 $v_{i,m+1} = v_{i,m}$; $v_{j,m+1} = v_{j,m}$.

else $p > \lambda$ if $v_{i,m} s_{r i i} \leq V_{\max}$,
 $a = 0$; $b = \lambda$.

else $V_{\max} v_{i,m} s_{r i, m} s_{r i i i i} + - < \lambda$, a $V_{\max} v_{i,m+1} s_{r i, m} s_{r i i i i} = -$; b $V_{\max} v_{i,m+1} s_{r i, m} s_{r i i i i} = - - () \lambda$. end.

Case II : $d d_{i, m} s_{r j, m} s_{r i i j j} < :$ Similarly, let $\lambda' = - () - - \lambda \tau v v d d_{i, m} s_{r j, m} s_{r i, m} s_{r j, m} s_{r i i j j i i j j}$, and a' , b' are obtained by solving the following minimization programming problem:

min $\lambda' + \lambda' - \lambda' a b \lambda$ (19.56)

subject to $v a V_{\max} j, m s_{r j, m} s_{r j j j j} + \lambda' \leq +$ (19.57)
 $v b V_{\min} i, m s_{r i, m} s_{r i i i i} - \lambda' \geq +1 i$ (19.58)
 $\lambda' \geq a \ 0$ (19.59) $\lambda' \geq b \ 0$ (19.60)

The rules for determining the values of a' and b' , to minimize TPTT, are same as those of Case I.

Flexible road-segment encounter scenario: The update rules for average running speed

adjustment, for the flexible road-segment encounter scenario, when the distances $d_{i, m} s_{r i i}$ and

$d_{j, m}$

$s_{r j j}$ are not equal, are same as for the fixed single-point encounter scenario. However, when

the distances $d_{i, m} s_{r i i}$ and $d_{j, m} s_{r j j}$ are equal, the update rules for travel speed adjustment are a bit

different. New rules for average running speed update are given as follows:

if $d d_{i, m} s_{r j, m} s_{r s i i j j} = >$ if $v v v v v v_{i, m} s_{r j, m} s_{r i, m} s_{r i, m} s_{r i, m} s_{r j, m} s_{i i j j i i i i i i} j > = - - +$, $\max 1 1 2 r_{i, m} s_{r j i i} V_{\min} () \tau \tau \tau \tau \tau \tau + , , 1 v v v v V_{\min} i, m s_{r i, m} s_{r i, m} s_{r j, m} s_{r i, m} s_{r i i i i i i i j j i i} + + = - - () 1 1 1 2 \max , \tau \tau \tau \tau \tau \tau \tau \tau ; v v v v V_{\max} j, m s_{r j, m} s_{r i, m} s_{r j, m} s_{r j, m} s_{r j j j j i i j j j j} + + = + - () 1 1 1 2 \min , \tau \tau \tau \tau \tau \tau \tau \tau$. else if $v v , v v v v_{i, m} s_{r j, m} s_{r i, m} s_{r i, m} s_{r j, m} s_{r i, m} s_{i i j j i i i i j j i} < = + - +1 1 2 \min r_{i, m} s_{r i i i} V_{\max} () \tau \tau \tau \tau \tau \tau \tau + , , 1 v v v v V_{\max} i, m s_{r i, m} s_{r j, m} s_{r i, m} s_{r i, m} s_{r i i i i i j j i i i i} + + = + - () 1 1 1 2 \min , \tau \tau \tau \tau \tau \tau \tau \tau ; v v v v V_{\min} j, m s_{r j, m} s_{r i, m} s_{r j, m} s_{r j j j j j j i i j j} + + = - - () 1 1 1 2 \max , \tau \tau \tau \tau \tau \tau \tau$.

```

else v v i, m s r j, m s r i i j j = , v v i, m s r i, m s r i
i i i + = 1 ; v v j, m s r j, m s r j j j j + = 1 . end,

else d d d i, m s r j, m s r s i i j j = ≤ , v Vmax i, m s r
i, m s r i i i i + + = 1 1 ; v Vmax j, m s r j, m s r j j j j
+ + = 1 1 .end.

```

The change of scenario occurs when the distances of the two vehicles to the end of their

shared road segment are equal and less than the length of the shared road segment. In next

intervals, the vehicles will run by their maximum speeds in order to minimize the TPTT.

Holding vehicles at transfer point: Holding vehicles at transfer points can improve the

opportunity of direct transfers. However, it will increase the dwell time. It has both positive

and negative impacts on the TPTT. There is a trade-off between the number of transferring

and onboard passengers.

Evaluation of performance: Evaluation of the performance of the proposed distance-based

dynamic speed-adjustment model, IVC scheme based, takes place using two main measures:

(1) total number of simultaneous arrivals and (2) $\Delta TPTT_i$. The total number of simultaneous

arrivals of an observation time horizon can be obtained directly by comparing the arrival

times of vehicles at a transfer point.

For the i th vehicle, the $\Delta TPTT_i$ includes four elements: the change of in-vehicle passenger

travel time (ΔPTT_i), the change of passenger waiting time (ΔPWT_i), the change of passenger

transfer waiting time ($\Delta PTWT_i$), and the holding time at transfer points (H_i). The $\Delta TPTT_i$

where h_{rj} is the headway of route j

k_i is a n i n t e g e r v a r i a b l e t h a t i s u s e d t o m i n i m i z e t h e P T W T

The h o l d i n g t i m e a t t r a n s f e r p o i n t s (H_i) is d e f i n e d b y $H_{p e l i q s r i q s r i q s r N s i r i s i r i i i N s i r i s i r i i i N s i r i s i r i i i} = + - \dots$ (19.67)

By substituting Equations 19.65 and 19.66 into Equation 19.64, and combining Equations

19.62, 19.63, a n d 19.67, t h e $\Delta T P T T_i$ is attained.

19.6.4 Case study

The effectiveness of the optimization model and its potential for implementation is assessed by

a case study of a real-life transit network of Beijing, China. The transit network is illustrated

in Figure 19.20; it comprises three bus routes with five transfer points. Routes 658 and 694

run from north to south and route 728 runs from west to east. Transfer points 1, 2, 3, and

4 are fixed single-point encounter transfers and transfer point 5 is a road-segment encounter

transfer.

19.6.4.1 Data collection

Data of the three routes were collected from the Beijing Public Transport Holdings, Ltd. The

data include information of routes, vehicles, and passengers.

Route information: Comprises route ID, route direction, number of stops, stop ID, stop latitude and longitude, distance between stops, and preplanned timetable.

Vehicle information: All vehicles are equipped with GPS device and can share their realtime speed and location information with the control center. The information was updated using intervals of 30 s. The collected vehicle

information comprises vehicle ID, driver ID, departure and arrival times at each stop, planned headway, vehicle location, and average running speed.

Passenger information: Collected from integrated circuit (IC) cards and comprises the number of passengers boarding and alighting at each stop and the number of passengers transferring at each transfer stop.

19.6.5 Results

The optimization model described (Liu et al., 2014) was applied to the case study with a

comparison between the operation without using tactics and the operation with tactics.

The optimization procedure for updating average running speeds, for all the

transfer points, was implemented using MATLAB®. The original time-distance diagram, of

the case study, without using any operational tactics is depicted on the left-hand side of

Figure 19.21 at all the transfer points. The adjusted time-distance diagram, using

the operational tactics (model outcome), is shown on the right-hand side of Figure 19.21.

The interval of speed adjustment, of the case study, was 5 min. The average transfer

walking time was assumed to be 1 min. The arrival time difference t is 1 min; thus, when

this difference is equal or less 1 min, holding tactics are employed to increase the oppor

tunity of direct transfers. The results of holding times of vehicles on each route at each Transfer point 2

Transfer point 3 Transfer point 4 Transfer point 5 Route 658 Route 694 Route 728 Transfer point 1

Figure 19.20 A bus network of Beijing with three bus routes and five transfer points. (Note: Dark-coloured lines represent the road network.) (a) 08:00:00 08:10:00 08:20:00

08:30:00 0 0.5 1 1.5 Distance (km) 2 2.5 3 3.5 4
Time Without tactics 08:00:00 08:10:00 08:20:00 08:30:00 0
0.5 1 1.5 2 2.5 3 3.5 4 Time Distance (km) With
tactics Vehicle on route 658 Vehicle on route 694 Vehicle
on route 658 Vehicle on route 694 Direct transfer Direct
transfer Missed direct transfer 08:15:00 08:30:00 08:45:00
09:00:00 0 1 2 3 4 5 6 7 8 Time Without tactics 08:15:00
08:30:00 08:45:00 09:00:00 0 1 2 3 4 5 6 7 8 Time Distance (km) Distance (km) With tactics Vehicle
on route 658 Vehicle on route 694 Vehicle on route 658
Vehicle on route 694 Missed direct transfer Direct transfer
Direct transfer (b)

Figure 19.21 Time-distance diagrams of the five transfer
points with and without tactics: (a) transfer point 1,
(b) transfer point 2. (Continued) 08:30:00 08:45:00
09:00:00 09:15:00 0 2 4 6 8 10 12 Time Without tactics
08:30:00 08:45:00 09:00:00 09:15:00 0 2 4 6 8 10 12 Time Distance (km) Distance (km) With tactics
Vehicle on route 658 Vehicle on route 728 Vehicle on route
658 Vehicle on route 728 Direct transfer (c) Direct
transfer Missed direct transfer (d) 08:45:00 09:00:00
09:15:00 0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5 Time Distance (km) Without tactics Vehicle on route 694 Vehicle on
route 728 Direct transfer 08:45:00 09:00:00 09:15:00 0 0.5
1 1.5 2 2.5 3 3.5 4 4.5 5 Time Distance (km) With
tactics Vehicle on route 694 Vehicle on route 728 Direct
transfer Direct transfer Missed direct transfer

Figure 19.21 (Continued) Time-distance diagrams of the five
transfer points with and without tactics: (c) transfer
point 3, (d) transfer point 4. (Continued)

transfer points are shown in Table 19.9 and it ranges
between zero (simultaneous arrivals

of vehicles) and 47 s.

The results of the two main measures of comparison are shown in Table 19.10. These

results indicate that without using any operational tactics, replanned direct transfers

do not always occur. There is only one direct transfer that occurred at transfer point 4

for passengers transferring from route 694 to route 728. By implementing the opera

tional tactics (model outcome), the number of successful direct transfers becomes 12, an

increase of 100%. We note that a transfer point 5 (see Figure 19.21), which is a road

segment encounter transfer, simultaneous arrivals of vehicles occur at a shared road segment

ment and contribute largely to this significantly improved result. The TPTT with tactics

is 547,340 s and reduced by 83,236 s, which is a reduction of 13.2% compared with the

Figure 630,576 s related to without using tactics. No doubt that both results will make

the service more attractive and reliable.

These encouraging results demonstrate the potential benefits existing for public transit

systems with the implementation of IVC scheme and online operational tactics. 08:30:00 09:00:00 09:30:00 0 1 1.4 2 3 4 5 6 7 Time Without tactics Distance (km) Vehicle on route 658 Vehicle on route 694 08:45:00 09:00:00 09:15:00 0 0.5 1 1.4 1.5 2 2.5 3 3.5 4 Time Distance (km) With tactics Vehicle on route 658 Vehicle on route 694 Direct transfer Direct transfer Direct transfer without transfer walking Missed direct transfer (e)

Figure 19.21 (Continued) Time-distance diagrams of the five transfer points with and without tactics: (e) transfer point 5. Table 19.9 Holding times of vehicles on each route at each transfer point Transfer point Route 658 Route 694 Route 728 17 s 38 s – 23 s 4 s – 334 s – 10 s 4 – 47 s 0 s 50 s 0 s –

19.7 LITERATURE REVIEW AND FURTHER READING

Different classes and types of measures are employed at each level of the transit operations

planning process for providing integrated, coordinated, and seamless transit service. At the

network design stage, measures of network integration such as integrated physical connection of transfers, optimized layout of transfer centers, and flexible road segment transfers are first used to reduce transfer walking time (Chowdhury and Ceder, 2013). In order to improve the reliability of scheduled transit service, various kinds of timetable strategies, such as creating maximal synchronized timetable (Domschke, 1989; Voss, 1992; Ceder et al., 2001), timed-transfer systems (Abkowitz et al., 1987; Maxwell, 1999), optimal slack time (Zhao et al., 2006), and time control point (Yan et al., 2012), are employed to improve the simultaneous arrival of vehicles. However, in practice, because of some uncertain and unexpected factors, such as traffic disturbances and disruptions, inaccurate transit driver behavior and actions, and random passenger demands, replanned scheduled transit service is not always materialized, especially the synchronized transfer service. Missed connection of transfer service will not only frustrate current passengers but also suffer a loss of potential new users. Therefore, it is important to control the randomness existing in the movement of transit vehicles.

Osuna and Newell (1972) discussed the holding strategy at a single service point. They considered an idealized transit system of one or two vehicles and formulated the optimal

control process as a DP problem. The objective is to minimize the average waiting time of

each passenger. Barnett (1974) also considered holding vehicles at a chosen control stop,

but with two objectives: to minimize both the average passenger waiting time and the control

stop and average in-vehicle passenger delay; the author provided a simple algorithm for constructing a

near-optimal dispatch strategy. Wirasinghe and Liu (1995) investigated

the determination of the number and location of time control points using DP. Eberlein

et al. (1999, 2001) studied the deadheading, expressing, and holding strategies both singly

and in combination under the assumption that real-time information is available. They

formulated the holding problem as a quadratic mathematical programming model using

heuristic algorithms to solve it. Dessouky et al. (1999) showed the potential benefits of

real-time control of timed transfers using ITS technologies. Dessouky et al. (2003) exam

ined simulated systems in which holding and dispatching strategies are used; they also

investigated the dependency between system performance and available technologies. The

analysis of Dessouky et al. (2003) addresses even different holding strategy combinations

for allowing transfers, each of which differs in its definition of holding and in its approach

to be used for connecting buses. The results show that advanced technology is most advan

tageous when there are many connecting buses;
the schedule slack is then lost to zero, and

the headway is large.

Hickman (2001) developed an analytic stochastic model and formulated the holding

problem as a control problem as a convex quadratic program. Sun and Hickman (2005) formu

lated the real-time skip-stop strategy as a nonlinear integer programming problem using Table 19.10 Results of the number of direct transfers and change in total passenger travel time without and with tactics Without tactics With tactics Number of direct transfers 1 12 Change of total passenger travel time 0 -83236 s

an exhaustive search method to solve it. Sun and Hickman (2008) further investigated

holding vehicles at multiple holding stations. They also formulated the problem as a con

vex quadratic program with convex objective function and linear constraints. Fuetal.

(2003) proposed a pairing dynamic-operation strategy in which the following vehicle, of

a pair of vehicles, was allowed to skip some stations. Zolfaghari et al. (2004) studied the

holding strategy with real-time bus-location information proposing the use of a simulated

annealing-based heuristic algorithm as a solution method. Yu and Yang (2009) investigated

the holding strategy considering the prediction of next stop departure time; in their study,

a support-vector machine is used to predict the departure time using genetic algorithm to

optimize the holding time.

Cats et al. (2011) proposed a dynamic transit simulation model, BusMezzo, to inves

investigate the holding strategy considering the interaction between passenger activity, transit

operations, and traffic dynamics. Delgado et al. (2012) investigated the holding and limited

boarding strategies; the authors used a mathematical model without binary variables for

real-time transit operations. Another way to examine the impact of control strategies on

transit operations is through a dynamic-control approach; it is widely used for situations

with uncertain dynamic characteristics. Daganzo (2009) and Daganzo and Pilachowski

(2011) used a adaptive-control approach to alleviate the irregular meetings of buses (bus

unching) at nontransfer stops. Xuan et al. (2011) further extended the dynamic-control

approach using a family of dynamic holding strategies to improve bus schedule adherence

and headway regularity.

Additional control-based approach to deal with the uncertainty of transit-vehicle move

ments, road-traffic disturbances, and fluctuations of passenger demand is predictive con

trol (PC). Deschutter et al. (2002) proposed a model PC (MPC) framework to coordinate

transfers of railway systems. The MPC was transferred to a nonlinear, nonconvex optima

tion problem. Negenborn et al. (2008) investigated the control of large-scale transportation

networks using multiagent MPC scheme; two schemes, parallel and serial schemes, were

studied and compared. The authors concluded that serial scheme performed better than

parallel scheme with regard to the convergence speed and the quality of the solution. Sáez

et al. (2012) developed a hybrid PC (HPC) framework to investigate the real-time control of

transit systems using dynamic-control strategies and online information of system behavior;

this is to reduce the total in-vehicle travel time and out-of-vehicle waiting time. The authors

proposed genetic algorithm to solve the optimization problem. Muñoz et al. (2013) further

compared the performance of the HPC and deterministic-optimization approaches under

different scenarios. The authors concluded that the HPC approach performs better than the

deterministic-optimization approach when vehicle capacity could not be reached frequently

along the route.

The rapid development of information and communication technology allows for new

real-time based modeling to attain direct/synchronized transfers in transit operations.

These modeling techniques can make the use of operational tactics (Ceder, 2007) of

lowering the methodologies and examples used in this chapter. The term “tactics” in this

chapter refers to real-time control actions of transit vehicles, usually taken place in a

short-time manner. This is in comparison with the term “strategies,” which is usually

referring to a wider perspective at the transit-network level. Recent studies by Hadas and

Ceder (2008a,b,c, 2010), Ceder et al. (2013), Liu et al. (2014), and Nesheli and Ceder

(2014) showed that by applying selected operational tactics (holding, skip-stop/segment,

speed change, short turn, flexible route (counter), the frequency of simultaneous

meeting occurrence between two or more transit vehicles can be significantly improved

along with a reduction of the TPTT.

The major characteristics of the models/methods reviewed are summarized in Table 19.11.

Table 19.11 Summary of the characteristics of the methods reviewed

Source Approach Strategies/tactics Comments

Osuna and Newell

(1972) Dynamic programming Holding An idealized transit system

Barnett (1974) Mathematical programming Holding A simple model

Wirasinghe and Liu

(1995) Dynamic programming Holding The determination of the number and location of time control points

Eberlein et al.

(1999, 2001) Quadratic programming Deadheading, expressing, and holding Real-time information available

Hickman (2001) Convex quadratic programming Holding An analytical and stochastic model

Dessouky et al.

(1999, 2003) Simulation Holding Real-time control using ITS

De Schutter et al.

(2002) Model predictive control Holding Transfers at railway systems

Fu et al. (2003) Nonlinear integer programming Skip-stop A pairing dynamicoperation strategy

Zolfaghari et al.

(2004) Mathematical modeling Holding Simulated annealing-based heuristic algorithm

Sun and Hickman

(2005) Nonlinear integer programming Skip-stop Exhaustive search method

Yu and Yang (2009) Mathematical modeling Holding Consider the prediction of next stop departure time

Sun and Hickman

(2008) Convex quadratic programming Holding Multiple holding stations

Hadas and Ceder

(2008a,b,c) Simulation Flexible route segment transfer

Hadas and Ceder

(2010) Dynamic programming Various operational tactics

Daganzo (2009),

Daganzo and

Pilachowski (2011) Adaptive control Speed change

Xuan et al. (2011) Adaptive control Holding

Cats et al. (2011) Dynamic simulation Holding BusMezzo

Delgado et al.

(2012) Mathematical programming Holding and limited boarding

Sáez et al. (2012) Hybrid predictive control Holding

Muñoz et al. (2013) Hybrid predictive control and

deterministic optimization Holding

Ceder et al. (2013) Modeling and simulation Holding,
skip-stop

Nesheli and Ceder

(2014) Modeling optimization and simulation Holding,
skip-stop/ segment Real-time control

Liu et al. (2014) Modeling and optimization
Vehicle-to-vehicle communication, speed change, holding
Real-time control

EXERCISE

Given is a small IVC system depicted graphically in Figure 19.17 with two transit routes,

routes 1 and 2. Suppose that the information about the number of passengers boarding,

alighting, and transferring at each transit stop, together with the running time in each route

section is known by using IC cards and GPS data; this information is shown in the following

table of input data. The initial number of onboard passengers of route 1 is 40. The number

of passengers boarding at stop 1B (see Figure 19.17) is 10 and alighting is 5. Thus, the

number of onboard passengers is 45 at stop 1B. The number of passengers boarding at stop

1C is 10 and alighting is 26 with 20 passengers wishing to transfer from route 1 to route 2.

Input data and calculated total passenger travel time of the example of the exercise

Route	1	1B	1C*	PTT (min)	PTWT (min)	TPTT (min)	10↑	5↓	10↑
26↓	5↖	5←	6↗	20→					

N1(pass) 40 45 29

T1(min) 0 10 15

Route 2 2A 2B* 5↑ 0↓ 25↑ 15↓ 5↖ 20← 10↘ 5→

N2(pass) 20 25 35

T2(min) 0 20 20

Note: ↑ = boarding, ↓ = alighting, ↖ = pure boarding, ← = transferring boarding, ↘ = pure alighting, → = transferring alighting, * = transfer point indicator.

(a) Draw the time-based load profile of the example for the condition without using any tactics. Calculate the passenger travel time (PTT) of routes 1 and 2, P TWT, and T PTT.

(b) Using the IVC system, the average running time of the first section of route 1 is changed from 10 to 15 min; the speed of the second section is unchanged. On route 2, the average running time of the two sections changes from 20 to 15 min, say because of changes of road traffic condition. Draw the new time-based load profile of the example using speed change tactic. Calculate the posttactic PTT of routes 1 and 2, P TWT, and T PTT.

(c) Calculate the Δ PTWT and Δ TPTT.

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20 Chapter 20: Future Developments in Transit Operations

Albert Einstein once said: “I never think of the future. It comes soon enough.” Indeed, the rate

at which the future seems to be arriving provides a good reason to plan and design ahead; this

should be an integral part of any routine transit operations planning undertaking. In this last

chapter, we will attempt to touch the near future while being fully aware that choice, and not

chance, determines destiny. Our times, especially the past decade, have known extreme revolutions, especially in commu

nications, and these have brought about a whole new culture, including changes in language and

behavior. The Internet and the cellular phone have altered the way we live. These two revolu

tions have far-reaching consequences, especially for the rapid development of technologies sup

porting them: fast switching, self-healing distributed wireless and mobile networks, proximity

and location technologies, nanotechnologies, motion control, mobile and miniscule power sup

plies, new user interfaces, and user experience. All these developments are gradually trickling

into the field of transit service. This chapter starts with an introductory section describing a new (future) concept of

automatic-passenger transfers along road segments rather than at transit stops and then dis

cusses technological issues in transit-system automation. In fact the novel transfer concept

described in the introduction fits some of the methodologies presented in Chapter 19. Section

20.2 moves to the technology perspective, describing

existing automated transit systems and

proposed future concepts, such as the dual-mode personal rapid transit system. Section 20.3

describes an innovative two-way elevated transit system. Section 20.4 provides a selected litera

ture review, and Section 20.5 ends the book with a concluding remark. Essentially, practitioners are encouraged to visit all sections of this chapter about possible future

development. The chapter presents challenges for devising alternative future transit solutions. Finally, one piece of advice for practitioners insofar as future developments are as follows:

(1) future solutions to existing transit problems are, in fact, existing solutions (some of which

may be found in this book) to future problems; (2) whatever you do, try to act like an optimistic

scientist, who will never cry over spilled milk, because 88% of it is just water in any case.

20.1 INTRODUCTION AND NEW DIRECT-TRANSFER CONCEPT

Niels Bohr famously cautioned: "Prediction is very difficult, especially of the future."

Nonetheless, future developments in transit operations are already relected in a vast

amount of articles. Although we cannot change the direction of the wind (evolution of life

styles, land-use patterns), we can adjust the sails (create attractive transit systems that will

naturally shift people from the automobile to public transit vehicles). This sail-adjustment

realization relies heavily on the ongoing development of new technologies.

Innovative technologies in transit operations have three major objectives:

- To increase the productivity of transit operations, particularly through the introduction of automation
- To improve safety, performance, and service capabilities and to achieve this in a cost-effective manner
- To support transit-related priorities at the national level, such as energy conservation, safety, central city revitalization, and environmental protection

Essentially, new technologies should pursue the goal of a prudent, well-connected transit

system as defined in Section 13.2: A n advanced, attractive transit system that operates

reliably and relatively rapidly, with smooth (ease of) synchronized transfers, part of the

door-to-door passenger chain. Interpretation of each component of this definition appears

in Section 13.2. What follows are introductory remarks of a new synchronized transfer

concept, a section on transit automation, a section on a two-way elevated transit system,

a section on selected literature review, and a closing remark.

20.1.1 New concept to reduce the uncertainty of synchronized transfers

Advanced transit systems, such as dial-a-ride (e.g., Dial, 1995) or other demand-responsive

systems, are not widely used and have never replaced existing conventional transit systems.

The flexible routing and scheduling approach can be combined into a new concept consisting

of elements suitable for a mass transit system. This new concept has the following main

characteristics: (a) the use of a multiagent s

ystem (MAS); (b) r eal-time, m ultilegged t rip

planning, based on each passenger's attributes; (c) transit stops independent of synchronized

t ransfers; a nd (d) d eployment o f o perational t actics (hold, sk ip s top, e tc.) t o o vercome t he

s tochastic nature of the transit system.

Models t hat t reat t he s tochastic n ature o f t he t ransit s ystem a pparently w ere n ot

d eveloped s peci cally f or p ublic t ransit, b ut f or v ehicle-routing m odels (e.g., G endreau

et al., 1996; Ghiani et al., 2003). A public transit model that uses variable demand was

d eveloped b y C hien e t al. (2001), b ut i t d oes n ot c ontain t he c ombination o f v ariable

t ravel t ime, p assenger d emand, a nd t ransfer p robability t hat w ill b e p resented i n t his

c hapter.

Following t he d escription o f s ynchronized t ransfers i n C hapter 1 9 a nd t he r eliability

problems in Chapter 18, the reader will recognize the uncertainty of a planned simultaneous

a rrival o f t wo v ehicles a t a n e xisting s top. I n o rder t o a lleviate t his u ncertainty o f s imultane

ous a rrivals, S ection 1 9.3 i ntroduces a n ew c oncept: t he e xtension o f t he c ommonly u sed

s ingle-point e ncounter (at a s ingle t ransit s top) t o a r oad-segment e ncounter (any p oint a long

t he r oad c onstitutes a p ossible e ncounter p oint). S uch a c hange i n c oncept w ill r educe t he

uncertainty of meeting at a certain point and will enable

more flexibility in deploying the

operational tactics of Chapter 19.

The new concept, which relies on online information, allows two vehicles to meet in time

and space so as to eliminate the need for waiting time at transfer points. Moreover, if we do

not permit conventional transit systems to obstruct our imagination, the possibility exists of

physically arranging for two vehicles that arrive at the meeting point to align together either

longitudinally or side-by-side in order to allow passengers an easy transfer. This is like the

meeting of satellites in space. Such a perfect transfer arrangement is illustrated schematically

in Figure 20.1.

Realization of the new concept can be based on MAS as described in Section 19.2. In this

section, it is mentioned that MAS is defined as collections of autonomous agents within an

environment that interact with each other for achieving internal and/or global objectives

(Van Dyke Parunak, 1997). Moreover, Minsky (1986) argued that an intelligent system

could emerge from non-intelligent parts. His definition of the Society of Mind makes use

of small, simple processes, called agents, each of which performs some simple action, and

the combination of all these agents forms an intelligent society (system). Zhao et al. (2003)

developed MAS for a bus-holding algorithm. The

authors treat each bus stop as a node;

the agents negotiate with each other, based on marginal cost calculations, to devise minimal

passenger waiting-time costs.

Last, as we noticed in Chapter 15, designing a public transit system produces conflicts

among the parties involved. The agency will usually prefer a small number of routes with

transfer centers in order to reduce fleet size. The passenger will prefer more routes and

fewer transfers because transfers naturally decrease comfort and reliability. This conflict

may intensify in areas with variable and sparse demand. The solution is to combine the Direction of travel Bus 1 Bus 2 ON-LI.. (a) Direction of travel ON-LINE VMS ON-LINE VMS Bus 1 Bus 2 Direction of travel (b) ON-LINE VMS

Figure 20.1 Schematic illustration of two buses aligning together longitudinally in part (a) and side-by-side in part (b), each allowing an easy transfer for passengers.

advantages of a transfer-based system (lower costs) with models that increase the comfort of

transfers and reduce waiting and travel time. Table 20.1 summarizes these options.

20.1.2 Transit-system automation

Emerging technologies, both those being implemented now and those to be implemented

in the near future, are surrounding us at an increased pace. The Internet and mobile phone

revolutions are creating a technological breakthrough from tradition that is affecting every

aspect of life; it is a revolution that cannot be stopped and so affects transit operations, as

well. One of the basic claims that is frequently made to delay or reject the introduction of a

new technology is this: "show us where these new systems are running in the world." The

answer to this question will be given shortly, once these systems are introduced. The revolution

is happening, and it is essential in all fields of conveyance: baggage and cargo handling,

elevator and walkway systems, industrial conveyors, and of course mass transit. The question

for what new technologies are searching is echoed in the advertising of many large logistics

companies, "We take Anything, Anywhere, Anytime." Even the European Union Voyager

project (EURport, 2005), the main research project on the future of transportation, con-

cluded that what is needed and what the public demands is a high-quality, door-to-door

mobility service.

The automation revolution has been realized in the past few years, especially in the

fields of computers, telephones, and mobile networking. The automation revolution has

brought technology to the point where creating an automated transit system is a work-

able option. Rosenbloom and Fielding (1998) in Transit Cooperative Research Program

Report 28 state that single-occupant drivership can only be reduced with reverse-transit

solutions, services to employers or universities, pooling incentives, and restructuring routes

ing and feeder services. These changes are actually being implemented with the aid of the

new technologies.

Another important aspect of transit-system automation is the development of special

information technologies. Two technologies deserve attention: radio frequency identification

(RFID) and proximity information. RFID utilizes a radio frequency transmitter and

receiver to identify items and their location. Usually, these are extremely small, low-cost

devices that use very little or no power and draw their needed power from the air; such

devices can be used to identify transit-vehicle locations and driving paths, prevent theft

of goods, and enable cashier-less supermarkets.

Proximity information, termed the location revolution, is a technology that started

around 2001 for finding locations, for instance, giving correct route and map directions,

creating personalized advertisements according to the types of vehicles or mobile phones

passing by, and even locating lost children. Together, these two technologies are paving the

way to produce automated transit vehicles and driverless systems, as well as online tactics

to improve transit reliability.

Table 20.1 Perception of transit options, by party Party

Option Passenger Agency Authority

More routes Better service More expensive Better coverage

More transfers Reduced comfort and reliability Less expensive More user complaints

Smart transfers Reduced waiting and travel time Less expensive and increased ridership Increased use of public transit

20.2 DEVELOPMENTS IN TRANSIT AUTOMATION

Several automatic transit systems have already been implemented, using such technologies

as magnetic location, distributed networking, and linear induction motors. The most common,

well-known automated transit systems are automated train operations (ATO), auto-

mated people movers (APM), elevated rail systems for automatic guided vehicle (AGV),

group rapid transit (GRT), and personal rapid transit (PRT).

Examples of known ATO systems are the Paris Metro, Bay Area Rapid Transit (BART,

California), Metrorail (Washington, D C), and Mass Transit Railway (MTR, Hong Kong).

Known APM systems are implemented especially in airports, such as John F. Kennedy

(JFK), Newark, Düsseldorf, Frankfurt, and Schiphol in Amsterdam. APM systems are also

available in light-rail metro systems in Toronto, Vancouver, Detroit, and New York.

20.2.1 Elevated rail systems

Elevated monorails have been in use for more than 120 years, or since 1890, with not one

known fatality. Outside the city, elevating s

systems benefit nature by not creating a new
ecologi

cal barrier. An elevated system cannot crash with cars; it consumes less land, and it is not

susceptible to traffic congestion. In some noted cases, elevated rail was the only system available

to work after heavy earthquakes that brought down bridges and freeways (e.g., Red Line

California 1994, Osaka 1999, Seattle 2001).

In urban areas, the problem with elevated rail systems usually concerns the right-of-way

for residents who live on floors at the level of travel. Certainly, this is not a problem at air

ports or business districts. Heavy monorails are a solution as still extremely expensive and

once installed have further difficulties.

The advanced systems—APM, AGV, and PRT—constantly failed to be realized over the

past 30 years. However, some progress has recently been made in that direction. In London,

Heathrow Airport has ordered a limited system that employs the Urban Light Transport

(ULTRA) concept, a PRT-prototype system developed in Wales for the European Union,

to take passengers from the long-term parking lot to the main airport terminal. In South

Korea, a system that evolved from Raytheon's PRT is being installed. At Uppsala University

in Stockholm, Sweden, the European Union is finally building a large test track for an auto

matic system, code-named Vectra. And in Dubai City, UAE, there is now a tender for PRT

to work a large-scale city and hotel loop. Furthermore, a research project is in progress in

Germany for an automatic urban railcar system (to operate on the NeoVAL system). Figure

20.2 illustrates several PRT systems and one AGV system related to these projects.

20.2.2 Elevator-PRT and dual-mode concepts

Often, a PRT system is seen as a limited solution because it does not take into consideration

heavy cargo or large groups of people, such as big families or school classes. In some city

transportation committee meetings that are debating light-rail solutions, PRT is usually

brought up to show how futuristic a solution it is and then discarded (PRT, 2004). One

possible, and faster, way to introduce PRT systems may be via the elevator industry, which

has been searching for a true door-to-door elevator, in other words, an elevator that can go

between buildings and reach parking lots. The joint elevator-PRT concept has been exhib

ited in several futuristic movies. Figure 20.3 illustrates such an example as visualized for the

movie Minority Report.

Finally, there is a dual-mode concept suggested by Reynolds (2006). This consists of envi

ronmentally clean transit vehicles and automobiles that use guideways at different speeds

(location based) as one (main) mode and ordinary driving as a second mode. The driving on

the guideways is done by an automatic computer-controlled system. Reynolds justified his

concept with the following argument: A hundred years ago most rural people got water into their houses by carrying it in buckets from a well in the yard. Those old buckets delivered very little water compared to modern plumbing, since a pipe can provide continuous flow. The same observation applies to transportation: Buses and trains are like one-at-a-time buckets. Cars are supposed to run or flow continuously on the highways, and they will flow continuously on the guideways. The big spaces we leave between cars on the highways are necessary because of the unsynchronized traffic, the limitations of human drivers, and the limitations of automobile braking and tire traction.

Figure 20.2 PRT (upper illustrations) and AGV systems.

20.3 EXAMPLE OF A TWO-WAY MONOBEAM ELEVATED TRANSIT

This section describes, as an example, an elevated small-group automated rapid transit

(ESGART) labeled as SkyCabs, designed in New Zealand (Chapman et al., 2011). SkyCabs

comprises lightweight, eight-seater, automated cabs that travel up to 80 km/h on a two

way, elevated monobeam (Chapman et al., 2011). The system is designed to run on an

on-demand basis with high frequencies resulting in fast, pollution-free, unimpeded travel.

Artistic impressions and schematics of SkyCabs are illustrated in Figures 20.4 and 20.5.

Chapman et al. (2011) provide in Table 20.2 a comparison between SkyCabs and other

modes concerning different characteristics. Most characteristics and values for other modes

Figure 20.3 Joint elevator-PRT concept envisioned for the movie, *Minority Report*.

Figure 20.4 Artist's impression of SkyCabs.

Table 20.2 SkyCabs comparison to other passenger travel modes Part busway and part bus lane Light rail Standard monorail PRT SkyCabs (ESGART)

Cost per km of

two-way track US \$15-\$25 million US \$31-\$43.5 million
Varies to US \$62 million US \$18.6- \$31 million Below US
\$20 million

Cost of stations

and vehicles Not included Included Included Included
Included

Distance

between

stations 200-2000 m off-lane 200-500 m In-line 1000-1500 m
In-line 500-1000 m Off-line 750 m Off-line

Speed of

construction Two opposing on ground lanes by motorway Two
on ground tracks in road Two large elevated tracks
required Two small elevated tracks required Single small
two-way elevated beam over footpath

Average speed

of travel (km/h) 30 24 42 42 60

Capacity seated 2400 people/h/ direction at 1 min
intervals 3000 people/h/ direction 3000 people/h/
direction 1200-4800 people/h/ direction 4800 people/h/
direction

Capacity seated

and standing Up to 7200/h crammed at 1 min intervals
6000/h, up to 9000/h crammed 6000/h, up to 9000/h
crammed 1200-4800 people/h/ direction 9000 people/h/
direction

Peak time

service 2-10 min (6-30 veh/h) 2-10 min (6-30 veh/h)
4-6 min wait (10-15 veh/h) Time to arrange passengers
30 s to max 4 min (15-120 veh/h)

Off-peak service 15-30 min wait (2-4 veh/h) 12 min wait
(5 veh/h) 12 min wait (5 veh/h) No waiting 30 s to max
4 min (15-120 veh/h) Luggage Bogie Bogie

Figure 20.5 Elevation view of SkyCabs.

are from Carnegie et al. (2007). The SkyCabs figures are
from design parameters and consul

tant and manufacturer estimates. Fiscally responsible
choice of public transit modes should

be governed by the selection of the most economic method of
adding capacity to arterial

roads to ensure congestion is significantly eased.

Chapman et al. (2011) examine the SkyCabs system
as a way to alleviate some of the

traffic problems of major cities around the world.
Architectural and engineering aspects of

eight-seater SkyCabs, including cab frequency, sta-
tions, and lines, are described in detail.

Furthermore, this study demonstrates initial con-
nectivity comparisons (using the transit con

nectivity measurement by various parameters and
components, as described in Chapter 13)

of two example lines within a SkyCabs network
to other comparable paths on the North

Shore busway in Auckland, New Zealand. These results
indicate favorable performance

of SkyCabs, mainly due to the faster, un-
impeded travel on elevated guideway and sh-
orter

waiting times and headways. The authors em-
phasize that future work should be carried out
ut

for a SkyCabs network within Auckland to determine the aspects of internal connectivity

and the connectivity of various SkyCabs routes to hubs/nodes of existing public passenger

transit systems.

Ceder et al. (2014a) describe the SkyCabs system concept based on three criteria. First,

a comparative analysis is undertaken by researching and comparing the system characteristics

of the SkyCabs system and other modes of transport available. Then, a computer

simulation has been carried out to investigate the operational feasibility of the system.

Finally, an economic analysis has been presented by calculating and comparing the ben

efit-cost ratios for each transport mode considered. The results obtained through model

elling and simulation exercises clearly illustrate that SkyCabs concept has significantly

lower dwell and wait times compared to other buses. Furthermore, it is revealed that the

expected cab occupancy levels are lower than their respective capacity levels for every

demand scenario tested. The results of the comparative analysis indicate that, overall,

SkyCabs concept possesses favorable system characteristics when compared to all other

modes of transport analyzed. Finally, the economic assessment indicates that SkyCabs is

the most economically viable option in comparison to PRT, light rail, and GRT systems.

The authors conclude that implementation of a SkyCabs system into a urban environment

ment with similar characteristics to that of Auckland's North Shore is feasible. As such,

SkyCabs concept offers a viable alternative to the traditional and other modern modes of

public transit within urban areas with similar economic, social, and geographical characteristics

to Auckland's North Shore.

Ceder et al. (2014b) continue to investigate the feasibility of implementing the SkyCabs

system to and from Auckland central business district and Auckland international airport

(in New Zealand) by examining four variables: different routes, different number of stops/

stations, different passenger-demand levels, and different number of cabs in the system. The

analysis made utilizes geographical information systems (GIS) and simulation tools for the

various scenarios considered. The comparative analysis involved obtaining information on

the following system characteristics:

- Maximum theoretical capacity
- Average travel speed
- Energy consumption
- Operation and maintenance (O&M) costs
- Average capital costs

Information on the same system characteristics was then determined by Ceder et al.

(2014b) by carrying out research for each of the following traditional and modern modes

of transport:

- Bus
- Light rail
- Private vehicles
- PRT
- GRT

As a basis for comparison, all the researched monetary values were converted to 2012 dollars

by applying an inflation factor. They were then converted to New Zealand dollars

(NZD) using the following exchange rates valid as of July 2012:

1NZD = 0.79 United States dollars

1NZD = 0.50 British pounds

1NZD = 0.78 Australian dollars

1NZD = 0.63 Euro

The results of the comparative analysis appear in Table 20.3. Generally speaking, it shows

that it is possible to assess the cost-benefit of alternative routes in terms of level of service

and rate of return on investment. Implementation of such a system is feasible because it has

competitive pricing, increases the public transit use, and has an attractive level of service.

Finally, the authors assert that due to the competitive fares and the sheer attractiveness of

this new system, it is likely to experience an increased public transit patronage.

Table 20.3 Results of the comparative analysis SkyCabs

Bus/bus lane Car/road lane Light rail PRT GRT

Capacity seated

(pass/h/dir) 4,800 3,000 2,160 3,600 2,400 7,200

Capacity

including

standees

(pass/h/dir) 9,600 4,800 2,160 6,000 2,400 9,000

Average travel

speed (km/h) 60 19.3 38.7 43 39 30

Energy

consumption

(MJ/pass-km) 0.7 3.15 2.38 2.08 0.55 2.16

Capital cost

(NZ\$/km) \$13,000,000 \$53,500,000 \$47,000,000 \$40,000,000
\$14,800,000 \$44,600,000

O&M costs

(NZ\$/

pass-km) \$0.33 \$0.63 \$0.29 \$0.48 \$0.35 \$0.39

20.4 LITERATURE REVIEW AND FURTHER READING

Recent years have seen intensive discussion in the literature about various issues relating

to transit systems of the future. Naturally, much of this discussion concerns technological

development; however, some conceptual issues, mode-specific issues, and others have

been analyzed, as well. This section reviews only a few concepts of these developments

including automated transit networks (ATN), PRT,

straddling buses, and some future trends

and advanced systems.

20.4.1 Automated transit networks

Larsen (2012) has presented a technical memorandum, to the Transportation and

Environment Committee, on the feasibility of a
new ATN system to provide a connection to the

airport from the city of San José (in California, United States). This report outlines
specific

technical and regulatory aspects of the
proposed ATN system. Potential connection points

between the airport and other locations within
the city have been discussed. Given that the

high-speed rail system is not expected to arrive in San
Jose until 2027, it has been concluded

that the ATN technology has merit and is worth continued
pursuit.

Recently, Furman et al. (2014) discuss and
present the concept of ATN in the context

of urban planning/design and public policy. Also,
follow-on projects on this concept have

been suggested for decision makers, in order
to move forward to sustainable urban trans

portation. The comprehensive report of Furman et al.
(2014) outlines and discusses unique

features of ATN, that is, (1) its ability to
provide direct origin to destination services
with

out the need for transfers or stops at
intermediate stations, (2) small vehicle size
that can

cater exclusively to an individual or a small

group, (3) demand-driven service functionality

(as opposed to fixed service schedules), (4) fully automated nature of service, and (5) other

aspects relating to vehicle guideways and stations within a fully connected network. The

authors assert that ATN implementations require close cooperation from many kinds of

local officials and most probably will be implemented through a carefully negotiated public

public-private partnership (PPP). Furthermore, the existing niche applications of ATN, such as

within the Heathrow Airport, have been presented and the extent to which the capabilities

of these existing technologies can be expanded has been discussed. Future work has been

recommended to identify the ultimate capabilities of high-speed/high-capacity ATN.

20.4.2 PRT system overview and operations

The history and a comprehensive overview of PRT systems have been presented by McDonald

(2013). This study traces the beginning PRT explorations all the way to the first implemen

tation for public use of the complete paradigm change for transit. The complex interrela

tionship between innovative thinking in technology, communication systems, hardware,

architecture, transportation, and urban design/planning spanning the globe has been discussed

with specific examples from various cities.

Raney et al. (2007) explore a number of system-design perspectives for new 47-station

PRT shuttle system for the Pleasanton's Hacienda business park (HBP) major activity center

and surrounding area. This system is deemed to significantly increase the attractiveness of

rapid transit by allowing carpools, vanpools, bicycles, and bus commuters. The PRT ser-

vice provides elevated, electric, demand-responsive, nonstop, medium capacity, no-wait,

30 mph service for the commuter's last 2 miles and services mid-day shopping and recreation

trips. The authors have investigated a number of design ideas for new transit-circulator

system, including horizontal mixed use; line-haul constraints; station placement; allocation

of stations to passengers; ideal real-estate characteristics; high-demand, multimodal hubs;

style choices; and new accessories. The subsequent insight gained from this research sug-

gests the need for detailed new technology product research on horizontal mixed use for

suburban resident customers. Furthermore, interview research followed by information

acceleration-style surveys is suggested as future work.

Schweizer et al. (2011) provide detailed comparisons and summaries of capacity estima-

tions and size characteristics of serial-type and sawtooth-type PRT stations. Station char-

acteristics have been modeled as a complex function by incorporating geometry, vehicle

dynamics, boarding strategies, and user behavior. Comparisons have been made between

the capacities and space efficiencies with various load assumptions, using analytical and

micro simulation techniques. The subsequent results of this study will be useful for selecting

the most suitable PRT station for a given space constraint and demand scenario.

Gustafsson (2009) describes the operational characteristics of Vectus, which is a t t he

forefront of PRT systems, currently operating in Korea and Sweden. A test track, which

has been built in Sweden for validation of the novel Vectus system and for obtaining safety

approval, has been considered in this study. This proposed system is able to transport a

large number of people by occupying a large number of vehicles running close to each

other—with headways of a round 3 s between vehicles. With such a short headway, line

capacities comparable to tram lines can be achieved by counteracting the limited vehicle

capacity by high-frequency services. The results obtained from the test track operations

have indicated that the Vectus concept for PRT fulfills all the required operating charac

teristics such as safety, reliability, and performance of the vehicles. This system is capable

of operating without any active components in the track to detect vehicle position or speed

therefore making the system more favorable in harsh conditions, in particular, with snow

and ice conditions.

Villagra et al. (2012) present a novel approach to planning smooth paths and speed profiles

for automated public transit vehicles in unstructured environments. The proposed approach

deals explicitly with efficiency and comfort of automated transit systems by developing a

planning algorithm. This algorithm comprises three layers: (1) an optimal local continuous

curvature planner for obstacle-free situations; (2) a global planner that finds intermedi-

ate points to connect the configuration space to the desired degree, taking obstacles into

account; and (3) a speed planner that uses the set of curves of the previous layer to compute

analytically a comfort-constrained profile of velocities and accelerations. This system will

therefore provide the public transit system with a priori knowledge on the shortest path

within a selected area that guarantees lateral accelerations and steering wheel speeds below

given preset thresholds. Furthermore, the speed profiler can utilize semantic information

from the path planner to set a continuous velocity reference that considers both bounds on

lateral and longitudinal accelerations consistent with comfort and also a bound on longitudi-

nal jerk. Real-life applications of the features mentioned earlier have been undertaken and

the results of the algorithms were found to be satisfactory in an automated public transit

vehicle when compared with real driving maneuvers performed by human drivers. Future

work has been proposed by the authors in order to include the path planner in an o-

verall

control scheme in which an adapted robust control algorithm can be used to track a closely

as possible both the planned path and the planned speed.

Muller et al. (2012) present a ride-sharing methodology that can be applied to any

PRT system with more than some 10 stations. The authors acknowledge that the typical PRT

systems with small driverless vehicles operating at larger than initially anticipated headways,

of around 30 or more seconds, have reduced capacity implications. Therefore, a ride-sharing

scheme associated with such a system, even if it requires some or most passengers to no longer

receive onstop service, can be effective in increasing the system capacity. A methodology

that is simple to implement with sufficient flexibility to accommodate changing demand in

each station (e.g., morning peak, off-peak, evening peak, late night) has been presented in

this study. Additional information and analysis have been undertaken to demonstrate the

overall ride-sharing methodology and its functionality. The results of the study indicate that

the overall-system capacity (in terms of passengers served) can be significantly improved

over a reasonably wide range of demand levels both between and at stations.

20.4.3 Straddling bus concept

Lee (2010) and Wassener and Deng (2010) describe the novel concept of straddling bus pro

posed in China. The straddling bus has been identified to have all of the advantages of bus

rapid transit (BRT) while offering a greener substitution for BRT and subway systems in the

future. The key innovation of this concept is that the proposed straddling buses run above

the other vehicles on the road and under the overpasses. Figure 20.6 illustrates this idea. The

straddling bus concept enables saving road space, one of key issues in urban transportation.

Furthermore, this concept provides an efficient service, because traffic delays, caused by other

vehicles on the road, can be avoided by the straddling bus. Preliminary findings have revealed

that this concept can reduce up to 25%-30% traffic jams on main routes. Operating at an

average speed of 40 km/h, each bus can provide high capacity of up to 1200 people at a time.

In comparison with other four main types of public transit modes in China, subway,

light-rail train, BRT, and regular buses, the proposed straddling bus concept offers major

advantages in a number of aspects: reductions in road space requirements, pollution levels,

noise levels, traffic congestion, and delays. Another key strength offered by the straddling

bus concept is its relatively short construction life cycle. It is estimated that within 1 year,

up to 40 km of the system can be constructed, compared to a 3-year construction period for

a 40 km long subway system. Furthermore, these buses will not require allocated bus stop

spaces such as those required by regular buses. They can be parked at the stops, without

affecting the passage of other vehicles on the road.

Construction of straddling buses involves two key steps:
First, the roads will require

remodeling by means of either laying rails on either side
or creating an auto pilot tech

nology for the buses that could simply follow a line
marking along the road pavement.

The second step involves provision of station platforms for
passengers. This can either be

Figure 20.6 The straddling bus concept. (From Lee, A.,
"Straddling" bus—a cheaper, greener and faster alternative
to commute, Retrieved from

through the sides of the bus or through a built-in ladder
through the ceiling door. Given the

typical dimensions of 6 m width by 4–4.5 m height
of these buses, careful emergency exit

planning strategies must be considered. The pro-
posed emergency evacuation of passengers

will be similar to those found in a
airplanes—where a dedicated ladder will be used
to slide

down and away from the vehicle in case of
an emergency. The proposed bus system will be

powered by both municipal electricity and solar energy,
thus allowing significant saving

of the amounts of fuel per year and sub-
sequently a reduction of the carbon emissions
by a

substantial amount.

As of 2013, Beijing's Mentougou district is test-
ing this technology and has planned to

build a 186 km straddling bus system. At these
initial stages, the construction cost of a

single bus including 40 km of route facilities is estimated at approximately US \$7.4 million.

This cost represents approximately 10% of the cost of building a subway system of the

same length.

20.4.4 Future trends: Transit related

Cuddy et al. (2014) describe smart/connected cities and discuss their potential for interfacing

with the emerging connected transportation environment. The authors aim to provide

a well-defined foundation for the US Department of Transportation's (USDOT) Connected

Vehicle Program, in order to identify and exploit opportunities to help ensure that connected

vehicles and connected transportation fulfill their potential to improve safety, mobility,

and environmental outcomes in a complexly interdependent and multimodal environment.

Furthermore, in order to understand the potential benefits of such a concept, this report

explains in detail the definition of a smart/connected city. The intelligent infrastructure

utilized by smart cities, such as the devices and equipment that can sense the environment

and/or their own status, send data, and often, receive commands, along with their potential

transportation implications, has been reviewed and outlined.

Jarašūniene (2007) describes the main intelligent transport systems (ITS) enabling technologies

with a new emphasis on involving human factors

ctor experts at an early stage of design of

ITS equipment and facilities. The author asserts that ITS technologies, such as loop detectors,

tors, are well known to transportation professionals; however, there are a number of less

familiar technologies and system concepts that are keys to ITS functions. These ITS services

have been conceptualized as an information chain that includes data acquisition (from the

transportation system), communications, data processing, information distribution, and

information utilization (for decision and control support for the ITS users).

Velaga et al. (2009) have developed a real-time, weight-based topological map-matching

algorithm by improving the limitations of existing algorithms. The main novel features of

this enhanced topological map-matching algorithm include (1) the selection of candidate

links in the initial map-matching process and the map matching at junctions; (2) the intro

duction of two additional weight parameters, connectivity and turn restriction; (3) use of

an optimization process to derive the relative importance of weights using data collected in

different operational environments; and (4) the implementation of two consistency checks to

reduce mismatches. The proposed improved map-matching algorithm has been tested using

real-world data collected in different operational environments. The results indicate

that all of the novel features incorporated w

within the proposed algorithm have contributed

to the improved performance, in that this algorithm has identified 96.36% of the road seg-

ments correctly in an urban area and 97.01% of the road segments with a horizontal accu-

racy of 8.9 m (95%) in a suburban area. Future work has been proposed to investigate the

optimization of weights using more positioning data from each operational environment to

ensure that these optimal values are transferable.

Biem et al. (2010) describe the use of IBM InfoSphere Streams, a component-based dis-

tributed stream processing platform, for tackling the challenges of scalability, extensibility,

and user interaction in the domain of intelligent transportation services. A prototype system

that generates dynamic, multifaceted views of transportation information for the city of

Stockholm, using real vehicle global positioning system (GPS) and road network data, has

been described in detail. This system is capable of continuously deriving current traf-

istics and provides useful value-added information such as shortest-time routes from real

time observed and inferred traffic conditions. The performance experiments undertaken

indicate that this system is scalable; thus, it can process over 120,000 incoming GPS points

per second, combine with a map containing over 600,000 links, continuously generate dif-

ferent kinds of traffic statistics, and answer user queries.

20.4.5 Advanced systems categorized by technology

A considerable number of papers focus on new technologies that are presently used in vari-

ous transit systems. Several more recent examples follow.

20.4.5.1 World Wide Web

Lu and Shi (2007) have applied the theory of complex networks to classify public transit

networks into route networks, transfer networks, and bus-station networks. The practical

significance of each of the network parameters has subsequently been analyzed. Real-life

application of this concept has been undertaken for the transit networks in Langfang, Jinan,

and Dalian in China. The results obtained have indicated that transit networks possess the

features of complex networks. Furthermore, it was found that all the urban transportation

network parameters significantly affect the accessibility, convenience, and terrorist security

capability of the urban public transportation network.

Mallat et al. (2008) present the results obtained from a study of mobile ticketing service

adoption in transit. The theoretical underpinnings of this study involve technology adoption

and trust theories, which are complemented with concepts of mobile use context and

mobility. The empirical findings from surveyed data analysis indicate that compatibility of

the mobile ticketing service with consumer behavior

behavior is a major determinant of a doption.

Furthermore, the adoption decision was strongly dependent on mobility and contextual

factors such as budget constraints, availability of other alternatives, and time pressure in

the service use situation. The authors conclude that contextual and mobile service-specific

features are important determinants of mobile service adoption and should therefore be

incorporated into the traditional adoption models.

20.4.5.2 Global positioning system

Liao et al. (2007) have described the foundations and experimental validation of a hierarchical

model that can learn and infer a user's daily movements and use of different modes

of transportation. The model uses multiple levels of abstraction in order to bridge the gap

between raw GPS sensor measurements and high-level information such as a user's destination

and mode of transportation. In order to achieve high levels of efficiency in the inference

process, the authors have applied Rao-Blackwellized particle filters at multiple levels of the

model hierarchy. The results indicate that the proposed approach can provide predictions

of movements to distant goals. Furthermore, a simple and effective strategy for detecting

novel events that may indicate user errors (e.g., taking a wrong bus) has been proposed by

explicitly modeling activities in the context of the user's

historical data. Finally, the authors

discuss an application named Opportunity Knocks that employs the proposed techniques to

assist cognitively impaired people use public transit safely.

Mazloumi et al. (2009) gave a good example of the use of GPS for transit operations

analysis. The authors recognize that a major obstacle to researching the adverse impacts of

day-to-day transit service reliability is the lack of comprehensive data sets on bus travel times.

The authors address this issue, by utilizing a GPS data set for a bus route in Melbourne,

Australia. The nature and shape of travel time distributions for different departure time

windows at different times of the day along with the factors causing travel time variability

of public transit vehicles has been explored using a linear regression analysis. The results

indicate that in narrower departure time windows, travel time distributions are best characterized

by normal distributions. In contrast, for wider departure time windows, peak-hour

travel times follow normal distributions, while off-peak travel times follow lognormal distributions.

The contributing factors toward the variability of travel times were identified as

land use, route length, number of traffic signals, number of bus stops, and departure delay

relative to the scheduled departure time. Furthermore, the travel time variability for the

data set analyzed was found to be higher in the AM peak and lower in the off-peak periods.

20.4.5.3 Geographical information systems

Yigitcanlar et al. (2007) describe a comprehensive accessibility indexing model—the Land

use and public transport accessibility indexing model (LUPTAI). The authors demonstrate

that it is possible to produce a viable accessibility model, apply the model to a major urban

area, and produce a mappable accessibility index. A real-life application of this concept has

been demonstrated within the city of Gold Coast in Australia. The results indicate that this

proposed model is accurate, straightforward, and transparent enough to be understood by

the general public and is able to be customized and replicated.

Currie (2010) identifies spatial gaps in public transit provision for people who are socially

disadvantaged. This study outlines the research context for measurement of public transit

supply and needs and then describes the methodology developed for an application within

Melbourne, Australia. The techniques described are relatively easy to develop and apply

and can be powerful in identifying spatial gaps in transit service provision in an open and

defensible quantitative manner. The results of the application indicate significant gaps

between services supplied and social needs for transit services. Finally, the author has out

lined relevant implications of this study in transport-policy development.

20.4.5.4 Web-based geographical information systems

Hoar (2008) describes a web-based geographic information system that disseminates the

same schedule information through intuitive GIS techniques. A map-based interface has been

created, using data from Calgary, Canada, to show the users the routes, stops, and moving

buses simultaneously. Zoom and pan controls as well as satellite imagery functions have been

embedded to allow users to apply their personal knowledge about the local geography to

achieve faster and more pertinent transit results. The users can utilize this application to add

or remove the buses and respective stops interactively without the need to wait for responses

to web-based requests; this is achieved through a synchronous requests to web services.

Pun-Cheng (2012) describes a web-map public transit inquiry system. This system is

capable of providing bilingual (English and Chinese) information on all transit runs in Hong

Kong in the form of interactive maps and texts. Furthermore, the users can obtain real-time

derivations of optimal traveling routes in terms of multiple criteria, such as preferred mode,

least changes, shortest traveling time, or lowest fare.

20.4.5.5 Automatic vehicle location

Milkovits (2008) develops and implements preprocessing techniques that estimate a well

time model and analyzes the impact of the secondary factors. The data used for analysis have

been obtained from automatic-passenger counting, automatic-fare counting, and automatic

vehicle location systems installed on Chicago Transit Authority buses. The estimates derived

from the proposed model demonstrate that the advantage of smart media fare cards over

magnetic stripe fare tickets is about 1.5-2 s in time savings per passenger. This advantage,

however, is not observed when there are more passengers onboard than the seating capacity

of the bus. Therefore, it is concluded that the advantages of the new fare media technology

in improving bus operations occur mostly when service levels do not experience crowding.

Furthermore, the large data set obtained enable a comparison of the residuals across time

of day, stop, operator, and route to look for other significant relationships. Future work has

been proposed to extend the ordinary least-squares regression model to include distributions

to account for the impacts of operator, time of day, stop, and other factors.

Pangilinan et al. (2008) demonstrate how to use the real-time automatic vehicle loca-

tion (AVL) information to benefit service reliability on high-frequency bus routes. The

decision-making process on deployment of operations control strategies can be enhanced

by the knowledge on real-time headways and bus locations.
The Chicago (Illinois) Transit

Authority's real-time AVL pilot project for Route 20 Madison has been used as a case study

for the evaluation of the effectiveness of real-time AVL to improve reliability. A simulation

model of the route has been developed using archived AVL data to predict the effects on

service reliability when real-time AVL information is used in bus supervision. The experiment

was undertaken over a period of 1 week, in order to verify the model and to address

the feasibility and scalability of the system. From the results obtained, the authors conclude

that real-time AVL does indeed have great potential to improve service reliability. Service

restoration strategies previously impossible to execute are now feasible because of this new

information stream. Future work has been proposed by the authors in the area of network

wide implementation of the system, including the supervision communications structure

and manpower deployment questions.

20.4.6 Advanced systems categorized by purpose

Some works place the main emphasis not on the type of technology used but on its purpose.

The following is a review of important publications of this sort.

20.4.6.1 Passenger information

Dziekan and Kottenhoff (2007) provide a comprehensive framework of the possible effects

that dynamic a t-stop real-time information displays can have on public transit customers.

The seven main effects identified in this study are (1) reduced wait time; (2) positive psycho

logical factors, such as reduced uncertainty, increased ease of use, and a greater feeling of

security; (3) increased willingness to pay; (4) adjusted travel behavior such as better use of

wait time or more efficient traveling; (5) mode choice effects; (6) higher customer satisfaction

tion; and finally, (7) better image. Two case studies have been presented in this work. The

first study proves that perceived wait times can be reduced by 20% by employing a before-

after implementation evaluation study with questionnaires on a timeline. The second study

illustrates the effects of real-time displays on passenger behavior in the form of a adjusted

walking speeds, by using a behavior observation method in a subway station. The authors

propose future work in order to account for interrelations between the effects identified.

Caulfield and O'Mahony (2007) describe provision of public transit information to

the customers in Dublin, Ireland. Examination of both existing and potential methods of

accessing information, with particular focus on the implementation of various ITS applica

tions, has been undertaken. The authors have c

categorized the decision-making process of a

passenger, when undertaking a transit trip, into four key stages: (1) pretrip to destination,

(2) at-stop, (3) onboard, and (4) pretrip to origin (the return journey). Data on passenger

passengers in Dublin would like to see improvements in their public transit information systems

20.4.6.2 Data collection

Pelletier et al. (2011) provide a comprehensive review of key aspects of smart card data use

in the public transit context. First, the authors present and address the available advanced

technologies such as the hardware and information systems required in operating these

tools and privacy concerns and legal issues related to the dissemination of smart card data,

data storage, and encryption. Second, the various uses of these data have been described

at three distinct levels: strategic (long-term planning), tactical (service adjustments and

network development), and operational (ridership statistics and performance indicators).

Finally, in light of the comprehensive review, the authors have identified the following potential

challenges for smart card transit operators and researchers: (1) technological improve

ments, (2) data validation, (3) economic feasibility, (4) journey validation, (5) new modeling

approaches, and (6) new methods of analysis.

20.4.6.3 Bus priority

Liu et al. (2008) present an analytical approach for the design and evaluation of a transit signaling

nal priority (TSP) system with two fundamental TSP strategies: early-green and extended

green operations. Through graphical illustration, it has been demonstrated that the impact

of early green and extended green on bus and general traffic could be quantified using

D/D/1 queuing models if the issue of concern is the induced delay. A numerical application

has also been presented and the results indicate that the proposed method is promising in

that it requires very limited information and provides fairly reasonable results. Therefore,

this analytical approach can be used independently or as a supplementary tool to simulation

models for complicated situations.

Ma et al. (2010) test a different bus-priority method referred to as coordinated and con

ditional bus priority (CCBP). The proposed approach considers the bus arrival time at the

downstream intersection and strategies are derived for early buses. Coordinated, signalized

intersection groups are adopted as control objects. Buses are detected one or more cycles

before their arrival at the first intersection of the control object. The proposed CCBP includes

two kinds of priority strategies: increasing and decreasing bus delay. A model has been devel

oped to generate the optimal combination of p

priority strategies for intersection groups in

order to match the real delay of buses closely with the permitted delay defined by the bus

operation system. A field application has been presented to illustrate the CCBP approach and

to compare it with other two options: no priority and unconditional priority. The authors

conclude that by using the CCBP approach, significant reductions on bus delay and headway

deviations can be achieved. Furthermore, the results indicate that using the CCBP approach

causes only minor increases in total average delay to other traffic. The authors concluded

that this approach could be used to decrease bus delay deviation and enhance the reliability

of bus service without significantly affecting the delay of other motor vehicles.

20.5 CONCLUDING REMARK

This is the last section of the book, and thus it deserves a final word, which will be given

by way of symbolism. There is a Swedish proverb: "Worry often gives a small thing a big

shadow," about which Marie Curie said: "Nothing in life is to be feared. It is only to be

understood." Understanding the nature of transit operations problems is already halfway

toward reaching adequate solutions. Lastly, an African proverb cautions: "Smooth seas do

not make skillful sailors." Indeed, experiencing problems in transit operations provides a n

important lesson that, together with the solutions and thoughts proposed in this book, can

hopefully end with a significantly improved transit service.

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